IndianOil Corporation Limited
Marketing (HO), Contract Cell

Minutes of Pre-Bid Meeting: Tender No - HCC/LUBE-21/PT-75/2021-22 (E-TENDER ID: 2021_MKTHO_138954_1)

Ref: LP/EPC Chennai/PT/Pre-bid Date: 19.08.2021

Name of the Work: Design, Engineering, Procurement, and Construction (EPC) and commissioning of Mechanical works for Storage Tanks, Tank Farm Equipments, Process Equipments, Utility Equipments, Pipeline works, Ware House Management system, Filling Lines, Installation of Free issued Core Process Equipments, etc. including AMC for 5 Years for setting up of Grassroot Lube Complex at Villages Amullavoyal and Vaikkadu, Chennai.

Mode of Pre-bid Meeting: Through Video Conference

Date of Pre-Bid Meeting : 11.08.2021

List of Participants:

<table>
<tr><td>From IOCL</td><td colspan="2">1. Mr. Sanjoy Bhattacharjee, CGM(LP)
2. Ms. Sanghamitra Samal, GM(LP)
3. Mr. Prabhat Kumar, DGM(LP)
4. Mr. MM Kadam, CM(LP)
5. Mr. Sabyasachi Roy, CM(Contracts)</td></tr>
<tr><td>Consultant PEC Ltd</td><td colspan="2">1. Mr. Lim Teng Sam
2. Mr. Gopalakrishnan
3. Mr. Annamalai
4. Mr. Sivakumar
5. Mr. Senthil Kumar
6. Mr. Vanchinathan
7. Mr. Dathatray Rao</td></tr>
<tr><td rowspan="7">Vendors</td><td colspan="2">Total 5 parties had attended the pre-bid meeting</td></tr>
<tr><td rowspan="2">1. M/s. G R Engineering Private Limited</td><td>Mr. Suresh Pillai</td></tr>
<tr><td>Mr. Prashant More</td></tr>
<tr><td>2. M/s. Offshore Infrastructures Limited</td><td>Mr. Mayank Singh</td></tr>
<tr><td>3. M/s. Afcons Infrastructures Limited</td><td>Mr. Yashwant Bhangale</td></tr>
<tr><td rowspan="2">4. M/s. Bridge and Roof Co. (India) Ltd</td><td>Mr. Souvik Kundu</td></tr>
<tr><td>Mr. Indradeep Das</td></tr>
<tr><td>5. M/s. Tuaman Engineering Limited</td><td>Mr. Dibyendu Saha</td></tr>
</table>

Response of IOCL to various queries raised by prospective bidders and additional clarifications are attached herewith. The clarification / revised requirement given in Pre-Bid Minutes supersede the requirement given in tender document.

Name of tendered job : Design, Engineering, Procurement, and Construction (EPC) and commissioning of Mechanical works for Storage Tanks, Tank Farm Equipments, Process Equipments, Utility Equipments, Pipeline works, Ware House Management system, Filling Lines, Installation of Free issued Core Process Equipments etc including AMC for 5 Years for setting up of Grassroot Lube Complex at Villages Amullavoyal and Vaikkadu, Chennai.

TENDER NO: HCC/LUBE-21/PT-75/2021-22

Tender ID: 2021_MKTHO_138954_1

Minutes of Pre-bid meeting held via VC on 11.08.2021 @ 11:00 hrs.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
A	**M/s. G R Engineering Private Limited, vide mail dtd 11.08.2021**			
1	General	Process plant	We understand that Core Process Equipment for Grease Plant, Lube Blending Plant, Bitumenous Lube Plant, BN Extract Plant, VII Disolution Plant, Rubber Extender Plant will be provided by Core Process EPC and not by Bidder, Further Process Guarantee will not be in Bidders Scope. Also kindly confirm all Process equipment for balance of Plant in Lube Complex, viz, Transformer Oil Plant, White Oil Plant Etc, will be in bidders scope. Also understand Process Guarantee for the foresaid Plant will not be in our scope.	Yes, Core Process Equipment's for Grease Plant, Lube Blending Plant, Bitumenous Lube Plant, BN Extract Plant, VII Disolution Plant, Rubber Extender Plant are excluded from Bidders Scope. However, Balance equipment as per Scope matrix provided in the tender document & as stipulated in BoQ / SoW for the above said plants will be in Bidders Scope. Process Guarantee for the above Core Process Plant is excluded from Bidders Scope. However, All Equipment's for balance of plant in Lube Complex, viz, Transformer Oil Plant, White Oil Plant, Coolant Plant, DEF Plant, Brake Fluid, etc., will be in Bidders Scope including its process gurantee.
2	General	Safety Studies	Please confirm HAZOP, HAZEN, SIL, other safety studies, etc as well PESO & OISD requirement of the layout will be in bidders scope.	HAZOP, HAZEN other safety studies, etc., will not be in Bidders scope. However, Bidder scope includes providing necessary assistance / documents to the HAZOP / HAZEN consultants appointed by IOCL / PMC. However, Bidders to follow PESO & OISD guidelines in respect of layout considerations.
3	General	Civil & Structural & Foundation	We understand that civil, strucutural & foundation will be in owners(IOCL) scope excluding that for OWS requirement. Incase any damage/repair to the foundation, civil works during erection /installation shall be taken care by IOCL.	Refer additional clarifications from Consultant & IOCL (as listed below). Minor civil works pertaining to foundations of some of the equipments as listed separately below will be in the scope of Mechanical Contractor.
4	General	Fabrication Yard / Land	Please confirm whether Graded and levelled land will be provided to the bidder, Also confirm Fabrication land with access will be provided by IOCL	Bidder understanding is correct. However Bidders are required to visit the site before bidding & submit the declaration of visit.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
5	General	Power, Water & Utilities	We understand Power, Water & Utilites will be in the scope of bidder. Please confirm	Bidder understanding is correct. Construction power through TNEB if available shall be provided to the bidder on chargable basis.
6	General	Appointment of TPI	We understand appointment of TPI for stagewise & inspection of boughtout will be in IOCL Scope. Please confirm	Yes. TPI (Third Party Inspectios) will be in IOCL Scope.
7	General	Review & approval of drawings and engineering	Please confirm whether IOCL will appoint Third Party for review & approval of Detailed engineering including Technical bid evaluation of all critical equipment/ system.	Detail Engineering deliverables from Bidder as well as Technical bid evaluation of all critical equipment / systems will reviewed by IOCL / PMC.
8	General	Pigging of Piping	Please confirm Pigging of Pipeline (all process & piggable lines) is in the scope of bidder.	Design, Supply of Piggable Pipelines,its fittings, related Equipments and Supervision of Erection & Installation works will be in CPE vendor scope. The mechanical package vendor appointed by IOCL shall erect/ Install the equipments supplied by process vendor. The scope is including preparation of Piping GA, Isometrics & obtaining approval from CPE (Process package contractor), fabrication & installation with necessary supports, painting & insulation, pre-commissioning, testing, flushing, cleaning & commissioning under the Supervision of CPE Vendor. The PFD & P&IDs along with other details like physical properties, etc., are enclosed in the tender document. So the review, vetting, design and submission of final PFD, P&ID, etc., shall be in the scope of process vendor for the plants for which equipments are to be supplied by CPE (Proces Package Contractor), commensurating the proposed capacity & grades for each plant. Mechanical package vendor (The successful tenderer against subject tender) shall provide the routing, pipeline GA and Isometrics. These details provided by Mechanical package vendor shall be vetted by process package vendor so as to establish seamless integration. The Onus of ensuring capacity and grades remains with process package vendor for the plants for which equipments are to be supplied by CPE vendor for Grease Plant, Lube Blending Plant, Bitumenous Lube Plant, BN Extract Plant, VII Disolution Plant, Rubber Extender Plant.The Onus of ensuring capacity and grades remains with Mechanical package vendor for the plants for which equipments are to be supplied by Mechanical pkg vendor for Transformer Oil Plant, White Oil Plant, Coolant Plant, DEF Plant, Brake Fluid, etc., Mechanical package vendor shall submit PFD, P&ID, Detailed Engineering documents for the plants (Transformer oil, White oil, Diesel Exhaust Fluid, Coolant, Brake Fluid) for which equipment are to be supplied by mechanical package vendor in line with the respective items of BOQ/SOW. The scope of vendor includes submission of PFD, P&ID for these plants in entirety including input & finish tank, utility & facilities. For more details please refer to the scope matrix provided alongwith the techncial bid.
9	Tender notice (NIT) / 06 of 22 / Clause 14	Completion period	Request you to extend the completion period as 24 months from the 10th day of LOI/handover of site	No change. Tender conditions shall prevail.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
B	**M/s. Offshore Infrastructure Limited, vide mail dtd 11.08.2021**			
10	Book 3 OF 8 / Page no. 13 of 444 / Chapter-II / Clause 2.6.11	Obtain the necessary NOC and permits required for the installation of the facility	Please elaborate what type of NOCs and Permits we have to take?	These may include approvals/ permits for operation of equipments viz. Filling lines & Weighbridges (Department of Legal metrology - weights & measures), Air receiver of compressor, Pressure Vessel testing (Factory inspectors), Lifts & lifting equipments (from Factory Inspectors / CPWD / any other statutory bodies), etc.,
11	Book 3 OF 8 / Page no. 20 of 444 / Clause 5.05 (E) & Book 3 OF 8 / Page no. 211 of 444	3D modeling of entire plant including civil, structural, electrical, instruments, firefighting, HVAC, CPE, CFE ate in contractor scope. Also, we read the same scope in Page no :211 of 444 where you have mentioned the Civil /Structural group shall be responsible for modeling of civil/structural/architectural items based on actual design. The discipline members shall model their elements under the respective sites assigned to them.	Please elaborate the 3D modeling scope for Mechanical Contractor.	Please refer to Sr. No. 4.02 in SoW. Bidder Scope includes 3D Modeling for the entire Plant Facilities in E3D (AVEVA) Software. Please ignore the comment on Page no. 211 of 444; Civil, Structural & Architectural modeling will also be in the Scope of Bidder. Mechanical pkg vendor shall obtain required inputs from Civil pkg vendor.
12	Book 4 OF 8	Integrated Control system with provision for communicating with the Main Plant wide Control System (PCS) for Demand-for-Product (i.e. Product top up in Product Tank),	We have considered potential free contact for plant wide control system (PCS) for automatic pail line filling system.	Bidder understanding is correct.
13	Book 4 OF 8 / Page no. 241 / Clause 2.1 (point no. 6) & Book 4 OF 8 / Page no. 242 / Clause 2.2	Palletizer with option for low level palletizer – that can be shared with other 20L Pail line for DEF; Possibility to be examined Palletizer will be shared with other 20 lit pail line for lube Possibility to be examined;	Please confirm the Option because material & automation cost will be changes accordingly.	The option of common palletizer for Pail filling lines stand deleted. The scope of work includes to provide individual palletizer for each pail filling line.
14	Book 4 OF 8 / Page no. 238-B & 242 / Clause 2.2	No. of Heads for Automatic Pail Filling Line for DEF	Please confirm the no of heads	Bidder to consider no. of heads to achieve rated capacity & functional requirements.
15	Book 4 OF 8 / Page no. 244 / Clause 3.1 (point no.3)	Design Optimization to ensure operability, efficiency and maintenance; provide user friendly menus and diagnostic features as well as remote access & support	Please elaborate – Does the bidder required to consider provision of dedicated MODEM ?	Yes, Bidder to consider dedicated MODEM for Filling lines. The no. of MODEM connections to be finalized with OEM & needs to be provided so as to meet the OEM protocol for remote access to the Filling line.
16	Book 4 OF 8 / Page no. 244 / Clause 3.2 (point no.9)	Deployment of manpower for 12 months at site from the date of commissioning of each line;	Does one person for each line OR one person for all 3 lines of pail filler?	Please ignore this condition/ Tender Clause. The deployment of manpower at Plant shall be as per line item stipulated in BOQ for AMC.
17	Book 4 OF 8 / Page no. 246 (point no.f)	where in you have mentioned efficiency is 98% for filling lines	We have considered the efficiency up to 90 to 92%. Your suggestion required on the same.	Bidder understanding is correct. 90% is minimum for Overall Equipment Efficiency (OEE), that is for the entire line as a whole.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
18	Book 4 OF 8 / Page no. 246 (point no.g)	Vendor shall provide procure to measure MTTR and MTB	Please elaborate. (As it can be developed from real time experience only).	Bidder to ignore this requirement provided in the tender.
19	Book 4 OF 8 / Page no. 253 (point no.2 & 4)	where in you have mentioned Changeover of product time and rinsing Oil quantity is 15 mins & 150 ML.	It depends on grade and characteristic of product – please elaborate.	Product / Pack change-over time shall be as per OEM Standard specification. Domain expertise being available with OEM, it is presumed that best equipment will be provided by vendor to have minimum change-over time.
20	Book 4 OF 8 / Page no. 254 (point no.5)	you have mentioned the max allowable waste is 0.2%	Max waste may also depend on inputs given by customer for production. Your suggestion is required on the same.	The wastage should be minimized & shall be as per OEM standard specifications.
21	Book 4 OF 8 / Page no. 259 / Clause 3.4	Material for conducting FAT- Strap for top layer strapping.	In scope of work strapping machine is not mention.	Strapping machine is not required. Bidders to ignore the requirement mentioned in FAT.
22	Book 4 OF 8 / Page no. 259 / Clause 4.2	Labeling for rectangular and circular buckets	Please confirm two side labeling required in the same inline? (Rectangular and circular)	Two side labelling is required for Rectangular cross-section buckets. For Circular x-section buckets, the labels shall be only wrap-around type. Bidder may refer to the details of packaging materials - which forms part of the Tender document.
23	Book 4 OF 8 / Page no. 260 / Table no. A4.3	Load cell capacity-100Kg each Head	we have decided load cell capacity as per weight to be filled and required accuracy.	Yes. Bidder may finalize the Load cell capacity as per OEM recommendation - so as to optimize the filling accuracy for the desired capacity.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
24	Book 4 OF 8 / Page no. 263 / Table no. A4.3	Interface with main Plantwide control system Min DI- ESD Min DO- Call Operator	Please elaborate.	Integration of the Filling line control system with the Plant wide Control system (DCS) is envisaged to enable the following : a) communicating for Demand-for-product (i.e., Product top up in Product tank), b) Rinsing of the Filling nozzles at filling startup at the beginning of the operation; c) Pigging of product line from Piggable Manifold to machine as well as Air-blow & clearing the line at the end of the filling operation; d) Emergency Shut down - so as to suspend any operation (viz. Product Top up etc.) in the case of any emergency; e) Exchange of Productivity related Data such as No. of cans filled per shift, product wise etc. that will be useful for monitoring the performance of the line. Please refer Book 04 of 08; Clause A-8.8.1 (c); Page no. 35, For details of Feedback signals required for the Integration. This clarification is applicable for all types of Filling lines. I/O provisions required for communicating the above with the main Plantwide control system need to be considered by Filling machine vendor accordingly. The Filling machine vendor to provide a control box for communicating these feedbacks to the main plant control system regarding status of the filling machine. The filling machine control box feedbacks will be hard wired to the RIO panel supplied by the CPE. The scope of supply and installation of cables from Control Box to RIO panel are in the scope of the CPE vendor. Bidders to note that Clause 1.1 of Section D of Scope Matrix for Core Filling Equipments is superceded by the above. As the name suggests, ESD is for Emergency Shut down to suspend further operations; Call operator - to enable the operator know a particular step has been completed and to proceed to the next step. (viz. during pigging operation - to confirm that the pig has been received in the pig receiver and to commence with the air blowing & line clearing etc.)

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
25	Book 4 OF 8 / Page no. 265 / Table no. A-6	Lid pressing machine- Continuous belt type/ Pressure plate	We are proposing powerised (VFD controlled) set of roller type Lid pressing machine.	Tender specifications are minimum and Bidders can consider any improvements / improvisations on the same for bettering the performance of the line so as to achieve quality Lid pressing system for the desired capacity of each filling line. Bidder may please note that this is very critical requirement for filling line and integrity of packaging the pails depends on Lid pressing machine. So suitable proven type of Lid pressing machine shall be provided by Bidder to meet the functional requirements.
26	Book 4 OF 8 / Page no. 270 / Table no. A-16	QR code scanning for containers	Please elaborate the requirements of QR coding	QR coding based Trace & Track system is proposed for all inputs and outputs from the plant as per the Specifications given in tender document (Book 4 OF 8 / Page no. 774 / Specifications for QR Code based Trace-N Track System). QR code generation will be by IOCL Server and printing will be done by the Bidder. Unique, QR Codes shall be generated by IOCL server on real time basis & the Printers to be provided by Bidder shall be capable of accepting these codes & printing the same on respective items. QR code printing shall be provided on all incoming Raw materials (Packed), Packaging materials, on packed products during Filling & packing operation in the plant before storage in respective ASRS based Warehouses. Tracking & Tracing system allows monitoring their usage / consumption / dispatch using inventory management based on WMS. QR Coding system shall also encompass the rejections, missing one, etc., by way of having provision for QR Coding by handheld equipment. e.g. It may happen that 1 no filled container got leaked & same needs to be replaced manually in any particular carton.
C	**M/s. Bridge and Roof Co. (India) Limited, vide mail dtd 12.08.2021**			
27	General	SoW	In case of variation in quantities given in SOW, will payment be given accordingly.	Please refer to the special terms and conditions on modality of payments, wherein it has been amply clarified.
28	General	Fire Fighting System Scope	Whether Fire Fighting systems is in scope of bidder	Fire fighting system is not part of this Tender.
29	General	Scope of cables	Whether Cables for Systems like motors and other electric driven equipment and instruments are in scope of bidder	Cables for motors are in the scope of electrical contractor, except in the case of skids or package equipment, where electrical contractor will bring the cable only upto the main control panel or main distribution panel of the skid or package equipment. The termination into the panel and distribution within the skid equipment will be in the scope of mechanical contractor. Cables & Terminations for Bulk instruments & Valves are in the scope of CPE vendor. However instruments within the skid or package equipment supplied by Mechanical contrator will be in the scope of Bidder. For more details, please refer to the Scope matrix provided in the tender document. Also Bidders to consider the Electrical Design Considerations for Package equipment & System as attached. **Refer Annexure - I**

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
30	General	Scope of Instruments	Whether instruments are in scope of bidder or not	Please refer to the Scope matrix provided in the tender document. All the Bulk instruments & Valves which are outside of skids or Package equipments will be supplied by CPE Vendor. However in case of skids or package equipment in the scope of mechanical vendor, instruments including the local control systems for the skid or package equipment are in the scope of mechanical vendor. Making suitable provision for integration of these package PLC systems by provision of control boxes for data exchange as required for integration with the main plant wide control system are also in the scope of mechanical vendor. However, scope of supply & installation of cables from these Control Box to RIO panels are in the scope of the CPE vendor.
31	General	Quantities & Scope of work in BoQ	Whether quantities and scope of Work mentioned in BOQ are to be considered as final quantities and final list of work to be executed by the bidder, as FEED have been finalised	The BOQ quantities are tentative. Bidders are expected to validate the same before finalising their bid offers. Please refer execution methodology stipulated in Special terms & conditions of tender document for further clarifications.
32	General	Residual design	Whether only residual design has to be considered as per scope of bidder	Scope of the bidder includes complete detail design and engineering taking off from the FEED documents.
33	General	Statutory liaison work	Statutory liaison work to be carried out at an additional cost.	No additional payment will be made for the liaison work. Only payment towards official statutory related charges / fees will be reimbursed / paid by IOCL at actuals against submission of payment receipts.
34	General	Rates for Payment or Billing break up	It is not clear regarding rates to be considered for payment or billing break up.	Please refer to the special terms and conditions on modality of payments.
35	General	Reverse Auction	Reverse Auction to be done considered LSTK and item and AMC rates? IOCL may please clarify?	Bidder understanding is correct.
36	General	List of inventories	Kindly furnish the list of inventories to be considered during CAMC.	CAMC is not envisaged in this tender. Bidder to provide AMC services as per the respective item of BOQ.
D	**Additional Informations (M/s. IOCL & M/s. PEC LTD)**			
37	Book 2 OF 8 / 535 / Clause 2.0 / Point no. 9	Submersible Pumps (Scope of Supply)	The pump manufacturer shall extend a warranty on the Equipment for a minimum of five (5) years from the date of purchase. No lesser manufacturer's warranty shall be accepted.	Warranty / Guarantees shall be as per the Special Terms & Conditions in the Tender document.

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38	Book 6 OF 8 / Page no. 163, 164 & 165 / PID for Filling	Filters F-500101/102/153/154/155/ 156	Typo Error in PIDs - MOC for Filters	Bidder to Consider Housing Material For the Filters (F-500101/102/153/154/155/156) as CS, as indicated in Datasheets & SOW.
39	Book 1 OF 8 / Page no. 28 / Clause 4.4 Pumps (2)	Pumps P-500001 / 02 / 05 / 06 / 09 / 10 / 13 / 14 / 21 / 25 / 27 / 31 / 33 / 35	Capacity indicated as 30 m3/hr.	Bidder to consider the Capacity as 50 m3/hr (as per datasheet & BOQ/SOW).
40	BOQ/SOW - PART B - STATIC MIXERS 3.033 (a)	Static Mixers for Transformer Oil plant	Static Mixer quantity indicated as 4 nos.	For SOW Sr. No. 3.033 (a), Bidder to consider the Qty for Static Mixer as 2 nos.
41	BOQ/SOW - PART B - STATIC MIXERS 3.044 (a)	Static Mixer for White Oil plant	Static Mixer quantity indicated as 3 nos.	For SOW Sr. No. 3.044 (a), Bidder to consider the Qty for Static Mixer as 1 no.
42	Book 1 Of 8 / Page no. 49 / Clause 6.1(J)	Weighing Scale	Tag No. for Weighing Scale	Weighing Scale Tag No. Z-307061 to be read as Tag No. Z-301706.
43	Book 7 OF 8 / Page no. 8 (Utility Layout drawing)	Thermic Fluid Heater Dump tank	Capacity & Configuration: Horizontal Dump Tank 22 m3 in Utility Flow Diagram (UFD) & Scope of Works. However in Utility layout drawing Dump Tank is indicated as Vertical 100 KL.	Bidder to consider the capacity of Dump tank as 22 m^3 & Horizontal configuration as given in SOW.
44	Book 7 of 8 / Page no. 11 (Plot Plan Layout)	Fire Water Tank dimensions in Plot Plan	Plot plan indicates capacity & dimensions of Fire Water Tanks as 600 MT (8 m Dia x 12 m ht)	Bidder to consider the Capacity & dimensions for Fire Water Tanks as 900 MT (10 m Dia x 12.5 m Ht) as given in SOW/BOQ & Datasheet.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
45	BOQ/SOW Sl No 2.012 (A) BOQ/SOW Sl No 2.022 (a) Book 1 of 8 / Page no. 255 Book 7 of 8 / Page no. 11 (Plot Plan Layout)	Pipe Line Receipt	1. Foundations for Slop Tanks:- refer BOQ/SOW Sl No 2.012 (A) 2. Foundations for Slop Tank Pumps:- refer BOQ/SOW Sl No 2.022 (a)	**1. For SOW Sr. No. 2.012 (A), Bidder to include the following additional scope without any extra cost to IOCL.** Slop Tanks:- Design, Engineering, Supply, Construction of Foundations & Installation of 2 nos of 50 MT Slop Tanks (underground) as per revised Data sheet No. 50-0RG2N-DD-042 for Slop oil Tank (Tag No. SO-100049/51). **Refer Annexure - A.** **2. For SOW Sr. No. 2.022 (a), Bidder to include the following additional scope without any extra cost to IOCL.** For Slop Tank Pumps:- Design, Engineering, Supply, Construction of Civil foundations for 2 nos of Pumps (above ground). Bidder may please note the following information which are provided for location of slop tanks, pumps & piperack. Bidder scope includes the laying of pipelines from pipeline receipt manifold to destination through indicated piperack. Please note that the indicated piperack shall be provided by IOCL & Bidder scope includes the allied structural & supports to be provided for laying of pipeline from pipeline receipt manifold to piperack without any extra cost to IOCL. **Refer Annexure - B**, indicating the location of 2 nos of 50 MT Slop Tanks (underground); Doc. No. 20-0RG2N-BE-040. **Refer Annexure - C**, provides the Civil design basis & Specification that provides the design standards to be followed for Civil foundations. **Refer Annexure - G**, indicating the location of 2 nos of Slop oil pumps & Piperack.
46	BOQ/SOW Sl No 2.016 (A) BOQ/SOW Sl No 2.026 (a) BOQ/SOW Sl No 2.054 (a) BOQ/SOW Sl No 2.054 (b) Book 7 of 8 / Page no. 11 (Plot Plan Layout)	Super Clean Tank area near Loading Gantry	Foundations for i) Super Clean Tanks Tag No. FT-500038 & 40:- refer BOQ/SOW Sl No 2.016 (A) ii) Super Clean Pumps Tag No. P-500038 & 40:- refer BOQ/SOW Sl No 2.026 (a) iii) Filters Tag No. F-500038 & 40:- refer BOQ/SOW Sl No 2.054 (a) iv) Filters Tag No. F-500101, 102, 153, 154:- refer BOQ/SOW Sl No 2.054 (b) v) Pipe supports around equipment listed above	**For SOW Sr. No. 2.016 (A), Bidder to include the following additional scope without any extra cost to IOCL.** i) For Super Clean Tanks:- Design, Engineering and Construction of Foundations for 2 nos Storage Tanks (Tag No. FT-500038 & 40) including 300mm high dyke enclosure wall & impervious membrane to avoid seepage of oil into the sub-soil. **For SOW Sr. No. 2.026 (a), Bidder to include the following additional scope without any extra cost to IOCL.** ii) For Super Clean Pumps:- Design, Engineering and Construction of Foundations for 2 nos Pumps (Tag No. P-500038 & 40). **For SOW Sr. No. 2.054 (a), Bidder to include the following additional scope without any extra cost to IOCL.** iii) For Filters:- Design, Engineering and Construction of Foundations for 2 nos Filters (Tag No. F-500038 & 40) / Systems and required Pedestals for Pipe Supports. **For SOW Sr. No. 2.054 (b), Bidder to include the following additional scope without any extra cost to IOCL.** iv) For Filters:- Design, Engineering and Construction of Foundations for 4 nos Filters (Tag No. F-500101, 102, 153, 154) / Systems and required Pedestals for Pipe Supports.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
47	Book 7 of 8 / Page no. 11 (Plot Plan Layout) BOQ Sl No 40	Dye Dissolution Skid (Mini Bulk) adjacent to FP TF-1	i) Foundations for Dye Dissolution Skids supplied by Process Vendor ii) Covered Shed for Dye Dissoution Skids including its Foundation	**For BOQ Sr. No. 40, Bidder to include the following additional scope without any extra cost to IOCL.** Design, Engineering and Construction of Roof cover over the Dye Dissolution skid (5 nos of 100 Ltr Dye Dissolution vessesls with Pumps, Dosing arrangements, etc.,) to be supplied by CPE vendor. (Approx. dimensions of Roof cover over Dye Dissolution skid to be considered as L=8.0 m x W=4.0m x H=4.0m). Mechanical pkg vendor scope also includes plinth work, access stairs & Foundation pedestals for the skid. **Refer Annexure - C** for Civil design basis & Specification. **Refer Annexure - G** for Overall Plot plan Layout.
48	Book 7 of 8 / Page no. 8 BOQ/SOW Sl No 3.17	Cooling Tower	Foundations for Cooling Tower Package	**For SOW Sr. No. 3.17, Bidder to include the following additional scope without any extra cost to IOCL.** Design, Engineering and Construction of Foundations for Cooling Tower, Cooling water pumps & Pipe support sleepers, etc., as required. **Refer Annexure - C** for Civil design basis & Specification. **Refer Annexure - G** for Overall Plot plan Layout.
49	Book 7 of 8 / Page no. 11 (Plot Plan Layout) BOQ Sl No 760	Demister Vessels in Finished Product Tank farm-1 & 2, Loading Gantry and around Process Building.	Supporting arrangement for Demister Vessels supplied by CPE vendor	**For BOQ Sr. No. 760, Bidder to include the following additional scope without any extra cost to IOCL.** Design, Engineering and Construction of Steel supporting structural arrangements for Demister Vessels (Qty-6 Nos) supplied by Process Vendor (approx. capacity 200 Ltrs with Gross weight of 300 kgs) including provision of suitable brackets that may be anchored to the floor or to the Masanory wall such that the demister can be drained into a IBC / Drum from time-to-time. In case support is not possible to be provided from adjoining available structurals, bidder scope includes (without any extra cost to IOCL) to provide suitable support arrangement from floor to the demister skid so as to have proper stability.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
50	Book 2 of 8 / Page no. 316 / Clause 1.1 BOQ/SOW Sl No 3.092	DM Water Production Unit	Scope of Work - DM Water Production Unit	**For SOW Sr. No. 3.092, Bidder to include the following additional scope without any extra cost to IOCL.** Design, Engineering, Supply, Installation & Commissioning of suitable tank & pumping arrangements for collection, storage & transfer of following waste streams to waste water storage tanks for water recovery (a) DM plant backwash & regeneration waste stream (b) Boiler blowdowns (c) Clear water from Scrubber Sumps from Grease plant & Lube plant etc., so as to ensure Zero Liquid discharge compliance as required by TNPCB. **Bidder shall provide the following:** i) Suitable sump near boiler to receive waste water from boiler blowdown. Suitable pump & pipelines shall also be provided for transfer of waste water from the said sump to waste water storage tanks. ii) Suitable sump near DEF Plant to receive waste water of Scrubber sump & DM Water plant of DEF plant. Since the scrubber of grease plant is at higher elevation, the transfer of waste water from Scrubber to sump can be done by gravity. The connecting pipelines from DM Water plant of DEF plant & Scrubber of grease plant are also included in the scope of vendor. Suitable pump & pipelines shall also be provided for transfer of waste water from the said sump to waste water storage tanks. **Refer Annexure - K** which indicates revised UFD for Treated Water (Doc. No. 90-0RG2N-UFD-002-02). **Refer Annexure - L** which indicates revised UFD for Steam Generator (Doc. No. 90-0RG2N-UFD-008-01). **Refer Annexure - M** which indicates revised P & ID for Lube Blending (Doc. No. 90-0RG2N-PID-001-11). **Refer Annexure - N** which indicates revised P & ID for Grease Process (Doc. No. 90-0RG2N-PID-008-32 & 33). **Refer Annexure - O** which indicates revised P & ID for DEF Plant (Doc. No. 90-0RG2N-PID-111-02).
51	BOQ/SOW Sl No 3.092	DM Water Production Unit	Clarifications on Pump capacity for 1st stage RO Plant	Vendor to consider pump of suitable higher capacity for the 1st stage RO plant so as to enable the transfer of treated water from 1st RO upto the treated water storage tanks located in DEF plant (Approx. distance 315 Mtrs), without the need for additional pumping.
52	BOQ/SOW Sl No 3.091	DEF Plant Skid	Clarifications on Pipeline & Filters Scope for Finished product (DEF)	Bidder to consider scope of piping battery limits upto outlet of DEF plant skid. Filters for finished product & pipelines for transfer to finished product storage & from Finished product storage to filling are covered separately in Part-B (For Filters) & Part-C (For Piping works) respectively.
53	BOQ/SOW Sl No 3.091 / point no. (xi)	DEF Plant Skid	Clarifications on Heater for Water Pre-Heater	Thermic Fluid Heated pre-heater has been considered in the process scheme for DEF. Mechanical pkg vendor can consider Electrically heated pre-heater for water pre-heating as an option if required for VDA license compliance.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
54	General	All Technical Specifications & Data sheets	Warranty Clause	Warranty Clause Mentioned In Individual Datasheets to be ignored; Bidders to note that Warranty & Guarantees Shall Be only as per the Special Terms & Conditions in the Tender Document
55	General	All Drawings, Technical Specifications & Data sheets	VTC (Vendors To Confirm) / VTS (Vendors To Specify) Etc. Requirements Mentioned in the Technical Datasheets.	Bidders to note that VTC (Vendors To Confirm) / VTS (Vendors To Specify) Etc. requirements mentioned in the Technical Datasheets - to be Ignored by the Bidder at the Bidding Stage. Same needs to be provided only at the time of Detailed Engineering by the Successful bidder. However, Bidder to submit the declaration about their compliance & NIL deviations in respect of Technical requirements specified in the tender document.
56	Book 2 Of 8 / Page no. 10 & 11	Datasheet For Raw Water Day Tanks RW-311001/02	Name Correction - Raw Water Day Tanks, RO Permeate Day Tank & Treated Water Day Tank	Equipment name to be read as -"Raw Water Day Tanks"
57	Book 2 Of 8 / Page no. 11	Datasheet For Raw Water Day Tanks RW-311001/02	Tank Dimensions For Plastic Tanks	Bidder to consider Tank Dimensions For Plastic Tanks (Tag No. RW-311001/02) - Shell ID-4800mm & Cyl Ht- 7200 Mm
58	Book 2 Of 8 / Page no. 86	Datasheet For RO Water Day Tanks	Name Correction - Treated Water & RO Day Tanks	Equipment Name to be read as -"RO Permeate Day Tanks "
59	Book 2 Of 8 / Page no. 109	Datasheet For Screw Pumps	Typo Error-Tag No. for Pumps	Pump Tag No. PK-301702 to be read as "P-301702"
60	Book 2 Of 8 / Page no. 111	Datasheet For Screw Pumps	Discharge Pressure of Pump	Discharge Pressure for Pump (Tag No. P-301907) to be read as 10 Bar
61	Book 2 Of 8 / Page no. 343-352	Utility Packages- Datasheet For Chiller CH-600901A/B	Typo Error-Tag No. for Chiller	Chiller Tag No. U-600901A/B to be read as CH-600901A/B
62	Book 2 Of 8 / Page no. 380-495	Technical Specification For all the Powder Handling System	Material of Construction (MOC) for Powder Handling System (PHS)	Bidder to consider the MOC's of contact parts for Powder Handling Systems as follows: 1. PHS-304001-MOC Will Be CS.(Transformer Oil) 2. PHS-301701-MOC Will Be CS. (Polyurea Grease) 3. PHS-301702- MOC Will Be CS (CA Sulphonate Grease) 4. PHS-301705-MOC Will Be CS (AL- Complex Grease) 5. PHS-301706-MOC Will Be CS (Clay) 6. PHS-311101- MOC Will Be SS (DEF) 7. PHS-301001/02//04- MOC Will Be CS (Lithium Grease) 8. PHS-301003-MOC Will Be SS (Lithium Grease) 9. PHS-309401- MOC Will Be SS (Coolant) Provision shown in PID & Datasheet are Indicative & Bidders to consider Fully Automated Powder Charging System.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
63	Book 2 Of 8 / Page no. 215-218	Datasheet For Centrifugal Pumps_90-0RG2N-EDD-01_Rev B	Density Of Water	Density of Water to be read as 1000 Kg/M3.
64	Book 2 Of 8 / Page no. 496-497 & Book 7 Of 8 / Page no. 6 & 7	Process Plant Layout	Updated Process Plant Layout	Refer the revised Process Plant Layout (Doc. no.20-0RG2N-BE-011). **Refer Annexure - D.**
65	BOQ/SOW Sl No 5.06 (a)	Automated empty drum feed conveyor for Grease & Bituminous Lube drums	Tentative Layout for empty drum feed conveyors for Grease & Bituminous Lube	**For SOW Sr. No. 5.06 (a), Bidder to note the following clarification.** Please Refer the updated Process Plant Layout as attached (Doc. no.20-0RG2N-BE-011) where the tentative layout for the drum feed conveyor has been indicated. **Refer Annexure - D.**
66	BOQ/SOW Sl No 3.142 Book 2 Of 8 / Page no.539 / Clause 5 / Point no. 13.4	Technical Specification For Vertical Submersible Pumps & Motor Area class	Clarifications for Submersible Pumps for HSD & LDO	**For SOW Sr. No. 3.142, Bidder to note the following clarification.** Pumps for HSD & LDO Transfer shall be Vertical Submersible Centrifugal Pumps with FLP Motor Conforming to IS-2148, having Capacity of 20m3/Hr at 6 Bar; Motor shall be with flame proof enclosure conforming to IS 2148
67	Book 1 Of 8 / Page no. 223-229	Datasheet For Jet Eductors	Flow range for Jet Eductors	Bidders to note the Following clarifications on Jet Eductors For Base Oil Storage Tanks: Jet Eductors shall be designed to operate effectively to cater the Flow Range of 150-440 m3/Hr at 3-4 Bar(G): JT-100001B; .JT-100004B, JT-100007B, JT-100008B, JT-100009B, JT-100025B, JT-100026B, JT-100031B, JT-100033B, JT-100035B, JT-100037B, JT-100039B, JT-100041B, JT-100012B, JT-100017B
68	Book 1 Of 8 / Page no. 223-229	Datasheet For Jet Eductors- For Extract Tanks	Flow range for Jet Eductors	Bidders to note the Following clarifications on Jet Eductors For Extract Oil Storage Tanks: Jet Eductors shall be designed to operate effectively to cater the Flow Ranges given below : Jt-100043b- 50 To 75 M3/Hr at 3-4 Bar Jt-100047a- 30 M3/Hr at 3-4 Bar Jt-100047b- 50 To 75 M3/Hr at 3-4 Bar Jt-100046b- 50 To 75 M3/Hr at 3-4 Bar Jt-100048b- 50 To 75 M3/Hr at 3-4 Bar
69	Book 6 Of 8 / Page no. 7	P & ID For Grease Manufacturing	Typo Error - Tag No. in P&ID Description Table	Tag No. P-301909 to be read as P-301909A
70	Book 6 Of 8 / Page no. 8	P & ID For Grease Manufacturing	Typo Error - Tag No. in P&ID Description Table	Tag No. P-301910 to be read as P-301910A
71	Book 6 Of 8 / Page no. 9	P & ID For Grease Manufacturing	Typo Error - Tag No. in P&ID Description Table	Tag No. P-301911 to be read as P-301911A
72	Book 6 Of 8 / Page no. 10	P & ID For Grease Manufacturing	Typo Error - Tag No. in P&ID Description Table	Tag No. P-301912 to be read as P-301912A

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
73	Book 6 Of 8 / Page no. 11	P & ID For Grease Manufacturing	Typo Error - Tag No. in P&ID Description Table	Tag No. P-301913 to be read as P-301913A
74	Book 6 Of 8 / Page no. 39	P & ID For Lube Blending	Typo Error - Tag No. in P&ID	Tag No. P-302113 to be read as P-302113A
75	Book 6 Of 8 / Page no. 210	UFD For Treated Water	Typo Error - Flow rate in UFD Description Table	Flow Rate of Pump (Tag No. P-311007/08) to be read as 30 M3/Hr
76	Book 6 Of 8 / Page no. 226	UFD For Cold Oil	Typo Error - Tag No. in UFD Description Table	Tag No. U-600801 to be read as HE-600801
77	Book 6 Of 8 / Page no. 230	UFD For Steam Generator	Typo Error - Tag No. in UFD Description Table	Tag No. FW-600302 to be read as FW-600301
78	Book 6 Of 8 / Page no. 233	UFD For Steam Generator	Typo Error - Tag No. in UFD Description Table	Tag No. T-600301 to be read as T-600304
79	Book 3 Of 8 / Page no. 267 & 271	Design pressure for 'A1B' pipe material specification	Design Pressure of 'A1B' Specification	Design Pressure of 'A1B' to be read as 20 kg/cm2(g)
80	Book 3 Of 8 / Page no. 346 / 348 / 376 / 400 / 426 / 428 / 432 / 444	Data Sheet for Manual Gate Valves, Threaded, A1B Data Sheet for Manual Gate Valves, Flanged, A1B Data Sheet for Manual Ball Valve, Threaded, A1B Data Sheet for Check Valve, Flanged, A1B Data Sheet for Manual Piston Valves, Threaded, A1B Data Sheet for Manual Piston Valves, Flanged, A1B Data Sheet of Basket/Y-Type/T-Type Strainer, A1B Data Sheet for Hose	Design Pressure and test of 'A1B' Specification	Design Pressure of 'A1B' to be read as 20 kg/cm2(g); accordingly the body and seat test pressures shall be revised to 30 kg/cm2(g) & 22 kg/cm2(g) respectively.
81	Book 4 Of 8 / Page no. 835	Datasheet for Goods Lift	Capacity of Goods Lift specified as 2 MT	Capacity of Goods Lift to be read as 3MT; as specified in BoQ/SoW 6.202 (a)
82	BOQ/SOW SL NO 3.041(b)	Percolation Columns C-305201 to 210	Clay Bag Charging & Discharge of Spent clay from Percolation columns	**For SOW Sr. No. 3.041 (b), Bidder to include the following additional scope without any extra cost to IOCL.** Design, engineering, supply and installation of Clay Bag Charging System complete with Motorised Chain Pulley Block of approximate Capacity 1.0 MT with a monorail suitable for clay bag charging into all the columns - 10 nos (Tag No. C-305201 to 210) as per attached Process plant Layout (Doc. No.20-0RG2N-BE-011). **Refer Annexure - D.** Bidder scope also includes Design, engineering, supply and installation of suitable arrangments for discharge of spent clay from the percolation columns including provision of suitable collection tray arrangement for disposal of clay. IOCL will dump these disposals suitably & complete the incineration or shall be disposed off to Third-party.
83	Book 2 OF 8	Exhaust Blower for VI Solublisation Plant	Exhaust Blower Specification is added	Refer attached Exhaust Blower specification (Doc. No. 90-0RG2N-IS-042). **Refer Annexure - E.**

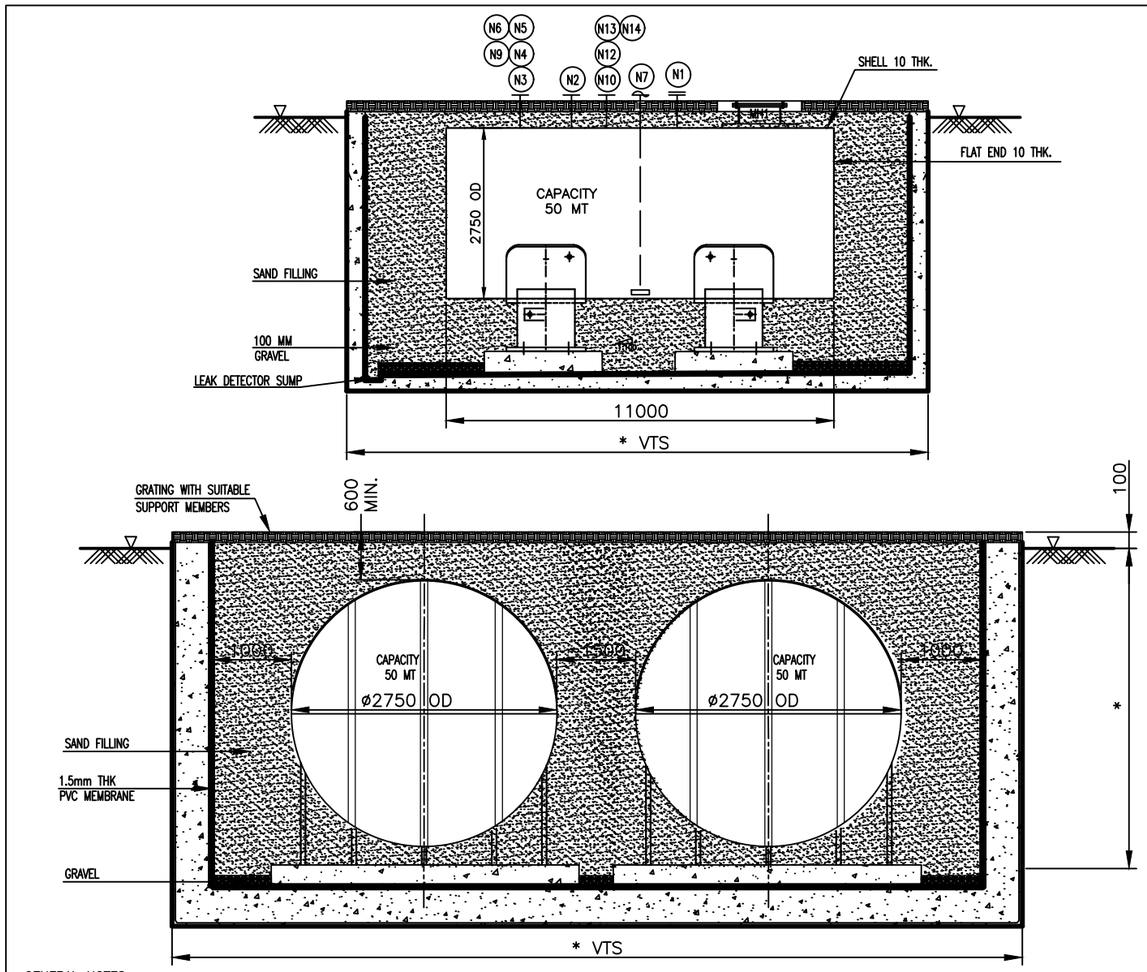
Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
84	Book 7 Of 8 / Page no. 276	Stress Critical Pipelines	Stress Critical Pipelines	Bidder to note that following additional clarifications: Stress critical lines & selection criteria have been mentioned in attached Line list (Doc. No. 90-0RG2N-LL-001) for performing Stress analysis by the successful bidder. **Refer Annexure - F.** For more details, please refer to the Piping design specifications provided in the tender document (Book 03 of 08; Clause 4.0; Page no. 115) to perform Stress analysis.
85	Book 7 Of 8 / Page no. 11	Overall Plot Plan	Updated Overall Plot Plan Layout	Refer the revised Overall Plot Plan Layout (Doc. No. 90-0RG2N-PL-001). **Refer Annexure - G.**
86	Book 4 of 8 / Page no. 151	5L Filling line for DEF filling (SCF-411001)	Type of Filling machine to be considered for DEF 5L filling	Bidder to note that following additional clarifications: Bidders to consider only Rotary, Volumetric, Piston Filler type Filling machine as specified & detailed in (Book 04 of 08; Section B-8.0; Page 157).
87	Book 4 of 8 / Page no. 586	Pail Filling line for Food Grade Greases (PFL-401901)	Common Filling machine for 20Kg, 1Kg & 0.5kg Cans;	Bidder to note that following additional clarifications: As the volume off-takes of Food Grade Greases is low, a common Weighmetric, Filling machine was proposed for 20kg, 1kg & 0.5kg Cans. However, on account of concerns about the accuracy of such a common filling machine while filling 0.5kg & 1kg Cans - in respect of compliance with legal metrology requirements, Bidders to provide a separate dedicated semi-automatic off-line filler that can be easily retrofitted into the line - while filling 1kg & 0.5kg Grease containers. All other line equipments may be shared with 20kg Pail line.
88	BOQ/SOW Sl No 6.206	Loading & Unloading Material Handling systems	Loading /Unloading material handling facilities to cover all 4 warehouses (viz. Additive warehouse, Packaging material warehouse, FP warehouse and Low flash material warehouse).	Bidder to note that following additional clarifications: Scope includes suitable Loading / Unloading material handling equipments for all 4 Warehouses (Additive, Packaging, Finished product & Low flash), with complete integration of unloading operations with Trace-N-Track system, Feeding arrangement to ASRS system - so as to ensure seamless integrated performance for Automated operations with minimum manual intervention.
89	General	Physical Properties for Base oils, Additives & Finished Products	Added Physical Properties for Base oils, Additives & Finished Products	Refer attached Physical Properties for Base oils, Additives & Finished Products. **Refer Annexure - H.**
90	Technical Bid / Annex-6 / Sr. No. 41	List of Recommended makes	Addition of vendor for Small Pack Filling Machines (Rotary - volumetric)	Please consider inclusion of SS Packaging Industries as recommended vendor for Small Pack Filling Machines (Rotary - volumetric)
91	Book 4 of 8	Clarifications on Screw-on Capping machine (Rotary type) for Small can Filling Lines	Vision Camera based On-line rejection system at the outlet of the Capper	Vision Camera based On-line rejection system at the outlet of the Capper shall be suitable for online rejection of Cans with improper capping or cross capping only. Same will not be suitable for improper torque. This clarification is applicable for all Small Can Filling lines.
92	Book 4 of 8	Clarifications on Screw-on Capping machine (Rotary type) for Small can Filling Lines	Location of Bulk Cap Hopper	Bulk Cap Hopper may be located near to the Capping machine on Ground Floor. This clarification is applicable for all Small Can Filling lines.

Sr. No.	Item/ Page No/ Section/ Clause	Description	Query	IOCL Reply
93	General	Type of Instruments & Recommended makes	Type of Instruments & Recommended makes	Attached Type of Instruments & recommended makes to be consider by bidder for skids as well as package units. **Refer Annexure - J**

INDEX (ANNEXURES)

(TECHNICAL SPECIFICATIONS / DATA SHEETS / DRAWINGS)

Sr. No.	Description	Document Number	Annexure	Page No.
1	Data Sheet for Slop oil Tank (Tag No. SO-100049 / 51)	50-0RG2N-DD-042	Annexure – A	2
2	Pipeline receipt & RTU Room (Architectural Layout)	20-0RG2N-BE-040	Annexure – B	3
3	**Civil - Design Basis & Specification** **NOTE:** Bidder shall refer to applicable sections of this Civil Design basis & spec. for Civil related Scope of works.	General	Annexure – C	4
4	Process Plant Layout	20-0RG2N-BE-011	Annexure – D	262
5	Technical Specification for Exhaust Blower	90-0RG2N-IS-042	Annexure – E	264
6	Line List for stress critical pipelines	90-0RG2N-LL-001	Annexure – F	273
7	Overall Plot Plan Layout	90-0RG2N-PL-001	Annexure – G	374
8	Physical Properties for Base oils, Additives & Finished Products	General	Annexure – H	375
9	Electrical Design Considerations for Package equipment & System	General	Annexure – I	406
10	Type of Instruments & recommended makes	General	Annexure – J	416
11	UFD for Treated Water	90-0RG2N-UFD-002-02	Annexure – K	417
12	UFD for Steam Generator	90-0RG2N-UFD-008-01	Annexure – L	418
13	P & ID for Lube Blending	90-0RG2N-PID-001-11	Annexure – M	419
14	P & ID for Grease Process	90-0RG2N-PID-008-32 90-0RG2N-PID-008-33	Annexure – N	420
15	P & ID for DEF Plant	90-0RG2N-PID-111-02	Annexure – O	422



GENERAL NOTES :

1. ALL DIMENSIONS ARE IN MM UNLESS OTHERWISE SPECIFIED.
2. REINFORCING PADS TO HAVE TELL TALE HOLES OF 6mm. THESE PADS SHALL BE AIR TESTED AT 1.05 kg/cm2(g) WITH SOAP SOLUTION AND THERE AFTER FILLED WITH HARD GREASE.
3. ALL BOLT HOLES IN NOZZLE FLANGES AND MANHOLES TO BE STRADDLE CENTER LINES OF TANK AND NOZZLE.
4. ALL ERECTION LUGS AND CLEATS TO BE REMOVED BEFORE HYDRO TEST.
5. PRJ. OF NOZZ. SHALL BE MEASURED FROM INTERSECTION OF NOZZ. AXIS & OUTER SURFACE OF SHELL / ROOF.
6. ALL WELD SPATTERS AND LARGE WELD BEADS SHOULD BE REMOVED BY GRINDING.
7. FLANGE GASKET SURFACES SHALL BE 125 AARH FINISH.
8. NOZZLES 50NB AND BELOW SHALL BE PROVIDED WITH 2 STIFFENER FLATS OF SIZE 50Wx6THK AT 90° APART. 80 NB & ABOVE NOZZLES TO BE PROVIDED WITH R.F. PADS.
9. ALL SHARP CORNERS TO BE ROUNDED OFF.
10. NOZZLES 250 NB AND ABOVE MAY BE FABRICATED FROM PLATE WITH 100 % RADIOGRAPHY
11. DIMENSIONS / DATA MARKED WITH '*' WILL BE SPECIFIED BY VENDOR.
12. THICKNESS & WEIGHT INDICATED ARE PRELIMINARY FOR ESTIMATION PURPOSE ONLY. EQUIPMENT VENDOR SHALL VERIFY THE THICKNESS & WEIGHTS.

TESTING REQUIREMENT:.

1. NOZZLE PAD ON SHELL ONLY : BY AIR PRESSURE AT 15 P.S.IG
2. END : AIR PRESSURE & SOAPY WATER
3. SHELL : BY WATER FILLED TEST / HYDROSTATIC TEST
4. UNLESS OTHERWISE SPECIFIED, SUPPLY AND DISPOSAL OF HYDRO-TEST WATER IS THE RESPONSIBILITY OF THE CONTRACTOR. AFTER HYDRO TEST, TANK TO BE COMPLETELY DRIED (PHYSICALLY) AND OIL TO BE APPLIED ON INSIDE AND BOTTOM OF TANK. PROCUREMENT OF OIL NOT IN SCOPE OF CONTRACTOR.
5. WHERE EARTHING LUG CANNOT BE PUT AT AN ELEVATION OF 400 mm ABOVE SADDLE BASE PLATE IT SHALL BE LOCATED AS HIGH AS POSSIBLE.

PLEASE NOTE :

CIVIL WORKS RELATED TO RCC SECONDARY CONTAINMENT WITH LEAK CONTAINMENT MEMBRANE, FOUNDATION FOR SADDLE SUPPORT WITH ANCHORING ARRANGEMENT SAND FILL, LEAK DETECTION PROVISION, TURNBUCKLE WITH ANCHORING ARRANGEMENT & GRATING ETC. – ARE IN THE SCOPE OF MECHANICAL CONTRACTOR. ABOVE DRAWING IS ONLY FOR INFORMATION TO THE FABRICATOR – OF THE PROPOSED INSTALLATION SCHEME.

SERVICE DATA	QTY . REQD. : TWO	PLANT SITE : CHENNAI	CAPACITY : 50 MT	WORKING : 58.8 M^3 ACTUAL : 65.3 M^3
	TYPE – HORIZONTAL CYL. TANK WITH FLAT ENDS & FULL BURRIED			
	DELIVERY : SINGLE PIECE / SHOP FABRICATED		TAG NO. SO-100049/51	
	CONTENTS		SLOP OIL	
	DENSITY	Kg/m3	900	
	NOM./MAX. WORKING PR.	(TANK) Kg/cm^2	ATM	
	NOM./MAX. WORKING TEMP.	(TANK)°C	AMBIENT	
DESIGN DATA	DESIGN CODE		IS-10987, ASME SEC-VIII DIV-1 & GEP.	
	OTHER APPLICABLE CODES	SEISMIC	IS-875, BASIC WIND SPEED 48M/S , IS-1983 ZONE-III	
			IS-10987 REAFFIRMED 2003; STATIC & MOBILE PRESSURE VESSEL CODE	
	DETAILS OF SUPPORT		SADDLE SUPPORTS	
	STATUTORY APPROVAL REQUIRED		(a) FROM CCE : NO ; (b) FROM IBR : NO	
	DESIGN TEMP.	(SHELL) °C	120°	
	DESIGN PRESSURE	(SHELL) Kg/cm^2 (g)	FULL OF WATER	
	TESTING		HYDRO TEST	
	VACUUM TEST PRESSURE	mm Hg (abs)	--	
	RADIOGRAPHY	(SHELL/END)	SPOT/SPOT	
	WELD JOINT EFFECIENCY	(SHELL/END)	0.85/0.85	
	CORROSION/EROSION ALLOWANCE	mm	1.5mm (INT) / 1.5mm (EXT)	
	SURFACE FINISH (INTERNAL)	mm	SMOOTH SURFACE FINISH OR CLEAN FINISH	
	INTERNALS		-----	
	DETAILS OF INSULATION : (NOT REQD.)			
			TYPE OF INSULATION : ----- : INSULATION THK. : -----	
	DETAILS OF VESSEL WEIGHTS , MT		EMPTY WT : * MT : OPERATING WT :* MT	
			HYD. TEST WT : * MT : MAX. ERECTION WT : *	
MATERIAL	SHELL : IS 2062 GR. B	END : IS 2062 GR. B	: END : IS 2062 GR. B	
	NOZZLE PIPES	SA-106 GR.B SCH 40 /IS 2062 GR.B		
	NOZZLE FLANGES	SA-105		
	MANWAY PIPE	IS 2062 GR. B	BOLTING FOR NOZZLES AND MANWAY:	SA193 B7G.I/SA194 2H G.I
	MANHOLE NECK,COVER, FLANGES :	IS 2062 GR. B		
	GASKETS : CNAF		BOLTS & NUTS : SA193 B7,SA194 2H	
	PADS : IS 2062 GR. B	: PADS FLANGES : ---	ANCHOR BOLTS: IS 1367 CL 4.6/4.0	
OTHER	WELDING PROCEDURE : AS PER ASME SEC IX			
	AREA CLASSIFICATION : HAZARDOUS	: BRACING : IS 2062 GR. B		
	PAINTING : AS PER PAINTING SPECIFICATION			

NOZZLE SCHEDULE

NOZZLES	SERVICE	NOM. BORE	SCH./ THK.	FLANGES STD.	FLANGES TYPE	FLANGES CLASS	NOZZLES REINF.	REMARKS
N1	ROOF SPARE NOZZLE	100	SCH #40	B16.5	WNRF	150#	ø210X8THK	WITH BLIND FLANGE
N2	OUTLET	100	SCH #40	B16.5	WNRF	150#	ø210X8THK	
N3	VENT	150	SCH #40	B16.5	WNRF	150#	PAD	GOOSE NECK VENT
N4	LT	100	SCH #40	B16.5	WNRF	150#	ø210X8THK	
N5	LS	50	SCH #40	B16.5	WNRF	150#	ø180x8THK.	WITH DIP PIPE
N7	DIP HATCH	150	SCH #40	B16.5	WNRF	150#	ø210X8THK	WITH STILLING WELL
N8	INLET	100	SCH #40	B16.5	WNRF	150#	ø210X8THK	
N9	TI	40	SCH #40	B16.5	WNRF	150#		WITH STILLING WELL
N10	OIL INLET	150	SCH #40	B16.5	WNRF	150#	ø210X8THK	
N12	OIL INLET	50	SCH #40	B16.5	WNRF	150#	ø210X8THK	
N13	OIL INLET	150	SCH #40	B16.5	WNRF	150#	ø210X8THK	
N14	OIL INLET	100	SCH #40	B16.5	WNRF	150#	ø210X8THK	
MH-1	MANHOLE-1	600	8 THK.	8 THK.FLANGE/8 THK.COVER			ø1255x8THK. AS REQ.	

REV	DATE	DESCRIPTION	DONE BY	CHK'D BY	TECH APP	PROJ APP
B	18/08/2021	RE-ISSUED FOR TENDER	K.S.N.	V.R/R.S.	D.R.	K.G.K
A	10/05/2021	ISSUED FOR TENDER	N.B.M.	S.B.S.	D.R.	K.G.K

CLIENT: INDIAN OIL CORPORATION LIMITED
MUMBAI-400051
INDIA

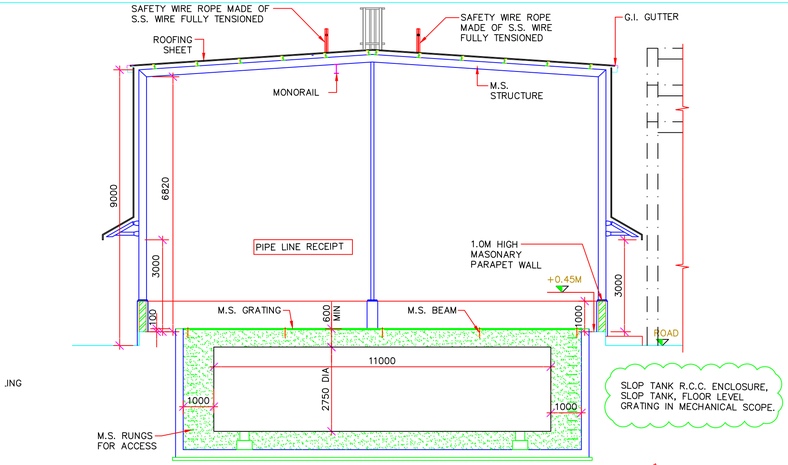
MAIN CONTRACTOR: PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT: GRASSROOT LUBE COMPLEX AT AMMULAVAYOL, CHENNAI

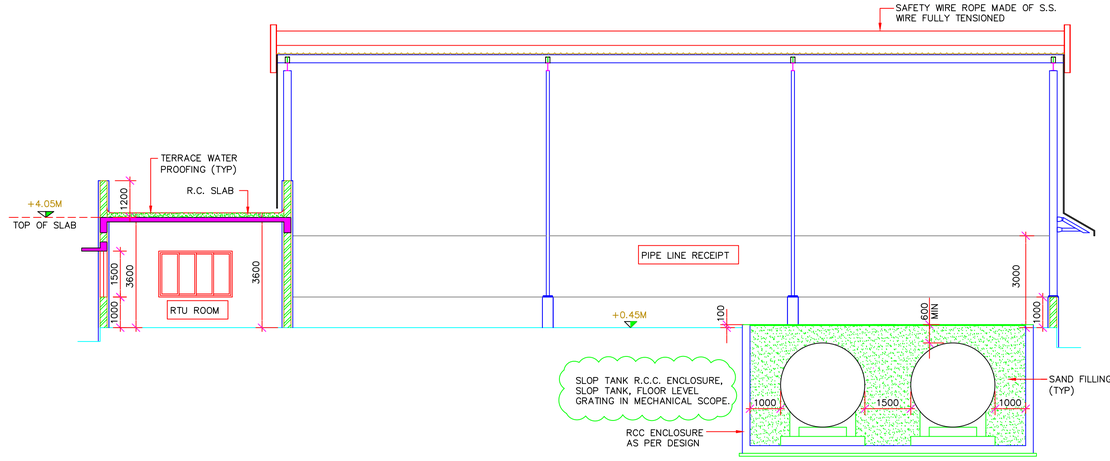
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DOC.NO.: 50-ORG2N-DD-042 SHEET 1 OF 1 REV: B

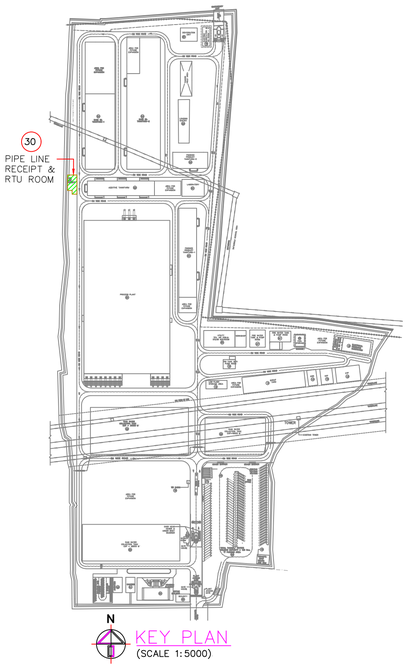
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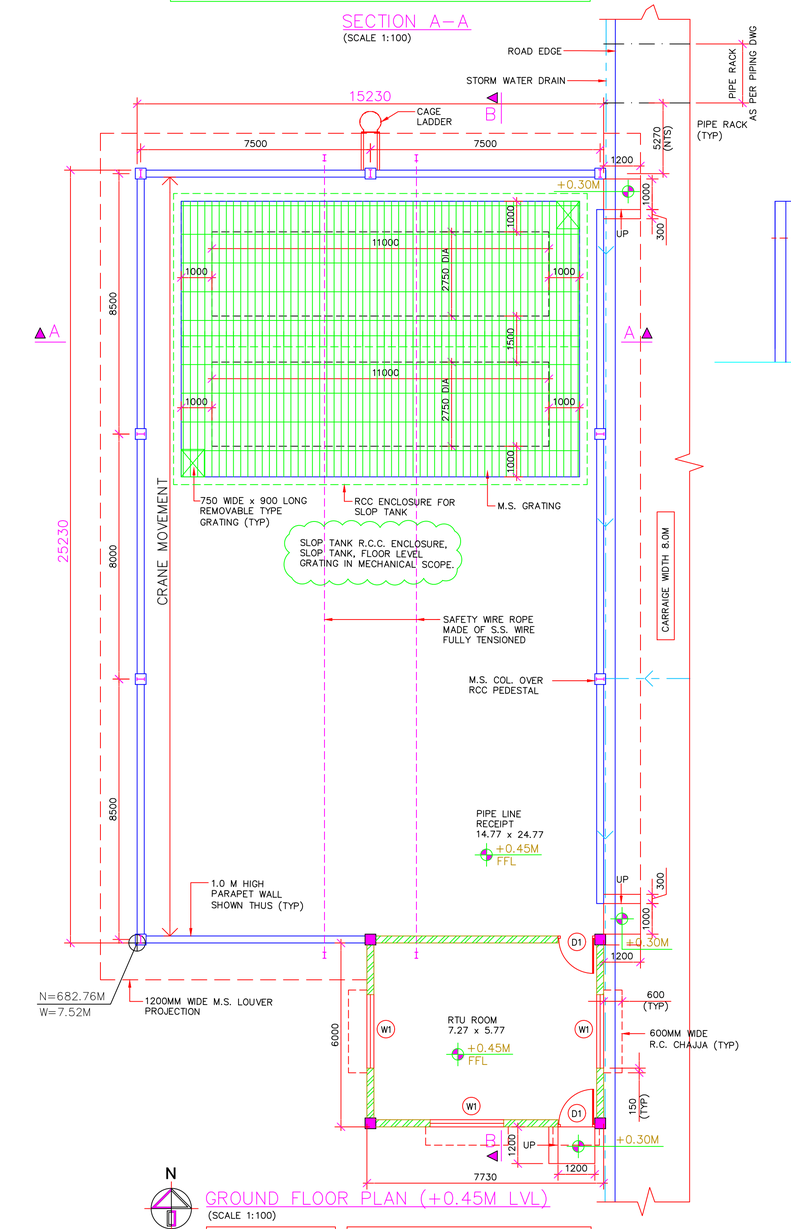
SECTION A–A
(SCALE 1:100)



SECTION B–B
(SCALE 1:100)



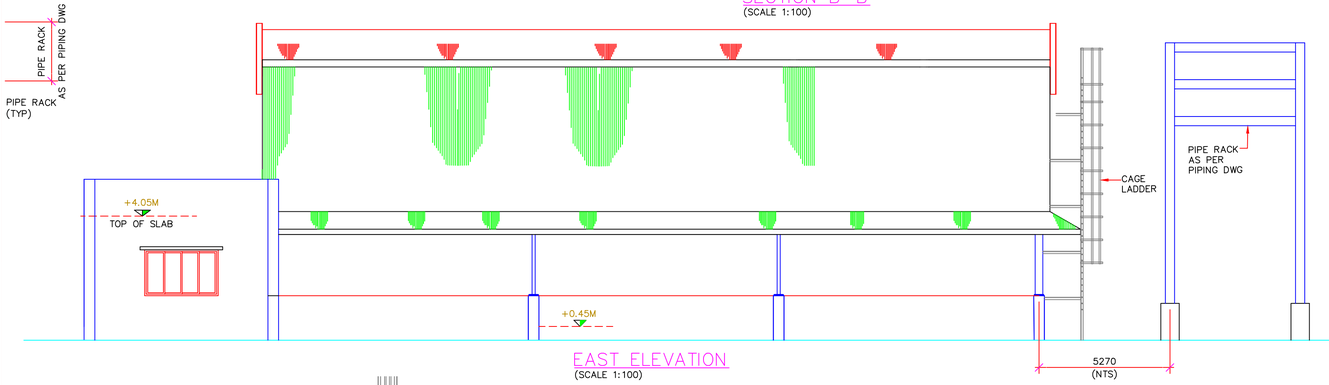
KEY PLAN
(SCALE 1:5000)



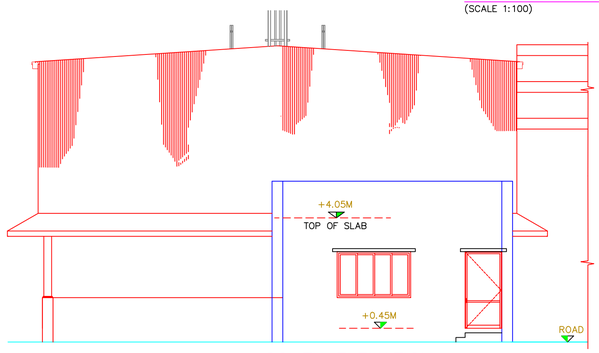
GROUND FLOOR PLAN (+0.45M LVL)
(SCALE 1:100)

±0.0M = F.G.L
+0.10M = TOP OF ROAD

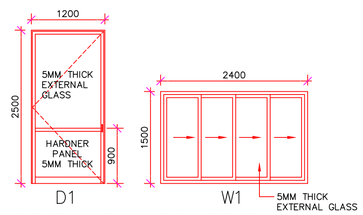
* PIPE RACK LAYOUT & SECTIONAL DETAILS , PIPE RACK COLUMNS LOCATIONS AS PER PIPE RACK DWG



EAST ELEVATION
(SCALE 1:100)



SOUTH ELEVATION
(SCALE 1:100)

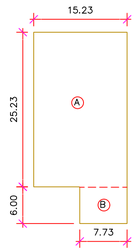


DOOR AND WINDOW PROFILES
(SCALE 1:50)

NOTE :–
* ALL EXTERNAL GLASS TO HAVE MAX. SHGC VALUE OF 0.42 WHEN WWR RATIO IS LESS THAN OR EQUAL TO 40% AND 0.36. WHEN WWR RATIO IS GREATER THAN 40%. THE MAX U VALUE SHALL BE 5.7.
* ALL INTERNAL GLASS NOT EXPOSED TO OUTSIDE RADIATION SHALL BE 5MM THICK CLEAR GLASS OR HEAT STRENGTHENED GLASS WITH THICKNESS 12MM OR AS SPECIFIED.

SCHEDULE OF DOORS & WINDOWS

TYPE	SIZE (mt.)	DESCRIPTION	CILL	JAMB/HEAD LINING	CILL LINING
D1	1.20 x 2.50	ALU. DOOR WITH DOOR CLOSER	NIL	GRANITE CILL IN STEP	NIL
W1	2.40 x 1.50	ALU. SLIDING WINDOW	1.00	GRANITE WITH STEP	GRANITE WITH STEP



AREA DIAGRAM FOR GROUND FLOOR
(SCALE 1:500)

AREA CALCULATIONS

AREA OF GROUND FLOOR =

A = 15.23 x 25.23 = 384.25
B = 7.73 x 6.00 = 46.38

AREA OF GROUND FLOOR = A + B = 430.63 SQM ----- 1

TOTAL BUILT–UP AREA = 1 = 430.63 SQM

STRUCTURE NO – 30

R E V	DATE	DESCRIPTION	DONE BY	CHK'D BY	TECH APPROJ A
B	11/08/21	SLOP TANK ADDED	Prabha	HNS	NNS	GK
A	20/01/21	ISSUED FOR TENDER	MADHAVI	HNS	NNS	GK

INDIAN OIL CORPORATION LIMITED
MUMBAI–400051
INDIA

PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT TITLE:
GRASS ROOT LUBE COMPLEX, CHENNAI

DRAWING TITLE:
PIPE LINE RECEIPT & RTU ROOM
ARCHITECTURAL LAYOUT

SCALE	DRAWING No.	SIZE	REVISION
AS SHOWN	20–ORG2N–BE–040	A1	B
DATE 22/12/2020			

CIVIL - DESIGN BASIS AND SPECIFICATION

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Architectural, Civil and Structural Design Basis

A. Architectural, Civil and Structural Design Basis

1 Structural design basis

a.CODES, STANDARDS & REFERENCE DOCUMENTS

Codes

I. S. CODE NO.	SUBJECT
IS – 455	Specification for Portland slag cement
IS - 2386	Method of test for aggregates for concrete
IS – 516	Methods of test for strength of concrete
IS - 3696	Safety Code for Scaffolds and ladders.
IS - 3674	Safety Code for Excavation
IS - 1498	Soil Classification
IS - 1200	Methods of Measurement of Building Works
IS – 456	Code of practice for plain and reinforced concrete.
IS - 2720	Method of test for soils
IS - 2185	Specification for hollow & solid concrete blocks
IS - 1661	Code of Practice for Application of cement plaster.
IS – 1905-1969	Safety of Masonry walls.
IS – 1597-1967	Rubble Stone Masonry.
IS – 1361-1978	Steel windows for Industrial building.
IS-3370-1965	Concrete Structures for storage of Liquid
IS-1892-1962	Site investigation for foundations.
IS-1786-1961	Cold Twisted Deformed Bars.
IS-1948	Aluminium doors, windows & ventilation
IS-6248	Metal Rolling Shutters and Grills.
IS-73	Paving Bitumen
IS-217	Cutback Bitumen
IS-783	Code of Practice for Laying of concrete pipe

IRC-19	Standard spec. and code of Practice for WBM.
IS 383,	Material soundness for GSB.
IRC-15	Standard Specification and code of Practice for Construction of Concrete Road
IRC-29	Spec. For Bituminous Concrete.
IS - 1077	Common burnt clay bricks
IS - 1786	Cold twisted re-bars for concrete reinforcement
IS – 800	Code of practice for General Construction in Steel.
IS: 3989	Centrifugal cast (spun) cast iron pipes
IS: 6163	Centrifugal cast low-pressure cast-iron pipes i) Spigot and socket type ii) Socket-less type
IS: 1536	Centrifugal cast iron pipes, spigot & socket or flanged type.
IS: 2720	CBR Test for Soil – Lab & Field
IS: 7215	Tolerances for fabrication of steel structures.
IS: 12843	Tolerances for erection of steel structures.
IS: 813-1986	Symbols for welding.
IS: 816-1986	Metal arc welding.
IS: 1538 (Part 1 to 24)	Cast iron fittings and specials.
IS: 10262	Design Mix.
IS: 651	Salt glazed stoneware pipes
IS: 458	Concrete pipes
IS: 4989	HDPE pipes for potable water supplies, sewage & industrial effluents.
IS: 985	PVC pipes for potable water supplies.
IS: 110221	Code of practice for coating and wrapping of underground mild steel pipelines.
IS: 3114	Code of practice for laying of cast iron pipes.
NBC	National building code – 2016*
CPCB	Central Pollution Control Board of India
DGMS	Directorate General of Mines Safety
IER	Indian Electricity Rules

IS: 1239	Mild steel tubes, tubular and other wrought steel fittings
IS: 1498	Classification and identifications of soils for general Engineering purposes.
IS: 1726	Cast iron manholes covers and frames
IS: 1729	Sand cast iron spigot and socket soil, waste and ventilating pipes, fittings and accessories
IS: 1893	Criteria for earthquake resistance design of structures
IS: 2116	Sand for masonry mortars Qualifications
IS: 2470	Code of practice for installation of septic tanks
IS: 269	Ordinary Portland cement, 33 grades
IS: 2720	Part XIV - Determination of Density index (relative density) of cohesion less soils.
IS: 2720	Part VIII - Determination of water content - dry density relation using heavy compaction.
IS: 3486	Cast iron spigot and socket drainpipes
IS: 3950	Surface boxes for sluice valves
IS: 4985	Unplasticized PVC pipes for potable water supplies
IS: 5455	Cast iron steps for manholes
IS: 554	Dimensions for pipe threads where pressure-tight joints are made on threads
IS: 651	Salt-glazed stoneware pipes and fittings
IS: 778	Copper alloy gate, globe and check valves for water works purposes
IS: 782	Caulking lead
IS: 875	Codes of practice for design loads (other than earthquake) for buildings and structures.
IS:2720 Part V	Methods of tests for soils-Part 5, Determination of liquid & plastic limit.
IS:2720 Part VII	Methods of tests for soils Part 6, Determination of shrinkage factors.
OISD	Oil Industry Safety Directorate
OMR	Oil Mines Regulations

P6EC-STD-CI-S-1002	Standard Specification for Road Works
P6EC-STD-CI-S-1010	Standard Specification for General Civil & Structural Works
IS:269	Specification for 33 grade ordinary Portland cement
IS:8112	Specification for 43 grade ordinary Portland cement
IS:455	Specification for Portland slag cement
IS:1489 Part 1	Specification for Portland Pozzolana cement (fly ash based)
IS:1489 Part 2	Specification for Portland Pozzolana cement (calcined clay based)
IS:3466	Specification for Masonry Cement
IS:6452	Specification for high alumina cement for structural use
IS:8041	Specification for rapid hardening Portland cement
IS:12269	Specification for 53 grade ordinary Portland cement
IS:12330	Specification for Sulphate Resisting Portland cement
IS:1785	Specification for plain hard-drawn steel wire for Prestressed concrete.
IS:1786	Specification for High strength deformed steel bars and wires for concrete reinforcement.
IS:6003	Specification for indented wire for Prestressed concrete.
IS:1161	Steel tubes for structural purposes.
IS:1239	Mild Steel Tubes, tubular and other wrought steel fittings.
IS:2062	Steel for general structural purposes (Grade A).
IS:4923	Hollow steel sections for structural use.
IS:412	Expanded Metal Steel Sheets for General purposes
IS:2062	Mild steel bars of anchor bolts, rungs, metal inserts, grating etc.)

IS:1363	Hexagonal head bolts, screws & nuts of product grade C
IS:811	Cold formed light gauge structural steel sections.
IS:1367	Technical supply conditions for threaded steel fasteners.
IS:2016	Plain washers
IS:2266	Steel wire ropes for general engineering purposes
IS:2315	Thimbles for wire ropes
IS:2361	Bulldog grips
IS:1239	Mild Steel Tubes, tubular and other wrought steel fillings. (For Hand, rail tubular sections).
IS:2485	Drop forged sockets for wire ropes for general engineering purposes.
IS:3502	Steel chequered plates
IS:3138	Hexagonal bolts and nuts (M42 to M150).
IS:1123	Method of identification of natural building stones.
IS:1127	Recommendations for dimensions and workmanship of natural building stones for masonry work.
IS:1129	Recommendations for dressing of natural building stones.
IS:73	Specification of Paving Bitumen
IS:1195	Specification for bitumen mastic for flooring.
IS:1322	Specification for Bitumen felts for water proofing and damp proofing
IS:1834	Specification for Bituminous compounds for water proofing and caulking purposes
IS:1838	Specification for preformed fillers for expansion joint in concrete pavements and structures
IS:3037	Specification for bitumen mastic for use in water proofing of roofs
IS:3384	Specification for bitumen primer for use in water proofing and damp proofing
IS:5871	Specification for Bitumen Mastic for Tanking and Damp proofing
IS:7193	Specification for Glass fibre base coal tar pitch & bitumen felts.

IS:7198	Code of practice for damp proofing using bitumen mastic
IS:8374	Specification for bitumen Mastic, Anti-Static and Electrically conducting grade.
IS:399	Classification of commercial timbers and their zonal distribution
IS:3337	Specification for ballies for general purposes
IS:4990	Specification for Plywood for concrete shuttering work

b. Abbreviations

FGL	Finished Ground Level
NGL	Natural Ground Level
FFL	Finished Floor level
TOS	Top of Steel
TOC	Top of Concrete
TOGR	Top of Grating
SBC	Safe bearing capacity of soil
TOG	Top of Grout
UNO	Unless Noted Otherwise
IL	Invert Level of drain
HPF	High Point on Floor
LPF	Low Point on Floor
TYP	Typical for details, dimension, thickness or material
MOC	Material of Construction
UG	Under ground
MSL	Mean Sea Level
RCC	Reinforced Cement Concrete
PCC	Plain Cement Concrete
UCR	Uncoursed Rubble Masonry

GENERAL STRUCTURAL DESIGN BASIS

a. Design Loads

- Dead Loads:

Density to be used for calculating dead loads of commonly used material would be as under. All other dead loads would be as per IS 875: Part 1

Material	Density
Concrete	25 kN/cum
Steel	78 kN/cum
Soil	20 kN/cum
Water	10 kN/cum
Roof sheets with purlins	0.3 kN/sqm
Brickwork	20 kN/cum
Light weight blocks	8 kN/cum
False floor or false ceiling	1 kN/sqm
Non plant area for light weight internal partitions.	1 kN/sqm
Non plant area for masonry internal walls upto 1.5m high at any location	2 kN/sqm

- **Live Loads**:

All live loads noted under are minimum value. Apart from the specified live loads, any other equipment load or possible overloading during construction/hydro-test/maintenance/erection shall also be considered in the design. If any specific live load is mentioned in dwg, that live load shall be taken into account for that component of that specific structure.

Area	Minimum Loads (kN/sqm)
Non-Plant Operating area suspended floor	5
Non-Plant Operating area grade slab floor	30 or a point load of 25 kN at any point of the slab
Mezzanine floor	3
Non- accessible roofs	1
Accessible roofs with no possibility of industrial use	3
Accessible roofs with possibility of industrial use	5
Non-plant area habitable floor	3
Non-plant area stairs, lobbies & exits	5
Non-plant area accessible terrace	3
Roads/Culverts/ Pipe crossing	20 (fire engine truck load) or 150 Kn of axle load

- **Imposed Loads**

Imposed loads shall include all equipment loads at operation condition.

- Equipment Load for Static Condition:-

Equipment Loads = Dead Load + Filling Load (for static condition)

- Equipment Load for Dynamic Condition:-
- Equipment Load= Dead Load + Filling Load + 25% of dead load if dynamic data not available
- The dead load of process equipment shall include all fixtures, platforms, ladders and attached piping. If piping weight is not indicated separately or included in the weight of the equipment, the same shall be taken as 10% of the weight of the equipment. For horizontal equipment, the weight on two supports shall be calculated based on the total weight and C.G. of the equipment.
- The weight of solar fixtures on the flat/sloping roof shall be considered during design.

Following special considerations for imposed loads shall be taken for Process Plant Building:

Ground Floor: Bitumen, Grease, Solubilization, General area without equipment pad	30kN/sqm or 200 kN static load tank standing on grade slab
Ground Floor: Coolant, Lube blending, Extract, Rex, Brake Fluid	30kN/sqm or 200 kN static load tank standing on grade slab
Ground Floor- White oil	30kN/sqm or 200 kN static load tank standing on grade slab
Ground Floor- Finished Product Warehouse, PM Storage, Additive Warehouse, Empty Drum Manufacturing area	40kN/sqm
First Floor- Grease, Solubilization, Bitumen	5kN/sqm + Imposed loads due to equipment.
First Floor- Coolant, BM Extract, Rex	5kN/sqm + Imposed loads due to equipment.
First floor- Control rooms, powder rooms, staging areas	5 kN/sqm
First Floor- White Oil	5kN/sqm + Imposed loads due to equipment.
First Floor- Plastic Can Manufacturing Area (Area between grids A, G, 10, 24)	15 kN/sqm
First Floor-All area except Plastic Cane Manufacturing area and not noted above	12.5 kN/sqm
Platforms above ground & first floor	5kN/sqm + Imposed loads due to equipment.

- EPC Contractor are advised to consider the equipment loads from the relevant attached mechanical layout drawings & equipment list for the design and type of the foundation to be provided.
- For design of RCC suspended slabs, a forklift carrying 3 ton load should be considered.

- **Wind Loads**

Parameter	Value
Code:	IS 875: 2015
Basic wind speed: Vb	44m/s
Factor k1: Probability factor	1.0 for permanent structure 0.7 for temporary structure
Factor k2:Terrain roughness & height factor	Terrain category 1, height as per structure
Factor k3: Topography factor	1
Factor k4: Importance factor	1.15 for all bldg, due to proximity to seashore
Design wind speed: Vd	Vb*k1*k2*k3*k4
Design wind pressure	0.6*Vd^2 N/sqm
Internal & external pressure coefficient:	As per IS 875: Part 3

- **Seismic Loads**

Parameter	Value
Code:	IS 1893: 2016 Part 1 & IS1893: 2015 Part 4
Seismic zone:	III
Zone factor:	0.16
Importance factor:	Process Plant, All Tank Farms, Pipe Racks=2 Utility, Low Flash Additive Storage, Laboratory =1.5 All other structures= 1
Time period:	As per IS 1893: 2016 (All Parts)
Response reduction factor:	As per IS 1893: 2016 (All Parts)
Diaphragm effect:	For RCC floor with several cut-out: flexible diaphragm For RCC floor with very few cut-out: rigid diaphragm. For steel floor (cheq plate or grating): no diaphragm
Nos of modes to be considered:	12 modes or 90% of modal mass inclusion which ever higher.
Peak response combination method:	CQC, combined quadratic combination.
Section modifiers:	For RC structures: Columns: 70% of I gross, Beams: 35% of I gross

	For steel structures: No section modifiers.
Critical damping:	5% for RCC 2% for steel structures
Seismic load distribution:	As per dynamic analysis.
Mass source for calculations of bass shear.	Dead loads : 100% Live loads : 50% Imposed loads: 75%

- Impact & Vibratory Loads

Structures subjected to impact or vibratory loads shall be designed as per the provisions of IS:875 & IS: 2974.

- Temperature Loads

Thermal Stress shall be considered based on the temperature variation in Chennai. Maximum temperature shall be considered as 40ºC and minimum temperature shall be considered as 20ºC. The ambient temperature shall be 30ºC. The casting temperature should be around 25ºC.

- **Other Loads**

Design of all structures shall also consider any other relevant and consequential load/stress imparted to the structure. All liquid retaining/storage structures shall be designed assuming liquid up to the full height of wall irrespective of provision of any overflow arrangement.

Pressure relief valves or similar pressure relieving devices shall not be made in underground water retaining/storage RCC structures.

- **Factor of Safety**

Type of Structure	Minimum FOS against Overturning		Minimum FOS against Sliding	
	With Wind/Seismic	Without Wind/Seismic	With Wind/Seismic	Without Wind/Seismic
All Buildings/Structures/ Equipment in Units	1.5	2	1.5	
All Buildings/Structures/ Equipment in Units	1.5	2	1.5	

Pipe racks	1.5	2	1.5	
Retaining Wall	1.5		1.5	
Overhead Tanks/ Tank Pads	1.5*		1.5	
Flare Supporting Structures	1.5		1.5	

*- In empty condition

b. Analysis and Design

- Load Combinations

Design shall be done to resist the worst combination of the loads as per code (IS 456, IS 800 & IS 1893) and all the structures shall be designed for the worst combination which gives maximum critical forces for all service conditions.

- Types of Analysis

For static loads:	Static analysis
For Dynamic loads	Modal analysis using Eigenvectors for determining modes of vibrations. Dynamic analysis to find out various responses
For special conditions with tension & compression limits and for steel structures	Non-linear static analysis & P-delta analysis
For long span RCC members slabs above 6.0m & RC beams above 10.0m.	Non-linear static analysis using time dependent properties
For conducting buckling check on steel structure	Buckling analysis
To understand vibration effect of steel members	Time history analysis.

c. Foundations Design

The Geo-technical report is enclosed herewith for reference for bidding purpose. EPC contractor shall design based on this report. Plant shall have pile foundation, rest all works shall have open foundation.

SITE DEVELOPMENT

Overall Plot Plan

- The Overall Plot Plan shall be developed by EPC CONTRACTOR based on indicative plot plan / equipment layout plan attached with the bid, as stipulated in Bid document, good engineering practice and meeting the requirements of various statutory bodies like OISD, OMR and local authorities, site conditions, process requirements etc. The details shown in the indicative Overall Plot Plan showing the sizes, location inter distance between equipment and direction of all equipment & facilities are indicative and for reference only. Shall be verified and firmed-up by the CONTRACTOR as per actual site condition.
- The Overall Plot Plan shall be submitted to the OWNER/PMC for review/approval after incorporating the location/co-ordinates in the drawing after necessary site verification. After receipt of the PMC/ OWNER's approval only, the Overall Plot Plan shall be valid for further engineering.

Site Grading

- The area within scope limit shall be handed over to EPC CONTRACTOR as is where basis. Land Under acquisition by client where the facilities are envisaged to be graded/developed by EPC CONTRACTOR along with the boundary wall.
- Ungraded site shall be handed over to EPC CONTRACTOR. All works related to the site grading involving cutting of earth in all types of soil, earthwork in filling up to desire levels with approved good earth as per specifications, free of clods, grass, vegetation etc shall be in EPC CONTRACTOR scope.
- Micro grading shall be done after completion of all works. Extra earth required to make up to finished grade levels shall be arranged by the EPC CONTRACTOR at his own cost from the approved borrow areas.
- Ordinary soil/mechanical sand/ murrum/ earth mix with gravel /Borrow soil/ (in case of no option due to non-availability of any kind of earth, Bottom fly ash can be used only after proper scheme of filling and approval by IOCL) used for filling shall meet the requirements from the Standard Specifications and approval of IOCL/PMC is required before finalization of material for filling.
- Borrow areas (including royalty or other fee) shall be arranged by the EPC CONTRACTOR anywhere outside the installations. Any non-expansive earth required for filling around

foundations/pits etc shall also be arranged by the EPC CONTRACTOR from borrow areas to be arranged by themselves from outside the installations.

- EPC contractor shall follow IGBC guidelines “Platinum” rating, for earth handling activities. All earth works shall be carried out as per specification laid by AP deputed at site.
- The survey / contour drawing showing spot levels shall be arranged by EPC CONTRACTOR to verify the same with the actual spot levels and firm-up the desired finished ground levels. Local cut & fill for the existing area to obtain desired level to be done by EPC CONTRACTOR. Any variation in quantities due to difference in the survey and actual levels shall not be considered as extra work and shall form a part of scope for site grading works without any extra time and cost implications.
- EPC CONTRACTOR shall construct benchmarks and grid pillars for defining coordinates and levels for proposed facilities scope battery limit. Benchmark levels shall be transferred from existing reference benchmark provided by the OWNER.
- The grading of the area shall be done by cutting and filling with the following:
    - Cutting area: Thoroughly rolled and compacted.
    - Filling area: Compacted in layers not exceeding 300mm in loose thickness to achieve minimum 95% of maximum Laboratory dry density as per IS: 2720 Part VII.
- All the FGL & FFL for different structures shall be as per the architectural drawings provided by the PMC.

Contour Survey Report

- Observations from the contour survey report.
    - The plot is almost flat with maximum & minimum contour level as -5.0 to -6.0 over entire plot.(Reference level = -4.57, Top of Wimco Nagar railway station tracks)
    - The plot has two roads, one on North Kossapur Andar Kuppam road, with closed drains on both sides, second on South Jawaharlal Nehru Salai.
    - The road on North is close to the plot boundary, whereas on road on South is 230.0m away from the plot boundary.
    - Top of north road is -4.0m level, top of South road is -3.4m level.
    - No major trees on the plot, two transmission lines cutting across the plot which are planned by IOCL to shift through TNSEB.

- Recommendations:
  - Fill up the plot & match with the North side road level top, so as to keep the plot at road level and avoid any chances of flooding in future.
  - The edge compound wall would need a retention wall of varying height, varying from 1.0m to max 2.0m, most of the plot would be having retention height of approx. 1.5m.

Roads

- Road Width: As per relevant drawings.
- Road Type: Concrete roads over Water Bound Macadam layer.
- Camber: As per IRC
- Side Slope: 1Vertical: 1.5Horizontal
- Radius of the curve: As per relevant drawings.
- CBR to be used for design: As per Geo-tech report.
- Design Standard: IRC 37 (latest revision)
- Traffic Data: 500 heavy vehicle per day (including in and out)
- Road to have road end curbs on edge as shown in the dwg.
- All roads
- Use of course sand in RCC for roads, may be permitted in the road.

Site Finish

Area	Concrete
Operating/Equipment Area	RCC Grade Slab
Roads & Approaches	RCC M30 150mm thick with Water Bound Macadam Layer below
Patrolling Walkway	Same as Road
Tank Farms (inside Dyke)	PCC M15, 150mm thick
Parking (including truck parking)	Road finish
Truck Loading/Unloading Area	Road finish

Storm Water Drainage System

- Storm water drainage system within proposed facilities shall consist of well-designed open surface drains so that all the storm water is efficiently drained off without any water logging/ flooding.

Trench levels of the drains shall be fixed such that there is no back flow from the drains into the proposed facilities.

- Storm water drains within scope shall be designed to cater the requirement of rainwater and all the storm water drains to be planned independently. Open Rectangular Storm water drains shall be provided on one side of the road. However, there shall be branches at intervals to collect the water from the other side. Wherever very crucial, the drain may have to be on both sides of the road. Material of construction of drains shall be RCC, walls & base minimum 150mm thk. Storm water shall be discharged at recharge well or rainwater collection tank as shown in the block plan. The recharge wells shall be designed with adequate capacity, based on water retention capacity of soil. The recharge well shall be designed by the specialist. In case, if required recharge well may be re-located based on recommendation by the specialist. Max Rainfall Intensity of storm water drains shall be considered as per the IMD data for past 5 years. Design Rainfall Intensity = Max Rainfall intensity. Duration of Max rainfall intensity
- would be considered as 15 minutes.
- Run-off Coefficient
- Hydraulic Design parameters
  - Velocity (Maximum) = 2.4m/s
  - Velocity (Minimum) = 0.6m/s

a	Paved Area	Concrete	1.0
		Bitumen	0.9
b	Compacted Area such as Tank Farms/ Offsite areas/ expansion areas		0.7
c	Unusable open area/ green belt area		0.4 to 0.6

- Drain slope.
  - Slope (Rectangular drain) = 1:500 (Preferable)
  - Slope (Trapezoidal drain) = 1:1000 (Preferable)

Water Supply (for domestic use)

- Max fixed population per shift : 220
- Total fixed population per day, inclusive of visitors: 400
- Total floating population (including TT) : 730
- Max water consumption per day : 100 KL per day.

- Meals to be prepared & served in canteen: Breakfast/ Lunch/Dinner.
- Fresh Water supply source : Raw water tank.
- Flushing Water supply source: STP/if not available from STP, raw water should be used.
- Water from Raw water tank shall be pumped using Hydro-Pneumatic system to the water outlets in Security, Office, Lab, Canteen, Amenity, North Gatehouse, Utility, Driver Toilet & Toilets(2 nos) in plant building. Water, for domestic use, required for entire plot shall be distributed by Hydro-Pneumatic system to all the points.. Stand-by water pump shall be provided, in case of failure of main pump of Hydro-Pneumatic system.
- All drinking water points, as indicated in dwg, shall have RO filter with provision to dispense water at room temperature, hot water, cold water. Drinking water requirement for fixed population shall be 10 lit per day, for floating population shall be 5 lit per day.
- The fresh water shall be supplied from the raw water tank to the overhead tank by means of pipes buried under ground or exposed as feasible. When underground it shall have chambers at not more than 50m centers for maintenance work. The chamber shall have a spool piece for the same
- The flushing water shall be supplied from the STP to Raw Water Tank. Raw water tank to have dedicated compartment for flushing water storage, dedicated pipelines from flushing tank compartment to be run into the flushing fixtures in any works.
- The system shall be such designed that a pressure of 0.3 - 0.5 bar is availed at all spouts.
- While construction is in progress, EPC CONTRACTOR shall supply potable water to IOCL/PMC site office in 20 lit cans, in required nos.

Domestic Drainage

- All domestic drainage shall be done using PVC sewer pipes of adequate diameter.
- Food waste from Kitchen to be used for making vermin-compost.
- Chambers to be provided over the drainage lines at distance not exceeding 12 m
- All domestic drainage shall be discharged in STP. EPC CONTRACTOR to decide nos of STP and its location based on discharge locations.
- STP shall be designed using MBBR technology. Performance to match with the requirements of Central Pollution Control Board

- The treated water from STP shall be used for gardening or pumped into the dedicated flushing tank. EPC CONTRACTOR shall lay pipes lines to pump treated water to dedicated flushing tank & for gardening purpose.

Compound Wall

- Compound Wall shall be as per standard drawing. It shall be constructed using brick walls. It shall have RCC Columns every 3.5 m. It shall have RCC wall up to 1200mm from FGL, overall height of compound wall shall be 3000mm + concertina. Expansion joints shall be provided every 30 m maximum. Concertina wire to be fixed over the boundary wall
- The design of the compound wall should include wind loads as per IS 875: Part 3 (2015) & seismic loads as per IS 1893: Part 1 (2016).
- Compound Wall shall be painted with Acrylic Emulsion Paint.

Culverts & Road Crossings

- Storm Water drains/ water supply lines shall be connected by Hume Pipes under roads. RCC Pipe Way Bridges shall be provided for crossing of process/utility pipelines under road.

Entry Arches

- Entry Arches shall be erected as per the dimensions given in the drawing. These arches should be built in such a manner that it reflects the culture and architectural heritage of Tamilnadu.Stone cladding to be done over these arches using locally available stone to further enhance the traditional Architecture.

Bollards & Boom Barrier

- Necessary civil works to be done for fixing of static Bollards and Automatic rising (Traffic Control) Bollards, as required by other discipline.
- Necessary civil works to be done for fixing boom barrier at all location as required by other discipline.

Speed Breaker

- Heavy duty speed breaker of suitable design & material to be provided convenient for all types of vehicles of black and yellow colour.

Road Traffic Signs

- Road traffic signs for easy navigation within installation, shall be placed overall roads and junction. Specification of road sign to be as per IRC standard.

Road Paint

- Road will be painted for the edge marking, zebra crossing, etc using Thermoplastic fast drying highly durable 100% acrylic retro-reflective water borne paints

2 Special considerations for RCC structures

Sr No	Parameter	Remarks
1	Exposure condition	Concrete works above ground: Severe as per IS 456:2000 (concrete surfaces exposed to severe wetting & drying cycle due to presence of coast) Concrete works below ground: Very Severe as per IS 456:2000 (concrete in contact with or buried under aggressive subsoil / groundwater)
2	Minimum cement content for exposure condition	As per IS 456:2000
3	Minimum concrete clear cover for exposure conditions:	As per IS 456:2000
4	Minimum bar diameter for various structural members	Footings : 10mm Columns longitudinal bars: 12mm Columns links: 8mm Beams longitudinal bars: 12mm Beam links: 8mm Slab: 8mm Minor equipment foundation: 10mm Major equipment foundation: 12mm
5	Concrete grade for RCC works	M30 and above
6	Concrete grade for PCC works	M20 for Severe Exposure Condition
7	Reinforcement	TMT Fe 500D grade
8	Minimum width or thickness of structural elements:	Footing : 350mm Column: 300mm Beam width: 230mm

		Grade or suspended slab: 150mm UG liquid retention structure wall: 200mm Above ground liquid retention structure wall: 150mm UG pit base slab: 300mm ETP drain walls: 150mm Storm waters drain walls : 125mm
9	Modulus of Elasticity of concrete for short term deflection	5000√Fck
10	Maximum crack width for Liquid retention structure	0.2mm
11	Expansion Joints	It shall be provided after maximum 80 m in either directions for buildings. It shall be provided every 30 m for compound wall.
12	Fire Rating	4 hours for plant bldg, MCC rooms & Sub-station. 2 hours for all other structures.
13	Long term deflection Parameters	Suspended Slabs & Beams: Long term deflection shall not exceed span/350 or 20 mm whichever is less. Critical equipment resting on pedestal or grade slab: 5 mm
14	Shuttering to RCC elements	Above ground: Form Finish Below ground: Wrought Form Finish
15	Piles	Minimum 500mm dia piles, with minimum 6 bars of 16mm dia. Minimum vertical reinforcement of 0.6% of gross pile cross section.
16	Cement	53 grade OPC for structural use of concrete 43 grade PPC for non-structural use of concrete/mortar

3 Special considerations for steel structures

Sr No	Parameter	Remarks
1	Steel grade	Hot Rolled Steel: IS 2062, E250A Cold Rolled Steel: IS 1079:2009, Grade HR5, minimum tensile strength 370 MPA (design shall be as per IS:801)
2	Hot rolled steel shapes	I, C, L, SHS, RHS, CHS, plates
3	Cold rolled shapes	Lipped channel
4	Cold rolled steel usage criteria	Only if the area & structure is non-corrosive type
5	Bolts grade for all structural connection	5.6 grade minimum for static loads, minimum 8.8 grade HSFG for dynamic loads & column splices
6	Deflection criteria for various structural members:	Lateral deflection of bldg: h/200 Vertical deflection of beam: L/325

		Vertical deflection of beams supporting equipment : L/500 or as per equipment requirement For Purlins: L/200
7	Beam to column connections:	Preferred Moment connection (may be altered to shear if so designed): Bolted connection for erection purpose, after erection full penetration butt weld to top & bottom flanges to column flange or web
8	Beam to beam connections:	Shear connection: Shear cleat welded to primary beam & bolted to secondary beam.
9	Equipment support beams	Shear connection: Shear cleat welded to primary beam & bolted to secondary beam. With provision to seating plate or seat cleat.
10	Bolted column splice	Bolted column splice to be provided at change in column section or col length exceeds maximum member length permitted.
11	Welded column splice	Welded column splice shall be provided for to join two pieces of same section, not more than one welded splice in maximum member length permitted. Full penetration butt weld shall be used.
12	Welded beam splice	Maximum one splice permitted in on span. No welded splice in equipment support beams. Full penetration butt weld shall be used.
13	Usage of decking sheet on floor	1.0 mm minimum decking sheet thickness. All decking sheets shall be fixed to steel beams using shear studs.
14	Surface protection for steel members	All structure to be painted with approved painting scheme as given below. Rates are inclusive of painting. a) Surface preparation by shot blasting conforming to SA 2.5 (to be done at shop) b) Prime coat of Epoxy Based Zinc Ethyle Silicate Primer (min. 60% vol. solids), one coat of 75 - 80 micron DFT. (to be done at shop c) Intermediate coat of TWO PACK EPOXY HIGH BUILD MIO COATING (min. 60% vol. solids). one coat of 100 - 125 micron DFT (to be done at field after touch up primer on welded surfaces). d) Final coats of TWO PACK ALIPHATIC ACRYLIC POLYURETHANE (min. 52% vol. solids), Two coats of combined 60 -70 micron DFT (to be done at field/Shop).

		Total DFT shall be min. 225 micron. Painting shall be done strictly as per the instruction of IOCL/CONSULTANT. Intermediate/Final Coat may be required to provide at Shop/field as directed by IOCL engineer in charge. (Rate shall include cost of supply of paint, primer, surface preparation, scaffolding, painting complete in all respect as per instructions of CONSULTANT/IOCL Engineer-in-Charge). Primer & paint shall be of approved make as specified by CONSULTANT/IOCL.
15	Steel painting after erection	c) Intermediate coat of TWO PACK EPOXY HIGH BUILD MIO COATING (min. 60% vol. solids). one coat of 100 - 125 micron DFT (to be done at field after touch up primer on welded surfaces). d) Final coats of TWO PACK ALIPHATIC ACRYLIC POLYURETHANE (min. 52% vol. solids), Two coats of combined 60 -70 micron DFT (to be done at field/Shop). Total DFT shall be min. 225 micron. Painting shall be done strictly as per the instruction of IOCL/CONSULTANT. Intermediate/Final Coat may be required to provide at Shop/field as directed by IOCL engineer in charge.
16	Painting to Handrails	Red Oxide Primer with two coats of Polyurethane paints
17	Hand-railing on steel (chequered plate / grating) floor	Height: 1.2m from finish floor Verticals & Top rail: 50mm dia GI pipes. Middle rail : 25mm dia GI pipe Toe plate: 100 x 6mm
18	Hand-railing on stairs	Height: 1.2m from finish floor Verticals & Top rail: 50mm dia GI pipes. Middle rail : 25mm dia GI pipe No Toe plate
19	Hand railing on concrete floors	Height: 1.2m from finish floor Vertical to be fixed on 200mm high RC curb wall. Verticals & Top rail : 40mm dia GI pipes. Middle rail : 25mm dia GI pipe No Toe plate
20	Chequered plate	6.0mm minimum, with paint specification same as structural steel on both surfaces
21	Grating- Electro Forged	All gratings shall be fabricated using Electro-forging process. & hot-dipped conforming to IS 4759: 1996 and IS: 2629:1985 Or MS grating with painting scheme same as for structural

		steel works. 25mm high x 5mm thick minimum for walkways & maintenance floor. 35mm x 5mm thick minimum for operating floor.
22	Grating for steps:	25mm x 5mm thick minimum. All grating for steps to have chequered plate on foot edge. Or MS grating with painting scheme same as for structural steel works.

4 Special considerations for masonry works

- All masonry works above plinth shall be designed in accordance with IS:1905, IS:1597, IS:2185, IS:4326, IS:12894 and other relevant IS Codes as applicable. All external and internal walls shall be made of 200 thick AAC blocks or with thickness as noted in the drawing.
- All masonry works below plinth shall be accordance with IS:1905, IS:1597, IS:2185, IS:4326, IS:12894 and other relevant IS Codes as applicable.
- Cement Mortar
    - All masonry work shall be constructed in cement sand mortar 1:5 for load bearing walls and 1:6 cement sand mortar for non-load bearing walls except half brick partition walls which shall be constructed in 1:4 cement sand mortar with two numbers of 6mm diameter MS bars provided at every fourth course properly anchored with cross walls or pillars.

5 Design criteria- specific applications

Machine Foundations

- Machine foundations shall satisfy the requirements of IS: 2974 and any other parameter as per machine vendor
- The minimum grade of concrete to be used shall be as per clause B.4
- Minimum reinforcement as per requirements of IS: 2974 shall be provided unless required otherwise by design.
- All units of the foundation system, except foundation raft shall be provided with symmetric reinforcement on opposite faces, even if not required by design considerations.

- The soil stress below foundations under dead loads shall not exceed 80% of the allowable soil bearing capacity for static loading.
- The combined centre of gravity of the machine and foundation system shall, as far as possible, pass through the centre of area of the foundation raft or centroid of the pile group. Where unavoidable, eccentricity shall be less than 5% for block foundations and 3% for frame foundations. However, in highly compressible soils, no eccentricity shall be permitted.
- Foundations shall be so designed that natural frequency of the foundation system shall not resonate with the following:
    - Operating speed of the motor.
    - Operating speed of the machine.
    - 2 x operating speed of the machine.
    - Critical speed of the machine (for centrifugal machines).
- Natural frequency of the foundation shall preferably be ±20% away from the above mentioned frequencies. However, amplitude of vibration of the foundation block shall always be checked to be within permissible limits.
- Amplitudes of vibration shall be less than values specified by the machine manufacturer. If not specified, provision of IS: 2974 shall be followed.
- The foundation and its superstructure shall be separated from adjacent foundations and platforms. Clear air gaps shall be provided in the superstructure to avoid transmission of vibration to adjacent structures. Special note shall be given on the drawing in this respect, and suitable details shall be shown as required
- Foundations resting partly on rock and partly on soil shall be avoided. If unavoidable, the soil area shall be replaced by lean concrete (1:4:8). However, IOCL/PMC‟s concurrence shall be obtained for such cases prior to taking up of design.
- In case location of foundations on previously back filled or sensitive soils becomes unavoidable, IOCL/PMC‟s concurrence shall be obtained for any soil treatment prior to taking up of design.
- For frame foundations, base raft shall be cast in a single concreting operation. A properly designed construction joint shall be provided between the base slab and columns. The entire superstructure of columns and upper deck shall be cast in a continuous concreting operation.
- If height of the frame columns above raft level exceeds 8.0m, an additional construction joint at the junction of columns/top-deck may be provided.

- Block foundations shall be cast in a single concreting operation. (i.e., Single pour)

Tank Pads

- The foundations for all tanks in all tank farms shall be optimized based on Geo-tech report. The foundation shown in dwg is only indicative. :
- Impermeable membrane, sand-witched between 75mm layer of sand bedding shall be installed under plinth to eliminate soil contamination.
- Ring walls for flat bottom tanks & skirt supported tanks shall be minimum 300mm thick, and foundation shall be laid at minimum 3000mm below the natural ground level.
- All flat bottom tank pads shall have 75mm thick Sand-Bitumen pad on top, with minimum 300mm sand layer below.
- All concrete works upto 1.0m height above FGL shall be considered as RCC works upto plinth for tank farms & pipe racks.
- For tank foundation with ring-wall foundation settlement analysis for testing conditions shall be carried out. During construction of foundation the provision for settlement shall be made as per settlement analysis carried out. No settlement of tank permitted under operating conditions.
- A stability analysis of tank foundation shall be carried out taking into account following aspects:
    - Wind Loads
    - Earthquake Loads
    - Initial height above adjacent ground level
    - Highest possible ground water level
    - Geo-technical properties of tank foundation materials and subsoil
    - Load of the tank and its content during hydro static testing, operation and maintenance.

Pipe Racks

- Structural Arrangement of Pipe Rack
    - The pipe racks shall be designed as a series of transverse portals or column with fixed at the base in the transverse direction. Beam to column connections shall be moment resistant. A knee bracing may be provided below the lowermost tier in the transverse direction so as to provide a clear passageway under the pipe rack.
    - Longitudinal beams shall be provided between the columns. TOS of longitudinal beams shall be same as TOS of transverse beams on lower tier. The longitudinal frames shall have shear

connections at beam/column junctions. At the base, the columns shall be assumed to be hinged in the longitudinal direction.

- Horizontal loads in the longitudinal direction shall be transferred to all foundations. Bracing shall be designed as tension only, to restrict longitudinal movement.
- Cable tray supports shall be provided at every 2 meters or less. Pipe supports shall be provided at 5.0 m or less.

- **Vertical Loads on Pipe Racks**
  - Pipe racks shall be designed in the empty and operating conditions assuming that the vertical load per tier is as under: (The pipe racks loads shall be validated after the loads are available from other discipline)
    Empty Condition: 1.5 kN/sqm
    Operating Condition: 3.0 kN/sqm
  - This is based on 150mm pipes at 350mm C/C.
  - Pipe rack junctions and crossings shall be designed for 1.5 times the above loads,
  - Cable tray loading shall be taken as provided by Electrical / Instruments (generally 1.5 kN/sqm per tray).

- **Piping Loads on Pipe Racks**
  - Pipes larger or equal than 300mm diameter shall be considered as concentrated loads in their actual locations.
  - When individual pipe loads are not available for racks with pipes exceeding 300mm diameter, but not exceeding 400mm diameter the piping may be considered as a distributed load of not less than 3.0 kN/m for the operating condition and 5.0 kN/m for test condition.
  - The general forces produced by the thermal expansion and contraction of pipes sliding on a pipe rack, shall be treated as internal forces in the structure (i.e., not transferred to the foundations).
  - Such internal forces shall be catered for in the design of the pipe support beam and its appurtenances by taking into consideration a horizontal force transverse to that beam and applied to the top flange, equal to 10% of the vertical load due to all pipes on that beam during the operating condition.

- If the pipelines are piggable lines, 20% of vertical load of piggable lines should be applied along longitudinal direction along with transverse direction of force.
- Transverse forces due to wind acting on piping shall be calculated in accordance with the expression: W=Dn + 0.1L where W is force due to wind in transverse direction, Dn is the greatest pipe diameter and L the width of the pipes deck.
- The actual loading shall be used if it exceeds the above minimum loading requirements.
- Note: When equipment producing important dynamic loads (i.e., air cooler) is erected on a structure, the vibration loads have be taken in account for the design of the structure.
- When the piping (operating) vertical loads are given in terms of area load, 30% of the vertical loads are to be used as horizontal forces for local member design. (0.3x0.707xFx + 0.3x0.707xFz)
- When the horizontal loads are used in combination with wind / seismic; t0% of the vertical loads are to be used as horizontal forces. (0.1x0.707xFx + 0.1x0.707xFz)

- Thermal Loads on Pipe Racks: Generally, the thermal loads shall be issued by piping stress analysis group. These loads shall be used for design of pipe supports/structures. The design of structure and foundation shall satisfy following minimum requirements.
  - The calculation of thermal loads (TL) shall be as per the requirements of the governing codes and specification. Thermal Loads caused by expansion and contraction due to a change in temperature from the erection condition shall be carefully considered. Included are forces due to anchorage of piping and equipment, sliding and rolling of equipment, and expansion and contraction of structures.
  - Thermal loads due to the constraints and frictional forces of piping shall be considered as follows:

    (a) A minimum of 10% (in both longitudinal & transverse directions) of the gravity load for the pipe supports / sleepers carrying 4 or more lines.

    (b) A minimum of 30% of the gravity load in longitudinal direction & 10% in transverse direction on pipe supports / sleepers carrying less than 4 lines. Equipment on structural supports shall be analysed for thermal loads to be resisted by the structure and provisions shall be made to relieve the forces too large for the equipment or the supporting structure.
  - The co-efficient for friction factors to be used are the ones defined here under. Steel to Steel = 0.3 Steel to PTFE Pad = 0.08 Steel to Concrete = 0.413

- All concrete works upto 1.0m height above FGL shall be considered as RCC works upto plinth for tank farms & pipe racks.
- All structural steel for pipe rack shall be painted as noted in painting scheme for PEB structural steel works in SOW.

Earth Retaining Structures (UG Tanks, Retention Walls etc)

- Earth pressure shall be considered for structural elements below ground level. Active Earth Pressure Coefficient shall be at least 0.35, higher if recommended by soil investigation report.
- All rafts below ground level should be designed for buoyancy forces. Water level shall be considered at 500 mm below FGL.
- Surcharge load of 10 kN/sqm shall be considered for all Retention walls, UG Tanks etc.
- Angle of fill for retention wall design to be considered as 0.0 degree

6 Construction requirements- general

- All construction works shall be carried out as noted in specification, as per relevant IS codes noted in specification. In case if EPC contractor wish to deviate from the noted standard, he shall obtain written approval from PMC for doing so. In case of any works that are not noted in specification and are required to be carried out, EPC contractor shall do the same, as per specification furnished or mutually agreed by EPC & PMC.
- All concrete works shall be carried out as per the provisions of IS: 456, IS: 3370, IS: 2974 and other relevant IS codes. Concrete Mix Proportioning and Design Mix; Sampling and Strength test of Concrete; Testing of Structures, Production and Control of Concrete tolerances and placing of reinforcement and for cover; Transporting, Placing, Compacting and Curing etc., shall be as specified in IS: 456.
- Continuous concreting shall be done for structures supporting dynamic equipment as per the provisions of IS: 2974.
- Form work and stripping of form work shall be as per the provisions of IS: 456.
- Assembly of reinforcement in RCC structures shall conform to IS: 456.

- All piles for main structures shall be driven piles, shall conform to IS 2911 . All testing procedure of piles shall be strictly followed as per provision made in the code. For minor works like compound wall, etc, under-ream piles would be permitted.
- All excavation works carried out for the building foundation works shall be as noted in specification
- Fabrication of all structural steel works shall be carried out as per the provisions of IS: 800/801/802/806 and other relevant IS codes. Fabrication shall include cleaning, straightening, cutting, bending, holding, bolting, welding, machining, painting, marking, assembling, erecting, inspecting and testing etc.
- Erection of all structural steel works including supply of plant & equipment, storing and handling, setting out, field connections, field welding and security during erection shall conform to IS: 800/801/802/806.
- All masonry works shall be carried out as per the provisions of IS: 1597/2212/4326 and other relevant IS codes.

7 Construction requirements- specific

- Apart from the conditions mentioned in the design requirements given in the document, the following shall be strictly adhered to.
- Cable/pipe trenches & precast slab covers shall be designed to withstand the load of hydra-crane. Seating surface of the slab shall be at least 100mm wide with structural ISA50x50x6 edge protection embedded throughout the length of the trench.
- Only steel shuttering in good condition shall be used for civil construction unless form finish is required.
- EPC CONTRACTOR shall make necessary arrangement for placing the anchor bolts in position before concreting. Whenever there are more than four foundation bolts, these shall be fixed by using template. In case bolts are not available at site at the time of casting of foundation, proper pockets shall be left as per direction of the Engineer-in-charge. The bolts after fixing shall be protected with plastic sheet covers, to prevent any damage to the threading
- EPC CONTRACTOR to ensure isolation of structures/equipment with difference of temperature for free expansion while providing interconnecting platform and for connection to the stair structure.
- Buildings shall be provided with anti-termite treatment.

- All designs, detailing & construction shall strictly conform to enclosed standards, specifications & drawings. However, drawings marked “Issued for Information only” are for only guidance to the EPC CONTRACTOR.
- Sequence of construction is to be shown on the AFC drawings by indicating construction joints wherever required.
- No equipment shall be directly supported on suspended floor slab. Suitable arrangement of beams shall be provided underneath to support the equipment.
- The foundation design shall be based on approved Geo-technical investigation recommendations. Geo-technical investigation shall be in scope of the EPC CONTRACTOR. However, Geo-technical report attached in bid for reference only.
- EPC CONTRACTOR shall depute his concerned Civil-Structural design engineer to IOCL /PMC review office as and when required for review of his documents.
- During such reviews involving computer aided analysis/design/drafting of structures, the EPC CONTRACTOR shall make his own arrangement of Personal Computer (PC) in the form of Lap-top in the premises of IOCL/PMC review office. This is required to expeditiously resolve all the comments including those involving the use of PC by EPC CONTRACTOR in his submission. The EPC CONTRACTOR shall ensure that these PC's are fully operational along with necessary software already loaded including the input/output/drawing files of the structures being reviewed. The EPC CONTRACTOR shall revise and re-submit the analysis/design and drawings as required.
- Care shall be taken during construction of UG Tanks during monsoon, so that they are not displaced due to water collection.
- In case of water retaining structures, concrete should be tested for leakages and necessary grouting should be done to plug leakages before plastering is done. Approval of PMC shall be taken after testing of leakages. Plastering works shall commence only after approval of PMC.
- EPC contractor to co-ordinate with other disciplines for all cut-outs and penetration required in concrete/masonry and execute the same.

8 Architectural design basis

REFERENCE PUBLICATIONS & STANDARDS

1. National Building Code of India - 2016

2. Factory Law/ State Govt. Factory Acts
3. Local Municipality or any other Authority's Bye-laws as applicable.
4. OISD Standards
5. BIS (Bureau of Indian Standards) Codes
6. Any other applicable Law, Rules, Standard as referred in respective clause.

Note: All codes & standards shall be latest revision with all amendments issued there to. However, effort shall be made for effective utilization of space and facilities for cost effective design.

GENERAL ARCHITECTURAL REQUIREMENTS

Spatial Requirements

- Spatial requirements of the buildings shall mainly be decided based on the activities to be performed in the building, equipment/panel layout and consequent occupancy pattern, clearances, maintenance, and safety requirements. The objective of spatial arrangement shall be to satisfy functional requirements, physical comfort, and safety regulations as well as aesthetics. Requirement of services/utilities such as air conditioning, LAN etc. shall be finalized as per Owner's specific requirement.
- Spatial requirements inside a building shall be decided based on activities to be classified as follows, which shall be provided in all the buildings/sheds.

Lighting

Established level of illumination shall be maintained for all parts of the buildings by means of windows, ventilators etc. Provision of referenced publications like NBC part-VIII; Section-1; IS-2440-1975; IS-3646 (part II-1996); IS-7662 (part I-1974), Factory rule, green building codes or other relevant rules etc. shall

- be adhered to in this regard. Openings shall be provided with shading devices to avoid glare. In general, all Sun-shade are 900mm wide. For the purpose of illumination, day lighting shall also be supplemented by artificial illumination.

Ventilation

Natural Ventilation

- Established level of ventilation in terms of air changes per hour shall be maintained for all spaces as per the provision of referenced publications like State Factory rules, NBC part-VIII Section-1,

IS:3101-1975 (Industrial building), IS:3362-1975 (Residential buildings), IS-:7662 (Part I-1974) or other relevant code/ rules. Natural ventilation shall also be supplemented by mechanical or electrical means of ventilation in all human occupied areas. Sufficient no. of Glazed/Louvered windows/Ventilators shall be provided and supplemented by exhaust fans.

Mechanical ventilation

- In addition to natural ventilation, if required Mechanical or electrical ventilation shall be provided depending on the type of building and its use.

Safety Requirements

- Safety from fire and like emergencies shall be taken into account in building design as per NBC-Part IV; State Factory Rules and other relevant code/ rules. The buildings shall be provided with exits sufficient to permit safe escape of occupants in case of emergency. The exits shall be in terms of doorway, corridors, etc. to internal/ external staircase or to areas having access to the outside. Railings shall be provided to steps outside the building whenever the plinth height exceeds 900mm. All emergency doors to open outside & shall have a view panel & grab bar to open the door, in emergency.

Aesthetics

- Apart from the fulfilment of functional & safety requirement, aesthetic requirement of the building shall be taken care of in the design. Preliminary drawings indicating Architectural scheme shall be submitted for Owner's approval.
- Architectural scheme shall be based on general principles of Aesthetics. Building facades shall reflect such principles like symmetry, balance, proportion, rhythm, light and shade etc.
- Building Elements like canopies, overhangs & shading devices, gutters, roof projections, parapets, door; window/ ventilator composition, External wall/ facade shall be considered as contributory elements to aesthetics. Local Climatic conditions shall be considered while designing the buildings.
- Architectural scheme including design of above mentioned elements shall be subjected to Owner/ PMC approval.

Steps/Ramps/Stairs

- Steps/ ramps shall be provided for access to the Building for pedestrian/ vehicular, equipment entry as per relevant code. Minimum 1200 mm wide platform shall be provided in between entrance door and steps/ramps. This extended platform and floor near the entrance shall have outward slope to avoid ingress of rainwater splash. It shall be 150mm below the plinth level, except in case of emergency exits. The external steps shall be finished with IPS and shall have edge angles unless mentioned otherwise.
- Following dimensions of the steps/ ramps shall be adhered to.
    - Stairs width 1500mm clear minimum for access to all working areas
    - 600/ 900/ 1000mm wide for access to non-working areas
    - Tread 300 mm minimum for access to working areas, 200mm minimum for access to non-working areas.
    - Riser 150 mm maximum for access to working areas, 200mm maximum for access to non-working areas.
    - Ratio of Tread & Riser 2 Riser + Tread= 600mmor as per requirement
    - Maximum no. of risers per flight 12 Nos. or as per requirement
    - Landing width 1500 mm minimum or as equal to flight width for dog legged stairs.
    - Mid-landing width of minimum 600 mm for single flight stairs.
    - Width of ramp 1500 mm minimum or as per requirement.
    - Slope of Ramp shall not be steeper than 1:10 or as per requirement
- Ramp access for physically challenged shall be provided for important buildings such as admin, Canteen, Security etc. or else specified. Design of ramps shall be as following:
    - Width: 1800mm minimum
    - Slope of Ramp: Greater than 1 in 20 for long distance or maximum of 1 in 12 for short distance up to 9000mm.
    - Railing: Ramp shall have G.I. handrails on at least one side, and preferably two sides, or if there is a wall a handrail shall be provided, that are 900 mm high, measured from the surface of the ramp, that are smooth, and that extend 300 mm beyond the top and bottom of the ramp. The railing shall be as shown in the drawing.
    - Ramp surface: Non-slip surface.

- Level Platform: Ramp shall have a level platform at the top which is at least 1800 mm long, if a door swings out onto the platform or toward the ramp. This platform shall extend at least 300 mm beyond each side of the doorway.
- Level Platform intervals: Ramp shall have level platforms at 10 m to 12 m intervals for purposes of rest and safety and shall have platforms minimum 1.5 m length wherever they turn.
- Straight clearance at the bottom: Ramp shall have at least 1800 mm of straight clearance at the bottom.

Canopy/ Overhangs/ Sun-shade

- Canopy/overhangs shall be provided at all entries & exits for rain & sun protection as shown in the drawing.
- Non RCC roof overhangs shall be provided beyond the face of the wall unless gable wall is shown. Size of the overhang minimum of 600 mm.
- Waterproofing & drainage provision shall be provided for the canopies & overhangs.
- Sun-shade over the openings and window shall be 900mm deep and shall project 150mm beyond the face of each opening. In case of a few buildings like security, weigh bridge cabin, the overhang is different and shall be as per the drawing.

Rainwater Pipes

- Rainwater pipes shall be provided for roof water drainage. Number of rainwater pipes shall be decided on the basis of roof area, slope and rainfall intensity as per NBC Part IX Section 2. RCC or GI spouts may be used for drainage of sun-shade/ small canopies of ground floor.
- All the rainwater generated should be collected through rainwater pipes and conveyed to SW drains, from which it should be conveyed into the recharge well or rain water collection tank. They shall be fixed using a clamp so that there is a distance of about 40mm between the pipe and the wall/ cladding face.

Parapet

- Parapets shall be of RCC for all buildings with minimum 300/600 mm (as per drawing) high for small non-approachable roof and 1200 mm high for approachable roof or roofs with water tank/ solar panels.

Railings

- G.I. railings of 1000 mm height shall be provided in stairs and in all unprotected openings in slabs as a safety device. Railings in loading/ unloading bay of shall be of open able type, wherever required. Railing for disabled ramp are specified earlier.

Plinth Protection

- Plinth protection 1000mm wide shall be provided around all structures to prevent ingress of water. It shall be made of PCC M10 laid on GSB filling laid on well compacted earth filling.

External Plaster & External Paint

- For all building with masonry walls, 20mm thick sand faced plaster shall be applied in 2 coats (12 + 8mm). All junctions of sun-shade and wall to be rounded and drip mould made on the bottom side of the sun-shade / canopies/ overhangs.
- The external paint shall be Acrylic Emulsion Paint shall be applied which is Anti Dust, Anti-Fungal & Anti Algal. The paint shall have a manufacturer's warranty of minimum 7 years.

Finish Floor Level

- The finish floor level shall be as mentioned in the drawing. In all exits, the outside open plinth shall be a step down to avoid water ingress. In case of emergency exits it shall be at the same level. In both the case they shall slope outwards to avoid water ingress.

Walls

- All walls below plinth shall be 230mm thick brickwork. Walls above plinth shall generally be 230mm thick AAC block work, or as noted in the drawing.

Internal Plaster

- 12mm thick cement plaster shall be applied to all internal surfaces except, slab bottom, beam bottom and internal beam sides. Chicken wire mesh shall be fixed at all flush joints between RCC and masonry, before plastering. Wall care putty shall be applied to all internal RCC surfaces having form finish shuttering.
- Before applying wall care putty, all concrete surfaces shall be made even using grinder, so that all surfaces look even after application of putty.

Waterproof Plaster

- For all water retaining structures, 20 mm thick waterproof plaster shall be applied after completion of testing leakages.

Door, Windows, Ventilators, Precast RCC Grills and Cills

- The doors/ windows/ ventilators/ precast RCC grill to be used shall be as per the drawing. Door / Window /Ventilator sizes & profile shall be matched, as far as possible.
- RCC grills shall have rain protective design, minimum overall width of 100 mm.
- The requirement of the cill with/ without step shall be as mentioned in the drawing. The joint between the door/ window and the stone to be made watertight with sealant. The exposed edges of the stone cills shall be given a moulding while the middle edge formed in case of door cills shall have edge polish. The stone to be used shall be mentioned in the drawing.
- The glass used for all doors, windows and ventilator shall be: External window glass, if WWR (wall area to window area ratio) is less than 40%, the glass SHGC (Solar Heat Gain Coefficent) shall be 0.42, when WWR is more than 40%, the glass SHGC shall be 0.36, max U value (U-Value: U-value indicates the rate of heat flow due to conduction, as a result of temperature difference between inside and outside. The lower the U-factor, lower the heat transferred through the glass. U-Value is measured in W/m2oK.).

Roofing Sheet & Cladding

- Roofing and cladding sheets shall be galvalume sheets, Double skin (with insulation) or Single skin (without insulation) with the minimum SRI value of 78, 0.55mm of Total coated thickness, 0.45mm of Bare metal thickness and a laid slope of minimum 3°. All sheets, single or double skin roof shall be non-screwing type profile (KLIP-LOK).
- Air extractors and poly carbonate light sheets shall also be provided in case of closed structures. Poly carbonate light roof sheeting shall not exceed 5% of roof plan area. Thickness of polycarbonate sheet shall be 2 mm. The profile of Polycarbonate sheet should match the profile of metal sheets.
- Air extractors and poly carbonate light sheets shall also be provided in case of closed structures. Poly carbonate light roof sheeting shall not exceed 5% of roof plan area. Thickness of polycarbonate sheet shall be 2 mm. The profile of Polycarbonate sheet should match the profile of metal sheets.
- Cage Ladder and safety wire rope shall be provided for access to the roof. In case of double side sloping roof, safety wire ropes will be provided on both sides. Safety wire ropes shall be SS, 0.5mm dia. GI Metal gutter matching to sheet profile shall be provided.
- Roof accessories like ridge, bends, flashing shall be provided wherever necessary with matching roof profile.

- All metal Roof to be designed for Solar panel load with maintenance walkway & handrail provided over the roof. Roofing for covered parking is noted separately. All roofing sheets shall be of profile, so as solar panel frames, walkway & hand-railing frames can be clipped to the sheet, without puncturing the sheet.

Terrace Finish

- China Mosaic chips made of broken glazed tiles procured from local dumping sites shall be laid over brickbat coba. All wall floor junctions shall have coving. Adequate slopes shall be given to collect the water into rain water pipes. The parapet wall plaster shall project beyond the start point of the coving to avoid cracking. A small depression shall be made at the point of discharge into the rain water pipe outlet. A metal screen shall be placed at that junction to avoid dirt going into the pipes.

Water Proofing

- Waterproofing shall be provided to all wet areas like the terrace, cooler and the toilet sunk slab as per the water proofing specifications.

Cage ladder/ Ladder

- All structures shall be provided with cage ladder to access the terrace/ roof, where access heights exceed 3.5m from FGL where there is no staircase. In structures where access height is less than 3500mm, ordinary wide M.S. ladder will be provided. All ladder, ordinary or cage shall be 600mm wide minimum.

Skirting/ Wall Tiles in Cooler Area

- No skirting shall be provided in areas with VDF, IPS finish, walls with dado tiling, external areas. In internal areas the skirting shall be 75mm high of the same finish as the flooring. Skirting will also be provided on the partitions. In areas with cooler, glazed tile dado on the wall behind and around cooler upto 2.1m height and on the sides and top of the curb wall around cooler.

Water Supply & Drainage

- CPVC pipes of adequate diameter shall be used for both external and internal building supply lines. The exposed lines shall be fixed using a clamp so that there is a distance of about 40mm between the pipe and the wall face. There shall be enough water force at all locations. PVC drainage pipes of adequate diameter shall be provided for both external and internal drainage lines. The exposed lines shall be fixed using a clamp so that there is a distance of about 40mm between the pipe and the wall

face. Stop cock shall be provided to each WC, urinal, shower, basin so that there is no disturbance during repairing. A master stop cock shall also be provided.

Anti-termite treatment.

- Anti-termite treatment as noted in specification shall be provided to the plinth of all covered spaces.

Architectural, Civil & Structural Specification

B. Architectural, Civil & Structural Specification

1 Specification for works

GENERAL NOTES

i. The detailed specifications given hereinafter are for the items of works described in the drawing & design basis, and shall be guidance for proper execution of work to the required standards.

ii. It may also be noted that the specifications are of generalised nature and these shall be read in conjunction with the design basis & drawings. The work also includes all minor details of construction which are obviously and fairly intended and which may not have been referred to in

these documents but are essential for the entire completion in accordance with standard Engineering practice.

iii. Unless specifically otherwise mentioned, all the applicable codes and standards published by the Bureau of Indian Standards and all other standards which may be published by them before the date of receipt of tenders, shall govern in all respects of design, workmanship, quality and properties of materials and methods of testing, method of measurements etc. Wherever any reference to any Indian Standard Specifications occurs in the documents relating to this contract, the same shall be inclusive of all amendments issued thereto or revisions thereof, if any, up to the date of receipt of tenders.

iv. In case there is no I.S. specification for the particular work, such work shall be carried out in accordance with the general guidelines of manufacturer/specialised agency subject to full satisfaction and approval of the Engineer-in-Charge.

v. The work shall be carried out in a manner complying in all respects with the requirements of relevant bye-laws of the Municipal Committee/Municipal Corporation/Development Authority/Improvement Trust under the jurisdiction of which the work is to be executed or as directed by the PMC and, unless otherwise mentioned, nothing extra shall be paid on this account.

vi. Samples of various materials, fittings etc. proposed to be incorporated in the work shall be submitted by the contractor for approval of the PMC before order for bulk supply is placed.

vii. The EPC CONTRACTOR shall take instructions from the PMC regarding collection and stacking of materials in any place. No excavated earth or building materials shall be stacked on areas where other buildings, roads, services, compound walls etc. are to be constructed.

viii. The EPC CONTRACTOR shall maintain in perfect condition all works executed till the completion of the entire work awarded to him. Where phased delivery is contemplated, this provision shall apply to each phase.

ix. The EPC CONTRACTOR shall give a performance test of the entire installation(s) as per standard specifications before the work is finally accepted and nothing extra whatsoever shall be payable to the contractor for the test.

x. The EPC CONTRACTOR shall clear the site thoroughly of all scaffolding materials and rubbish etc. left out of his work and dress the site around the building to the satisfaction of the PMC before the work is considered as complete.

xi. Post construction inspection and testing: After completion of the work and during maintenance period liability of the contractor, the work shall also be subjected to 'Post construction inspection and testing'. In case the materials or articles incorporated in the work are found to be inferior, though the sample collected for the same might have been passed at the time of execution, it shall be the responsibility of the EPC CONTRCATOR to replace the same at his own cost, failing which the PMC may rectify the same at the risk and cost of the EPC CONTRACTOR or PMC /OWNER

may accept the work as sub-standard, and cost be adjusted from the outstanding security deposit, as per the terms and conditions of the contract for the work.

xii. The PMC/OWNER, shall be the sole deciding authority as to the meaning, interpretations and implications for various provisions of the specifications and his decision in writing shall be final and binding on all concerned.

xiii. In case any difference or discrepancy between the specifications and the design basis, design basis shall take precedence.

2 List of standards codes

Following are the Indian Standards, relevant to building works	
(All Latest Versions of I.S. code shall be referred)	
I. S. CODE NO.	**SUBJECT**
1. CARRIAGE OF MATERIALS	
4082-1996	Recommendations on stacking & storage of construction. materials and components at site.
2. EARTH WORK	
632-1978 (Reaffirmed 2002)	Specification for Gamma-BHC (Lindane) Emulsifiable Concentrate.
1200 (Part-1) 1992	Method of measurement of Earth work.
4081-1986	Safety code for Blasting and related drilling operations.
6313 (Part 2) – 2001	Anti-Termite Measures in Buildings –Part-2- Pre-constructional chemical treatment Measures.
6313 (Part 3) – 2001	Anti-Termite Measures in Buildings – Part-3- Treatment for existing buildings
6940-1982	Methods of Test of Pesticides and their formulation.
8944-2005	Specification for Chlorpyrifos Emulsifiable Concentrate.
8963-2006	Technical Specification- Chlorpyrifos
3. MORTAR	
196-1966	Atmospheric conditions for testing (Reaffirmed - 1990)

269-2013	Ordinary Portland cement, 33 Grade
383-1970	Coarse and fine aggregates from natural sources for concrete.
455-1989	Portland blast furnace slag cement
650-1991	Standard sand for testing of cement
1344-1981	Specifications for calcined clay pozzolana
1489-1991	Portland pozzolana cement Fly ash based
1542-1992	Sand for Plastering.
1727-1967	Methods of tests for pozzolanic materials
2116-1980	Sand for Masonry mortar
2250-1981	Code of practice for preparation and use of masonry mortar. (Reaffirm-1990)
2386-1963	Methods of Test for Aggregates for Concrete
2386 Pt.I-1963	Particle size and shape
2386 Pt. II-1963	Estimation of deleterious materials and organic impurities
2386 Pt.III-1963	Specific gravity, density, voids, absorption and bulking
3025-1964	Methods of sampling & test (Physical & Chemical) water used in industry. (Reaffirmed-2003)
3812 (Part-1)-2013	Pulverised Fuel ash: For use as pozzolana in cement, cement mortar and concrete.
3812 (Part-2)-2013	Pulverised Fuel ash: For use as admixture in cement mortar and concrete
4031 (Part-1 to 13) - 1996	Methods of physical tests for hydraulic cement
4032-1985.	Method of chemical analysis of hydraulic cement (Reaffirmed - 1990)
8041-1990	Rapid hardening Portland cement
8042-1989	Specifications for white cement
8043-1991	Hydrophobic Portland cement
8112-2013	Specification 43 grade ordinary Portland cement

12269-2013	Specification 53 grade ordinary Portland cement
4. CONCRETE WORK	
383-1970	Coarse and fine aggregate from natural sources for concrete (Reaffirmed - 1990)
456-2000	Code of practice for plain and reinforced concrete
516-1959	Method of test for strength of concrete (Reaffirmed in 2004)
1199-1959	Method of sampling and analysis of concrete
1200 (Part-2)-1987	Methods of measurements of cement concrete work. (Reaffirmed - 1992)
1322-1993	Bitumen felts for water proofing and damp proofing. (Reaffirmed - 1998)
2386 (Part-1 to 8) - 1977	Methods of test for aggregate for concrete
2505-1992	Concrete vibrators- Immersion type-General requirements
2506-1985	General requirements of screed type vibrators
2645-2003	Integral water proofing compounds for Cement Mortar and Concrete - Specification
7861-1975 (Part-1)	Hot weather concreting. .(Reaffirmed -1990)
7861-1981 (Part-2)	Cold weather concreting. .(Reaffirmed -1992)
9103-1999	Admixture for concrete.
5. R.C.C. WORK	
432-(Part-1)-1982	Mild steel & medium tensile steel bars and hard drawn steel wire for concrete reinforcement.- Mild steel & medium tensile steel bars
432 (Part-2)-1982	Mild steel & medium tensile steel bars and hard drawn steel wire for concrete reinforcement - Hard drawn steel wire
456-2000	Code of practice for plain and reinforced concrete
457-1957	Code of practice for general const. of plain & reinforced concrete for dams & other massive structure.
516-1959	Methods of test for strength of concrete
1161-1998	Steel tubes for structural purposes

1199-1959	Methods of sampling and analysis of concrete. (Reaffirmed - 1999)
1200 (Part-2)-1974	Method of measurement of cement concrete work
1200 (Part-5)-1982	Method of measurement of form work. (Reaffirmed - 1989)
1343-2012	Pre-stressed Concrete - Code of practice
1566-1982	Hard drawn steel wire fabric for concrete reinforcements (II Rev.) (Reff.1998)
1608-2005	Method of tensile testing of steel products
1785-1983 (Part-I& II)	Specifications for plain hard drawn steel wire for pre-stressed concrete
1786-2008	High Strength Deformed Steel Bars and Wires for Concrete Reinforcement - Specification
1893 (Part-1)-2002	Criteria for Earthquake resistant design of Structures(General provisions and Building)
1893 (Part-4)-2005	Criteria for Earthquake resistant design of Structures (Industrial structures including stack like structures)
2090-1983	Specifications for high tensile steel bars used in prestressed concrete.
2204-1962	Code of practice for construction of reinforced concrete shell roof. (Reaffirmed - 1990)
2210-1988	Criteria for the design of shell structure and folded plates (Reaffirmed - 1998)
2314 -1986	Steel Sheet Piling Sections.
2502-1963	Code of practice for bending and fixing of bars for concrete reinforcement. (Reaffirmed - 1999)
2750-1964	Specifications for steel scaffoldings
2751-1979(Reaf-1992)	Code of practice for welding of mild steel bars used for reinforced concrete construction.
2911-(Part-1)-(Section-1 to 3)2010	Code of practice for design & Construction of pile foundations
2911(Part-1-Section-4)-2010	Design & construction of Pile Foundations – Bored/driven precast concrete piles.
2911 (Part-3)-1980	Under reamed pile foundations

2911 (Part-4)-1985	Load test on Piles
3201-1988	Criteria for design and construction of precast concrete trusses. (Reaffirmed - 1995)
3370. (Part I to IV)	Code of practice for concrete structures for storage of liquids. (Reaffirmed 1999)
3414-1968	Code of practice for design and installation of joints in buildings. (Reaffirmed - 1990)
3558-1983(Reaf-91)	Code of practice for use of immersion vibrators for consolidating concrete
3696 (Part 1 & 2)	I-1987: Safety code of scaffolds; II-1991: Safety code of ladders
3935-1966	Code of practice for composite construction. (Reaffirmed – 1998)
4014-1967(Part1 & 2)	COP for steel tubular scaffolding (I: Definition/Material; II: Safety Resolutions) (Reaffirmed 1999)
4326-2013	Earthquake resistant design and construction of Buildings – Code of Practice
4926-2003	Code of practice for Ready Mix Concrete
4990-2011	Plywood for concrete shuttering work - Specifications
5121-1969	Safety code for piling and other deep foundation
9716-1981	Guide for lateral dynamic load test on piles
10262-2009	Guidelines for Concrete Mix Design Proportioning
13311(Part-1)-1992	Non-destructive testing of concrete-Method of test for ultrasound pulse velocity.
13311(Part-2)-1992	Non-destructive testing of concrete-Method of test by rebound hammer.
13920-1993(Reaffirmed 2003)	Code of Practice-Ductile detailing of Reinforced concrete structures subject to seismic forces.
14362-1996	Pile boring equipment-General requirements.
6. EQUIPMENTS	
460-1985(Pt-I,II& III)	Specification for test sieves. (Reaffirmed - 1998)

1791-1985	Specification for batch type concrete mixer. (Reaffirmed – 1990)
2430-1986	Methods for sampling of Aggregates for concrete.
2505-1992	General requirement for concrete vibrators, immersion type
2506-1985	General requirements for screed board concrete vibrators
2514-1963	Specification for concrete vibrating tables. (Reaffirmed - 1991)
2722-1964(Reaf-95)	Specification for portable swing weigh batchers for concrete (single and double bucket type).
2750-1964	Specification for steel scaffolding. (Reaffirmed – 1991)
7. BRICK WORK:	
1077-1992	Common burnt clay building bricks
1200 (Pt.III)-19920	Method of measurements of brick work. (Reaffirmed - 1992)
2116-1980	Sand for masonry mortars. (Reaffirmed - 1998)
2212-1991	Code of practice for brick work
2222-1991	Burnt clay perforated building bricks.
2849-1983	Specification for non-load bearing gypsum partition blocks(solid and hollow types)
3495 (Part 1 to 4)-1992	Method for test for burnt clay building brick
4139-1989	Specification for calcium silicate bricks
5454-1978	Method for sampling of clay building bricks. (Reaffirmed - 1995)
12894-2002	Pulverized fuel ash- lime bricks specification
13757-1993	Specification of burnt clay fly ash bricks
8. STONE WORK:	
1121(Part-1to4)	Methods for determination of compressive, transverse, tensile & shear strengths of natural building stones
1122-1974	Methods for determination of specific gravity and porosity of natural building stones
1123-1975	Methods for identification examination of natural building stones

1124-1974	Methods of test for water absorption of natural building stones
1125-2013	Determination of weathering of natural building stones – Method of Test
1126-2013	Determination of durability of natural building stones – Method of Test
1129-1972	Dressing of natural building stones
1200 (Part-4)-1992	Method of measurement of stone masonry.
1597. (Part-1)-1992	Code of practice for construction of Rubble stone masonry. (Reaffirmed - 1996)
1597 (Part-.2)-1992	Code of practice for construction of Ashlar masonry (Reaffirmed - 1996)
1805-1973	Glossary of Terms relating to stone Quarrying and dressing. (Reaffirmed - 1993)
4101 (Part-1)-1967	Stone facing. (Reaffirmed - 1990)
9. MARBLE WORK:	
1122-1974	Methods for determination of specific gravity and porosity of natural building stones
1124-1974	Methods of test for water absorption of natural building stones
1130-1969	Marble (blocks, slabs and tiles)
3316-1974	Structural granite
14223(Part-1)-1995	Polishing building stones-Granite.
10. WOOD WORK:	
204-1991/92	Tower bolts (Part I-1991: ferrous metals; Part II - 1992 : Nonferrous metals).
205-1992	Non-ferrous metal butt hinges
1734 - 1983	Methods of tests for plywood (IIR) (Ref 1993)
206-2010	Tee and strap hinges - Specification
207-1964	Gate and shutter hooks and eyes. (Reaffirmed - 1996)
208-1996	Door handles - Specification
281-2009	Mild steel sliding door bolts for use with padlocks - Specification
287-1993	Permissible moisture contents of timber used for Different purposes –

	Recommendations.
303-1989	Plywood for general purpose
362-1991	Parliament hinges
363-1993	Hasps and staples
364-1993	Fanlight catch
401-2001	Preservation of timber Code of practice
419 - 1967	Putty for use on window frame (I Rv.) (and out 3)
451-1999	Technical supply condition for wood screws
452-1973	Door springs, rat-tail type(II Rev.) (Reaffirmed 1990)
453-1993	Double acting spring hinges. (Reaffirmed – 1999)
707-2011	Glossary of term applicable to timber technology and utilization.
710-2010	Specifications for marine ply.
723-1972	Steel counter sunk head wire nails. (Reaffirmed - 1996)
729-1979	Drawer locks, cupboard locks, and box locks (III Rev.) (Reaffirmed 1992)
848-2006	Specification for Synthetic resin adhesive for plywood (phenolic and aminoplastic)
851-1978	Synthetic resin adhesive for construction work (Non-structural) in wood (I-Rev.) (amt) (Reaffirmed 1990)
852-1994	Specifications for animal glue for general wood working purposes.
875 (Part-1) - 1987	Dead loads – Unit weight of building materials & stored materials
883-1994	Design of structural timber in Building.
1003(Part-1I)-2003	Timber panelled and glazed shutters (Door shutters)
1003 (Part-2)-1994	Timber panelled and glazed shutters(windows and ventilator shutters)
1019-1974	Rim latches. (Reaffirmed - 1991)
1141-1993	Code of practice for seasoning of timber (II Rev.)

1200	Method of measurement of Building and Civil Engineering works
1200 (Part-14)-1984	Glazing. (Reaffirmed - 1990)
1200 (Part-21)-1973	Wood work and joinery. (Reaffirmed - 1992)
1328-1996	Veneered decorative plywood
1341-1992	Steel Butt hinges (VI Rev.)
1378-1987	Oxidized copper finishes. (Reaffirmed - 1998)
1568-1970	Wire cloth for general purposes. (Reaffirmed - 1998)
1658-2006	Fibre hard board - Specifications
1659-2004	Block boards
1708-1986	Method of testing of clear specimen of timber (II Rev) (Q.1) (Reaffirmed 1990)
1823-1980	Floor door stoppers. (Reaffirmed - 1992)
1837 – 1966	Fanlight pivots (I Rev.) (Reaffirmed 1990)
1868-1996	Anodic coating on Aluminium & its alloy (II Rev.) (Reaffirmed 1991)
2046-1995	decorative thermosetting synthetic resins bonded laminated sheet.
2095 (Part 1) – 2011	Gypsum plaster board – Part 1 – Plain Gypsum Plaster Board.
2095 (Part 2) – 2001	Gypsum plaster board – Part 2 – Coated/ Laminated Gypsum Plaster Board.
2096-1992	Specification for A.C. flat sheet.
2191(Part1)-1983	Wooden flush door shutter (cellular and hollow core type)- Plywood face panels
2191 (Part2)-1983	Particle board face panels and hard board face panels. (Reaffirmed-1991)
2202-(Part1)1999	Wooden flush door shutters (solid core type)- Plywood face panels
2202 (Part2)-1983	Particle board face panels for wooden flush door shutters. (Reaffirmed - 1991)

2209(Part1)-1976	Mortise locks (vertical type) (Reaffirmed 1992)
2380-1981	Method of test for wood particle board and boards from lignocellulosic materials (Reaf.1993)
2547-1976	Specification for gypsum plaster
2681-1993	Non ferrous metal sliding door bolts (aldrop) for use with pad locks
2753-1991	Method for estimation of preservatives in treated timber and its treating solutions.
2835-1987	Flat transparent sheet glass (3rd Revision). (Reaffirmed - 1992)
3087- 2005	Particle boards of wood and other lignocellulosic materials (medium density) for general purposes - Specification
3097- 2006	Veneered particle boards - Specification
3400 (Part I)-2012	Method of test for vulcanized rubbers Part 1: Tensile Stress – Strain Properties
3400-(Part2)-2003	-Hardness
3400-(Part4)-1987	-Accelerated aging
3400 (Part9)-2003	-Relative density and density.
3564-1995	Door closers (Hydraulically regulated)
3618-1966	Phosphate treatment of iron and steel for protection against corrosion. (Reaffirmed - 1991)
3818-1992	Continuous (Piano) hinges
3828 – 1968	Ventilator chains (Reaffirmed 1990)
3847-1992	Mortise Night Latches
4020-1998 (1 to 16)	Methods of tests for wooden flush Doors .
4021-1995	Timber door, window and ventilator frames
4835 -1979	Polyvinyl acetate dispersion base adhesive for wood (1990)
4948-2002	Welded steel wire fabric for general use. (Reaffirmed - 1992)
4992-1975	Door Handles for mortise locks (vertical type). (Reaffirmed - 1990)

5187-1972	Flush bolts (1990)
5509-2000	Specification for Fire Retardant Plywood
5523-1983	Method of testing anodic coating on aluminium & its alloys. (Reaffirmed - 1991)
5930-1970	Mortise latch (vertical types) (1991)
6318-1971	Plastic window stays & fasteners
6607-1972	Rebated mortise locks (vertical type)
6760-1972	Slotted countersunk head wood screws. (Reaffirmed - 1988)
7196-1974	Hold fasts (1992)
7197-1974	Double action floor springs (without oil check) for heavy doors
7534-1985	Sliding locking bolt for use with padlocks. (Reaffirmed – 1991)
7638-1999	Wood/lignocellulosic based panel products-Method for sampling
8756 - 1978	Mortise ball catches for use in wooden almirah (1992)
11215-1991	Moisture content of timber and timber products-Method of determination.
12406 - 2003	Medium density fibre board for general purpose - Specification
12817-2013	Stainless steel Butt hinges
12823-1990	Specification for wood products-Pre-laminated particle boards
14856-2000	Glass fibre reinforced plastic (GRP) panel type door shutters for internal use - Specifications
14900-2000	Specification for transparent float glass.
11. STEEL WORK:	
63-2006	Whiting for paints
198-1978	Varnish, gold size. (Reaffirmed - 1991)
228--1987	Methods of chemical analysis of Structural steel

277-2003	Specification for galvanised steel sheets (plain and corrugated)
278-2009	Galvanised steel barbed wire for fencing - Specification
800-2007	General Construction in Steel - Code of practice
806-1968	Code of practice for use of steel tube in general building construction
813-1986	Scheme of symbols for welding. (Reaffirmed – 2003).
814-2004	Covered electrodes for manual metal arc welding of carbon and carbon manganese steel
817-1966	Code of practice for training and testing of metal arc welders. (Reaffirmed – 2003)
818-1968 (Reaf-03)	COP for safety & healthy requirements in electric & gas welding & cutting operation.
822-1970	Code of practice for inspection of welds..
1038-1983	Steel doors, windows and ventilators
1081-1960(Reaf-91)	COP for fixing & glazing of metal (steel & aluminium) doors, windows & ventilators
1148-2009	Steel rivet bars (Medium and High Strength) for structural purposes
1161-1998	Steel tubes for structural purposes
1182-1983(Reaf-00)	Recommended practice for radiographic examination of fusion welded butt joints in steel plates.
1200 (Part8)-1993	Method of measurements of steel work and iron works
1363-2002 (Part 1 to 3)	Hexagon head bolts, screws and nuts of product grade C
1599-2012	Metallic Materials – Bend Test
1608-2005	Mechanical testing of materials - tensile testing
1821-1987	Dimensions for clearance holes for metric bolts. (Reaffirmed - 2003)
1852-1985	Rolling and cutting tolerance for hot rolled steel products. (Reaffirmed - 1991)

2062-2011	Hot rolled medium and high tensile structural steel
4351-2003	Steel door frames. (Reaffirmed – 1991)
4736-1986	Hot-dip zinc coatings on steel tubes. (Reaffirmed – 2001)
4923-1997	Hollow steel sections for structural use-specification
6248-1979	Metal rolling shutters and rolling grills.
6911-1992(Reaf-1998)	Stainless Steel plates, Sheet and Strip- Specification.
7452-1990	Hot rolled steel sections for doors, windows & ventilators.
12. FLOORING:	
210-2009	Grey iron casting
653-1992	Sheet linoleum
IS: 802 Part 2 :	Code of Practice for use of Structural Steel in Overhead Transmission
809-1992	Rubber flooring materials for general purpose
IS: 808 : 1989	Dimensions for Hot Rolled Steel Beam, Column, Channel and Angle
IS:816 : 1969	Code of Practice for use of Metal Arc Welding for General Construction in Mild steel
1122-1974	Methods for determination of specific gravity (*and porosity of natural building stones)
1124-1974	Method of test for water absorption of natural building stones
1130-1969	Marble (blocks, slabs and tiles). (Reaffirmed – 1993)
1197-1970	Code of practice for laying of rubber floors. (Reaffirmed – 1990)
1198-1982	Code of practice for laying and maintenance of linoleum floors
1200 (Part11)-1977	Method of measurements of paving and floor finishes.
1237-2012	Cement concrete flooring tiles - Specifications
IS:1367 Part 1 To 3	Technical Supply Conditions for Threaded Steel Fasteners

& 5 To 7 : 2002	
1443-1972	Code of practice for laying and finishing of cement concrete flooring tiles
1661-1972	Code of practice for application of cement and cement lime plaster finishes
2114-1984	Code of practice for laying in situ terrazzo floor finish
2571-1970	Code of practice for laying in situ cement concrete flooring
3462-1986	Flexible P.V.C. Flooring. (Reaffirmed – 1991)
IS:3502 : 1994	Specification For Steel Chequered Plates
IS:3658 : 1999	Code of Practice for Liquid Penetrant Flaw Detection
3670	Code of practice for construction of timber floor
IS:3757 : 1985	High Strength Friction Grip Structural Bolts
IS:4260 : 2004	Recommended Practice for Ultrasonic Testing of Butt Welds in Ferritic Steel
4457	Acid and/or alkali resistant tiles.
4631-1986	Code of practice for laying of resin floor toppings (Reaffirmed – 2001)
5318-1969	Code of practice for laying of flexible P.V.C. sheet & tiles flooring
IS:5334 : 2003	Code of Practice for Magnetic Flaw Detection of Welds
IS:5369 : 1975	General Requirements for Plain Washers and Lock Washers.
IS:5372 : 1975	Specification – Taper Washers for Channels.
IS:5374 : 1975	Specification – Taper Washers for I – Beams
5389-1969	Code of practice for laying of hardwood parquet and wood block floors. (Reaffirmed – 1998)
IS:6639 : 2005	Specification for Hexagon Bolts for Steel Structures
IS:7205 : 1974	Safety Code for Erection of Structural Steel Work
IS:7215 : 1974	Tolerances for Fabrication Of Steel Structures

IS:7307 Part 1 : 1974	Approval Tests for Welding Procedures Part 1- Fusion Welding of Steel
IS:7310 Part 1 : 1974	Approval Tests for Welders working to Approved Welding Procedures. Part1Fusion Welding of Steel
IS:7318 Part 1 : 1974	Approval Tests for Welders when Welding Procedure Approval is not required. Part 1- Fusion Welding of Steel
IS:8500 : 1991	Weldable structural steel (Medium and High Strength qualities)
9197-1979	Specifications for epoxy resin, hardeners and epoxy resin compositions for floor topping (Reaffirmed – 2001)
IS:9595 : 1996	Recommendation for metal arc welding of carbon manganese steel
IS:12843	Tolerances for Erection Of Steel Structures 1989
13630 (Part1to15) - 2006	Methods of tests for ceramic tiles .
13712-2006	Specification for ceramic tiles; definition, classification ,characteristic and marking.
15622-2006	Specification for pressed ceramic tile
15685-2006	Precast Concrete Blocks for paving - Specification
13. ROOFING	
73-2013	Paving Bitumen
277-2003	Galvanised steel sheets (plain and corrugated)
458-2003	Concrete pipes (with and without reinforcement)
459-1992	Unreinforced corrugated and semi-corrugated asbestos cement sheets
651-2007	Glazed stone ware pipes and fittings
702-1988	Industrial Bitumen
1199-1959	Method of Sampling & Analysis of concrete. (Reaffirmed - 1991)
1200 (Part9)-1973	Method of measurements of roof covering (including cladding)

1200 (Part10)-1973	Method of measurements of ceiling and lining
1322-1993	Bitumen felts for water proofing and damp-proofing. (Reaffirmed -1988)
1346-1991	Code of practice for waterproofing of roof with bitumen felts
1609-1991	Code of practice for laying damp proof treatment using bitumen felts
1626-1994(Part 1-3)	Asbestos cement building pipes, gutters and fittings (Spigot and socket types)
1834-1984	Specification for hot applied sealing compounds for joints in concrete. (Reaffirmed - 1990)
1838-(Part 1 & 2)-1983	Preformed filler for expansion joints in concrete- non-extruding and resilient type Bitumen impregnated fibre). (Reaffirmed - 1990)
2115-1980	Code of practice for flat roof finish: mud phuska. (Reaffirmed - 1998)
2633-1986	Method of testing uniformity of coating on zinc coated articles. (Reaffirmed – 2001)
3007-(Part1)-1999	Code of practice for laying of corrugated asbestos cement sheets. (Reaffirmed – 1991)
3007-(Part2)-1999	Code of practice for laying of semi corrugated asbestos cement sheet. (Reaffirmed - 1991)
3348-1965	Fibre insulation boards. (Reaffirmed - 1990)
3384	Specification for bitumen primer for water proofing and damp proofing.
3607-1979	Magnesite for chemical Industry. (Reaffirmed – 2003)
4671	Expanded polystyrene for thermal insulation purposes.
7193-2013	Glass fibre base Bitumen felts
8183-1993	Bonded mineral wool. (Reaffirmed 2004)
14862	Fibre cement flat sheets- specifications.
14871	Specifications for products in fibre reinforced cement-Long corrugated or Asymmetrical section sheets and fittings for roofing and cladding.

14. FINISHING	
75-1973	Linseed oil, raw and refined. (Reaffirmed – 2003)
77-1976	Linseed oil, boiled, for paints. (Reaffirmed - 1999)
104-1979	Specification for ready mixed paint, brushing, zinc chrome, priming. (Reaffirmed - 1999)
109-1968	Specification for ready mixed paint, brushing priming plaster to Indian Standard colour, white and off white.
133-2004	Enamel, interior (a) under coating (b) finishing colour as required
137-1965	Ready mixed paint, brushing, matt or egg-shell flat, finishing, interior, to Indian Standard Colour, as required. (Reaffirmed – 1999)
158-1981	Ready mixed paint, brushing, bituminous, black lead free acid alkali, water and heat resisting for general purposes. (Reaffirmed – 1999)
168-1993	Ready mixed paint, air drying for general purpose.(Reaffirmed 2002)
217-1988	Cut back bitumen (reaffirmed 1999)
218-1983	Creosote and anthracene oil for use as wood preservatives (Reaffirmed 1998)
337-1975	Varnish, finishing interior. (Reaffirmed – 2001)
341-1973	Black Japan, types A, B, and C (Reaffirmed 2002)
347-1975	Varnish, shellac for general purpose. (Reaffirmed – 2001)
348-1968	French polish. (Reaffirmed – 2001)
419-1967	Putty for use of window frames. (Reaffirmed – 2001)
427-2013	Distemper, dry, colour as required.
428-2013	Washable distemper - Specification
524-1983	Varnish, finishing, exterior, synthetic air drying. (Reaffirmed – 2000)
525-1968	Varnish, finishing, exterior and general purposes. (Reaffirmed –2001)

533-2007	Gum spirit of turpentine (oil of turpentine)
712-1984	Specification for building limes. (Reaffirmed - 1995)
1200 (Part12)-1976	Method of measurements of plastering and pointing
1200 (Part13)-1994	Method of measurements of white washing, colour washing, distempering and other finishes
1200 (Partt.15)-1987	Methods of measurements of painting, polishing & varnishing.
2095 (Part1-3)	Gypsum plaster boards
2096-1992	Asbestos cement flat sheets.
2339-2013	Aluminium paint for general purposes – Specification
2547-1976 (Part2)	Gypsum building plaster-premixed light weight plasters.
2932-2003	Enamel synthetic, exterior (a) Under coating (b) Finishing.
2933-1975	Enamel, Exterior (a) Under coating (b) Finishing
5410-2002	Cement paint - Specification
6278-1971	Code of practice for white washing & colour washing. (Reaffirmed -1991)
13607 – 1992	Ready Mixed Paint, Finishing, General Purposes, Synthetic (Reaffirmed 2002)
15489-2013	Paint, Plastic Emulsion - Specification
15. ALUMINIUM:	
733-1983	Wrought Aluminium and Aluminium alloys, bars, rods and sections(For General Engineering purpose)-Specification.
737-2008	Wrought Aluminium and Aluminium alloys sheet and strip for General Engineering purpose)-Specification.
1285-2002	Wrought Aluminium and Aluminium alloys, Extruded round tube and Hollow section and sections(For General Engineering purposes))-Specification.
1868-1996	Anodic coating on Aluminium and its alloys- Specification.

1948-1961	Specification for Aluminium Doors, windows and ventilators.
2553 – 1990 (Part 1)	Safety Glass- General Purpose (For toughened and laminated glass)
3908-1986	Specification for Aluminium equal leg angles.
3909-1986	Specification for Aluminium unequal leg angles.
5523-1983	Method of testing anodic coating on aluminium and its alloy.
6477-1983	Dimensions of extruded hollow section and tolerances.
14900-2000	Transparent Float glass-Specification.
16. WATER PROOFING:	
1322-1993	Specifications for bitumen felts for water proofing and damp proofing
2645-2003	Specifications for Integral cement water proofing compounds.
3370-(Part1)2009	Code of practice for concrete structures for the storage of liquid-General requirements
3370-(Part2)2009	Code of practice for concrete structures for the storage of liquid-Reinforced concrete structures.
3384-1986	Specifications for bitumen primer for water proofing and damp proofing.
7193-2013	Glass fibre base bitumen felts.
12200-2001	Provision of water stops at transfers construction joints in masonry and concrete dams-code of practice.
12432(Part3)-2002	Application for spray applied insulation- Code of practice: Polyurethane/ polyisocyarurate.
17. DEMOLITION AND DISMANTLING:	
1200(Part18)-1974	Method of measurements of building- demolition and dismantling.
4130-1991	Demolition of buildings - Code of Safety

18. ROAD WORKS:	
73-2013	Paving Bitumen
164-1981	Ready mixed paint for road marking
217-1988	Cut Back bitumen
334-2002	Glossary of terms relating to bitumen and tar.
702-1988	Specification for industrial bitumen.
1195-2002	Specification for l bitumen mastic for flooring
1834-1984	Specification for hot applied sealing compound for joints in concrete.
2720(Part5)-1985	Method of test for soil: determination of liquid limit and plastic limit.
2720(Part7)-1980	Method of test for soil: determination of water content - dry density relation using light compaction.
2720(Part27)-1974	Method of test for soil: determination of dry density of soils in place, by sand replacement method.
5640-1970	Method of test for determining aggregate impact value of soft coarse aggregates.
IRC 10	Recommended practice for borrow pits for road embankments constructed by manual operation.
IRC 29	Specification for bituminous concrete for road pavements
IRC 36	Recommended practice for construction of earth embankments for road works.
IRC 60	Tentative guidelines for the use of lime fly ash concrete as pavement base of sub base.
IRC 88	Recommended practice for lime fly ash stabilized soil base/sub base in pavement construction.
IRC 107	Tentative Specification for bitumen mastic wearing courses.
19. SAFETY CODES	

818-1968 (Reaffirm-2003)	Safety and healthy requirements in Electric and gas welding and cutting operations.
3696 (Part-I)-1987	Safety code for scaffolds
3696 (Part-II)-1991	Safety code for ladders
3764-1992	Safety code for Excavation works
4081-1986	Safety code for blasting and related drilling operation
4130-1991	Safety code for Demolition of Building
5916-1970	Safety code for construction involving use of hot bituminous materials
6922-1973(Reaffirm-2003)	Criteria for safety & Design of structures subject to underground blasts.
7293-1974	Safety code for working with construction machinery.
20. OISD CODES	
CPCB	Central Pollution Control Board of India
DGMS	Directorate General of Mines Safety
IER	Indian Electricity Rules
OISD	Oil Industry Safety Directorate
OMR	Oil Mines Regulations
IRC	Indian Road Congress
IS:3370	Code of practice for concrete structures for storage of liquids.
IS:4326	Code of practice for earthquake resistant design and construction of buildings

3 Mandatory tests

NOTES:

The mandatory tests shall be carried out when the quantity of materials to be used in the work exceeds the minimum quantity specified.

Optional tests specified or any other tests shall be carried out in case of specialised works or important structures as per direction of the PMC.

Testing charges, including incidental charges and cost of sample for testing shall be borne by the EPC Contractor for all mandatory tests.

In case of non-IS materials, it shall be the responsibility of the EPC Contractor to establish the conformity of material with similar IS specification by carrying out necessary tests. Testing charges including

incidental charges and cost of sample for testing shall be borne by the EPC Contractor for all such tests.

THE MANDATORY TESTS SHALL BE AS FOLLOWS

Material	**Test**	**Field / laboratory test**	**Test procedure**	**Minimum quantity of material / Work for carrying out the test**	**Frequency of testing**
	b) cube test	Lab	IS: 516	5 Cum.	Same as Per Frequency of Sampling guidelines of IS 456-2000
Ready mixed cement concrete (At RMC Plant) (IS-4926)	Cube test	Lab (At RMC Plant Lab)	IS-516 and as per para 6.3.2 of IS-4926-2003	50 Cum	One for every 50 Cum of production or every 50 batches, whichever is the greater frequency
Note: For all other small items and where RCC done in a day is less than 5 cum, Cube test may be carried out as required by Engineer-in-Charge.					
Sand	Bulking of Sand	Field	IS 2386 – Part 3	dec	Every 20 Cum or part there of or more frequently as decided by Engineer-in-Charge

	Silt content	Field	IS 2386 – Part 2	20 Cum	Every 20 Cum or part there of or more frequently as decided by Engineer-in-Charge
	Particle size and distribution	Field or Laboratory as decided by the Engineer -in-Charge	IS 2386 – Part 1	40 Cum.	Every 40 Cum of fine aggregate / sand required in RCC. Work only
	Organic Impurities	Field	IS 2386 – Part 2	20 Cum	Every 20 Cum or part thereof or more frequently as decided by Engineer-in-Charge
Cement	Fineness (m2/kg)	Lab.	IS 4031 (Part-2)	Each fresh lot	Every 50 MT or part thereof
	Normal consistency	Lab.	IS 4031 (Part-4)	-Do-	-Do-
	Setting time (minutes) (a)Initial b) Final	Lab.	IS 4031 (Part-5)	-Do-	-Do-

Material	Test	Field / laboratory test	Test procedure	Minimum quantity of material / Work for carrying out the test	Frequency of testing

	Soundness Le-Chat expansion (mm) Auto clave (%)	Lab.	IS 4031 (Part-3)	-Do-	-Do-
	Compressive strength (MPa) a) 72 ± 1 hour b)168 ± 2hr	Lab.	IS 4031 (Part-6)	-Do-	-Do-
Fly Ash	Total Chloride in Percentage by Mass	Lab	IS 12423	10 Cum	Every 10 Cum or part thereof
	Loss on Ignition in percentage by mass (Max)	Lab	IS 1727	10 Cum	Frequency as decided by EIC
	Fineness, Specific Surface in (m2/kg)	Lab/ Field	IS 4031 (Part- 1 & 2)	10 Cum	Frequency as decided by EIC
	Compressive Strength at 28 days in N/mm2 (Min)	Lab	IS 4031 (Part-6)	10 Cum	Only in cases when fly ash is used as pozzolana in cement.
Stone Aggregate	a) Percentage of soft or deleterious materials	General visual inspection/ Lab test where required by the EIC	IS 2386 - Part 2	One test for each source	One test for each source
	b) Particle size distribution	Field / Lab	IS 2386 - Part 1	10 Cum	Every 40 Cum Or part thereof and Once in three months for each source of coarse aggregates (RCC works)

	c) Specific Gravity	Field / Lab	IS 2386 – part 3	10 Cum	Once in three months for each source of coarse aggregates (RCC works)
	d) Estimation of Organic Impurities	Field / Lab	IS 2386 – Part 2	10 Cum	-do-
	e) Bulk Density	Field / Lab	IS 2386 – Part 3	10 Cum	-do-
	f) Aggregate crushing strength	Field / Lab	IS 2386 – part 4	10 Cum	-do-
	g) Aggregate impact value	Field / Lab	IS 2386 – part 4	10 Cum	-do-
Material	Test	Field / laboratory test	Test procedure	Minimum quantity of material / Work for carrying out the test	Frequency of testing
Timber	Moisture	Field (by moisture meter) Laboratory test as required by Engineer- in-Ch.		1 Cum	Every one Cum or part thereof
Flush Door	End immersion test Knife test Adhesion test	Laboratory	IS: 2202 (Part 1) & Part II	26 shutters	As per sampling and testing as instructed by the Engineer-in-Charge.
Aluminum sections	i) Thickness of anodic coating/ Powder coating ii)	Field/Lab	IS: 5523/ IS: 13871	Field: 50kg. Lab:200kg.	Field: Each section of lot as directed by Engineer-in Charge. Lab: Three

	Weight				specimen of each section from every lot at PDI
Bricks	Testing of bricks / brick tiles for dimensions, Compressive strength, Water absorption, Efflorescence.	Laboratory	IS 3495 Part I to IV	2000 Nos	No of bricks to be selected from brick lot (For each of the characteristic) 5: 2001 to 10000 10: 10001 to 35000. 15: 35001 to 50000 5: for every addl. 50000 or part thereof Provided Make/Brand remains same. If < 2000, As per decision of the EIC
Steel for RCC	Physical tests Tensile strength Retest Re-bound test Nominal mass Bend test Elongation test Proof stress	Lab / field	IS 1608 IS 1786 IS 1786 IS 1786 IS 1599 IS 1786 IS 1786	Each lot from each source from each diameter of bar	Below 100 Tonnes Dia< 10 mm one sample for each 25 tonnes or part thereof If dia is >10 mm but less than 16 mm: One sample each 35 tonnes or part thereof. If dia>16 mm one sample for each 45 tonnes

Material	Test	Field / laboratory test	Test procedure	Minimum quantity of material / Work for carrying out the test	Frequency of testing
	Chemical Tests: 1. Carbon Constituent 2. Sulphur 3. Phosphorus 4. Phosphorus & Sulphur	Lab	IS 1786		Once for every fresh lot of one or more trucks as directed by the Engineer-in-Charge.
Soil core test	OMC Proctor density		As per IS 12175	-	Two for every 50 Sqm
Ceramic tiles/ Vitrified tiles			As per IS 13630		3000 tiles and more for each manufacturer and thereafter for every 3000 tiles or part thereof.

OTHER MATERIALS FOR MANDATORY TEST (AS PER CODAL PROVISIONS)

Civil: Mechanical couplers, Insulating Materials(Glass Wool, Phenotherm), Metallic Roofing Sheets, Mangalore Tiles, Physical Tests for Structural Steel including Tubular & Hollow Sections, Zinc Coating on Galvanized Iron Door Frame, Physical & Chemical Tests for Stainless Steel Sheets and Pipes, Chemical Admixtures, Chemical Polymers, Test for Preservatives in Chemically Treated Wood Items, G.I Pipes and RCC Hume Pipes,

ROAD WORK: Soil Core Tests, Grading of Metal for WBM, Crushing and Impact Value Test on Aggregates, Bitumen Grade, Bitumen Content, Precast Concrete Gratings, Paver Blocks and Tiles. OPTIONAL TESTS: Testing Aggregate-Surface Moisture, Petrography Test on Aggregates, Alkali-Aggregate Reaction, Steel Door/ Windows Test for Chemical and Physical Properties, Chemical Testing of Structural Steel, K-Value for Thermal Insulation.

TESTING, TOLERANCE AND ACCEPTANCE

The material should pass all tests and tolerance in dimensional, chemical and physical properties.

In case of non-standard materials, materials not covered under any IS Specifications, such as aluminium sections, the payment shall be made based on the actual unit weight basis as determined by testing at random sampling.

4 Earth work and allied works

a. EARTH WORK

SCOPE OF WORK

The scope of work covered under this specifications pertains to excavation of foundations, trenches, pits and over areas, in all sorts of soils, soft and hard rock, correct to dimensions given in the drawing including shoring, protections of existing underground utilities if any, such as water lines, electric cables etc., dewatering and shoring if necessary, stacking the useful materials as directed within the lead specified, refilling around the foundation and into the plinth with selected useful excavated earth and disposing off the surplus earth/materials within specified lead and finishing the surface to proper levels, slopes and camber etc. all complete.

SITE CLEARANCE

Before the earth work is started the area coming under cutting and filling shall be cleared of all obstructions, loose stones, shrubs, rank vegetation, grass, brush-wood, trees and saplings of girth upto 30 cm. measured at a height of one metre above ground and rubbish removed upto a distance of 150 metres outside the periphery of the area under clearance. The roots of trees shall be removed to a minimum depth of 60 cm. below ground level, or a minimum of 30cm. below formation level whichever is lower, and the hollows filled up with earth, levelled and rammed. This work is deemed to be included in the earth work items and no separate payment will be admissible for the work.

The trees of girth above 30 cm. measured at a height of one meter above ground, shall only be cut after permission of the PMC is obtained in writing. The roots shall also be removed as described in the preceding sub-para. Payment for cutting and removing roots of such trees shall be made separately. Any material obtained from the site will be the property of the OWNER/PMC and the useful materials as decided by the PMC will be conveyed and properly stacked as directed within the lead specified.

SETTING OUT AND MAKING PROFILES

Masonry or concrete pillars will be erected at suitable points in the area to serve as bench marks for the execution of the work. These bench marks shall be connected with G. T. S. or any other permanent bench mark approved by the PMC. Necessary profiles with pegs, bamboos and strings or Burjis shall be made to show the correct formation levels before the work is started. The EPC Contractor shall supply labour and materials for setting out and making profiles and Burjis for the work at his own cost and the same shall be maintained during the excavation work. The OWNER/PMC will show grid co-ordinates or other reference points. It shall be the responsibility of the EPC Contractor to set out centre lines correctly with reference to the drawings and install substantial reference marks. Checking of such alignment by the OWNER/PMC

will not absolve the EPC Contractor from his responsibility to execute the work strictly in accordance with the drawings.

EXCAVATION

The EPC Contractor shall notify the PMC before starting excavation and before the ground is disturbed, to enable him to take existing levels for the purpose of measurements. The ground levels shall be taken at 5 to 15 metres intervals in uniformly sloping ground and at closer distance where local mounts, pits or undulations are met with, as directed by the PMC. The ground levels shall be recorded in field books and plotted on plans, which shall be signed by the EPC Contractor and the PMC, before the earth work is actually started. The labour required for taking levels, shall be supplied by the EPC Contractor at his own cost. The EPC Contractor shall perform excavation in all types of soils, murrum, soft and hard rock, boulders etc. in foundation, over areas and in trenches to widths, lines, levels, grades and curves as shown in the drawing or lesser widths, lines and levels as directed by the PMC and as per items in the schedule of quantities.

All excavated earth shall be classified into fertile and non-fertile earth at site. Both types of earth shall be stockpiled at different location.

All fertile earth shall be used for gardening purpose in the plot, if fertile earth is surplus, same shall be given for gardening purpose. The proof of same shall be handed over to PMC. Proof shall be in form of document of hand-over & pictures.

Non-fertile earth may be used for land-filling & back-filling of pits, if same approved by PMC.

CLASSIFICATION OF EARTH WORK

The earth work shall be classified under the following main categories and measured separately for each category.

All types of soils, murum, boulders.

Soft rock.

Hard rock.

All types of Soils, Murum, Boulders : Generally any strata, such as sand, gravel, loam, clay, mud, black cotton murum, shingle, river or nallah bed boulders, siding of roads, paths etc. and hard core, macadam surface of any description (water bound, grouted tarmac etc.), lime concrete, mud (disintegrated) concrete and their mixtures which for excavation yields to application of picks, shovels, jumper, scarifier, ripper and other manual digging implements.

Soft Rock : Generally any rock which can be excavated by splitting with crow bars or picks and does not require blasting, wedging or similar means for excavation such as lime stone, sand stone, hard laterite, hard conglomerate and un-reinforced cement concrete below ground level.

If required light blasting may be resorted to for loosening the materials but this will not in any way entitle the material to be classified as „Hard rock".

Hard Rock : Generally any rock or boulder for the excavation of which blasting is required such as quartzite, granite, basalt, reinforced cement concrete (reinforcement to be cut through but not separated from concrete) below ground level and the like.

Hard rock (blasting prohibited): Hard rock requiring blasting as described above but where the blasting is prohibited for any reason and excavation has to be carried out by chiselling, wedging, use of rock hammers and cutters or any other agreed method.

Excavation in Hard Rock by Blasting : Where blasting is permitted the excavation in rock shall be done by means of blasting. No heavy blasting will be permitted and only controlled/muffled blasting will be permitted at the discretion of the PMC. The EPC Contractor shall be governed by the relevant statutory laws, rules and regulations on explosives, pertaining to the acquisition, transport, storage, handling and use of explosive which shall be rigidly followed and shall obtain himself all necessary materials and equipment for blasting. Blasting shall be executed through a licensed blaster with prior permission from police authorities. Prior to blasting sufficient notice shall be given to concerned parties to avoid danger to people, materials and nearby structures. All the damages caused by careless blasting if any shall be made good by the EPC Contractor at his own expenses.

Excavation in Hard Rock by Chiselling and Wedging : Where blasting is not permitted and if the PMC so desires, the excavation shall be done by chiselling and wedging or any other agreed method.

NOTE : All the excavated hard rock obtained shall be stacked properly and neatly within the specified lead by the EPC Contractor as directed by the PMC.

EXCAVATION PARAMETERS

Excavation where directed by the PMC shall be securely barricaded and provided with proper caution signs, conspicuously displayed during the day and properly illuminated with red lights and/or written using fluorescent reflective paint as directed by PMC during the night to avoid accident(s).

The EPC Contractor shall take adequate protective measures to see that the excavation operations do not damage the adjoining structures or dislocate the services. Water supply pipes, sluice valve chambers, sewerage pipes, manholes, drainage pipes and chambers, communication cables, power supply cables etc. met within the course of excavation shall be properly supported and adequately protected, so that these services remain functional. However, if any service is damaged during excavation shall be restored in reasonable time.

The excavation shall be done manually or by mechanical means as desired by PMC considering feasibility, urgency of work, availability of labour /mechanical equipment and other factors involved. EPC Contractor shall ensure every safety measures for the workers. Neither any deduction will be made nor any extra payment will be made on this account.

“Earth work by mechanical means” involves careful planning keeping in view site conditions i.e. type of soil, nature of excavation, distances through which excavated soil is to be transported and working space available for employing these machines. The earth moving equipment should be accordingly selected.

The excavation under all classifications in areas in trenches or in pits shall be carried out systematically. Cutting shall be done from top to bottom and no under-pining or undercutting will be allowed. The bottom and sides of excavation shall be dressed to proper level, slopes, steps, camber etc. by removing high spots, and ramming thoroughly as directed by the PMC.

All the excavation shall be carried out strictly to the dimensions given in the drawing. The width shall generally be of the width of mud-mat concrete and depth as shown in drawing or as directed by the PMC, according to availability of the desired bearing capacity of soil below. Any excavation if taken below the specified depths and levels, the EPC Contractor shall at his own cost fill up such overcut to the specified level with cement concrete 1:4:8 in case of excavation in all types of soils and with cement concrete 1:2:4 in case of excavation in soft and hard rock.

After the excavation is completed, the EPC Contractor shall notify the PMC to that effect and no further work shall be taken up until the PMC has approved the depth and dimensions and also the nature of foundation materials. Levels and measurements shall also be recorded prior to taking up any further work.

SHORING

Unless separately provided for in the schedule of quantities, the quoted rate for excavation shall include excavation of slopes to prevent falling in soil by providing and/or fixing, maintaining and removing of shoring, bracing etc. The EPC Contractor would be responsible for the design of shoring for proper retaining of sides of trenches, pits etc. with due consideration to the traffic, superimposed loads, etc. Shoring shall be of sufficient strength to resist the pressure and ensure safety from slips and to prevent damage to work and property and injury to persons. It shall be removed as directed after items for which it is required are completed. Should the slips occur, the slipped material shall be removed and slope dressed to a modified stable slope. Removal of the slipped earth will not be measured for payment.

DEWATERING

Unless specifically provided for as a separate item in the schedule of quantities, rate shall also include bailing or pumping out all water which may accumulate in the excavation during the progress of further works such as mud mat concrete, R.C. footings, shuttering etc. either due to seepage, springs, rain or any other cause and diverting surface flow by bunds or other means. Care shall be taken to ensure that the water discharged sufficiently away from the foundations to keep it free from nuisance to other works in the neighbourhood.

EARTH WORK FOR MAJOR WORKS

Excavation shall be undertaken to the width of the Basement/Retaining wall footing including necessary margins for construction operation as per drawing(s) or directed otherwise. Where the nature of soil or the depth of the trench and season of the year, do not permit vertical sides, the EPC Contractor at his own expense shall put up the necessary shoring, strutting and planking or cut slopes with or without steps, to a safer angle or both with due regard to the safety of personnel and works and to the satisfaction of the PMC.

All the major excavation shall be carried out by mechanical excavator. No extra payment shall be made for that.

The EPC Contractor shall make at his own cost all necessary arrangements for maintaining water level, in the area where works are under execution low enough so as not to cause any harm to the work shall be considered as inclusive of pumping out or bailing out water, if required, for which no extra payment shall be made. This will include water coming from any source, such as rains, accumulated rain water, floods, leakages from sewer and water mains, subsoil water table being high or due to any other cause whatsoever. The EPC Contractor shall make necessary provision of pumping, dredging bailing out water coming from all above sources and excavation and other works shall be kept free of water by providing suitable system approved by the PMC. Sub-soil water table at work site is observed low in the month of April/May(pre-monsoon month). The water level is likely to rise significantly during rainy season. In order to avoid possibility of basement floor of main building being getting uplifted/damaged due to water pressure, the EPC Contractor shall lower the ground water table below the proposed foundation level by boring tube wells all around the proposed building using well point sinking method or any suitable method as approved by PMC. Sub soil water table shall be maintained at least 50 cm. below the P.C.C. level during laying of P.C.C. water proofing treatment, laying of basement raft and beams including filling of earth/sand under the basement floor. The water table shall not be allowed to rise above base of raft level until completion of outer retaining walls including water proofing of vertical surface of walls and back filling along the walls upto ground level and until the structure attains such height to counter balance the uplift pressure. However, the EPC Contractor should inspect the site and make his own assessment about sub-soil water level likely to be encountered at the time of execution and quote his rates accordingly. Rate of all items are inclusive of pumping out or bailing out water, if required. Nothing extra on this account whatsoever shall be paid . The sequence of construction shall be got approved by the PMC.

The EPC Contractor shall take all necessary measures for the safety of traffic during construction and provide, erect and maintain such barricades including signs, markings, flags, lights and flagman, as necessary at either end of the excavation/embankment and at such intermediate points as directed by the PMC for the proper identification of construction area. The EPC Contractor shall be responsible for all damages and accidents caused due to negligence on his part.

The EPC Contractor shall provide suitable barricading with suitably painted single row of G.I. Sheets about 3"- 0" wide (90 cms.) nailed or bolted with wooden poles spaced 2 to 3 metre apart and each pole 1.6 m to 2 m long 8 cm. to 10 cm. dia. The poles will be embedded in mobile iron pedestal rings suitably framed for giving stable support as per direction of the PMC. All management (including watch and ward) of barricades shall be the full responsibility of the EPC Contractor. The barricades shall be removed only after completion of the work or part of the work. The EPC Contractors rate shall include all above items of work and nothing extra shall be paid to the EPC Contractor over and above his quoted rates.

DISPOSAL OF EXCAVATED MATERIALS

Antiquities: Any finds of archaeological interest such as relics of antiquity, coins, fossils or other articles of value shall be delivered to the PMC and shall be the property of the Government.

Useful materials : Any material obtained from the excavation which in the opinion of the PMC is useful, shall be stacked separately in regular stacks as directed by the PMC and shall be the property of the Government.

No material excavated from foundation trenches of whatever kind they may be are to be placed even temporarily nearer than about 3 m. from the outer edge of excavation. Discretion of the PMC in such cases is final. All materials excavated will remain the property of the OWNER/PMC. Rate for excavation includes sorting out of the useful materials and stacking them separately as directed within the specific lead.

Materials suitable and useful for backfilling or other use shall be stacked in convenient place but not in such a way as to obstruct free movement of materials, workers and vehicles or encroach on the area required for constructional purposes. It shall be used to the extent required to completely backfill the structure to original ground level or other elevation shown on the plan or as directed by the PMC. Materials not useful in anyway shall be disposed off, levelled and compacted as directed by the PMC within a specified lead. The site shall be left clean of all debris and levelled on completion.

BACKFILLING IN SIDES OF FOUNDATIONS, PLINTH, UNDER FLOOR ETC.

The back filling shall be done after the concrete or masonry has fully set and shall be done in such a way as not to cause under-thrust on any part of the structure. Where suitable excavated material is to be used for back filling, it shall be brought from the place where it was temporarily deposited and shall be used in backfilling. The scope of work for back filling/filling in foundation, plinth, under floors etc. shall include filling for all the buildings covered under the contract. Surplus earth available from one building, if required, shall be used for backfilling/filling for other buildings also within the specified lead mentioned in the item.

All timber shoring and form work left in the trenches, pits, floors etc. shall be removed after their necessity ceases and trash of any sort shall be cleared out from the excavation. All the space between foundation masonry or concrete and the sides of excavation shall be backfilled to the original surface with approved materials in layers not exceeding 200 mm. in thickness, watered and well consolidated by means of rammers to atleast 90% of the consolidation obtainable at optimum moisture content (Proctor density). Flooding with water for consolidation will not be allowed. Areas inaccessible to mechanical equipment such as areas adjacent to walls and columns etc. shall be tamped by hand rammer or by hand held power rammers to the required density. The backfill shall be uniform in character and free from large lumps, stones, shingle or boulder not larger than 75 mm. in any direction, salt, clods, organic or other foreign materials which might rot. The backfilling in plinth and under floors shall be done in similar way inlayers not exceeding 200 mm. thick and shall be well consolidated by means of mechanical or hand operated rammers as specified to achieve the required density.

Test to establish proper consolidation as required will be carried out by the OWNER/PMC at rates specified. Two tests per 50 Sqm. will be taken to ascertain the proper consolidation. The cost of tests carried out will be borne by the EPC Contractors.

Earth filling done for land-filling shall be placed in horizontal layer of not exceeding 600mm, unless otherwise authorized by the PMC, well compacted by mechanical means, to the satisfaction of the PMC. If the total filling height exceeds 900mm, the top 600mm filling Earth filling done for land-filling shall be placed in horizontal layer of not exceeding 600mm, unless otherwise authorized by the PMC, well compacted by mechanical means, to the satisfaction of the PMC. If the total filling height exceeds 900mm, the top 600mm filling shall be done with well graded soil with size not exceeding 25mm, the height below 600mm from top may be done with well graded soil with size not exceeding 100mm.

FILLING IN PLINTH AND UNDER FLOORS

After the available suitable excavated materials are exhausted as backfilling, the EPC Contractor shall notify the PMC, of the fact and levels taken jointly with PMC. The earth, murum, sand, gravel etc. or such materials suitable for filling proposed to be filled under floors and so mentioned in the item of schedule of quantities shall then be brought to site from approved locations and sources.

Earth Filling : The earth, soft murum etc. so brought shall be filled up in layers of 20 cm depth, each layer being well watered and consolidated by approved hand or mechanical tampers or other suitable means to achieve the required density.

Gravel or Sand Filling : Gravel if required to be filled under floors, shall be single washed gravel of approved quality and of size varying from 12 mm. to 20 mm. it shall be uniformly blinded with approved type of soil and/or sand to obtain full compaction. Gravel shall be filled in specified thickness and shall be well watered and rammed entirely to the satisfaction of the PMC.

If sand is required to be filled under floors, it shall be clean, medium grained and free from impurities. The filled in sand shall be kept flooded with water for 24 hrs. to ensure maximum consolidation. Any temporary work required to maintain sand under flooded condition shall be done by the EPC Contractor at his own cost. The surface shall then be well dressed and got approved from PMC before any other work is taken over the fill.

Sand used shall be sourced from distance within 400km of radius from the site, and as approved by the project authority.

SURFACE DRESSING

Surface dressing shall include cutting and filling upto a depth of 15 cm and clearing of shrubs, rank vegetation, grass, brushwood, trees and saplings of girth upto 30 cm measured at a height of one metre above the ground level and removal of rubbish and other excavated material upto a distance of 50metres or as specified outside the periphery of the area under surface dressing. High portions of the ground shall be cut down and hollows depression filled upto the required level with the excavated earth so as to give an even, neat and tidy look.

LEAD & LIFT

Lead: The lead for disposal/deposition of excavated materials shall be as specified in the respective item of work. For the purpose of measurements of lead, the area to be excavated or filled or area on which excavated material is to be deposited/disposed off shall be divided in suitable blocks and for each of the block, the distance between centre lines shall be taken as the lead which shall be measured by the shortest straight line route on the plan and not the actual route adopted.

Lift: Lift shall be measured from ground level. Excavation up to 1.5 m depth below ground level and depositing excavated material on the ground shall be included in the item of earthwork for various kinds of soil. Extra lift shall be measured in unit of 1.5 m or part thereof. Obvious lift shall only be measured; that is lifts inherent in the lead due to ground slope shall not be measured except for lead upto 250 m. All excavation shall be measured in successive stages of 1.5 m stating the commencing level. This shall not apply to cases where no lift is involved as in hill side cutting.

b. ANTITERMITETREATMENT

GENERAL

Pre-constructional anti-termite treatment is a process in which soil treatment is applied to a building in early stages of its construction. The purpose of anti-termite treatment is to provide the building with a chemical barrier against the sub-terranean termites.

Antitermite treatment being a specialized job, calls for thorough knowledge of the chemicals, soils, termite to be dealt with and the environmental conditions, in order to give effective treatment and lasting protection to the property undergoing treatment. It is, therefore, imperative that the works of antitermite treatment should be got executed through specialized agencies only. The specialized agency should be a member of the Indian Pest Control Association and shall have sufficient experience of carrying out similar works of magnitude envisaged in this tender.

The pre-constructional soil treatment is required to be applied during the construction stages of the sub-structure upto plinth level. The EPC Contractor has to be watchful of the various stages of sub-structure works and arrange to carry out the soil treatment in time after proper co-ordination with OWNER/PMC and other EPC Contractors if any, working at site.

SCOPE

The scope of pre-constructional anti-termite treatment covers the soil treatment with approved chemicals in water emulsion in foundation trenches for columns, plinth beams, pile caps, brick walls, service trenches, lift pits, steps, ramps etc. in top surfaces of plinth filling, at junction of walls and floor, in expansion joints etc. in stages as detailed in this specifications and drawings. Unless otherwise stipulated, the antitermite treatment will be carried out as per I.S.6313 (part 2) - 2001 and/or as per direction of the PMC.

The work shall be executed through approved specialized agencies who are a member of Indian Pest Control Association, approved by Central Insecticides Board.

SITE PREPARATION

In order to ensure uniform distribution of the chemical emulsion and to assist penetration, the following site preparation shall be carried out:

Remove all trees, stumps, logs or roots from the building site.

Remove all concrete form work if left anywhere, levelling pegs, timber off-cuts and other builders debris from the area to be treated.

If the soil to be treated is sandy or porous, preliminary moistening will be required to fill capillary spaces in soil in order to prevent the loss of emulsion through piping or excessive percolations.

In the event of water logging of foundation, the water shall be pumped out before application of chemical emulsion and it should be applied only when the soil is absorbent.

On clays and other heavy soils where penetration is likely to be slow and on sloping sites, where run-off of the treating solution is likely to occur, the surface of the soil should be scarified to a depth of 75 mm. atleast.

All sub-floor levelling and grading should be completed, all cutting, trenches and excavations should be completed with backfilling in place, borrowed fill must be free from organic debris and shall be well compacted. If this is not done supplementary treatments should be made to complete the barrier.

CHEMICAL TO BE USED

Chemicals approved by Central Insecticides Board conforming to relevant Indian Standards (IS 8944-Chloripyrifos E.C./ IS 632-Lindane E.C.) in water emulsion or as specified in item may be used for soil treatment in order to protect a building from termite attack. The chemical solutions or emulsions are required to be dispersed uniformly in the soil and to the required strength so as to form an effective chemical barrier which is lethal and repellent to termites.

Soil Treatment: Treatment shall be carried through a specialized Agency who is a member of the Indian Pest Control Association (IPCA) and shall be engaged with prior approval of the PMC.

The EPC Contractor should produce voucher(s) for the chemical purchased and should get verified the sealed container(s) of the specified chemical from the PMC before preparing the emulsion for the treatment.

PRECAUTIONS DURING TREATMENT

Utmost care shall be taken to see that the chemical barrier is complete and continuous. Each part of the area shall receive the prescribed dosage of chemical emulsion.

The treatment should not be carried out when it is raining or when the soil is wet with rain or sub-soil water.

Once formed, the treated soil barrier shall not be disturbed. If by chance, treated soil barriers are disturbed, immediate steps shall be taken to restore the continuity and completeness of the barrier system.

PRECAUTIONS FOR HEALTH HAZARDS AND SAFETY MEASURES

All the chemicals mentioned above are poisonous and hazardous to health. These chemicals can have an adverse effect upon health when absorbed through the skin, inhaled as vapour or spray mist or swallowed. Persons handling or using these chemicals should be warned of these dangers and advised that absorption through the skin is the most likely source of accidental poisoning. They should be cautioned to observe carefully the safety precautions given below particularly when handling these chemicals in the form of concentrates.

These chemicals are usually brought to the site in the form of emulsifiable concentrates. The containers should be clearly labelled and should be stored carefully so that children and pets cannot get at them. They should be kept securely closed.

Particular care should be taken to prevent skin contact with concentrates. Prolonged exposure to dilute emulsions should also be avoided. Workers should wear clean clothing and should wash thoroughly with soap and water especially before eating and smoking. In the event of severe contamination, clothing should be removed at once and the skin washed with soap and water. If chemicals splash into the eyes they shall be flushed with plenty of soap and water and immediate medical attention should be sought.

The concentrates are oil solutions and present a fire hazard owing to the use of petroleum solvents.

Flames should not be allowed during mixing.

Care should be taken in the application of chemicals/soil-toxicants to see that they are not allowed to contaminate wells or springs which serve as source of drinking water.

GUARANTEE

The EPC Contractor has to furnish the guarantee for 10 (ten) years from the date of completion of work, stating that in case of reappearance of termites within the building area due to defective materials or workmanship or due to any other reasons, the EPC Contractor will carry out the necessary post constructional treatment to keep the entire area free from termite, once again, without any extra cost to the OWNER/PMC during the guarantee period.

POST CONSTRUCTION ANTI-TERMITE TREATMENT

Once the termites have an ingress into the building, they keep on multiplying and destroy the wooden and cellulosic materials, and as such it becomes essential to take measures for protection against termites. Chemical treatment for the eradication and control of sub-terranean termites in existing building shall be done as per IS 6313 (Part 3) only through the approved specialized agencies using the approved chemical as directed by PMC.

1.8 TREATMENT

Anti termite measures described below are necessary for the eradication and control of termites in existing building. To facilitate proper penetrations of chemical in to the surface to be treated, hand operated pressure pump shall be used. To have proper check for uniform penetration of chemical, graduated containers shall be used. Proper check should be kept so that the specified quantity of chemical is used for the required area during the operation. Chemical treatment for the eradication and control of sub-terranean termites in existing building shall be carried out as per IS 6313 (Part III).All other precautions during treatment and safety measures shall generally be as described above under pre-constructional treatment.

Treatment along outside of foundations: The soil in contact with the external wall of the building shall be treated with chemical emulsion at the rate of 7.5 litres per square metre of vertical surface of the sub-structure to a depth of 300 mm. To facilitate this treatment, a shallow channel shall be excavated along and close to the wall face. The chemical emulsion shall be directed towards the wall at 1.75 litres per running metre of the channel. Roding with 12 mm diameter mild steel rods at 150 mm apart shall be done in the channel. If necessary, for uniform dispersal of the chemical to 300 mm depth from the ground level. The balance chemical of 0.5 litre per running metre shall then be used to treat the backfill earth as it is returned to the channel directing the spray towards the wall surface.

If there is a concrete or masonry apron around the building, approximately 12 mm diameter holes shall be drilled as close as possible to the plinth wall about 300 mm apart, deep enough to reach the soil below and the chemical emulsion pumped into these holes to soak the soil below at the rate of 2.25 litres per linear metre.

Treatment of Soil under Floors: The points where the termites are likely to seek entry through the floor are the cracks at the following locations.

At the junction of the floor and walls as result of shrinkage of the concrete

On the floor surface owing to construction defects

At construction joints in a concrete floor, cracks in sections

Expansion joints in the floor

Chemical treatment shall be provided in the plinth area of ground floor of the structure, wherever such cracks are noticed by drilling 12 mm holes at the junction of floor and walls along the cracks on the floor and along the construction and expansion joints at the interval of 300 mm to reach the soil below. Chemical emulsion shall be squirted into these holes using a hand operated pressure pump to soak the soil below until refusal or upto a maximum of one litre per hole. The holes shall then be sealed properly

with cement mortar 1:2 (1 cement: 2 coarse sand) finished to match the existing floors. The cement mortar applied shall be cured for at least 10 days as per instruction of PMC.

Treatment of Voids in Masonry: The movement of termites through the masonry wall may be arrested by drilling holes in masonry wall at plinth level and squirting chemical emulsions into the holes to soak the masonry. The holes shall be drilled at an angle of 45 degree from both sides of the plinth wall at 300 mm intervals and emulsion squirted through these holes to soak the masonry using a hand operated pump. This treatment shall also be extended to internal walls having foundations in the soil. Holes shall also be drilled at wall corners and where door and window frames are embedded in the masonry or floor at ground. Emulsion shall be squirted through the holes till refusal or to a maximum of one litre per hole. Care shall be taken to seal the holes after the treatment.

Treatment at Points of Contact of Wood Work: The wood work which has already been damaged beyond repairs by termites shall be replaced. The new timber shall be dipped or liberally brushed at least twice with chemical in oil or kerosene. All existing wood work in the building which is in contact with the floor or walls and which is infested by termites, shall be treated by spraying at the points of contacts with the adjoining masonry with the chemical emulsion by drilling 6 mm holes at a downward angle of about 45 degree at junction of wood work and masonry and squirting chemical emulsion into these holes till refusal or to a maximum of half a litre per hole. The treated holes shall then be sealed.

Infested wood work in chaukhats, shelves, joints, purlins etc., in contact with the floor or the walls shall be provided with protective treatment by drilling holes of about 3 mm diameter with a downward slant to the core of the wood work on the inconspicuous surface of the frame. These holes should be at least 150 mm centre to centre and should cover in entire frame work. Chemicals shall be liberally infused in these holes. If the wood is not protected by paint or varnish two coats of the chemicals shall be given on all the surfaces and crevices adjoining the masonry.

GUARANTEE

The EPC Contractor has to furnish the guarantee for 10 (ten) years from the date of completion of work, stating that in case of reappearance of termites within the building area due to defective materials or workmanship or due to any other reasons, the EPC Contractor will carry out the necessary post constructional treatment to keep the entire area free from termite, once again, without any extra cost to the OWNER/PMC during the guarantee period.

c. HARD CORE / SOLING UNDER FLOORS / FOUNDATIONS

SCOPE OF WORK

The work covered under this specification includes all type of soling work either by bricks or by rubble stones laid under floors/foundations, hand packed, complete as per under mentioned specification and applicable drawings.

RUBBLE STONE SOLING

The rubble stone shall be of best variety of black trap/granite/basalt or other approved variety of stone available locally. The stone shall be hard, durable, free from defects and of required size and shall be approved by the PMC before incorporation in the work.

Preparation of Surface : The bed on which rubble soling is to be laid shall be cleared of all loose materials, levelled, watered and compacted and got approved by the PMC before laying rubble soling. Cable or pipe trenches if shown in the drawing and as required by the PMC shall be got done before the soling is started.

Workmanship: Over the prepared surface, the stone shall be set as closely as possible and well packed and firmly set. The stones shall be of full height and shall be laid so as to have their bases of the largest area resting on the sub-grade. Soling shall be laid in one layer of 230 mm. or 150 mm. or other specified thickness and no stones shall be less than 230 mm. or 150 mm. depth or specified thickness of soling with a tolerance of 25mm.

After packing the stones properly in position, the interstices between them shall be carefully filled with quarry spoils or stone chips of larger size possible, to obtain a hard, compact surface. Spreading of loose spoils or stone chips is prohibited.

The entire surface shall be examined for any protrusions and the same shall be knocked off by a hammer and all interstices shall be filled with approved murum/ stone dust as specified. Excess murum/ stone dust if any over the surfaces shall be removed. Unless otherwise specified, the murum/ stone dust shall be supplied by the EPC Contractor at his own cost from the selected areas. The surfaces shall then be watered and consolidated with mechanical or sufficiently heavy wooden tampers and log-rammers as approved by the PMC to give the required slope or level and dense sub-base. After compaction, the surface shall present clean look. Adequate care shall be taken by the EPC Contractor while laying and compacting the rubble soling to see that concrete surfaces in contact with soling are not damaged.

d. PLAIN / REINFORCED CONCRETE AND ALLIED WORKS

SCOPE

Scope of the specification deals with the structural and general use of plain and reinforced cement concrete.. The specifications cover the qualitative and quantitative requirements in respect of selection of ingredients, proportioning, manufacture of concrete, transport, placing, consolidation, curing, finishing, acceptance criteria etc.,.

BATCHING

The manufacture and placing of concrete shall be done by weigh batching. However, Volume batching may be allowed only where weigh-batching is not practicable and provided accurate used in concrete have earlier been established. Allowance for bulking shall be made in accordance with IS 2386 (Part 3). The mass volume relationship should be checked as frequently as necessary, the frequency for the given job being determined by PMC to ensure that the specified grading is maintained.

CEMENT CONCRETE (PLAIN AND REINFORCED)

The quality of materials and method and control of manufacture and transportation of all concrete work in respect of mix, whether reinforced or otherwise, shall conform to the applicable portions of these specifications.

Mandatory tests: The PMC shall have the right to inspect the sources of materials, the layout and operation of procurement and storage of materials, the concrete batching and mixing equipment and the quality control system. The EPC Contractor shall arrange such an inspection and the PMC approval shall be obtained prior to starting the concrete work. Mandatory test and their frequency of testing for RCC and their constituent materials shall be carried out as per provisions mentioned in Chapter-III.

Materials for Concrete: The constituent materials used in the manufacture of concrete shall be asper IS456-2000.

Cement: Unless otherwise specified or called for in the contract specifications, cement shall beany of the following and the type selected should be appropriate for the intended use and as per the contract conditions, specifications and drawings.

33 Grade Ordinary Portland cement conforming to IS 269

43 Grade ordinary Portland cement conforming to IS 8112

53 Grade ordinary Portland Cement conforming to IS 12269

Portland slag cement conforming to IS 455.

Portland pozzolana cement (fly ash based) conforming to IS 1489 (Part I)

Portland pozzolana cement (calcined clay based conforming to IS 1489 (Part 2)

Sulphate resisting Portland cement conforming to IS 12330.

In case the job requires specific use of any of the following cements the same shall be used with the prior approval of the EIC and necessary precautions with regard to their setting and hardening time, time required for de-shuttering, curing etc., shall be taken after carefully complying with specific literature with regard to those types.

High Alumina cement - conforming to IS 6452

Low heat cement - conforming to IS 12600

Super sulphate cement - conforming to IS 6909

Rapid Hardening cement - conforming to IS 8041

Blended Cement for finishing work as below

Other combinations of Portland Cement with mineral admixtures of quality conforming to relevant Indian Standards laid down may also be used in the manufacture of concrete provided that there are satisfactory data on their suitability, such as performance test on concrete containing them and only in such case where in specifically called for in the contract.

Mineral Admixtures

Pozzolana: Pozzolanic materials conforming to relevant Indian Standards may be used with the permission of PMC, provided uniform blending with cement is ensured.

Fly ash (pulverized fuel ash): Fly ash conforming to Grade 1 of IS 3812 may be used as part replacement of ordinary Portland cement provided uniform blending with cement is ensured.

Silica fume: Silica fume conforming to a standard approved by the deciding authority can be used as part replacement of cement provided uniform blending with the cement is ensured.

Note: The silica fume (very fine non–crystalline silicon dioxide)is a by-product of the manufacture of silicon, ferrosilicon or the like, from quartz and carbon in electric arc furnace. It is usually used in proportion of 5 to 10 percent of the cement content of a mix.

Rice husk ash: Rice husk ash giving required performance and uniformity characteristics may be used with the approval of the deciding authority.

Note: Rice husk ash is produced by burning rice husk and contain large proportion of silica. To achieve amorphous state, rice husk may be burnt at controlled temperature. It is necessary to evaluate the product from a particular source for performance and uniformity since it can range from being as deleterious as silt when incorporated in concrete. Water demand and drying shrinkage should be studied before using rice husk.

Metakaoline: Metakaoline having fineness between 700 to 900m2/kg may be used as pozzolanic material in concrete.

Note: Metakaoline is obtained by calcination of pure or refined kaolintic clay at a temperature between 6500C and 8500C, followed by grinding to achieve a fineness of 700 to 900 m2/kg. The resulting material has high pozzolanicity.

Ground Granulated Blast Furnace Slag: Ground granulated blast Furnace slag obtained by grinding granulated blast furnace slag conforming to IS 12089 may be used as part replacement of ordinary Portland cement provided uniform blending with cement is assured.

A certified report attesting to the conformity of the cement to I.S. specifications by the cement manufacturer’s chemist shall be furnished to the PMC, if demanded. The EPC Contractor, shall make his own arrangements for the storage of adequate quantity of cement at no extra cost at the site of work as per instructions and approval of the PMC.

Note: Specification for Storage: Stacking and storage of cement and other construction material(s) shall be as per IS 4082-1996. Cement in bags shall be stored and stacked in a shed, which is dry, leak-proof and moisture proof as far as possible. Storage under tarpaulins will not be permitted. Flooring of the shed shall consists of the two layers of dry bricks laid on well consolidated earth to avoid contact of cement bags with the floor. Stacking shall be done about 150 to 200 mm clear above the floor using wooden planks. Cement bags shall be stacked at least 450 mm clear off the walls and in rows of two bags leaving

in a space of at least 600 mm between two consecutive rows. In each row the cement bags shall be kept closed together so as to reduce air circulation. Stacking shall not be more than ten bags high to avoid lumping under pressure. In stacks more than eight bags high, the cement bags shall be arranged in header and stretcher fashion i.e. alternately lengthwise and crosswise so as to tie the stacks together and minimize the danger of toppling over.

Damaged or reclaimed or partly set cement will not be permitted to be used and shall be removed from the site. The storage arrangements shall be such that there is no dead storage consignments in cement shall be stored as received and shall be consumed in the order of their delivery.

Cement held in store for a period of ninety (90) days or longer shall be retested before use in work. Should at any time the PMC have reasons to consider that any cement is defective, then irrespective of its origin and/or manufacturers test certificate, such cement shall be tested immediately at a National Test Laboratory/OWNER/PMC at Laboratory or such approved laboratory, and until the results of such tests are found satisfactory, it shall not be used in any work.

Aggregates

—Aggregate in general designates both fine and coarse inert materials used in the manufacture of concrete.

—Fine Aggregate is aggregate most of which passes through 4.75 mm I.S. sieve

—Coarse Aggregate is aggregate most of which is retained on 4.75 mm I.S. sieve. Aggregate shall comply with requirement of IS 383. As far as possible preference shall be given to machine broken and graded aggregate.

All fine and coarse aggregates proposed for use in the work shall be subject to the PMC approval and after specific materials have been accepted, the source of supply of such materials shall not be changed without prior approval of the PMC.

Aggregate shall, except as noted above, consists of natural sand, crushed stone and gravel from a source known to produce satisfactory aggregate for concrete and shall be chemically inert, strong, hard, durable against weathering, of limited porosity and free from deleterious materials that may cause corrosion

to the reinforcement or may impair the strength	**and/or durability**	**of concrete. The grading of aggregates**	
shall be such as to produce a dense concrete of specified strength and consistency			**that will work**
readily into position without	**shall be**	**on the “mix**	**and preliminary**

segregation and	**based**	**design”**	
test on concrete specified herein-after.			

Sampling and testing : Sampling of the aggregates for mix design and determination of suitability shall be taken under the supervision of the PMC and delivered to the laboratory, well in advance of the schedule for placing of concrete. Records of tests which have been made on proposed aggregates and on concrete made from this source of aggregates shall be furnished to the PMC in advance of the work for use, in determining suitability of the proposed aggregate.

Storage of aggregates : All coarse and fine aggregates shall be stacked separately in stock piles in the material yard near the work site in bins properly constructed to avoid inter mixing of different aggregates. Contamination with foreign materials and earth during storage and while heaping the materials shall be avoided. The aggregate must be of specified quality not only at the time of receiving at site but also at the time of loading into mixer. Rakers shall be used for lifting the coarse aggregate from bins or stock piles. Coarse aggregate shall be piled in layers not exceeding 1.00 metres in height to prevent conning or segregation. Each layer shall cover the entire area of the stock pile before succeeding layers are started. Aggregates that have become segregated shall be rejected. Rejected material after re-mixing may be accepted, if subsequent tests demonstrate conformity with required gradation.

Specific Gravity : Aggregates having a specific gravity below 2.6 (saturated surface dry basis) shall not be used without special permission of the PMC.

Fine Aggregate: Fine aggregate except as noted above, and for other than light weight concrete shall consist of natural or crushed sand conforming to I.S. 383. The sand shall be clean, sharp, hard, strong and durable and shall be free from dust, vegetable substances, adherent coating, clay, loam, alkali, organic

matter, mica, salt or other deleterious substances which can be injurious to the setting qualities/strength/durability of concrete.

Machine Made/Manufactured Sand : Machine made sand will be acceptable, provided the constituent rock composition is sound, hard, dense, non-organic, uncoated and durable against weathering. Machine made sand shall be accepted provided grading & finer particle limits conform to IS 383.

Screening and Washing : Sand shall be prepared for use by such screening or washing or both, as necessary, to remove all objectionable foreign matter while separating the sand grains to the required size fractions. Sand with silt content more than 3 percent will not be permitted to be used unless the same is washed and silt content is brought within 3% by weight.

Foreign Material Limitations : The percentages of deleterious substances in sand, delivered to the mixer shall not exceed the following:

Table – Percentage of Foreign Material Limitation

Sl.No	Substances	Percent by weight	
		Uncrushed	Crushed
(i)	Material finer than 75 micron I.S. Sieve	3.00	15.00
(ii)	Shale	1.00	--
(iii)	Coal and lignite	1.00	1.00
(iv)	Clay lumps	1.00	1.00
(v)	Total of all above substances including	5.00	2.00
	items (if) to (iv) for uncrushed sand and		
	items (iii) and (iv) for crushed sand.		

Gradation: Unless otherwise directed or approved, the grading of sand shall be within the limits indicated here under

Table – Standard Grading of Sand

I.S. Sieve	Percentage passing for			
Designation	Grading	Grading	Grading	Grading
	Zone -I	Zone -II	Zone -III	Zone -IV
9.5 mm	100	100	100	100
4.75mm	90-100	90-100	90-100	95-100
2.36mm	60-95	75-100	85-100	95-100
1.18 mm	30-70	55-90	75-100	90-100
600 micron	15-34	35-59	60-79	80-100
300 micron	5-20	8-30	8-30	20-65
150 micron	0-10	0-10	0-10	0-15

Where the grading falls outside the limits of any particular grading zone of sieves, other than 600 micron (I.S.) sieve by not more than 5%, it shall be regarded as falling within that grading zone. This tolerance

shall not be applied to percentage passing the 600 micron (I.S.) sieve or to percentage passing any other sieve size on the coarser limit of grading zone I or the finer limit of grading zone IV. Fine aggregates conforming to Grading Zone IV shall not be used unless mix designs and preliminary tests have shown its suitability for producing concrete of specified strength and workability.

Fineness Modulus : The sand shall have a fineness modulus of not less than 2.2 or more than 3.2. The fineness modulus is determined by adding the cumulative percentages retained on the following I.S. sieve sizes (4.75 mm, 2.36 mm, 1.18 mm, 600 micron, 300 micron and 150 micron) and dividing the sum by 100.

Coarse Aggregate: Coarse aggregate for concrete, except as noted above and for other than light weight concrete shall conform to I.S. 383. This shall consist of natural or crushed stone and gravel, and shall be clean and free from elongated, flaky or laminated pieces, adhering coatings, clay lumps, coal residue, clinkers, sag, alkali, mica, organic matter or other deleterious matter.

The coarse aggregate and fine aggregate shall be tested from time to time as required by the PMC to ascertain its suitability for use in construction and the charges for testing aggregate shall be born by the EPC Contractor as specified herein after.

Screening and Washing : Crushed rock shall be screened and/or washed for the removal of dirt or dust coating, if so demanded by PMC.

Grading : Coarse aggregates shall be either in single or graded, in both the cases. The grading shall be within the following limits:

Table – Standard Grading Limits of Course Aggregate

I.S. Sieve	**Percentage passing for single sized aggregates of nominal size**						**Percentage passing for graded aggregates of nominal size**			
Designation	**63 mm**	**40 mm**	**20 mm**	**16mm**	**12 .5mm**	**10 mm**	**40 mm**	**20 mm**	**16 mm**	**12.5 mm**
75 mm	100	-	-	-	-	-	-	-	-	-
53 mm	85-100	100	-	-	-	-	100	-	-	-
37.5 mm	0-30	85-100	100	-	-	-	95-100	100	-	-

19 mm	0-5	0-20	85-100	100	-	-	30-70	95-100	100	100
13.2 mm	-	-	-	85-100	100	-	-	-	90-100	-
11.2 mm	-	-	-	-	85-100	100	-	-	-	90-100
9.5 mm	-	0-5	0-20	0-30	0-45	85-100	10-35	25-55	30-70	40-85
4.75 mm	-	0-5	0-5	0-10	0-20	0-20	0-5	0-10	0-10	0-10
2.36 mm	-	-	-	-	0-5	0-5	-	-	-	-

The pieces shall be angular in shape and shall have granular or crystalline surfaces. Friable, flaky and laminated pieces, mica and shale, if present, shall be only in such quantities that will not, in the opinion of PMC, affect adversely the strength and/or durability of concrete, the maximum size of coarse aggregate shall be the maximum size specified above, but in no case greater than 1/4 of the minimum thickness of the member, provided that the concrete can be placed without difficulty so as to surround all reinforcement thoroughly and fill the corners of form.

Plums above 160 mm. and upto any reasonable size can be used in plain mass concrete work of large dimensions upto a maximum limit of 20% by volume of concrete when specifically approved by PMC.

For heavily reinforced concrete members, the nominal maximum size of the aggregate shall be 5 mm. less than the minimum clear distance between the reinforcing main bars or 5mm less than the minimum cover to the reinforcement whichever is smaller.

The amount of fine particles occurring in the Free State or as loose adherent shall not exceed 1% when determined by laboratory sedimentation tests as per I.S. 2386.

After 24 hours immersion in water, a previously dried sample shall not have gained more than 10% of its oven dry weight in air, as determined by I.S.2386.

Foreign Material Limitations : The percentages of deleterious substances in the coarse aggregate delivered to the mixer shall not exceed the following.

Table : Foreign Material Limitations in Course Aggregate

Sl.No.	Substances	Percentage by weight of aggregates

		Uncrushed	Crushed
i)	Material finer than 75 micron I.S. Sieve	3.00	3.00
ii)	Coal and lignite	1.00	1.00
iii)	Clay lumps	1.00	1.00
iv)	Soft fragments	3.00	--
v)	Total of all the above substances	5.00	5.00

WATER

Water used for both mixing and curing shall be clean and free from injurious amounts of deleterious materials. Viz oils, acids, alkalis, salts, sugar, organic materials or other substances that may be deleterious to concrete or steel. Potable waters are generally satisfactory for mixing and curing concrete. In case of doubt, the suitability of water for making concrete shall be ascertained by the compressive strength and initial setting time test specified in I.S. 456 - 2000. The sample of water taken for testing shall be typical of the

water proposed to be used for concreting, due account being paid to seasonal variation. The samples shall not receive any treatment before testing other than that envisaged in the regular supply of water proposed for use in concrete. The sample shall be stored in a clean container previously rinsed out with similar water.

Average 28 days compressive strength of at least three 150 mm. concrete cubes prepared with water proposed to be used shall not be less than 90% of the average strength of three similar concrete cubes prepared with distilled water as per IS - 516.

The initial setting time of test block made with the appropriate cement and the water proposed to be used shall not be less than 30 minutes and shall not differ by more than (±) 30 minutes from the initial setting time of control test block prepared with the same cement and distilled water. The test blocks shall be prepared and tested in accordance with the requirements of I.S. 4031(Part 5).

Where water can be shown to contain an excess of acid, alkali, sugar or salt, PMC may refuse to permit its use. As a guide, the following concentrations represent the maximum permissible values:

Limits of acidity : To neutralize 100 ml sample of water, using phenolphthalein as an indicator, it should not require more than 5 ml. of 0.02 normal NaOH. The details of test shall be as per I.S. 3025 (Part 22)

Limits of alkalinity : To neutralize 100 ml sample of water, using mixed indicator , it should not require more than 25 ml. of 0.02 normal H_2SO_4. The details of test shall be as per I.S. 3025 (Part 23).

Permissible limits for solids shall be as under (water)

Table : Permissible limits for solids in Water

Sl.No.	Type of solid	Tested as per	Permissible limit (Max.)
i)	Organic	IS 3025 (Part 18)	200 mg / l
ii)	Inorganic	IS 3025 (Part 18)	3000 mg / l
Iii)	Sulphates (as SO2)	IS 3025 (Part 24)	400 mg / l
iv)	Chlorides (as Cl)	IS 3025 (Part 32)	2000 mg / l for concrete not containing embedded steel and 500 mg/l for reinforced cement concrete work.
v)	Suspended matter	IS 3025 (Part 17)	2000mg / l

The pH value of water shall be not less than 6.

DESIGN MIX CONCRETE

All reinforced concrete in the works shall be “Design Mix Concrete” as defined in I.S. 456-2000.

Mix Design

Design mix concrete is that concrete in which the design of mix i.e. the determination of proportions of cement, aggregate & water is arrived as to have target mean strength for specified grade of concrete. The proportions of the mix shall be determined by weight. Mix proportioning shall be carried out as per I.S. 10262 – 2009 or other standard method(s) approved by PMC. The cost of design mix shall be borne by the EPC Contractor.

Mix proportions shall be designed to ensure that the workability of fresh concrete is suitable for conditions of handling and placing, so that after compaction it surrounds all reinforcement and completely fills the formwork. When concrete is hardened, it shall have the stipulated strength, durability and impermeability.

All concrete proportions for various grades of concrete shall be designed separately and the mix proportions established keeping in view the workability for various structural elements, methods of placing and compacting.

Design mix concrete is preferred to nominal mix. If design mix cannot be used for any reason on the work such as soon after award of the work, if so desired by the EPC Contractor, he/they may be allowed by the PMC, till the designed mix is obtained, to carry out the reinforced concrete work in foundation and plinth as per equivalent nominal mix against the specified design mix concrete. However, all other specification for design mix shall govern for nominal mix also and nothing extra shall be paid for use of extra cement or else on this account whether the cement is supplied by the OWNER/PMC or procured by the EPC

Contractor. Where the quantity of RCC is very small, under such circumstance equivalent nominal mix can also be permitted by the PMC.

Table : Grades of Concrete (The concrete shall be in grades designated as below)

Group	Grade Designation	Specified Characteristic compressive strength of 150 mm cube at 28 days (N/mm2)
Ordinary concrete	M 10	10
	M 15	15
	M 20	20
Standard Concrete	M 25	25
	M 30	30
	M 35	35
	M 40	40
	M 45	45
	M 50	50
	M 55	55
High strength concrete	M 60	60
	M 65	65
	M 70	70
	M 75	75
	M 80	80

NOTE :

The characteristic strength is defined as the strength of material below which not more than 5% of the test results are expected to fall.

In the designation of a concrete mix, letter "M" refers to the mix and the number to the specified characteristic compressive strength of 150 mm. size cubes at 28 days expressed in N/ mm2.

Minimum Cement Content, Maximum Water Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum size are as follows.

Table : Minimum Cement Content, Maximum Water Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum size

		Plain concrete			Reinforced concrete		
SINo.	**Exposure**	**Minimum cement content Kg/m3**	**Maximum Free Water Cement Ratio**	**Minimum grade of concrete**	**Minimum cement content Kg/m3**	**Maximum Free Water Cement Ratio**	**Minimum grade of concrete**
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(i)	Mild	220	0.60	-	300	0.55	M-20
(ii)	Moderate	240	0.60	M-15	300	0.50	M-25
(iii)	Severe	250	0.50	M-20	320	0.45	M-30
(iv)	Very severe	260	0.45	M-20	340	0.45	M-35
(v)	Extreme	280	0.40	M-25	360	0.40	M-40

NOTE

Cement content prescribed in this table is irrespective of the grades of cement and it is inclusive of additions mentioned in mineral admixtures. The additions such as fly ash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and water-cement ratio if the suitability is established and as long as the maximum amounts taken into account do not exceed the limit of pozzolana and slag specified in IS 1489 (Part 1) and IS 455 respectively.

Minimum grade for plain concrete under mild exposure condition is not specified.

Table : Nominal cover to meet Durability Requirements

Exposure	Nominal concrete cover in mm not less than
Mild	20
Moderate	30
Severe	45

Very severe	50
Extreme	75

NOTES

1. For main reinforcement up to 12 mm diameter bar for mild exposure the nominal cover may be reduced by 5 mm.
2. Unless specified otherwise, actual concrete cover should not deviate from the required nominal cover by + 10 mm
3. For exposure conditions „severe" and „ very severe" , reduction of 5 mm may be made, where concrete grade is M 35 and above.

Table : Nominal cover to meet specified period of fire resistance

Fire resistance	Nominal cover						
	BEAMS		SLABS		RIBS		COLUMNS
	Simply Supported	Continuous	Simply Supported	Continuous	Simply Supported	Continuous	
Hr.	mm	mm	mm	mm	mm	mm	mm
0.5	20	20	20	20	20	20	40
1	20	20	20	20	20	20	40
1.5	20	20	25	20	35	20	40
2	40	30	35	25	45	35	40
3	60	40	45	35	55	45	40
4	70	50	55	45	65	55	40

NOTES

1. The nominal covers given relate specifically to the minimum member dimensions as per drawing

2. Cases that lie below the bold line require attention to the additional measures necessary to reduce the risks of spalling.

Table 4.10 Adjustments to Minimum cement contents for Aggregates other than 20 mm

Nominal Maximum size

Sl.No	Nominal maximum Aggregate size mm	Adjustments to Minimum cement content kg/ m3
(1)	(2)	(3)
i)	10	+40
ii)	20	0
iii)	40	-30

For concrete of compressive strength greater than M55 design parameters given in the standard may not be applicable and the values may be obtained from specialized Literatures and experimental results.

The mix shall be designed to produce the grade of concrete having the required workability and characteristic strength not less than appropriate values given in the table above.

Approval of Design Mix: The calculated mix proportions shall be checked by means of trial batches as per IS 10262-2009. The EPC Contractor shall submit details of each trial mix of each grade of concrete designed for various workability conditions to the PMC for his comments and approval. Concrete of any particular design mix and grade shall be produced/ manufactured for works only on obtaining written approval of the PMC.

For any change in quality/ quantity in the ingredients of a particular concrete, for which mix has been designed earlier and approved by the PMC, the mix has to be redesigned and approval obtained again. Cost of such mix design shall be borne by the EPC Contractor.

Whenever there is a change either in required strength of concrete or water cement ratio or workability or the source of aggregates and/or cement, fresh trial mix shall be carried out to determine the revised proportion of the mix to suit the altered conditions.

Standard Deviation: Standard deviation calculations of test results based on tests conducted on the same mix design for a particular grade designation shall be done in accordance with IS 456.

Standard Deviation Based on Test Results

Number of test results - The total number of test results required to constitute and acceptable record for calculation of standard deviation shall be not less than 30. Attempts should be made to obtain the 30 test results, as early as possible, when a mix is used for the first time.

Standard deviation to be brought up to date - The calculation of the standard deviation shall be brought upto date after every change of mix design and at least once a month.

Determination of standard deviation

Concrete of each grade shall be analysed separately to determine its standard deviation.

The standard deviation of concrete of a given grade shall be calculated using the following formula from the results of individual tests of concrete of that grade obtained as specified for test strength of sample:

Estimated standard deviation

Where □ = Deviation of the individual test strength from the average strength of a sample and n = Number of sample test results.

When significant changes are made in the production of concrete (for example changes in the materials used, mix design, equipment or technical control), the standard deviation value shall be separately calculated for such batches of concrete.

Assumed Standard Deviation

Where sufficient test results for a particular grade of concrete are not available, the value of standard deviation given in table below may be assumed for design of mix in the first instance. As soon as the results of samples are available, actual calculated standard deviation shall be used and the mix designed properly. However, when adequate past records for a similar grade exist and justify to the designer a value of standard deviation different from that shown in table below, it shall be permissible to use that value.

Table: Assumed Standard Deviation

Grade of Concrete	Assumed Standard Deviation N/ mm2
M 10	3.5
M 15	
M 20	4.0
M25	
M30	
M35	
$S = \sqrt{\frac{\sum \Delta^2}{(n-1)}}$	5.0
M40	

M45	
M50	

Note: The above values correspond to the site control having proper storage of cement: weigh batching of all materials: controlled addition of water: regular checking of all materials: aggregate grading and moisture contents : and periodical checking of workability: and strength. Where there is deviation from the above the values given in the above table shall be increased by 1 N / mm.2

Proportioning, Consistency, Batching and Mixing of Concrete

Proportioning

Aggregate : The proportions which shall be decided by conducting preliminary tests shall be by weight. These proportions of cement, fine and coarse aggregates shall be maintained during subsequent concrete batching by means of weigh batchers conforming to I.S. 2722, capable of controlling the weights within one percent of the desired value. Except where it can be shown to the satisfaction of the PMC that supply of properly graded aggregate of uniform quality can be maintained over the period of work, the grading of aggregate shall be controlled by obtaining the coarse aggregate in different sizes and blending them in the right proportions. The different sizes shall be stacked in separate stock piles. The grading of coarse and fine aggregates shall be checked as frequently as possible, as determined by the PMC, to ensure maintaining of grading in accordance with samples used in preliminary mix design. The materials shall be stock piled well in advance of use.

Cement : The cement shall be measured by weight. Every facility should be provided to the PMC for sampling and inspection of stored cement at site of work.

Table - Exposure conditions

S.No:	Environment	Exposure Conditions
(1)	(2)	(3)
i)	Mild	Concrete surfaces protected against weather or aggressive conditions, except those situated in coastal area.
ii)	Moderate	Concrete surfaces sheltered from severe rain or freezing whilst wet Concrete exposed to condensation and rain Concrete continuously under water Concrete in contact or buried under non-aggressive soil/ground water Concrete surfaces sheltered from saturated salt air in coastal area

iii)	Severe	Concrete surfaces exposed to severe rain, alternate wetting and drying or occasional freezing whilst wet or severe condensation. Concrete completely immersed in sea water Concrete exposed to coastal environment.
iv)	Very severe	Concrete surfaces exposed to seawater spray, corrosive fumes or severe freezing conditions whilst wet. Concrete in contact with or buried under aggressive sub-soil/ground water .
v)	Extreme	Surface of members in tidal zone Members in direct contact with liquid/solid aggressive chemicals.

WATER : Only such quantity of water shall be added to the cement and aggregate in the concrete mix as to ensure dense concrete, specified surface finish, satisfactory workability, consistent with strength stipulated for each class of concrete. The water added to the mix shall be such as not to cause segregation of materials or the collection of excessive free water on the surface of the concrete.

Definition of water cement ratio: The water cement (W/C) ratio is defined as the weight of water in mix(including the surface moisture of the aggregates) divided by the weight of the cement in the mix.

Water cement ratio :The actual water cement ratio to be adopted shall be determined in each instance by EPC Contractor and approved by the PMC.

Proportioning by water-cement ratio: The W/C ratio specified for use by the PMC shall be maintained. EPC Contractor shall determine the water content of the aggregate as frequently as directed by the PMC as the work progresses and as specified in I.S. 2386 part III and the amount of mixing water added at the mixer shall be adjusted as directed by the PMC so as to maintain the specified W/C ratio. To allow for the variation in their moisture content, suitable adjustments in the weights of aggregates shall also be made.

Consistency and slump: Concrete shall be of a consistency and workability suitable for the conditions of the job. After the amount of water required is determined, the consistency of mix shall be maintained throughout the progress of the corresponding parts of the work and approved tests e.g. slump tests, compacting factor tests etc. in accordance with I.S. 1199, shall be conducted from time to time to ensure the maintenance of such consistency.

The following tabulation gives a range of workability which shall generally be used for various types of construction unless other wise instructed by the PMC.

Workability of concrete

Placing condition	**Degree of workability**	**Slump (mm)**
(1)	(2)	(3)
Blinding Concrete; Shallow Sections; Pavement using pavers	Very low	See note 1.
Mass concrete; Lightly reinforced Sections in slabs, Beams, walls, columns; Floors; Hand placed pavements; Canal lining; Strip footings	Low	25-75
Heavily Reinforced sections In slabs, beams, walls, columns, slip form work; pumped concrete	Medium	50-100 75- 100
Trench fill, In-situ piling	High	100-150
Tremie Concrete	Very High	See note 2.

Notes

1. For most of the placing conditions, internal vibrators (needle vibrators) are suitable. The diameter of the needle shall be determined based on the density and spacing of reinforcement bars and thickness of sections. For tremie concrete, vibrators are not required to be used.
2. the "very low" category of workability where strict control is necessary, for example pavement quality concrete, measurement of workability by determination of compacting factor will be more appropriate than slump (see IS 1199) and a value of compacting factor of 0.75 to 0.80 is suggested.
3. In the „Very high" category of workability, measurement of workability by determination of flow will be appropriate (see IS 9103).

Production of concrete

Quality assurance measures

In order that the properties of the completed structure be consistent with the requirements and the assumptions made during the planning and the design, adequate quality assurance measures shall be taken. The construction should result in satisfactory strength, serviceability and long term durability so as to lower the overall life-cycle cost. Quality assurance in construction activity relates to proper design use of adequate materials and components to be supplied by the producers, proper workmanship in the execution of works by the EPC Contractor and ultimately proper care during the use of structure including timely maintenance and repair by the owner.

Quality assurance measures are both technical and organizational. Some common cases should be specified in a general Quality Assurance Plan which shall identify the key elements necessary to provide fitness of the structure and the means by which they are to be provided and measured with the overall purpose to provide confidence that the realized project will work satisfactorily in service fulfilling intended needs. The job of quality control and quality assurance would involve quality audit of both the inputs as well as the outputs. Inputs are in the form of materials for concrete; workmanship in all stages of batching, mixing, transportation; placing, compaction and curing; and the related plant, machinery and equipment; resulting in the output in the form of concrete in place. To ensure proper performance, it is necessary that each step in concreting which will be covered by the next step is inspected as the work proceeds.

Each party involved in the realization of a project should establish and implement a Quality Assurance Plan, for its participation the project. Suppliers and EPC Contractors activities shall be covered in the plan. The individual quality assurance plans shall fit into the general Quality Assurance Plan. A quality assurance plan shall define the tasks and responsibilities of all persons involved, adequate control and checking procedures, and the organization maintaining adequate documentation of building process and its results. Such documentation should generally include

a) test reports and manufacturers certificate for materials, concrete mix design details;

b) pour cards for site organization and clearance for concrete placement;

c) record of site inspection of workmanship, field tests

d) non-conformance reports, change orders;

e) quality control charts;

f) statistical analysis.

NOTE – Quality control charts are recommended wherever the concrete is in continuous production over considerable period.

Batching and mixing of concrete

To avoid confusion and error in batching, consideration should be given to using the smallest practical number of different concrete mixes on any site or in any one plant. In batching concrete, the quantity of both cement and aggregate shall be determined by mass; admixture, if solid, by mass; liquid admixture may however be measured in volume or mass; water shall be weighed or measured by volume in a calibrated tank (see also IS4925)

Ready mixed concrete supplied by ready-mixed concrete plant shall be preferred. For large and medium project sites the concrete shall be sourced from ready-mixed concrete plants or from on site or off site batching and mixing plants (see IS 4926)

Except where it can be shown to the satisfaction of the PMC that supply of properly graded aggregate of uniform quality can be maintained over a period of work, the grading of aggregate should be controlled by obtaining the coarse aggregate in different sizes and blending them in the right proportions when required, the different sizes being stocked in separate stock – piles. The material should be stock-piled for several hours preferably a day before use. The grading of coarse and fine aggregate should be checked as frequently as possible, the frequency for a given job being determined by the PMC to ensure that the specified grading is maintained.

The accuracy of the measuring equipment shall be within ± 2 % of the quantity of cement being measured and within ± 3 percent of the quantity of aggregate, admixtures and water being measured.

Proportion / Type and grading of aggregate shall be made by trial in such a way so as to obtain densest possible concrete. All ingredients of the concrete should be used by mass only.

Volume batching may be allowed only where weigh-batching is not practical and provided accurate bulk densities of materials to be actually used in concrete have earlier been established. Allowance for bulking shall be made in accordance with IS 2386 (Part 3). The mass volume relationship should be checked as frequently as necessary, the frequency for the given job being determined by PMC to ensure that the specified grading is maintained.

It is important to maintain the water-cement ratio constant at its correct value. To this end determination of moisture contents in both fine and coarse aggregates shall be made as frequently as possible the frequency for a given job being determined by the PMC according to weather conditions. The amount of the added water shall be adjusted to compensate for any observed variations in the moisture content. For the determination of moisture content in the aggregates IS 2386 (Part 3) may be referred to. To allow for the variation in mass of aggregate due to variation in their moisture content, suitable adjustments in the masses of aggregates shall also be made. In the absence of exact data only in the case of nominal mixes the amount of surface water may be estimated from the values given in table below.

Table - Surface water carried by aggregate

Sr. No.	Aggregate	Approximate quantity of surface water	
		Percent	l/m3

		by mass	
(1)	(2)	(3)	(4)
	Very wet sand	7.5	120
	Moderately wet sand	5.0	80
	Moist sand	2.5	40
	Moist gravel or crushed rock	1.25 –2.5	20–40
	(Coarser the aggregate less water it will carry)		

No substitutions in materials used on the work or alterations in the established propositions except as permitted as above shall be made without additional tests to show that the quality and strength of concrete are satisfactory.

Mixing: Concrete shall be mixed in a mechanical mixer. The mixer should with IS 1791 and IS12119. The mixers shall be fitted with water measuring (metering) devices. The mixing shall be continued until there is a uniform distribution of the materials and the mass is uniform in colour and consistency. If there is segregation after unloading from the mixer, the concrete should be remixed.

For guidance, the mixing time shall be at least 2 min. For other types of more efficient mixers, manufacturer's recommendations shall be followed; for hydrophobic cement it may be decided by the PMC.

Workability should be checked at frequent intervals.

Dosage of retards, plasticizers and super plasticizers shall be restricted to 0.50, 1.0 and 2.0 % respectively by weight of cementitious materials and unless higher value is agreed upon between the manufacturer and constructor based on performance test.

Each time the work stops, the mixer shall be cleaned out and when next commencing the mixing, the first batch shall have 10% additional cement to allow for sticking in the drum.

SAMPLING AND TESTING CONCRETE IN THE FIELD

Facilities required for sampling materials and concrete in the field, if PMC so desires, shall be provided by EPC Contractor at no extra cost. The following equipment with operator shall be made available at Engineers request (all must be in serviceable condition)

Table : Testing equipment required at site

1.	Cast iron cube moulds 15 cm. Size	12 Nos.(min.)

2.	Slump cone complete with tamping rod	1 Set
3.	Lab. balance to weigh upto 5 kg. with sensitivity of 10 gm.	1 No.
4.	Laboratory balance of 2 kg. capacity and of sensitivity of 1 gm.	1 No.
5.	I.S. sieves for coarse and fine aggregates.	1 Set
6.	A set of measures from 5 litres. to 0.1 ltr.	1 Set
7.	Electric oven with thermostat upto 120 0C.	1 No.
8.	Pycnometer	1 No.
9.	Calibrated glass jar 1 ltr. Capacity	2 Nos.
10	Glass flasks and metal containers	As required
11	Concrete cube testing machine (optional)	1 No.

TESTING CHARGES :Different tests required to be carried out for concrete works including the mix design, cube tested as per the above specifications shall be got done by the contractor at his own cost in one of the approved laboratories. The choice of laboratory shall rest with the Department. All incidental charges / cost shall be borne by the contractor

SAMPLING AND STRENGTH TEST OF CONCRETE

General: Samples from fresh concrete shall be taken as per I.S. 1199 and cubes shall be made, cured and tested at 28 days in accordance with I.S. 516 .

In order to get a relatively quicker idea of the quality of concrete, option tests on beams for modulus of rupture at 72 (±) 2 hours or at 7 days or compressive strength tests at 7 days may be carried out in addition to 28 days compressive strength tests. For this purpose, the values given in table below may be taken for general guidance in the case of concrete made with ordinary cement. In all cases, the 28 days compressive strength specified shall alone be the criterion for acceptance or rejection of the concrete. If however, from test carried out in particular job over a reasonably long period, it has been established to the satisfaction of the PMC that a suitable ratio between 28 days compressive strength and modulus of rupture at 72 (±) 2 hours or 7 days or compressive strength at 7 days may be accepted, the PMC may suitably relax the frequency of 28 days compressive strength, provided the expected strength values at the specified early age are consistently met.

Table - Optional Tests Requirements of Concrete

Grade of Concrete	Compressive strength on 15 cm cubes min. at 7 days	Modulus of rupture by beam test min.	
		At 72 (±) 2 hours N / mm2	At 7 days N / mm2
M 10	7.0	1.2	1.7
M 15	10.0	1.5	2.1
M 20	13.5	1.7	2.4
M 25	17.0	1.9	2.7
M 30	20.0	2.1	3.0
M 35	23.5	2.3	3.2
M 40	27.0	2.5	3.4

Frequency of Sampling : Sampling Procedure : A random sampling procedure shall be adopted to ensure that each concrete batch shall have a reasonable chance of being tested, i.e. the sampling should be spread over the entire period of concreting and cover all mixing units.

Frequency : The minimum frequency of sampling of concrete of each grade shall be in accordance with following

Table: Frequency of Sampling for Concrete

Quantity of Concrete in the work (in cum.)	Number of samples
1-5	1
6-15	2
16-30	3
31-50	4
51 and above	4 plus one additional for each additional

	50 cum. or part there of

At least one sample shall be taken from each shift. Where concrete is continuous production unit, such as ready - mixed concrete plant, frequency of sampling may be agreed upon mutually by suppliers and purchasers.

Test Specimen: Three test specimen shall be made from each sample for testing at 28 days. Additional cubes may be required for various purposes such as to determine the strength of concrete at 7days or at the time of striking the form work or to determine the duration of curing or to check the testing error. Additional cubes may also be required for testing cubes cured by accelerated methods as described in I.S. 9013 - 1978. The specimen shall be tested as described in I.S. 516 - 1959.

Test Strength of Samples: The test strength of the samples shall be the average of the strength of three specimen. The individual variation should not be more than (±) 15 percent of the average.

Consistency: Slump test shall be carried out as often as demanded by the PMC and invariably from the same batch of concrete from which the test cubes are made. Slump tests shall be done immediately after sampling

Standard Deviation : Vide clause 4.4.3 of this specification.

ACCEPTANCE CRITERIA

The concrete shall be deemed to comply with the strength requirements when both the following conditions are met

The mean strength determined from any group of four consecutive test results complies with the appropriate limits in column 2 of Table below

Any individual test result complies with the appropriate limits in column of Table below.

Table - Characteristic Compressive Strength Compliance Requirement

Specified grade	Mean of the Group of 4 Non-overlapping consecutive test results in N / mm2	Individual test results in N/ mm2
(1)	(2)	(3)

M 15	≥ fck + 0.825 x established standard deviation (rounded off to nearest 0.5 N/mm2 or ≥ fck + 3 N / mm2 whichever is greater	≥fck – 3 N/ mm2
M 20 Or Above	≥ fck + 0.825 x established standard deviation (rounded off to nearest 0.5 N/ mm2 Or ≥ fck + 4 N / mm2 , whichever is greater	≥fck – 3 N/ mm 2

Note – In the absence of established value of standard deviation, the values given in (assumed standard deviation) may be assumed, and attempt should be made to obtain results of 30 samples as early as possible to establish the value of standard deviation.

Flexural Strength

When both the following conditions are met, the concrete complies with the specified flexural strength.

The mean strength determined from any group of four consecutive test results exceeds the specified characteristic strength by at least 0.3 N/mm2

The strength determined from any test result is not less than the specified characteristic strength less 0.3N/mm2

Quantity of Concrete Represented by Strength Test Results

The quantity of concrete represented by a group of four consecutive test results shall include the batches from which the first and last samples were taken together with all intervening batches.

For the individual test result requirements given in column 2 of above table or in item (b) of flexural strength, only the particular batch from which the sample was taken shall be at risk.

Where the mean rate of sampling is not specified the maximum quantity of concrete that four consecutive test results represent shall be limited to 60m3

If the concrete is deemed not to comply pursuant to above the structural adequacy of the parts affected shall be investigated and any consequential action as needed shall be taken.

Concrete of each grade shall be assessed separately

Concrete is liable to be rejected if it is porous or honey-combed, its placing has been interrupted without providing a proper construction joint, the reinforcement has been displaced beyond the tolerances specified, or construction tolerances have not been met. However, the hardened concrete may be accepted after carrying out suitable remedial measures to the satisfaction of the PMC.

ADMIXTURES

Admixtures, if used shall comply with IS 9103. Previous experience with and data on such materials should be considered in relation to the likely standards of supervision & workmanship to the work being specified. Admixtures should not impair durability of the concrete not combined with the constituent to form harmful compounds nor increase the risk of corrosion of reinforcement.

The workability, compressive strength and the slump loss of concrete with & without the use of admixtures shall be established during the trial mixes before use of admixtures.

The relative density of liquid admixtures shall be checked for such drum containing admixtures and compared with the specified value before acceptance.

The chloride content of the admixtures shall be independently tested for each batch before acceptance. If two or more admixtures are used simultaneously in the same concrete mix data should be obtained to assess their interaction and to ensure their compatibility.

General

Admixture may be used in concrete only with the approval of PMC based upon evidence that, with the passage of time neither the compressive strength nor its durability reduced. When admixtures are used, the designed concrete mix shall be corrected accordingly. Admixtures shall be used as per manufacturer's instructions and in the manner and with the control specified by PMC.

Calcium Chloride : Calcium chloride shall not be used for accelerating set of the cement for any concrete containing reinforcement or embedded steel parts. When calcium chloride is permitted to be used, such as in mass concrete works, it shall be dissolved in water and added to the mixing water in an amount not to exceed 1.5 percent of the weight of cement in each batch of concrete.

Air Entraining Agents : Where specified and approved by PMC, neutralized vinsol resin or any other approved air entraining agent may be used to produce the specified amount of air in the concrete mix and these agents shall conform to the requirements of ASTM standard 6.260, Air Entraining admixtures for concrete. The recommended total air content of the concrete is 4% (±)1%. The method of measuring air content shall be as per I.S.1199.

Retarding Admixtures: Where specified and approved by PMC, retarding agents shall be added to the concrete mix in quantities specified by PMC.

Water Reducing Admixtures: Where specified and approved by PMC, water reducing lignosulfonate mixture shall be added in quantities specified by PMC. The admixtures shall be added in the form of a solution.

Water Proofing Agents: Where specified and approved by PMC, chloride and sulphate free water proofing agents shall be added in quantities specified by PMC.

Other Admixtures: PMC may at his discretion, instruct EPC Contractor to use any other admixture in the concrete.

INSPECTION OF STRUCTURES

Immediately after stripping the form work, all concrete shall be carefully inspected and any defective work or small defects, either removed or made good before concrete has thoroughly hardened, as instructed by PMC.

In case of doubt regarding the grade of concrete used or results of cube strength are observed to be lower than the designed strength as per specifications at 28 days, compressive strength test of concrete based on core test, ultrasonic test and/or load test shall be carried out by the digital ultrasonic concrete tester by an approved agency as directed by the PMC all at the cost of the EPC Contractor. In case these tests do not satisfy the requirements, the OWNER/PMC will be at liberty to reject the concrete, and the EPC Contractor, at his own cost, has to dismantle and re-do the same or carry out such remedial measures as approved by the OWNER/PMC.

TESTING OF STRUCTURES:

Optional Tests : PMC, if he so desires, may order for tests to be carried out on cement, sand, coarse aggregate, water etc. in accordance with the relevant Indian Standards.

Tests on cement will be carried out by OWNER/PMC and shall include (i) fineness test, (ii) test for normal consistency, (iii) test for setting time, (iv) test for soundness, (v) test for compressive strength, (vi) test for heat of hydration (by experiment and by calculations) in accordance with I.S.269.

Tests on sand shall include (i) sieve test, (ii) test for organic impurities, (iii) decantation test for determining clay and silt content, (iv) specific gravity test, (v) test for unit weight and bulkage factor, (vi) test for sieve analysis and fineness modulus.

Tests on coarse aggregate shall include (i) sieve analysis, (ii) specific gravity and unit weight of dry, loose and rodded aggregate, (iii) soundness and alkali aggregate reactivity, (iv) petro-graphic examination, (v) deleterious materials and organic impurities, (vi) test for aggregate crushing value.

The test on aggregates would normally be ordered to be carried out only if PMC feels the materials are not in accordance with the specifications or if the specified concrete strengths are not obtained and shall be performed by EPC Contractor at an approved test laboratory. EPC Contractor shall have to pay all the testing charges. If the work cubes do not give the stipulated strengths, PMC reserves the right to ask EPC Contractor to dismantle such portions of the work, which in his opinion are unacceptable and re-do the work to the standards stipulated at EPC Contractors cost.

The unit rate for concrete shall be all inclusive including making preliminary mix design and test cubes, works cubes, testing them as per specifications, slump tests, etc. complete.

In case of doubt regarding the grade of concrete used, either due to poor workmanship or based on cube strength tests, tests such as rebound hammer test, ultrasonic pulse test, core test, load test etc. shall be carried out to prove the suitability of concrete as decided by PMC. The cost of the such tests shall be borne by the EPC Contractor.

Core Test : The points from which cores are to be taken and the number of cores required, shall be at the discretion of the PMC and shall be representative of the whole of concrete concerned. In no case, however, shall fewer than three cores be tested. Cores shall be prepared and tested as described in I.S. 516

Concrete in the member represented by a core test shall be considered acceptable if the average equivalent cube strength of the cores is equal to at least 85% of the cube strength of the grade of concrete specified for the corresponding age and no individual core has a strength less than 75%.

In case the core test results do not satisfy the requirements as above or where such tests have not been done, load test (see 4:11:3) may be resorted to.

Load Tests on Parts of Structure

Load tests should be carried out as soon as possible after expiry of 28 days from the time of placing of concrete.

The structure should be subjected to a load equal to full dead load of the structure plus 1.25 times the imposed load for a period of 24 hours and then the imposed load shall be removed.

Note: Dead load includes weight of the structural members plus weight of finishes and walls or partitions, if any, as considered in the design.

The deflection due to imposed load only shall be recorded. If within 24 hours of removal of the imposed load, the structure does not recover at least 75% of the deflection under super imposed load, the test may be repeated after a lapse of 72 hours. If the recovery is less than 80%, the structure shall be deemed to be unacceptable.

If the maximum deflection in mm. shown during 24 hours under load is less than $40L^2/D$, where L is the effective span in M. and D the overall depth of the section in mm, it is not necessary for recovery to be measured and the recovery provision as above will not apply.

Other non-destructive test methods may be adopted, in which case the acceptance criteria shall be agreed upon between the PMC and the EPC Contractor and the test shall be done under expert guidance.

Members Other than Flexural Members

Members other than flexural members should be preferably investigated by analysis.

Non – destructive tests

Non-destructive tests are used to obtain estimation of properties of concrete in the structure. The methods adopted include ultrasonic pulse velocity [see IS 13311 (Part 1)] and rebound hammer [IS 13311 (Part 2)], probe penetration, pull out and maturity. Non destructive tests provide alternatives to core tests for estimating the strength of concrete in a structure, or can supplement the data obtained from a limited number of cores. These methods are based on measuring a concrete property that bears some relationship to strength/ the accuracy of these methods, in part is determined by the degree of correlation between strength and the physical quality measured by the non-destructive tests.

Any of these methods may be adopted, in which case the acceptance criteria shall be agreed upon prior to testing.

Testing of Underground Water Tank/Septic Tank/Underground Structures

The tank will be tested after the completion according to the procedure laid down here
The middle compartment shall be filled first to its full height and the leakage if any shall be checked on its outer surfaces and if found, the same shall be examined carefully and defects rectified/grouted if necessary. The drop in level of surface of water shall also be recorded for 48 hours. If this drop in level exceeds 20 mm. and shows any leakage in the said walls, necessary steps shall be taken in consultation with the PMC.

After this compartment is tested to the satisfaction of the PMC, all water from middle compartment shall be pumped into side compartment to the full height and checked for water leakages from the outer surfaces of the tank as well as inner surface of the middle compartment. The drop in level of surface of water shall also be checked as stated above and defects rectified.

The external surface of the tank shall then be plastered and cured as per the specifications and back filling shall be taken up thereafter. The water from the compartments shall then be pumped out and the inner surface of the tank in all compartments then be checked and defects rectified.
After satisfactory completion of checks, internal plaster shall be taken up as specified in the specifications.

The EPC Contractor shall be responsible for carrying out the complete test, rectifying the leakages if any. The cost of providing all equipment, labour for carrying out tests shall be borne by the EPC Contractor. The rates quoted for concreting items for constructing under ground water tank shall be inclusive of testing of RCC tank for water tightness as per above specifications. The EPC Contractor shall make his own arrangement to tap the water from the OWNER/PMC at supply line for filling the tank, if supply of water stipulated under Schedule „A". If supply of water not stipulated under Schedule „A", the EPC Contractor shall make his own arrangement as per contract conditions at his own cost.

Unsatisfactory Tests

Should the results of any test prove unsatisfactory, or the structure shows signs of weakness, undue deflection or faulty construction, EPC Contractor shall remove and rebuild the member or members involved or carry out such other remedial measures as may be required by PMC. EPC Contractor shall bear the cost of so doing, unless the failure of the member or members to fulfil the test conditions is proved to be solely due to faulty design. The cost of load and other tests shall be borne by EPC Contractor if the tests show unsatisfactory results; otherwise such costs will be borne by the OWNER/PMC.

CONCRETE IN ALKALI SOILS WATER & AGGREGATES

Some aggregates containing particular varieties of silica may be susceptible to attack by alkalis (Na20 and K20) originating from cement and other sources, producing an expansive reaction which can cause cracking and disruption of concrete. Damage to concrete from this reaction will normally only occur when all the following are present together.

a) A high moisture level, within the concrete
b) A cement with high alkali content, or another source of alkali
c) Aggregate containing an alkali reactive constituent

Where the service records of particular cement / aggregate combination are well established, and do not include any instances of cracking due to alkali-aggregate reaction, no further precautions should be necessary. When the materials are unfamiliar, precautions should take one or more of the following forms:

a) Use of non-reactive aggregate from alternate sources

b) Use of low alkali ordinary Portland cement having total alkali content not more than 0.6 percent (as Na2O equivalent).

c) Further advantage can be obtained by use of fly ash (Grade I) conforming to IS 3812 or granulated blast furnace slag conforming to IS 12089 as part replacement of ordinary Portland cement (having total alkali content as Na2O equivalent not more than 0.6 percent) provided fly ash content is at least 20 % or slag content is at least 50 percent.

d) Measures to reduce the degree of saturation of the concrete during service such as use of impermeable membranes

e) Limiting the cement content in the concrete mix and thereby limiting total alkali content in the concrete mix as per approval of Engineer- in-charge.

Chlorides in the concrete: Whenever there is chlorides in concrete there is an increased risk of corrosion of embedded metal. The higher the chloride content or if subsequently exposed to warm moist conditions, the greater the risk of corrosion. All constituents may contain chlorides and concrete may be contaminated by chlorides from the external environment. To minimise the chance of deterioration of concrete from harmful chemical salts, the levels of such harmful salts in concrete materials, that is, cement, aggregates, water and admixtures, as well as by diffusion from the environment should be limited. The total amount of chloride content (as Cl) in the concrete at the time of placing shall be as given below in the table.

Table - Limits of Chloride Content of Concrete

Sl.No	Type or Use of Concrete	Maximum Total Acid soluble Chloride Content Expressed as kg/m3 of Concrete.
i)	Concrete containing metal and steam cured at elevated temperature and pre-stressed concrete	0.4
ii)	Reinforced concrete or plain concrete containing embedded metal	0.6
iii)	Concrete not containing embedded metal or any material requiring protection form chloride	3.0

The total acid soluble chloride content should be calculated from the mix proportions and the major chloride contents of each of the constituents. Whenever possible the total chloride content of the concrete should be determined as per the approval of the PMC.

Sulphates in concrete: Sulphates are present in most cements and in some aggregates; excessive amounts of water-soluble sulphate from these or other mix constituents can cause expansion and disruption of concrete. To prevent this, the total water-soluble sulphate content of the concrete mix, expressed as SO3, should not exceed 4 per cent by mass of the cement in the mix. The sulphate content should be calculated as the total from the various constituents of the mix as per the approval of the PMC.

The 4 percent limit does not applied to concrete made with super sulphated cement complying with IS 6909 or as approved by the PMC.

PREPARATION PRIOR TO CONCRETE PLACEMENT, FINAL INSPECTION & APPROVAL

Before the concrete is actually placed in position, the inside of the form work shall be inspected to see that they have been cleaned and oiled. Temporary openings shall be provided to facilitate inspection, especially at bottom of columns and wall forms, to permit removal of saw dust, wood shavings, binding wire, rubbish, dirt etc. Opening shall be placed or holes drilled so that these materials and water can be removed easily. Such openings / holes shall be later suitably plugged.

The various traders shall be permitted ample time to install drainage and plumbing lines, floor and trench drain, conduits, hangers, anchors, inserts, sleeves, bolts, frames and other miscellaneous embedment to be cast in the concrete as indicated on the drawing or as necessary for the proper execution of the work All such embedment shall be correctly positioned and securely held in the forms to prevent displacement during depositing and vibrating of concrete.

Slots, openings, holes, pockets etc. shall be provided in the concrete work in the positions indicated in the drawings or as directed by the PMC.

Reinforcement and other items to be cast in concrete shall have clean surfaces that will not impair bond.

Prior to concrete placement, all works shall be inspected and approved by PMC, and if found unsatisfactory, concrete shall not be poured until all defects have been corrected at EPC Contractors cost.

Approval by PMC of any and all materials and work as required herein shall not relieve EPC Contractor from his obligations to produce finished concrete in accordance with the drawings and specifications.

Rain or Wash Water: No concrete shall be placed in wet weather or on a water covered surface. Any concrete that has been washed by heavy rain shall be entirely removed, if there is any sign of cement and sand having been washed away from the concrete mixture. To guard against damage which may be caused by rain, the works shall be covered with tarpaulins immediately after the concrete has been placed and compacted before leaving the work unattended. Any water accumulating on the surface of the newly placed concrete shall be removed by approved means and no further concrete shall be placed thereon until such water is removed. To avoid flow of water over/around freshly placed concrete, suitable drains and sumps shall be provided.

Bonding Mortar: Immediately before concrete placement begins, prepared surfaces, except formwork, which will come in contact with concrete to be placed, shall be covered with a bonding mortar of same strength of concrete.

Transportation: All buckets, containers or conveyers used for transporting concrete shall be mortar-tight. All means of conveyance shall be adopted to deliver concrete of the required consistency and plasticity without segregation or loss of slump whatever method of transportation is employed. Chute shall not be used to transport the concrete without the written permission of the PMC and concrete shall not be re-handled before placing.

Re-tempered or Contaminated Concrete: Concrete must be placed in its final position before it becomes too stiff to work. On no account water shall be added after the initial mixing. Concrete which has become stiff or has been contaminated with foreign materials and which has not been placed within half an hour of mixing water with cement shall be rejected.

Cleaning of Equipment: All equipment used for mixing, transporting and placing of concrete shall be maintained in clean condition. All pans, buckets, hoppers, chutes, pipe lines and other equipment shall be thoroughly cleaned after each period of placement.

Procedure for Placing of Concrete

Engineers Approval of Equipment and Methods : Before any concrete is placed, the entire placing programme, consisting of equipment, layout proposed procedures and methods shall be submitted to PMC and no concrete shall be of such size and design to ensure a practically continuous flow of concrete during depositing without segregation of materials, considering the size of the job and placement location.

Time Interval Between Mixing and Placing : Concrete shall be placed in its final position before the cement reaches its initial set and concrete shall normally be compacted in its final position within thirty minutes of leaving the mixer and once compacted it shall not be disturbed.

Avoiding Segregation : Concrete shall in all the cases be deposited as nearly as practicable directly in its final position and shall not be re-handled or caused to flow in a manner which will cause segregation, loss of materials, displacement of reinforcement, shuttering or embedded inserts or impair its strength. For locations where direct placement is not possible, and in narrow forms, EPC Contractor shall provide suitable prop and Elephant Trunks to confine the movement of concrete. Special care shall be taken when concrete is dropped from a height, especially if reinforcement is in the way, particularly in columns and thin walls.

Placing by Manual Labour : Except when otherwise approved by PMC, concrete shall be placed in the shuttering by shovels or other approved implements and shall not be dropped from a height more than 1.0 m. or handled in a manner which will cause segregation.

Placing by Mechanical Equipment : The following specifications shall apply when placing of concrete by use of mechanical equipment is specially called for while inviting bids or is warranted, considering the nature of work involved.

The control of placing shall begin at the mixer discharge. Concrete shall be discharged by a vertical drop into the middle of the bucket of hopper and this principle of a vertical discharge of concrete shall be adhered-to through out all stages of delivery until the concrete comes to rest in its final position.

Type of buckets : All concrete shall be conveyed from the mixer to the place of final deposit in suitable buckets, dumpers, containers which shall be leak-tight. All means of conveyance shall be adopted for delivering concrete to the required consistency/ workability and plasticity without segregation.

Central bottom dump buckets of a type that provides for positive regulation of the amount and rate deposition of concrete in all dumping position shall be employed.

Operation of Bucket : In placing concrete in large open areas, the bucket shall be spotted directly over the position designated and then lowered for dumping. The open bucket shall clear the concrete already in place and the height of drop shall not exceed 1.0 m. The bucket shall be opened slowly to avoid high vertical bounce. Dumping of buckets on the swing or in any manner which results in separation of ingredients or disturbance of previously placed concrete will not be permitted.

Placement in Restricted Forms : Concrete placed in restricted forms by borrows, buggies, cars, short chutes or hand shovelling shall be subject to the requirement for vertical delivery of limited height to avoid segregation and shall be deposited as nearly as practicable in its final position.

Chuting: Where it is necessary to use transfer chutes, specific approval of PMC must be obtained to type, length, slopes, baffles, vertical terminal and timing of operations. These shall be so arranged that almost continuous flow of concrete is obtained at the discharge and without segregation. To allow for the loss of mortar against the sides of the chutes, the first mixes shall have less coarse aggregate. During cleaning of chutes, the waste water shall be kept clear of the forms. The concrete shall not be permitted to fall from the end of the chutes by more than 1.0 m. Chutes, when approved for use, shall have slopes not flatter than 1 vertical to 3 horizontal and not steeper than 1 vertical to 2 horizontal, chutes shall be of metal or metal line and of rounded cross section. The slopes of all chute sections

shall be approximately same. The discharge end of the chutes shall be maintained above the surfaces of the concrete in the forms.

Placing by Pumping/ Pneumatic Placers : Concrete may be conveyed and placed by mechanically operated equipment e.g. pumps or pneumatic placers, only with the written permission of PMC. The slump shall be held to the minimum, necessary for conveying concrete by this method.

When pumping is adopted, before pumping of concrete is started, the pipelines shall be lubricated with one or two batches of mortar composed of one part cement and two parts sand. The concrete mix shall specially designed to suit pumping. Care shall be taken to avoid stoppages in work once pumping has started.

When pneumatic placer is used, the manufacturers advice on layout of pipeline shall be followed to avoid blockages and excessive wear. Restraint shall be provided at the discharge box to cater for the reaction at the end.

Manufacturers advice shall be followed regarding concrete quality and all other related matters when pumping/ pneumatic placing equipment are used.

Concrete in Layers : Concreting, once started, shall be continuous until the pour is completed. Concrete shall be placed in successive horizontal layers of uniform thickness ranging from 15 cm. to 90 cm. as directed by PMC. These shall be placed as rapidly as practicable to prevent the formation of cold joints or planes of weakness between each succeeding layers within the pour. The thickness of each layer shall be such that it can be deposited before the previous layer has stiffened. The bucket loads or other units of deposit, shall spotted progressively along the face of the layer with such overlap as will facilitate spreading the layer to uniform depth and texture with a minimum of shovelling stones into mortar rather than mortar on to stones. Such a condition shall be corrected by redesign of mix or other means, as directed by PMC.

Bedding of Layers : The top surface of each pour and bedding planes shall be approximately horizontal unless otherwise instructed.

Compaction : Concrete shall be compacted during placing, with approved vibrating equipment, until the concrete has been consolidated to the maximum practicable density, is free of pockets of coarse aggregate and fits tightly against all form surfaces, reinforcement and embedded fixtures. Particular care shall be taken to ensure that all concrete placed against the form faces and into corners of forms or against hardened concrete at joints is free from voids or cavities. The use of vibrators shall be consistent with the concrete mix and caution is to be exercised not to over vibrate the concrete to the point of segregation.

Type of Vibrators : Vibrators shall conform to I.S. specifications. Type of vibrators to be used shall depend upon the structure where concrete is to be placed. Shutter vibrators, to be effective, shall be firmly secured to the form work which must be sufficiently rigid to transmit the vibrations and strong enough not to be damaged by it. Immersion vibrators shall have No load frequency amplitude and acceleration as per I.S.2505 depending on the size of the vibrator. Immersion vibrators in sufficient numbers and each of adequate size shall be used to properly consolidate all concrete. Tapping or external vibrating of forms by hand tools or immersion vibrators will not be permitted.

Use of Vibrators : The exact manner application and the most suitable machines for the purpose must be carefully considered and operated by experienced men. Immersion vibrators shall be inserted vertically at points not more than 450 mm. apart and withdrawn when air bubbles cease to come to the surface. Immersion vibrators shall be withdrawn very slowly. In no case shall immersion vibrators be used to transport concrete inside the forms. Particular attention shall be paid to vibration at the top of lift, e.g. in a column or wall.

Melding successive batches : When placing concrete in layers, which are advancing horizontally as the work progress, great care shall be exercised to ensure adequate vibration, blending and melding of the concrete between the successive layers.

Penetration of vibrators: The immersion vibrator shall penetrate the layer being placed and also penetrate the layer below while the under layer is still plastic to ensure good bond and homogeneity between the two layers and prevent the formation of cold joints.

Vibrating against reinforcement : Care shall be taken to prevent contact of immersion vibrators against reinforcement steel. Immersion vibrators shall not be allowed to come in contact with reinforcement steel after start of initial set. They shall also not be allowed to come in contact with forms or finished surfaces.

Use of Form Attached Vibrators : Form attached vibrators shall be used only with specific authorisation of PMC.

Use of surface vibrators : The use of surface vibrators will not be permitted under normal conditions. However, for thin slabs, such as highways, runways and similar construction, surface vibrations by specifically designed vibrators may be permitted, upon approval of PMC.

Stone Pockets and Mortar Pondages : Formation of stone pockets or mortar pond ages in corners and against faces of forms shall not be permitted. Should these occur, they shall be dug out, reformed and refilled to a sufficient depth and shape for thorough bounding as directed by PMC.

Placement Interval : Except when placing with slip forms, each placement of concrete in multiple lift work, shall be allowed to set for at least 24 hours after the final set of concrete and before the start of a subsequent placement.

Special Provision in Placing: When placing concrete in walls with openings, in floors of integral slab and beam construction and other similar conditions, the placing shall stop when the concrete reaches the top of the opening in walls or bottom horizontal surface of the slabs as the case may be. Placing shall be resumed before the concrete in place takes initial set, but not until it has had time to settle as determined by PMC.

Placing Concrete Through Reinforcing Steel : While placing concrete through reinforcing steel, care shall be taken to prevent segregation of the coarse aggregate. Where the congregation of steel make placing difficult, it may be necessary to temporarily move the top steel aside to get proper placement and restore reinforcing steel to design position.

Bleeding : Bleeding or free water on top of concrete being deposited in to the forms, shall because to stop the concrete pour and the conditions causing this defect corrected before any further concreting is resumed.

CONSTRUCTION JOINTS AND KEYS

Concrete shall be placed without interruption until completion of the part of the work between predetermined construction joints, as specified therein after. Time lapse between the pouring of adjoining units shall be as specified in the drawings or as directed by PMC.

Construction joints and cold joints: Joints are a common source of weakness and therefore it is desirable to avoid them. If this is not possible, their number shall be minimized. Concreting shall be carried out continuously up to construction joints the position and arrangement of which shall be indicated by the designer. Construction joints should comply with IS 11817.

Construction joints shall be placed at accessible locations to permit cleaning out of laitance, cement slurry and unsound concrete, in order to create rough/uneven surface. It is recommended to clean out laitance and cement slurry by using wire brush on the surface of joint immediately after initial setting of concrete and to clean at the same immediately thereafter. The prepared surface should be in a clean saturated surface dry condition when fresh concrete is placed, against it. In the case of construction joints at locations where the previous pour has been cast against shuttering the recommended method of obtaining a rough surface for the previously poured concrete is to expose the aggregate with a high pressure water jet or any other appropriate means. At high pressure with a lower flow, water jet shall remove all loose laitance, cement slurry and unsound concrete without damaging the sound concrete below.

Fresh concrete should be thoroughly vibrated near construction joints so that mortar from the new concrete flows between large aggregates and develop proper bond with old concrete.
Where high shear resistance is required at the construction joints, shear keys may be provided.
Sprayed curing membranes and release agents should be thoroughly removed from joint surfaces.

If stopping of concreting becomes unavoidable anywhere, a properly formed construction joint shall be made where the work is stopped. Joints shall be either vertical or horizontal, unless shown otherwise in drawing. In case of an inclined or curved member, the joints shall be at right angles to the axis of the member. Vertical joints in walls shall be kept to a minimum. Vertical joints shall be formed against a stop board, horizontal joints shall be level and wherever possible, arranged, so that the joint lines coincide with the architectural features of the finished work. Battens, shall be nailed to the form work to ensure a horizontal line and if directed, shall also be used to form a grooved joint. For tank walls, similar work joints shall be formed as per I.S. 3370. Concrete that is in the process of setting shall not be disturbed or shaken by traffic either on the concrete itself or upon the shuttering. Horizontal and vertical construction joints and shear keys shall be located and shall conform in detail to the requirements of the plans unless otherwise directed by PMC. Where not described, the joints shall be in accordance with the following:

Column Joints : In a column, the joint shall be formed 75 mm. below the lowest soffit of the beams including haunches if any. In flat slab construction the joint shall be 75 mm. below the soffit of column capital. At least 2 hours shall elapse after depositing concrete in column, piers or walls, before depositing in beams, girders or slabs supported thereon.

Beam and Slab Joints: Concrete in a beam shall be placed throughout without a joint but if the provision of a joint is unavoidable, the joint shall be vertical and at the centre or within the middle third of the span unless otherwise shown in drawing. Where a beam intersects a girder, the joints in the girder shall be offset a distance equal to twice the width of the beam and additional reinforcement provided for shear. The joints shall be vertical throughout the full thickness of the concrete member. A joint in a slab shall be vertical and parallel to the principal reinforcement. Where it is unavoidable at right angles to the principle reinforcement, the joint shall be vertical and at the middle of span.

Joints in Liquid Retaining Structures: Vertical construction joints in watertight construction will not be permitted unless indicated on the drawings. Where a horizontal construction joint is required to resist water pressure, special care shall be taken in all phases of its construction to ensure maximum water tightness.

Dowels: Dowels for concrete work, not likely to be taken up in the near future, shall be wrapped in tarpaper and burlap.

Mass Foundations: Mass foundations shall be poured in lifts not exceeding 1.5 m. in height unless, otherwise indicated on the drawings and approved by PMC.

Treatment of Construction Joints on Resuming Concreting: Drier shall be used for the top lift or horizontal pours to avoid a laitance. All laitance and loose stones shall be thoroughly and carefully removed by wire brushing/ hacking and surface washed.

Just before concreting is resumed, the roughened joint surface shall be thoroughly cleaned and loose matter removed and then treated with a thin layer of cement grout of proportion specified by PMC and worked well into the surface. The new concrete shall be well worked specially against the prepared face before the grout mortar sets. Special care shall be taken to obtain thorough compaction and to avoid segregation of the concrete along the joint plane.

CURING, PROTECTING, REPAIRING, AND FINISHING

Curing : All concrete shall be cured by keeping it continuously damp for the period of time required for complete hydration and hardening to take place. Preference shall be given to the use of continuous sprays or ponded water, continuously saturated covering of sacking, canvas, hessian or other absorbent materials, or approved effective curing compounds applied with spraying equipment capable of producing a smooth, even textured coat. Extra precautions shall be exercised in curing concrete during cold and hot weather as outlined hereinafter. The quality of curing water shall be the same as that used for mixing concrete.

Certain types of finish or preparation for overlaying concrete must be done at certain stages of curing process and special treatment may be required for specific concrete surface finish.
Curing of concrete made of high alumina cement and super sulphated cement shall be carried out as directed by PMC.

Curing with Water : Fresh concrete shall be kept continuously wet for a minimum period of 14days from the date of placing of concrete, following a lapse of 12 to 24 hours after laying concrete. The curing of horizontal surfaces exposed to the drying winds shall however begin immediately the concrete has hardened. Water shall be applied to unformed concrete surfaces within 1 hour after concrete has set. Water shall be applied to formed surfaces immediately upon removal of forms. Quantity of water applied shall be controlled so as to prevent erosion of freshly placed concrete.

Continuous Spraying : Curing shall be assured by use of an ample water supply under pressure in pipes, with all necessary appliances of hose sprinklers and spraying devices. Continuous fine mist spraying or sprinkling shall be used, unless otherwise specified or approved by PMC.

Alternate Curing Methods: Whenever in the judgment of PMC, it may be necessary to omit the continuous spray method, covering of clean sand or other approved means such as wet gunny bags, which will prevent loss of moisture from the concrete, may be used. Any type of covering which would stain or damage the concrete during or after the curing period, will not be permitted. Covering shall be kept continuously wet during the curing period.

For curing of concrete in pavements, side-walks, floors, flat roofs or other level surfaces, the ponding method of curing is preferred. The method of containing the ponded water shall be approved by PMC. Special attention shall be given to edges and corners of the slab to ensure proper protection to these areas. The ponded areas shall be kept continuously filled with water during the curing period.

Curing Compounds : Surface coating type curing compound shall be used only on special permission of PMC. Curing compounds shall be liquid type while pigmented, conforming to U.S. Bureau of Reclamation Specification. No curing compound shall be used on surface where future blending with concrete, water or acid proof membrane or painting is specified.

Curing Equipment: All equipment and materials required for curing shall be on hand and ready for use before concrete is placed.

Moist Curing: Exposed surfaces of concrete shall be kept continuously in a damp or wet condition by ponding or by covering with a layer of sacking , canvas, hessian or similar materials and kept constantly wet for at least seven days from the date of placing concrete. In case of ordinary Portland cement and at least ten days where mineral admixtures or blended cement are used. The period of curing shall not be less than ten days of concrete exposed to dry and hot weather conditions. In the case of concrete the mineral admixtures or blended cement are used, it is recommended that the above minimum periods may be extended to fourteen days as per the approval of the PMC.

Membrane Curing: Approved curing compounds may be used in lieu of moist curing with the permission of PMC. Such compounds shall be applied to all exposed surfaces of the concrete as soon as possible after the concrete has set. Impermeable membranes such as poly ethylene sheeting covering, closely the concrete surface may also be used to provide effective barrier against evaporation.

For the concrete containing Portland pozzolana cement, Portland slag cement or mineral admixtures increased period of curing may be decided by PMC.
The rate of strength development at early age of concrete made with super sulphated cement is significantly reduced at lower temperatures. Super sulphated cement concrete is seriously affected by inadequate curing and the surface has to be kept moist for at least seven days or more as per the approval of the PMC.

PROTECTING FRESH CONCRETE

Fresh concrete shall be protected from the elements, from defacements and damage due to construction operations by leaving forms in place for ample period as specified later in this specification. Newly placed concrete shall be protected by approved means such as tarpaulins from rain, sun and winds. Steps as approved by PMC, shall also be taken to protect immature concrete from damage by debris, excessive loading, vibrations, abrasion or contact with other materials etc. that may be warned against and prevented from disturbing green concrete during its setting period. If it is

necessary that workmen enter the area of freshly placed concrete, PMC may require that bridges be placed over the area.

REPAIR AND REPLACEMENT OF UNSATISFACTORY CONCRETE

Immediately after the shuttering is removed, the surface of concrete shall be very carefully gone over and all defective areas called to the attention of PMC who may permit patching of the defective areas or else reject the concrete unit either partially or entirely. Rejected concrete shall be removed and replaced by EPC Contractor at no additional expense to the OWNER/PMC. Holes left by form bolts etc. shall be filled up and made good with mortar composed of one part of cement to one and half parts of sand passing through 2.36 mm. I.S. sieve after removing any loose stones adhering to the concrete. Mortar filling shall be struck off flush at the face of the concrete. Concrete surface shall be finished as described under the particular item of work.

Superficial honey combed surfaces and rough patches shall be similarly made good immediately after removal of shuttering, in the presence of PMC and superficial water and air holes shall be filled in. The mortar shall be well worked into the surface with wooden float. Excess water shall be avoided. Unless instructed otherwise by PMC, the surface of the exposed concrete placed against shuttering shall be rubbed down immediately on removal of shuttering to remove fine or other irregularities, care being taken to avoid damaging the surfaces. Surface irregularities shall be removed by grinding.

If reinforcement is exposes or the honey combing occurs at vulnerable position e.g. ends of beams or columns, it may be necessary to cut out the member completely or in part and reconstruct. The decision of PMC shall be final in this regard. If only patching is necessary, the defective concrete shall be cut out till solid concrete is reached (or to a minimum depth of 25 mm.), the edges being cut perpendicular to the affected surface or with a small undercut if possible, anchors, tees or dowels shall be provided in slots whenever necessary to attach the new concrete securely in place. An area extending several centimetres beyond the edges and the surfaces of the prepared voids shall be saturated with water for 24 hours immediately before the patching material is placed.

Use of Epoxy: The use of epoxy for bonding fresh concrete used for repairs will be permitted upon written approval of PMC. Epoxies shall be applied in strict accordance with the instruction of the manufacturer.

Method of Repair: Small size holes having surface dimensions about equal to the depth of the hole, holes left after removal of form bolts, grout insert holes and slots cut for repair of cracks shall be repaired as follows:

The hole to be patched shall be roughened and thoroughly soaked with clean water until absorption stops.

A 5 mm. thick layer of grout of equal parts of cement and sand shall be well brushed into the surface to be patched followed immediately by the patching concrete which shall be well consolidated with a wooden float and left slightly proud of the surrounding surface. The concrete patch shall be built up in 10 mm. thick layers. After an hour or more, depending upon weather conditions, it shall be worked off flush with a wooden float and a smooth finish obtained by wiping with hessian. Steel trowel shall not be used for this purpose. The mix for patching shall be of the same materials and in the same proportions as that used in the concrete being repaired, although some reduction in the maximum size of the coarse aggregates may be necessary and the mix shall be kept as dry as possible.

Mortar filling by air pressure (guniting) shall be used for repair of areas to large and/ or too shallow for patching with mortar. Patched surfaces shall be given a final treatment to match the colour and texture of the surrounding concrete. White cement shall be substituted for ordinary cement, if so directed by Engineering-in-Charge, to match the shade of the patch with the original concrete.

Curing of Patched Work: The patched area shall be covered immediately with an approved non-staining water-saturated material such as gunny bags, which shall be kept continuously wet and protected against sun and wind for a period of 24 hours. Thereafter, the patched area shall be kept wet continuously by a fine spray of sprinkling water for not less than 10 days.

Approval by PMC: All materials, procedures and operations used in the repair of concrete and also the finished repair work shall be subject to the approval of PMC. All fillings shall tightly bonded to the concrete and shall be sound, free from shrinkage cracks after the fillings have been cured and dried.

FINISHING

This specification is intended to cover the treatment of concrete surfaces of all structures.

Finish for Formed Surfaces: The type of finish for formed concrete surfaces shall be as follows, unless otherwise specified by the PMC:

For surfaces against which backfill or concrete is to be placed, no treatment is required except repair of defective areas.

For surfaces below grade, which will receive waterproofing treatment, the concrete shall be free of surface irregularities which would interfere with proper application of the water-proofing materials which is specified for use.

Unless specified, surfaces which will be exposed when the structure is in service shall receive no special finish, except repair of damaged or defective concrete, removal of fins and abrupt irregularities, filling of holes left by form ties and rods and clean up of loose or adhering debris.

Surfaces which will be exposed to the weather and which would normally be levelled, shall be sloped for drainage. Unless the drawing specify a horizontal surface or shows the slope required, the tops of narrow surfaces such as staircase treads, walls, curbs and parapets shall be sloped across the width approx. as 1 in 30. Broader surfaces such as walkways, roads, parking areas and platforms shall be sloped about 1 in 50. Surfaces that will be covered by backfill or concrete, sub floors to be covered with concrete topping, terrazzo or quarry tile and similar surfaces shall be smooth, screeded and levelled to produce even surfaces. Surface irregularities shall not exceed 6mm. Surfaces which will not be covered by backfill, concrete or tile topping such as outside decks, floors of galleries and sumps, parapets, gutters, sidewalks, floors and slabs shall be consolidated, screeded and floated.

Excess water and laitance shall be removed before final finishing. Floating may be done with hand or power tools and started as soon as the screeded surface has attained a stiffness to permit finishing

operations and these shall be the minimum required to produce a surface uniform in texture and free from screed marks or other imperfections. Joints and edges shall be tooled as called for on the drawings or as directed by PMC.

Standard Finish for Exposed Concrete: Exposed concrete shall mean any concrete other than floors or slabs exposed to view upon completion of the job. Unless otherwise specified on the drawings, the standard finish for exposed concrete shall be of smooth finish.

smooth finish shall be obtained with use of lined or plywood forms having smooth and even surfaces and edges. Panels and form linings shall be of uniform size and be as large as practicable and installed with closed joints. Upon removal of forms, the joint marks shall be smoothed off and all blemishes, projections etc. removed, leaving the surfaces reasonably smooth and unmarred.

Integral Cement Concrete Finish: When specified on the drawings, an integral cement concrete finish of specified thickness for floors and slabs shall be applied either monolithic or bonded, as specified in the drawings and as per I.S.2571. The surface shall be compacted and then floated with a wooden float or power floating machine. The surface shall be tested with a straight edge and any high and low spots eliminated. Floating or trowelling of the finish shall be permitted only after all surface water has evaporated. Dry cement or a mixture of dry cement and sand shall not be sprinkled directly on the surface of the cement finish to absorb moisture or to stiffen the mix.

Rubbed Finish: A rubbed finish shall be provided only on exposed concrete surfaces as specified on the drawings. Upon removal of forms, all fins and other projections on the surfaces shall be carefully removed, off sets levelled and voids and/ or damaged sections immediately saturated with water and repaired by filling with concrete or mortar of the same composition as was used in the surfaces. The surfaces shall then be thoroughly wetted and rubbed with carborundum or other abrasive. Cement mortar may be used in the rubbing, but the finished surfaces shall not be brush coated with either cement or grout after rubbing. The finished surfaces shall present a uniform and smooth appearance.

PROTECTION

All concrete shall be protected against damage until final acceptance by PMC.

FOUNDATION BEDDING, BONDING AND JOINTING

All surfaces upon or against which concrete will be placed shall be suitably prepared by thoroughly cleaning, washing and dewatering as may be indicated in the plans or as PMC may direct to meet the various situations encountered in the work.

Soft or spongy areas shall be cleaned out and back filled with either a soil cement mixture, lean concrete or clean sand fill compacted to minimum density of 90% Modified Proctor, unless otherwise mentioned in schedule of quantities.

Prior to construction of form work for any item where soil will not act as bottom form, approval shall be obtained from PMC as to the suitability of the soil.

PREPARATION OF ROCK STRATA OF FOUNDATIONS

To provide tight bond with rock foundations, the rock surface shall be prepared and the following general requirements shall be observed:

Concrete shall not be deposited on large sloping rock surface. Where required by PMC or as indicated on the plans, the rock shall be cut to form rough steps or benches to provide roughness or a more suitable bearing surface.

Rock foundation stratum shall be prepared by picking, barring, wedging and similar methods which will leave the rock in an entirely sound and unshattered condition.

Shortly before concrete is placed, the rock surface shall be cleaned with high pressure water and air jet even though it may have been previously cleaned in that manner.

Prior to placing concrete, the rock surface shall be kept wet for a period of 2 to 4 hours unless otherwise directed by the PMC.

Before placing concrete on rock surfaces, all water shall be removed from depressions to permit through inspection and proper bonding of the concrete to the rock.

PREPARATION OF EARTH STRATA OF FOUNDATIONS

All earth surfaces upon which or against which concrete is to be placed, shall be well compacted and free from standing water, mud or debris. Soft, yielding soils shall be removed and replaced with suitable earth and well compacted as directed by the PMC. Where specified, lean concrete shall be provided in the earth stratum for receiving concrete. The surface of absorptive soil against which concrete is to be placed shall be moistened thoroughly so that no moisture will be drawn form the freshly placed concrete and later shall help to cure the concrete.

PREPARATION OF CONCRETE SURFACES

Preparation of concrete surface upon which additional concrete is to be placed later, shall preferably be done by scarifying and cleaning while the concrete is between its initial and final set. This method shall be used wherever practicable and shall consist of cutting the surface with picks and stiff brooms and by use of an approved combination of air and water jet as directed by PMC. Great care shall be taken in performing this work to avoid removal of too much mortar and the weakening of the surface by loosening of aggregate. When it is not practicable to follow the above method, it will be necessary to employ air tools to remove laitance and roughen the surface.

The final required result shall be a pitted surface from which all dirt, unsound concrete, laitance and glazed mortar have been removed.

BONDING TREATMENT (MORTAR)

After rock or concrete surfaces upon which new concrete is to be placed have been scarified, cleaned and wetted as specified herein, it shall receive a bonding treatment, immediately before placement of the concrete.

The bonding medium shall be a coat of cement sand mortar. The mortar shall have the same cement-sand proportion as the concrete which shall be placed on it. The water cement ratio shall be determined by placing conditions and as approved by PMC.

Bonding mortar shall be placed in sufficient quantity to completely cover the surface about 10 mm. thick for rock surface and about 5 mm. thick for concrete surfaces. It shall be brushed or broomed over the surface and worked thoroughly into all cracks, crevices and depressions. Accumulations or puddles of mortar shall not be allowed to settle in depressions and shall be brushed out to a satisfactory degree as determined by PMC.

Mortar shall be placed at such a rate that it can be brushed over the surface just in advance of placement of concrete. Only as much area shall be covered with mortar as can be covered with concrete before initial set in the mortar takes place. The amount of mortar that will be permitted to be placed at any one-time, or the area which is to cover, shall be in accordance with PMC.

CLEANING AND BONDING OF FORMED CONSTRUCTION JOINTS

Vertical construction joints shall be cleaned as specified above or by other methods approved by PMC. In placing concrete against formed construction joints, the surfaces of the joints, where accessible, shall be coated thoroughly with the specified bed-joint bonding mortar immediately before they are covered with concrete or by scrubbing with wire brooms, dipped into the fresh concrete. Where it is impracticable to apply such a mortar coating, special precautions shall be taken to ensure that the new concrete is brought into intimate contact with the surface of the joint by carefully puddling and spading with aid of vibrators and suitable tools.

EXPANSION AND CONTRACTION

Provision shall be made for expansion and contraction in concrete by use of special type joints located as shown in the drawings. Construction joint surfaces shall be treated as specified in the specifications, shown in the drawings or as directed by PMC.

HOT WEATHER REQUIREMENTS

All concrete work performed in hot weather shall be in accordance with I.S. 456, except as herein modified.

Admixtures may be used only when approved by PMC.

Adequate provision shall be made to lower concrete temperatures by cool ingredients, eliminating excessive mixing, preventing exposure of mixers and conveyers to direct sunlight and the use of reflective paint, on mixers etc. The temperature of the freshly placed concrete shall not be permitted to exceed 30^0c.

Consideration shall be given to shading aggregate stock piles from direct rays of the sun and spraying stock piles with water, use of cold water available and burying, insulation, shading and/ or painting white the pipe lines and water storage tanks and conveyances.

In order to reduce loss of mixing water, the aggregates, wooden forms, subgrade, adjacent concrete and other moisture absorbing surfaces, shall be well wetted prior to concreting. Placement and finishing shall be done as quickly as possible.

Extra precautions shall be taken for the protection and curing of concrete. Consideration shall be given to continuous water curing and protection against high temperatures and drying hot wind for a period of at least 7 days immediately after concrete has set and after which normal curing procedures may be resumed.

PLACING CONCRETE UNDER WATER

Under all ordinary conditions all foundations shall be completely dewatered and concrete placed in the dry. However, when concrete placement under water is necessary, all work shall conform to I.S.456 and procedure shall be as follows:

Method of Placement : Concrete shall be deposited under water by means of tremies or drop bottom buckets of approved type.

DIRECTION, INSPECTION AND APPROVAL

All work requiring placement of concrete underwater shall be designed, directed and inspected with regard to the local circumstances and purposes. All under water concrete shall be placed according to the plans or specifications and as directed and approved by PMC.

PRECAST CONCRETE & PRECAST REINFORCED CONCRETE

Precast concrete & precast reinforced concrete shall comply with I.S. 456, and with the following requirements:

General requirements : Precast reinforced concrete units such as columns, fencing posts, doors and window frames, lintels, chajjas, copings, sills, shelves, slabs, louvers etc. shall be of grade of mix as specified and cast in forms or moulds. The forms / moulds shall be of fibre glass or of steel sections for better finish. Provision shall be made in the forms and moulds to accommodate fixing devices such as nibs, clips, hooks, bolts and forming of notches and holes. Precast concrete shall be cast on suitable bed or platform with firm foundation and free from wind. The EPC Contractor may precast the units on a cement or steel platform which shall be adequately oiled provided the surface finish is of the same standard as obtained in the forms. Each unit shall be cast in one operation. EPC Contractor shall be responsible for the accuracy of the level or shape of the bed or platform. A suitable serial number and the date of casting shall be impressed or painted on each unit.

Concrete used for precasting the units should be well proportioned as specified, mixed, placed and thoroughly compacted by vibrations or tamping to give a dense concrete free from voids and honeycombing.

Precast articles shall have a dense surface finish showing no coarse aggregate and shall have no cracks or crevices likely to assist in disintegration of concrete or rusting of steel or other defects that would interfere with the proper placing of the units. All angles of the precast units with the exception of the angles resulting from the splayed or chamfered faces shall be true right angles. The a rises shall be clean and sharp except those specified or shown to be rounded. The wearing surface shall be true

to the lines. On being fractured, the interior of the units should present a clean homogenous appearance.

The longitudinal reinforcement shall have a minimum cover of 12 mm or twice the diameter of the main bar, whichever is more, unless otherwise directed in respect of all items except fencing posts or electric posts where the minimum cover shall be 25 mm.

CURING

After having been cast in the mould or form the concrete shall be adequately protected during setting in the first stages of hardening from shocks and from harmful effects of frost, sunshine, drying winds and cold.
All precast work shall be protected from the direct rays of the sun for at least 7 days after casting and during that period each units shall be kept constantly watered or preferably by completely immersed in water if the size of unit so permits. Otherwise curing practices as given in clauses stated earlier shall be followed.

The precast articles shall be matured for 28 days before erection or being built in so that the concrete shall have sufficient strength to prevent damage to units when first handled. Side shutters shall not be struck in less than 24 hours after depositing concrete and no precast unit shall be lifted until the concrete reaches a strength of at least twice the stress to which the concrete may be subjected at the time of lifting.

Marking: Precast units shall be clearly marked to indicate the top of member and its location and orientation in the structure.

Precast units shall be stored, transported and placed in position in such a manner that they will not be overstressed or damaged. The lifting and removal of precast units shall be undertaken without causing shocks, vibration or under bending stresses to or in the units. Before lifting and removal takes place, EPC Contractor shall satisfy PMC or his representative that the methods he proposes to adopt for

these operations will not over-stress or otherwise affect seriously the strength of the precast units. The reinforced side of the units shall be distinctly marked.

SLOTS, OPENINGS ETC.

Slots, openings or holes, pockets etc. shall be provided in the concrete work in the positions indicated in the drawings or as directed by the PMC. Any deviation from the approved drawings shall be made good by EPC Contractor at his own expenses without damaging any other work. Sleeves, bolts, inserts etc. shall also be provided in concrete work where so specified.

GROUTING

Standard Grout : Grout shall be provided as specified in the drawing.

The proportions of grout shall be such as to produce a flowable mixture consistent with minimum water content and shrinkage. The grout proportions shall be limited as follows

Table: Grout Properties

Use	Grout thickness	Mix. proportions	W/C. Ratio in (Max.)
a) Fluid mix	Under 25 mm.	One part Portland cement to one part sand.	0.44
b) General	25 mm. and over but less than 50 mm.	One part Portland cement to 2 parts of sand.	0.53
c) Stiff mix.	50 mm. and over	One part Portland cement to 3 parts of sand.	0.53

Sand shall be such as to produce a flowable grout without any tendency to segregate. Sand, for general grouting purposes, shall be graded within the following limits:

Table– Sand for General Grouting

Passing I.S. sieve 2.36 mm.	95 to100%

Passing I.S. sieve 1.18 mm.	65 to95%
Passing I.S. sieve 300 micron above	10 to30%
Passing I.S. sieve 150 micron above	3 to10%

Sand for fluid grouts shall have the fine material passing the 300 and 150 micron sieves at the upper limits specified above.

Sand for stiff grouts, shall meet the usual grading specifications and concrete.

Surface to be grouted shall be thoroughly roughened and cleaned of all foreign matter and laitance.

Anchor bolts, anchor bolt holes and bottom of equipment and column base plates shall be cleaned of all oil, grease, dirt and loose material. The use of hot, strong, caustic solution for this purpose will be permitted.

Prior to grouting, the hardened concrete surfaces to be grouted, shall be saturated with water.
Water in anchor bolt holes shall be removed before grouting is started.
Forms around base plates shall be, reasonably, tightened to prevent leakage of the grout.
Adequate clearance shall be provided between forms and base plate to permit grout to be worked properly into place.

Grouting, once started shall be done quickly and continuously to prevent segregation, bleeding and breakdown of initial set. Grout shall be worked from one side of one end to the other to prevent entrapment of air. To distribute the grout and to ensure more release from entrapped air, link chains can be used to work the grout into place.

Grouting through holes in base plate shall be by pressure grouting.
Variations in grout mixes and procedures shall be permitted if approved by the PMC.

Special Grout : Special grout where specified on the drawing shall be provided in strict accordance with the manufacturers instructions/ specifications on the drawings.

INSPECTION

All materials, workmanship and finished construction shall be subject to the continuous inspection and approval of PMC.

All rejected materials supplied by EPC Contractor and all rejected work or construction performed by EPC Contractor, as is not in conformance with the specifications and drawings, shall immediately be replaced at no additional expense to the OWNER/PMC.

Approval of any preliminary material or phase of work shall in no way relieve the EPC Contractor from the responsibility of supplying concrete and/ or producing finished concrete in accordance with the specifications and drawings.

All concrete shall be protected against damage until final acceptance by the OWNER/PMC or its representatives.

CLEAN UP

Upon the completion of concrete work, all forms, equipment, construction tools, protective coverings and any debris resulting from the work shall be removed from the premises.

All debris i.e. empty containers, scrap wood etc. shall be removed to dump daily or as directed by the PMC.

The finished concrete surfaces shall be left in a clean condition to the satisfaction of the PMC.

PLAIN CEMENT CONCRETE FOR GENERAL WORK

For plain cement concrete work, the specification for materials viz. cement, sand, fine and coarse aggregates and water shall be the same as that specified in reinforced concrete work specification.

But the proportion of mix will be nominal and the ratio of fine and coarse aggregate may be slightly adjusted within limits, keeping the total value of aggregates to a given volumes of cement constant to

suit the sieve analysis of both the aggregates. Cement shall on no account be measured by volume, but it shall always be used directly from the bags (i.e. 50 kg/bag).

The proportion of cement, sand, aggregate and water for concrete of proportion 1:5:10, 1:4:8, 1:3:6 & 1:2:4 by volume shall generally consist of quantities as given below :

Table : The proportion of cement, sand, aggregate and by volume

Proportion of Ingredients	Cement	Quantity of materials used per bag of cement			
		Fine aggregate (sand)	Coarse aggregate	Total of fine sand coarse aggregates	Water
1:5:10	1	175 litres.	350 litres.	800 kgs.	60 litres.
1:4:8	1	140 litres.	280 litres.	625 kgs.	45 litres.
1:3:6	1	105 litres.	210 litres.	480 kgs.	34 litres.
1:2:4	1	70 litres.	140 litres.	330 kgs.	32 litres.

The quantity of water used shall be such as to produce concrete of consistency required by the particular class of work and shall be decided by the use of a slump cone. Sufficient care should be taken to see that no excess quantity of water is used. The final proportion of the aggregate and quantity of water shall be decided by the PMC on the basis of test in each case.

Table: Proportion of The Aggregate And Quantity of Water

Mix proportion	Cement	Sand	Coarse Aggregate in CUM			Water
Ordinary mix in volume			40 mm	20 mm	12 mm	
1:5:10	2.60	0.475	0.6623	0.2583	-	156
1:4:8	3.40	0.500	0.6883	0.6883	-	153
1:3:6 (with 40mm aggregates)	4.4	0.485	0.672	0.672	0.262	176
1:3:6 (with 20 mm aggregates)	4.4	0.485	-	0.727	0.242	162.5
1:2:4 (with 20 mm aggregates)	6.4	0.47	-	0.705	0.235	205
1:2:4 (with 40 mm	6.4	0.47	0.544	0.241	0.126	235

aggregates)						
1:1.5:3	8.0	0.441	-	0.6615	0.2205	240
1:1:2	12.20	0.45	-	0.675	0.225	330

The slump shall be specified for each class of work and shall in general be as follows:

Table : Maximum Slump in Concrete

Type of concrete	**Max. slump (in mm.)**
Mass concrete	50
Concrete below water proofing treatment	50
Coping	25
Floor paving	50

All plain concrete should be preferably mixed in a drum type powder driven machine with a loading hopper which will permit the accurate measure of various ingredients. If hand mixing is authorised, it should be done on a water tight platform.

The mixing of each batch in the concrete mixer shall continue for not less than 1.5 minutes after the materials and water are in the mixer. The volume is mixed materials per batch shall not exceed the manufacturers rated capacity of the mixer. The mixer shall rotate at a peripheral speed of about 60 metres per minute.

PLUM CONCRETEAND PLUM MASONRY:-

The proportion of plums should not exceed 50 per cent of total volume of plum concrete /plum masonry. The stone size to be used for plum shall be between 200 to 300 mm. The stone shall be basalt ,trap or any other approved locally available stone with minimum crushing strength of 100 kg/sq.cm obtained from quarries approved by Engineering -in-charge.

The stone shall be hard durable and tough .The length of stone shall not exceed 3 time its height.

it will be sufficient to make up the total volume with plums and fill all the interstices, which should not be less than 150cm, with cement concrete. The cement concrete shall be in 1:4:8 (cement, sand, coarse aggregate). The coarse aggregate shall be 25 mm down. The water cement ratio shall be adjusted at site to maintain the flowability of concrete to fill all the interstices properly. The plums are laid in layers using the cement concrete as mortar.

The plum shall be raised uniformly, and no part, at any time shall be raised more than 900 mm above adjoining work.

Double scaffolding shall be provided for construction and piercing of walls for scaffolding shall not be permitted. The contactor shall be responsible for any damage or injury resulting from poor scaffolding.

All plum concrete/ plum masonry shall be maintained wet for at least 7 days

Weep holes shall be provided by HDPE pipes buried in plum masonry. The end of the weep holes, on the filled up side, shall be covered by geo-membrane and filled by gravels of sizes 40 mm.

Form work shall be of plywood or steel.
Concrete shall be poured and consolidated in its final position within half an hour of mixing. The retempering of concrete which has partially hardened, that is remixing with or without additional cement aggregate or water shall not be permitted. Concrete of mix 1:3:6 and 1:2:4 will be required to be vibrated if specified and directed by the Engineer. In case of the thickness of concrete is more than 150 mm., it may be vibrated as directed by the Engineer.

The concrete shall be cured for 10 days in ordinary weather and 15 days in hot weather.

Measurements for the work done shall be exact length, breadth and depth shown in figures on the drawings or as directed by the PMC and after the concrete is consolidated. No extra shall be paid for excess quantity resulting from faulty workmanship.

FINISHING OF CONCRETE

General : Unless otherwise specified, concrete finishes shall conform to the following specifications:

Finish F1, F2 and F3 shall describe formed surface.

Finish U1, U2 and U3 shall describe un-formed surface.

Off sets or fins caused by disposed or misplaced form sheathing lining or form sections or by defective form lumber shall be referred to as abrupt irregularities. All other irregularities shall be referred to as gradual irregularities. Gradual irregularities shall be measured as deviation from a plane surface with a template 1.5 m. long for formed surface and 3 m. long for unformed surfaces.

Formed Surfaces

Finish F1—shall apply to all formed surfaces for which finish F2, F3 or any other special finish is not specified and shall include filling up all form tie holes.

Finish F2—shall apply to all formed surfaces so shown on the drawings or specified by the PMC. This shall include filling all form tie-holes, repair of gradual irregularities exceeding 6 mm., removal of ridges and abrupt irregularities by grinding.

Finish F3—shall apply to all formed surfaces exposed to view or where shown in the drawings or specified by the PMC. Finish F3 - shall include all measures specified for Finish-F2 and in addition, Filling air holes with mortar and treatment of the entire surface with sack rubbed finish. It shall also include clean up of loose and adhering debris. Where a sack rubbed finish is specified, the surfaces shall be prepared within two days after removal of the forms.

The surface shall be wetted and allowed to dry slightly before mortar is applied by sack rubbing. The mortar used shall consist of one part cement to one and half parts by volume of fine (minus No. 16 mesh) sand. Only sufficient mixing water to give the mortar a workable consistency shall be used. The mortar shall then be rubbed over the surface with a fine burlap or linen cloth so as to fill all the surface voids. The mortar rubbed in the voids shall be allowed to stiffen and solidify after which the whole surface shall be wiped clean so that the surface presents a uniform appearance without air holes, irregularities etc. Curing of the surface shall be continued for a period of ten (10) days.

Unformed Surfaces

Finish U1—shall apply to all unformed surfaces for which the finish U2, U3 or any other special finish is not specified and shall include screeding the surface of the concrete to the required slope and grade. Unless the drawing specifies a horizontal surface or shows the slope required, the tops of narrow surfaces such as stair, treads, walls, curbs and parapets shall be sloped approximately 10 mm. per 300 mm. width. Surfaces to be covered by backfill or concrete sub-floors to be covered with concrete topping, terrazzo and similar surfaces shall be smooth screeded and levelled to produce even surface, irregularities not exceeding 6 mm.

Finish U2—shall apply to all unformed surfaces as shown in the drawing or specified by the PMC and shall include screeding and applying a wood float finish to the surface of the concrete to the required slopes and grade.

Repair of abrupt irregularities unless a roughened texture is specified. Repair of gradual irregularities exceeding 6 mm.

Finish U3—shall apply to unformed surfaces for which a high degree of surface smoothness is required, where shown on the drawing or specified by the PMC. This shall include screeding, floating and applying a steel trowel finish to the surface of the concrete to the required slopes and grade.

Repair of abrupt irregularities: Repair of gradual irregularities exceeding 6 mm., finishing joints and edges of concrete with edging tools.

ANCHORS / REBARS IN CONCRTE

This specification deals with the use of various type of anchors e.g. Mechanical Expansion Fasteners, Resin grouted chemical fasteners, Resin grouted rebars etc. for providing fixtures to concrete surfaces.

Expansion Type Anchors:

Material grade shall be as per design requirements. The outer diameter of this type of fasteners shall be larger than the diameter of the mounting hole of the fixture. This will enable to be installed at a marked location and expanded before the fixture can be placed in position and secured through its mounting holes by means of bolts.

Chemical Type:

Resin grouted fasteners shall be as per manufacturer's specification and resin grout shall be as per design requirements depending on various loadings conditions and position etc. Selection of Fastener e.g. HIT-V A307 Steel, HAS-E -5.8 Steel or HAS-E-B High Str. Steel of approved make and the resin grout e.g. Hilti HIT-HY 200, HIT-RE 500 V3, HIT-RE 100, HIT-HY 100,or HIT-ICE etc. or approved make shall be as per design requirements and as indicated on the drawings.

Resin grouted Rebars

Rebars material shall be same as per reinforcement grade (Fe-415/Fe-500 etc, and resin grout shall be as per design requirements depending on various loadings, position etc. As indicated in 'B' above. The EPC Contractor shall submit to the PMC for his approval, manufacturer's specification and samples of specified expansion fasteners.

The EPC Contractor shall install the fasteners/rebars at the specified places according to the approved manufacturer's specification with the help of necessary tools specified by the manufacturers with the prior approval of PMC. The EPC Contractor shall obtain the approval of PMC for every installation the type of fasteners in regard to its correct location and strength. The EPC Contractor shall get the resin re-bar & resin anchor job through a competent vendor, approved by the resin vendor. If desired, by the PMC in charge EPC Contractor shall get a sample test done on existing concrete, to establish its load carrying capacity of the anchors.

e. FORM WORK

GENERAL

The form work shall consist of shores, bracings, sides of beams and columns, bottom of slabs etc. including ties, anchors, hangers, inserts etc. complete which shall be properly designed and planned for the work. The false work shall be so constructed that up and down vertical adjustment can be made smoothly. Wedges may be used at the top or bottom of timber shores, but not at both ends, to facilitate vertical adjustment or dismantling of form work.

DESIGN OF FORM WORK

The design and engineering of form work as well as its construction shall be the responsibility of EPC Contractor. If so instructed, the drawings and calculations for the design of the form work shall be submitted well in advance to the PMC for approval before proceeding with work, at no extra cost to the OWNER/PMC. PMCs approval shall not however, relieve EPC Contractor of the full responsibility for the design and construction of the form work. The design shall take into account all the loads vertical as well as lateral that the forms will be carrying including live and vibration leadings.

TOLERANCES

Tolerances are a specified permissible variation from lines, grade or dimensions given in drawings. No tolerances specified for horizontal or vertical building lines or footings shall be constructed to permit encroachment beyond the legal boundaries. Unless otherwise specified, the following tolerances will be permitted

Tolerances for R.C. Buildings

i. Variation from the plumb

a) In the line and surfaces of columns, piers, walls and in buttresses: 5 mm. per 2.5 m., but not more than 25 mm.

b) For exposed corner columns and other conspicuous lines.

In any bay or 5 m. maximum:(±) 5 mm.

In 10 m. or more:(±) 10 mm.

ii. Variation from the level or from the grades indicated on the drawings.

a) In slab soffits, ceilings, beam soffits and in arises.

In 2.5 m.:(±) 5 mm.

In any bay or 5 m. maximum:(±) 8 mm.

In 10 m. or more:(±) 15 mm.

b) For exposed lintels, sills, parapets, horizontal grooves and other conspicuous lines.

In any bay or 5 m. maximum:(±) 15 mm.

In 10 m or more:(±) 10 mm.

iii. Variation of the linear building lines from established position in plan and related position of columns, walls and partitions.

In any bay or 5 m. maximum:(±) 10 mm.

In 10 m. or more:(±) 20 mm.

iv. Variations in the sizes and locations of sleeves, openings in walls and floors except in the case of and

for anchor bolts:(±) 5 mm.

v. **Variation in cross-sectional dimensions of columns and beams and in the thickness of slabs and walls:**

(+)10 mm./(-) 5 mm.

vi. Footings

a) Variation in dimensions in plan:(+) 50 mm./(-) 5 mm.

b) Misplacement or eccentricity: 2% of footing within the direction of misplacement but not more than 50 mm.

c) Reduction in thickness: (-) 5% of specified thickness subject to maximum of 50 mm.

vii. Variation in steps.

a) In a flight of stairs.

Rise: (±) 3.0 mm.

Tread: (±) 5.0 mm.

b) In consecutive steps.

Rise: (±) 1.5 mm..

Tread: (±) 3.0 mm.

Tolerances in other Concrete Structure

A. All structures:

i) Variation of the constructed linear outline from established position in plan.

In 5 m.: (±) 10 mm.

In 10 m. or more: (±) 15 mm.

ii) Variation of dimensions to individual structure features from established positions in plan.

In 20 m. or more: (±) 25 mm.

In buried constructions: (±)150 mm.

iii) Variation from plumb, from specified batter or from curved surfaces of all structures.

In 2.5 m.: (±) 10 mm.

In 5.0 m.: (±) 15 mm.

In 10.0 m. or more: (±) 25 mm.

In buried constructions: (±) Twice the above limits.

iv) Variation from level or grade indicated on drawings in slabs, beams, soffits, horizontal grooves and visible arises.

In 2.5 m.: (±) 5 mm.

In 7.5 m. or more: (±) 10 mm.

In buried constructions: (±) Twice the above limits.

v) Variation in cross-sectional dimensions of columns, beams, buttresses, piers and similar members.

(+)12 mm./(-) 6 mm.

vi) Variation in the thickness of slabs, walls, arch sections and similar members.(+)12 mm./(-) 6 mm.

B. Footings for columns, piers, walls, buttresses and similar members:

i) Variation of dimensions in plan:(+)50 mm./(-)12 mm.

ii) Misplacement or eccentricity: 2% of footing within the direction of misplacement but not more than 50 mm.

iii) Reduction in thickness: 5% of specified thickness subject to a maximum of 50 mm.

Tolerances in other types of structures shall generally conform to those given in Clause 2.4 of Recommended Practice for concrete form work (ACI 347)

TYPE OF FORMWORK

Form work may be of timber, plywood, metal, plastic or concrete. For special finishes, the formwork may be lined with plywood, steel sheets, oil tempered hard board etc. Sliding forms and slip forms may be used with the approval of Engineer-in- Charge.

FORMWORK REQUIREMENTS

Forms shall conform to the shapes, lines, grades and dimensions including camber of the concrete as called for in the drawings. Ample studs, waler braces, straps, shores etc. shall be used to hold the forms in proper position without any distortion whatsoever until the concrete has set sufficiently to permit removal of forms. Forms shall be strong enough to permit the use of immersion vibrators. In special cases, from vibrators may also be used. The shuttering shall be close boarded. Timber shall be well seasoned, free form sap, shakes, loose knots, worm holes, warps or other surface defects in contact with concrete. Faces coming in contact with concrete shall be free from adhering grout, plaster, paint, projecting nails, splits or other defects. Joints shall be sufficiently tight to prevent loss of water and fine material from concrete.

Plywood shall be used for exposed concrete surfaces, where called for. Sawn and wrought timber may be used for unexposed surfaces. Inside faces of forms for concrete surfaces which are to be rubbed finished shall be planed to remove irregularities or unevenness in the face. Form work with lining will be permitted.

All new and used from lumber shall be maintained in a good condition with respect to shape, strength, rigidity, water tightness, smoothness and cleanliness of surfaces. Form lumber unsatisfactory in any respect shall not be used and if rejected by PMC shall be removed from the site.

Shores supporting successive stories shall be placed directly over those below or be so designed and placed that the load will be transmitted directly to them. Trussed supports shall be provided for shores that cannot be secured on adequate foundation.

Formwork, during any stage of construction showing signs of distortion or distorted to such a degree that the intended concrete work will not conform to the exact contours indicated on the drawings, shall

be repositioned and strengthened. Poured concrete affected by the faulty formwork, shall be entirely removed and the formwork corrected prior to placing new concrete.

Excessive construction camber to compensate for shrinkage settlement etc. that may impair the structural strength of members will not be permitted.

Forms for substructure concrete may be omitted when, in the opinion of PMC, the open excavation is firm enough to act as the form. Such excavations shall be slightly larger than required by the drawings to compensate for irregularities in excavation and to ensure the design requirement.

Forms shall be so designed and constructed that they can be stripped in the order required and their removal do not damage the concrete. Face formwork shall provide true vertical and horizontal joints, conforming to the architectural features of the structure as to location of joints and be as directed by PMC.

Where exposed smooth or rubbed concrete finishes are required, the forms shall be constructed with special care so that the desired concrete surfaces could be obtained which require a minimum finish.

Bracings, Struts and Props: Shuttering shall be braced, strutted, propped and so supported that it shall not deform under weight and pressure of the concrete and also due to the movement of men and
other materials. Bamboos shall not be used as props or cross bracings.
The shuttering for beams and slabs shall be so erected that the shuttering on the sides of beams and under the soffit of slab can be removed without disturbing the beam bottoms.

Re-propping of beams shall not be done except when props have to be reinstated to take care of construction loads anticipated to be in excess of the design load. Vertical props shall be supported on wedges or other measures shall be taken whereby the props can be gently lowered vertically while striking the shuttering.

If the shuttering for a column is erected for the full height of the column, one side shall be left open and built upon sections as placing of concrete proceeds, or windows may be left for pouring concrete from the sides to limit the drop of concrete to 1.0 m. or as directed by PMC.

INSPECTION OF FORM WORK

Following points shall be borne in mind while checking during erection of form work and form work got approved by the PMC before placing of reinforcement bars :

a) Any member which is to remain in position after the general dismantling is done, should be clearly marked.

b) Material used should be checked to ensure that, wrong items / rejects are not used.

c) If there are any excavations nearby which may influence the safety of form works, corrective and strengthening action must be taken.

d)

i. The bearing soil must be sound and well prepared and the sole plates shall bear well on the ground.

ii. Sole plates shall be properly seated on their bearing pads or sleepers.

iii. The bearing plates of steel props shall not be distorted.

iv. The steel parts on the bearing members shall have adequate bearing areas.

e) Safety measures to prevent impact of traffic, scour due to water etc. should be taken. Adequate precautionary measures shall be taken to prevent accidental impacts etc.

f) Bracing, struts and ties shall be installed along with the progress of form work to ensure strength and stability of form work at intermediate stage. Steel sections (especially deep sections) shall be adequately restrained against tilting, over turning and form work should be restrained against horizontal loads. All the securing devices and bracing shall be tightened.

g) The stacked materials shall be placed as catered for, in the design.

h) When adjustable steel props are used, they should :

i. be undamaged and not visibly bent.

ii. have the steel pins provided by the manufacturers for use.

iii. be restrained laterally near each end.

iv. have means for centralising beams placed in the fork-heads.

i) Screw adjustment of adjustable props shall not be over extended.

j) Double wedges shall be provided for adjustment of the form to the required position wherever any settlement / elastic shortening of props occurs. Wedges should be used only at the bottom end of single prop. Wedges should not be too steep and one of the pair should be tightened / clamped down after adjustment to prevent their shifting.

k) No member shall be eccentric upon vertical member.

l) The number of nuts and bolts shall be adequate.

m) All provisions of the design and / or drawings shall be complied with.

n) Cantilever supports shall be adequate.

o) Props shall be directly under one another in multistage constructions as far as possible.

p) Guy ropes or stays shall be tensioned properly.

q) There shall be adequate provision for the movement and operation of vibrators and other construction plant and equipment.

r) Required camber shall be provided over long spans.

s) Supports shall be adequate, and in plumb within the specified tolerances.

FORM OIL

Use of form oil shall not be permitted on the surface which require painting. If the EPC Contractor desire to use form oil on the inside of formwork of the other concrete structures, a non staining mineral oil or other approved oil CEMOL-35 of Ms. Hindustan Petroleum Co. Ltd. may be used, provided it is applied before placing reinforcing steel and embedded parts. All excess oil on the form surfaces and any oil on metal or other parts to be embedded in the concrete shall be carefully removed. Before treatment with oil, forms shall be thoroughly cleared of dried splatter of concrete from placement of previous lift.

CHAMFERS AND FILLERS

All corners and angles exposed in the finished structure shall be formed with mouldings to form chamfers or fillers on the finished concrete. The standard dimensions of chamfers and fillets, unless otherwise specified, shall be 20 x 20 mm. Care shall be exercised to ensure accurate mouldings. The diagonal face of the moulding shall be planed or surfaced to the same texture as the forms to which it is attached.

VERTICAL CONSTRUCTION JOINT CHAMFERS

Vertical construction joints on faces which will be exposed at the completion of the work shall be chamfered as above except where not permitted by PMC for structural or hydraulic reasons.

WALL TIES

Wire ties passing through the walls, shall not be allowed. Also through bolts shall not be permitted. For fixing of formwork, alternate arrangements such as coil nuts shall be adopted at the EPC Contractors cost.

REUSE OF FORMS

Before reuse, all forms shall be thoroughly scraped, cleaned, nails removed, holes that may leak suitably plugged and joints examined and when necessary repaired and the inside retreated to prevent adhesion, to the satisfaction of PMC. Warped lumber shall be resized. EPC Contractor shall equip himself with enough shuttering to complete the job in the stipulated time.

REMOVAL OF FORMS

EPC Contractors shall record on the drawings or a special register, the date upon which the concrete is placed in each part of the work and the date on which the shuttering is removed there from. In no circumstances shall forms be struck until the concrete reaches a strength of the at least twice the stress due to self weight

and any construction erection loading to which the concrete may be subjected at the time of striking formwork.

In normal circumstances (generally where temperatures are above 15 0C) forms may be struck after expiry of the following periods

Table - Stripping time of Formwork

S.No	Part of Structure	Earliest concrete age at stripping	
		Ordinary Portland cement concrete	Portland Pozzolana / C.C Fly Ash Based
a)	Walls, columns and vertical sides of	16 to 24 hours	24 to 36 hours

	beams		
b)	Slabs(Props left Under)	3 days	4 days
c)	Beam, soffits (Props left under)	7 days	10 days
d)	Removal of props i)Spanning upto 4.5m ii) Spanning over 4.5m	7 days 14 days	10 days 20 days
e)	Removal of props in beams & arches i) Spanning upto 6m ii) Spanning over 6m	14 days 21days	20 days 30 days

Note: For other types of cement, the stripping time recommended for ordinary Portland cement may be suitably modified. Generally If Portland pozzolana or low heat cement or OPC with direct addition of fly ash has been used for concrete, the stripping time will be 10/7 of the period stated for OPC with 43 grade cement above.

The number of props left under the sizes and the position shall be such as to able to safely carry the full dead load of the slab, beam or arch., as the case may be together with any live load likely to occur during curing or further constructions.

Where the shape of the element is such that the form work has the re-entrant angles the form work shall be removed as soon as possible. After the concrete has set, to avoid shrinkage cracking occurring due to the restraint imposed.

Striking shall be done slowly with utmost care to avoid damage to arise and projection and without shock or vibration, by gently easing the wedges. If after removing the formwork, it is found that timber has been embedded in the concrete, it shall be removed and made good as specified earlier.

Reinforced temporary openings shall be provided, as directed by PMC, to facilitate removal of formwork which otherwise may be in-accessible.

Tie rods, clamps, form bolts etc. which must be entirely removed from walls or similar structures shall be loosened not sooner than 24 hours nor later than 40 hours after concrete has been deposited. Ties, except those required to hold forms in place, may be removed at the same time. Ties withdrawn from walls and grade beams shall be pulled towards the inside face. Cutting ties back from the faces of walls and grade beams will not be permitted. Work damaged due to premature or careless removal of forms shall be re-constructed at EPC Contractors cost.

CIRCULAR SHUTTERING: For all structural elements circular/ with arc the shuttering shall be done accordingly. The shutter width shall be so selected so as the shape of structural element could be correctly achieved.

HACKING OF CONCRETE: Other than for requirement of form finish concrete surface, all the concrete surface immediately after removal of formwork shall be closely hacked with hand chiselled hacking tool in order to make it rough for proper bond/grip to receive plaster (wherever specified) finish over concrete surface. As a guideline for close hacking, hacking strokes shall be approx. 15 to 20 mm centre to centre and 3 to 4 mm deep.

f. STEEL FOR CONCRETE REINFORCEMENT

SCOPE OF MATERIAL

The EPC Contractor shall make his own arrangement for procurement of Reinforcement steel bars and wires for use in Reinforced Cement Concrete works. Unless otherwise specified in drawings / Schedule of quantities, the steel bars shall be of "High strength deformed steel bars and wires" conforming to the IS1786-2008 (Amendment 1)), in the following strength grades:

a) Fe 415, Fe 415D, Fe415S
b) Fe 500, Fe 500D, Fe500S
c) Fe 550, Fe 550D, and
d) Fe 600.

i) Where **—Fe** stands for specified minimum 0.2% proof / yield stress in N/mm2 and

ii) **—D and S** following the strength grade indicates the categories with same specified minimum 0.2% proof/ yield stress but with enhanced and additional requirements.

Special coating **to steel Reinforcement:** For important structures under aggressive environment, if structural requirement specifies for additional protection against corrosion to reinforcement bars, the above mentioned high strength deformed bars with a protective coating as specified in item, may be used. These coatings may be in the form of „Galvanized reinforcement, or „Fusion Bonded Epoxy coated" reinforcement". Both these process is a specialized job carried out in a factory and care must be taken for site treatment of bar after cutting, bending and placement. Touch up on the galvanized/epoxy coating have to be carefully done to the satisfaction of PMC.

i) Galvanized Reinforcement: Galvanizing of reinforcement consists of dipping the steel bars in molten zinc. The zinc surface reacts with calcium hydroxide in the concrete to form a passive layer and prevents corrosion. Hot Dip Galvanize (HDG) coating is highly impact and abrasion resistant with alloy layers harder than the underlying steel. Galvanized steel bars for concrete reinforcement shall conform to ASTM A767. The thickness of coating shall be minimum 150microns.

ii) Fusion Bonded Epoxy coating: The Fusion bonded Epoxy coating provides excellent protection against corrosion under very aggressive condition. Fusion bonded Epoxy coated reinforcement bars shall conform to ASTM specification:A775/775M.

Initially the bar is shot blasted to remove all mill scale and to give kind of surface finish required for adequate bonding between epoxy and steel. The bar is then heated to a carefully controlled temperature(200-250deg.C) chamber and then passed through a spray booth where electro statically charged epoxy powder particles are deposited evenly(fused) on the surface of bar. The thickness of coating shall be minimum 150 microns(0.005 to 0.009inch) for effective protection against chloride and adequate flexibility during bending.

Stainless Steel Rebars: Under special functional requirement such as use in the vicinity of magnetic sensitive equipment (MRI-Magnetic Resonance Imaging), to prevent magnetic foot print non-magnetic stainless steel rebars of grade S-304/S-316(preferred) conforming to BS 6744 shall be used. The yield stress(0.02% proof stress) of plain SS Rebars shall be 500N/mm2 (U.T.S=550N/mm2) and elongation shall be more than 14% as per BS 6744. High strength and high bond strength stainless steel rebars (also known as Grip-Rib Bar) having yield strength of 650N/mm2 and UTS of 750N/mm2 conforming to BS-6744 may be used if specified. Binding wires used for tying shall be 1.2mm dia. S.S wire of similar grade stainless steel. Chemical Composition of Stainless Steel grade S-304/S-316 shall be as per IS 6911-1992.

TERMINOLOGY

Elongation: The increase in length of a tensile test piece under stress, expressed as a percentage of the original gauge of a standard piece.

Longitudinal Rib: A uniform continuous protrusion, parallel to the axis of the bar/wire (before cold-working, if any).

Nominal Diameter or Size: The diameter of a plain round bar/wire having the same mass per metre length as the deformed bar/wire.

Nominal Mass: The mass of the bar/wire of nominal diameter and of density 0.00785 kg/mm2per meter.

Nominal Perimeter: 3.14 times the nominal diameter of a deformed bar/Wire.

Percent Proof Stress: The stress at which a non-proportional elongation equal to 0.2% of the original gauge length takes place.

Uniform elongation: The elongation corresponding to the maximum load reached in a tensile test (also termed as percentage total elongation at maximum force).

Tensile Strength: The maximum load reached in a tensile test divided by the effective cross-sectional area of the gauge length portion of the test piece (also termed as ultimate tensile stress).

Transverse Rib: Any rib on the surface of a bar/wire other than a longitudinal rib

Yield Stress: Stress (that is, load per unit cross sectional area) at which elongation first occurs in the test piece without increasing the load during the tensile test. In the case of steels with no such definite yield point, proof stress shall be applicable.

The high strength deformed steel bars and wires for concrete reinforcement shall be hot rolled steel without subsequent treatment or hot rolled steel with controlled cooling and tempering and cold worked steel, and reinforcing bars and wires which may be subsequently coated.

Steel bars shall be supplied from M/s. Steel Authority of India Ltd. (SAIL) or M/s. TATA Steel (TISCO) or M/s. RashtriyaIspat Nigam Ltd (RINL) from their own plants rolled from virgin material, and shall be

procured directly or from their authorised dealers and not from re-rollers or conversion agents. The EPC Contractor shall supply copy of Documentary evidence of purchase of steel from the specified manufacturers.

TESTS

The EPC Contractor shall submit the test certificate of manufacturer. Regular tests on steel supplied by the EPC Contractor shall be performed by the EPC Contractor at the approved lab, in presence of the OWNER/PMC Engineers as per relevant Indian Standards. PMC may require EPC Contractor to perform necessary tests of samples at random as per relevant B.I.S. All cost of such tests and incidentals to such tests shall be borne by the EPC Contractor. The quality, grade, colour coding embossing marks etc. all shall be to the entire satisfaction of the PMC. Steel not conforming to above test criteria shall be rejected.

The Chemical, Physical & Mechanical properties of the steel reinforcement bars shall be as per IS 1786. Unless otherwise specified, Selection and Preparation of Test Sample shall be as per the requirements of IS 2062.

All test pieces shall be selected either from the cuttings of bars / wires; or from any bar/wire after it has been cut to the required or specified size and the test piece taken from any part of it. In neither case, the test piece shall be detached from the bar/wire except in the presence of the EIC or his authorized representative.

The test pieces shall be full sections of the bars/wires and shall be subjected to physical tests without any further modifications. No reduction in size by machining or otherwise shall be permissible, except in case of bars of size 28 mm and above. No test piece shall be annealed or otherwise subjected to heat treatment. Any straightening which a test piece may require shall be done cold.

For the purpose of carrying out tests for tensile strength, proof stress, percentage elongation and percentage elongation at maximum force for bars 28 mm in diameter and above, deformations of the bars only may be machined. For such bars, the physical properties shall be calculated using the actual area obtained after machining. The following IS codes shall be referred for test methods:

Table - Indian Standards for Steel Tests

SN	Title	IS No	ISO No.
i	Mechanical testing of metals -Tensile testing	1608	6892
ii	Methods for bend test	1599 7438 & 1786	15630-1
iii	Method for re-bend test for metallic wires & bars	1786	15630-1

THE PROPERTIES AS PER IS 1786 – 2008 ARE REPRODUCED BELOW

Table - Chemical Composition of the bars

Constituents	Maximum Permissible Percent							Permissible max. Variation
	Fe 415	Fe 415D/ Fe 415S	Fe 500	Fe 500D/ Fe 500S	Fe 550	Fe 550D	Fe 600	
Carbon	0.300	0.250	0.300	0.250	0.300	0.250	0.300	0.020%
Sulphur	0.060	0.045	0.055	0.040	0.055	0.040	0.040	0.005%
Phosphorus	0.060	0.045	0.055	0.040	0.050	0.040	0.040	0.005%
Sulphur & Phosphorus	0.110	0.085	0.105	0.075	0.100	0.075	0.075	0.010%

Notes:

i. For welding of deformed bars, the recommendations of IS 9417 shall be followed.

ii. In case of deviations from the specified maximum, two additional test samples shall be taken from the same batch and subjected to the test or tests in which the original sample failed. Should both additional test samples pass the test, the batch from which they were taken shall be deemed to comply

with this standard. Should either of them fail, the batch shall be deemed not to comply with this standard.

Table : Mechanical Properties of High Strength Deformed Bars and Wires

Sl. No.	Property	Grades Of High strength Deformed Bars and Wires						
		Fe 415	Fe 415D / Fe 415S	Fe 500	Fe 500D / Fe 500S	Fe 550	Fe 550D	Fe 600
1	2	3	4	5	6	7	8	9
I	0.2 percent proof stress / yield stress, Min, N/mm2	415.0	415.0	500.0	500.0	550.0	550.0	600.0
ii.	0.2 percent proof stress / yield stress, Max, N/mm2	--	---/ 540.0 for Fe 415 S	---	---/ 625.0 for Fe 500S	---	---	---
Ii	Elongation, percent, Min. on gauge length 5.65 √A, where A is the Cross – sectional area of the test piece	14.5	18.0	12.0	16.0	10.0	14.5	10.0
Iii	Tensile strength, Min	10% more than the actual 0.2% proof stress/yield stress but not less than 485.0 N/mm2	12% more than the actual 0.2% proof stress/yield stress but not less than 500.0 N/mm2	8% more than the actual 0.2% proof stress/yield stress but not less than 545.0 N/mm2	10% more than the actual 0.2% proof stress/yield stress but not less than 565.0 N/mm2	6% more than the actual 0.2% proof stress/yield stress but not less than 585.0 N/mm2	8% more than the actual 0.2% proof stress/yield stress but not less than 600.0 N/mm2	6% more than the actual 0.2% proof stress/yield stress but not less than 660.0 N/mm2

iv	Total elongation at maximum force, percent, Min of gauge length 5.65 √A, where A is the cross sectional area of the test piece	-	5	-	5	-	5	-

Note: To satisfy Clause 26 of IS 456 -2000, no mixing of different types of grades of bars shall be allowed in the same structural members as main reinforcement, without prior written approval of the PMC.

STACKING & STORAGE

Steel for reinforcement shall be stored in such a way as to prevent distorting and corrosion. The steel for reinforcement shall not be kept in direct contact with ground. Fresh / Fabricated reinforcement shall be carefully stored to prevent damage, distortion, corrosion and deteriorations. Care shall be taken to protect steel from exposure to saline atmosphere during storage, fabrication and use. It may be achieved by treating the surface of reinforcement with cement wash or by suitable methods. Bars of different classifications, sizes and lengths shall be stored separately to facilitate issue in such sizes and lengths to cause minimum wastage in cutting from standard length.

QUALITY

Steel not conforming to specifications shall be rejected. All reinforcement shall be clean, free from grease, oil, paint, dirt, loose mill, scale, loose rust, dust, bituminous material or any other substances that will destroy or reduce the bond. All rods shall be thoroughly cleaned before being fabricated. Pitted and defective rods shall not be used. All bars shall be rigidly held in position before concreting. No welding of rods to obtain continuity shall be allowed unless approved by the PMC. If welding is approved, the work shall be carried as per I.S. 2751, according to best modern practices and as directed by the PMC. In all cases of important connections, tests shall be made to prove that the joints

are of the full strength of bars welded. Substitution of reinforcement will not be permitted except upon written approval from PMC.

NOMINAL SIZES

The nominal sizes of bars/wires shall be 4mm, 5mm, 6mm, 8mm, 10mm, 12mm, 16mm, 20mm, 25 mm, 28mm, 32mm, 36mm and 40 mm.

NOMINAL MASS

For the purpose of checking the nominal mass, the density of steel shall be taken as 0.00785 kg/mm3 of the cross-sectional area per metre. Unless otherwise specified, the tolerances on nominal mass shall be as per following Table.

Table 6.4: Tolerances on Nominal Mass

SN	Nominal Size in mm	Tolerance on the nominal mass in Percent		
		Batch	Individual sample	Individual sample for coils only
1	2	3	4	5
i)	Up to and including 10	± 7	- 8	± 8
ii)	Over 10 up to and including 16	± 5	- 6	± 6
iii)	Over 16	± 3	- 4	± 4

ASSEMBLY OF REINFORCEMENT

Reinforcement shall be bent and fixed in accordance with procedure specified inIS2502.The high strength deformed steel bars should not be re-bent or straightened without the approval of Engineer-in- Charge.

Bar bending schedules shall be prepared for all reinforcement work and shall be get approved by Engineer-in- Charge.

All reinforcement shall be placed and maintained in position shown in the drawings by providing proper cover blocks, spacers, supporting bars etc.
Crossing bars should not be tack-welded for assembly of reinforcement unless permitted by PMC.

PLACING AND TOLERANCE ON PLACING

Reinforcement shall be accurately fixed by any approved means maintained in the correct position as shown in the drawings by the use of blocks, spacers and chairs as per I.S. 2502 to prevent displacement during placing and compaction of concrete. Bars intended to be in contact at crossing point shall be securely bound together at all such points with number 16 gauge annealed soft iron wire. The vertical distances required between successive layers of bars in beams or similar members shall be maintained by the provision of spacer bars at such intervals that the main bars do not perceptibly sag between adjacent spacer bars.

Tolerance on Placing of Reinforcement: Unless otherwise specified, reinforcement shall be placed within the following tolerances :
Tolerance in spacing
a) For effective depth, 200 mm or less + /- 10 mm
b) For effective depth, more than 200 mm+ /- 15 mm

COVER TO REINFORCEMENT

Nominal cover is the design depth of concrete cover to all steel reinforcements, including links. It is the dimension used in design and indicated in the drawings. It shall be not less than the diameter of the bar. Unless otherwise specified, cover to reinforcement shall be provided generally as per guidelines of IS 456.
Nominal cover to meet durability requirement: Minimum values for the nominal cover of normal weight aggregate concrete which should be provided to all reinforcement, including links depending on the condition of exposure described in 4.4 above and as per (nominal cover to meet durability requirements).

However for a longitudinal reinforcing bar in a column nominal cover shall in any case not be less than 40mm or less than the diameter of such bar. In the case of columns of minimum dimension of 200 mm or under, whose reinforcing bar do not exceed 12 mm , a nominal cover of 25 mm may be used.

For footings minimum cover shall be 50 mm.

Nominal cover to meet specified period of fire resistance

Minimum values of nominal cover of normal-weight aggregate concrete to be provided to all reinforcement including links to meet specified period of the resistance as per the tables given under clause 4.4.1 of this specifications.

The cover shall in no case be reduced by more than one third of specified cover or 5 mm whichever is less.

Unless indicated otherwise on the drawings, clear concrete cover for reinforcement (exclusive of plaster or other decorative finish shall be as follows:

a. At each end of reinforcing bar not less than 25mm., nor less than twice the diameter of such bar.

b. For a longitudinal reinforcing bar not less than 25 mm., nor more than 40 mm., nor less than the diameter of such bar. In the case of column of maximum dimensions of 200 mm. or under, whose reinforcing bars do not exceed 12 mm., a cover of 25 mm. may be used.

c. For longitudinal reinforcing bar in a beam, not less than 25mm., nor less than diameter of such bar.

d. For tensile, compressive, shear, or other reinforcement in a slab, not less than 25 mm, nor less than the diameter of such bar, and

e. For any other reinforcement not less than 15 mm., nor less than the diameter of such bar.

f. Increased cover thickness may be provided when surfaces of concrete members are exposed to the action of harmful chemicals (as in the case of concrete in contact with earth faces contaminated with such chemicals), acid, vapour, saline atmosphere, sulphurous smoke (as in the case of steam- operated railways) etc. and such increase of cover may be between 15 mm. and 50 mm. beyond the figures given in (a to e) above as may be specified by the PMC.

g. For reinforced concrete members, totally immersed in sea water, the cover shall be 40 mm. more than specified (a to e) above.

h. For reinforced concrete members, periodically immersed in sea water or subject to sea spray, the cover of concrete shall be 50 mm. more than that specified (a to e) above.

i. For concrete of grade M 25 and above, the additional thickness of cover specified in (f), (g) and (h) above may be reduced to half. In all such cases the cover should not exceed 75 mm.

j. Protection to reinforcement in case of concrete exposed to harmful surroundings may also be given by providing dense impermeable concrete with approved protective coating, as specified on the drawings. In such case the extra cover, mentioned in (h) and (i) above, may be reduced by the PMC, to those shown on the drawing.

k. The correct cover shall be maintained by cement mortar briquettes or other approved means. Reinforcement for footings, grade beams and slabs on subgrade shall be supported on precast concrete blocks as approved by the PMC. The use of pebbles or stones shall not be permitted.

l. The minimum clear distance between reinforcing bars shall be in accordance with I.S. 456 or as shown in drawing.

LAPS

Reinforcement splicing, if required shall be done as per provisions in IS 456. Laps and splices for reinforcement shall as far as possible be away from the sections of maximum stress and be staggered and shall be as shown in the drawings. Splices in adjacent bars shall be staggered and the locations of all splices, except those specified in the drawings, shall be approved by the PMC. The bars shall not be lapped unless the length required exceeds the maximum available lengths of bars at site. Lap splices shall not be used for bars larger than 32mm; for larger diameters, bars may be welded.

WELDED JOINTS OR MECHANICAL CONNECTION

Welded joints or mechanical connections in reinforcement may be used provided tests shall be made to ensure that the joints are of full strength of bars connected.

Welded Splicing: Welding of reinforcements shall be done in accordance with there commendations of IS 2751 and IS 9417.Welded splices shall not be used within a distance equal to twice the member depth from the column or beam face for special moment frames or from sections where yielding of the reinforcement is likely to occur as a result of inelastic lateral displacements.

Welding of stirrups, ties, inserts or other similar elements to longitudinal reinforcement shall not be permitted.

Mechanical splicing: Mechanical splicing (coupler) system shall conform to International codesACI-318-08 or BS8110, BS5400 and shall be of approved manufacturer. The Couplers shall be properly fixed at site by specialized workman as per manufacturer's specification. Manufacturer shall submit all test certificates of the couplers supplied tested as per relevant codes.

Mechanical splicing shall be of standard threaded mechanical splice coupler. Splicing procedure and qualification, testing shall be done as per relevant Indian/ International standards. The cutting, Cold forging and threading of reinforcement bars shall be done as per manufacturer specifications. Reduction of nominal cross section area of the parent bar is not allowed. This type of splicing shall be carried out using necessary type threaded couplers including locking arrangements wherever required as per approved manufacturers specification. The coupler shall be such that there should not be any requirement of rotating the rebars for connection. The nominal cross section of the bar at the root of the thread shall not be less than nominal cross section of the parent bar. Actual performance of the coupler joint shall be at least equivalent to the characteristic strength of the parent bar i.e tensile failure should occur in the bar and shall be satisfactorily away from the joint. Elongation after failure must be equivalent to the parent bar. Permanent elongation after loading upto 0.6 fy shall not exceed 0.1mm.

Technical Requirement

In case of mechanical connection, a full mechanical splice shall develop in tension or compression, as required, at 1.25 times specified yield strength (fy). As per ACI-318 Mechanical couplers shall be of the two types:

Type-1: Type-1 Couplers are where minimum load in tension/compression should be 1.25fy of the bar.Type-1 splices shall not be used within a distance equal to twice the member depth from the column or beam face for special moment frames or from sections where yielding of the reinforcement is likely to occur as a result of inelastic lateral displacements

Type-2: Type-2 Couplers must achieve under tension/compression 1.25 times fy of the bar and also achieve the ultimate tensile strength of the parent bar.Type-2 mechanical splices shall be permitted to

be used at any location including in yielding regions where the tensile stresses in reinforcement may approach the tensile strength of the reinforcement.

Based on functional requirement mechanical coupler shall be of following type;

i) Bar break coupler :For joining two same diameter bars(standard coupler).
ii) Bar reducer coupler: For joining two different diameter bars(Reducer or Transition coupler).
Some other types on functional basis are anchor couplers, crimpers, positioner couplers etc.

Procedure for splicing

The reinforcement rod to be spliced shall be cut by a hacksaw to get a perfectly plane end surface. The portion of the bar being engaged with the splice sleeve shall be cleaned with the power brush. The inner surface of the sleeve shall be cleaned with acetone or any other approved solvent.

The splicing shall be carried out either by manufacturers technician or personnel of EPC Contractor having been trained by the manufacturer. Some of the guidelines in coupler manufacturing are as follows :
i) The system involves cold forging of the ends of the bar for enlarging the size, cutting square threads such that root thread diameter is slightly bigger than parent bar. The bars are joined at site with coupler. Unless otherwise specified the coupler shall be such that there should not be any requirement of rotating the rebars for connection
ii) The system should be capable in using effectively for following elements of the following structural locations.

a) Main column/ Wall reinforcement.
b) Column beam junction – Column can be taken up independent of beam by providing L – bars with threaded couplers.
c) Shear wall slab junction – shear wall can be taken up independent of beam providing L – bars with threaded couplers.
d) Creating temporary access for final cleaning of pour.

e) Avoiding laps in slab pours/ Raft pours/ Circumferential reinforcement and thereby reducing congestion of reinforcement.

iii. The system should be capable for joining bars from 16mm dia to 40mm dia.

iv. Reducers/transition Couplers are to be used for joining bars at different diameter.

v. Instructions for preparing joints at site are as under.

a) Only hand tightening of couplers to be done. No wrench to be used. Use of wrench may damage the threads and weaken the joint.

b) The bars are to be aligned properly by holding the bar by two labourers as close to each other as possible(not more than 3mm) and the coupler is to be hand tightened .Assembly until the shorter length thread is fully engaged in the position coupler and the longer length rebar thread is engaged approximately half of its overall threaded length.

c) To be assured that the position setting bar threads have been fully engaged into the Bar splicer Position coupler, use a pipe wrench or chain wrench to snug and tighten the setting bar threads (Note: Splice bar threads will remain loose within coupler.)

d) Inspection to be visual i.e to ensure that all the threads are covered by a coupler. However, there should be adequate margin in designing the joints such that even if 4 (four) threads are covered by coupler, guaranteed bar break is ensured.

e) Ensure that same type joints are not provided fully at any joint.

f) For mixed construction where reinforcement bars are to be welded to steel structure, weldable couplers are to be used.

Quality checks for threads and joints

i. Double forging is not allowed. In case of improper forging, such as bucking, re-cut the bar and repeat forging.

ii. Check the threaded end of the bar with „go" and „no go" gauge. The „go" gauge must go over the full thread length. The „no-go" gauge must not go over more than half of the full thread length.

However this check is required to be done after every setting of machine for particular type of couplers

Testing Requirements: Tensile test of joint is to be conducted on 3 specimens per diameter at start of work/ project. Then minimum one specimen is to be tested per 100 joints for the particular diameter upto 1000 joints. Afterwards, one specimen is to be tested per 1000 joints for the particular diameter. However the PMC at his discretion can increase the number of samples for testing to ensure the

quality at no extra cost to the OWNER/PMC. Acceptance criteria are that bar braking shall be away from the coupler joint and also away from the forged area. Mechanical joints with Couplers of each type and each diameters shall be tested for strength, slip test, bend and re-bend test etc. as per relevant standards.

BENDING

All bars shall be accurately bent according to the sizes and shapes shown on the detailed working drawing bar bending schedules. They shall be bent gradually by machine or other approved means. Reinforcing bars shall not be straightened and re-bent in a manner that will injure the materials. Bars containing cracks or splits shall be rejected. They shall be bent cold, except bars of over 25 mm. in diameter which may be bent hot if specifically approved by the PMC. Bars that depend for their strength on cold working shall not be bent hot. Bars bent hot shall not be heated beyond cherry red colour (not exceeding 6450C) and after bending shall be allowed to cool slowly without quenching. Bars incorrectly bent shall be used only after straightening and re-bending be such as shall not, in the opinion of the PMC, injure the material. No reinforcement bar shall be bent when in position in the work without approval, whether or not it is partially embedded in hardened concrete. Bars having kinks or bends other than those required by design shall not be used.

Bending at Construction Joints: Where reinforcement bars are bent aside at construction joints and afterwards bent back into their original position, care should be taken to ensure that at no time the radius of the bend is less than 4 bar diameters for plain mild steel or 6 bar diameters for deformed bars. Care shall also be taken when bending back bars to ensure that the concrete around the bar is not damaged.

HE BARS SHALL BE KEPT IN CORRECT POSITION BY THE FOLLOWING METHODS

a) In case of beam and slab construction precast cover blocks in cement mortar 1:2 (1 cement : 2 coarse sand) about 4 x 4 cm section and of thickness equal to the specified cover shall be placed between the bars and shuttering, so as to secure and maintain the requisite cover of concrete over reinforcement.

b) In case of cantilevered and doubly reinforced beams or slabs, the vertical distance between the horizontal bars shall be maintained by introducing chairs, spacers or support bars of steel at 1.0 metre or at shorter spacing to avoid sagging.

c) In case of columns and walls, the vertical bars shall be kept in position by means of timber templates with slots accurately cut in them; or with block of cement mortar 1:2 (1 cement : 2 coarse sand) of required size suitably tied to the reinforcement to ensure that they are in correct position during concreting.

d) In case of other R.C.C. structure such as arches, domes, shells, storage tanks etc. a combination of cover blocks, spacers and templates shall be used as directed by PMC.

INSPECTION

Erected and secured reinforcement shall be inspected and approved by PMC prior to placement of concrete.

PILE FOUNDATION

SPECIFICATIONS FOR BORED/DRIVEN CAST-IN-SITU R.C.C. PILES

Scope: This specification covers the design and construction of load bearing concrete bored cast-in-situ piles of appropriate diameter which can transmit the load of the structure to the soil by both resistance developed at the tip by end bearing and along the surface of the pile shaft by friction.

This specification also covers the design and construction of R.C.C. pile caps and plinth beams appropriate for the piles offered and structure to be supported.

LST Contractor shall be responsible for the construction of the entire pile foundation system and shall guarantee the stability of the pile foundation system offered against the risks of settlement and other type of damage to the structure.

Equipment & Accessories: The equipment and accessories shall depend on the type of bored cast-in-situ piles chosen for the job and shall be selected giving due consideration to the sub-soil strata, ground water conditions, type of founding materials and the required penetration, manner of operation etc. For Bored cast-in-situ piles percussion boring by suitable drilling rigs using direct mud circulation (DMC) methods is to be adopted and the size of the cutting tool shall not be less than the diameter of

the pile by more than 75mm Bentonite shall be used as drilling mud and its basic properties shall conform to Annex D of I.S. 2911 (Part-I/Section-2) 2010. Tremie shall be used for placing concrete into the bore holes.

General Considerations: The construction of pile foundation shall be in such a way that the load from the structure it supports, can be transmitted to the soil without causing any soil failure & without causing such settlement, differential or total under permanent transient loading as may result in structural damage and/or functional distress to the buildings.

When working near the existing structures, any damage to such structures shall be made good at no extra cost to the OWNER/PMC. The EPC Contractor shall, therefore, take adequate care to avoid any damage to the existing structures.

In case of deep excavations adjacent to piles proper shoring or other suitable arrangement shall be done at no extra cost to the OWNER/PMC to guard against the lateral movement of soil or releasing the confining soil stress.

As per loading details, piles will be required to withstand vertical load axial or otherwise and horizontal load associated with moments. Axial load from a pile should be transmitted to the soil through skin friction along the shaft and end-bearing at its tip. A horizontal load shall be transmitted to the sub-soil by horizontal sub-grade reaction generated in the upper part of the pile shaft.

Coarse Aggregate, Fine Aggregate & Water: Coarse aggregate fine aggregate and water shall conforming to IS 456 (L.R) and I.S 383 (L.R)

Concrete: Concrete to be used for the pile shaft shall be M 25. Materials and method of manufacture for cement concrete shall, in general, be in accordance with the method of concrete under the condition of pile installation. Consistency of concrete mix for cast-in-situ piles shall be suitable to the method of installation of piles. Concrete mix shall be so designed as to have a homogeneous mix having a flowable character consistent with the method of concreting of pile. The slump of concrete shall range between 150 to 180 mm depending on the method/manner of concreting. Minimum cement

content shall be 400 kg per cubic metre. In case of piles where concreting is done under water or drilling mud using methods other than tremie 10% extra cement over that required for the design grade of concrete at specified slump shall be used subject to a minimum quantities of cement specified above. Cost of excess cement over the minimum quantity specified shall be borne by the EPC Contractor. For mix design, manufacture, placing etc. specification for cement concrete given hereinafter shall be referred.

Workmanship

Control of piling installation: Bored cast-in-situ piles shall be installed by employing suitable drilling rigs using a combination of bailer and a suitable chisel with DMC method. Bore hole shall be stabilised by bentonite as drilling mud and concreting shall be done by use of tremie.

Control of Alignment : Piles shall be installed as accurately as possible as per the designs and drawings. Greater care should be exercised in respect of installation of single pile or piles in two pile groups. The piles shall be installed vertically with tolerance as stipulated by I.S 2911 (Part-I/Section-2) (Latest revision). Piles shall not deviate more than 75mm or D/6 whichever is less in case of piles having diameter less than 600mm, 75 mm or D/10 whichever is more in the case of piles having diameters more than 600mm from their designed positions at the working level. In the case of a single pile in a column, positional tolerance should not be more than 50mm or D/6 which ever is less (100 mm in case of piles having diameter more than 600 mm). In case of piles deviating beyond these limits, EPC Contractor shall carry out necessary remedial measures duly approved by the OWNER/PMC at no extra cost. Piles that are deviated to such an extent that the resulting eccentricity cannot be taken care of by a redesign of the pile cap or plinth beams, the piles should be re-placed or supplemented by one or more additional piles at no extra cost to the OWNER/PMC.

A minimum length of two/three metres of temporary guide casing upto piling platform shall be inserted in each bored pile after completion of bailer driving. Additional length or temporary casing may be used depending on the condition of the strata, ground water level etc.

Founding Level : The bore hole shall be advanced by chisel and direct mud circulation method after installation of guide casing till the required founding level is reached. The founding level shall be as per the drawings and as directed by the PMC.

In case, drilling mud within the hole stabilizes a bored pile, the bottom of the hole shall be cleaned very carefully before concreting work is taken up. The cleaning of the hole shall be ensured by careful operation of boring tool and/or flushing of the drilling mud through the bottom of the hole by tremie for half an hour minimum.

In case, a hole is bored by use of drilling mud, the specific gravity of the mud suspension near about the bottom of the hole shall, wherever practicable, be determined by suitable, slurry sampler and recorded. Consistency of the drilling mud suspension shall be controlled throughout the boring as well as concreting operations in order to keep the hole stabilized as well as to avoid concrete getting mixed up with the thicker suspension of the mud. The concreting operations should not be taken up when the specific gravity of bentonite slurry is more than 1.2.

In addition to the normal precautions to be taken in tremie concreting, the following requirements shall be applicable to the use of tremie concrete in piles:

i. The concrete should be coherent, rich in cement (not less than 400 kg/cum) & of slump not less than 150mm.
ii. When concreting is carried out under water, a temporary casing should be installed to the full depth of the bore hole or 2 to 3 M into the top stratum, so that fragments of ground cannot drop from the sides of the hole into the concrete as it is placed.
iii. The hopper & tremie should be a closed system embedded in the placed concrete, through which water cannot pass.
iv. The first charge of concrete should be placed with a sliding plug pushed down the tube ahead of it or with a steel plate of adequate charge to prevent mixing of concrete and water. However, the plug should not be left in the concrete as a lump.
v. The tremie pipe should always penetrate well into the concrete with an adequate margin of safety against accidental withdrawal of the pipe is surged to discharge the concrete.
vi. The pile should be concreted wholly by tremie and the method of deposition should not be changed part way up the pile, to prevent the laitance from being entrapped within the piles.
vii. All tremie tubes should be scrupulously cleaned after use.

Normally concreting of the piles should be uninterrupted. In the exceptional case of interruption of concreting, but which can be resumed within 1 or 2 hours, the tremie shall not be taken out of the concrete. Instead it shall be raised and lowered slowly, from time to time to prevent the concrete around the tremie from setting. Concreting should be resumed by introducing a little richer concrete with a slump of about 200 mm for easy displacement of the partly set concrete.

If the concreting cannot be resumed before final set of concrete already placed, the pile so cast may be rejected.

In case of withdrawal of tremie out of the concrete either accidentally or to remove a choke in the tremie, the tremie may be reintroduced in the following manner to prevent impregnation of laitance or scum lying on the top of the concrete already deposited in the bore.

The tremie shall be gently lowered on to the old concrete with very little penetration initially. A vermiculite plug should be introduced in the tremie. Fresh concrete of slump between 150mm and 175 mm should be filled in the tremie, which will push the plug forward and will emerge out of the tremie displacing the laitance/scum. The tremie will be pushed further in steps making fresh concrete sweep away laitance/scum in its way. When tremie is buried by about 60 to 100 cm, concreting may be resumed.

During installation bored cast-in-situ piles, the convenience of installation may be taken into account while determining the sequence of piling in a group.

The top of concrete in a pile shall generally be brought above the cut-off level, upto ground level to permit removal of all laitance and weak concrete before capping and to ensure good concrete at the cut-off level for proper embedment into the pile cap. When concrete is placed by tremie method, concrete shall be cast to the piling platform level at ground level to permit overflow of concrete for visual inspection.

Defective Pile: In case defective piles are formed, they shall be removed or left in place whichever is convenient without affecting, performance of the adjacent piles or the cap as a whole without any extra cost to the OWNER/PMC. Additional piles shall be provided to replace them as directed.

Any deviation from the designed location alignment or load capacity of any pile shall be noted and adequate measures taken well before the concreting of the pile cap and plinth beam if the deviations are beyond the permissible limit.

During chipping of the pile top manual chipping maybe permitted after three days of pile casting, pneumatic tools for chipping shall not be used before seven days after pile casting. Before chipping/ breaking the pile top, a 40mm deep groove shall be made manually all around the pile at the required cut off level.

After concreting the actual quantity of concrete shall be compared with the average obtained from observations actually made in the case of a few piles initially cast. If the actual quantity is found to be considerably less, special investigations shall be conducted and appropriate measures taken.

Routine Load Test: The EPC Contractor shall be required to carry out routine load tests as directed by the PMC on an individual pile or on a group of piles or on both. The routine load tests shall be carried out generally as per IS 2911 (Part-IV) – 1985. Report on routine load tests shall be submitted in an approved format for OWNER/PMC approval at no extra cost. In case the tests on the routine piles reveal safe capacity less than specified, the EPC Contractor shall, at his own cost, provide suitable modifications to the pile or other remedial measures after obtaining approval of the PMC. In case of an unsatisfactory results being revealed on any routine tests it shall be the EPC Contractor's responsibility to carry out additional routine tests, at his own cost till the criteria laid down are fulfilled.

Rate for routine load test shall be inclusive of providing kentledges, making other arrangements for the test loading platforms, providing tools and plants, equipment like hydraulic jack, dial gauges etc. other measuring instruments and all labour involved in carrying out tests. Cost of pile shall, however, be paid for by the OWNER/PMC at the rates accepted in the tender since the piles are working piles.

Guarantee for the Pile Foundation System (Applicable in case the pile foundation system design is under the scope of EPC Contractor)

The EPC Contractor shall guarantee the stability of the pile foundation system and shall indemnify the OWNER/PMC against risks of settlement and against all damages caused to building erected on the pile foundation system for a period of 5 (five) years from the certified date of completion. Certificate of completion shall be issued by the PMC after the satisfactory completion of the job.

NOTE: Pro-forma for guarantee to be executed by EPC Contractors for stability of pile foundation system against risk of settlement & other type of distress to the building(s) is attached in this book as **Appendix– —DII.**

g. READY MIX CONCRETE

INTRODUCTION

Concrete mixed in a stationery mixer in a central batching and mixing plant or in a truck mixer and supplied in fresh condition to purchaser (EPC Contractor) either at site or into purchaser"s vehicle. RMC produced and supplied shall conform to provision of IS : 4926-2003 (or latest revision). Plant of RMC Producer shall be first got approved from PMC. The EPC Contractor shall arrange with RMC producer for facilitating thorough inspection of RMC plant for the purpose of approval.

MATERIALS

Materials used should satisfy the requirements for the safety, structural performance durability and appearance of the finished structure, taking full account of the environment to which it will be subjected. The selection and use of materials shall be in accordance with IS 456. Materials used shall conform to the relevant Indian Standards applicable. Where materials are used which are not covered by the provisions of the relevant Indian Standard, there should be satisfactory data on their suitability and assurance of quality control. Records and details of performance of such materials should be maintained. Account should be taken of possible interactions and compatibility between IS 4926 and materials used. Also, prior permission of the PMC shall be obtained before use of such materials.

Cement: cement for concrete shall be as specified and shall be in accordance with the requirementsof IS 456-2000.

Mineral Admixtures : Use of mineral admixtures shall be permitted in accordance with theprovisions of IS 456.

Fly ash when used for partial replacement of cement, shall conform to Grade I of IS-3812 and meet the physical and chemical requirements as per IS 3812(Part-I)-2013. Percentage of fly ash as part replacement of OPC shall be guided as per IS 1489-2005(Part 1). Blending operation of fly ash with cement should be properly designed for uniform and intimate blending.

Specific care shall be taken in terms of curing, protecting, repairing, finishing, de-shuttering etc. as detailed in the relevant Chapter of “Fly Ash Concrete ”and “Formwork”.

Aggregates shall comply with the requirements of IS: 383. Natural aggregates shall be givenpreference. Use of other types of aggregate meeting suitability requirements, may be permitted by PMC, if satisfied with the data on the properties of concrete made with them.

Water used for concrete shall conform to the requirement of IS 456-2000.

The Chemical admixtures may be permitted in accordance with the provisions of IS: 456-2000 andIS: 9103.

It shall be the responsibility of the producer to establish compatibility and suitability of any admixture with the other ingredients of the mix and determine the dosage required to give the desired effect.

The amount of admixture added to mix shall be recorded in the production record. In special circumstances, if necessary, additional dose of admixture may be added at project site to regain the workability of concrete with the mutual agreement between the producer and the purchaser (EPC Contractor).

SUPPLY

The ready-mixed concrete shall be manufactured and supplied on either of the following basis

i. Specified strength based on 28-day compressive strength of 15 -cm cubes tested in accordance with IS : 456-2000.

ii. Specified mix proportion.

NOTE : Under special circumstances and as specified the strength of concrete in (a) above may be based on 28-day or 7-day flexural strength of concrete instead of compressive strength of 15-cm cube tested in accordance with IS : 456-2000.

When the concrete is manufactured and supplied on the basis of specified strength, the responsibility for the design of mix shall be that of the manufacturer and the concrete shall conform to the requirements specified.

When the concrete is manufactured and supplied on the basis of specified mix proportions, the responsibility for the design of the mix shall be that of the purchaser and the concrete shall conform to the requirements specified.

GENERAL REQUIREMENTS

The manufacturer shall keep batch records of the quantities by mass of all the solid materials, total amount of water used in mixing and results of all ingredients of concrete. If required/insisted, the manufacturer shall furnish certificates, at agreed intervals, giving these information's. It shall be obligatory on part of RMC manufacturer to allow inspection of their plant including control room during supply of RMC to site.

Transport of Concrete: Ready-mixed concrete shall be transported from the mixer to the point of placing as rapidly as practicable by methods that will maintain the required workability and will prevent segregation, loss of any constituents or ingress of foreign matter or water. The concrete shall be placed as soon as possible after delivery, as close as is practicable to its final position to avoid re-handling or moving the concrete horizontally by vibration. If required by the purchaser the producer can utilize admixtures to slow down the rate of workability loss, however this does not remove the need for the EPC Contractor to place the concrete as rapidly as possible. The EPC Contractor should plan his arrangements so as to enable a full load of concrete to be discharged within 30 minutes of arrival on site.

Time in Transport: The general requirement is that concrete shall be discharged from the truck-mixer within 2 hour of the time of loading. However, a longer period may be permitted if retarding admixtures

are used or in cool humid weather or when chilled concrete is produced. The time of loading shall start from adding the mixing water to the dry mix of cement and aggregate or of adding the cement to the wet aggregate whichever is applicable.

Workability: The test for acceptance is to be performed upon the producer's delivery vehicle discharge on site or upon discharge into the purchaser's vehicle. If discharge from the producers vehicle is delayed on site due to lack of preparedness on behalf of the purchaser then the responsibility passes to the purchaser after a delay of more than 30 min.

The workability shall be within the following limits on the specified value as appropriate:

Slump: ± 25 mm or 1/3 of the specified value, whichever is less.

Compacting factor : ± 0.03, where the specified value is 0.90 or greater,

± 0.04, where the specified value is less than 0.90 but more than 0.80,

± 0.05, where the specified value is 0.80 or less.

Flow table test may be specified for concrete, for very high workability (IS 9103)

The compressive strength and flexural strength tests shall be carried out in accordance with the requirement of IS: 516 and the acceptance criteria for concrete whether supplied on the basis of specified strength or on the basis of mix proportion, shall conform to the requirements and other related requirements of IS: 456 -2000.

QUALITY CONTROL

Quality control of ready-mixed concrete may be divided into three components, forward control, immediate control and retrospective control.

Forward control: Forward control and consequent corrective action are essential aspects of qualitycontrol. Forward control includes the following.

i. Control of purchased material Quality
ii. Control of Materials storage
iii. Mix design and mix design modification
iv. Transfer and Weighing Equipment : The producer shall be able to demonstrate that a documented calibration procedure is in place. The use of electro-mechanical weighing and metering systems, that is, load cells, flow meters, magmeters, etc, is preferable over purely mechanical system, that is, knife edge and lever systems.

v. Plant mixers where present and truck mixers used shall be in an operational condition.

Immediate Control: Immediate control is concerned with instant action to control the quality of the concrete being produced or that of deliveries closely following. It includes the production control and product control.

(i)Production Control : The production of concrete at each plant shall be systematically controlled. This is to ensure that all the concrete supplied shall be in accordance with these requirements and with the specifications that has formed the basis of the agreement between the producer and purchaser.

Each load of mixed concrete shall be inspected before dispatch and prior to discharge. The workability of the concrete shall be controlled on a continuous basis during production and any corrective action necessary taken.

For each load, written, printed or graphical records shall be made of the mass of the materials batched, the estimated slump, the total amount of water added to the load, the delivery ticket number for that load, and the time the concrete was loaded into the truck.

Regular routine inspections shall be carried out on the condition of plant and equipment including delivery vehicles.

(ii) Product Control : Concrete mixes shall be randomly sampled and tested for workability and where appropriate, plastic density, temperature and air content. Where significant variations from target values are detected, corrective action shall be taken.

It is important to maintain the water cement ratio constant at its correct value. The amount of added water shall be adjusted to compensate for any observed variations in the moisture contents in the aggregates. Suitable adjustments should also be made in masses of the aggregates due to this variation (As per IS 456). Any change in water content due to change in aggregate grading shall be taken care of by forward control by suitable modifications to mix design.

Retrospective Control: Retrospective control is concerned with those factors that influence the control of production. Retrospective control may cover any property of materials or concrete, such as aggregate grading, slump, or air content, but is particularly associated with 28-day cube strength because by its very nature it is not property which can be measured ahead of, or at the time of, manufacture.

Mix Performance: The producer shall be responsible for ensuring that suitable control procedures are in place ensure the following.

(i) Design Mixes : A quality control system shall be operated to control the strength of design mixes to the levels required as per IS 456 and shall be based on random tests of mixes which form the major proportion of production. The system shall include continuous analysis of results from cube tests to compare actual with target values together with procedures for modifying mix proportions to correct for observed differences. Compressive strength testing shall be carried out using a machine that meets the requirements of IS 14858.

(ii) Prescribed Mixes : Periodic and systematic checks shall be made to ensure that the cementitious material contents of prescribed mixes comply with their mix descriptions.

Stock Control of Materials: The producer shall operate a materials stock control procedure to enable verification of total quantities used and to confirm that only approved materials have been received.
Complaints: The producer shall have a procedure in place to enable the diagnosis and correction of faults identified from complaints.

h. FLY ASH CONCRETE AND OTHER SPECIAL TYPE CONCRETE(S)

FLY ASH CONCRETE

Introduction: In environment subjected to aggressive chloride or sulphate attach in particular, use of fly ash admixed or PPC based concrete is recommended. Fly Ash admixed cement concrete (FACC) and fly ash Blended cements conforming to IS-1489 (Part I) may be used in RCC structures as per guidelines given below

i) The fly ash should have consistent quality satisfying the requirements of Grade-I of IS 3812 and Class-F of ASTM C-618.
ii) The source of fly ash should be so selected that test results of fly ash samples collected from these sources during last one year at frequency of maximum one month interval should satisfy the requirements of above codes.

iii) The characterization of fly ash proposed to be used, should be done as per above two codes for each batch of fly ash.
iv) The fly ash should be stored in bins at the plant.
v) All concrete should be manufactured at RMC plant to ensure uniform blending of fly ash.
vi) The mix proportion should conform to provisions of IS codes.
vii) Samples to be taken, at each time of casting, for 7 days and 28 days. Sampling shall be as per relevant IS provisions.
viii) All ingredients of concrete should satisfy the requirements of relevant IS codes and specifications.

Curing, protecting, repairing and finishing

Curing: All concrete shall be cured by keeping it continuously damp for the period of time required for complete hydration and hardening to take place. Preference shall be given to the use of continuous sprays or ponded water, continuously saturated covering of sacking, canvas, hessian or other absorbent materials, or approved effective curing compounds applied with spraying equipment capable of producing a smooth, even textured coat. Extra precautions shall be exercised in curing concrete during cold and hot weather as outlined hereinafter the quality of curing water shall be the same as that used for mixing concrete.

Certain types of finish or preparation for overlaying concrete must be done at certain stages of curing process and special treatment may be required for specific concrete surface finish.

Curing of concrete made of high alumina cement and super sulphated cement shall be carried out as directed by PMC. Curing of concrete with low water binder ratios having partial replacement of cement by pozzolanic materials.

The structural elements with concrete having water binder ratio less than or equal to 0.4 or partial replacement of cement by pozzolanic materials (5% or above replacement by silica fume or high reactivity metakaoline, or 15% or above by fly ash) shall be cured in two stages, initial curing and final curing.

i) The initial curing should be started not later than 3 hours or initial setting time, whichever is lower, after placement of concrete. The concrete surface exposed to environment should be covered by plastic sheet or other type of impermeable covers. The initial curing should be continued upto a minimum period of 12 hours or 2 hours plus final setting time of concrete, whichever is higher.

ii) Final curing should be done with water. It should commence immediately after initial curing and continue upto a minimum period of 14 days.

Curing with Water

Continuous Spraying

Alternate Curing Methods

Curing Compounds

Curing Equipment

Form Work

i. DAMP PROOF COURSE

SCOPE OF WORK

The work covered under this specifications consists supplying and laying plain cement concrete of specified nominal mix and thickness at plinth level of building as damp proof course with approved waterproofing admixture in accordance with this specification and applicable drawings.

WORKMANSHIP

Surface to receive damp proof course shall be cleaned and carefully swept to remove all dust, laitance etc. and shall be approved by the PMC. Damp proof course shown shall be cement concrete of proportion and thickness as indicated in the schedule. Approved water proofing compound@ 2% by weight of cement or as directed by the manufacturer shall be mixed in cement mortar for this concrete. The damp proof course shall be laid to the full width of the peripherals walls including shaft walls and the edges shall be straight, even and truly vertical. Wooden forms shall be used to obtain good edges. A coat of hot bitumen of grade 80/100 @ 1.7Kg/Sqm is applied over the laid concrete course after 24 hours.

j. PRECAST REINFORCED CEMENT CONCRETE JALLI

GENERAL

The item refers to manufacturing, supplying and fixing in position reinforced cement concrete jalli with thickness, proportion, design and pattern as specified in the description of item of schedule of quantities.

MATERIALS

The concrete shall consist of a mix as specified in the item and specifications for cement concrete shall be similar to that specified under cement concrete. Reinforcement shall be as specified in the item description or as shown in the drawing or as directed by the PMC.

MANUFACTURE

Precast jalli of approved design and thickness as specified in description of item shall be cast at site and fixed in position at all floors and levels by the EPC Contractor. Steel moulds to manufacture approved size, design and thickness of reinforced jalli as per drawings shall be manufactured and got approved before starting large scale casting. Reinforced jalli shall be cast at site. Reinforcement of specified dia. shall be bent to the exact shape of jalli to run vertically and horizontally along centre of all walls and suitably kept in position. Use of teak wood/shuttering plywood moulds can be allowed at the discretion of the PMC. Precast Jalli shall be made by filling the moulds of the approved design, pattern with complete batch of 1:2:2 or as specified (using 10 mm. down graded stone aggregate) cement concrete so that the entire work may set in one time. If one batch is sufficient to complete a unit, each succeeding batch should follow the preceding batch as quickly as possible. The specifications for reinforced cement concrete shall conform to I.S. 456-2000 and as specified earlier. The jalli shall be cast according to drawing, design and as approved by the PMC. Necessary holes and keys for pins/ dowels to house in the jalli shall be left for fixing. Jallies shall be cured in a curing tank for a minimum period of 14 days. All the imperfection in surfaces and edges shall be repaired with neat cement paste to finish with sharp and smooth edges. The jalli so manufactured shall be perfectly at right angles when checked with straight edge. Jallies which are not approved due to defective pattern, shape, finish etc. shall be removed forthwith from the site of work.

FIXING

The jalli shall be fixed in position in cement mortar 1:2 in preformed opening true to line, level and plumb with 6 mm. M.S. pins, all as per drawing or as directed by the PMC. Jalli with damaged edges of surfaces shall not be **fixed. The joints shall be cured for a minimum period of 7 days. After grouting the sides with cement mortar etc.,** the jalli shall be rechecked for its levels and alignment. Finally the jambs, sills and soffits shall be plastered embedding the jalli uniformly on sides. The jalli shall be cleaned off all the mortar splashes, dirt etc.

5 Masonry works

STRUCTURAL STEEL

Scope

This specification covers the requirements for material, storage, preparation of fabrication drawings, fabrication, assembly, tests/examinations, transportation, erection and painting of all types of bolted and/or welded structural steel works for general construction work. Fabrication of structures shall also include fabricating:

a) Built up sections/plate girders made out of rolled section and/or plates.
b) Compound sections made out of rolled sections.

Materials

Structural Steel

1. All structural steel shall be of tested quality. The material of all Indian rolled section and plates shall conform to IS 2062 Grade – A designation Fe410WA specified therein.
2. Wherever the material is procured by the EPC Contractor, the EPC Contractor shall submit the test certificates conforming to the relevant Indian standards of all steel materials used for fabrication. All structural steel shall be free from blisters, rust, scales, seams, lamination, cracks, fissures and other surface defects.

3. Carbon steel pipes/tubes for structural purposes shall be as per IS: 1161 grade YST-25. In case mild steel is available according to latest IS: 2062, steel to grade A, B & C will be used as applicable.

Bolts and Nuts

Bolts and nuts shall conform to IS: 1363 or IS: 1364: 2002 as applicable and as shown in the drawing. Unless otherwise specified nuts and bolt heads shall be hexagonal. Property class of nuts and bolts shall be compatible. The contractor shall submit test certificates when called for. Wherever shown in the drawing high strength friction grip bolts (HSFG bolts) and nuts conform to IS: 3757 and IS: 6623: 2004 respectively shall be used. Nuts shall be of at least the strength grade appropriate to the grade of bolts or other threaded elements with which they are used.

Washers

Plain washers shall be made of mild steel conforming to IS: 5369 unless noted otherwise. Minimum one washer shall be supplied for each bolt and in case of special types of bolts more than one washer as required for the purpose shall be supplied. Helical spring washer conforming to IS: 6755: 1980 shall be provided for bolts carrying dynamic or fluctuating loads and those in direct tension. Tapered washers conforming to IS: 5372 & IS: 5374 shall be used for channels & beams respectively. Washers for high strength friction bolts shall conform to IS: 6649: 1985

Welding Consumables

1. Covered electrodes (for metal arc welding of structural steel) shall conform to IS: 814 & IS: 2062.
2. Filler rods & wires for gas welding shall conform to IS: 1278: 1972
3. Base wire electrodes (in submerged arc welding of structural steel) shall conform to IS: 7280: 1974 The combination of wire and flux shall comply with the requirements of IS: 3613: 1974
4. Filler rods & base electrodes (for gas shield arc welding of structural steel) shall conform to IS: 6419: 1996
5. Welding consumables & procedures shall be such that the mechanical properties of deposited weld metal are not less than the respective minimum values for the parent metal being welded.

Steel Gratings, Stair Treads & Chequered Plates

1. All gratings used for floor, stairs & open platform shall be Hot-dipped Galavanized grating, designed for load specified in design basis.

Gratings shall be of rectangular grid type. Hot Dip galvanizing shall confirm to IS 2629 & IS 2633. Gratings shall be fabricated using electro-forging process. Cross bars (secondary bars) shall be fixed into bearing bars by electro-forging method, every fifth cross bar shall be manually welded from one side. All grating cutout above 250mm, any of the cutout dimension shall be done while fabricating the grating, cutouts smaller than 250mm, shall be done at site.

Permissible tolerance in grating dimensions:

Span width : + or - 5mm

Diagonal: +or - 5mm

Distance between main bars: + or - 3mm

Distance between cross bars : + or - 5mm.

Load bearing bars to confirm to IS 2062 & cross bar shall conform to IS : 7887

2. All cheq plates shall conform to IS 3502. Cheq plate when laid on the floor, shall be free of any sag or bends. Cheq plate of size 6/7 shall not span more than 750mm. Any steel member, main, secondary or tertiary shall be provided under all the joints cheq plate. No transparency acceptable between the upper & lower section of cheq plate.

Receipt & Storing of Materials

1. Each section shall be marked for identification and each lot shall be accompanied by manufacturer's test/quality certificate, chemical analysis and mechanical characteristics.

2. All sections shall be checked, sorted out and arranged by grade and quality in the store. Any instruction given by the PMC in this respect shall be strictly followed.

3. All material shall be free from surface defects such as pitting, cracks, laminations, twists etc. Defective material shall not be used and all such rejected material shall be immediately removed from the store/site. The decision of the PMC in this regard shall be final and binding.

4. Welding wires and electrodes (packed in their original cartons) shall be stored separately by quality and lots inside a dry and enclosed room in compliance with IS: 9595 and as per the instructions

given by the PMC. Electrodes shall be kept perfectly dry to ensure satisfactory operation and weld metal soundness.

5. Each lot of electrodes, bolts, nuts etc. shall be accompanied by manufacturer's quality/test certificates.

6. All bolts (including nuts & washers) shall be checked, sorted out and arranged diameter-wise by grade and quality in the store.

Material Tests

1. The EPC Contractor shall submit manufacturers' quality certificates for all the materials supplied by him. In case, quality certificates are not available or are incomplete or when material quality differs from standard specifications, such materials shall not be used in the construction. However, the EPC Contractor shall get all appropriate tests conducted in approved test houses for such materials as directed by the PMC, at no extra cost, and submit the same to PMC for his approval. The PMC may approve the use of such materials entirely at his discretion.

2. The EPC Contractor shall ensure that all materials brought to site are duly approved by the PMC. Rejected materials shall not be used and shall be removed from site forthwith. Any material of doubtful quality for which specific tests are to be carried out as per the instruction of the PMC shall be separately stacked and properly identified and shall not be used. These shall be removed from site forthwith.

Fabrication Drawings

Fabrication and erection drawings shall be prepared by the EPC Contractor on the basis of "Approved for Construction (AFC)" design drawings, PMC Standards issued to the EPC Contractor. These drawings conforming to IS: 800 shall be prepared by the EPC Contractor or by an agency appointed by the EPC Contractor and approved by the PMC.

Fabrication and erection drawings shall be thoroughly checked, stamped "Approved for Construction" and signed by the EPC Contractor's own responsible PMC irrespective of the fact that such drawings

are prepared by the EPC Contractor or his approved agency, to ensure accuracy and correctness of the drawings. Unchecked and unsigned drawings shall not be used for the purpose of proceeding with the work. The EPC Contractor shall proceed with the fabrication and erection work only after thoroughly satisfying himself in this regard.

All fabrication and erection drawings shall be issued for construction by the EPC Contractor directly to his work- site. Six copies of such drawings shall simultaneously be submitted to the PMC who may check/ review some or all such drawings at his sole discretion and offer his comments for incorporation in these drawings by the EPC Contractor.

However, the EPC Contractor shall not proceed with the fabrication of such structures whose fabrication drawings are required to be reviewed before taking up the fabrication work as noted on "Approved for Construction (AFC)" design drawings issued to the Contractor or as conveyed by the PMC. The fabrication of such structures shall be done only as per the reviewed fabrication drawings.

The review of such drawings by PMC shall be restricted to the checking of the following only:

i) Structural layout, orientation and elevation of structures members,
ii) Sizes of members,
iii) Critical joint details.

Fabrication drawings shall be drawn to scale and shall convey the information clearly and adequately. Following information shall be furnished on such drawings:

- Reference to design drawing number (along with revision number) based on which fabrication drawing has been prepared.
- Structural layout, elevations & sections (with distinct erection marking of all members).
- Framing plans, member sizes, orientation and elevations.
- Layout and detailing of rain water pipes and gutters showing all necessary levels, connections and provisions wherever required.
- Detailing of shop/field joints, connections, splices, for required strength and erection. Location, type, size and dimensions of welds and bolts.

- Shapes and sizes of edge preparation for welding.
- Details of shop and field joints/welds.
- Bill of materials/D.O.D. Lists.
- Quality of structural steel, plates etc., welding electrodes, bolts, nuts and washers to be used.
- Erection assemblies identifying all transportable parts and sub-assemblies with special erection instructions, if required.
- Method of erection and special precautions to be taken during erection as required.
- The design of ladders, platforms, handrails, stairways & the like shall be as per standard drawings.

The EPC Contractor shall additionally ensure accuracy of the following and shall be solely responsible for the same:

i) Provision for erection and erection clearances.
ii) Marking of members
iii) Cut length of members
iv) Matching of joints and holes.
v) Provision kept in the members for other interconnected members.
vi) Bill of materials.

Connections, splices and other details where not shown on the design drawings shall be suitably designed and shown on the fabrication drawings based on good engineering practice developing full member strength. Design calculations for such connections/splices shall be submitted to the PMC along with the fabrication drawings.

Any substitution or change in section shall be allowed only when prior written approval of the PMC has been obtained. Fabrication drawings shall be updated incorporating all such substitutions/changes by the EPC Contractor at no extra cost to the Owner.

In case during execution of the work, the PMC on review of drawings considers any modifications/substitutions necessary to meet the design parameters good engineering practice, these shall be brought to the notice of the EPC Contractor who shall incorporate the same in the drawings and works without any extra cost to the owner. The EPC Contactor will be totally responsible for the correctness of the detailed fabrication drawings and execution of the work.

EPC Contractor shall incorporate all the revisions made in the design drawings during the course of execution of work in his fabrication drawings, and resubmit the drawings at no extra cost to the Owner. All fabrication shall be carried out only as per the latest AFC design drawings and corresponding fabrication drawings.

The Contractor shall supply two prints each of the final/as built drawings along with their transparencies to PMC for reference and record. The rates quoted shall include for the same.

Fabrication

General

1. Fabrication of structures shall be done strictly as per "Approved for Construction" fabrication drawings (prepared by the Contractor based on the latest design drawings) and in accordance with IS: 800, 9595 & other relevant BIS Codes and BIS Hand Book SP: 6(1).
2. Prior to commencement of structural fabrication, undulations in the fabrication yard, if any, shall be removed and area leveled and paved by the EPC Contractor.
3. Any defective material used in the work shall be replaced by the EPC Contractor at his own expense. Necessary care and precautions shall be taken so as not to cause any damage to the structure during any such removal and replacement.
4. Any faulty fabrication pointed out at any stage of work by the PMC, shall be made good or replaced by the Contractor at his own cost.
5. Tolerances for fabrication of steel structures shall be as per IS: 7215.

Fabrication Procedure

Straightening & Bending

1. All materials shall be straight and if necessary, before being worked shall be straightened and/or flattened (unless required to be of curvilinear form) and shall be free from twists.

2. Bending of rolled sections and plates shall be done by cold process to shape/s as shown on drawings.

Clearances

The erection clearance for cleated ends of members shall be not greater than 2mm at each end. The erection clearance at ends of beams without web cleats and end plates shall be not more than 3mm at each end but where for practical reasons, greater clearance is necessary, suitably designed seatings approved by the MMCI/HPCL PMC shall be provided.

Cutting

1. Prior to cutting, all members shall be properly marked showing the requisite cut length/width, connection provisions e.g. location and dimensions of holes, welds, cleats etc. Marking for cutting shall be done judiciously so as to avoid wastages or unnecessary joints as far as practicable. Marking shall be done by placing the members on horizontal supports/pads in order to ensure accuracy. Marking accuracy shall be limited to + 1mm.

2. Cutting may be affected by shearing, cropping or sawing. Gas cutting by mechanically controlled torch shall be permitted for mild steel. Hand flame cutting may be permitted subject to the approval of the MMCI/HPCL PMC.

3. Except where the material is subsequently joined by welding, no loads shall be transmitted into metal through a gas cut surface.

4. Shearing, cropping and gas cutting shall be clean, square, free from any distortion & burrs, and should the MMCI/HPCL Engineer- in-Charge find it necessary, the edges shall be ground afterwards, to make the same straight and uniform at no extra cost to the Owner.

Holing

1. Holes for bolts shall not be formed by gas cutting process.
2. Holes through more than one thickness of material of members such as compound stanchions and girder flanges shall, where possible, be drilled after the members are assembled and tightly clamped/bolted together. Punching may be permitted before assembly, provided the thickness of metal is less than 16 mm and the holes are punched 3 mm less in diameter than the required size and reamed, after assembly, to the full diameter. Punching shall not be adopted for dynamically loaded structures.
3. Holes may be drilled in one operation through two or more separable parts and burrs removed from each part after drilling.
4. Holes in connecting angles and plates, other than splices, also in roof members and light framing, may be punched full size through material not over 12 mm thick, except where required for close tolerance bolts or barrel bolts.
5. All matching holes for black bolts shall register with each other so that a gauge of 2 mm less in diameter than the diameter of hole shall pass freely through the assembled members in the direction at right angle to such members. Finished holes shall be not more than 2 mm in diameter larger than the diameter of the black bolt passing through them, unless otherwise specified by the PMC.
6. Holes for turned and fitted bolts shall be drilled to a diameter equal to the nominal diameter of the shank or barrel subject to H8 tolerance specified in IS: 9I9. Parts to be connected with close tolerance or barrel bolts shall be firmly held together by tacking bolts or clamps and the holes drilled through all the thicknesses in one operation and subsequently reamed to size. Holes not drilled through all the thicknesses in one operation shall be drilled to a smaller size and reamed out after assembly. Where this is not possible, the parts shall be drilled and reamed separately.
7. To facilitate grouting, holes shall be provided in column bases or seating plates exceeding 300mm in width for the escape of air.
8. To avoid accumulation of water in gusseted column bases of laced, battened or box type stanchions, suitable reverse U-type holes shall be provided at the junction of base plate and column section in the vertical gussets for draining out of any water.

Assembly

The component parts shall be assembled and aligned in such a manner that they are neither twisted nor otherwise damaged, and shall be so prepared that the required camber, if any, is provided. Proper clamps, clips, jigs and other fasteners (bolts and welds) shall be placed in a balanced pattern to avoid any distortion in the members and to ensure their correct positioning (i.e. angles, axes, nodes etc.). Any force fitting, pulling/stretching of members to join them shall be avoided. Proper care shall be taken for welding shrinkage & distortion so as to attain the finished dimensions of the structure shown on the drawings.

Welding

General

a) All joints shall be welded unless noted otherwise on the design drawings.
b) Welding shall be in accordance with IS: 816, IS: 819: 1957, IS: 1024: 1999, IS: 1261: 1959, IS: 1323: 1982 and IS: 9595 as appropriate.
c) The EPC Contractor shall make necessary arrangement for providing sufficient number of welding sets of the required capacity, all consumables, cutting and grinding equipment with requisite accessories/ auxiliaries, equipment & materials required for carrying out various tests such as dye penetration, magnetic particle, ultrasonic etc.
d) Adequate protection against rain, dust, snow & strong winds shall be provided to the welding personnel and the structural members during welding operation. In the absence of such a protection no welding shall be carried out. It shall be the responsibility of the Contractor to ensure that all welding is carried out in accordance with the terms of this specification and relevant BIS codes. The Contractor shall provide all the supervision to fulfill this requirement.

Preparation of Member for Welding

a) Edge Preparation

Edge preparation/beveling of fusion faces for welding shall be done strictly as per the dimensions shown in the drawings. In case, the same are not indicated, edges shall be prepared (depending on

the type of weld indicated in the drawing) as per the details given in IS: 9595. Bevelling of fusion faces shall be got checked and approved by the PMC. The tolerances on limits of gap, root face & included angle shall be as stipulated in IS: 9595.

b) Cleaning

Welding edges and the adjacent areas of the members (extending up to 20mm) shall be thoroughly cleaned of all oil, grease, scale and rust and made completely dry. Gaps between the members to be welded shall be kept free from all foreign matter.

c) Preheating

Preheating of members shall be carried out as per IS: 9595 when the base metal temperature is below the requisite temperature for the welding process being used. Preheating shall be done in such a manner that the parts, on which the weld metal is being deposited, are above the specified minimum temperature for a distance of not less than 75mm on each side of the weld line. The temperature shall be measured on the face opposite to that being heated. However, when there is access to only one face, the heat source shall be removed to allow for temperature equalization (1 minute for each 25mm of plate thickness) before measuring the temperature.

d) Grinding

i) Column splices & butt joints of struts and compression members (depending on contact for load transmission) shall be accurately ground and close-butted over the whole section with a tolerance not exceeding 0.2mm locally at any place. In column caps & bases the ends of shafts together with the attached gussets, angles, channels etc., shall be accurately ground so that the parts connected butt over minimum 90% surface of contact. In case of connecting angles or channels, care shall be taken so that these are fixed with such accuracy that they are not reduced in thickness by grinding by more than 2 mm.

ii) Ends of all bearing stiffeners shall be ground to fit tightly at both top and bottom. Similarly bottom of the knife edge supports along with the top surface of column brackets shall be accurately ground to provide effective bearing with a tolerance not exceeding 0.2 mm locally at any place.

iii) Slab bases and caps shall be accurately ground over the bearing surfaces and shall have effective contact with the ends of stanchions. Bearing faces which are to be grouted direct to foundations need not be ground if such faces are true & parallel to the upper faces.

Welding Processes

Welding of various materials under this specification shall be carried out using one or more of the following processes.

- Manual Metal Arc Welding Process (MMAW)
- Submerge Arc Welding Process (SAW)
- Gas Metal Arc Welding Process (GMAW)
- Flux Cored Arc Welding Process (FCA W)

The welding procedure adopted and consumables used shall be specifically approved by the PMC. A combination of different welding processes or a combination of electrodes of different classes/makes may be employed for a particular joint only after qualifying the welding procedures to be adopted and obtaining the written approval of the PMC. Welding procedure shall cover the following:

i) Type and size of electrodes
ii) Position of welding
iii) Current and arc voltage
iv) Length of run per electrode or (for automatic welding) speed of travel of electrode
v) Number of run in multipass welds and arrangement
vi) Preparation of the parts
vii) Welding sequence
viii) Pre or post heating

Approval & Testing of Welders

The EPC Contractor shall satisfy the PMC that the welders are suitable for the work upon which they will be employed. For this purpose the welders shall have satisfied the relevant requirements of IS: 7318. If the welders will be working to approved welding procedures, they shall have satisfied the relevant requirements of 1S: 7310.

Adequate means of identification shall be provided to enable each weld to be traced to the welder by whom it was made. The EPC Contractor shall intimate the PMC sufficiently in advance, the commencement of tests, to enable him to be present to witness the same.

Approval & Testing of Welding Procedures

The EPC Contractor shall carry out procedure tests in accordance with IS: 7307 to demonstrate by means of a specimen weld of adequate length on steel representative of that to be used, that he can make welds with the welding procedure to be used for the work to the complete satisfaction of the PMC. The test weld shall include weld details from the actual construction and it shall be welded in a manner simulating the most unfavourable instances of fit-up, electrode condition etc., which are anticipated to occur on the particular fabrication. Where material analysis is available, the welding procedure shall be carried out on material with the highest carbon equivalent values.

After welding, but before the relevant tests given in IS: 7307 are carried out, the test weld shall be held as long as possible at room temperature, but in any case not less than 72 hours, and shall then be examined for cracking. The examination procedure shall be sufficiently rigorous to be capable of revealing significant defects in both parent metal and weld metal.

After establishing the welding method, the EPC Contractor shall finally submit to the PMC for his approval the welding procedure specification in standard format given in IS: 9595 before starting the fabrication.

Sequence of Welding

(i) As far as practicable, all welds shall be made in a sequence that will balance the applied heat of welding while the welding progresses.

(ii) The direction of the general progression in welding on a member shall be from points where the parts are relatively fixed in position with respect to each other towards points where they have a greater relative freedom of movement.

(iii) All splices in each component part of a cover-plated beam or built up member shall be made before the component part is welded to other component parts of the member.

(iv) Joints expected to have significant shrinkage shall be welded before joints expected to have lesser shrinkage.

(v) Welding shall be carried continuously to completion with correct number of runs.

(vi) The EPC Contractor shall choose the welding sequence after carefully studying each case such as to minimize distortion and shrinkage & submit the same to the PMC for comments and approval.

(vii) The welding seams shall be left to cool slowly. The EPC CONTRACTOR shall not be allowed to cool the welds quickly by any other method.

(viii) For multipass welding, before welding the following layer the formerly welded layer shall be cleaned metal bright by light and wire brushing.

Welding Technique

a) After the fusion faces are carefully aligned and set with proper gaps, the root pass of butt joints shall be executed properly so as to achieve full penetration with complete fusion of the root edges.

b) On completion of each run all slag and spatters shall be removed and the weld and the adjacent base metal shall be cleaned by wire brushing and light chipping. Visible defects such are cracks, cavities and other deposition faults, if any, shall be removed to sound metal before depositing subsequent run of weld.

c) All full penetration butt welds shall be completed by chipping/gouging to sound metal and then depositing a sealing run of weld metal on the back of the joints. Where butt welding is practicable

from one side only, suitable backing steel strip shall be used and joint shall be arranged in such a way as to ensure that complete fusion of all the parts is readily obtained.

d) While welding is in progress care shall be taken to avoid any kind of movement of the components, shocks, vibrations to prevent occurrence of weld cracks.

e) Any deviation desired from the recommended welding technique and electrodes shall be adopted only after obtaining written approval of the PMC.

Inspection & Testing of Welds

The method of inspection shall be according to IS: 822: 1970 and extent of inspection and testing shall be in accordance with the relevant applicable standard or, in the absence of such a standard, as specified by the PMC. Welds shall not be painted or otherwise obscured until they have been inspected, approved and accepted.

The PMC or his representative shall have access to the EPC Contractor's work at all reasonable times and the EPC Contractor shall provide him with all facilities necessary for inspection during all stages of fabrication and erection with, but not limited to, the following objectives.

i) To check the conformity with the relevant standards and suitability of various welding equipments and their performance.
ii) To witness/approve the welding procedure qualification.
iii) To witness/approve the welders performance qualification.
iv) To check whether shop/field welding being executed is in conformity with the relevant specifications and codes of practice.

Inspection and testing of all fabricated structures shall be carried out by the EPC Contractor by any, or, a combination of all the following methods as directed by the PMC and no separate payment shall be made, unless otherwise mentioned, for inspection and testing of welds/fabricated structures:

A. Visual Inspection

i) All finished welds (i.e. 100 percent) shall be visually inspected for identification of the following types of weld defects & faults.
ii) Weld defects occurring at the surface such as blow holes, exposed porosity, unfused welds etc.
iii) Surface cracks in the weld metal or in the parent metal adjacent to it.
iv) Damages to the parent metal such as undercuts, burning, overheating etc.
v) Profile defects such as excessive convexity or concavity, overlapping, unequal leg lengths, excessive reinforcement, incompletely filled grooves, excessive penetration beads, root grooves etc.
vi) Distortion due to welding i.e., local shrinkage, camber, bowing, twisting, rotation, wariness etc.
vii) Linear eccentric, angular and rotational misalignment of parts.
viii) Dimensional errors.

B. Mechanical Tests

The mechanical testing (such as tensile load tests, bend tests, impact tests etc.) shall be done in accordance with the relevant standards and as per the instructions of the PMC.

C. Magnetic Particle/Dye Penetration/Ultrasonic Examination:

The examination shall be done at random as directed by the PMC. Whenever such tests are directed, the tests shall be carried out on joints chosen by him. The tests shall be carried out by employing approved testing procedure in accordance with IS: 822.

D. Radiographic Examination

Radiographic examination shall be carried out only in special cases for random joints as directed by the PMC. The Contractor shall not be paid extra for such examination the cost shall be borne by him. The EPC Contractor shall make necessary arrangement at his own expense for providing the radiographic equipment, films and all other necessary materials required for carrying out the examination. The tests shall be carried in the presence of the PMC by employing approved testing procedure in accordance with IS: 822. The Contractor shall fulfill all the statutory safety requirements while handling X-ray and Gamma-ray equipment and provide the PMC all the necessary facilities at site such as dark room, film viewer etc., to enable him to examine the radiographs.

Repair of Faulty Welds

No repair of defective welds shall be carried out without proper permission of the PMC and his approval for the corrective procedure.

Welds not complying with the acceptance requirements (as specified by BIS Codes & the PMC), as revealed during inspection & testing of welds or erection or in-situ condition shall be corrected either by removing & replacing or as follows:

a)	Excessive convexity	Reduced to size by removal of excess weld metal.
b)	Shrinkage cracks, cracks in parent	Defective portions removed down to sound
	plates and craters	metal and rewelded
c)	Under cutting.	Additional weld metal deposited.
d)	Improperly fitted/ misaligned parts.	Welding cut & edges suitably prepared and parts.
e)	Members distorted by heat of welding	Member straightened by mechanical means or careful application of limited amount of heat.
		Temperature of such area shall be less than 650 degree Centigrade (dull red heat).

In removing defective parts of a weld, gouging, chipping, oxygen cutting or grinding shall not extend into the parent metal to any substantial amount beyond the depth of weld penetration,
unless cracks or other defects exist in the parent metal. The weld or parent metal shall not be undercut in chipping, grinding, gouging or oxygen cutting.

Any fabricated structure or its component which in the opinion of PMC is defective and/or beyond any corrective action shall be removed forthwith from the site as instructed by the PMC without any extra claim. The owner reserves the right to recover any compensation due to any loss arising out of such rejections.

Bolting

1. All bolts shall be provided such that no part of the threaded portion of the bolts is within the thickness of the parts bolted together. Washers of suitable thickness shall be used under the nuts to avoid any threaded portion of the bolt being within the thickness of parts bolted together.
2. The threaded portion of each bolt shall project through the nut at least one thread.
3. Flat washers shall be circular and of suitable thickness. However, where bolt heads/nuts bear upon the bevelled surfaces, they shall be provided with square tapered washers of suitable thickness to afford a seating square with the axis of the bolt.

Splicing

1. Splicing of built up/compound/latticed sections shall be done in such a fashion that each component of the section is joined in a staggered manner.
2. Where no butt weld is used for splicing, the meeting ends of two pieces of joist/channel/built up section shall be ground flush for bearing on each other and suitable flange and web splice plates shall be designed and provided for the full strength of the flange/ web of the section and welds designed accordingly.
3. Where full strength butt weld is used for splicing (after proper edge preparation of the web and flange plates) of members fabricated out of joist/ channel/ angles/ built up section, additional flange and web plates shall be provided, over and above the full strength butt welds, to have 40% strength of the flange and web.
4. Where a cover plate is used over a joist/channel section the splicing of the cover plate and channel/joist sections shall be staggered by minimum 500mm. Extra splice plate shall be used for the cover plate and joist/channel section as per clause 5.2.8.2 or 5.2.8.3.
5. Prior approval shall be obtained by the EPC Contractor for locations of splices where not shown on design drawings. Only a single splice at approved location shall be allowed for members up to a length of 6 to 7m. Maximum two numbers of splices shall be allowed for members exceeding this length.

Machining & Grinding

1. All slab bases and slab caps shall be accurately machined over the bearing surfaces and shall be in effective contact with the ends of column sections (shafts).
2. For slab bases and slab caps, ends of column shafts shall be accurately machined. However, for gusseted bases and caps, the column shafts shall be ground flush for effective contact with parts connected together.
3. Gusseted bases and caps shall be ground flush for effective contact with ends of column sections.
4. End of all bearing stiffeners shall be machined or ground to fit tightly at top and bottom without any air gap.
5. While machining or grinding care shall be taken so that the length or thickness of any part does not get reduced by more than 2.0 mm.
6. For all machining or grinding works for gusseted base and cap plates, the clearance between the parts joined shall not exceed 0.2 mm at any location.

Marking for Identification

Each component shall be distinctly marked (with paint) before delivery in accordance with the marking diagrams and shall bear such other marks as will facilitate erection.

For small members which are delivered in bundles or crates, the required marking shall be done on small metal tags securely tied to the bundle.

Shop Erection

The steel work shall be temporarily shop erected complete or as directed by the PMC, so that the accuracy of fit may be checked before dispatch.

Inspection & Testing of Structures

The PMC (or his authorized representative) shall have free access at all times to those parts of the EPC Contractor's works which are concerned with the fabrication of the steel work and shall be provided with all reasonable facilities for satisfying himself that the fabrication is being undertaken in accordance with the provisions of these specifications & other relevant BIS Codes.

Should any structure or part of a structure be found not to comply with any of the provisions of this specification (or relevant BIS Codes as referred to), it shall be liable to rejection. No structure or part of the structure, once rejected shall be resubmitted for inspection, exception cases where the PMC or his authorized representative considers the defect as rectifiable.

Defects which may appear during/after fabrication/ erection shall be made good only with the consent of the PMC and procedure laid down by him.

All necessary gauges and templates shall be supplied free to the PMC by the EPC Contractor whenever asked for during inspection. The PMC may at his discretion, check the test results obtained at the EPC Contractor's works by independent tests at a test house, and the cost of such tests shall be borne by the EPC Contractor.

Shop Painting

All components and members of steel work shall be given one shop coat of primer as per design basis, , immediately after the surfaces have been properly prepared (i.e. degreased, derusted, descaled & cleaned) in accordance with IS: 1477: 1971. The primer coat shall be applied over completely dry surfaces (using brushes of good quality) in a manner so as to ensure a continuous and uniform film without "holidaying". Special care shall be taken to cover all the crevices, corners, edges etc. However, in areas which are difficult to reach by brushing, daubers/mops shall be used by dipping the same in paint and then pulling/ pushing them through the narrow spaces. The primer coat shall be air dried and shall have a minimum film thickness as noted in design basis.

Surfaces which are inaccessible after shop assembly, shall receive the full specified protective treatment before assembly (this shall not apply to the interior of sealed hollow sections).

Steel surfaces shall not be painted within a suitable distance of any edges to be welded if the paint specified would be harmful to welders or impair the quality of the welds.

Welds and adjacent parent metal shall not be painted prior to deslagging, inspection and approval by the PMC.

Parts to be encased in concrete shall have only one coat of primer and shall not be painted after erection.

Packing

All items shall be suitably packed in case these are to be dispatched from the fabrication shop to the actual site of erection so as to protect them from any damage/distortion or falling during transit. Where necessary, slender projecting parts shall be temporarily braced to avoid warping during transportation.

Small parts such as gussets, cleats etc., shall be securely wired on to their respective main members. Bolts, nuts washers etc. shall be packed in crates.

Transportation

Loading and transportation shall be done in compliance with transportation rules. In case, certain parts can not be transported in the lengths stipulated on the drawings, the position details of such additional splice joints shall be got approved by the PMC.

Site (Field) Erection

Plant & Equipment

The suitability and capacity of all plant and equipment used shall be to the complete satisfaction of the PMC.

Storing & Handling

All steel work shall be so stored and handled at site so that the members are not subjected to excessive stresses and any damage.

Setting Out

One set of reference axes and one bench mark level shall be furnished to the Contractor. These shall be used for setting out of structures.

The EPC Contractor shall assume complete responsibility for correct setting out of all steel work, erecting it correctly as per alignment / levels shown in the drawings and plumb (verticality) of vertical members.

Tolerances

Tolerances for erection of steel structures shall be as per Annexure 'A'

Safety & Security during Erection

1 The EPC Contractor shall comply with IS: 7205 for necessary safety and adhere to safe erection practices and guard against hazardous as well as unsafe working conditions during all stages of erection.

2 During erection, the steel work shall be securely bolted or otherwise fastened and when necessary, temporarily braced/guyed to provide for all loads to be carried by the structure during erection till the completion, including those due to the wind, erection equipment & its operation etc. at no extra cost to the owner. For the purpose of guying, the Contractor shall not use other structure in the vicinity without prior written permission of the PMC.

3 No permanent bolting or welding shall be done until proper alignment has been achieved.

4 Proper access, platform and safety arrangement shall be provided for working and inspection, (at no extra cost to the owner) whenever required.

Field Connections

Field Bolting

Field bolting shall be carried out with the same care as required for shop bolting.

Field Welding

All field assembly and welding shall be executed in accordance with the requirements for shop assembly and welding. Holes made for all erection bolts- where removed after final erection shall be plugged by welding. Alternatively erection bolts may be left and secured.

Scheme and Sequence of Erection

The Contractor shall furnish the detailed scheme and sequence of erection to match with the project schedule and get the same approved by the PMC. All necessary coordination and synchronization shall be done with the civil contractor where civil works are not included in the scope of structural contractor at no extra cost so as to match with the project schedule.

Painting after Erection

General

- The scope of painting after erection shall be at the sole discretion of the PMC and the Contractor shall obtain written instruction in this regard sufficiently prior to taking up any procurement of paint and execution of painting work after erection of steel structures.
- The EPC Contractor shall carry out the painting work in all respects with the best quality of approved materials (conforming to relevant BIS Codes) and workmanship in accordance with the best engineering practice. The Contractor shall furnish characteristics of paints (to be used) indicating the suitability for the required service conditions. The paint manufacturer's instructions supplemented by PMC direction if any shall be followed at all times. Particular attention shall be paid to the following:

- Proper storage to avoid exposure & extremes of temperature,
- Surface preparation prior to painting.
- Mixing & thinning.
- Application of paint and the recommended limit on time intervals between consecutive coats.

- Painting shall not be done in frost or foggy weather, or when humidity is such as to cause condensation on the surfaces to be painted.

- Surface which shall be inaccessible after site assembly shall receive the full specified protective treatment before assembly.

- Primers & finish coat paints shall be from the same manufacturer in order to ensure compatibility. Painting colour code shall be as per Annexure-'B'.

Rub Down & Primer Application

The shop coated surfaces shall be rubbed down thoroughly with emery/abrasive paper to remove dust, rust, other foreign matters and degreased, if required, in accordance with IS: 1477, cleaned with warm fresh water and air dried. The portions, from where the shop coat has peeled off, shall be touched up and allowed to dry.

Primer coat of red oxide zinc chromate primer (conforming to IS: 2074: 1992), shall be applied by brushing/ spraying over the shop coat in a manner so as to ensure a continuous and uniform film throughout. Special care shall be taken to cover all the crevices, corners, edges etc. The final primer coat shall be air dried and shall have a minimum film thickness of 25 microns after drying, as applicable.

In case a different cleaning procedure & primer specifications are specified in the drawing/Tender, the same shall be adopted.

Final Paint Application

After the primer is hard dry, the surfaces shall be dusted off and one coat of synthetic enamel paint of approved colour & shade (conforming to IS: 2932: 2003) as specified by PMC, shall be applied by brushing/spraying so that a film free from "holidaying" is obtained. The colour & shade of first coat of paint shall be slightly lighter than the second coat in order to identify the application of each coat. The second coat of paint shall be applied after the first coat is hard dry. The minimum thickness of each film shall be 20 microns (+ 10% tolerance) after drying. In case a different type of paint & painting procedure are specified in the drawing tender, the same shall be adopted.

Inspection & Testing of Painting Works

1. All painting materials including primers & thinners brought to site by the EPC Contractor for application shall be procured directly from reputed and approved manufacturers and shall be accompanied by manufacturer's test certificates. Paint formulations without certificates shall not be accepted.
2. The PMC at his discretion may call for additional tests for paint formulations. The EPC Contractor shall arrange to have such tests performance including batch wise test of wet paints for physical & chemical analysis. All costs shall be borne by the EPC Contractor.
3. The painting work shall be subject to inspection by the PMC at all times. In particular, the stage inspection will be performed and EPC Contractor shall offer the work for inspection and approval at every stage before proceeding with the next stage. The record of inspection shall be maintained. Stages of inspection are as follows:

- Surface preparation
- Primer application
- Each coat of paint

4. Any defect noticed during the various stages of inspection shall be rectified by the EPC Contractor to the entire satisfaction of the PMC before proceeding further.

Irrespective of the inspection, repair and approval at intermediate stages of work the EPC Contractor shall be responsible for making good any defects found during final inspection/guarantee period/defect liability period, as defined in General Conditions of Contract. Dry film thickness (DFT) shall be checked and recorded after application of each coat. The thickness shall be measured at as many locations as decided by the PMC. The EPC Contractor shall provide standard thickness measuring instrument such as Elko meter (with appropriate range for measuring dry film thickness of each coat) free of cost to the PMC whenever asked for.

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Annexure A: Maximum Permissible Erection Tolerances

A. Columns

1. Deviation of column axes at foundation top level with respect to true axes.

i) In longitudinal direction	±5 mm
ii) In lateral direction	±5 mm

2. Deviation in the level of bearing surface of columns at foundation top with respect to true level ±5 mm
3. Out of plumb (Verticality) of column axis from true vertical axis, as measured at top:
   i) Up to and including 30m height ±H/1000 or ±25 mm Whichever is less
   ii) Over 30m height ±H/1200 or +35 mm Whichever is less.
4. Deviation in straightness in longitudinal & transverse planes of column throughout the height.±H/1500 or ±10 mm Whichever is less.
5. Difference in the erected positions of adjacent pairs of columns along length or across width of building prior to connecting trusses/beams with respect to true distance. ±5 mm
6. Deviation in any bearing or seating level with respect to true level ±5 mm
7. Deviation in difference in bearing levels of a member on adjacent pair of columns both across & along the building. ±5 mm

Note: 1. Tolerance specified under 3 should be read in conjunction with 4 & 5.

Note: 2. 'H' is the column height in mm.

B. Trusses

1. Shift at the centre of span of top chord member with respect to the vertical plane passing through the centre of bottom chord. ±1/250 of height of truss in mm at centre of span or ±15 m whichever is less.
2. Lateral shift of top chord of truss at the centre of span from the vertical plane passing through the centre of supports of the truss. ±10mm ±5mm ±1/1200 of span of truss in mm or 20mm whichever is less.
3. Lateral shift in location of truss from its true position.
4. Lateral shift in location of purlin from its true position.
5. Deviation in difference of bearing levels of truss from the true level.

C. Gantry Girders & Rails

1. Shift in the centre line of crane rail with respect to centre ±[web thickness(mm) +2mm] line of web of gantry girder.

2. Shift of alignment of crane rail (in plan) with respect to ±5mm true axis of crane rail at any point.

3. Deviation in crane track gauge with respect to true gauge
   i) For track gauge up to and including 15 m. ±5mm
   ii) For track gauge more than 15m. ± [5+0.25 (S-15)] mm Subject to maximum ± 10 mm, where S in metres is true track gauge.
4. Deviation in the crane rail level at any point from true level. ±10mm
5. Difference in levels between crane track rails (across the bay) at Supports of gantry girders 15 mm Mid span of gantry girders 20 mm
6. Relative shift of crane rail surfaces at a joint in plan and 2 mm subject to grinding of **Elevation surfaces for smooth Transition**

Annexure B: Painting Colour Code for Structural Steel

1.	STRUCTURAL FRAME (includes columns, beams, tie beams, rakers& purlins) IN CLOSED AREA	DARK GREY
2.	STRUCTURAL FRAME (includes columns, beam & tie beams) IN OPEN AREA	LIGHT GREY
3.	PIPE RACK FRAMES (includes columns, beams & sleepers)	LIGHT GREY
4.	LADDERS, includes cage ladder	BLACK
5.	CHEQUERED PLATE (BOTH FACES)	BLACK
6.	HAND RAILING	
	HANDRAIL, MIDDLE RAIL, TOE PLATE	BLACK & YELLOW STRIPES
	VERTICAL POST	BLACK

BRICK WORK

SCOPE OF WORK

8.

The work covered under this specification pertains to masonry work using various types of bricks and its workmanship in building walls of various thickness, in compliance with the relevant specifications and applicable drawings.

MATERIALS

Bricks used in the masonry shall be of the following type

The Common Burnt Clay Brick : Brick shall be well burnt clay bricks of designated class and shall satisfy the strength criteria and shall be got approved by the PMC before incorporation in the work. The bricks shall be hand moulded or machine moulded and shall be free from nodules of free lime, visible cracks, flaws, warpage and organic matter.

In general, the nominal size of bricks shall be 22.9 x 11.4 x 7 cm. (9”x4.5”x2.75”). Permissible tolerance on dimensions shall not be more than (±) 8%. The EPC Contractor shall get approved the sample and source of bricks from PMC before procurement on large scale and shall maintain the same for the entire work. The bricks shall have smooth rectangular faces with sharp corner and shall be uniform in colour.

In case the size of bricks used in the work is found lesser than the specified one but within the permissible tolerance i.e. {-} 8% , the following shall apply:

i. Extra cement consumed due to more number of joints and due to additional thickness of plaster than the specified in the tender to match with adjoining columns and beams, shall be borne by the EPC Contractor without any extra cost to the OWNER/PMC.
ii. If the plastering to be done is more than the specified thickness to maintain the plaster surface to perfect line, level and plumb with adjoining columns, beams, walls etc., the EPC Contractor shall be responsible to provide more thickness of plaster at his own cost and nothing extra will be paid on this account.

In case the size of bricks used in the work is found more than the permissible, the EPC Contractor shall chip out the exposed edges of bricks upto the required level of wall to receive specified thickness of plaster.

Bricks shall generally conform to I.S. 1077-1992. In any case minimum crushing strength shall not be less than 35 kg/cm2 and water absorption shall not be more than 25% by weight. The PMC shall have the right to reject bricks obtained from any field where the soil has an appreciable quantity of sulphates and chlorides. The specifications for cement, sand and water shall be same as described herein before under cement concrete. Bricks shall be thoroughly soaked in water before using till the bubbles ceases to come up. No half or quarter brick shall be used except as closer. The closer shall be cut to required size and used near the end of the walls. The walls shall be raised truly to plumb. The type of bond to be adopted shall be decided by the PMC, but vertical joints shall be laid staggered.

Fly Ash Lime Bricks (FALG Bricks) : The Fly Ash Lime Bricks (FALG Bricks) shall conform to IS12894. Visually the bricks shall be sound, compact and uniform in shape free from visible cracks, warpage, flaws and organic matter. The bricks shall be solid and with or without frog on one of its flat side.

Fly Ash: Fly ash shall conform to IS 3812.

Note: This item will be operated only for load bearing structure upto 2 storeys and for non-load bearing walls 23 cms thick for multi-storied building's.

Bottom ash used as replacement of sand shall not have more than 12% loss on ignition when tested.

Sand: Deleterious materials, such as clay and silt in the sand shall preferably be less than 5%.

Lime: Lime shall conform to class „C" hydrated lime of IS 712.

Additives: Any suitable additive considered not detrimental to the durability of bricks may be used.

Mechanized Autoclave Fly Ash Lime Brick: These bricks shall be machine moulded and prepared in plant by appropriate proportion of fly ash and lime. The autoclave fly ash bricks shall conform to IS 12894. Visually, the bricks shall be sound, compact and uniform shape, free from visible cracks,

warpage and organic matters. The brick shall be solid with frog, and of 100/80 mm in length, 40 mm width and 10 to 20 mm deep on one of its flat side as per IS 12894. The brick shall have smooth rectangular faces with sharp corners and shall be uniform in shape and colour. Fly ash shall conform to IS 3812 and lime shall conform to class „C" hydrated lime of IS 712.

Clay Fly Ash Bricks: The clay fly ash bricks shall conform to IS 13757. The bricks shall be sound, compact and uniform in shape and colour. Bricks shall have smooth rectangular faces with sharp and square corners. The bricks shall be free from visible cracks, flaws, warpage, nodules of free lime and organic matter, the bricks shall be hand or machine moulded. The bricks shall have frog of 100 mm in length 40 mm width and 10 to 20 mm deep on one of its flat sides. If made by extrusion process may not be provided with frogs. Fly Ash shall conform to grade I or grade II of IS 3812.

Autoclaved Aerated Concrete Block (AAC)

AAC blocks shall be first class having uniform cut size of 625mm/650mmx230mmx 75/100/125/150/200/225 mm thk., it shall be clean, whole and free from cracks, it shall have sharp edges & oven dry density 550 to 650 kg/m3 & minimum compressive strength > 3N/mm2(MPa), it shall have maximum water absorption 8% at equilibrium state.

a. WORKMANSHIP

Four courses of brick work with four joints should not exceed by more than 40 mm., the same bricks piled one over the other without mortar.

Brick work shall not be raised more than 10 courses a day unless otherwise approved by the PMC. The brick work shall be kept wet for at least 7 days. Brick work shall be uniformly raised alround and no part shall be raised more than 1.0 metre above another at any time.

All joints shall be thoroughly flushed with mortar of mix as specified in the schedule of quantities, at every courses. Care shall be taken to see that the bricks are bedded effectively and all joints completely filled to the full depth.

The joints of brick work to be plastered shall be raked out to a depth not less than 10 mm. as the work proceeds. The surface of brick work shall be cleaned down and watered properly before the mortar sets.

The adhesion between the brick masonry surface and the concrete surface of columns, beams, chajjas, lintels etc. should be proper by ensuring that the concrete surface coming in contact with brick masonry is hacked/ chipped/ keyed, cleaned and cement slurry is applied so that a proper bond is achieved between the two dissimilar materials. It is the responsibility of the EPC Contractors to ensure that there will not be any cracks/ fissures anywhere in the brick masonry.

In case the cracks appear subsequently in those areas, they should be made good by cement grouting or epoxy putty grouting/ poly sulphide compound grouting or as per standard modern specifications/ methods with the prior approval of the Engineer-in-Charge, at the cost of the contractor.

All the courses shall be laid truly horizontal and all vertical joints shall be truly vertical. Specified mortar of good and approved quality shall be used. Lime shall not be used where reinforcement is provided in brick work. The mortar should completely cover the bed and sides of the bricks. Proper care should be taken to obtain uniform mortar joint throughout the construction. The walls should be raised uniformly in proper, approved bond. In construction of the wall, first of all two end corners are carefully laid to line and level and then in between portion is built, with a cord stretching along the headers or stretchers held in position at the ends. This helps in keeping the alignment of the courses and maintaining them in level. Similarly all other courses are built. Care shall be taken to keep the perpends properly aligned within following maximum permissible tolerances:

a) Deviation from vertical within a storey shall not exceed 6 mm per 3 m height.
b) Deviation in verticality in total height of any wall of building more than one storey in height shall not exceed 12.5mm.
c) Deviation from position shown on plan of any brick work shall not exceed 12.5 mm.
d) Relative displacement between load bearing wall in adjacent storeys intended to be vertical alignments shall not exceed 6 mm.

e) A set of tools comprising of wooden straight edge, masonic spirit levels, square, 1 meter rule line and plumb shall be kept on the site of work for every 3 masons for proper check during the progress of work.

No brick work shall be carried on during frosty weather except with the written permission of the PMC, who will give special directions as to the manner in which the work is to be performed. All brick work laid during the day, shall, in seasons liable to frost, be properly covered up at night as directed by the PMC. Should any brick work be damaged by frost, the brick work shall, at the discretion of the PMC, be pulled down and made good, at the cost of the EPC Contractor.

Concrete surfaces of columns, beams, lintels, chajjas etc. coming in contact with masonry work shall be properly chipped, washed and given a thick coat of cement slurry before start of work. The rate quoted shall include wire brushing and cleaning brickwork covered with fungus or deleterious materials.

Brick work shall be well watered/ cured throughout the day for at least a week from the date of building and the work shall be protected from sun and rain.

HALF BRICK WORK

Materials and workmanship for a half brick or brick on edge partition wall shall be as specified above. The wall shall be stiffened by R.C.C. stiffeners of size 115 mm. wide x 80 mm. thickness to the full length of wall and shall be provided with 2 Nos. 8 mm. diameter HYSD. Bars. These bars shall be securely anchored at their end where the partition end. The free ends of the reinforcement shall be keyed into the mortar of the main brick work to which the half brick work is joined. Overlaps in reinforcement, if any, shall not be less than 30 cm.

SAMPLING AND TESTS

Samples of bricks shall be subjected to the following mandatory tests

a) Dimensional tolerance b) Water absorption c) Efflorescence d) Compressive strength

Note :

1) Cost of above tests shall be borne by the EPC Contractor and the quoted rate shall be inclusive of such testing.

2)Sampling and frequency of test shall be as mentioned under mandatory tests.

STONE MASONRY

SCOPE OF WORK

The work covered under this specifications consists of supplying and erecting stone masonry walls with available best quality of stone in strict compliance with this specifications and applicable drawings.

RANDOM RUBBLE MASONRY

14.

Material: The rubble shall be of the best quality trap/granite/basalt/quartzite stones obtained fromthe approved quarry. The sample of the stone, to be used shall be got approved from the PMC. All stones shall, generally, be freshly quarried and shall be sound, dense, hard, free from segregation, cracks, weathered portions and other structural defects or imperfections, tending to off set soundness and strength. The percentage of water absorption shall generally not exceed 5% by weight.

For laterite this percentage shall not exceed 12%. All stones shall be wetted before use. Stones shall be neatly worked to requisite sections and forms and shall have fully dressed beds and joints. At least 50% of the stones shall be 0.015 cum in content when reckoned individually. The length of stones for stone masonry shall not exceed three times the height and the breadth or base shall not be greater than three fourth the thickness of wall, or not less than 15 cm .Stones having sharp edges or round surfaces shall not be used. The height of stone may be upto 30 cm. Stones shall be laid on the natural beds and shall run sufficiently inside the wall thickness. No hollow space shall be left out and inter spaces of stones being filled with mortar and stone chips, driven hard & not with mortar only.

All mortar to be used shall be of the type and proportion mentioned in the item. Cement, sand and water to be used shall conform to their relevant specifications as described under cement concrete.

The masonry shall be laid to plumb, lines levels, curves, shapes as shown in drawings. All required holes for passage of water or pipes are to be embedded during construction as specified.

All stones shall be wetted before laying in masonry. Concrete surfaces of columns, beams, lintels, chajjas etc. coming in contact with masonry shall be properly chipped, washed and wetted before start of masonry work. The concrete surface coming in contact of masonry shall be given a thick coat of cement slurry as the masonry work progresses in height. Clean chips and sprawls carefully selected to fit in the space shall be wedged into the mortar joints and beds wherever necessary to avoid thick beds or joints or mortar. However, proper shaping and dressing of stones shall be done prior to their laying in masonry and hammering shall not be resorted to often after the stones are laid in position. The Bond stones/Through stone shall be properly laid into the work(one in every square meter area of masonry wall) and shall extend from front to back to thin walls having width of 600mm and shall overlap by at least 150mm in walls having thickness more than 600mm when laid from both sides. PMC may permit cement concrete 1:2:4, in-situ / pre-cast, to act as bond stone, as required. Bond stone shall be marked by the EPC Contractor with white lead paint. When the work has to be started on the old or the one completed a long while ago or in the previous working seasons, care shall be taken to roughen and clean old surface satisfactorily without disturbing the masonry before laying the new. It shall be wetted before laying the bedding mortar.

When practicable, the whole masonry in any structure shall be carried out upto a uniform level throughout. But when breaks are unavoidable in carrying the work continuously in uniform level, sufficiently long steps shall be left. All junction of walls shall be formed at the time when walls are being built. Cross walls should be carefully bonded into the main walls. All masonry built in cement mortar shall be kept continuously wet for 14 days from the date of laying. Should the mortar perish i.e. becomes dry, white or powder through neglect of watering and if the masonry shows hollow joints or non adherence of mortar to the stones or if the work does not conform to drawings and specifications, the work shall be pulled down and rebuilt by the EPC Contractor at his own cost and risk. All masonry shall be thoroughly cleaned and washed down on completion and all stains, adhering mortar removed from the surface and raking of joints carried out as the scaffolding is being lowered and removed. Holes left in masonry for supporting scaffolding shall be filled and made good before pointing/ plastering.

RUBBLE (KHANDKI FACING) STONE MASONRY- COURSED

The specifications for Random rubble masonry as described above under 14.2 shall generally apply to these for quality of stones, workmanship etc. except for the following:

The face of the stones shall be hammer dressed on all beds and joints so as to give them square/rectangular block shape in order to set on proper bases and render uniform joints. The stones may have bushing projected not more than 40 mm on an exposed face and 10mm on a face to be plastered. The external faces shall be laid in alternate courses of headers and stretchers. The height of each course shall not be less than 150mm nor more than 300mm.The quoins stone shall be of the same height as the course and shall be at least 450mm long and shall be laid stretchers and headers alternatively. Bond stones shall be provided and marked as specified above under 14.2.

The other specifications, mode of measurements etc. shall be same as per specifications for R.R. Masonry mentioned above.

DRY STONE CLADDING

Material: Stone shall be of the type as specified in the item. It shall be hard, sound durable andtough free from cracks, decay and weathering and defects like cavities cracks, flaws, holes, veins, patches of soft or loose materials etc. Thickness of stone shall be as specified.

Stone shall be cut with the gang saw to the required size and shape on all beds and joints so as to free from any waviness and to give truly vertical horizontal surface as required. The exposed face and sides of stones forming joints shall be such that the straight edge laid along the face of the stone is in contact with every point on it. All the visible angle and edges shall be square and free from chipping. The dressed stone shall be of the thickness specified with permissible tolerance of ± 2 mm

Before starting the work, the EPC Contractor shall get the samples of stone approved by PMC. Approved sample shall be kept in custody of PMC and stones supplied and used on the work shall conform to sample with regard to soundness, colour, veining and general texture. The stone shall be cut by gang saw into slabs of required thickness along the places parallel to the natural bed. When necessary double scaffolding for fixing the stone at greater heights, jib crane or other mechanical

appliances shall be used to hoist the heavy pieces of stone and placed them into correct positions. Care shall have to be taken that corners of the stone are not damaged. Stone shall be covered with gunny bags before tying chain or rope is passed over and it shall be handled carefully. No pieces which has been damaged shall be used in work.

Stacking and Storing: Stone slabs are thin and brittle and should never be stacked flat across timber supports. They should therefore, be stacked on edge on timber or like runners. Packing pieces inserted between the slabs may be rope or timber. Slabs shall be well covered with plastic sheeting to protect them from any possible staining.

Scaffolding: Double scaffolding having two sets of vertical supports shall be provided. The supports shall be sound and strong, tied together with horizontal pieces over which scaffolding planks shall be fixed.

Fixing: The size & shape of the cramps shall be as per drawing and as per directions of PMC. The samples of steel cramps should be approved in advance before starting the stone cladding work. The cramp shall be attached to top and bottom of the stone. The cramps shall have inbuilt adjustment for vertical and horizontal alignment. The cramps used to hold support and transfer the load of stone unit to the supporting structured steel shall be designed by the manufacturer and approval of the same shall be obtained from the PMC.
The minimum number of clamps required shall be as per requirement of design to carry the load of individual stone slabs. The cramps shall be spaced not more than 60 cm horizontally and vertically along the stone side for insertion of pins / bolt attached with the steel cramps. Adequate cutting in stone shall be made with precision instrument to hold the cramps pins at the joints.

Stone shall be secured with clamps with high quality workmanship. The walls shall be carried up truly plumb. All the courses shall be laid truly horizontal and all the vertical joints truly vertical. The sequence of execution for cladding work shall be approved by the PMC.

Jointing: Joints horizontal and vertical shall be filled with weather sealant of make as approved by PMC with the help of pouring gun for filling the sealant. Before filling the joint with sealant, masking tape are required to be fixed on stones surface on both edges of joints of the stones, so that sealant

may not spoil the surface of the stone. When all the joints are filled and sealant has dried, the masking tape may be removed.

Protection: Work shall be protected from rain by suitable covering. The work shall also be suitably protected from damage and rain during construction involved in all operations described above including cost of support scaffolding staging, sealant, pouring guns but excluding the cost of steel cramps drilling holes / making recesses in stones which shall be paid for separately.

STRUCTURAL STEEL FRAME WORK FOR DRY STONE CLADDING

Specification for structural frame work for dry stone cladding are same specifications as for steel work in built up sections (welded or bolted).

Fixing of Frame: The properly designed structural frame for withstanding the weight of stone slabare fixed/supported on wall surface with the help of M.S. brackets/lugs of angle iron/flat etc. which is welded at each junctions of member of frame and also embedded in cement concrete block 1:2:4 (1 cement: 2 coarse sand: 4 graded stone aggregate 20 mm nominal size) of size 300 x 230 x 300 mm. The concrete block can be made by cutting the hole of size as mentioned in brick wall and filling the hole with cement concrete including provision of necessary centring/ shuttering for holding of concrete. The frame can also be supported on RCC surface with the help of approved expansion hold fastener by drilling the holes in RCC surface. Steel cramps are either welded or bolted to the frame (by making necessary holes in frame work) for holding of stone.

ADJUSTABLE STAINLESS STEEL CRAMPS

- The cramps shall be stainless steel of make approved by the PMC.
- The weight of the stainless steel clamp (including weight of nut and washer) shall not be less than 260 gms.
- Necessary holes at suitable locations are to be done on steel frame work for dry stone cladding to be fixed.
- Necessary recessed are required to be done in stone slab which is required to be supported by clamps.
- The one end of steel clamp is fixed on frame with nut and bolt and other end is inserted into recesses/hole for fixing the dry cladding stone on frame.

- The rate includes cost of materials and other operations mentioned as above.

STRUCTURAL ALUMINIUM GRID SYSTEM FOR DRY STONE CLADDING

Prefabricated Aluminium extrusions grid system of mullion and transom, of alloy 6063 T6 in anodized finish fixed with SS-316 grade anchor bolts ,Hilti clamps of SS-316 and fasteners as per approved design & shop drawing may be provided if specified, for dry stone cladding. The system shall be designed to accommodate surface temperature upto 80 degree Celsius and expansion-contraction resulting from temp. differential of 25 degree Celsius along with wind load, dead load and thermal loads.

6 Flooring works

PRECAST CEMENT CONCRETE SOLID BLOCK MASONRY

SCOPE OF WORK

The work covered under this specifications pertains to procurement of best quality locally available or locally manufactured precast cement concrete solid block and workmanship in building walls of various thickness in strict compliance with the specifications and applicable drawings.

MATERIAL

15.

Precast cement concrete solid blocks shall be of best quality locally available/ manufactured at site and should be approved by the PMC before incorporation in the work. The ingredient and the cement concrete used shall conform to relevant I.S. as stipulated in specification for cement concrete works herein before.

Minimum crushing strength of the solid blocks shall be of M-5 grade. The type of the bond to be adopted will be decided by the PMC but vertical joints shall be staggered. The size of the blocks shall be 290 x 190 x 100 mm. and 290 x 140 x 100 mm or sizes as available locally and the proportion used in making the blocks shall be minimum 1:11 (1 cement : 11 fine and coarse aggregates) to satisfy the strength requirement . The blocks shall be cured well at least for 14 days before incorporation in to the

work. The cement mortar for concrete blocks masonry shall be 1:4 and joints shall not be more than 10 mm. thick.

CEMENT CONCRETE FLOORING (IPS)

SCOPE OF WORK

The work covered under this specification consists of providing and laying at all levels and floors, flooring of different types, strictly in accordance with these specifications and relevant drawings.

CEMENT CONCRETE FLOORING (INDIAN PATENT STONE)

Materials: The specifications for materials, grading, mixing and the quantity of water to be added shall generally conform to their relevant specifications described under plain and reinforced concrete. The maximum size of coarse aggregate shall be 10 mm. The fine aggregate shall consist of properly graded sand. Concrete shall be mixed preferably by machine, and hand mixing shall be avoided as far as practicable.

Preparation of Base: The base concrete surface shall be thoroughly chipped to remove laitance, caked mortar, loose sand, dirt etc. cleaned with wire brush and washed clean and watered until no more water is absorbed. Where the base concrete has hardened so much that roughening the surface by wire brushes is not possible, the same shall be roughened by chipping or hacking at close intervals. The surface shall be soaked with water for at least 12 hours and surface water removed and dried before laying the topping. Before laying the concrete, cement slurry at 2.00 kg./ sqm. of surface shall be applied for better bond. Concrete flooring shall then be laid in alternate bays in pattern and joints, wide/ flush as per drawing. The edge of each panel into which the floor is divided shall be supported by wooden or metal strips duly oiled to prevent sticking. The panels shall be of uniform size and, unless otherwise specified, no dimension of panel shall exceed 2 m. and the area of a panel shall not be more than 2.00 sqm. However, the exact size of panel shall be decided by the Engineer-in-Charge to suit the size of the room. The joints in the floor finish shall extend through the borders and skirting/dado. The border shall have mitred joints at the corners of the room. Where glass/ aluminium dividing strips are proposed to be provided, the same shall be fixed over thiyyas of cement mortar 1:2 to form panel of size as specified.

The depth and thickness of dividing strips shall be as specified in the item. In the case of flush joints, alternate panels only may be cast on same day. At least 48 hours shall elapse before the concreting of adjacent bay is commenced.

Mixing: The topping concrete shall be of mix of one part of cement, two parts of sand and 4 parts of well graded stone chips of 10 mm. maximum size. The ingredients shall be thoroughly mixed with just sufficient water to the required plasticity, having water cement ratio not more than 0.4.

Laying: The free water on the surface of the base shall be removed and a coat of cement slurry to the consistency of thick cream shall be brushed on the surface. On this fresh grouted base, the prepared cement concrete shall be laid immediately after mixing. The concrete shall be spread evenly and laid immediately after mixing. The concrete shall be spread and levelled carefully. The concrete shall be completed and brought to the specified levels by means of a heavy straight edge resting on the side forms and down ahead with a sawing motion in combination with a series of lifts and drops alternatively with small lateral shifts, either mechanically or manually as directed by the PMC.

While concreting the adjacent bays, care shall be taken to ensure that the edges of the previously laid bays are not broken by carelessness or hand tamping. Immediately after laying the concrete, the surface shall be inspected for high or low spots and any needed correction made up by adding or removing the concrete and whole surface is again levelled. When the layer is made even, the surface shall be completed by ramming or beating and then screed to a uniform line and level. Before the initial set commences, the surface shall be trowelled to smooth and even surface free from defects and blemishes and tested with straight edges. No dry cement or mixture of dry cement and sand shall be sprinkled directly or empty gunny bags spread over the surface of the concrete to absorb excess water coming on top due to floating.

Finishing the Surface : After the concrete has been fully compacted, it shall be finished by trowelling or floating. Finishing operations shall start shortly after the compaction of concrete and shall be spread over a period of one to six hours depending upon the temperature and atmospheric conditions. The surface shall be trowelled intermittently at intervals for several times so as to produce a uniform and

hard surface. The satisfactory resistance of floor to wear depends largely upon the care with which trowelling is carried out. The object of trowelling is to produce as hard and close knit a surface as possible. The time interval allowed between successive trowelling is very important. Immediately after laying only just sufficient trowelling shall be done to give a level surface. Excessive trowelling in the earlier stages shall be avoided as this tends to work a layer rich in cement to the surface, some time. After the first trowelling, the duration depending upon the temperature, atmospheric conditions and the rate of setting of cement used, the surface shall be retrowelled many times at intervals to close any pores in the surface, and to bring to surface and scrap off any excess water in concrete or laitance (it shall not be trowelled back into the topping). The final trowelling shall be done well before the concrete has become too hard but at such a time that considerable pressure is required to make any impression on the surface. Trowelling of rich mix of dry cement and fine aggregate on to the surface shall not be permitted. Trowel marks should not be seen on the finished surface.

Where broom finish is specified, after the concrete has been thoroughly compacted, and when most of the surface water has disappeared, the surface shall be given broom finish with an approved type of brass or M.S. fiber. The broom shall be pulled gently over the surface from edge to edge in such a manner that corrugation shall be uniform in width and depth, the depth shall be not more than 1.5 mm. Brooming shall be done when the concrete is in such a condition that the surface will not be torn or unduly roughened by the operation. Coarse or long bristles which cause irregularities or deep corrugation shall be trimmed out. Brooms which are worn or otherwise unsatisfactory shall be discarded.

After the concrete in the bays has set, the joints of the panels should be filled with cement cream and neatly floated smooth or jointed. Care should be taken that just the minimum quantity of cream for joint is used and excess spilling over the already finished surface shall be removed when the cream is still green.

In case of wide joints the same shall be filled with pigmented cement concrete (1:2:4) using approved pigment and the joint shall be finished in perfectly straight line.

Steel Trowel Finish: Areas where PVC Flooring/ Self leveling Epoxy Flooring are proposed to be used are required to have base concrete finished smooth by steel trowel.

Curing: The completed flooring shall be protected from sun, wind and rain for the first two days and movement of persons over the floor is prohibited during this period. The finished surface shall be covered and cured continuously form the next day after finishing, at least for a period of 7 days. Bunding with murum for curing is prohibited as it will leave permanent stain on the finished floor.

Curing shall be done by spreading sand and kept damp throughout the curing period of seven days minimum. The surface shall be protected from any damage to it whatsoever. The surface shall then be allowed to dry slowly. All corners, junctions of floor with plastered wall surface shall be rounded off when required at no extra cost.

IRONITE (OR HARDER) FLOORING

GENERAL

To withstand heavy wear and tear, concrete flooring with metallic concrete hardening compound such as Ironite/ hardonate shall be laid as wearing layer as detailed below

METALLIC CONCRETE HARDENING COMPOUND

The metallic compound shall be Ironite/ Hardner of approved quality consisting of uniformly graded iron particles, free from non-ferrous metal particles, oil, grease and soluble alkaline compound.

CEMENT CONCRETE UNDER LAYER

Cement concrete flooring of specified thickness and mix shall be laid as specified and generally conforming to specifications laid down for cement concrete flooring. The top surface shall be roughened with brushes while the concrete is still green and the form shall be kept projecting up 12 mm. over the concrete surfaces, to receive the metallic hardening compound topping.

METALLIC CONCRETE HARDENER TOPPING

This shall consist of 12mm. thick layer of mix 1:2 (1 part of cement mixed with hardener: 2 parts of stone aggregate of 6 mm. nominal size by volume). The metallic concrete hardener compound being

mixed with cement in the ratio of 1:4 (1 metallic concrete hardener: 4 cement used by weight) or as specified by the manufacturer. Concrete hardener shall be dry mixed thoroughly with cement on a clean dry pucca platform. This dry mixture shall then mixed with stone aggregate 6mm. nominal size or as otherwise specified in the ratio of 1:2 (1 cement mixed with hardener: 2 stone aggregate) by volume, and well turned over. Just enough water shall then be added to this dry mix as required for floor concrete, water cement ratio not exceeding 0.40 .

The mixture so obtained shall be laid in 12mm. thickness, on cement concrete floor within 1 to 4 hours of its laying. The topping shall be laid true to provide a uniform and even surface. It shall be firmly pressed into the bottom concrete so as to have good bond with it. The concrete shall be compacted well mechanically. Manual compaction will not be permitted unless approved by the PMC. After the initial set has started, the surface shall be finished smooth and true to slope with steel floats.

CONCRETE FLOORING & EPOXY FLOORING

GENERAL

The Tremix Vacuum de-watering method is monolithic Industrial flooring. The advantage of this method is that a plastic and easily workable concrete can be placed with light equipment, and after that great part of the surplus water is extracted from the concrete desirable improvement in quality of concrete is achieved. Through vacuum dewatering process shrinkage is reduced significantly (>50%) as compared to normal concrete and so joint-less bay of larger size is possible.

Base: Generally base of flooring at ground floor shall be same as described for IPS flooring or other type of flooring i.e. sub-base PCC above rubble/brick soling of specified thickness over well compacted plinth filling. A separation layer of plastic sheet (125micron expanded polythene foam sheet), if specified, is placed over for protection against humidity and water which also act as a sliding layer to prevent floor concrete from sticking to sub base in order to avoid cracking.

Installations: The position of trenches and pipes for services such as water and drainage, heating, electric supply and telephones etc. must be fixed before floor concreting starts.

18.1Floor concrete: Floor concrete shall be of M20 grade or as specified.

Floor Thickness: Floor concrete thickness generally shall be 200mm or as specified.

Joints: Expansion/contraction/dummy(working) joints shall be as per drawing and/or as directed by PMC.

Reinforcement: Reinforcement shall be provided as per drawing. Reinforcement at the interface of expansion joint shall also be provided as shown in drawing. Measurement and payment of reinforcement steel shall be as per relevant item of work.

Placement method: Vacuum dewatered concrete should be made according to standard DIN 1045. Normally flooring shall be done forming bays of size not exceeding 6.0mx4.5m. Concrete of specified strength and workability is spread in specified thickness along the stretch of two parallel slotted M.S. channels acting as formwork placed approx.3.0 to 4.0m apart and well compacted using Tremix surface vibrator of suitable length. After compaction surplus excess water is sucked out placing filler pad consisting of two layers on wet concrete. The bottom layer of filler pad in contact with concrete is fine nylon cloth to act as filter whereas the upper layer consist of a special type of plastic net which acts as a water duct. The suction mat (filler pad) is attached to suction pump through suction hose(vacuum de-watering system).Vacuuming time is approx. 1 to 2 minutes per centimeter thickness of concrete. The aim is to reduce initial w/c ratio by 10 to 15% and this also makes concrete denser through static compaction. Care should be taken to ensure that optimum water required for hydration reaction must be there in concrete during vacuum treatment of concrete. The surface then shall be troweled with the help of power troweling floater to get neat finish .If specified in item chemical hardener @ specified quantity per unit area is spread while power troweling in progress, to get hard top wearing surface. The same process is repeated in alternate stretches of concrete bay of specified size. Provisions of Dummy or working joints shall depend on daily normal concrete production. Construction/ Contraction joints upto 1/4th of floor thickness and of width (4mm to 6mm) are cut using mechanical saw afterwards and filled with suitable sealants.

Technical specifications for Ordinary VDF flooring Works

Wear Resistance flooring by use of Vacuum Dewatering Method and power troweling method with skim floater:

Concrete shall be laid between the steel forms and a precast concrete rail acting, as a stop end and also 'rail to be used for surface vibration.

The concrete thus laid shall be vibrated with poker vibrator. During poker vibration, proper compaction of coarse aggregates, fine aggregates and cement shall be obtained. The surface will be then finished in level with the help of surface vibrator to give a dense level surface of concrete.

Vacuum dewatering method will be used to remove excess water from the laid concrete and filler pad and suction mat shall be laid on the freshly laid concrete which will not allow cement paste to flow out, and the suction pump are then started immediately to remove the excess water. The suction time normally is 20 to 30 minutes. This vacuum process will enable to remove 15 to 25 % of water content and making the surface hard enough to- enable to carry the floating operations.

The top surface after removal of mat shall be floated with a mechanical skim floater with trowelling blade to enable the top surface to grind and give a uniform water resistance surface on top. Under no circumstances neat Cement be sprinkled directly on concrete surface to absorb bleed water as surface scaling may occur later. Similarly water should not be applied between trowelling operation as it may cause surface weakness. Minimum two passes shall be carried out. The surface shall then be watered and cured as per clause.

The vacuum dewatering process consists of leveling, compacting and vacuum dewatering the concrete flooring by using vibrating screed, vacuum pumps, suction mats, filter pads, accessories etc. The sequence of operation shall be placing of concrete, vibration, vacuum treatment and floating and the operation shall follow immediately behind each other.

The contractor shall have persons well experienced in the vacuum dewatering process, and in the operation of all related equipments. All process equipment to be used shall be in good working condition and shall be subject to the approval of the Engineer.

The work should be planned well in advance with a view to determine areas to be concreted daily, the required number of equipment, size of vacuum mats, length of vacuum hoses, arrangement of rails, screeds etc.

The area to be concreted shall be thoroughly cleaned, reinforced checked and got approved by the Engineer. Then the specified grade of concrete shall be placed in position without any segregation and properly vibrated.

Immediately after placement of concrete, the vibrating screed, fixed at the proper position to achieve the required specified finished level, shall be allowed to run over the concrete on a true surface to level the concrete. For better consolidation proper surcharge of concrete should be maintained in front of the leading edge of the screed and the vibrating screed shall be allowed to move forward rapidly. The concrete surface shall be screeded high by 2 of the slab's thickness to compensate for the compaction caused by the vacuum dewatering process. (Slabs which have an aggregate hardener shall have compensation made to maintain elevation.)

Immediately after leveling the concrete shall be covered with filter pads and suction mats in strict accordance with the recommendation of the, Manufacturer to' have the slab fully dewatered. The suction mat shall extend 100 mm beyond the edge of the filter pad on all sides. The pads shall extend to within 100 mm of the edges of concrete slab, and the mats shall cover entire slab. Before connecting the hose on the suction mat to the vacuum pump, the edges of the mat shall be smoothed to enable, an airtight seal to be created. A vacuum shall then be applied to the mat. After a minute the gauge on the vacuum pump should indicate a minimum vacuum of 0.70 atmospheres (24.0 in Hg) and if not, the mat must be checked for leakage. For concrete that dewaters readily the vacuum should then be maintained at 0.70 - 0.80 atmosphere (24.0-25.5 in Hg). For concrete which dewaters less efficiently (eg. Air-entrained concrete) the vacuum shall then be reduced to 0.50-0.60 atmospheres (15.0-18.0 in Hg). After approximately 10 minutes the vacuum can then be increased to 0.80 atmospheres.

The vacuum shall be maintained for at least 3 minutes per 25 mm of concrete thickness at 0.80 atmospheres. (Where aggregate hardeners are specified, sufficient moisture shall be maintained to

meet Manufacturer's requirements). The suction mats and filter pads shall then be removed and move to the next section in a leapfrog manner. The vacuum dewatering can be stopped when light footprints only are left m the concrete when stepped upon. A suitable suction time can also be checked with. a Proctor-apparatus, which should show 1.5-2 Kg / sq cm. Upon removal of the suction mats and filter pads the concrete shall be power floated with out delay until all imprints from the vacuum process are removed. If crusting occurs, the floating operation must be delayed till the concrete carries the machine. The higher speed is recommended for the floating operation. The passes with the floating disc should be made in the junction of two mats in order to avoid risk of cracking. .

The waiting time after the floating operation depends on concrete temperature and humidity and varies from 10 minutes to 2 hours.

The trowelling operation cannot take place before the concrete has hardened enough to carry the machine i.e., the trowelling blades will not leave any marks on the concrete. Repeated trowelling, with intervals between the passes, which are adapted to the setting of the concrete, greatly improves the surface characteristics. The surface will be more wear resistant and less dusty.

Atleast two passes are recommended for floors, which are not to be covered.
Vacuum dewatered concrete should be cured like any quality concrete in order to achieve a good final result. Use curing compounds, plastic sheets or wet burlap.

The contractor has the responsibility for achieving the quality of concrete specified by controlling the concrete mixes, placing, vacuum process, finishing and curing. The concrete technician in charge must be present at the site when work is in progress.

The contractor shall be responsible for mix adjustments, performing necessary tests, correcting deficiencies and trouble shooting in general.

The contractor shall be required to maintain control charts showing individual test results for aggregate graduation, slump, air content and compressivestrength.

Joints in Concrete floor (using sealing compound)

1. General

Joints shall be of the types and dimensions as indicated and shall be located as indicated. For ordinary VDF flooring, joints shall not exceed more than 3.5m to 4.0m in both direction.

2. Dummy Joints

The dummy joints shall be 5 mm wide and shall extend vertically from the surface of the slab to a depth equal to 1/3 to 1/4 of the thickness of the slab. The joint may be formed by depressing into the soft but compacted concrete a high tensile mild steel. 'Tee' or flat bar of depth not less than the required depth of the joint plus 25 mm. The bar used for forming the groove shall be coated with soft seal or other suitable lubricant and have built in handles rigidly fixed to facilitate its removal without spalling or crumbling the edges. When the steel bar is removed, joints shall be nearly reformed immediately with proper tools and with mortar/fine material from the slab itself. No additional cement mortar shall be used. Alternatively the slot may be formed by sawing the concrete with a joint cutting machine (diamond cutter) of approved design within 6 hours of placing under moderate climatic conditions and when the concrete has sufficiently hardened. Under extreme cold conditions, this period may be suitably increased based on experience. In all cases, except where cutting is done with saw, the joint edges shall be bull nosed. Care shall be taken that the edges of the joints are not damaged. The edge shall not stand proud of the concrete slabs.

3. Construction Joints

The construction joints shall be 10 mm wide and straight and vertical though the full thickness of the slab. The vertical edge of the concrete on the side of the joint shall be treated with a coat of lime.wash or bituminous paint before the adjacent bay is concreted. A groove 2.5 cm deep and 1 cm wide shall be formed at the top surface of the joint to receive the sealing compound. The groove shall be formed in the same manner as that for a dummy joint. The edges of the groove shall be bull-nosed and not stand proud of the concrete surface.

4. Expansion Joints

The expansion joints shall be straight and shall extend through the full thickness of the slab and shall be of the shape and dimensions shown on the drawings. The slab edges adjacent to the joint shall be

formed truly vertical. The joints shall be filled with a 2 cm thick filler board. Cold applied joint sealant (polysulphide or polyurethane) of approved make with minimum ten years of performance guarantee will be used as joint sealant. The guarantee should be taken in writing from the contractor before the approval of joint sealants. The technical specifications of the cold applied joint sealants (irrespective of whether polysulphide or polyurethane) should meet the requirements mentioned in BS 5212, BS 4254 and EN-141875-2003 (for hydrolysis/water resistance test). In addition, the Movement Accommodation Factor (MAF) of the sealant should be minimum ± 30 %. The groove to receive the sealing compound may be formed by cutting the extra filler board to the required depth.

5. Sealing of Joints

All joints shall be sealed as soon as practicable after 28 days of placing of the slabs. The joints shall be finished flush with the finished concrete surface if the sealing of joints is done in summer and 3 mm below the finished concrete surface, if the sealing of joints is done in winter. After the sealing compound has hardened, the excess sealing compound, if any, adhering to the slab outside the joints shall be removed by scraping or otherwise and the surface left clean.The pavement shall be opened to traffic only after the completion of joint sealing over the entire pavement.

6. Cleaning of Joints

All foreign materials in the joints shall be removed with pneumatic blower. The joints shall, thereafter, be cleaned with a coir brush. Fine particles clinging to the concrete faces shall be removed with the help of an air compressor only to avoid
damage to the edges. The joints shall be cleaned and surface dried before the application of primer.

7. Application of Primer

The cleaned joint shall be primed with a 20-25 mm side painter's brush, while painting, light pressure shall be applied so that the primer penetrates into the pores of concrete. The primer shall be applied twice on one side (i.e. by forward and reverse movement of brush) The primer shall be applied in the thinnest possible complete film and then left for some till the primer feels "tacky" soon after the primer is applied, the joint is covered with 10-15 cm wide paper strips so that no dust is deposited on the primer.

8. Sealant joints

Only cold applied Polysulphide or Polyurethane joint sealant of approved make will be used in rigid pavements. The criteria for selection of the joint sealant will depend on the minimum performance guarantee of ten years offered by the firm. This performance-based selection would accrue better cost effective results rather than using a joint sealant without any guarantee. The technical specifications of the cold applied joint sealants (irrespective of whether polysulphide or polyurethane) should meet all the requirements mentioned in BS-5212, BS 4254 and EN- 141875-2003 (for hydrolysis/water resistance test).In addition, the movement accommodation factor (MAF)of the sealant should be minimum ± 30 %. While the joints shall be sealed flush with the adjacent pavement surface in summer, in winter they shall be filled to a depth of 3-4 mm below the surface. This procedure will reduce the possibility of ingress of grit and other foreign matter into the sealing compound as well as dislodging of the hardened sealing compound under traffic.

Joints in pavement concrete, (using polysulphide polyurethane material)

1. General

Joint shall be of the types and dimensions specified and be located in all as directed by Gamson Engineer. The edges of the groove/joints shall be bull nosed & not stand proud of the concrete surface.

2. Dummy Joints

The size of joints shall be as indicated/ specified. The joint shall be formed using mechanical equipment (diamond cutter) within 6 hour of placing of concrete under moderate climatic conditions and when the concrete has sufficiently hardened. Cutting or sawing by a sawing mounted at movable frame and driven mechanically will also be permitted as a method for making the joint. Care shall be taken that the edge of the joints are not damaged.

In case of sudden rain or storm, the work can be concluded at the dummy joint but the latter will then be formed into a constructionjoint.

3. Construction Joints

Construction joints shall also be provided at places where concreting is stopped due to unforeseen circumstances. The size of joints shall be as specified and as shown on drawings.

Construction joints shall be straight and vertical through the full thickness of the slab. The vertical edge of the concrete of the side of the joint shall be treated with a coat of lime wash or bituminous paint before the adjacent bay is concreted.

A groove of dimension as specified in contract shall be formed. The groove shall be formed in the same manner as that for a dummy joint.

4. Expansion Joints

The expansion joints shall consist of a joint filler board as detailed in the drawing. The depth of the non extruding filler pad Point filler board) shall be cut by 25 mm from top to prepare the joint.

Joints shall be straight and shall extend through the full thickness of the slab and shall be of the shape and dimensions shown on the drawings. The slab edge adjacent to the joint shall be formed truly vertical. The joints shall be filled with approved joint filler as per clause 20.B.7.10.4.

Before the provision of expansion joint, the face of the already laid concrete slab shall be painted with the approved primer at the rate of 2.6 liters per 10 square meters. The expansion pad shall be properly cut to shape. Bond breaker tape shall be applied on the top face of the pad before inserting the dosed cell backup rod. It shall then be placed in position abutting the painted face of the already laid concrete slab. The adjacent slab shall then be concreted. The faces of the pad against which the new concrete slab is to be laid shall also be painted with the approved primer before laying the concrete. While concreting a neat groove as per drawing shall be formed on top of the pad taking care that the edges are absolutely straight and that the groove so made does not get filled with any material like concrete, mortar and other rubbish.

The groove to receive the sealant may be formed by cutting the excess filler board material to the required depth.

Expansion joints shall be provided both longitudinally and transverse direction at spacing as shown on drawing or as directed by PMC.

5. Procedure Of Joint Filling With Polysulphide / Polyurethane Sealant

The technical specifications of the cold applied joint sealants (irrespective of whether polysulphide or polyurethane) should meet all the requirements mentioned in BS-5212,BS 4254 and EN-141875-2003(for hydrolysis/water resistance test). In addition, the Movement Accommodation Factor (MAF)of the sealant should be minimum :f: 30 %.The criteria for selection of the joint sealant of approved make will be a minimum performance guarantee of ten years offered by the firm. This performance-based selection would accrue better cost effective results rather than using a joint sealant without any guarantee. Contractor shall not procure the materials required for joint filling unless the samples are approved by the GE and a ten years of guarantee is given in writing by the contractor. The primer and sealant shall got tested by GE from a reputed testing laboratory who has the NABL accreditation's like Indian Rubber Manufacturer's Research Association's Laboratory /CRRI/IIT/NIT/SEMT Wing, CME Pune before approval. The test certificate shall be obtained for every 5 MT of material incorporated in the work.

6. Joint Preparation

Before commencing joint sealing operations, the following shall be ensured:-

(a) The groove extends fully across the bay between consecutive longitudinal joints in the case of transverse joints and is continuous in the case of longitudinal-joints.

(b) No concrete and foreign matter shall be left in the groove.

(c) In case of expansion joint, the filled materials is exposed to the full length of the joint and expansion joint filler is tightly packed.

(d) Joint surface must be dry, free from dust, coaling, bituminous mastics, concrete curing agencies, mould release agents, oil, grease and loose particles.

(e) All joints shall be thoroughly cleaned out by compressed air and sanding with emery paper or other approved means and shall be approved by Engineer-in- Charge before they are sealed.

(f) The cleaned and prepared joints shall be primed on the sides of the joints up to the depth where sealing component is to be provided and filled with approved sealant.

(g) Wipe out oil and grease by solvent soaked cloth (such as Xylene, Toluene or Acetone or Gardoclean).

(h) Before sealing, insert a bond breaker tape and bond breaker(closed cell polyethylene frame rod) caulked tightly into the base of the sealing groove to prevent the sealant from adhering to the base of slot. Width of bond breaker (backup rod) shall be 20 to 25% more than the joint so that there is no seepage of sealant through joints edges.

7. Fixing/Masking Tape

Fix masking tape to prevent edges of joints becoming dirty due to spillage of sealant at the time of pouring.

Application Of Primer

i. Suitable primer shall be first applied to the. vertical faces of the concrete joint before pouring cold applied sealant confirming to specifications given at clause 20.B.7.11.5.1, in order to improve the adhesive qualities of the product. Primer as specified by manufactures shall be applied to the joint vertical surfaces and allowed to dry for 30 minutes to 2 hours depending on the climatic condition. The surfaces shall be primed twice @0.075 liter (minimum) primer per square meter.

ii. If the primer film has become completely tack free, the surface must be re-primed before applying the sealant.

iii. If the primed areas are left unsealed overnight the primer film must be removed by grit blasting or grinding and the joints interfaces shall be re-primed.

Mixing

Mixing and application of sealant will be through mechanical means to avoid any human error. The mixed product shall be used immediately as per manufacturer's instructions within its pot life i.e. within 30 minute. The mixed sealant should be kept for a few minutes to allow air to escape before commencement of jointing filling.

8. Application or Pouring of Sealant

Mixed sealant shall be applied by mechanical means (machine application) based on the manufacturer's instructions and shall be executed by his approved and trained applicator. Application

temperature of sealant should be in between 5 to 45 degree Celsius (or as per manufacturer's instructions). It should be filled to a level 2 to 3 mm below the top of the pavement leaving a recess to protect the sealant from damage. To prevent accidental spillage of sealant on the top surface and to give a neat finish masking tape should be applied on front edges of joint in such a manner that the material will not be spilled on the exposed surface of the concrete. Any excess filler on the surface of the pavement shall be removed immediately and the pavement surface cleaned. All necessary precautions as per the manufacturer's recommendations shall be taken. The sealant should be immediately tooled either with stainless steel or wooden spatula of the size of the joint to give a smooth finish before it begins to set. Masking tape shall be removed immediately after the sealant has been tooled. The sealant should be allowed to cure as per manufacturer's instructions during which period no traffic should be allowed on the pavement. The test certificate shall be obtained for every 5 MT of material incorporated in the work.

9. Some Miscellaneous Aspects

(i) The sealant are sensitive to temperature while in storage as well as during mixing. Temperatures for all phase of handling of the sealing compounds viz mixing, placing and curing conditions must be in accordance with manufacturer's recommendations.

(ii) All safety precautions during handing and application of these sealants as prescribed by the manufacturers shall also be strictly adhered to.

10. Precautions

(a) Some people are sensitive to resins, hardeners, vapor etc. Therefore it is advisable to use hand gloves / goggles and suitable protective clothing.

(b) Avoid application below 10 degree temperature.

(c) Avoid application on damp or Moist surfaces.

(d) Do not expose primer to naked flames or other sources of ignition.

(e) Materials to be kept in no smoking area.

(f) Containers should be tightly sealed when not in use.

(g) In the event of fire, extinguish with carbon dioxide or foam.

(h) Should accident skin contact occurs, remove immediately with a resin removing cream, followed by soap and water. Do not use solvent.

(i) In case of contact with eyes, rinse immediately with plenty of clean water and seek medical advise.

(j) Use only in well ventilated areas.
(k) All consumables (masking tape, empty cartridges etc) should be removed and disposed off safely.

Technical specifications for Laser screed VDF flooring Works.

The laser screed VDF flooring the procedure for laying concrete is same as ordinary VDF flooring. Towelling shall be done using Laser screed machine to almost 0.0 level accuracy. For Laser screed panel size would be as per machine manufacturer's specification, but more than 4.0m, Laser screed shall simultaneously cut, vibrate & level all in single pass, with all stage operation control using laser transmitters, receivers connected to a computer. This would reduce any manual error, thereby producing a super-flat floor.

18.3.1.IPS on Steps with angle nosing:

Angle L 50 x 50 x 6 with hold-fast shall be screwed to the step in upright position in proper line & level. Fixity of angle to the steps shall be ensured. After ensuring proper fixity of nosing angle, IPS 50mm thk shall be laid over the step, as described in specs for IPS flooring.

Technical Specification for Self-Leveling Epoxy Polyurethane Flooring

Surface Preparation for Epoxy Primer -

1. Surface Preparation:

It is essential that surface should be sound, clean and dry surfaces in order that maximum bond strength is achieved between the substrate and the flooring system. All dust and debris should be removed prior to application of the product.All cracks, crevices and undulations shall be filled with epoxy based mortar.

Primer coat on the naked surface as a bonding coat and proper drying time should be given to carry out the Epoxy Flooring

2. Epoxy Underlay

Generally material is supplied in pre-weighed packs (base and hardener & aggregate), which are ready for immediate on-site use. Part mixing of these components is not acceptable and will affect both performance and appearance of the finished floor. Mixing should be carried out using either a

forced action mixer, or a heavy duty, slow-speed drill with proprietary mixing paddle attachment. The components should be mixed in a suitably sized mixing vessel. The base component should be added to the mixing vessel first and followed by the hardener and these two components mixed together for approximately 2 minutes until an even colour and texture is obtained. Thereafter, the contents of the graded aggregate pack should be slowly added and mixing carried out for a further 3 minutes until a completely homogenous material is obtained.

3. Application of Epoxy Polyurethene

Mixing Epoxy polyurethane coating is supplied in pre-weighed packs ready to use on site. Solvent or thinners should not be added. A forced action mixer with a paddle fitted into a heavy duty, slow speed electric hand drill is recommended for mixing. Hardener component is mixed with Base resin in a suitable mixing vessel. The full colour paste is then added and mixed until an even colour is obtained. Finally the filler as supplied is added and mixed further for three minutes until homogenous lump free slurry is obtained. Laying The material is poured onto the primed substrate and spread to the required thickness with a steel trowel. Alternatively, a serrated trowel can be used. The resin floor should not be overworked but spread slowly and evenly. Immediately after spreading, the floor should be firmly rolled with a spiked roller to help release any entrapped air in the material and level any slight trowel marks. The floor should now self- smooth to an even coloured dense, impervious floor. 2 | P a g e Rate The rate shall include the cost of all labour and materials involved in all the above operations.

RAISED ACCESS FLOORING

RAISED ACCESS FLOORING (FALSE FLOORING)

General: A raised access floor is used to provide a means of creating a void below floor level whichis capable of ensuring building services are available at their required destination. These services will typically include electrical power, data cables, telecom, air-conditioning, fire detection, security, water and drainage etc. as per functional requirements. The use of raised access floor also allow quick and easy access to these services for maintenance reasons.

Key Components and Material: The key components of a raised access flooring system are

Floor Panel: This is the horizontal load bearing component of a raised floor. It is normally 600mm*600mm square (industry standard module size) fabricated out of press hollow CRCA sheet in filled with cementitious fillers either bare finished top surface to receive a carpet tile or with a factory bonded finish such as anti static vinyl, high pressure plain/decorative laminates etc. to suit the functional application as specified in item. If specified, the Floor panels can also be made of steel-clad particleboard with top surface finish as mentioned above.

Pedestal: Pedestals are gridded metal framework of adjustable height supports to floor panels. This is the complete vertical, adjustable supporting structure to the raised floor panels. The pedestals are normally bonded to the sub floor using a epoxy resin based adhesive with mechanical fixings also provided if required. The pedestal assembly provides vertical adjustment of 1½" to allow the raised floor to be installed flat and level despite undulations in the sub floor. The pedestal head provides panel location and also when required a means of fixing the panel to the pedestal head.

The height of the legs/pedestals is dictated by the volume of cables and other services provided beneath. The height of raised access floor from finished floor level shall generally be not less than150mm.For height of raised floor more than 600mm. additional care shall be taken for grid/framework design.

Stringer: This is a horizontal component that connects pedestals together. It connects to the pedestal head and is used to provide additional lateral support at greater floor height and/or increase the structural performance of the raised floor system. Stringers can be „snap on" or „bolt on" stringers as per site requirement. Perimeter stringers shall be provided if required, as an additional support to cut panels along perimeters.

Note:

(i) The raised access floor system shall be designed and tested for the uniform distributed load(U.D.L) and Point Load as specified in item. Generally the commonly fabricated False flooring systems by leading manufacturer are for capacity as (a)1080kg/m2UDL and 360kg point load, (b)1350kg/m2UDL and 450kg point load & (c)1680kg/m2 and 560kg point load.

(ii) The complete false flooring system shall be of specialized manufacturer and shall be got approved by PMC before procurement.

Laying/Installations: The complete raised access floor system (of specified capacity) as described above shall be installed by the specialized/skilled workers under the guidance of approved manufacturer and as per approved design. The sub floor beneath shall be structurally sound and surface shall be flat and in clean condition.

The Tolerance in level of top finish surface of raised floor shall not be more than ±1.5mm.The installed flooring system shall be tested for load capacity as per standard practice of testing.

KOTAH STONE FLOORING/ SKIRTING/ STEPS

MATERIALS

The stone shall be hard, sound, durable, homogeneous in texture and resistant to wear. These shall be without any soft veins, cracks or flaws and shall have uniform colour. They shall have natural surface free from broken flakes on top. Hand cut/ machine cut for exposed edges and machine polished. Kotah stone shall be of the best quality and of the specified thickness, size and the shade, which shall be got approved by the PMC.

The slabs / tiles shall be rectangular or square in shape or as per pattern shown in drawing and as directed by the PMC. The sizes given in schedule of quantities are tentative and can vary only slightly as per the availability in the market. The thickness of the slab after it is dressed shall be 20, 25, 30 or 40 mm as specified in the item. Tolerance of (±) 2 mm shall be allowed for the thickness. In respect of length & width, tolerance in length & width shall be permissible upto (±) 5 mm for hand cut slabs & (±) 2 mm for machine cut slabs.

Uniformity of size and colour / shade shall generally be maintained for the stones used in any one room. The exposed surface shall be machine polished to a smooth, even and true plane and the edges hand cut and dressed true and squares. The evenness of the surface of slabs and edges of the slab shall not be marred by careless dressing or handling and no patching up shall be allowed for the slab. The edges shall be quite straight. The under face may be left as required or rough dressed.

Before taking up the work, samples of stone slabs to be used and their dressing and polishing shall be got approved by the PMC and kept in his office as approved sample and the stone slabs to be used shall conform to the same.

BEDDING/ BACKING COAT

In case of flooring / skirting / dado, the mortar bedding / backing shall be of cement mortar of thickness and mix specified in the schedule of work. In case required the bedding thickness will be increased to allow for electrical services.

CEMENT MORTAR

Cement mortar bedding shall be as specified under relevant specification for terrazzo/ plain cement tile flooring.

CONSTRUCTION DETAILS

Cement mortar as specified for bedding shall be uniformly mixed. The amount of water added shall be the minimum necessary to give just sufficient plasticity for laying and satisfactory bedding. Care shall be taken in preparing the mortar to ensure that there are no hard lumps that would interfere with the even bedding of the stones. Before spreading the mortar, the sub-floor or base shall be cleaned of all dirt, set mortar scum or laitance and of loose materials by hacking and brought to original levels and then well wetted without forming pool of water on surfaces.

FIXING THE STONE SLAB/ TILE

Before laying, the stone shall be thoroughly wetted with clean water, neat cement grout (2.75 kg/ sqm.) of honey like consistency shall be spread on the mortar bed over as much areas as could be covered with the slabs within half an hour. The specified type of stone shall be laid on the neat cement float and shall be evenly and firmly bedded to the required level and slope in the mortar bed. Each stone shall be gently tapped with wooden mallet till it is firmly and properly bedded.

There shall be no hollows left. If there is a hollow sound on gently tapping off the slab, such slab shall be removed and reset properly. The joints shall be grouted with matching cement slurry. Approved pigment shall be used in cement slurry to match with shade of stone. Pigment required to match the

shade of stone shall be supplied by the EPC Contractor at no extra cost. The stone adjoining the wall shall go about 12mm. under the plaster, skirting or dado for the wall. All stone slabs, tiles shall be so laid as to have continuous lines from various rooms to the corridors. No change of lines shall be permitted at junction between rooms and corridors. Only one piece machine cut, Kotah stone shall be used for treads and risers, unless otherwise specified in the tender schedule.

CURING

The work shall be kept well wetted with damp sand or water for seven days.

POLISHING AND CLEANING

The stone polishing process in general consists of following steps

Grinding: This method is a very aggressive approach to refinishing stone. It uses a metal-bonded and diamond grit heavy weighted floor buffer to remove deep scratches and lippage (Lippage is a condition where one edge of a stone is higher than adjacent edges, giving the finished surface an uneven appearance). The objective of grinding is to flatten the floor and make it dust free.

Honing: Honing process follows the grinding process. This process is less aggressive approach than grinding. Although similar to grinding, the materials and grits used are not as coarse as they are used for the grinding process. If the stone surface to be polished is having minor to moderate scratches and etch marks, the process of floor finishing may start with honing and skip the grinding phase. This method shares the same objective as grinding, ending with a flat floor.

Polishing: After the honing phase, a higher grit and sometimes combinations of higher gritcompounds are used to bring out a shine in stone during this process.

Method: When the bedding and joints have completely set and attained required strength, the surface shall be machine polished to give smooth, even and true plane to the flooring. All flooring shall be thoroughly cleaned and handed over free from any mortar stains etc. The floor shall then be kept wet for a minimum period of 7 days.

The polishing shall be carried out using polishing machine fitted with high performance resin bonded diamond tools/pads starting with coarser grit pad then medium grit pad, fine grit pad and very fine grit pad in that sequence to get the required polished surface.

The surface shall be ground evenly with machine fitted with coarse grade diamond grit pad (grit size. 100). Water shall be used profusely with grinding. After grinding, the surface shall be thoroughly washed, remove all grindings, mud cleaned and mopped, and the joints opened out during grinding shall be grouted once again wherever necessary with matching cement. The surface shall be again cured. The second grinding shall then be carried out with machine fitted with grit pad (grit size200) and shall be grouted again the opened out joints with matching cement. The process is repeated using medium grit(size400) and then with fine grit diamond pad (Grit 800) for normal shine polish finish .For mirror polished finish(i.e. high gloss finish) the process of polishing is carried out using very fine grit pad/ tool (grit size 1500). After the final polish, oxalic acid crystals powder(or other approved floor polishing chemical) shall be dusted over the surface (@ 32.5 gm. per m2, sprinkled water and rubbed hard with Namdah block (pad of woolen rags). The following day the floor shall be wiped with a moist rag and dried with a softcloth and finished clean. If any tile is disturbed or damaged, it shall be refitted or replaced, properly jointed and polished.

For small areas or where circumstances so required, hand polishing using smaller size grit pads of medium to fine grit grades may be permitted in lieu of machine polishing after laying, entirely at the discretion of the PMC.

In all other respects, the process shall be similar as for machine polishing. The finished floor shall not sound hollow when tapped with a wooden mallet.

SKIRTING AND DADO/ FACIA

The quality and type of stone shall be same as mentioned for flooring except of their height and thickness or backing coat which shall be as mentioned in item schedule. The backing shall conform to the specifications for cement mortar specified for item. EPC Contractor should take into consideration the fact that touching up of the plaster at the junction of skirting / dado is invariably done after the

skirting/ dado/ facia work is completed and quote rates accordingly. Nothing extra for the same shall be entertained.

The skirting tiles shall be true in plane, line, level and plumb or in slope. The vertical lines of skirting tiles should be in line with that of flooring tile lines. The colour of the skirting tile and floor tile shall match. The undone portion of plaster work left above the skirting work shall be finished round or as directed by the PMC in the matching plaster. The item of plastering shall be inclusive of this plaster finishing above the skirting tiles, required to be done after laying of skirting tiles. No additional payment will be admissible for this extra operation.

Polishing may be done by hand, but a smooth surface and fine polishing shall be obtained as specified. Joints shall be finished in neat matching cement slurry. The junction of plaster and the upper edges of the dado/ skirting shall be finished smoothly as directed by the PMC without any extra cost.

MOULDING AND DHAR POLISH

Providing full/ half round/stepped MOULDING to stone edges. The edges should be checked with a plywood profile to check the accuracy. The molding should be polished so that the polish is the same as on the surface. The workmanship shall be to the satisfaction of the PMC.

Providing DHAR POLISH to the stone edges. The polish on the edge should be equivalent to the polish on the surface. The workmanship shall be to the satisfaction of the PMC

TANDUR STONE / CUDDAPPA STONE / POLISHED SHAHABAD STONE / BLUE WADI STONE FLOORING / SKIRTING / DADO

The specifications for these items shall be similar to those for Kotah stone as above.

MARBLE STONE FLOORING, TREADS, RISERS, SILLS, CLADDING, DADO ETC

MARBLE STONE SLABS

The colour and quality of marble slabs shall be of the kind of marble specified in item/drawings/as directed by the PMC. The marble from which the slabs are made, shall be of selected quality, hard, sound, dense and homogenous in texture, free from cracks, decay, weathering and flaws. Before starting the work, the EPC Contractor shall get the samples of marble slabs approved by the PMC. All slabs which goes into work shall strictly conform to the samples, failing which the entire materials are likely to be rejected.

The slabs shall be machine polished and machine cut to the dimensions specified in items of schedules of quantities/drawings and as directed by the PMC.

DRESSING OF SLABS

Every stone shall be cut to the required size and shape, fine dressed on all sides to the full depth so that a straight edge laid along the side of the stone is full in contact with it. The top surface shall also be fine dressed to remove all waviness. The top surface of slabs shall be machine polished and exposed edges machine cut, or as specified in the item and as directed by the PMC. All visible angles and edges of the slabs shall be true, square or as required, and free from chippings and the surface shall be true and plane.

The thickness of the slabs shall be 25 mm. or as specified in the description of item. The minimum size of stone to be used for various items shall be as mentioned in the schedule of quantities/drawings of this tender. Marble stones of approved smaller sizes other than mentioned in the schedule of quantities, if required for bands, borders, flooring etc. shall be provided and laid as directed by the PMC.

Any opening of required size and shape at any desired place in flooring, bands, borders etc. shall be made in such a way that marble bounded by number of marble stones/slabs. No broken or defaced stone shall be permitted in the work.

BEDDING/BACKING MORTAR

The bedding/backing shall be of cement mortar/lime mortar of mix and thickness as specified in the description of the item. In case required the bedding thickness will be increased to allow for electrical services.

Mixing : The mixing of mortar shall be done in mechanical mixer or hand mixing as specified/as directed by the PMC.

Mixing in Mechanical Mixer : Cement and sand in the specified proportion shall be mixed dry thoroughly in a mixer. Water shall then be added gradually and wet mixing continued for at least one minute. Care shall be taken not to add more water than that which shall bring the mortar to the consistency of stiff paste.

Only the quantity of mortar, which can be used within 30 minutes of its mixing shall be prepared at a time.

Mixer shall be cleaned with water each time, before suspending the work.

Hand Mixing : If approved by PMC, hand mixing shall be allowed. The measured quantity of sand shall be levelled on clean masonry platform and cement bags emptied on top. In hand mixing, the quantity of cement shall be increased by 5% over the approved constant, with no extra cost to the OWNER/PMC. The cement and sand shall be thoroughly mixed dry by being turned over and over, backwards and forwards, several times till the mixture gives an uniform colour. The quantity or dry mix which can be used within 30 minutes shall then be mixed on masonry through with just sufficient quantity of water to bring the mortar to the consistency of stiff paste.

General : Mortar shall be used as soon as possible after mixing and before it has begun to set,and in any case within 30 minutes after the water is added to the dry mixture. Mortar unused for more than 30 minutes shall be rejected and removed from the site of work immediately.

LAYING - FLOORING

Before laying the cement mortar bedding/backing, the concrete/brick, floor/wall surfaces shall be thoroughly hacked, cleaned of all mortar scales, concrete lumps etc., brushed, washed with water to

remove mud, dirt etc. from the surface and shall be thoroughly wetted. Until and unless the surface is approved by the PMC, the flooring shall not be started. A bedding of cement mortar of 20 mm. average thickness with the minimum thickness at any place under the slab not less than 13mm. shall be laid evenly and to the required slopes as directed. The marble slabs shall be thoroughly washed and cleaned and then be laid on the bedding/ backing with cement floating at the rate of 4.39 kg./sqm. All slabs shall be truly and evenly set in a thick cement slurry or paste like consistency applied to the sides and bottom and over the prepared base. The slabs shall then be tamped down with a wooden mallet until they are exactly in true plane and line with adjacent slabs. All slabs shall be extended upto the unplastered surface of masonry walls/RCC columns/RCC walls. The slabs shall be close jointed in matching cement slurry and the cement slurry coming out through the thin joints shall be immediately wiped clean. The grains of marble stone shall be matched as shown in drawing or as directed by the PMC. All slabs shall be so laid as to have continuous lines from various rooms to the corridors. No change of lines shall be permitted at junction between rooms and corridor, if the same flooring is specified in both the places.

MARBLE SILLS, TREADS ETC.

Marble stone for sills/ treads/ risers, etc shall be of approved quality. Dressing of stone slab, mortar mix. for bedding/backing, laying etc. shall be similar to as described above as far as applicable. Marble slabs of specified thickness and width shall only be provided. Normally the length of the sill shall be equal to the width of the window. It shall be screwed to the masonary if essential. The sill shall project beyond the face of external masonary by 12mm and internal as directed by PMC. The sill should be laid in step where directed so that there is no leakage or in case of a door. The joint between the stones and the stone and the door/ window should be sealed and made water tight. The internal edges of the sills shall be given a moulded edge. In case of doors, the edge between the stones shall be given dhar polish. All the tread edges shall be given a moulding to the exposed edges.

The above specification shall be valid for all types of stone sill/ tread/ riser other than Marble also.

MARBLE STONE DADO & CLADDING

Only machine cut and machine polished marble stone shall be used. Brass cramps and brass pins ofapproved quality, size and make shall be provided. The brass pins shall be provided at the meeting of two marble slabs both ways horizontally and vertically. The brass cramps shall be provided at the

places approved by the Engineer-in-Charge. Marble to be used shall be of approved size, colour, type of veins and laid as specified in schedule of quantities or to the pattern shown in drawings or as directed by the Engineer-in-Charge. Laying of marble stone shall be similar as stated above as far as applicable. All exposed edges shall be moulded as specified / shown in drawing and polished to match with the adjoining surfaces.

POLISHING AND FINISHING

Slight unevenness at the meeting edges of slabs shall be removed by fine chiseling and finished in the same manner as specified under kotah stone polishing works

MOULDING AND DHAR POLISH

The specifications will be similar to the moulding and dhar polish mentioned in the earlier chapter.

MIRROR POLISHED GRANITE STONE FLOORING, TREADS, RISERS, SILLS, CLADDING, DADO

Specification for mirror polished granite stone in flooring, dado, cladding, treads, risers, sills etc. shall be similar to marble specification except polishing as pre-mirror polished granite is used in work. In case of exposed edge, edge polishing/edge moldings shall be done at site by skilled stone polisher as specified/ shown in drawing and polished to match with the adjoining surface.

Note: Agency has to procure stone slab of specified thickness and sizes only. However a tolerances inthickness of all types of natural stones upto ± 2mm may be accepted considering market availability by PMC.

TILE FLOORING/SKIRTING/DADO

PRESSED CERAMIC TILES/ VITRIFIED TILES/ GLAZED TILES/ GRANITE TILES/ NON SLIP TILE

The tiles shall be of approved make and shall generally conform to IS 15622. They shall be flat and true to shape and free from blisters, cracks, crazing, spots, chips, welts, crawling or other imperfections detracting from their appearance. The tiles shall be tested as per IS 13630 (various parts). Definition, Classification and Characteristics of pressed ceramic tiles shall be as per IS 13712. Sampling and basis for acceptance shall be in accordance with IS 13630 (Part-15). Shapes, Sizes & Requirements of tiles shall be as under

Wall Tiles: Wall tiles for Skirting/Dado/Facia etc. shall conform to IS 15622, having water absorptionE >10% (Group B-III).

Floor Tiles: Floor tiles shall be as specified below:
Ceramic tiles: Tiles conforming to IS 15622, having water absorption 3%<E ≤ 6% (Group B-II).
Vitrified tiles: Tiles conforming to IS 15622, having water absorption E ≤ 0.08% (Group B-I).

PREPARATION OF SURFACE AND LAYING OF TILES FOR FLOORING

Preparation of Surface: Base concrete or the RCC slab on which the tiles are to be laid shall becleaned, wetted and mopped. The bedding for the tile shall be with cement mortar 1:6 (1 cement : 6 sand) or as specified. The average thickness of the bedding shall be 20 mm or as specified while the thickness under any portion of the tiles shall not be less than 10 mm. In case required the bedding thickness will be increased to allow for electrical services.

Mortar shall be spread, tamped and corrected to proper slope and levels. Over this mortar bedding neat grey cement slurry of honey like consistency shall be spread @ 3.3 kg of cement/sqm. over an area upto one square metre and combed using suitable combing trowel, to receive tiles. Tiles shall be soaked in water, washed clean and shall be fixed on the prepared mortar bed one after another following the arrow mark on back of the tiles. Each tile gently being tapped with a wooden mallet till it is properly bedded and in level with the adjoining tiles. The joints shall be kept as thin as possible and in straight lines or to suit the required pattern, unless specified use of specific spacers. After tiles have been laid, surplus cement slurry shall be cleaned off.

The surface of the flooring during laying shall be frequently checked with a straight edge of about 2m long, so as to obtain a true surface with the required slope. In bath, toilet, W.C., kitchen and balcony/verandah flooring, suitable tile drop or as shown in drawing shall be given in addition to required slope to avoid spread of water.

Where full size tiles cannot be fixed, these shall be cut/sawn to the required size, and their edge rubbed smooth to ensure straight and true joints. Tiles which are fixed in the floor adjoining the wall shall enter not less than 10 mm under the plaster, skirting or dado.

Pointing and Finishing: The joints shall be cleaned off the grey cement slurry with wire/coir brushor trowel to a depth of 2 to 3 mm and all dust and loose mortar removed. Joints shall then be flush pointed with white cement added with pigment as required to match the colour of tiles. Where tiles spacer lugs are provided, the half the depth of joint shall be filled with cement grout and the remaining top half of the joint shall be filled with approved tile grouting compound or as specified without the lugs remaining exposed. After necessary curing, the finished surface shall be washed and cleaned. The finished floor shall not sound hollow when tapped with a wooden mallet.

PREPARATION OF SURFACE AND CLADDING OF TILES FOR SKIRTING/DADO ETC.

Preparation of Surfaces: The joints shall be raked out to a depth of at least 15 mm in masonry walls. In case of concrete walls, the surface shall be hacked and roughened with wire brushes. The surface shall be cleaned thoroughly, washed with water and kept wet before commencing the work.

Cladding: 12 mm thick plaster of cement mortar 1:4 (1 cement : 4 sand) mix or as specified shall be applied and roughened with wire brushes or by scratching diagonal at closed intervals.

The tiles should be soaked in water, washed clean, and a coat of cement paste applied fully at the back of tiles and set in the bedding mortar. The tiles shall be tamped and corrected to proper plane and lines. The tiles shall be set in the required pattern and jointed. The joints shall be as fine as possible. Top of skirting or dado shall be truly horizontal and joints truly vertical except where otherwise indicated. Odd/cut size of tile shall be adjusted at bottom to take care of slope of the flooring.

Skirting and dado shall rest on the top of the flooring. Where full size tiles cannot be fixed, these shall be cut/sawn to the required size and their edges rubbed smooth. Skirting/dado shall not project from

the finished “surface of wall” by more than the tile thickness; undulations if any shall be adjusted in wall.

Finishing & Curing: The joints shall be cleaned off the grey cement grout with wire/coir brush ortrowel to a depth of 2 to 3 mm and all dust and loose mortar removed. Joints shall then be filled with white cement added with pigments as required to match the colour of tiles. After necessary curing, the surface shall be washed and finished clean. The finished work shall not sound hollow when tapped with a wooden mallet.

skirting or dado including curves where specials such as coves, internal and external angles and beads are used.

Where cornices are used, the area of dado shall be measured excluding the cornices. Nothing extra shall be paid for cutting of the tiles to required size to suit the site condition and/or to complete the work as per the drawings.

FIXING OF TILE FLOORING USING QUICK SET TILE ADHESIVE

When tile flooring is to be laid over the existing flooring without dismantling old flooring, it can be laid with adhesive. Process shall be as under:

Surface Preparation: The old flooring shall be thoroughly checked for hollowness if any. Hollow sounding tiles shall be dismantled and the gap shall be filled with CC 1:2:4 correct to the existing floor level. Final surface of the floor shall be cleaned off all dirt, dust, mortar splashes, greasy materials etc. and level checked. Undulations, if any, shall be rectified with tile adhesive.

Fixing of Tiles: High polymer modified quick set approved tile adhesive (conforming to IS 15477)shall be thoroughly mixed as per Manufacturer’s specifications. The adhesive shall have a maximum VOC content of 65 g/L in accordance with IGBC rules. It shall be prepared in such limited quantities so that it can be used within 1.5 to 2 hours. It shall be spread over the prepared surface to get uniform top level for an area not more than one sqm at one time, keeping 3 mm average thickness of the adhesive. The adhesive so spreaded shall be combed using suitable combing trowel. Tiles shall be pressed firmly into the position with slight twisting action checking it simultaneously to ensure good contact gently being tapped with wooden mallet till it is properly backed with adjoining tile following the

arrow mark on back of the tiles. The tiles shall be fixed within 20 minutes of application of adhesive. The surplus adhesive from the joints and surface of the tiles shall be immediately cleaned.

The surface of the flooring shall be frequently checked during laying with straight edge of about 2m long so as to attain a true surface with required slope. Where spacer lugs tiles are provided these shall be filled with grout without lugs remaining exposed.

Where full size tile cannot be fixed, these shall be cut/sawn to the required size, and edges rubbed smooth to ensure straight and true joints. Tiles which are fixed in floor adjoining to wall/concrete surface shall abutt the vertical surface.

Finishing / Measurements / Rates: Specifications as per Pressed Ceramic Tile Flooring / Skirting /Dado / Facia etc., as above shall apply mutatis-mutandis Item shall include rectification of existing floor undulations, levelling concrete, additional thickness of tile adhesive etc.

ACID OR ALKALI RESISTANT TILES

Manufacture and Finish: The tiles shall be of vitreous ware and free from deleterious substances. The iron oxide content allowable in the raw material shall not exceed two percent. The tiles shall be vitrified at the temperature of 11000C and above and shall be kept unglazed. The finished, tile, when fractured shall appear fine grained in texture, dense and homogenous. The tiles shall be sound, true to shape, flat and free from flows and manufacturing defects affecting their utility.

The tiles shall be conforming to IS 4457-2007. The tiles to be tested for water absorption, compressive strength, acid resistance as per IS 4457. Sampling procedure for acceptance tests and criteria for conformity shall be as per IS 4457.

Dimensions and Tolerances: Ceramic unglazed vitreous acid-resistant tiles shall be as specified in the Schedule of items. The thickness shall be as per approved manufacturer's specifications or as specified in the Schedule of items conforming to relevant standards. The depth of the grooves on the underside of the tile shall not exceed 3 mm. Tolerance on length, breadth and thickness of tiles shall be +2 percent.

Shape: The tiles shall be square shaped. Half tiles rectangular in shape shall also be available. Halftiles for use with full tiles shall have dimensions which shall be such as to make two half tiles, when joined together, match with the dimension of full tile. The shape of tiles other than square shall be as agreed to between the purchaser and the manufacturer. Tiles shall be checked for squareness and warp as per IS 4457-2007.

Performance Requirements: The tiles when tested in accordance with method given in IS 4457,shall conform to be requirement specified in the code.

Loss in Abrasion: The maximum percentage of loss in abrasion of the ceramic unglazed vitreous acid resistant tiles determined in accordance with the procedure laid down in IS 1237, shall be as mentioned in IS 4457.

Marking: Tiles shall be legibly marked on the back with the name of the manufacturer or his trademark, manufacturer's batch number and year of manufacture. Each tile may also be marked with the ISI certification mark.

Preparation of Surface and Laying: Preparation of surface and laying shall be as mentioned in preceding previous 22.2.1 and 22.2.2 except the bedding used to be acid and or alkali resistant epoxy mortar(5to8mm) over a bed of bitumastic compound(8to10mm) over two coats of acid resistant bitumastic primer as specified and approved by PMC. Thickness of backing of mortar for dado/skirting to be 12 mm or specified on item.

Pointing and Finishing: As per Para 22.2.2 except that mortar/grout used for pointing to be acidand or alkali resistant as specified.

PVC SHEET / TILES FLOORING

The thermoplastic binder shall consist substantially of one or both of a) Vinyl chloride polymer and b) Vinyl chloride copolymer The polymetric material shall be compounded with suitable plasticizers and stabilizers.

PVC flooring material shall be of approved manufacturer and shall be any of the following types meeting the required specification:

i. Acoustic / Flexible compact (reinforced with glass fiber grid layer with RD quartz crystal chips) PVC vinyl flooring sheets/tile with high abrasion and scratch resistant properties having anti bacterial and anti fungi quality.

ii. Homogeneous(single-layer composition which increase s scratch-resistance) PVC sheet/tile

iii. Hetrogeneous multi-layer PVC vinyl flooring sheet/tile,

iv. Antistatic & Conductive PVC sheet/tile: These are specialty PVC floorings used at operation theatres, clinics, computer/server rooms, electronic manufacturing facilities, supermarkets, schools public offices, and commercial offices where static electricity needs to be avoided. Anti-Static sheet material is designed to reduce the buildup of static electricity that is created when objects and people come in contact with the material. Flat clear anti-static sheet material is formulated to have a surface that is more resistant to static electricity, this means that a 5000-volt charge would be dissipated to effectively zero in less than ¼ of a second when in contact with anti-static PVC.

All of the above products of approved manufacturer shall be tested and certified as per relevant standard codes as directed by the EIC.

Thickness: The preferred thickness of PVC tiles for normal floor covering shall be 1.5, 2.0, 2.5, 3.0 or 4.0 mm. Thickness of PVC sheets shall be measured with micrometer or Ratchet type or a dial gauge graduated to 0.02 mm. The micrometer shall have flat bearing surfaces of at least 6.5 mm diameter at both contact points.

For sheets and rolls the thickness of the specimen shall be measured at twenty scattered points.

For polystyrene wall tiles, the cavity depth of the test specimen shall be measured at five points taken at random on the rear surface of each tile with a suitable depth gauge.

Size: The width of flooring sheets and rolling in continuous length shall be 1000, 1500 and 2000 mm When supplied in rolls the length of the rolls shall not be less than 10 metre. The measurement shall be carried out with a travelling microscope or suitable scale graduated to 0.02 mm. Each tile shall be measured for length and width at the three quarter point in each direction.

Tolerance: Tolerance limits shall be as given in the Table

Table: Tolerance Limits

(a)	**In Thickness**	± 0.15 mm
(b)	**In Width as under**	
(i)	300 mm square tiles	± 0.2mm
(ii)	600 mm square tiles	±0.4mm
(iii)	900 mm square tiles	±0.6mm
(iv)	Sheets and rolls	± 0.1 per cent

Adhesive: Rubber based adhesives are suitable for fixing PVC flooring over concrete, wooden and metal sub-floors. PVA based adhesives shall be used for concrete and wooden sub floors. PVA based adhesives are not suitable for metallic surfaces and also for locations where there is constant spillage of water. All adhesives used must be compliant with IGBC rules.

PREPARATION OF SUB-FLOORS

Before laying PVC sheets / tiles, it is essential to ensure that the base is thoroughly dry and damp proof as evaporation of moisture can not take place once the PVC flooring is laid. Moisture slowly damages the adhesive resulting in PVC sheet / tiles being separated from the base and curled up. In case of new work a period of 4 to 8 weeks shall be allowed for drying the sub-floor under normal conditions. Concrete sub-floors on the ground floor shall be laid in two layers. The top of the lower layer of concrete shall be painted with two coats of A-90 grade (conforming to IS: 1580) applied at the rate of 1.5 kg/sqm. The top surface of the lower layer shall be finished smooth while laying the concrete so that the bitumen can be applied uniformly. The bitumen shall be applied after the concrete has set and is sufficiently hard. Bitumen felt conforming to IS : 1322 shall be sandwiched in the sub-floor laid in two layers.

In new concrete floor, the smooth finish required shall be produced by using cement slurry spread on fresh concrete floor and finished smooth. If the concrete floor is old and surface not even, the surface should be made smooth by first cleaning it free of all foreign material and then a layer of cement mortar 1:2 of average thickness of 6 mm shall be applied on the surface finishing the surface smooth.

The finished surface shall be cured for 7 days and then allowed to dry thoroughly. Alternatively, if specified, small undulations may be rectified with plaster of Paris.

Where it is expected that the dampness may find its way from the surrounding walls, the same shall also be effectively damp-proofed upto atleast 150 mm above the level of the sub-floor and the damp proof treatment below the floor shall be extended over the walls.

LAYING AND FIXING

Prior to laying, the flooring tiles / rolls / sheets shall be brought to the temperature of the area in which it is to be laid by stacking in a suitable manner within or near the laying area for a period of about 24 hours.
Where air-conditioning is installed, the flooring shall not be laid on the sub-floor until the conditioning units have been in operation for at least seven days. During this period the temperature shall neither fall below 20OC nor exceed 300C. These conditions shall be maintained during laying and for 48 hours, thereafter.

Before commencing the laying operations, the sub-floor shall be examined for evenness and dryness. The sub-floor shall then be cleaned with a dry cloth. The PVC flooring shall not be laid on a sub-floor unless the sub-floor is perfectly dry. Dryness of the sub-floor shall be tested conforming to relevant IS codes and manufacturers recommendations as directed by the PMC.

The layout of the PVC flooring on the sub-floor to be covered should be marked with guidelines. The PVC flooring shall be first laid for trial, without using the adhesive, according to the required layout.

The adhesive shall be applied by using a notched trowel to the sub-floor and to the back side of the PVC sheet tile flooring. When set sufficiently for laying, the adhesive will be sticky to touch, but will not mark the fingers. In general,
the adhesive will require about half an hour for setting. It should not be left after setting for too long a period as the adhesive properties will be lost owing to dust films and other causes.

Care should be taken while laying the flooring under high humidity conditions so that condensation does not take place of the adhesive. It is preferable to avoid laying under high humidity conditions.

The area of adhesive to be spread at one time on the sub-floor depends entirely upon local circumstances. In case of a small room, adhesive may be spread over the entire area but relatively small areas of tiles/sheets flooring should be treated in a larger room.

When the adhesive is just tack free the PVC flooring sheet shall be carefully taken and placed in position from one end onwards slowly so that the air will be completely squeezed out between the sheet and the background surface. After laying the sheet in position, it shall be pressed with suitable roller weighing about 5 kg to develop proper contract with the sub-floor. The next sheet with its back side applied with the adhesive shall be `[laid edge to edge] with the sheet already laid and fixed in exactly the same manner as the first sheet was fixed. The sheets shall be laid edge to edge so that there is minimum gap between joints.
The alignment should be checked after laying of each row of sheet is completed. If the alignment is not perfect, the sheets may be trimmed by using a straight edge.

The tiles shall be fixed in exactly the same manner as for the sheets. It is preferable to start laying of the tiles from the centre of the area. Care should be taken that the tiles are laid close to each other with minimum gap between joints. The tiles should always be lowered in position and pressed firmly on to the adhesive. Care should be taken not to slide them as this may result in adhesive being squeeze up between the joints. PVC tiles after laying shall be rolled with a light wooden roller weighing about 5 kg to ensure full contact with the under layer. Any undulations noticed on the PVC surface shall be rectified by removing and relaying the tiles after thorough cleaning of the underside of the affected tiles. The adhesives applied earlier in such places shall be thoroughly removed by using proper solvents and the surface shall be cleaned to remove the traces of solvents used. Work should be constantly checked against guidelines in order to ensure that all the four edges of adjacent tiles meet accurately.

Any adhesive which may squeeze up between sheets or tiles should be wiped off immediately with a wet cloth before the adhesive hardens. If, by chance, adhesive dries up and hardens on the surface of

the sheet or tile, it should be removed with a suitable solvent. A solution of one part of commercial butyleacetate and three parts of turpentine oil is a suitable solvent for the purpose.
A minimum period of 24 hours shall be given after laying the flooring for developing proper bond of the adhesive. During this period, the flooring shall not be put to service. It is preferable to lay the PVC flooring after completion of plastering, painting and other PMCorative finish works so as to avoid any accidental damage to the flooring.

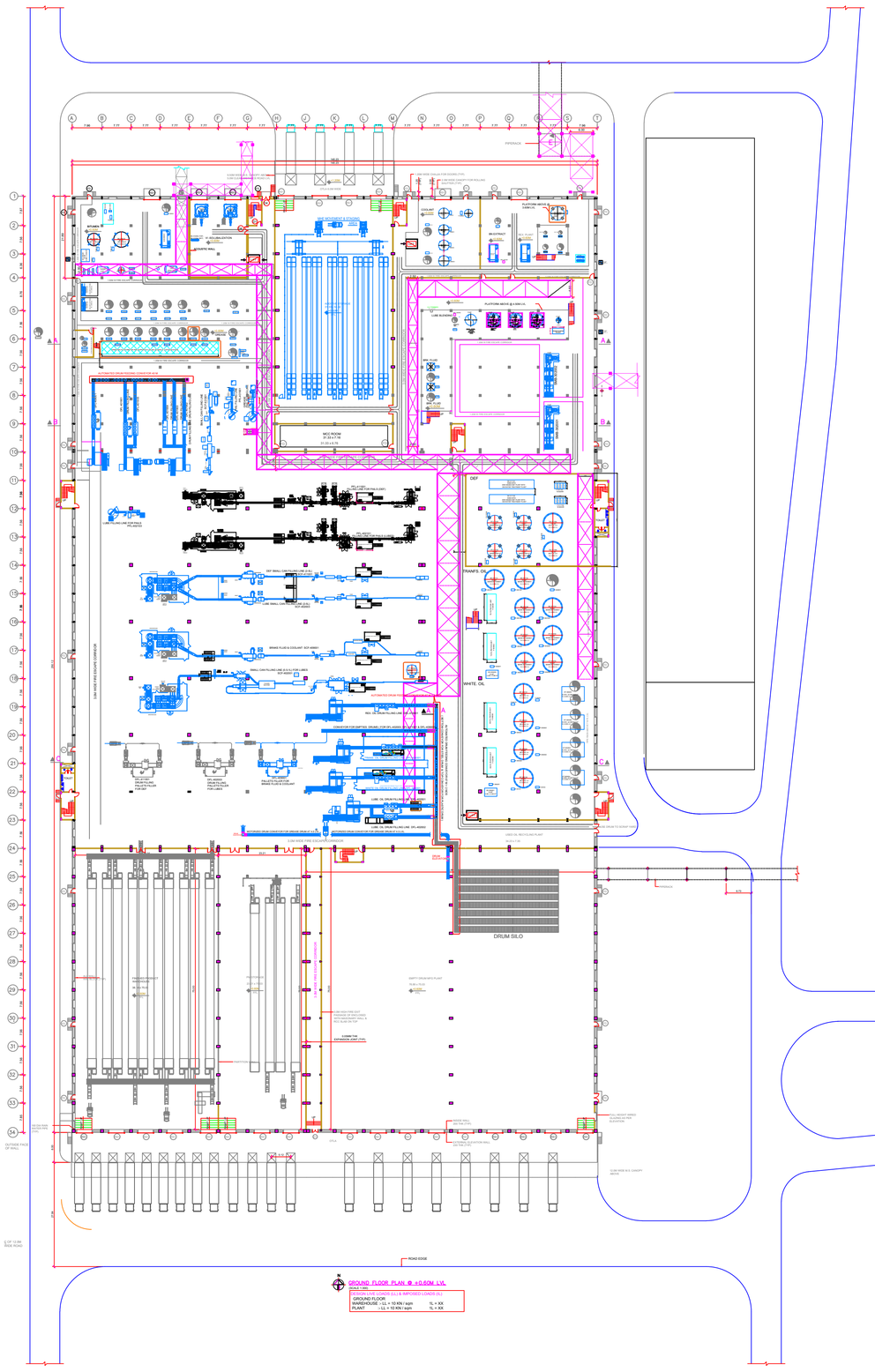
When the flooring has been securely, fixed, it shall be cleaned with a wet cloth soaked in warm soap solution (two spoons of soap in 5 litres of warm water).
When the edges of the PVC sheets or tiles are exposed, as for example, in doorways and on stair treads, it is important to provide protection against damage of flooring materials. Metallic edge strips may be used and should be securely fastened to the sub floor to protect edges of the flooring.

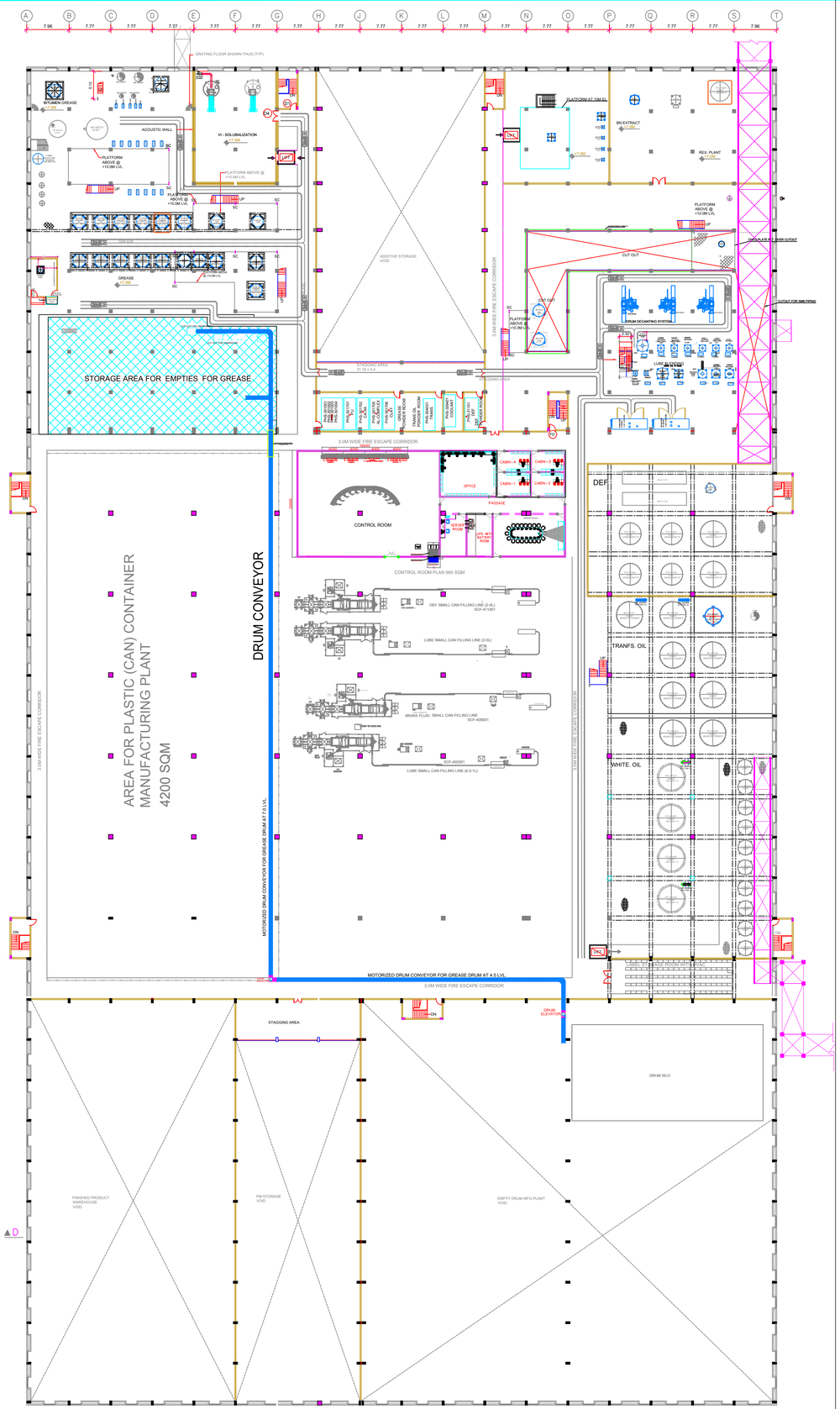
PRECAUTION FOR MAINTENANCE

PVC flooring subject to normal usage may be kept clean by mopping with soap solution using a clean damp cloth. Water shall not be poured on the PVC flooring for cleaning purpose as the water may tend to seep through the joints and cause the adhesive to fail. To maintain a good wearing surface a good appearance, the flooring may be periodically polished. When polish is applied frequently, a thick layer builds up which collects dirt and dust and is tacky to walk on.

If the traffic is light, the floor shall be given frequent brushing regular polishing by an application of new polish every 4 to 6 weeks. Under moderate traffic conditions the floor shall be given an occasional wash with a wet mop but no detergents shall be used so that the polish is not removed.

Application of polish may be done every one to three weeks. PVC flooring should not be over waxed. When this condition develops, the coatings should be cleared off with white spirit or paraffin and a light even coat of polish applied. When the PVC flooring has been polished, it will remain bright for a considerable period if dry mop is applied each day. It is this daily `dry polish` that maintains the glossy surface. After exceptionally heavy traffic PVC flooring should be swept with a hair broom, rubbed with a mop or cloth frequently rinsed in clean water and finally rubbed dry.


MCC ROOM
VOID
DRUM SILO
WHITE OIL
TRANFS. OIL
DRUM FILLING LINE (4.0 L/L FOR LUBES)
SCF 400/ISO
LUBE OIL DRUM FILLING LINE (IPL-400/ISO
BRAKE FLUID & COOLANT SCF 400/ISO
LUBE OIL DRUM FILLING LINE (IPL-400/ISO
DRUM FILLING LINE (4.0 L/L FOR LUBES)
SCF 400/ISO
BRAKE FLUID & COOLANT
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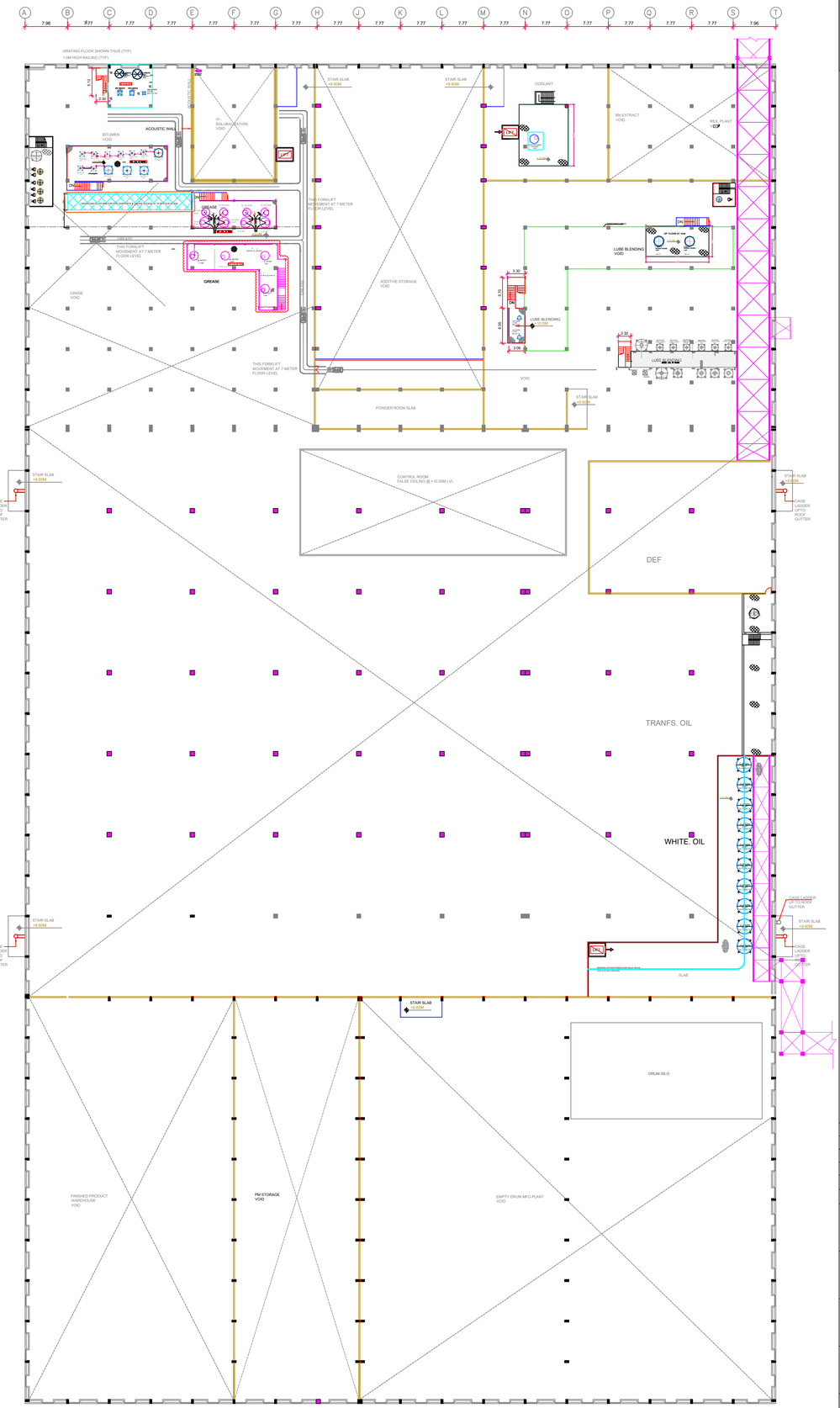


FIRST FLOOR PLAN @ +7.0M
(SCALE 1:300)

DESIGN LIVE LOADS (LL) & IMPOSED LOADS (IL)

FIRST FLOOR RCC SLAB
PLANT AREA :- LL = 5 KN / sqm IL = XX

FIRST FLOOR STEEL



PLAN BETWEEN FIRST FLOOR & ROOF FLOOR PLAN @ (+10.0M LVL)
(SCALE 1:300)



REV.	DATE	DESCRIPTION	DRW'N	CHK'D	APP'D
E	14.08.21	REVIEWED AND REISSUED FOR TENDER	NBM	SBS	DR
D	22.07.21	ISSUED FOR REVIEW	NBM	SBS	DR
C	12.12.20	ISSUED FOR REVIEW	NBM	SBS	DR
B	30.09.20	ISSUED FOR REVIEW	NBM	SBS	DR
A	30.08.20	ISSUED FOR REVIEW	NBM	SBS	DR

INDIAN OIL CORPORATION LIMITED
MUMBAI–400051
INDIA

PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT TITLE:
GRASS ROOT LUBE COMPLEX

DRAWING TITLE: PROCESS PLANT BLDG.
FIRST FLOOR PLAN @ +7.0M
PLAN BETWEEN GR. FLOOR & FIRST FLOOR PLAN @ (+3.6M & 5.0M LVL)

SCALE	DRAWING No.	SIZE	REVISION
1:100	20–ORG2N–BE–011	A1	E
DATE 31/08/2020			

PEC LTD.
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DISTRIBUTION	
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SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1

PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

TECHNICAL SPECIFICATION FOR EXHAUST BLOWER

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

	Total No. of Sheets	Page
	9 (Incl, cover sheet)	Page 1 of 9

REV.	DESCRIPTION	PREP BY	CHK BY	TECH APP	PROJ APP	DATE
A	ISSUED FOR TENDER	HTO	SBS	DR	KGK	17/07/21

TITLE: TECHNICAL SPECIFICATION FOR EXHAUST BLOWER

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV.
0RG2N0410000	0RG2N0410000	HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)	90-0RG2N-IS-042	A4	A

1.0. GENERAL

Exhaust system is proposed for VI Dissolution vessel VD-302101 to evacuate the fumes generated during the POLYMER Dissolution process.

1.1. SCOPE OF WORK

Design, detailed engineering, procurement, supply, fabrication, assembly, inspection, testing, including packing, forwarding, loading at shop & unloading at site, installation, pre-commissioning, running trials and commissioning of Exhaust Blower with associated ducting for VII Dissolver unit

1.2. SCOPE OF SUPPLY

Exhaust blower with drive motor of suitable capacity – including all the mechanical, electrical and instrumentation items as per the details given in the specifications

1.3. OWNERS (IOC) TO FURNISH:

The owner shall furnish the following:

- Site preparation
- Supporting structure for the Exhaust Blower system
- Instrument air connection
- Power supply connection to the unit

1.4. SITE DATA

- **Utilities:**
  - Motor Circuit: 415V /3 ph /50 Hz.

- ❖ Control Circuit: 240V /1 ph /50 Hz.
- ❖ Pneumatic: 4 kg/cm2 (g), compressed air at -40°C dew point.

- ➢ **Site Elevation:** < 100 Mtr above Sea level.
- ➢ **Ambient Temperature:** 15°C min. to 45 °C max.
- ➢ **Area Electrical Classification:** SAFE, Non-flameproof.

1.5. CODES

The Exhaust system shall be designed in accordance with the latest edition of following specifications:

IS 3103	:	Code of practice for Industrial Ventilation
IS 4894	:	Specifications for Centrifugal fans

For provisions not covered by the above codes & standards, applicable engineering practices & norms shall govern.

1.6. TECHNICAL SPECIFICATIONS

This specification and accompanying data sheet constitute the minimum requirement under which the Vendor/ Manufacturer shall design, manufacture, supply, erect, test and commission the Exhaust Blower Unit complete with all associated duct work &accessories/ auxiliaries.

1.6.1 Exhaust Blower:

Exhaust blower shall be direct driven, axial flow type fan with the motor remote from the Air stream so as to enable fan to handle steam, fumes or gases. Fan shall be suitable for air & oil contaminated vapors at suction temperatures as high as 80 deg C with suitable Materials of construction. Exhaust blower shall be complete with motor, with impeller and should be a

direct driven fan and capable of developing static discharge pressure of 25-30mm Water gauge (VTC actual pressure drop before finalizing the order) and shall be designed for continuous duty during the Polymer charging sequence (duration - 3 hrs approx. per batch). The motor shall be suitable for indoor application. The motor should have maintenance free bearings. The direction of rotation, discharge direction shall be marked on the fan assembly. The fans shall be suitably installed as per the locations shown in the drawings with direction of air flow towards outside of the building to a safe location. The impeller of the fan shall be forward / backward curved (depending on the static pressure requirement). The whole fan assembly shall be statically and dynamically balanced. Ball bearings should be completely maintenance free. The bearing lubricant should be suitable for application at maximum ambient temperature and shall be with a life expectancy of minimum of 40000 hours. The fan motor, which will be direct driven, should have a built-in thermal contact. At the critical high temperature point the thermal contact shall open and break the power supply to the fan. The fan motor should have class B insulation and should be rated for IP 54.

1.6.2 Performance Data:

Fan shall be selected for the lowest operating noise level. Capacity ratings, power consumption, with operating points clearly indicated, shall be submitted and verified at the time of testing and commissioning of the installation.

1.6.3 Testing:

Capacity of fan shall be measured by an anemometer. Measured air flow capacities shall conform to the specified capacities and quoted ratings. Power consumption shall be computed from measurements of incoming voltage and input current.

1.7. VENDOR DATA REQUIREMENT

1.7.1 WITH PROPOSAL

- Complete Technical details with salient features of the machines & accessories offered with leaflets, brochures, catalogs, photographs etc.

- Comments on our technical specifications, terms & conditions clearly indicating acceptance or deviations if any.

- The Bidder shall provide with his bid an overall" standard" inspection and quality assurance plan for all major equipment parts, including the following:
  - Test and checks on major and sub-components.
  - Certifying documents for raw materials, material tests and inspections, including certification level.
  - Acceptance criteria.
  - Inspection levels (witnessed/unwitnessed/documentation review, etc.)

- All data sheets, drawings and technical documentation for the package unit components – duly filled in with vendor data as required;

- Control schematic and control narrative

- List of recommended commissioning spares included in offer.

- List of Vendor's standard spares parts for 'two' years normal operation. This list shall be made separately for each item including auxiliaries and drivers in the form of table and show: -Part name, description and number. -Quantity installed in one unit -Quantity

recommended per unit for two years normal operation.

- An itemised list of special tools included in the offer.
- Any start up, shut down or operating restrictions required to protect the integrity of the equipment.
- A specific statement that the scope of supply, the offered equipment, system and all its components are in strict accordance with data sheets, job specifications, this specification and all documents, except for specific deviations as listed in the proposal.
- Utility requirements (such as power, lubricating oil and any other utility requirements) shall be clearly stated indicating qualities required, specifications and supply and return conditions.
- A statement of the rate for furnishing a competent erection / commissioning supervisor plus an estimate of the length of time the supervisor's services might be required as well as the desired skilled / unskilled labour and lifting tackles.
- Guarantees and warranties shall be as specified in the Purchaser's requisition.
- Duly filled in experience record Performa giving details of Models, operating parameters, name of purchaser, consultant, date of commissioning and number of operating hours completed.

2.0 DATASHEET FOR EXHAUST BLOWER

Sr. No.	ITEM	DESCRIPTION
1	**TAG NO.**	**B-312101**
2	**QUANTITY NO.**	1 no Exhaust Blower system with Drive Motor of suitable rating, Inlet & Outlet Dampers, Anti-vibration pads complete and in all respect;
3	**DUTY**	For removal of exhaust fumes / oil contaminated gases from vessel VD-312101
4	**FLUID**	Air & Oil Contaminated vapours
5	**TYPE**	Direct driven, Axial flow fan with the Motor remote from the Air stream; so as to enable fan to handle steam, fumes or gases,
6	**SUCTION PRESSURE**	Atmospheric
7	**SUCTION TEMPERATURE**	80 Deg C max
8	**FAN STATIC DISCHARGE PRESSURE**	40 mm Water gauge at Operating Temp;

9	**CLASS OF SERVICE**	Continuous during charging of polymer bales (~ 3 hrs) per each batch; No. of Batches per day = 3-4;
10	**NO. OF BATCHES/DAY**	Normal – 3; Design Max - 4
11	**CAPACITY**	2,000 m3/hr (FAD);
12	**FAN IMPELLER DIAMETER**	Vendor to specify * mm
13	**SUCTION DUCT SIZE**	200 NB shall be provided sloping upwards and Drain line of 25NB with a Ball valve shall be provided at a convenient point before the Exhaust Blower – so as to facilitate drainage of any condensate.
14	**FAN SUCTION**	Vendor to specify * mm
15	**FAN OUTLET**	Vendor to specify * mm
16	**FAN MOC**	**1)** Casing: PP + FRP or MS with Hot dip galvanized; **2)** Impeller: SS 304 **3)** Pedestal: MS **4)** Hub: SS 304 casting
17	**SUCTION AND DISCHARGE DAMPER**	Manual damper suitable for the blower inlet and outlet duct sizes to be provided
18	**DAMPER MOC**	SS 304, 4 mm thick plates
19	**SOUND LEVEL**	85 DBA max at 1 mtr distance or better
20	**DRIVE**	* HP (* kw) / * rpm, TEFC motor, foot

		mounted, IP-55, 3-ph, 50Hz, 415±6%V, AC supply; Approved Make: BBL / Crompton greaves/ Siemens / ABB or equivalent;
21	**DISCHARGE**	Right hand side Top Horizontal discharge viewed from suction
22	**ANTI VIBRATION PAD**	* No's Anti Vibration pad to be provided for base of the blower
23	**ACCESSORIES**	Blower Stand with Stainless steel mounting bolts and epoxy painted.
24	**BALANCING**	The impeller with motor shall be statically and dynamically balanced as per IS: 11342 Standard and balancing data should be provided at the time of supply.
25	**DIMENSIONS**	Vendor to specify * mm
26	**DIRECTION/ ROTATION**	Vendor to specify
27	**SPECIAL CONDITIONS**	**1)** Operation of the Exhaust Blower shall be integrated with operation of VI Dissolver unit VD-312101 and shall be controlled ON-OFF by the main PLC system through the RIO panel to be mounted near VD-312101; Vendor to facilitate this seamless integration for smooth operation of the unit;

PEC LTD.
21 Shipyard Road, Singapore 628 144
Tel: 6268 9788 Fax: 6268 9488
URL: http://www.peceng.com

TITLE:
LINE LIST - WITH STRESS CRITICAL LINES

Project:

GRASSROOT LUBE COMPLEX PROJECT AT CHENNAI

Contractor:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

DISTRIBUTION	
INTERNAL	
HEAD OFFICE	
SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1
PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

	Total No. of Sheets	Page
	3 (Incl, cover sheet)	Page 1 of 3

REV.	DESCRIPTION	PREP'D	CHK'D	APP'D	DATE
D	FOR APPROVAL	HTO	SDU	DR	13-08-2021
C	FOR APPROVAL	HTO	SDU	DR	12-05-2021
B	FOR APPROVAL	HTO	SDU	DR	13-01-2021
A	FOR APPROVAL	HTO	SDU	DR	14-10-2020

TITLE:
PIPING MATERIAL SPECIFICATION

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV. D
			90-0RG2N-LL-001		

STRESS CRITICAL PIPING

1.1 Piping Systems

Piping systems are subjected to various loadings such as internal pressure, weight of pipe, valves, fittings, insulation and process fluids flow velocity, wind loads, thermal expansion /contraction etc. These loadings must be taken into account to ensure that the piping system meets the requirements of ASME B31.3 & ASME Section VIII Div 2 and does not impose excessive loadings on equipment nozzles or the structure.

1.2 Thermal Expansion / Contraction

A. Expansion / contraction of piping should be accommodated wherever possible by the natural flexibility of the pipe work. If necessary, the route of the piping should be modified, or expansion loops should be incorporated, to obtain sufficient flexibility. The use of expansion bellows is to be avoided and may only be considered with the express permission of the COMPANY / ENGINEER.

B.Expansion computations shall be based on minimum and maximum operating temperatures. Where the minimum design temperature is higher than the minimum ambient temperature, the minimum ambient temperature shall be used.

1.3 Criticality Category

The line is deemed as stress critical based upon following factors

a) Maximum operating temperature for the line.

b) The line connected to strain sensitive equipment.

c) Line Size and its wall thickness.

1.4 Formal pipe stress analyses are required to be performed and documented for lines falling any one or more category:

a) Alloy piping 3" and larger

b) Piping 3" TO 12" with Maximum operating temperature exceeding **150°C**

c) Lines subject to dynamic and cyclic loading such as PSV,Hot Thermic Oil & Cold Thermic Oil connections to vessel

d) Lines classified as "lethal service"

e) All sizes of non-metallic piping

f) Lines connected to strain sensitive equipment (glass lined vessels, graphite exchangers, plastic or FRP equipment, etc.)

1.5 The following means of analysis are acceptable:

a) Statement that a line is duplicate of a previous line with a successful operating record.

b) ASME B31.3 Section 319.4.1 Equation 16 where applicable and with no more than two points of fixation and no intermediate restraints.

c) CAESAR II analysis or other computer analysis.

1.6 No formal stress analysis shall be submitted for such lines as below:

The lines which possess adequate flexibility by virtue of small size, near ambient temperature (not connected to strain sensitive equipment) or very flexible configuration can be reviewed / inspected visually.

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1	80	DM	002	01	01	D5N	-	-	WATER	L	P-309001	TW-309001	6	8	40	60	*	20	1000	44	10	60	Water	*	0	10Kg/cm2 PR.RATING/ TO SUIT	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
2	80	DM	002	01	02	D5N	-	-	WATER	L	TW-309001	P-309001	0.6	1	40	60	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
3	80	DM	002	01	03	D5N	-	-	WATER	L	TW-309001	P-309001	0.6	1	40	60	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
4	50	DM	002	01	04	D5N	-	-	WATER	L	P-309001	AD-309003	6	8	40	60	*	20	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
5	80	DM	002	01	05	B4C	-	-	WATER	L	AT-309002	P-309002	0.6	1	40	60	*	*	1000	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
6	50	DM	002	01	06	B4C	-	-	WATER	L	P-309002	BK-309102	6	8	40	60	*	20	1000	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
7	50	HOR(M)	002	02	14	A1A	IH	-	HOT OIL	L	HEADER	BK-309101	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
8	50	HOS(M)	002	02	15	A1A	IH	-	HOT OIL	L	HEADER	BK-309101	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
9	50	DM	002	02	03	D5N	-	-	WATER	L	TW-309001	BK-309101	ATM	FULL OF LIQUID	AMB	60	*	*	1000	44	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
10	50	AD	002	02	02	B4C	-	-	ADDITIVE	L	AT-309002	BK-309101	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	1.5	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
11	25	PA	002	02	04	G6C	-	-	PLANT AIR	V	*	BK-309101	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
12	50	AD	002	02	01	A1A	-	-	ADDITIVE	L	AT-200004/06	BK-309101	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
13	40	AD	002	02	07	B4C	-	-	ADDITIVE	L	AD-309003	BK-309101	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
14	50	AD	002	02	05	A1A	-	-	ADDITIVE	L	BK-309101	BK-309102	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	0	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
15	50	CL	002	02	06	B4C	-	-	COOLANT	L	P-309101	BK-309101	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
16	50	AD	002	02	08	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	BK-309101	ATM	FULL OF LIQUID	AMB	60	*	3	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
17	80	CL	002	02	09	B4C	-	-	COOLANT	L	BK-309101	P-309101	0.6	1.2	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
18	50	CL	002	02	10	B4C	-	-	COOLANT	L	P-309101	FP-309203	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
19	50	CL	002	02	11	B4C	-	-	COOLANT	L	P-309101	FP-309202	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
20	50	CL	002	02	12	B4C	-	-	COOLANT	L	P-309101	FP-309201	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
21	50	CL	002	02	13	B4C	-	-	COOLANT	L	P-309101	AT-309002	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
22	80	HOS(M)	002	02	16	A1A	IH	-	HOT OIL	L	HEADER	AD-309003	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
23	80	HOR(M)	002	02	17	A1A	IH	-	HOT OIL	L	AD-309003	HEADER	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
24	50	AD	002	03	01	A1A	-	-	ADDITIVE	L	TAPPING FROM AT	BK-309102	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
25	25	PA	002	03	02	G6C	-	-	PLANT AIR	V	*	BK-309102	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
26	50	HOS(M)	002	03	03	A1A	IH	-	HOT OIL	L	HEADER	BK-309102	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
27	50	HOR(M)	002	03	04	A1A	IH	-	HOT OIL	L	HEADER	BK-309102	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
28	50	CL	002	03	05	B4C	-	-	COOLANT	L	P-309102	BK-309102	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
29	80	CL	002	03	06	B4C	-	-	COOLANT	L	BK-309102	P-309102	0.6	1.2	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
30	50	CL	002	03	07	B4C	-	-	COOLANT	L	P-309102	FP-309203	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
31	50	CL	002	03	08	B4C	-	-	COOLANT	L	P-309102	FP-309202	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
32	50	CL	002	03	09	B4C	-	-	COOLANT	L	P-309102	FP-309201	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
33	50	CL	002	03	10	B4C	-	-	COOLANT	L	P-309102	AT-309002	6	8	AMB	60	*	20	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
34	50	AD	002	03	11	B4C	-	-	ADDITIVE	L	TW-309001	BK-309102	6	8	AMB	60	*	*	900	44	15	150	WATER	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
35	50	AD	002	04	01	A1A	-	-	ADDITIVE	L	AT-200004/06 BYP	MB-309303	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
36	50	AD	002	04	02	A1A	-	-	ADDITIVE	L	AT-200004/06 BYP	AD-309003	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
37	40	AD	002	04	03	B4C	-	-	ADDITIVE	L	P-309003	AD-309003	6	8	AMB	60	*	6	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
38	50	AD	002	04	04	A1A	-	-	ADDITIVE	L	AD-309003	P-309003	0.6	1.2	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
39	40	AD	002	04	05	B4C	-	-	ADDITIVE	L	P-309003	BK-309101	6	8	AMB	60	*	6	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
40	50	COS	002	04	06	A1A	IH	-	COLD OIL	L	HEADER	AD-309003	0.6	1.2	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
41	50	COR	002	04	07	A1A	IH	-	COLD OIL	L	HEADER	AD-309003	0.6	1.2	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
42	50	AD	002	05	01	A1A	-	-	ADDITIVE	L	TAPPING FROM AT	MB-309303	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
43	50	AD	002	05	02	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	MB-309303	ATM	FULL OF LIQUID	AMB	60	*	3	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
44	40	AD	002	05	03	A1A	-	-	ADDITIVE	L	P-309303	MB-309303	6	8	AMB	60	*	6	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
45	50	AD	002	05	04	A1A	-	-	ADDITIVE	L	MB-309303	P-309003	0.6	1.2	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
46	40	AD	002	05	05	A1A	-	-	ADDITIVE	L	MB-309303	BK-309102	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
47	40	AD	002	05	06	A1A	-	-	ADDITIVE	L	MB-309303	BK-309101	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
49	50	PD	002	06	02	B4C	-	-	POWDER	S	SL-309401	SC-309408	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
51	80	PD	002	06	15	B4C	-	-	POWDER	S	SC-309408	AD-309003	ATM	*	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
53	50	AD	002	07	01	A1A	-	-	ADDITIVE	L	TAPPING FROM AT	FP-309201	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
55	50	CL	002	07	02	B4C	-	-	COOLANT	L	P-309201	FP-309201	6	8	AMB	60	*	24	900	44	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
57	80	CL	002	07	03	B4C	-	-	COOLANT	L	FP-309201	P-309201	0.6	1.2	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
59	50	CL	002	07	04	B4C	-	-	COOLANT	L	F-309201A	MANIFOLD	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
61	80	CL	002	07	05	B4C	-	-	COOLANT	L	P-309201	BULK FILLING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
63	50	CL	002	07	06	B4C	-	-	COOLANT	L	P-309201	DRUM FILLING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
65	50	CL	002	07	07	B4C	-	-	COOLANT	L	P-309201	SMALL CAN FI	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
67	25	CL	002	07	08	B4C	-	-	COOLANT	L	P-309201	RINSING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
69	25	AD	002	07	09	A1A	-	-	ADDITIVE	L	TAPPING FROM AT	FP-309201	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
71	50	AD	002	08	01	A1A	-	-	ADDITIVE	L	TAPPING FROM AT	FP-309202	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
73	25	AD	002	08	02	A1A	-	-	ADDITIVE	L	BYPASS LINE TAPP	FP-309202	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
75	50	CL	002	08	03	B4C	-	-	COOLANT	L	P-309202	FP-309202	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
77	80	CL	002	08	04	B4C	-	-	COOLANT	L	FP-309202	P-309202	ATM	*	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
79	50	CL	002	08	05	B4C	-	-	COOLANT	L	F-309202	MANIFOLD	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
81	80	CL	002	08	06	B4C	-	-	COOLANT	L	F-309202	BULK FILLING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
83	50	CL	002	08	07	B4C	-	-	COOLANT	L	F-309202	SMALL CAN FI	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
85	50	CL	002	08	08	B4C	-	-	COOLANT	L	F-309202	DRUM FILLING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
87	25	CL	002	08	09	B4C	-	-	COOLANT	L	F-309202	RINSING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
89	50	AD	002	09	01	A1A	-	-	ADDITIVE	L	TAPPING FROM AT	FP-309203	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
91	25	AD	002	09	02	A1A	-	-	ADDITIVE	L	BYPASS LINE TAPP	FP-309203	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
93	50	CL	002	09	03	B4C	-	-	COOLANT	L	P-309203	FP-309203	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
95	80	CL	002	09	04	B4C	-	-	COOLANT	L	FP-309203	P-309203	0.6	1.2	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
97	50	CL	002	09	05	B4C	-	-	COOLANT	L	F-309203	MANIFOLD	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
99	80	CL	002	09	06	B4C	-	-	COOLANT	L	P-309203	BULK FILLING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
101	50	CL	002	09	07	B4C	-	-	COOLANT	L	P-309203	SMALL CAN FI	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
103	50	CL	002	09	08	B4C	-	-	COOLANT	L	P-309203	DRUM FILLING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
105	25	CL	002	09	09	B4C	-	-	COOLANT	L	P-309203	RINSING	6	8	AMB	60	*	24	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
107	50	AD	002	09	10	A1A	-	-	ADDITIVE	L	AT-200004/06	MB-309305	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
109	50	AD	002	09	11	A1A	-	-	ADDITIVE	L	AT-200004/06	MB-309304	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
111	50	AD	002	09	12	A1A	-	-	ADDITIVE	L	AT-200004/06	MB-309302	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
113	50	AD	002	09	13	A1A	-	-	ADDITIVE	L	AT-200004/06	MB-309301	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		

S.NO		LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
	PIPE SIZE (DN)	FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
115	50	PD	002	10	01	B4C	-	-	POWDER	L	MANUALLY	MB-309301	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
117	25	AD	002	10	02	A1A	-	-	ADDITIVE	L	P-309301	MB-309301	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
119	40	AD	002	10	03	A1A	-	-	ADDITIVE	L	MB-309301	P-309301	3	5	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
121	25	AD	002	10	04	A1A	-	-	ADDITIVE	L	P-309301	FP-309203	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
123	25	AD	002	10	05	A1A	-	-	ADDITIVE	L	P-309301	FP-309202	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
106	25	AD	002	10	06	A1A	-	-	ADDITIVE	L	P-309301	FP-309201	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
107	50	PD	002	11	01	B4C	-	-	POWDER	L	MANUALLY	MB-309302	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
108	25	AD	002	11	02	A1A	-	-	ADDITIVE	L	P-309302	MB-309302	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
109	40	AD	002	11	03	A1A	-	-	ADDITIVE	L	MB-309302	P-309302	3	5	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
110	25	AD	002	11	04	A1A	-	-	ADDITIVE	L	P-309302	FP-309203	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
111	25	AD	002	11	05	A1A	-	-	ADDITIVE	L	P-309302	FP-309202	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
112	25	AD	002	11	06	A1A	-	-	ADDITIVE	L	P-309302	FP-309201	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
113	50	PD	002	12	01	B4C	-	-	POWDER	L	MANUALLY	MB-309304	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
114	25	AD	002	12	02	A1A	-	-	ADDITIVE	L	P-309304	MB-309304	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
115	40	AD	002	12	03	A1A	-	-	ADDITIVE	L	MB-309304	P-309304	3	5	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
116	25	AD	002	12	04	A1A	-	-	ADDITIVE	L	P-309304	FP-309203	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
117	25	AD	002	12	05	A1A	-	-	ADDITIVE	L	P-309304	FP-309202	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
118	25	AD	002	12	06	A1A	-	-	ADDITIVE	L	P-309304	FP-309201	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
119	50	PD	002	13	01	B4C	-	-	POWDER	L	MANUALLY	MB-309305	ATM	FULL OF LIQUID	AMB	60	*	*	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
120	25	AD	002	13	02	A1A	-	-	ADDITIVE	L	P-309305	MB-309305	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
121	40	AD	002	13	03	A1A	-	-	ADDITIVE	L	MB-309305	P-309305	3	5	AMB	60	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
122	25	AD	002	13	04	A1A	-	-	ADDITIVE	L	P-309305	FP-309203	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
123	25	AD	002	13	05	A1A	-	-	ADDITIVE	L	P-309305	FP-309202	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
124	25	AD	002	13	06	A1A	-	-	ADDITIVE	L	P-309305	FP-309201	5	7	AMB	60	*	1	900	44	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-002	H1	P&ID FOR COOLANT PLANT		
125	80	BO	005	01	01	A1A	-	-	BASE OIL	L	BYPASS LINE FROM	HE-304003	6	8	60	100	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
126	80	BO	005	01	02	A1A	IH	-	BASE OIL	L	HE-304003	BK-304104	6	8	60	100	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
127	80	BO	005	01	03	A1A	-	-	BASE OIL	L	BYPASS LINE FROM	HE-304002	6	8	60	100	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
128	80	BO	005	01	04	A1A	IH	-	BASE OIL	L	HE-304002	BK-304104	6	8	60	100	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
129	80	HOS(M)	005	01	05	A1A	IH	-	HOT OIL	L	HEADER	HE-304003	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
130	80	HOR(M)	005	01	06	A1A	IH	-	HOT OIL	L	HEADER	HE-304003	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
131	80	HOS(M)	005	01	07	A1A	IH	-	HOT OIL	L	HEADER	HE-304002	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
132	80	HOR(M)	005	01	08	A1A	IH	-	HOT OIL	L	HEADER	HE-304002	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
133	80	BO	005	01	10	A1A	-	-	BASE OIL	L	ST-100002	HE-304002	6	8	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
134	150	HOS(M)	005	01	09	A1A	IH	-	HOT OIL	L	HEADER	BK-304104	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
135	80	BO	005	02	01	A1A	IH	-	HOT OIL	L	HE-304003	AD-304001	6	8	AMB	60	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
136	80	BO	005	02	02	A1A	IH	-	HOT OIL	L	HE-304002	AD-304001	6	8	AMB	60	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
137	80	AD	005	02	03	A1A	-	-	ADDITIVE	L	AD-304001	P-304001	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
138	50	AD	005	02	04	A1A	-	-	ADDITIVE	L	BYPASS LINE FROM	P-304001	6	8	AMB	60	*	20	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
153	50	AD	005	02	05	A1A	-	-	ADDITIVE	L	P-304001	AD-304001	6	8	AMB	60	*	20	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
154	50	AD	005	02	06	A1A	-	-	ADDITIVE	L	P-304001	BYPASS LINE V	6	8	AMB	60	*	20	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
155	80	HOR(M)	005	02	07	A1A	IH	-	HOT OIL	L	HEADER	AD-304001	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
156	80	HOS(M)	005	02	08	A1A	IH	-	HOT OIL	L	HEADER	AD-304001	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
157	50	AD	005	02	09	A1A	-	-	ADDITIVE	L	P-304001	BK-304104	6	8	AMB	60	*	*	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
158	50	AD	005	02	10	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	BK-304104	6	8	AMB	60	*	20	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
159	50	TO	005	03	01	B4C	-	-	ANSFORMER C	L	SLOP TANK	BK-304101	6	8	AMB	60	*	*	900	15	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
160	50	AD	005	03	02	A1A	-	-	ADDITIVE	L	P-304001	BK-304101	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
161	80	HOS(M)	005	03	03	A1A	IH	-	HOT OIL	L	HEADER	BK-304101	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
162	80	HOR(M)	005	03	04	A1A	IH	-	HOT OIL	L	HEADER	BK-304101	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
163	80	BO	005	03	05	A1A	IH	-	BASE OIL	L	HE-304003	BK-304101	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
164	80	BO	005	03	06	A1A	IH	-	BASE OIL	L	HE-304002	BK-304101	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
165	100	TO	005	03	07	A1A	-	-	ANSFORMER C	L	BK-304101	P-304101	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
166	80	TO	005	03	08	A1A	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304101	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
167	100	TO	005	03	09	A1A	-	-	ANSFORMER C	L	P-304101	BK-304101	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
168	80	TO	005	03	10	A1A	-	-	ANSFORMER C	L	P-304101	F-304301	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
169	80	TO	005	03	11	A1A	-	-	ANSFORMER C	L	P-304101	F-304302	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
170	100	TO	005	03	12	A1A	-	-	ANSFORMER C	L	P-304101	BK-304101	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
171	50	TO	005	04	01	B4C	-	-	ANSFORMER C	L	SLOP TANK	BK-304102	6	8	AMB	60	*	*	900	15	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
172	50	AD	005	04	02	A1A	-	-	ADDITIVE	L	P-304001	BK-304102	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
173	80	HOS(M)	005	04	03	A1A	IH	-	HOT OIL	L	HEADER	BK-304102	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
174	80	HOR(M)	005	04	04	A1A	IH	-	HOT OIL	L	HEADER	BK-304102	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
175	80	BO	005	04	05	A1A	IH	-	BASE OIL	L	HE-304003	BK-304102	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
176	80	BO	005	04	06	A1A	IH	-	BASE OIL	L	HE-304002	BK-304102	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
177	100	TO	005	04	07	A1A	-	-	ANSFORMER C	L	BK-304102	P-304102	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
178	80	TO	005	04	08	A1A	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304102	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
179	100	TO	005	04	09	A1A	-	-	ANSFORMER C	L	P-304102	BK-304102	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
180	80	TO	005	04	10	A1A	-	-	ANSFORMER C	L	P-304102	F-304301	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
181	80	TO	005	04	11	A1A	-	-	ANSFORMER C	L	P-304102	F-304302	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
182	100	TO	005	04	12	A1A	-	-	ANSFORMER C	L	P-304102	BK-304102	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
183	50	TO	005	05	01	B4C	-	-	ANSFORMER C	L	SLOP TANK	BK-304103	0.6	1.2	AMB	60	*	*	900	15	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
184	50	AD	005	05	02	A1A	-	-	ADDITIVE	L	P-304001	BK-304103	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
185	80	BO	005	05	03	A1A	IH	-	BASE OIL	L	HE-304003	BK-304103	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
186	80	BO	005	05	04	A1A	IH	-	BASE OIL	L	HE-304002	BK-304103	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
187	80	HOS(M)	005	05	05	A1A	IH	-	HOT OIL	L	HEADER	BK-304103	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
188	80	HOR(M)	005	05	06	A1A	IH	-	HOT OIL	L	HEADER	BK-304103	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
189	100	TO	005	05	07	A1A	-	-	ANSFORMER C	L	BK-304103	P-304103	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
190	80	TO	005	05	08	A1A	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304103	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
191	100	TO	005	05	09	A1A	-	-	ANSFORMER C	L	P-304103	BK-304103	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
192	80	TO	005	05	10	A1A	-	-	ANSFORMER C	L	P-304103	F-304301	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
193	80	TO	005	05	11	A1A	-	-	ANSFORMER C	L	P-304103	F-304301	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
194	100	TO	005	05	12	A1A	-	-	ANSFORMER C	L	P-304103	BK-304103	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
195	80	HOR(M)	005	06	14	A1A	IH	-	HOT OIL	L	BK-304104	HEADER	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
196	80	HOS(M)	005	06	13	A1A	IH	-	HOT OIL	L	HEADER	BK-304104	0.6	1.2	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
197	100	TO	005	06	07	A1A	-	-	ANSFORMER C	L	BK-304104	P-304104	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
198	80	TO	005	06	08	B4C	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304104	6	8	AMB	60	*	50	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
199	100	TO	005	06	09	A1A	-	-	ANSFORMER C	L	P-304104	BK-304104	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
200	80	TO	005	06	10	A1A	-	-	ANSFORMER C	L	P-304104	F-304302	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
201	80	TO	005	06	11	A1A	-	-	ANSFORMER C	L	P-304104	F-304301	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
202	100	TO	005	06	12	A1A	-	-	ANSFORMER C	L	P-304104	BK-304104	6	8	AMB	60	*	50	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
203	150	HOR(M)	005	06	01	A1A	IH	-	HOT OIL	L	BK-304104	HEADER	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
204	80	TO	005	07	01	B4C	-	-	ANSFORMER C	L	F-304301	FP-304201	6	8	AMB	60	*	50	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
205	80	TO	005	07	02	B4C	-	-	ANSFORMER C	L	F-304301	FP-304202	6	8	AMB	60	*	50	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
206	80	TO	005	07	03	A1A	-	-	ANSFORMER C	L	FP-304201	P-304201	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
207	80	TO	005	07	04	A1A	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304201	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
208	80	TO	005	07	05	A1A	-	-	ANSFORMER C	L	BYPASS LINE WITH	FP-304201 TO	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
209	80	TO	005	07	06	A1A	-	-	ANSFORMER C	L	FP-304202	FP-304201 OU	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
210	80	TO	005	07	07	A1A	-	-	ANSFORMER C	L	P-304201	F-304301	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
211	80	TO	005	07	08	A1A	-	-	ANSFORMER C	L	P-304201	FILLING	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
212	80	TO	005	07	09	A1A	-	-	ANSFORMER C	L	P-304201	F-304302	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
213	80	TO	005	08	01	B4C	-	-	ANSFORMER C	L	F-304302	FP-304203	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
214	80	TO	005	08	02	B4C	-	-	ANSFORMER C	L	F-304302	FP-304204	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
215	80	TO	005	08	03	A1A	-	-	ANSFORMER C	L	FP-304203	P-304203	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
216	80	TO	005	08	04	A1A	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304203	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
217	80	TO	005	08	05	A1A	-	-	ANSFORMER C	L	BYPASS LINE WITH	FP-304203 TO	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
218	80	TO	005	08	06	A1A	-	-	ANSFORMER C	L	FP-304204	FP-304204 OU	0.6	1.2	AMB	60	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
219	80	TO	005	08	07	A1A	-	-	ANSFORMER C	L	P-304203	F-304302	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
220	80	TO	005	08	08	A1A	-	-	ANSFORMER C	L	P-304203	FILLING	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
221	80	TO	005	08	09	A1A	-	-	ANSFORMER C	L	P-304203	F-304301	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
222	80	TO	005	09	01	B4C	-	-	ANSFORMER C	L	F-304301	FP-304201	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
223	80	TO	005	09	02	B4C	-	-	ANSFORMER C	L	F-304301	BULK / DRUM	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
224	80	TO	005	09	29	A1A	-	-	ANSFORMER C	L	MANIFOLD	F-304301	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
225	80	TO	005	09	27	B4C	-	-	ANSFORMER C	L	F-304301	DRUM FILLING	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
226	80	TO	005	09	05	B4C	-	-	ANSFORMER C	L	F-304301	BULK FILLING	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
227	80	TO	005	09	30	A1A	-	-	ANSFORMER C	L	MANIFOLD	F-304302	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
228	50	HOR(M)	005	09	07	A1A	IH	-	HOT OIL	L	F-304301	*	6	8	160	180	*	30	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
229	50	COR	005	09	08	A1A	IH	-	COLD OIL	L	F-304301	*	6	8	AMB	25	*	30	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
230	80	TO	005	09	09	B4C	-	-	ANSFORMER C	L	F-304302	FP-304204	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
231	80	TO	005	09	10	B4C	-	-	ANSFORMER C	L	F-304302	BULK / DRUM	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
232	80	TO	005	09	12	B4C	-	-	ANSFORMER C	L	F-304302	BULK FILLING	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
233	80	TO	005	09	28	B4C	-	-	ANSFORMER C	L	MANIFOLD	DRUM FILLING	6	8	AMB	60	*	30	900	15	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
234	50	HOR(M)	005	09	15	A1A	IH	-	HOT OIL	L	F-304302	*	6	8	160	180	*	30	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
235	50	COR	005	09	16	A1A	IH	-	COLD OIL	L	F-304302	*	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
236	50	HOS(M)	005	09	17	A1A	IH	-	HOT OIL	L	*	F-304301	6	8	160	180	*	30	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
237	50	COS	005	09	18	A1A	IH	-	COLD OIL	L	*	F-304301	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
238	25	AC	005	09	19	G6C	-	-	OMPRESSED AI	V	*	F-304301	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
239	25	AC	005	09	20	G6C	-	-	OMPRESSED AI	V	*	F-304301	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
240	50	HOS(M)	005	09	21	A1A	IH	-	HOT OIL	L	*	F-304302	6	8	160	180	*	30	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
241	50	COS	005	09	22	A1A	IH	-	COLD OIL	L	*	F-304302	6	8	AMB	60	*	30	900	15	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
242	25	AC	005	09	23	G6C	-	-	OMPRESSED AI	V	*	F-304302	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
243	25	AC	005	09	24	G6C	-	-	OMPRESSED AI	V	*	F-304302	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
244	80	TO	005	10	01	B4C	-	-	ANSFORMER C	L	SO-304403	P-304403	0.6	1.2	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
245	50	TO	005	10	02	B4C	-	-	ANSFORMER C	L	P-304403	BK-304101	6	8	60	100	*	10	900	15	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
246	50	TO	005	10	03	B4C	-	-	ANSFORMER C	L	P-304403	SO-304403	6	8	60	100	*	10	900	15	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
247	50	TO	005	10	04	B4C	-	-	ANSFORMER C	L	BYPASS LINE FROM	P-304403	6	8	60	100	*	10	900	15	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
248	50	PD	005	11	01	B4C	-	-	POWDER	S	PHS-304001	SL-304001	*	*	*	*	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
249	150	PD	005	11	02	B4C	-	-	POWDER	S	SC-304001	AD-304001	*	*	*	*	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
250	25	IA	005	11	03	G6C	-	-	NSTRUMENT AI	V	*	PHS-304001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
251	25	PA	005	11	04	G6C	-	-	PLANT AIR	V	*	PHS-304001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-005	H1	P&ID FOR TRANSFORMER OIL PLANT		
252	80	BO	004	01	04	A1A	IH	IE	BASE OIL	L	LEG-4 BYPAAS LINE	PV-306301	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
253	80	BO	004	01	05	A1A	IH	IE	BASE OIL	L	LEG-4 (ILB-306301	AD-306001	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
254	50	BO	004	01	06	A1A	-	-	BASE OIL	L	LEG-2 (ILB-306301	AD-306001	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
255	50	BO	004	01	07	A1A	-	-	BASE OIL	L	LEG-2 BYPAAS LINE	PV-306301	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
256	80	BI	004	01	01	P2A	-	-	BITUMEN	L	HLDV-306301 TYPE	BK-306102	6	8	60	80	*	10	900	51000	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
257	25	CA	004	01	09	G6C	-	-	OMPRESSED AI	V	*	HLDV-306301	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
258	25	V	004	01	10	G6C	-	-	VAPOR	V	HLDV-306301 TYPE	DEMISTER	6	8	40	60	*	10	900	51000	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
259	25	CA	004	01	08	G6C	-	-	OMPRESSED AI	V	*	HLDV-306301	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
260	80	BI	004	01	02	A1A	-	-	BITUMEN	L	HLDV-306301 TYPE	SM-306002	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
261	80	BI	004	01	03	A1A	-	-	BITUMEN	L	SM-306002	HLDV-306301	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
262	25	IA	004	01	11	G6C	-	-	NSTRUMENT AI	V	*	SKID	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
263	25	PA	004	01	12	G6C	-	-	PLANT AIR	V	*	SKID	6	8	40	60	*	10	900	51000	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
264	80	BI	004	01	13	A1A	-	-	BITUMEN	L	P-306002	HLDV-306301	6	8	40	60	*	*	1.25	0.00001	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
265	80	BI	004	01	14	A1A	-	-	BITUMEN	L	BYPASS LINE OF P-	P-306002	6	8	120	180	*	10	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
266	80	HOS(H)	004	02	01	A1B	IH	-	BASE OIL	L	HEADER	PV-306301	6	8	200	220	*	*	900	51000	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
267	80	HOR(H)	004	02	02	A1B	IH	-	BASE OIL	L	HEADER	PV-306301	6	8	200	220	*	*	900	51000	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
268	80	AD	004	02	03	A1A	IH	IE	ADDITIVE	L	AD-306001	P-306001	0.6	1.2	120	180	*	*	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
269	50	AD	004	02	04	A1A	-	-	ADDITIVE	L	BYPASS LINE (P-30	P-306001	6	8	120	180	*	12	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
270	50	AD	004	02	05	A1A	IH	IE	ADDITIVE	L	P-306001	LEG-4 (ILB-306	6	8	120	180	*	12	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
271	50	HOS(H)	004	02	06	A1B	IH	-	HOT OIL	L	HEADER	AD-306001	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
272	50	HOR(H)	004	02	07	A1B	IH	-	HOT OIL	L	HEADER	AD-306001	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
273	50	AD	004	02	08	A1A	IH	IE	ADDITIVE	L	P-306001	AD-306001	6	8	120	180	*	12	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
274	50	AD	004	02	09	A1A	IH	IE	ADDITIVE	L	P-306001	OUTLET OF AD	6	8	120	180	*	12	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
275	80	COS	004	03	01	A1A	IH	-	COLD OIL	L	HEADER	HE-306102	6	8	25	10	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
276	80	COR	004	03	02	A1A	IH	-	COLD OIL	L	HEADER	HE-306102	6	8	25	10	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
277	80	HOS(H)	004	03	03	A1B	IH	-	HOT OIL	L	HEADER	BK-306101	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
278	80	COS	004	03	04	A1A	IH	-	COLD OIL	L	HEADER	BK-306101	6	8	25	10	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
279	80	BI	004	03	05	A1A	IH	IE	BITUMEN	L	P-306101	BK-306101	6	8	120	180	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
280	80	HOS(H)	004	03	06	A1B	IH	-	HOT OIL	L	HEADER	HE-306102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
281	80	COR	004	03	07	A1A	IH	-	HOT OIL	L	HEADER	BK-306101	6	8	120	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
282	80	HOR(H)	004	03	08	A1B	IH	-	HOT OIL	L	HEADER	BK-306101	6	8	200	220	*	*	900	600	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
283	80	HOR(H)	004	03	09	A1B	IH	-	HOT OIL	L	HEADER	BK-306101	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
284	100	BI	004	03	10	A1A	IH	IE	BITUMEN	L	BK-306101	P-306101	0.6	1.2	120	180	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
285	80	BI	004	03	11	A1A	IH	IE	BITUMEN	L	BYPASS FROM P-3	P-306101	6	8	120	180	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
286	80	BI	004	03	12	A1A	IH	IE	BITUMEN	L	P-306101	F-306101	6	8	60	90	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
287	50	BI	004	03	13	A1A	IH	IE	BITUMEN	L	F-306101	SO-306201	6	8	60	90	*	24	900	26000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
288	80	BI	004	03	14	A1A	IH	IE	BITUMEN	L	F-306101	DRUM FILLING	6	8	60	90	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
289	80	BI	004	03	15	A1A	-	-	BITUMEN	L	PIG HEADER TYPE	BK-306101	0.6	1.2	120	180	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
290	80	BI	004	04	02	A1A	IH	IE	BITUMEN	L	P-306102	BK-306102	6	8	120	180	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
291	80	BI	004	04	03	A1A	-	-	BITUMEN	L	PIG HEADER TYPE	BK-306102	0.6	1.2	120	180	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
292	80	COR	004	04	04	A1A	IH	-	COLD OIL	L	HEADER	BK-306102	6	8	25	10	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
293	80	HOR(H)	004	04	05	A1B	IH	-	HOT OIL	L	HEADER	BK-306102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
294	80	HOR(H)	004	04	06	A1B	IH	-	HOT OIL	L	HEADER	BK-306102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
295	100	BI	004	04	07	A1A	IH	IE	BITUMEN	L	BK-306102	P-306102	0.6	1.2	120	180	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
296	80	BI	004	04	08	A1A	IH	IE	BITUMEN	L	BYPASS FROM P-3	P-306102	6	8	120	180	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
297	80	BI	004	04	09	A1A	IH	IE	BITUMEN	L	P-306102	F-306102	6	8	60	90	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
298	50	BI	004	04	10	A1A	IH	IE	BITUMEN	L	F-306102	SO-306201	6	8	60	90	*	24	900	26000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
299	80	BI	004	04	11	A1A	IH	IE	BITUMEN	L	F-306102	DRUM FILLING	6	8	60	90	*	24	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
300	50	AD	004	04	01	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	P-306102	0.6	1.2	120	180	*	6	900	26000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
301	80	HOS(H)	004	04	14	A1B	IH	-	HOT OIL	L	HEADER	BK-306102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
302	80	COS	004	04	13	A1A	IH	-	COLD OIL	L	HEADER	BK-306102	6	8	25	10	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
303	80	HOS(H)	004	04	15	A1B	IH	-	HOT OIL	L	HEADER	BK-306102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
304	80	HOS(H)	004	04	12	A1B	IH	-	HOT OIL	L	HEADER	BK-306102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
305	25	CA	004	04	16	G6C	-	-	OMPRESSED AI	V	*	AIR OPERATED	6	8	40	60	*	6	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
306	50	AD	004	04	17	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	AIR OPERATED	6	8	60	90	*	6	900	26000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
307	80	V	004	05	01	A6C	-	-	VENT	V	HE-306102	SAFE LOCATIO	6	8	120	180	*	*	900	600	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
308	100	BN	004	06	01	A1A	-	-	BN	L	SO-306401	P-306401	ATM	FULL OF LIQUID	70	120	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
309	80	BN	004	06	02	A1A	-	-	BN	L	BYPASS LINE P-306	P-306401	6	8	70	120	*	10	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
310	80	BI	004	06	03	A1A	-	-	BITUMEN	L	P-306401	ILB-306301	6	8	70	120	*	10	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
311	80	BI	004	06	04	A1A	-	-	BITUMEN	L	P-306401	RUBEER EXTEI	6	8	70	120	*	10	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
312	80	BI	004	06	05	A1A	-	-	BITUMEN	L	P-306401	SO-306401	6	8	70	120	*	10	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
313	80	BN	004	06	06	A1A	-	-	BITUMEN	L	TYPE 9	SO-306401	ATM	FULL OF LIQUID	70	120	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
314	25	CA	004	06	07	G6C	-	-	OMPRESSED AI	V	*	SO-306401	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
315	25	V	004	06	08	A1A	-	-	VAPOR	V	SO-306401	DEMISTER	ATM	FULL OF LIQUID	70	120	*	*	900	26000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
316	80	BO	004	06	09	A1A	-	-	BASE OIL	L	WASTE OIL	SO-306401	ATM	FULL OF LIQUID	70	120	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
317	80	BO	004	06	10	A1A	-	-	BASE OIL	L	RECYCLING PLANT	SO-306401	ATM	FULL OF LIQUID	70	120	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
318	50	V	004	07	01	A1A	-	-	VAPOR	V	MB-306201	VENT	ATM	FULL OF LIQUID	70	120	*	*	900	870	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
319	40	AD	004	07	02	A1A	IH	-	ADDITIVE	L	P-306201	MB-306201	6	8	70	120	*	3	900	870	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
320	80	AD	004	07	03	A1A	IH	-	ADDITIVE	L	MB-306201	P-306201	0.6	1.2	70	120	*	*	900	870	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
321	40	AD	004	07	04	A1A	IH	-	ADDITIVE	L	P-306201	OUTGOING LI	6	8	70	120	*	3	900	870	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
322	40	AD	004	07	05	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-306201	0.6	1.2	70	120	*	3	900	870	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
323	40	AD	004	07	06	A1A	IH	-	ADDITIVE	L	P-306201	AD-306001	6	8	70	120	*	3	900	870	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
324	40	AD	004	07	07	A1A	IH	-	ADDITIVE	L	BYPASS LINE FROM	P-306201	6	8	70	120	*	3	900	870	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
325	80	HOR(H)	004	07	08	A1B	IH	-	HOT OIL	L	HEADER	MB-306201	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
326	80	HOS(H)	004	07	09	A1B	IH	-	HOT OIL	L	HEADER	MB-306201	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
327	50	V	004	08	01	A1A	-	-	VAPOR	V	MB-306202	SAFE LOCATIO	ATM	FULL OF LIQUID	70	120	*	*	900	51000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
328	40	AD	004	08	02	A1A	IH	-	ADDITIVE	L	P-306202	MB-306202	6	8	70	120	*	1.5	900	51000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
329	150	AD	004	08	03	A1A	IH	-	ADDITIVE	L	MB-306202	P-306202	0.6	1.2	70	120	*	*	900	51000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
330	80	AD	004	08	04	A1A	IH	-	ADDITIVE	L	P-306202	OUTGOING LI	6	8	70	120	*	1.5	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
331	40	AD	004	08	05	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-306202	0.6	1.2	70	120	*	*	900	51000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
332	80	AD	004	08	06	A1A	IH	-	ADDITIVE	L	P-306202	AD-306001	6	8	70	120	*	1.5	900	51000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
333	40	AD	004	08	09	A1A	IH	-	ADDITIVE	L	BYPASS LINE FROM	P-306202	6	8	70	120	*	1.5	900	51000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
334	50	HOR(H)	004	08	07	A1B	IH	-	HOT OIL	L	HEADER	MB-306202	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
335	50	HOS(H)	004	08	08	A1B	IH	-	HOT OIL	L	HEADER	MB-306202	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
336	50	V	004	09	01	A1A	-	-	VAPOR	V	MB-306203	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
337	25	AD	004	09	02	A1A	-	-	ADDITIVE	L	P-306203	MB-306203	6	8	60	100	*	0.5	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
338	40	AD	004	09	03	A1A	-	-	ADDITIVE	L	MB-306203	P-306203	0.6	1.2	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
339	25	AD	004	09	04	A1A	-	-	ADDITIVE	L	P-306203	OUTGOING LI	6	8	60	100	*	0.5	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
340	40	AD	004	09	05	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-306203	0.6	1.2	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
341	25	AD	004	09	06	A1A	-	-	ADDITIVE	L	P-306203	AD-306001	6	8	60	100	*	0.5	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
342	25	AD	004	09	07	A1A	-	-	ADDITIVE	L	BYPASS LINE FROM	P-306203	6	8	60	100	*	0.5	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
343	50	V	004	10	01	A1A	-	-	VAPOR	V	MB-306205	SAFE LOCATIO	ATM	FULL OF LIQUID	120	180	*	*	900	1155	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
344	50	AD	004	10	02	A1A	IH	-	ADDITIVE	L	P-306205	MB-306205	6	8	120	180	*	9	900	1155	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
345	100	AD	004	10	03	A1A	IH	-	ADDITIVE	L	MB-306205	P-306205	0.6	1.2	120	180	*	*	900	1155	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
346	50	AD	004	10	04	A1A	IH	-	ADDITIVE	L	P-306205	OUTGOING LI	6	8	120	180	*	9	900	1155	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
347	40	AD	004	10	05	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-306205	0.6	1.2	60	100	*	*	900	1155	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
348	50	AD	004	10	06	A1A	IH	-	ADDITIVE	L	P-306205	AD-306001	6	8	120	180	*	9	900	1155	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
349	40	AD	004	10	07	A1A	IH	-	ADDITIVE	L	BYPASS LINE FROM	P-306205	6	8	120	180	*	9	900	1155	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
350	50	HOR(H)	004	10	08	A1B	IH	-	HOT OIL	L	HEADER	MB-306205	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
351	50	HOS(H)	004	10	09	A1B	IH	-	HOT OIL	L	HEADER	MB-306205	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
352	80	V	004	11	01	G6C	-	-	VAPOR	V	DV-306301	SAFE LOCATIO	6	8	40	60	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
353	50	V	004	11	02	G6C	-	-	VAPOR	V	MANIFOLD	DV-306301	6	8	40	60	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
354	50	V	004	11	03	G6C	-	-	VAPOR	V	DV-306301	DRAIN	6	8	120	150	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
355	100	V	004	12	01	A6C	-	-	VAPOR	V	PV-306301	VACUUM	8	10	150	200	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
356	100	GE	004	12	02	A3C	IH	IE	GREASE	L	P-306301A	PV-306301	8	10	150	200	*	6	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
357	100	GE	004	12	03	A3C	IH	IE	GREASE	L	P-306301	PV-306301	8	10	150	200	*	9	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
358	50	GE	004	12	04	A1A	IH	-	GREASE	L	DRUM ADDITIVE P	PV-306301	8	10	150	200	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
359	80	COR	004	12	05	A1A	IH	-	COLD OIL	L	PV-306301	*	8	10	25	10	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
360	80	HOR(H)	004	12	06	A1B	IH	-	HOT OIL	L	PV-306301	*	6	8	200	220	*	*	900	2.5	10	260	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
361	80	HOR(H)	004	12	07	A1B	IH	-	HOT OIL	L	PV-306301	*	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
362	80	COS	004	12	08	A1A	IH	-	COLD OIL	L	HEADER	PV-306301	8	10	25	10	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
363	80	HOS(H)	004	12	09	A1B	IH	-	HOT OIL	L	HEADER	PV-306301	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
364	150	GE	004	12	10	A3C	IH	IE	GREASE	L	PV-306301	P-306301	6	8	120	180	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
365	100	GE	004	12	11	A3C	IH	IE	GREASE	L	P-306301	CM-306301	8	10	150	200	*	9	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
366	25	CWS	004	12	12	A1A	IH	-	COLD WATER	L	P-306301	CM-306301	8	10	150	200	*	9	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
367	25	CWR	004	12	13	A1A	IH	-	COLD WATER	L	CM-306301	HEADER RETU	8	10	150	200	*	9	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
368	100	GE	004	12	14	A3C	IH	IE	GREASE	L	CM-306301	P-306301A	8	10	150	200	*	9	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
369	100	GE	004	12	15	A3C	IH	IE	GREASE	L	CM-306301	DRUM FILLING	8	10	150	200	*	9	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-004	H1	P&ID FOR BITUMEN PLANT		
370	150	HOS(M)	006	01	01	A1A	IH	-	HOT OIL	L	HEADER	MT-305102	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
371	80	HOS(M)	006	01	03	A1A	IH	-	HOT OIL	L	HEADER	HE-305001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
372	80	HOR(M)	006	01	04	A1A	IH	-	HOT OIL	L	HEADER	HE-305001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
373	80	BO	006	01	05	A1A	IH	-	BASE OIL	L	HE-305001	MT-305101	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
374	80	HOS(M)	006	01	06	A1A	IH	-	HOT OIL	L	HEADER	HE-305002	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
375	80	HOR(M)	006	01	07	A1A	IH	-	HOT OIL	L	HEADER	HE-305002	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
376	80	BO	006	01	08	A1A	IH	-	BASE OIL	L	HE-305002	MT-305102	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
377	80	BO	006	01	09	A1A	-	-	BASE OIL	L	ST-100037	HE-305001	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
378	80	BO	006	01	10	A1A	-	-	BASE OIL	L	BYPASS LINE OF ST	HE-305001	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
379	80	BO	006	01	11	A1A	-	-	BASE OIL	L	BYPASS LINE OF ST	HE-305002	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
380	80	BO	006	01	12	A1A	-	-	BASE OIL	L	ST-100002	HE-305002	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
381	150	HOR(M)	006	02	01	A1A	IH	-	HOT OIL	L	HEADER	HE-305001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
382	80	BO	006	02	02	A1A	IH	-	BASE OIL	L	HE-305001	MT-305101	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
383	80	BO	006	02	03	A1A	IH	-	BASE OIL	L	HE-305002	MT-305101	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
384	80	HOS(M)	006	02	04	A1A	IH	-	HOT OIL	L	HEADER	MT-305101	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
385	80	HOR(M)	006	02	05	A1A	IH	-	HOT OIL	L	HEADER	MT-305101	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
386	100	WO	006	02	06	A1A	-	-	WHITE OIL	L	MT-305101	P-305101	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
387	100	WO	006	02	07	A1A	-	-	WHITE OIL	L	P-305101 LINE WIT	MT-305101	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
388	100	WO	006	02	08	A1A	-	-	WHITE OIL	L	P-305101	MT-305101 O	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
389	80	WO	006	02	09	A1A	-	-	WHITE OIL	L	BYPASS LINE OF P-	P-305101	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
390	80	WO	006	02	10	A1A	-	-	WHITE OIL	L	P-305101	C-305201	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
391	80	HOS(M)	006	02	11	A1A	IH	-	HOT OIL	L	HEADER	MT-305102	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
392	80	HOR(M)	006	02	12	A1A	IH	-	HOT OIL	L	HEADER	MT-305102	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
393	100	WO	006	02	13	A1A	-	-	WHITE OIL	L	MT-305102	P-305102	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
394	80	WO	006	02	14	A1A	-	-	WHITE OIL	L	BYPASS LINE OF P-	P-305102	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
395	80	WO	006	02	15	A1A	-	-	WHITE OIL	L	P-305102	C-305202	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
396	100	WO	006	02	16	A1A	-	-	WHITE OIL	L	P-305102	MT-305102 O	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
397	100	WO	006	02	17	A1A	-	-	WHITE OIL	L	P-305102 LINE WIT	MT-305102	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
398	40	CA	006	03	01	G6C	-	-	OMPRESSED AI	V	HEADER	C-305210	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
399	25	CA	006	03	02	G6C	-	-	OMPRESSED AI	V	HEADER	C-305201	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
400	80	WO	006	03	03	A1A	-	-	WHITE OIL	L	P-305102	C-305201	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
401	80	WO	006	03	04	A1A	-	-	WHITE OIL	L	C-305201	F-305201	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
402	50	WO	006	03	05	A1A	-	-	WHITE OIL	L	F-305201	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
403	25	CA	006	03	06	G6C	-	-	OMPRESSED AI	V	HEADER	C-305202	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
404	80	WO	006	03	07	A1A	-	-	WHITE OIL	L	P-305102	C-305202	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
405	80	WO	006	03	08	A1A	-	-	WHITE OIL	L	C-305202	F-305202	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
406	50	WO	006	03	09	A1A	-	-	WHITE OIL	L	F-305202	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
407	50	WO	006	03	10	A1A	-	-	WHITE OIL	L	F-305202	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
408	50	WO	006	03	11	A1A	-	-	WHITE OIL	L	F-305201	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
409	80	WO	006	03	12	A1A	-	-	WHITE OIL	L	P-305101	C-305201	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
410	80	WO	006	03	13	A1A	-	-	WHITE OIL	L	C-305201	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
411	80	WO	006	03	14	A1A	-	-	WHITE OIL	L	P-305101	C-305202	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
412	80	WO	006	03	15	A1A	-	-	WHITE OIL	L	C-305202	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
413	80	WO	006	03	16	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
414	80	WO	006	03	17	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
415	25	CA	006	04	01	G6C	-	-	OMPRESSED AI	V	HEADER	C-305203	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
416	80	WO	006	04	02	A1A	-	-	WHITE OIL	V	C-305202	C-305203	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
417	80	WO	006	04	03	A1A	-	-	WHITE OIL	L	C-305203	F-305203	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
418	80	WO	006	04	04	A1A	-	-	WHITE OIL	L	F-305203	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
419	25	CA	006	04	05	G6C	-	-	OMPRESSED AI	L	HEADER	C-305204	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
420	80	WO	006	04	06	A1A	-	-	WHITE OIL	V	C-305202	C-305204	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
421	80	WO	006	04	07	A1A	-	-	WHITE OIL	L	C-305204	F-305204	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
422	80	WO	006	04	08	A1A	-	-	WHITE OIL	L	F-305204	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
423	50	WO	006	04	10	A1A	-	-	WHITE OIL	L	F-305204	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
424	50	WO	006	04	11	A1A	-	-	WHITE OIL	L	F-305203	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
425	80	WO	006	04	12	A1A	-	-	WHITE OIL	L	C-305201	C-305203	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
426	80	WO	006	04	13	A1A	-	-	WHITE OIL	L	C-305203	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
427	80	WO	006	04	14	A1A	-	-	WHITE OIL	L	C-305201	C-305204	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
428	80	WO	006	04	15	A1A	-	-	WHITE OIL	L	C-305204	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
429	80	WO	006	04	16	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
430	80	WO	006	04	09	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
431	25	CA	006	05	01	G6C	-	-	OMPRESSED AI	V	HEADER	C-305205	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
432	80	WO	006	05	02	A1A	-	-	WHITE OIL	V	C-305204	C-305205	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
433	80	WO	006	05	03	A1A	-	-	WHITE OIL	L	C-305205	F-305205	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
434	80	WO	006	05	04	A1A	-	-	WHITE OIL	L	F-305205	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
435	25	CA	006	05	05	G6C	-	-	OMPRESSED AI	L	HEADER	C-305206	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
436	80	WO	006	05	06	A1A	-	-	WHITE OIL	V	C-305204	C-305206	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
437	80	WO	006	05	07	A1A	-	-	WHITE OIL	L	C-305206	F-305206	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
438	80	WO	006	05	08	A1A	-	-	WHITE OIL	L	F-305206	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
439	50	WO	006	05	10	A1A	-	-	WHITE OIL	L	F-305206	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
440	50	WO	006	05	11	A1A	-	-	WHITE OIL	L	F-305205	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
441	80	WO	006	05	12	A1A	-	-	WHITE OIL	L	C-305203	C-305205	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
442	80	WO	006	05	13	A1A	-	-	WHITE OIL	L	C-305205	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
443	80	WO	006	05	14	A1A	-	-	WHITE OIL	L	C-305203	C-305206	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
444	80	WO	006	05	15	A1A	-	-	WHITE OIL	L	C-305206	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
445	80	WO	006	05	16	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
446	80	WO	006	05	09	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
447	25	CA	006	06	01	G6C	-	-	OMPRESSED AI	V	HEADER	C-305207	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
448	80	WO	006	06	02	A1A	-	-	WHITE OIL	V	C-305206	C-305207	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
449	80	WO	006	06	03	A1A	-	-	WHITE OIL	L	C-305207	F-305207	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
450	80	WO	006	06	04	A1A	-	-	WHITE OIL	L	F-305207	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
451	25	CA	006	06	05	G6C	-	-	OMPRESSED AI	L	HEADER	C-305208	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
452	80	WO	006	06	06	A1A	-	-	WHITE OIL	V	C-305206	C-305208	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
453	80	WO	006	06	07	A1A	-	-	WHITE OIL	L	C-305208	F-305208	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
454	80	WO	006	06	08	A1A	-	-	WHITE OIL	L	F-305208	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
455	50	WO	006	06	10	A1A	-	-	WHITE OIL	L	F-305208	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
456	50	WO	006	06	11	A1A	-	-	WHITE OIL	L	F-305207	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
457	80	WO	006	06	12	A1A	-	-	WHITE OIL	L	C-305205	C-305207	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
458	80	WO	006	06	13	A1A	-	-	WHITE OIL	L	C-305207	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
459	80	WO	006	06	14	A1A	-	-	WHITE OIL	L	C-305205	C-305208	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
460	80	WO	006	06	15	A1A	-	-	WHITE OIL	L	C-305208	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
461	80	WO	006	06	16	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
462	80	WO	006	06	09	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
463	25	CA	006	07	01	G6C	-	-	OMPRESSED AI	V	HEADER	C-305209	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
464	80	WO	006	07	02	A1A	-	-	WHITE OIL	V	C-305207	C-305209	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
465	80	WO	006	07	03	A1A	-	-	WHITE OIL	L	C-305209	F-305209	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
466	80	WO	006	07	04	A1A	-	-	WHITE OIL	L	F-305209	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
467	25	CA	006	07	05	G6C	-	-	OMPRESSED AI	L	HEADER	C-305210	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
468	80	WO	006	07	06	A1A	-	-	WHITE OIL	V	C-305207	C-305210	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
469	80	WO	006	07	07	A1A	-	-	WHITE OIL	L	C-305210	F-305210	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
470	80	WO	006	07	08	A1A	-	-	WHITE OIL	L	F-305210	HEADER 1	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
471	50	WO	006	07	10	A1A	-	-	WHITE OIL	L	F-305210	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
472	50	WO	006	07	11	A1A	-	-	WHITE OIL	L	F-305209	HEADER 2	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
473	80	WO	006	07	12	A1A	-	-	WHITE OIL	L	C-305208	C-305209	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
474	80	WO	006	07	13	A1A	-	-	WHITE OIL	L	C-305209	SAFE LOCATIC	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
475	80	WO	006	07	14	A1A	-	-	WHITE OIL	L	C-305208	C-305208	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
476	80	WO	006	07	15	A1A	-	-	WHITE OIL	L	C-305210	SAFE LOCATIO	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
477	80	WO	006	07	16	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIO	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
478	80	WO	006	07	09	A1A	-	-	WHITE OIL	L	BYPASS LINE OF C-	SAFE LOCATIO	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
479	80	WO	006	08	01	A1A	-	-	WHITE OIL	L	COLUMNS MANIF(	HT-305301	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
480	100	WO	006	08	05	A1A	-	-	WHITE OIL	L	HT-305301	P-305301	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
481	80	WO	006	08	06	A1A	-	-	WHITE OIL	L	BYPASS LINE P-305	P-305301	6	8	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
482	80	WO	006	08	07	A1A	-	-	WHITE OIL	L	P-305301	HT-305301	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
483	80	WO	006	08	08	A1A	-	-	WHITE OIL	L	P-305301	F-305305	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
484	80	WO	006	08	03	B4C	-	-	WHITE OIL	L	COLUMNS MANIF(	HT-305302	6	8	60	100	*	*	900	12	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
485	80	WO	006	08	13	A1A	-	-	WHITE OIL	L	P-305302	F-305306	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
486	80	WO	006	08	04	B4C	-	-	WHITE OIL	L	BYPASS LINE FROM	HT-305302	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
487	100	WO	006	08	09	A1A	-	-	WHITE OIL	L	HT-305302	P-305302	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
488	80	WO	006	08	10	A1A	-	-	WHITE OIL	L	BYPASS LINE FROM	P-305302	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
489	80	WO	006	08	11	A1A	-	-	WHITE OIL	L	P-305302	HT-305302	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
490	80	WO	006	08	12	A1A	-	-	WHITE OIL	L	P-305302	F-305305	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
491	80	WO	006	08	14	A1A	-	-	WHITE OIL	L	P-305301	F-305306	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
492	80	WO	006	08	15	A1A	-	-	WHITE OIL	L	COLUMNS MANIF(	HT-305302	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
493	80	WO	006	09	01	A1A	-	-	WHITE OIL	L	HT-305302	HT-305303	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
494	80	WO	006	09	02	A1A	-	-	WHITE OIL	L	HT-305302	HT-305303	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
495	100	WO	006	09	03	A1A	-	-	WHITE OIL	L	HT-305303	P-305303	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
496	80	WO	006	09	04	A1A	-	-	WHITE OIL	L	BYPASS LINE FROM	P-305303	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
497	80	WO	006	09	05	A1A	-	-	WHITE OIL	L	P-305303	HT-305303	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
498	80	WO	006	09	06	A1A	-	-	WHITE OIL	L	P-305303	F-305305	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
499	80	WO	006	09	07	A1A	-	-	WHITE OIL	L	HT-305302	HT-305304	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
500	80	WO	006	09	15	A1A	-	-	WHITE OIL	L	HT-305301	HT-305304	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
501	80	WO	006	09	08	B4C	-	-	WHITE OIL	L	HT-305302	HT-305304	ATM	FULL OF LIQUID	60	100	*	*	900	12	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
502	100	WO	006	09	09	A1A	-	-	WHITE OIL	L	HT-305304	P-305304	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
503	80	WO	006	09	10	A1A	-	-	WHITE OIL	L	BYPASS LINE FROM	P-305304	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
504	80	WO	006	09	11	A1A	-	-	WHITE OIL	L	P-305304	HT-305304	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
505	80	WO	006	09	12	A1A	-	-	WHITE OIL	L	P-305304	F-305306	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
506	80	WO	006	09	13	A1A	-	-	WHITE OIL	L	P-305304	F-305305	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
507	80	WO	006	09	14	A1A	-	-	WHITE OIL	L	P-305303	F-305306	6	8	60	100	*	50	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
508	80	WO	006	10	01	B4C	-	-	WHITE OIL	L	F-305305	3-WAY VALVE	6	8	60	100	*	50	900	12	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
509	80	WO	006	10	02	B4C	-	-	WHITE OIL	L	3-WAY VALVE	FP TANK	6	8	60	100	*	50	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
510	80	WO	006	10	03	B4C	-	-	WHITE OIL	L	3-WAY VALVE	HT-305302	6	8	60	100	*	50	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
511	80	WO	006	10	04	B4C	-	-	WHITE OIL	L	F-305306	3-WAY VALVE	6	8	60	100	*	50	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
512	80	WO	006	10	05	B4C	-	-	WHITE OIL	L	3-WAY VALVE	FP TANK	6	8	60	100	*	50	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
513	80	WO	006	10	06	B4C	-	-	WHITE OIL	L	3-WAY VALVE	HT-305304	6	8	60	100	*	50	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
514	50	HOR(M)	006	10	07	A1A	IH	-	WHITE OIL	L	F-305305	*	6	8	160	180	*	50	900	12	15	200	Water	*	0	SCH 40	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
515	50	HOS(M)	006	10	08	A1A	IH	-	WHITE OIL	L	*	F-305305	6	8	160	180	*	50	900	12	15	200	Water	*	0	SCH 40	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
516	50	COR	006	10	09	A1A	IH	-	WHITE OIL	L	F-305305	*	6	8	60	100	*	50	900	12	15	200	Water	*	0	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
517	50	COS	006	10	11	A1A	IH	-	WHITE OIL	L	*	F-305305	6	8	60	100	*	50	900	12	15	200	Water	*	0	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
518	25	IA	006	10	12	G6C	-	-	WHITE OIL	L	*	F-305305	6	8	60	100	*	50	900	12	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
519	25	PA	006	10	13	G6C	-	-	WHITE OIL	L	*	F-305305	6	8	60	100	*	50	900	12	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
520	50	HOR(M)	006	10	14	A1A	IH	-	WHITE OIL	L	F-305306	*	6	8	160	180	*	50	900	12	15	200	Water	*	0	SCH 40	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
521	50	HOS(M)	006	10	15	A1A	IH	-	WHITE OIL	L	*	F-305306	6	8	160	180	*	50	900	12	15	200	Water	*	0	SCH 40	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
522	50	COR	006	10	16	A1A	IH	-	WHITE OIL	L	F-305306	*	6	8	60	100	*	50	900	12	15	200	Water	*	0	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
523	50	COS	006	10	17	A1A	IH	-	WHITE OIL	L	*	F-305306	6	8	60	100	*	50	900	12	15	200	Water	*	0	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
524	25	IA	006	10	18	G6C	-	-	WHITE OIL	L	*	F-305306	6	8	60	100	*	50	900	12	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
525	25	PA	006	10	19	G6C	-	-	WHITE OIL	L	*	F-305306	6	8	60	100	*	50	900	12	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
526	25	N2	006	11	01	B4C	-	-	NITROGEN	V	*	FP-305401	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
527	100	WO	006	11	02	B4C	-	-	WHITE OIL	L	FP-305401	P-305401	0.6	1.2	60	100	*	*	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
528	80	WO	006	11	03	B4C	-	-	WHITE OIL	L	BYPASS LINE FROM	P-305401	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
529	80	WO	006	11	04	B4C	-	-	WHITE OIL	L	P-305401	FP-305401	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
530	80	WO	006	11	05	B4C	-	-	WHITE OIL	L	P-305401	FILLING HEAD	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
531	80	WO	006	11	06	B4C	-	-	WHITE OIL	L	P-305401	FILLING HEAD	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
532	80	WO	006	11	07	B4C	-	-	WHITE OIL	L	P-305401	FILLING HEAD	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
533	25	N2	006	12	01	B4C	-	-	NITROGEN	V	*	FP-305402	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
534	100	WO	006	12	02	B4C	-	-	WHITE OIL	L	FP-305402	P-305402	0.6	1.2	60	100	*	*	900	12	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
535	80	WO	006	12	03	B4C	-	-	WHITE OIL	L	BYPASS LINE FROM	P-305402	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
536	80	WO	006	12	04	B4C	-	-	WHITE OIL	L	P-305402	FP-305402	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
537	80	WO	006	12	05	B4C	-	-	WHITE OIL	L	P-305402	FILLING HEAD	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
538	80	WO	006	12	06	B4C	-	-	WHITE OIL	L	P-305402	FILLING HEAD	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
539	80	WO	006	12	07	B4C	-	-	WHITE OIL	L	P-305402	FILLING HEAD	6	8	60	100	*	30	900	12	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
540	80	WO	006	14	01	A1A	-	-	WHITE OIL	L	SO-305501	P-305501	0.6	1.2	60	100	*	*	900	12	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
541	50	WO	006	14	02	A1A	-	-	WHITE OIL	L	BYPASS LINE FROM	P-305501	6	8	60	100	*	10	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
542	50	WO	006	14	03	A1A	-	-	WHITE OIL	L	P-305501	HE-305102	6	8	60	100	*	10	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
543	50	WO	006	14	04	A1A	-	-	WHITE OIL	L	P-305501	SO-305501	6	8	60	100	*	10	900	12	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-006	H1	P&ID FOR WHITE OIL PLANT		
544	50	TW	111	01	07	D5N	-	-	WATER	L	P-311004	SK-311101/02	6	8	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
545	50	TW	111	01	06	D5N	-	-	WATER	L	BYPASS LINE OF P-	SK-311101/02	6	8	60	100	*	30	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
546	100	TW	111	01	08	D5N	-	-	WATER	L	TW-311004	P-311004	0.6	1.2	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
547	80	TW	111	01	09	D5N	-	-	WATER	L	BOTTOM LINE OF T	P-311004	0.6	1.2	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
548	80	TW	111	01	01	D5N	-	-	WATER	L	F-311005	TW-311009	0.6	1.2	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
549	80	TW	111	01	13	D5N	-	-	WATER	L	BOTTOM LINE OF T	TW-311004	0.6	1.2	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
550	80	TW	111	01	02	D5N	-	-	WATER	L	BOTTOM LINE OF T	P-311004	0.6	1.2	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
551	100	TW	111	01	03	D5N	-	-	WATER	L	TW-311009	P-311004	6	8	60	100	*	30	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
552	80	TW	111	01	12	D5N	-	-	WATER	L	F-311101	FP-311204	6	8	60	100	*	*	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
553	80	TW	111	01	11	D5N	-	-	WATER	L	BYPASS LINE F-311	FP-311202	6	8	60	100	*	18	1000	44	10	60	Water	*	0	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
554	32	DE	111	02	02	B4C	-	-	DIESEL EXHAUS	L	TREATED WATER D	SK-311101	6	8	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
555	80	DE	111	02	05	B4C	-	-	DIESEL EXHAUS	L	SK-311101	F-311101	6	8	60	100	*	18	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
556	80	DE	111	02	08	B4C	-	-	DIESEL EXHAUS	L	F-311101	FP-311202	6	8	60	100	*	18	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
557	80	DE	111	02	07	B4C	-	-	DIESEL EXHAUS	L	SK-311102	F-311102	6	8	60	100	*	18	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
558	80	DE	111	02	10	B4C	-	-	DIESEL EXHAUS	L	EQUALISATION	EQUALISATIO	6	8	60	100	*	18	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
559	80	DE	111	03	01	B4C	-	-	DIESEL EXHAUS	L	BT-311102	FP-311201	ATM	FULL OF LIQUID	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
560	80	DE	111	03	02	B4C	-	-	DIESEL EXHAUS	L	BT-311102	FP-311202	ATM	FULL OF LIQUID	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
557	80	DE	111	03	03	B4C	-	-	DIESEL EXHAUS	L	P-311201	FP-311201	6	8	60	100	*	24	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
558	100	DE	111	03	04	B4C	-	-	DIESEL EXHAUS	L	FP-311201	P-311201	0.6	1.2	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
559	100	DE	111	03	05	B4C	-	-	DIESEL EXHAUS	L	P-311201	F-311301	6	8	60	100	*	24	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
560	80	DE	111	03	06	B4C	-	-	DIESEL EXHAUS	L	F-311301	HEADER	6	8	60	100	*	24	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
561	80	DE	111	03	07	B4C	-	-	DIESEL EXHAUS	L	P-311202	FP-311202	6	8	60	100	*	24	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
562	100	DE	111	03	08	B4C	-	-	DIESEL EXHAUS	L	FP-311202	P-311202	0.6	1.2	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
563	100	DE	111	03	09	B4C	-	-	DIESEL EXHAUS	L	TAPPING FROM OI	P-311202	0.6	1.2	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
564	80	DE	111	03	10	B4C	-	-	DIESEL EXHAUS	L	P-311202	F-311302	6	8	60	100	*	24	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
565	80	DE	111	03	11	B4C	-	-	DIESEL EXHAUS	L	F-311302	HEADER	6	8	60	100	*	24	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
566	80	DE	111	04	01	B4C	-	-	DIESEL EXHAUS	L	P-311203	FP-311203	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
567	80	DE	111	04	02	B4C	-	-	DIESEL EXHAUS	L	BT-311203	FP-311203	ATM	FULL OF LIQUID	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
568	100	DE	111	04	03	B4C	-	-	DIESEL EXHAUS	L	FP-311203	P-311203	0.6	1.2	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
569	100	DE	111	04	04	B4C	-	-	DIESEL EXHAUS	L	FP-311204	P-311204	0.6	1.2	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
570	100	DE	111	04	05	B4C	-	-	DIESEL EXHAUS	L	TAPPING FROM OI	P-311204	0.6	1.2	60	100	*	*	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
571	80	DE	111	04	06	B4C	-	-	DIESEL EXHAUS	L	P-311204	FP-311204	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
572	80	DE	111	04	07	B4C	-	-	DIESEL EXHAUS	L	P-311204	F-311304	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
573	80	DE	111	04	08	B4C	-	-	DIESEL EXHAUS	L	F-311304	MANIFOLD	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
574	100	DE	111	04	09	B4C	-	-	DIESEL EXHAUS	L	MANIFOLD	MANIFOLD	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
575	80	DE	111	04	10	B4C	-	-	DIESEL EXHAUS	L	P-311203	F-311303	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
576	80	DE	111	04	11	B4C	-	-	DIESEL EXHAUS	L	F-311303	MANIFOLD	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
577	80	DE	111	04	12	B4C	-	-	DIESEL EXHAUS	L	MANIFOLD	SMALL CAN FI	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
578	80	DE	111	04	13	B4C	-	-	DIESEL EXHAUS	L	MANIFOLD	DRUM FILLING	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
579	80	DE	111	04	14	B4C	-	-	DIESEL EXHAUS	L	MANIFOLD	PAIL FILLING	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
580	80	DE	111	04	15	B4C	-	-	DIESEL EXHAUS	L	MANIFOLD	BULK FILLING	6	8	60	100	*	50	900	400	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
581	100	PD	111	05	01	B4C	-	-	POWDER	S	PHS-311101	SL-311101	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
582	100	PD	111	05	02	B4C	-	-	POWDER	S	SC-311101A	SK-311101	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
583	100	PD	111	05	03	B4C	-	-	POWDER	S	SC-311101B	SK-311102	*	*	60	100	70	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
584	25	PA	111	05	04	G6C	-	-	PLANT AIR	V	*	PHS-304001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
585	25	IA	111	05	05	G6C	-	-	NSTRUMENT AI	V	*	PHS-304001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
586	25	CA	111	05	06	G6C	-	-	OMPRESSED AI	V	*	PHS-304001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-111	H1	P&ID FOR DIESEL EXHAUST PLANT		
587	80	BO	116	01	01	A1A	-	-	BASE OIL	L	SMB-307501	AD-307001	6	8	60	100	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
588	100	BN	116	01	02	P2A	-	-	BN EXTRACT	L	SMB-307501	PIG HEADER	6	8	60	100	*	30	900	467	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
589	25	V	116	01	03	G6C	-	-	VAPOR	V	HLDV-307501	DEMISTER	6	8	60	100	*	30	900	467	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
590	25	CA	116	01	04	G6C	-	-	OMPRESSED AI	V	*	HLDV-307501	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
591	150	LO	116	01	05	A1A	-	-	LUBE OIL	L	TYPE 5	P-307501	0.6	1.2	60	100	*	*	900	467	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
592	100	LO	116	01	06	A1A	-	-	LUBE OIL	L	P-307501	OUTGOING LI	6	8	60	100	*	40	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
593	100	LO	116	01	07	A1A	-	-	LUBE OIL	L	BYPASS LINE OF P-	P-307501	6	8	60	100	*	40	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
594	100	LO	116	01	08	A1A	-	-	LUBE OIL	L	P-307501	TYPE 10	6	8	60	100	*	40	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
595	25	IA	116	01	09	G6C	-	-	NSTRUMENT AI	V	*	SKID	6	8	60	100	*	30	900	467	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
596	25	PA	116	01	10	G6C	-	-	PLANT AIR	V	*	SKID	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
597	200	HOS(M)	116	02	01	A1A	IH	-	HOT OIL	L	HEADER	FP-307202	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
598	200	HOR(M)	116	02	02	A1A	IH	-	HOT OIL	L	HEADER	AD-307001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
599	50	HOS(M)	116	02	03	A1A	IH	-	HOT OIL	L	HEADER	AD-307001	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
600	50	HOR(M)	116	02	04	A1A	IH	-	HOT OIL	L	HEADER	AD-307001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
601	50	BO	116	02	05	A1A	-	-	BASE OIL	L	SMB-307501	AD-307001	6	8	60	100	*	30	900	467	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
602	150	AD	116	02	06	A1A	IH	IE	ADDITIVE	L	AD-307001	P-307001	0.6	1.2	60	100	*	*	900	467	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
603	50	AD	116	02	07	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-307001	6	8	60	100	*	1	900	467	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
604	50	AD	116	02	08	A1A	IH	IE	ADDITIVE	L	P-307001	SMB-307501	6	8	60	100	*	1	900	467	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
605	100	BN	116	02	10	A1A	IH	IE	BN EXTRACT	L	ST-100047	HE-307101	ATM	FULL OF LIQUID	60	100	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
606	100	BN	116	02	11	A1A	IH	IE	BN EXTRACT	L	ST-100047	P-307002	ATM	FULL OF LIQUID	60	100	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
607	40	BO	116	02	12	A1A	-	-	BASE OIL	L	BYPASS LINE FROM	P-307002	6	8	60	100	*	30	900	467	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
608	100	BN	116	02	13	A1A	IH	IE	BN EXTRACT	L	P-307002	TANK TRUCK	6	8	60	100	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
609	100	BN	116	02	14	A1A	IH	IE	BN EXTRACT	L	HE-307101	BN EXT. 11150	6	8	60	100	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
610	80	HOS(M)	116	02	15	A1A	IH	-	HOT OIL	L	HEADER	HE-307101	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
611	80	HOR(M)	116	02	16	A1A	IH	-	HOT OIL	L	HEADER	HE-307101	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
612	50	AD	116	02	17	A1A	IH	IE	ADDITIVE	L	P-307001	AD-307001	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
613	100	BN	116	02	18	A1A	IH	IE	BN EXTRACT	L	HE-307101	ILB-313401	6	8	60	150	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
614	80	HOS(M)	116	03	01	A1A	IH	-	HOT OIL	L	HEADER	FP-307201	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
615	80	HOR(M)	116	03	02	A1A	IH	-	HOT OIL	L	HEADER	FP-307201	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
616	100	BN	116	03	03	A1A	-	-	BN EXTRACT	L	PIG HEADER TYPE	FP-307201	6	8	60	100	*	*	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
617	80	BN	116	03	04	A1A	IH	IE	BN EXTRACT	L	P-307201	FP-307201	6	8	60	100	*	30	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
618	100	BN	116	03	05	A1A	IH	-	BN EXTRACT	L	OUTGOING LINE FI	P-307201	0.6	1.2	60	100	*	*	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
619	150	BN	116	03	06	A1A	IH	IE	BN EXTRACT	L	FP-307201	P-307201	0.6	1.2	60	100	*	*	900	3600	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
620	80	BN	116	03	07	A1A	IH	-	BN EXTRACT	L	BYPASS LINE FROM	P-307201	6	8	60	100	*	30	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
621	80	BN	116	03	08	A1A	IH	IE	BN EXTRACT	L	F-307201	HLDV-307201	6	8	60	100	*	30	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
622	80	BN	116	03	09	P2A	-	-	BN EXTRACT	L	HLDV-307201	MANIFOLD	6	8	60	100	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
623	25	CA	116	03	11	G6C	-	-	OMPRESSED AI	V	*	HLDV-307201	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
624	25	V	116	03	10	G6C	-	-	VAPOR	V	HLDV-307201	DEMISTER	6	8	40	60	*	30	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
625	80	HOS(M)	116	04	01	A1A	IH	-	HOT OIL	L	HEADER	FP-307202	ATM	FULL OF LIQUID	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
626	80	HOR(M)	116	04	02	A1A	IH	-	HOT OIL	L	HEADER	FP-307202	ATM	FULL OF LIQUID	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
627	100	BN	116	04	03	A1A	-	-	BN EXTRACT	L	TYPE9	FP-307202	ATM	FULL OF LIQUID	60	100	*	*	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
628	80	BN	116	04	04	A1A	IH	IE	BN EXTRACT	L	P-307202	FP-307202	6	8	60	100	*	30	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
629	150	BN	116	04	05	A1A	IH	IE	BN EXTRACT	L	FP-307202	P-307202	0.6	1.2	60	100	*	*	900	650	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
630	25	CA	116	04	06	G6C	-	-	OMPRESSED AI	V	*	FP-307202	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
631	80	BN	116	04	07	A1A	IH	IE	BN EXTRACT	L	BYPASS LINE FROM	P-307202	6	8	60	100	*	30	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
632	80	BN	116	04	08	A1A	IH	IE	BN EXTRACT	L	F-307202	HLDV-307202	6	8	60	100	*	30	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
633	80	BN	116	04	09	P2A	-	-	BN EXTRACT	L	HLDV-307202 TYPE	MANIFOLD	6	8	60	100	*	*	900	650	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
634	25	V	116	04	10	G6C	-	-	VAPOR	V	*	HLDV-307202	6	8	40	60	*	*	900	650	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
635	25	CA	116	04	11	G6C	-	-	OMPRESSED AI	V	HLDV-307202 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
636	25	CA	116	04	12	G6C	-	-	OMPRESSED AI	V	*	TYPE9	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
637	25	V	116	04	13	G6C	-	-	VAPOR	V	TYPE9	DEMISTER	6	8	40	60	*	*	900	650	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
638	80	BO	116	05	01	A1A	-	-	BASE OIL	L	150 N GI	PM-307401	6	8	60	100	*	*	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
639	80	BN	116	05	02	P2A	-	-	BN EXTRACT	L	PM-307401	LOADING ARM	6	8	60	100	*	*	900	650	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
640	80	BN	116	05	03	P2A	-	-	BN EXTRACT	L	PM-307401	LOADING ARM	6	8	60	100	*	*	900	650	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
641	80	BN	116	05	04	P2A	-	-	BN EXTRACT	L	PM-307401	PIG RECEIVER	6	8	60	100	*	*	900	650	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
642	25	V	116	05	05	G6C	-	-	VAPOR	V	PM-307401	DEMISTER	6	8	40	60	*	*	900	650	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
643	25	PA	116	05	06	G6C	-	-	PLANT AIR	V	*	PM-307401	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
644	25	IA	116	05	07	G6C	-	-	NSTRUMENT AI	V	*	PM-307401	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
645	80	BO	116	05	08	A1A	-	-	BASE OIL	L	150 N GI	PM-307401	6	8	60	100	*	*	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
646	80	BN	116	05	09	A1A	-	-	BN EXTRACT	L	PIG RECEIVER TYPE	SLOP TANK	6	8	60	100	*	*	900	650	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
647	25	CA	116	05	10	G6C	-	-	OMPRESSED AI	V	PLANT AIR	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
648	100	BN	116	06	01	A1A	-	-	BN EXTRACT	L	SO-307301	P-307301	0.6	1.2	60	100	*	*	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
649	80	BN	116	06	02	A1A	-	-	BN EXTRACT	L	P-307301	SMB-307501	6	8	60	100	*	15	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
650	100	BN	116	06	03	A1A	-	-	BN EXTRACT	L	P-307301	BITUMEN	6	8	60	100	*	15	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
651	80	BN	116	06	04	A1A	-	-	BN EXTRACT	L	P-307301	REX PLANT	6	8	60	100	*	15	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
652	80	BN	116	06	06	A1A	-	-	BN EXTRACT	L	P-307301	SO-307301	6	8	60	100	*	15	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
653	80	BN	116	06	05	A1A	-	-	BN EXTRACT	L	BYPASS LINE FROM	P-307301	6	8	60	100	*	15	900	3600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-116	H1	P&ID FOR BN EXTRACT PLANT		
657	50	AD	115	01	01	A1A	IH	IE	ADDITIVE	L	MB-302001	P-302001	0.6	1.2	60	120	*	*	900	2100	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
658	25	AD	115	01	03	A1A	IH	IE	ADDITIVE	L	P-302001	MB-302001	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
659	25	AD	115	01	04	A1A	IH	IE	ADDITIVE	L	BYPASS LINE OF P-	P-302001	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
660	50	AD	115	01	05	A1A	IH	IE	ADDITIVE	L	HOSE CONNECTIO	P-302001	0.6	1.2	60	120	*	*	900	2100	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
661	25	AD	115	01	06	A1A	IH	IE	ADDITIVE	L	P-302001	SMB-302002	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
662	25	AD	115	01	07	A1A	IH	IE	ADDITIVE	L	P-302001	SMB-302001	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
663	25	AD	115	01	08	A1A	IH	IE	ADDITIVE	L	P-302001	V-302112	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
664	25	AD	115	01	09	A1A	IH	IE	ADDITIVE	L	P-302001	V-302111	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
665	25	AD	115	01	10	A1A	IH	IE	ADDITIVE	L	P-302001	AD-307001	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
666	25	AD	115	01	11	A1A	IH	IE	ADDITIVE	L	P-302001	AD-313101	6	8	60	120	*	1.5	900	2100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
667	50	HOS(M)	115	01	12	A1A	IH	-	HOT OIL	L	HEADER	MB-302001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
668	50	HOR(M)	115	01	13	A1A	IH	-	HOT OIL	L	MB-302014	HEADER	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
669	50	HOR(M)	115	01	14	A1A	IH	-	HOT OIL	L	HEADER	MB-302001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
670	50	HOS(M)	115	01	15	A1A	IH	-	HOT OIL	L	HEADER	MB-302001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
671	80	V	115	01	16	A1A	-	-	VAPOR	V	MB-302001	SAFE LOCATIO	ATM	FULL OF LIQUID	60	120	*	*	900	2100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
672	50	AD	115	02	01	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	MB-302002	ATM	FULL OF LIQUID	60	100	*	3	900	4	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
673	25	AD	115	02	02	A1A	-	-	ADDITIVE	L	P-302002	MB-302002	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
674	40	AD	115	02	04	A1A	-	-	ADDITIVE	L	MB-302002	P-302002	0.6	1.2	60	100	*	*	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
675	25	AD	115	02	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302002	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
676	25	AD	115	02	06	A1A	-	-	ADDITIVE	L	P-302002	SMB-302002	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
677	25	AD	115	02	07	A1A	-	-	ADDITIVE	L	P-302002	V-302103	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
678	25	AD	115	02	08	A1A	-	-	ADDITIVE	L	P-302002	V-302112	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
679	25	AD	115	02	09	A1A	-	-	ADDITIVE	L	P-302002	V-302111	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
680	25	AD	115	02	10	A1A	-	-	ADDITIVE	L	P-302002	SMB-302001	6	8	60	100	*	0.1	900	4	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
681	80	AD	115	02	11	A1A	-	-	ADDITIVE	L	MB-302002	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
682	50	AD	115	03	01	A1A	IH	IE	ADDITIVE	L	DRUM ADDITIVE P	MB-302003	ATM	FULL OF LIQUID	60	100	*	3	900	530	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
683	25	AD	115	03	02	A1A	-	-	ADDITIVE	L	P-302003	MB-302003	6	8	60	100	*	0.1	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
684	40	AD	115	03	04	A1A	-	-	ADDITIVE	L	MB-302003	P-302003	0.6	1.2	60	100	*	*	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
685	25	AD	115	03	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302003	6	8	60	100	*	0.1	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
686	25	AD	115	03	06	A1A	-	-	ADDITIVE	L	P-302003	SMB-302002	6	8	60	100	*	0.1	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
687	25	AD	115	03	07	A1A	-	-	ADDITIVE	L	P-302003	SMB-302001	6	8	60	100	*	0.1	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
688	25	AD	115	03	08	A1A	-	-	ADDITIVE	L	P-302003	V-302112	6	8	60	100	*	0.1	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
689	25	AD	115	03	09	A1A	-	-	ADDITIVE	L	P-302003	V-302111	6	8	60	100	*	0.1	900	530	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
690	80	V	115	03	10	A1A	-	-	VAPOR	V	MB-302003	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	530	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
691	50	AD	115	04	01	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	MB-302004	ATM	FULL OF LIQUID	60	100	*	3	900	6	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
692	25	AD	115	04	02	A1A	-	-	ADDITIVE	L	P-302004	MB-302004	6	8	60	100	*	0.5	900	6	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
693	40	AD	115	04	04	A1A	-	-	ADDITIVE	L	MB-302004	P-302004	0.6	1.2	60	100	*	*	900	6	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
694	25	AD	115	04	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302004	6	8	60	100	*	0.5	900	6	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
695	25	AD	115	04	06	A1A	-	-	ADDITIVE	L	P-302004	SMB-302002	6	8	60	100	*	0.5	900	6	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
696	25	AD	115	04	07	A1A	-	-	ADDITIVE	L	P-302004	V-302103	6	8	60	100	*	0.5	900	6	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
697	25	AD	115	04	08	A1A	-	-	ADDITIVE	L	P-302004	V-302112	6	8	60	100	*	0.5	900	6	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
698	80	V	115	04	09	A1A	-	-	VAPOR	V	MB-302004	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	6	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
699	25	AD	115	05	01	A1A	-	-	ADDITIVE	L	P-302005	MB-302005	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
700	40	AD	115	05	02	A1A	-	-	ADDITIVE	L	MB-302005	P-302005	0.6	1.2	60	100	*	*	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
701	25	AD	115	05	04	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302005	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
702	25	AD	115	05	05	A1A	-	-	ADDITIVE	L	P-302005	PV-301711	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
703	25	AD	115	05	06	A1A	-	-	ADDITIVE	L	P-302005	A-301701	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
704	25	AD	115	05	07	A1A	-	-	ADDITIVE	L	P-302005	A-301702	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
705	25	AD	115	05	08	A1A	-	-	ADDITIVE	L	P-302005	PK-301705	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
706	25	AD	115	05	09	A1A	-	-	ADDITIVE	L	P-302005	PV-301712	6	8	60	100	*	1	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
707	80	V	115	05	11	A1A	-	-	VAPOR	V	MB-302005	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
708	40	AD	115	05	10	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-302005	0.6	1.2	60	100	*	*	900	88	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
709	50	AD	115	06	01	A1A	IH	IE	ADDITIVE	L	DRUM ADDITIVE P	MB-302006	ATM	FULL OF LIQUID	60	100	*	3	900	620	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
710	25	AD	115	06	02	A1A	-	-	ADDITIVE	L	P-302006	MB-302006	6	8	60	100	*	0.1	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
711	40	AD	115	06	04	A1A	-	-	ADDITIVE	L	MB-302006	P-302006	0.6	1.2	60	100	*	*	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
712	25	AD	115	06	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302006	6	8	60	100	*	0.1	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
713	25	AD	115	06	06	A1A	-	-	ADDITIVE	L	P-302006	SMB-302001	6	8	60	100	*	0.1	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
714	25	AD	115	06	07	A1A	-	-	ADDITIVE	L	P-302006	SMB-302002	6	8	60	100	*	0.1	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
715	25	AD	115	06	08	A1A	-	-	ADDITIVE	L	P-302006	V-302111	6	8	60	100	*	0.1	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
716	25	AD	115	06	09	A1A	-	-	ADDITIVE	L	P-302006	V-302112	6	8	60	100	*	0.1	900	620	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
717	80	V	115	06	10	A1A	-	-	VAPOR	V	MB-302006	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	620	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
718	25	AD	115	07	01	A1A	-	-	ADDITIVE	L	P-302007	MB-302007	6	8	60	100	*	1	900	80	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
719	40	AD	115	07	02	A1A	-	-	ADDITIVE	L	MB-302007	P-302007	0.6	1.2	60	100	*	*	900	80	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
720	40	AD	115	07	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-302007	0.6	1.2	60	100	*	*	900	80	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
721	25	AD	115	07	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302007	6	8	60	100	*	1	900	80	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
722	25	AD	115	07	06	A1A	-	-	ADDITIVE	L	P-302007	V-302112	6	8	60	100	*	1	900	80	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
723	25	AD	115	07	07	A1A	-	-	ADDITIVE	L	P-302007	SMB-302002	6	8	60	100	*	1	900	80	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
724	80	V	115	07	08	A1A	-	-	VAPOR	V	MB-302007	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	80	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
725	50	AD	115	08	01	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	MB-302008	ATM	FULL OF LIQUID	60	100	*	3	900	55	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
726	25	AD	115	08	02	A1A	-	-	ADDITIVE	L	P-302008	MB-302008	6	8	60	100	*	0.1	900	55	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
727	40	AD	115	08	03	A1A	-	-	ADDITIVE	L	MB-302008	P-302008	0.6	1.2	60	100	*	*	900	55	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
728	25	AD	115	08	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302008	6	8	60	100	*	0.1	900	55	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
729	25	AD	115	08	06	A1A	-	-	ADDITIVE	L	P-302008	V-302112	6	8	60	100	*	0.1	900	55	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
730	25	AD	115	08	07	A1A	-	-	ADDITIVE	L	P-302008	V-302111	6	8	60	100	*	0.1	900	55	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
731	25	AD	115	08	08	A1A	-	-	ADDITIVE	L	P-302008	SMB-302001	6	8	60	100	*	0.1	900	55	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
732	80	V	115	08	09	A1A	-	-	VAPOR	V	MB-302008	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	55	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
733	25	AD	115	09	01	A1A	-	-	ADDITIVE	L	P-302009	MB-302009	6	8	60	100	*	3	900	75	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
734	40	AD	115	09	02	A1A	-	-	ADDITIVE	L	MB-302009	P-302009	0.6	1.2	60	100	*	*	900	75	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
735	40	AD	115	09	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-302009	0.6	1.2	60	100	*	*	900	75	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
736	25	AD	115	09	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302009	6	8	60	100	*	3	900	75	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
737	25	AD	115	09	06	A1A	-	-	ADDITIVE	L	P-302009	V-302112	6	8	60	100	*	3	900	75	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
738	25	AD	115	09	07	A1A	-	-	ADDITIVE	L	P-302009	SMB-302002	6	8	60	100	*	3	900	75	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
739	80	V	115	09	08	A1A	-	-	VAPOR	V	MB-302009	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	75	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
740	40	AD	115	10	01	A1A	-	-	ADDITIVE	L	P-302010	MB-302010	6	8	60	100	*	1	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
741	50	AD	115	10	02	A1A	-	-	ADDITIVE	L	MB-302010	P-302010	0.6	1.2	60	100	*	*	900	347	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
742	50	AD	115	10	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-302010	0.6	1.2	60	100	*	*	900	347	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
743	25	AD	115	10	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302010	6	8	60	100	*	1	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
744	40	AD	115	10	06	A1A	-	-	ADDITIVE	L	P-302010	V-302112	6	8	60	100	*	1	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
745	40	AD	115	10	07	A1A	-	-	ADDITIVE	L	P-302010	V-302111	6	8	60	100	*	1	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
746	80	V	115	10	08	A1A	-	-	VAPOR	V	MB-302010	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	347	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
747	40	AD	115	11	01	A1A	-	-	ADDITIVE	L	P-302011	MB-302011	6	8	60	100	*	1	900	130	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
748	40	AD	115	11	02	A1A	-	-	ADDITIVE	L	MB-302011	P-302011	0.6	1.2	60	100	*	*	900	130	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
749	40	AD	115	11	03	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-302011	0.6	1.2	60	100	*	*	900	130	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
750	25	AD	115	11	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302011	6	8	60	100	*	1	900	130	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
751	40	AD	115	11	06	A1A	-	-	ADDITIVE	L	P-302011	SMB-302002	6	8	60	100	*	1	900	130	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
752	40	AD	115	11	07	A1A	-	-	ADDITIVE	L	P-302011	V-302112	6	8	60	100	*	1	900	130	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
753	80	V	115	11	08	A1A	-	-	VAPOR	V	MB-302011	SAFE LOCATIO	ATM	FULL OF LIQUID	60	100	*	*	900	130	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
754	50	AD	115	12	01	A1A	IH	IE	ADDITIVE	L	DRUM ADDITIVE P	MB-3021012	ATM	FULL OF LIQUID	60	120	*	3	900	4200	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
755	40	AD	115	12	02	A1A	IH	IE	ADDITIVE	L	P-302012	MB-3021012	6	8	60	120	*	0.5	900	4200	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
756	40	AD	115	12	03	A1A	IH	IE	ADDITIVE	L	MB-302012	P-302012	0.6	1.2	60	120	*	*	900	4200	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
757	25	AD	115	12	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE OF P-	P-302012	6	8	60	120	*	0.5	900	4200	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
758	40	AD	115	12	06	A1A	IH	IE	ADDITIVE	L	P-302012	SMB-302001	6	8	60	120	*	0.5	900	4200	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
759	40	AD	115	12	07	A1A	IH	IE	ADDITIVE	L	P-302012	V-302112	6	8	60	120	*	0.5	900	4200	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
760	40	AD	115	12	08	A1A	IH	IE	ADDITIVE	L	P-302012	V-302111	6	8	60	120	*	0.5	900	4200	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
761	50	HOS(M)	115	12	13	A1A	IH	-	ADDITIVE	L	HEADER	MB-302012	6	8	160	180	*	*	900	4200	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
762	50	HOR(M)	115	12	14	A1A	IH	-	ADDITIVE	L	HEADER	MB-302012	6	8	160	180	*	*	900	4200	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
763	80	V	115	12	15	A1A	-	-	VAPOR	V	MB-302012	SAFE LOCATIO	ATM	FULL OF LIQUID	60	120	*	*	900	4200	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
764	50	AD	115	13	01	A1A	IH	IE	ADDITIVE	L	DRUM ADDITIVE P	MB-302013	ATM	FULL OF LIQUID	60	120	*	3	900	1100	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
765	40	AD	115	13	02	A1A	IH	IE	ADDITIVE	L	P-302013	MB-302013	6	8	60	120	*	0.1	900	1100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
766	80	AD	115	13	03	A1A	IH	IE	ADDITIVE	L	MB-302013	P-302013	0.6	1.2	60	120	*	*	900	1100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
767	25	AD	115	13	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE OF P-	P-302013	6	8	60	120	*	0.1	900	1100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
768	40	AD	115	13	06	A1A	IH	IE	ADDITIVE	L	P-302013	V-302112	6	8	60	120	*	0.1	900	1100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
769	40	AD	115	13	07	A1A	IH	IE	ADDITIVE	L	P-302013	V-302111	6	8	60	120	*	0.1	900	1100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
770	40	AD	115	13	08	A1A	IH	IE	ADDITIVE	L	P-302013	SMB-302002	6	8	60	120	*	0.1	900	1100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
771	50	HOS(M)	115	13	09	A1A	IH	-	ADDITIVE	L	HEADER	MB-302013	6	8	160	180	*	*	900	1100	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
772	50	HOR(M)	115	13	10	A1A	IH	-	ADDITIVE	L	HEADER	MB-302013	6	8	160	180	*	*	900	1100	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
773	100	V	115	13	11	A1A	-	-	VAPOR	V	MB-302013	SAFE LOCATIO	ATM	FULL OF LIQUID	60	120	*	*	900	1100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
774	100	PD	115	14	01	A1A	-	-	POWDER	L	SC-302014	MB-302014	ATM	FULL OF LIQUID	60	120	*	*	900	120	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
775	25	AD	115	14	02	A1A	-	-	ADDITIVE	L	P-302014	MB-302014	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
776	40	AD	115	14	03	A1A	-	-	ADDITIVE	L	MB-302014	P-302014	0.6	1.2	60	120	*	*	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
777	25	AD	115	14	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302014	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
778	25	AD	115	14	06	A1A	-	-	ADDITIVE	L	P-302014	SMB-302002	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
779	25	AD	115	14	07	A1A	-	-	ADDITIVE	L	P-302014	V-302111	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
780	25	AD	115	14	08	A1A	-	-	ADDITIVE	L	P-302014	A-301702	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
781	25	AD	115	14	09	A1A	-	-	ADDITIVE	L	P-302014	V-302112	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
782	25	AD	115	14	10	A1A	-	-	ADDITIVE	L	P-302014	FV-301201	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
783	25	AD	115	14	11	A1A	-	-	ADDITIVE	L	P-302014	FV-301202	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
784	25	AD	115	14	12	A1A	-	-	ADDITIVE	L	P-302014	FV-301203	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
785	25	AD	115	14	13	A1A	-	-	ADDITIVE	L	P-302014	FV-301204	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
786	25	AD	115	14	14	A1A	-	-	ADDITIVE	L	P-302014	FV-301205	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
787	25	AD	115	14	15	A1A	-	-	ADDITIVE	L	P-302014	FV-301206	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
788	25	AD	115	14	16	A1A	-	-	ADDITIVE	L	P-302014	PK-301705	6	8	60	120	*	3	900	120	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
789	80	V	115	14	17	A1A	-	-	VAPOR	V	MB-302014	SAFE LOCATIO	ATM	FULL OF LIQUID	60	120	*	*	900	120	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-115	H1	P&ID FOR MINIBULK TANKFARM		
790	50	AD	003	01	01	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	BK-310101	ATM	FULL OF LIQUID	60	100	*	3	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
791	50	AD	003	01	02	A1A	-	-	ADDITIVE	L	TAPPING FROMDR	BK-310101	ATM	FULL OF LIQUID	60	100	*	3	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
792	50	AD	003	01	03	A1A	-	-	ADDITIVE	L	TAPPING FROMDR	BK-310101	ATM	FULL OF LIQUID	60	100	*	3	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
793	25	N2	003	01	04	B4C	-	-	NITROGEN	V	NITROGEN	BK-310101	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
794	25	N2	003	01	05	B4C	-	-	NITROGEN	V	NITROGEN	BK-310101	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
795	25	AD	003	01	06	B4C	-	-	ADDITIVE	L	AD-310306	BK-310101	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
796	40	BR	003	01	07	B4C	-	-	BRAKE FLUID	L	P-310101	BK-310101	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
797	40	BR	003	01	08	B4C	-	-	BRAKE FLUID	L	HOSE CONNECTIO	P-310101	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
798	80	BR	003	01	09	B4C	-	-	BRAKE FLUID	L	BK-310101	P-310101	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
799	40	BR	003	01	10	B4C	-	-	BRAKE FLUID	L	P-310101	FP-310202	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
800	40	BR	003	01	11	B4C	-	-	BRAKE FLUID	L	P-310101	FP-310201	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
801	25	CA	003	01	12	G6C	-	-	OMPRESSED AI	V	*	AIR OPERATED	6	8	40	60	*	3	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
802	25	AD	003	01	13	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	AIR OPERATED	6	8	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
803	50	AD	003	02	01	A1A	-	-	ADDITIVE	L	BK-310101	AD-310306	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
802	50	AD	003	02	02	A1A	-	-	ADDITIVE	L	BK-310101	BK-310102	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
803	50	AD	003	02	03	A1A	-	-	ADDITIVE	L	BK-310101	BK-310102	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
804	25	N2	003	02	04	B4C	-	-	NITROGEN	V	NITROGEN	BK-310102	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
805	25	N2	003	02	05	B4C	-	-	NITROGEN	V	NITROGEN	BK-310102	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
806	25	AD	003	02	06	B4C	-	-	ADDITIVE	L	AD-310306	BK-310102	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
807	40	BR	003	02	07	B4C	-	-	BRAKE FLUID	L	P-310102	BK-310102	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
808	80	BR	003	02	08	B4C	-	-	BRAKE FLUID	L	BK-310102	P-310102	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
809	40	BR	003	02	09	B4C	-	-	BRAKE FLUID	L	HOSE CONNECTIO	P-310102	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
810	40	BR	003	02	10	B4C	-	-	BRAKE FLUID	L	P-310102	FP-310202	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
811	40	BR	003	02	11	B4C	-	-	BRAKE FLUID	L	P-310102	FP-310201	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
812	80	HOS(M)	003	03	01	A1A	IH	-	HOT OIL	L	HEADER	AD-310306	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
813	80	HOR(M)	003	03	02	A1A	IH	-	HOT OIL	L	AD-310306	HEADER	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
814	25	N2	003	03	03	B4C	-	-	NITROGEN	V	NITROGEN	FILTER	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
815	25	N2	003	03	04	B4C	-	-	NITROGEN	V	NITROGEN	AD-310306	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
816	50	AD	003	03	05	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	AD-310306	ATM	FULL OF LIQUID	60	120	*	1.8	900	347	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
817	50	AD	003	03	13	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	AIR OPERATED	6	8	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
818	25	CA	003	03	14	G6C	-	-	OMPRESSED AI	V	*	AIR OPERATED	6	8	40	60	*	3	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
819	25	BR	003	03	07	B4C	-	-	BRAKE FLUID	L	P-310306	AD-310306	6	8	60	120	*	1.2	900	347	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
819	25	HOS(M)	003	03	08	A1A	IH	-	HOT OIL	L	HEADER	AD-310306	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
820	25	HOR(M)	003	03	09	A1A	IH	-	HOT OIL	L	HEADER	AD-310306	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
821	25	BR	003	03	10	B4C	-	-	BRAKE FLUID	L	AD-310306	P-310306	0.6	1.2	60	120	*	*	900	347	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
822	25	BR	003	03	11	B4C	-	-	BRAKE FLUID	L	P-310306	BK-310101	6	8	60	120	*	1.2	900	347	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
823	25	BR	003	03	12	B4C	-	-	BRAKE FLUID	L	P-310306	BK-310102	6	8	60	120	*	1.2	900	347	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
824	25	N2	003	04	01	B4C	-	-	NITROGEN	V	NITROGEN	AD-310001	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
825	25	N2	003	04	02	B4C	-	-	NITROGEN	V	NITROGEN	AD-310001	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
826	200	AD	003	04	03	A1A	-	-	ADDITIVE	L	DYE HAND ADD 18	AD-310001	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
827	25	AD	003	04	04	A1A	-	-	ADDITIVE	L	P-310001	AD-310001	6	8	60	100	*	0.6	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
828	40	AD	003	04	05	A1A	-	-	ADDITIVE	L	AD-310001	P-310001	0.6	1.2	60	100	*	*	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
829	25	AD	003	04	06	A1A	-	-	ADDITIVE	L	P-310001	FP-310202	6	8	60	100	*	0.6	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
830	25	AD	003	04	07	A1A	-	-	ADDITIVE	L	P-310001	FP-310201	6	8	60	100	*	0.6	900	44	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
831	25	N2	003	05	01	B4C	-	-	NITROGEN	V	NITROGEN	FP-310201	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
832	25	N2	003	05	02	B4C	-	-	NITROGEN	V	NITROGEN	FILTER	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
833	40	BR	003	05	03	B4C	-	-	BRAKE FLUID	L	TAPPING FROM BK	FP-310201	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
834	50	BR	003	05	04	B4C	-	-	BRAKE FLUID	L	P-310201	FP-310201	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
835	80	BR	003	05	05	B4C	-	-	BRAKE FLUID	L	FP-310201	P-310201	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
836	40	BR	003	05	06	B4C	-	-	BRAKE FLUID	L	HOSE CONNECTIO	P-310201	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
837	50	BR	003	05	07	B4C	-	-	BRAKE FLUID	L	F-310301	MANIFOLD	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
838	50	BR	003	05	08	B4C	-	-	BRAKE FLUID	L	MANIFOLD	SMALL CAN FI	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
839	50	BR	003	05	09	B4C	-	-	BRAKE FLUID	L	MANIFOLD	DRUM FILLING	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
840	25	N2	003	06	01	B4C	-	-	NITROGEN	V	NITROGEN	FP-310202	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
841	25	N2	003	06	02	B4C	-	-	NITROGEN	V	NITROGEN	FILTER	6	8	60	100	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
842	40	BR	003	06	03	B4C	-	-	BRAKE FLUID	L	TAPPING FROM BK	FP-310202	ATM	FULL OF LIQUID	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
843	50	BR	003	06	04	B4C	-	-	BRAKE FLUID	L	P-310202	FP-310202	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
844	80	BR	003	06	05	B4C	-	-	BRAKE FLUID	L	FP-310202	P-310202	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
845	40	BR	003	06	06	B4C	-	-	BRAKE FLUID	L	HOSE CONNECTIO	P-310202	0.6	1.2	60	100	*	*	900	44	15	150	Water	*	0	SCH 40S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
846	50	BR	003	06	07	B4C	-	-	BRAKE FLUID	L	F-310302	MANIFOLD	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
847	50	BR	003	06	08	B4C	-	-	BRAKE FLUID	L	MANIFOLD	DRUM FILLING	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
848	50	BR	003	06	09	B4C	-	-	BRAKE FLUID	L	MANIFOLD	SMALL CAN FI	6	8	60	100	*	12	900	44	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-003	H1	P&ID FOR BRAKE FLUID PLANT		
849	80	BO	009	01	01	A1A	IH	-	BASE OIL	L	HE-312001	VD-312101	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
850	80	BO	009	01	02	A1A	IH	-	BASE OIL	L	HE-312002	VD-312101	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
851	80	BO	009	01	03	A1A	IH	-	BASE OIL	L	HE-312002	VD-312101	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
852	80	HOR(H)	009	01	09	A1B	IH	-	HOT OIL	L	HEADER	HE-312001	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
853	80	HOS(H)	009	01	10	A1B	IH	-	HOT OIL	L	HEADER	HE-312001	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
854	80	HOS(H)	009	01	11	A1B	IH	-	HOT OIL	L	HEADER	HE-312002	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
855	80	HOR(H)	009	01	12	A1B	IH	-	HOT OIL	L	HEADER	HE-312002	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
856	80	HOR(H)	009	01	15	A1B	IH	-	HOT OIL	L	HE-312001	HEADER	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
857	80	HOS(H)	009	01	16	A1B	IH	-	HOT OIL	L	HEADER	VD-312101	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
858	80	BO	009	02	01	A1A	IH	-	BASE OIL	L	BYPASS LINE OF HE	VD-312101	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
859	200	V	009	02	02	A6C	IH	-	VAPOR	V	VD-312101	B-312101	6	8	60	120	*	*	900	600	15	150	Water	*	1.5	6.3 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
860	100	AD	009	02	04	A1A	IH	IE	ADDITIVE	L	M-312103	VD-312101	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
861	200	PD	009	02	05	A1A	-	-	POWDER	S	BC-312101	VD-312101	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
862	100	AD	009	02	06	A1A	IH	IE	ADDITIVE	L	VD-312101	M-312103	6	8	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
863	150	AD	009	02	07	A1A	IH	IE	ADDITIVE	L	VD-312101	P-312101	0.6	1.2	60	150	*	*	900	600	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
864	100	AD	009	02	08	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-312101	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
865	100	AD	009	02	09	A1A	IH	IE	ADDITIVE	L	P-312101	HLDV-312101	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
866	100	AD	009	02	18	A1A	IH	IE	ADDITIVE	L	OUTGOING LINE O	BYPASS LINE (	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
867	25	CA	009	02	10	G6C	-	-	OMPRESSED AI	V	*	VD-312101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
868	25	PA	009	02	11	G6C	-	-	PLANT AIR	V	*	VD-312101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
869	100	AD	009	02	12	A1A	IH	IE	ADDITIVE	L	P-312101	AT-200012	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
870	100	AD	009	02	13	A1A	IH	IE	ADDITIVE	L	P-312101	AT-200016	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		

S.NO		LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
	PIPE SIZE (DN)	FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
871	100	AD	009	02	14	A1A	IH	IE	ADDITIVE	L	P-312101	AT-200014	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
872	100	AD	009	02	15	A1A	IH	IE	ADDITIVE	L	P-312101	AT-200013	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
873	100	AD	009	02	16	A1A	IH	IE	ADDITIVE	L	P-312101	S-312101	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
874	25	AD	009	02	17	A1A	IH	IE	ADDITIVE	L	S-312101	TANK TRUCK	6	8	60	150	*	30	900	600	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
875	25	CA	009	02	19	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	60	150	*	30	900	600	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
876	25	CHWS	009	02	20	A1A	-	-	CHILLED WATEF	L	*	SKID	6	8	60	150	*	30	1000	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
877	25	CHWR	009	02	21	A1A	-	-	CHILLED WATEF	L	*	SKID	6	8	60	150	*	30	1000	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-009	H1	P&ID FOR VI SOLUBILIZATION PLANT		
878	150	BO	010	01	01	A1A	-	-	BASE OIL	L	ST-100006	P-100006	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
879	100	BO	010	01	02	A1A	-	-	BASE OIL	L	TAPPING FROM BC	P-100006	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
880	100	BO	010	01	03	A1A	-	-	BASE OIL	L	P-100006	ST-100006	6	8	60	100	*	60	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
881	100	BO	010	01	04	A1A	-	-	BASE OIL	L	P-100006	INPUT LINE OI	6	8	60	100	*	60	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
882	100	BO	010	01	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100006	6	8	60	100	*	60	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
883	100	BO	010	01	06	A1A	-	-	BASE OIL	L	P-100006	BULK LOADIN	6	8	60	100	*	60	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
884	100	BO	010	01	07	A1A	-	-	BASE OIL	L	P-100006	ST-100008	6	8	60	100	*	60	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
885	100	BO	010	01	08	A1A	-	-	BASE OIL	L	P-100006	ST-100010	6	8	60	100	*	60	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
886	150	BO	010	02	01	A1A	-	-	BASE OIL	L	ST-100012	P-100012	0.6	1.2	60	100	*	*	900	15	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
887	100	BO	010	02	02	A1A	-	-	BASE OIL	L	TAPPING FROM BC	P-100012	0.6	1.2	60	100	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
888	100	BO	010	02	03	A1A	-	-	BASE OIL	L	P-100012	ST-100012	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
889	100	BO	010	02	04	A1A	-	-	BASE OIL	L	P-100012	INPUT LINE OI	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
890	100	BO	010	02	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100012	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
891	100	BO	010	02	06	A1A	-	-	BASE OIL	L	P-100012	BULK LOADIN	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
892	100	BO	010	02	07	A1A	-	-	BASE OIL	L	P-100012	ST-100014	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
893	150	BO	010	03	01	A1A	-	-	BASE OIL	L	ST-100013	P-100013	0.6	1.2	60	100	*	*	900	20	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
894	100	BO	010	03	02	A1A	-	-	BASE OIL	L	TAPPING FROM BC	P-100013	0.6	1.2	60	100	*	*	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
895	100	BO	010	03	04	A1A	-	-	BASE OIL	L	P-100013	ST-100013	6	8	60	100	*	60	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
896	100	BO	010	03	05	A1A	-	-	BASE OIL	L	P-100013	INPUT LINE OI	6	8	60	100	*	60	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
897	100	BO	010	03	06	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100013	6	8	60	100	*	60	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
898	100	BO	010	03	07	A1A	-	-	BASE OIL	L	P-100013	BULK LOADIN	6	8	60	100	*	60	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
899	100	BO	010	03	08	A1A	-	-	BASE OIL	L	P-100013	ST-100009	6	8	60	100	*	60	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
900	100	BO	010	03	09	A1A	-	-	BASE OIL	L	P-100013	ST-100011	6	8	60	100	*	60	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
901	150	BO	010	04	01	A1A	-	-	BASE OIL	L	ST-100017	P-100017	0.6	1.2	60	100	*	*	900	15	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
902	100	BO	010	04	02	A1A	-	-	BASE OIL	L	TAPPING FROM BC	P-100017	0.6	1.2	60	100	*	*	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
903	100	BO	010	04	03	A1A	-	-	BASE OIL	L	P-100017	ST-100017	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
904	100	BO	010	04	04	A1A	-	-	BASE OIL	L	P-100017	INPUT LINE OI	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
905	100	BO	010	04	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100017	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
906	100	BO	010	04	06	A1A	-	-	BASE OIL	L	P-100017	BULK LOADIN	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
907	100	BO	010	04	07	A1A	-	-	BASE OIL	L	P-100017	ST-100019	6	8	60	100	*	60	900	15	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
908	150	BO	010	05	01	A1A	IH	-	BASE OIL	L	ST-100016	P-100016	0.6	1.2	60	120	*	*	900	17	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
909	100	BO	010	05	02	A1A	-	-	BASE OIL	L	TAPPING FROM BC	P-100016	0.6	1.2	60	120	*	*	900	17	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
910	100	BO	010	05	03	A1A	IH	-	BASE OIL	L	P-100016	ST-100016	6	8	60	120	*	60	900	17	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
911	100	BO	010	05	04	A1A	IH	-	BASE OIL	L	P-100016	INPUT LINE OI	6	8	60	120	*	60	900	17	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
912	100	BO	010	05	05	A1A	IH	-	BASE OIL	L	BYPASS LINE OF P-	P-100016	6	8	60	120	*	60	900	17	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
913	100	BO	010	05	06	A1A	-	-	BASE OIL	L	P-100016	BULK LOADIN	6	8	60	120	*	60	900	17	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
914	100	BO	010	05	07	A1A	-	-	BASE OIL	L	P-100016	ST-100018	6	8	60	120	*	60	900	17	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
915	80	HOS(M)	010	05	09	A1A	IH	-	HOT OIL	L	HEADER	ST-100016	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
916	80	HOS(M)	010	05	10	A1A	IH	-	HOT OIL	L	ST-100016	HEADER	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-010	H1	P&ID FOR LOBS LOADING PLANT		
917	80	BO	007	01	01	A1A	-	-	BASE OIL	L	ILB-313401 (LEG-2	150N GI	6	8	60	100	*	50	900	5000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
918	80	BO	007	01	03	A1A	IH	IE	BASE OIL	L	ILB-313401 (LEG-3	AD-313101	6	8	60	100	*	50	900	5000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
919	100	RX	007	01	05	P2A	-	-	UBBER EXTEND	L	TYPE 1	FP TANKS	6	8	60	80	*	50	900	5000	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
920	25	CA	007	01	07	G6C	-	-	OMPRESSED AI	V	*	TYPE 1	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
921	25	V	007	01	08	G6C	-	-	VAPOR	V	TYPE 1	DEMISTER	6	8	40	60	*	50	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
922	150	LO	007	01	10	A1A	-	-	LUBE OIL	L	TYPE 5	P-313102	0.6	1.2	60	100	*	50	900	5000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
923	100	LO	007	01	11	A1A	-	-	LUBE OIL	L	P-313102	TYPE 5	6	8	60	100	*	60	900	5000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
924	100	LO	007	01	12	A1A	-	-	LUBE OIL	L	BYPASS LINE OF P-	P-313102	6	8	60	100	*	60	900	5000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
925	150	LO	007	01	13	A1A	-	-	LUBE OIL	L	P-313102	SM-313102	6	8	60	100	*	60	900	5000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
926	80	BO	007	02	01	A1A	-	-	BASE OIL	L	ILB-313401	RINSING	6	8	60	100	*	50	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
927	80	BO	007	02	03	A1A	IH	IE	BASE OIL	L	ILB-313401	AD-313101	6	8	60	100	*	50	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
928	200	HOS(M)	007	02	04	A1A	IH	-	HOT OIL	L	HEADER	FP-313206	6	8	160	180	*	50	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
929	200	HOR(M)	007	02	05	A1A	IH	-	HOT OIL	L	FP-313206	HEADER	6	8	160	180	*	50	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
930	50	HOS(M)	007	02	06	A1A	IH	-	HOT OIL	L	HEADER	AD-313101	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
931	50	HOR(M)	007	02	07	A1A	IH	-	HOT OIL	L	HEADER	AD-313101	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
932	80	HOS(M)	007	02	08	A1A	IH	-	HOT OIL	L	HEADER	HE-313002	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
933	80	HOR(M)	007	02	09	A1A	IH	-	HOT OIL	L	HEADER	HE-313002	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
934	50	AD	007	02	12	A1A	IH	IE	ADDITIVE	L	AD-313101	P-313101	0.6	1.2	60	120	*	*	900	5800	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
935	50	AD	007	02	13	A1A	IH	IE	ADDITIVE	L	P-313101	LEG-4	6	8	60	120	*	5	900	5800	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
936	50	AD	007	02	14	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-313101	6	8	60	120	*	5	900	5800	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
937	100	BO	007	02	17	A1A	IH	IE	BASE OIL	L	HE-313002	ILB-313401	6	8	60	120	*	*	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
938	50	BO	007	02	18	A1A	-	-	BASE OIL	L	ILB-313401	AD-313101	6	8	60	120	*	50	900	5800	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
939	50	AD	007	02	19	A1A	IH	IE	ADDITIVE	L	P-313101	OUTGOING LI	6	8	60	120	*	50	900	5800	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
940	50	BO	007	03	01	A1A	-	-	BASE OIL	L	ST-100019/12	FP-313201	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
941	100	RX	007	03	02	A1A	-	-	UBBER EXTEND	L	PIG HEADER TYPE	FP-313201	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
942	100	RX	007	03	03	A1A	IH	IE	UBBER EXTEND	L	FP-313201	FP-313202	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
943	80	RX	007	03	05	A1A	-	-	UBBER EXTEND	L	FP-313202	P-313201	0.6	1.2	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
944	100	RX	007	03	04	A1A	IH	IE	UBBER EXTEND	L	TAPPING LINE FRO	P-313201	0.6	1.2	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
945	80	RX	007	03	06	A1A	IH	IE	UBBER EXTEND	L	BYPASS LINE OF IN	P-313202	0.6	1.2	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
946	80	RX	007	03	07	A1A	IH	IE	UBBER EXTEND	L	BYPASS LINE OF P-	P-313201	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
947	80	RX	007	03	08	A1A	IH	IE	UBBER EXTEND	L	F-313201	BULK FILLING	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
948	100	RX	007	03	09	A1A	IH	IE	UBBER EXTEND	L	P-313201	FP-313202	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
949	25	CA	007	03	10	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
950	80	HOS(M)	007	03	11	A1A	IH	-	HOT OIL	L	HEADER	FP-313201	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
951	80	HOR(M)	007	03	12	A1A	IH	-	HOT OIL	L	HEADER	FP-313201	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
952	40	CA	007	03	13	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313201	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
953	40	CA	007	03	14	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313201	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
954	40	CA	007	03	15	G6C	-	-	OMPRESSED AI	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
955	80	RX	007	03	16	A1A	IH	IE	UBBER EXTEND	L	P-313201	F-313201	6	8	60	120	*	40	900	770	15	200	Water	*	2.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR RUBBER EXTENDER PLANT		
956	50	BO	007	04	01	A1A	-	-	BASE OIL	L	150N G II	FP-313202	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
957	100	RX	007	04	02	A1A	-	-	UBBER EXTEND	L	PIG HEADER TYPE	FP-313202	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
958	100	RX	007	04	03	A1A	IH	IE	UBBER EXTEND	L	FP-313201	FP-313202	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
959	80	RX	007	04	04	A1A	-	-	UBBER EXTEND	L	BOTTOM LINE OF I	FP-313201	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
960	100	RX	007	04	05	A1A	IH	IE	UBBER EXTEND	L	FP-313202	FP-313201	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
961	25	CA	007	04	06	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
962	40	CA	007	04	07	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313202	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
963	40	CA	007	04	08	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313202	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
964	40	CA	007	04	09	G6C	-	-	OMPRESSED AI	V	*	FP-313202	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
965	80	HOS(M)	007	04	10	A1A	IH	-	HOT OIL	L	HEADER	FP-313202	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
966	80	HOR(M)	007	04	11	A1A	IH	-	HOT OIL	L	HEADER	FP-313202	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
967	50	BO	007	05	01	A1A	-	-	BASE OIL	L	FP-313202	FP-313203	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
968	100	RX	007	05	02	A1A	-	-	UBBER EXTEND	L	FP-313202 TYPE 5	FP-313203	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
969	100	RX	007	05	03	A1A	IH	IE	UBBER EXTEND	L	FP-313203	FP-313204	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
970	100	RX	007	05	04	A1A	IH	IE	UBBER EXTEND	L	FP-313204	P-313203	0.6	1.2	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
971	80	RX	007	05	05	A1A	IH	IE	UBBER EXTEND	L	BYPASS LINE FOR I	P-313203	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
972	80	RX	007	05	06	A1A	-	-	UBBER EXTEND	L	BYPASS LINE OF P-	P-313203	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
973	80	RX	007	05	07	A1A	IH	IE	UBBER EXTEND	L	F-313203	TYPE 1	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
974	80	RX	007	05	08	P2A	-	-	UBBER EXTEND	L	TYPE 1	PIGGED MANI	6	8	60	120	*	40	900	770	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
975	100	RX	007	05	09	A1A	IH	IE	UBBER EXTEND	L	P-313203	FP-313204	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
976	25	CA	007	05	10	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
977	80	HOS(M)	007	05	11	A1A	IH	-	HOT OIL	L	HEADER	FP-313203	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
978	80	HOR(M)	007	05	12	A1A	IH	-	HOT OIL	L	HEADER	FP-313203	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
979	40	CA	007	05	13	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313203	6	8	40	60	*	*	1.25	0.00001	10	200	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
980	40	CA	007	05	14	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313203	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
981	40	CA	007	05	15	G6C	-	-	OMPRESSED AI	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
982	25	CA	007	05	16	G6C	-	-	OMPRESSED AI	V	*	TYPE 1	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
983	25	V	007	05	17	G6C	-	-	VAPOR	V	TYPE 1	DEMISTER	6	8	60	120	*	*	900	770	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
984	50	BO	007	06	01	A1A	-	-	BASE OIL	L	FP-313203	FP-313204	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
985	100	RX	007	06	02	A1A	-	-	UBBER EXTEND	L	FP-313203 TYPE 5	FP-313204	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
986	100	RX	007	06	03	A1A	IH	IE	UBBER EXTEND	L	FP-313204	FP-313203	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
987	25	CA	007	06	04	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
988	100	RX	007	06	05	A1A	IH	IE	UBBER EXTEND	L	FP-313203	FP-313204	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
989	80	HOS(M)	007	06	06	A1A	IH	-	HOT OIL	L	HEADER	FP-313204	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
990	80	HOR(M)	007	06	07	A1A	IH	-	HOT OIL	L	HEADER	FP-313204	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
991	40	CA	007	06	08	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313204	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
992	40	CA	007	06	09	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313204	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
993	40	CA	007	06	10	G6C	-	-	OMPRESSED AI	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
994	50	BO	007	07	01	A1A	-	-	BASE OIL	L	FP-313204	FP-313205	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
995	100	RX	007	07	02	A1A	-	-	UBBER EXTEND	L	FP-313204 TYPE 5	FP-313205	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
996	100	RX	007	07	03	A1A	IH	IE	UBBER EXTEND	L	FP-313206	P-313205	0.6	1.2	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
997	80	RX	007	07	04	A1A	IH	IE	UBBER EXTEND	L	BYPASS LINE FOR I	P-313205	0.6	1.2	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
998	80	RX	007	07	05	A1A	-	-	UBBER EXTEND	L	BYPASS LINE OF P-	P-313205	6	8	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
999	80	RX	007	07	06	A1A	IH	IE	UBBER EXTEND	L	F-313205	TYPE 9	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1000	80	RX	007	07	07	P2A	-	-	UBBER EXTEND	L	TYPE 9	PIGGED MANI	6	8	60	120	*	40	900	770	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1001	100	RX	007	07	08	A1A	IH	IE	UBBER EXTEND	L	P-313205	FP-313206	6	8	60	120	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1002	100	RX	007	07	09	A1A	IH	IE	UBBER EXTEND	L	FP-313205	FP-313206	ATM	FULL OF LIQUID	60	120	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1003	80	HOS(M)	007	07	10	A1A	IH	-	HOT OIL	L	HEADER	FP-313205	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1004	80	HOR(M)	007	07	11	A1A	IH	-	HOT OIL	L	HEADER	FP-313205	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1005	40	CA	007	07	12	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313205	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1006	40	CA	007	07	13	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313205	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1007	40	CA	007	07	14	G6C	-	-	OMPRESSED AI	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1008	25	CA	007	07	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1009	25	CA	007	07	16	G6C	-	-	OMPRESSED AI	V	*	TYPE 9	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1010	25	V	007	07	17	G6C	-	-	VAPOR	V	TYPE 9	DEMISTER	6	8	40	60	*	*	900	770	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1011	50	BO	007	08	01	A1A	-	-	BASE OIL	L	FP-313205	FP-313206	ATM	FULL OF LIQUID	60	120	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1012	100	RX	007	08	02	A1A	-	-	UBBER EXTEND	L	FP-313205 TYPE 5	FP-313206	ATM	FULL OF LIQUID	60	120	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1013	100	RX	007	08	03	A1A	IH	IE	UBBER EXTEND	L	FP-313206	FP-313205	ATM	FULL OF LIQUID	60	120	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1014	100	RX	007	08	04	A1A	-	-	UBBER EXTEND	L	FP-313205	FP-313206	ATM	FULL OF LIQUID	60	120	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1015	25	CA	007	08	05	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1016	40	CA	007	08	06	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313206	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1017	40	CA	007	08	07	G6C	-	-	OMPRESSED AI	V	CONTROL BOX	FP-313206	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1018	40	CA	007	08	08	G6C	-	-	OMPRESSED AI	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1019	80	HOR(M)	007	08	09	A1A	IH	-	HOT OIL	L	HEADER	FP-313206	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1020	80	HOS(M)	007	08	10	A1A	IH	-	HOT OIL	L	HEADER	FP-313206	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1021	40	AD	007	09	01	A1A	-	-	ADDITIVE	L	MB-313301	P-313301	0.6	1.2	60	100	*	*	900	18	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1022	25	AD	007	09	02	A1A	-	-	ADDITIVE	L	P-313301	OUTGOING LI	6	8	60	100	*	3	900	18	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1023	25	AD	007	09	03	A1A	-	-	ADDITIVE	L	P-313301	MB-313301	6	8	60	100	*	3	900	18	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1024	40	AD	007	09	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-313301	0.6	1.2	60	100	*	*	900	18	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1025	25	AD	007	09	05	A1A	-	-	ADDITIVE	L	P-313301	AD-313101	6	8	60	100	*	3	900	18	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1026	25	AD	007	09	06	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-313301	6	8	60	100	*	3	900	18	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1027	50	V	007	09	07	A1A	-	-	VAPOR	V	MB-313301	SAFE LOCATIO	6	8	60	100	*	*	900	18	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1028	80	AD	007	10	01	A1A	-	-	ADDITIVE	L	MB-313302	P-313302	0.6	1.2	60	100	*	*	900	60	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1029	40	AD	007	10	02	A1A	-	-	ADDITIVE	L	P-313302	OUTGOING LI	6	8	60	100	*	10	900	60	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1030	25	AD	007	10	03	A1A	-	-	ADDITIVE	L	P-313302	MB-313302	6	8	60	100	*	10	900	60	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1031	40	AD	007	10	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-313302	0.6	1.2	60	100	*	*	900	60	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1032	40	AD	007	10	05	A1A	-	-	ADDITIVE	L	P-313302	AD-313101	6	8	60	100	*	10	900	60	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1033	40	AD	007	10	06	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-313302	6	8	60	100	*	10	900	60	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1034	50	V	007	10	07	A1A	-	-	VAPOR	V	MB-313302	SAFE LOCATIC	6	8	60	100	*	*	900	18	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1035	50	AD	007	11	01	A1A	-	-	ADDITIVE	L	MB-313304	P-313304	0.6	1.2	60	100	*	*	900	347	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1036	40	AD	007	11	02	A1A	-	-	ADDITIVE	L	P-313304	OUTGOING LI	6	8	60	100	*	0.3	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1037	25	AD	007	11	03	A1A	-	-	ADDITIVE	L	P-313304	MB-313304	6	8	60	100	*	0.3	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1038	40	AD	007	11	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-313304	0.6	1.2	60	100	*	*	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1039	40	AD	007	11	05	A1A	-	-	ADDITIVE	L	P-313304	AD-313101	6	8	60	100	*	0.3	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1040	40	AD	007	11	06	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-313304	6	8	60	100	*	0.3	900	347	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1041	50	V	007	11	07	A1A	-	-	VAPOR	V	MB-313303	SAFE LOCATIC	6	8	60	100	*	*	900	18	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1042	25	V	007	12	01	G6C	-	-	VAPOR	V	PM-313501	DEMISTER	6	8	40	60	*	*	900	347	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1043	80	RX	007	12	02	P2A	-	-	UBBER EXTEND	L	PM-313501	LOADING ARM	6	8	60	80	*	*	900	347	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1044	80	RX	007	12	03	A1A	-	-	UBBER EXTEND	L	FP-313207	PIG TEE TYPE	6	8	60	100	*	*	900	347	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1045	80	RX	007	12	04	P2A	-	-	UBBER EXTEND	L	PM-313501 TYPE 1	DRUM FILLIN	6	8	60	80	*	*	900	347	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1046	25	PA	007	12	05	G6C	-	-	PLANT AIR	V	*	PM-313501	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1047	25	IA	007	12	06	G6C	-	-	OMPRESSED AI	V	*	PM-313501	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1048	100	BN	007	13	01	A1A	-	-	BN EXTRACT	L	SO-313301	P-313501	0.6	1.2	60	100	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1049	80	BN	007	13	02	A1A	-	-	BN EXTRACT	L	BYPASS LINE OF P-	P-313501	6	8	60	100	*	20	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1050	80	BN	007	13	03	A1A	-	-	BN EXTRACT	L	P-313501	BITUMEN	6	8	60	100	*	20	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1051	80	BN	007	13	04	A1A	-	-	BN EXTRACT	L	P-313501	PIG HEADER	6	8	60	100	*	20	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1052	80	BN	007	13	05	A1A	-	-	BN EXTRACT	L	P-313501	SO-313501	6	8	60	100	*	20	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1053	25	V	007	13	06	G6C	-	-	VAPOR	V	TYPE 9	DEMISTER	ATM	FULL OF LIQUID	40	60	*	*	900	770	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1054	100	BN	007	13	07	A1A	-	-	BN EXTRACT	L	ILB HEADER TYPE 9	SO-313501	ATM	FULL OF LIQUID	60	100	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1055	25	CA	007	13	08	G6C	-	-	OMPRESSED AI	V	*	TYPE 9	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1056	100	RX	007	14	01	A1A	-	-	UBBER EXTEND	L	FP-313207	P-313207	0.6	1.2	60	100	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1057	80	RX	007	14	02	A1A	-	-	UBBER EXTEND	L	BYPASS LINE OF P-	P-313207	6	8	60	100	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1058	80	RX	007	14	03	A1A	-	-	UBBER EXTEND	L	P-313207	OUTGOING LI	6	8	60	100	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1059	80	RX	007	14	04	A1A	-	-	UBBER EXTEND	L	P-313207	FP-313207	6	8	60	100	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1060	80	RX	007	14	05	A1A	-	-	UBBER EXTEND	L	F-313207	DRUM FILLIN	6	8	60	100	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1061	100	RX	007	14	06	A1A	-	-	UBBER EXTEND	L	TANK TRUCK	P-313207	0.6	1.2	60	100	*	*	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1062	80	RX	007	14	07	A1A	-	-	UBBER EXTEND	L	P-313207	F-313207	6	8	60	100	*	40	900	770	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1063	80	V	007	15	01	G6C	-	-	VAPOR	V	DV-313601	SAFE LOCATIC	6	8	40	60	*	*	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1064	50	V	007	15	02	G6C	-	-	VAPOR	V	DV-313601	DRAIN	6	8	40	60	*	*	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1065	50	V	007	15	03	G6C	-	-	VAPOR	V	MANIFOLD	DV-313601	6	8	40	60	*	*	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1066	80	V	007	16	01	G6C	-	-	VAPOR	V	DV-313602	SAFE LOCATIC	6	8	40	60	*	*	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1067	50	V	007	16	02	G6C	-	-	VAPOR	V	DV-313602	DRAIN	6	8	40	60	*	*	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1068	50	V	007	16	03	G6C	-	-	VAPOR	V	MANIFOLD	DV-313602	6	8	40	60	*	*	900	1	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-007	H1	P&ID FOR RUBBER EXTENDER PLANT		
1069	150	AD	113	01	01	A1A	IH	IE	ADDITIVE	L	SH-200001	P-200001	0.6	1.2	60	120	*	*	900	1481	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1070	100	AD	113	01	02	A1A	IH	IE	ADDITIVE	L	AT-200001	OUTGOING LI	ATM	FULL OF LIQUID	60	120	*	*	900	1481	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1071	100	AD	113	01	03	A1A	IH	IE	ADDITIVE	L	P-200001	AT-200001	6	8	60	120	*	20	900	1481	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1072	100	AD	113	01	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200001	0.6	1.2	60	120	*	*	900	1481	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1073	80	AD	113	01	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200001	6	8	60	120	*	20	900	1481	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1074	100	AD	113	01	06	A1A	IH	IE	ADDITIVE	L	P-200001	SMB-302001	6	8	60	120	*	20	900	1481	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1075	80	HOS(S)	113	01	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200001	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1076	80	HOR(S)	113	01	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200001	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1077	80	HOS(M)	113	01	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1078	80	HOS(M)	113	01	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200001	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1079	200	HOS(M)	113	01	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	AT-200029	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1080	200	HOS(M)	113	01	12	A1A	IH	-	HOT OIL	L	AT-200001	HEADER (PRIN	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1081	100	HOS(M)	113	01	13	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200029	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1082	100	HOS(M)	113	01	14	A1A	IH	-	HOT OIL	L	AT-200001	HEADER (SEC	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1083	100	AD	113	01	15	A1A	IH	IE	ADDITIVE	L	P-200001	OUTGOING LI	6	8	60	120	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1084	150	AD	113	02	01	A1A	IH	IE	ADDITIVE	L	SH-200002	P-200002	0.6	1.2	60	120	*	*	900	2678	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1085	100	AD	113	02	02	A1A	IH	IE	ADDITIVE	L	AT-200002	OUTGOING LI	6	8	60	120	*	*	900	2678	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1086	100	AD	113	02	03	A1A	IH	IE	ADDITIVE	L	P-200002	AT-200002	6	8	60	120	*	20	900	2678	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1087	100	AD	113	02	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200002	0.6	1.2	60	120	*	*	900	2678	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1088	80	AD	113	02	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200002	6	8	60	120	*	20	900	2678	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1089	100	AD	113	02	06	A1A	IH	IE	ADDITIVE	L	P-200002	SMB-302001	6	8	60	120	*	20	900	2678	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1090	100	AD	113	02	07	A1A	IH	IE	ADDITIVE	L	P-200002	V-302103	6	8	60	120	*	20	900	2678	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1091	80	HOS(S)	113	02	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200002	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1092	80	HOR(S)	113	02	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200002	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1093	80	HOS(M)	113	02	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200002	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1094	80	HOS(M)	113	02	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200002	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1095	100	AD	113	02	12	A1A	IH	IE	ADDITIVE	L	P-200002	OUTGOING LI	6	8	60	120	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1096	150	AD	113	03	01	A1A	IH	IE	ADDITIVE	L	SH-200003	P-200003	0.6	1.2	60	120	*	*	900	2205	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1097	100	AD	113	03	02	A1A	IH	IE	ADDITIVE	L	AT-200003	OUTGOING LI	6	8	60	120	*	*	900	2205	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1098	100	AD	113	03	03	A1A	IH	IE	ADDITIVE	L	P-200003	AT-200003	6	8	60	120	*	20	900	2205	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1099	100	AD	113	03	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200003	0.6	1.2	60	120	*	*	900	2205	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1100	80	AD	113	03	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200003	6	8	60	120	*	20	900	2205	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1101	100	AD	113	03	06	A1A	IH	IE	ADDITIVE	L	P-200003	SMB-302001	6	8	60	120	*	20	900	2205	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1102	80	HOS(S)	113	03	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200003	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1103	80	HOR(S)	113	03	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200003	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1104	80	HOS(M)	113	03	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200003	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1105	80	HOS(M)	113	03	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200003	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1106	100	AD	113	03	11	A1A	IH	IE	ADDITIVE	L	P-200003	OUTGOING LI	6	8	60	120	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1107	80	AD	113	04	01	A1A	-	-	ADDITIVE	L	AT-200004	P-200004	0.6	1.2	60	100	*	*	900	3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1108	50	AD	113	04	02	A1A	-	-	ADDITIVE	L	BOTTOM OF AT-20	P-200004	0.6	1.2	60	100	*	*	900	3	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1109	80	AD	113	04	03	A1A	-	-	ADDITIVE	L	TANK TRUCK	P-200004	0.6	1.2	60	100	*	*	900	3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1110	50	AD	113	04	04	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-200004	6	8	60	100	*	24	900	3	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1111	50	AD	113	04	05	A1A	-	-	ADDITIVE	L	P-200004	COOLANT	6	8	60	100	*	24	900	3	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1112	50	AD	113	04	06	A1A	-	-	ADDITIVE	L	AT-200004	CIRCULATION	ATM	FULL OF LIQUID	60	100	*	*	900	3	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1113	80	AD	113	04	07	A1A	-	-	ADDITIVE	L	AT-200006	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1114	50	AD	113	04	09	A1A	-	-	ADDITIVE	L	P-200004	AT-200006	6	8	60	100	*	24	900	3	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1115	150	AD	113	05	01	A1A	IH	IE	ADDITIVE	L	SH-200005	P-200005	0.6	1.2	60	120	*	*	900	1680	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1116	100	AD	113	05	02	A1A	IH	IE	ADDITIVE	L	AT-200005	OUTGOING LI	6	8	60	120	*	*	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1117	100	AD	113	05	03	A1A	IH	IE	ADDITIVE	L	P-200005	AT-200005	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1118	100	AD	113	05	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200005	0.6	1.2	60	120	*	*	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1119	80	AD	113	05	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200005	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1120	100	AD	113	05	06	A1A	IH	IE	ADDITIVE	L	P-200005	V-302101	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1121	100	AD	113	05	07	A1A	IH	IE	ADDITIVE	L	P-200005	SMB-302001	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1122	80	HOS(S)	113	05	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200005	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1123	80	HOR(S)	113	05	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200005	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1124	80	HOS(M)	113	05	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200005	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1125	80	HOS(M)	113	05	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200005	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1126	100	AD	113	05	12	A1A	IH	IE	ADDITIVE	L	P-200005	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1127	150	AD	113	06	01	A1A	IH	IE	ADDITIVE	L	SH-200007	P-200007	0.6	1.2	60	120	*	*	900	1365	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1128	100	AD	113	06	02	A1A	IH	IE	ADDITIVE	L	AT-200007	OUTGOING LI	6	8	60	120	*	*	900	1365	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1129	100	AD	113	06	03	A1A	IH	IE	ADDITIVE	L	P-200007	AT-200007	6	8	60	120	*	20	900	1365	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1130	100	AD	113	06	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200007	0.6	1.2	60	120	*	*	900	1365	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1131	80	AD	113	06	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200007	6	8	60	120	*	20	900	1365	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1132	100	AD	113	06	06	A1A	IH	IE	ADDITIVE	L	P-200007	SMB-302001	6	8	60	120	*	20	900	1365	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1133	80	HOS(S)	113	06	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200007	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1134	80	HOR(S)	113	06	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200007	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1135	80	HOS(M)	113	06	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200007	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1136	80	HOS(M)	113	06	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200007	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1137	100	AD	113	06	11	A1A	IH	IE	ADDITIVE	L	P-200007	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1138	150	AD	113	07	01	A1A	IH	IE	ADDITIVE	L	SH-200008	P-200008	0.6	1.2	60	120	*	*	900	1470	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1139	100	AD	113	07	02	A1A	IH	IE	ADDITIVE	L	AT-200008	OUTGOING LI	6	8	60	120	*	*	900	1470	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1140	100	AD	113	07	03	A1A	IH	IE	ADDITIVE	L	P-200008	AT-200008	6	8	60	120	*	20	900	1470	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1141	100	AD	113	07	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200008	0.6	1.2	60	120	*	*	900	1470	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1142	80	AD	113	07	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200008	6	8	60	120	*	20	900	1470	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1143	100	AD	113	07	06	A1A	IH	IE	ADDITIVE	L	P-200008	SMB-302001	6	8	60	120	*	20	900	1470	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1144	100	AD	113	07	07	A1A	IH	IE	ADDITIVE	L	P-200008	V-302101	6	8	60	120	*	20	900	1470	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1145	80	HOS(S)	113	07	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200008	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1146	80	HOR(S)	113	07	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200008	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1147	80	HOS(M)	113	07	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200008	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1148	80	HOS(M)	113	07	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200008	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1149	100	AD	113	07	12	A1A	IH	IE	ADDITIVE	L	P-200008	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1150	150	AD	113	08	01	A1A	IH	IE	ADDITIVE	L	SH-200009	P-200009	0.6	1.2	60	120	*	*	900	1733	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1151	100	AD	113	08	02	A1A	IH	IE	ADDITIVE	L	AT-200009	OUTGOING LI	6	8	60	120	*	*	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1152	100	AD	113	08	03	A1A	IH	IE	ADDITIVE	L	P-200009	AT-200009	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1153	100	AD	113	08	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200009	0.6	1.2	60	120	*	*	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1154	80	AD	113	08	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200009	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1155	100	AD	113	08	06	A1A	IH	IE	ADDITIVE	L	P-200009	SMB-302001	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1156	100	AD	113	08	07	A1A	IH	IE	ADDITIVE	L	P-200009	V-302101	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1157	80	HOS(S)	113	08	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200009	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1158	80	HOR(S)	113	08	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200009	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1159	80	HOS(M)	113	08	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200009	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1160	80	HOS(M)	113	08	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200009	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1161	100	AD	113	08	12	A1A	IH	IE	ADDITIVE	L	P-200009	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1162	100	AD	113	09	01	A1A	-	-	ADDITIVE	L	AT-200010	P-200010	0.6	1.2	60	100	*	*	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1163	100	AD	113	09	02	A1A	-	-	ADDITIVE	L	P-200010	AT-200010	6	8	60	100	*	24	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1164	80	AD	113	09	03	A1A	-	-	ADDITIVE	L	TANK TRUCK	P-200010	0.6	1.2	60	100	*	*	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1165	80	AD	113	09	04	A1A	-	-	ADDITIVE	L	BYPASS LINE FROM	P-200010	6	8	60	100	*	24	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1166	100	AD	113	09	05	A1A	-	-	ADDITIVE	L	P-200010	SMB-302002	6	8	60	100	*	24	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1167	100	AD	113	09	06	A1A	-	-	ADDITIVE	L	P-200010	GREASE	6	8	60	100	*	24	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1168	100	AD	113	09	07	A1A	-	-	ADDITIVE	L	P-200010	V-302102	6	8	60	100	*	24	900	16	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1169	100	AD	113	09	08	A1A	IH	IE	ADDITIVE	L	P-200010	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1170	150	AD	113	10	01	A1A	IH	IE	ADDITIVE	L	SH-200011	P-200011	0.6	1.2	60	120	*	*	900	49350	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1171	100	AD	113	10	02	A1A	IH	IE	ADDITIVE	L	AT-200011	OUTGOING LI	6	8	60	120	*	*	900	49350	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1172	150	AD	113	10	03	A1A	IH	IE	ADDITIVE	L	P-200011	AT-200011	6	8	60	120	*	15	900	49350	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1173	100	AD	113	10	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200011	0.6	1.2	60	120	*	*	900	49350	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1174	100	AD	113	10	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200011	6	8	60	120	*	15	900	49350	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1175	150	AD	113	10	06	A1A	IH	IE	ADDITIVE	L	P-200011	SMB-302002	6	8	60	120	*	15	900	49350	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1176	100	AD	113	10	07	A1A	IH	IE	ADDITIVE	L	P-200011	V-302102	6	8	60	120	*	15	900	49350	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1177	100	AD	113	10	08	A1A	IH	IE	ADDITIVE	L	P-200011	V-302101	6	8	60	120	*	15	900	49350	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1178	80	HOS(S)	113	10	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200011	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1179	80	HOR(S)	113	10	10	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200011	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1180	80	HOS(M)	113	10	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200011	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1181	80	HOS(M)	113	10	12	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200011	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1182	100	AD	113	10	13	A1A	IH	IE	ADDITIVE	L	P-200011	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1183	150	AD	113	11	01	A1A	IH	IE	ADDITIVE	L	SH-200012	P-200012	0.6	1.2	60	120	*	*	900	11050	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1184	100	AD	113	11	02	A1A	IH	IE	ADDITIVE	L	AT-200012	OUTGOING LI	6	8	60	120	*	*	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1185	100	AD	113	11	03	A1A	IH	IE	ADDITIVE	L	P-200012	AT-200012	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1186	100	AD	113	11	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200012	0.6	1.2	60	120	*	*	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1187	80	AD	113	11	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200012	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1188	100	AD	113	11	06	A1A	IH	IE	ADDITIVE	L	P-200012	SMB-302001	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1189	100	AD	113	11	07	A1A	IH	IE	ADDITIVE	L	P-200012	DRUM FILLING	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1190	100	AD	113	11	08	A1A	IH	IE	ADDITIVE	L	P-200012	BULK FILLING	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1191	100	AD	113	11	09	A1A	IH	IE	ADDITIVE	L	P-200012	V-302111	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1192	100	AD	113	11	10	A1A	IH	IE	ADDITIVE	L	P-200012	AD-313301	6	8	60	120	*	15	900	11050	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1193	80	HOS(S)	113	11	11	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200012	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1194	80	HOR(S)	113	11	12	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200012	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1195	80	HOS(M)	113	11	13	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200012	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1196	80	HOS(M)	113	11	14	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200012	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1197	25	CA	113	11	15	G6C	-	-	OMPRESSED AI	V	*	P-200012	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1198	100	AD	113	11	16	A1A	IH	IE	ADDITIVE	L	P-200012	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1199	150	AD	113	12	01	A1A	IH	IE	ADDITIVE	L	SH-200013	P-200013	0.6	1.2	60	120	*	*	900	10500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1200	100	AD	113	12	02	A1A	IH	IE	ADDITIVE	L	AT-200013	OUTGOING LI	6	8	60	120	*	*	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1201	100	AD	113	12	03	A1A	IH	IE	ADDITIVE	L	P-200013	AT-200013	6	8	60	120	*	15	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1202	100	AD	113	12	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200013	0.6	1.2	60	120	*	*	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1203	80	AD	113	12	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200013	6	8	60	120	*	15	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1204	100	AD	113	12	06	A1A	IH	IE	ADDITIVE	L	P-200013	V-302111	6	8	60	120	*	15	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1205	100	AD	113	12	07	A1A	IH	IE	ADDITIVE	L	P-200013	SMB-302111	6	8	60	120	*	15	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1206	80	HOS(S)	113	12	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200013	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1207	80	HOR(S)	113	12	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200013	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1208	80	HOS(M)	113	12	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200013	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1209	80	HOS(M)	113	12	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200013	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1210	25	CA	113	12	12	G6C	-	-	OMPRESSED AI	V	*	P-200013	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1211	100	AD	113	12	13	A1A	IH	IE	ADDITIVE	L	P-200013	DRUM FILLING	6	8	60	120	*	15	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1212	100	AD	113	12	14	A1A	IH	IE	ADDITIVE	L	P-200013	BULK FILLING	6	8	60	120	*	15	900	10500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1213	100	AD	113	12	15	A1A	IH	IE	ADDITIVE	L	P-200013	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1214	150	AD	113	13	01	A1A	IH	IE	ADDITIVE	L	SH-200014	P-200014	0.6	1.2	60	120	*	*	900	11813	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1215	100	AD	113	13	02	A1A	IH	IE	ADDITIVE	L	AT-200014	OUTGOING LI	6	8	60	120	*	*	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1216	100	AD	113	13	03	A1A	IH	IE	ADDITIVE	L	P-200014	AT-200014	6	8	60	120	*	15	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1217	100	AD	113	13	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200014	0.6	1.2	60	120	*	*	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1218	80	AD	113	13	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200014	6	8	60	120	*	15	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1219	100	AD	113	13	06	A1A	IH	IE	ADDITIVE	L	P-200014	SMB-302001	6	8	60	120	*	15	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1220	100	AD	113	13	07	A1A	IH	IE	ADDITIVE	L	P-200014	V-302111	6	8	60	120	*	15	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1221	100	AD	113	13	08	A1A	IH	IE	ADDITIVE	L	P-200014	DRUM FILLING	6	8	60	120	*	15	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1222	100	AD	113	13	09	A1A	IH	IE	ADDITIVE	L	P-200014	BULK FILLING	6	8	60	120	*	15	900	11813	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1223	80	HOS(S)	113	13	10	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200014	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1224	80	HOR(S)	113	13	11	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200014	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1225	80	HOS(M)	113	13	13	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200014	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1226	80	HOS(M)	113	13	14	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200014	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1227	25	CA	113	13	12	G6C	-	-	OMPRESSED AI	V	*	P-200014	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1228	100	AD	113	13	15	A1A	IH	IE	ADDITIVE	L	P-200014	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1229	150	AD	113	14	01	A1A	IH	IE	ADDITIVE	L	SH-200015	P-200015	0.6	1.2	60	120	*	*	900	2100	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1230	100	AD	113	14	02	A1A	IH	IE	ADDITIVE	L	AT-200015	OUTGOING LI	6	8	60	120	*	*	900	2100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1231	100	AD	113	14	03	A1A	IH	IE	ADDITIVE	L	P-200015	AT-200015	6	8	60	120	*	20	900	2100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1232	100	AD	113	14	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200015	0.6	1.2	60	120	*	*	900	2100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1233	80	AD	113	14	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200015	6	8	60	120	*	20	900	2100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1234	100	AD	113	14	06	A1A	IH	IE	ADDITIVE	L	P-200015	SMB-302001	6	8	60	120	*	20	900	2100	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1235	80	HOS(S)	113	14	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200015	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1236	80	HOR(S)	113	14	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200015	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1237	80	HOS(M)	113	14	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200015	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1238	80	HOS(M)	113	14	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200015	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1239	100	AD	113	14	11	A1A	IH	IE	ADDITIVE	L	P-200015	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1240	150	AD	113	15	01	A1A	IH	IE	ADDITIVE	L	SH-200016	P-200016	0.6	1.2	60	120	*	*	900	11970	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1241	100	AD	113	15	02	A1A	IH	IE	ADDITIVE	L	AT-200016	OUTGOING LI	6	8	60	120	*	*	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1242	100	AD	113	15	03	A1A	IH	IE	ADDITIVE	L	P-200016	AT-200016	6	8	60	120		15	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1243	100	AD	113	15	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200016	0.6	1.2	60	120	*	*	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1244	80	AD	113	15	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200016	6	8	60	120	*	15	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1245	100	AD	113	15	06	A1A	IH	IE	ADDITIVE	L	P-200016	V-302111	6	8	60	120	*	15	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1246	100	AD	113	15	07	A1A	IH	IE	ADDITIVE	L	P-200016	SMB-302001	6	8	60	120	*	15	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1247	100	AD	113	15	08	A1A	IH	IE	ADDITIVE	L	P-200016	DRUM FILLING	6	8	60	120	*	15	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1248	100	AD	113	15	09	A1A	IH	IE	ADDITIVE	L	P-200016	BULK FILLING	6	8	60	120	*	15	900	11970	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1249	80	HOS(S)	113	15	10	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200016	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1250	80	HOR(S)	113	15	11	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200016	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1251	80	HOS(M)	113	15	12	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200016	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1252	80	HOS(M)	113	15	13	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200016	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1253	25	CA	113	15	14	G6C	-	-	OMPRESSED AI	V	*	P-200016	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1254	100	AD	113	15	15	A1A	IH	IE	ADDITIVE	L	P-200016	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1255	150	AD	113	16	01	A1A	IH	IE	ADDITIVE	L	SH-200017	P-200017	0.6	1.2	60	120	*	*	900	45675	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1256	100	AD	113	16	02	A1A	IH	IE	ADDITIVE	L	AT-200017	OUTGOING LI	6	8	60	120	*	*	900	45675	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1257	150	AD	113	16	03	A1A	IH	IE	ADDITIVE	L	P-200017	AT-200017	6	8	60	120	*	15	900	45675	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1258	100	AD	113	16	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200017	0.6	1.2	60	120	*	*	900	45675	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1259	100	AD	113	16	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200017	6	8	60	120	*	15	900	45675	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1260	150	AD	113	16	06	A1A	IH	IE	ADDITIVE	L	P-200017	V-302101	6	8	60	120	*	15	900	45675	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1261	100	AD	113	16	07	A1A	IH	IE	ADDITIVE	L	P-200017	V-302102	6	8	60	120	*	15	900	45675	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1262	100	AD	113	16	08	A1A	IH	IE	ADDITIVE	L	P-200017	SMB-302002	6	8	60	120	*	15	900	45675	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1263	100	AD	113	16	09	A1A	IH	IE	ADDITIVE	L	P-200017	AL COMPLEX	6	8	60	120	*	15	900	45675	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1264	80	HOS(S)	113	16	13	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200017	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1265	80	HOR(S)	113	16	10	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200017	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1266	80	HOS(M)	113	16	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200017	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1267	80	HOS(M)	113	16	12	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200017	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1268	100	AD	113	16	14	A1A	IH	IE	ADDITIVE	L	P-200017	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1269	150	AD	113	17	01	A1A	IH	IE	ADDITIVE	L	SH-200018	P-200018	0.6	1.2	60	120	*	*	900	26000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1270	100	AD	113	17	02	A1A	IH	IE	ADDITIVE	L	AT-200018	OUTGOING LI	6	8	60	120	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1271	150	AD	113	17	03	A1A	IH	IE	ADDITIVE	L	P-200018	AT-200018	6	8	60	120	*	15	900	26000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1272	100	AD	113	17	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200018	0.6	1.2	60	120	*	*	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1273	100	AD	113	17	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200018	6	8	60	120	*	15	900	26000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1274	150	AD	113	17	06	A1A	IH	IE	ADDITIVE	L	P-200018	BITUMEN	6	8	60	120	*	15	900	26000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1275	80	HOS(S)	113	17	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200018	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1276	80	HOR(S)	113	17	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200018	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1277	80	HOS(M)	113	17	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200018	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1278	80	HOS(M)	113	17	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200018	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1279	100	AD	113	17	11	A1A	IH	IE	ADDITIVE	L	P-200018	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1280	150	AD	113	18	01	A1A	IH	IE	ADDITIVE	L	SH-200019	P-200019	0.6	1.2	60	120	*	*	900	6825	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1281	100	AD	113	18	02	A1A	IH	IE	ADDITIVE	L	AT-200019	OUTGOING LI	6	8	60	120	*	*	900	6825	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1282	100	AD	113	18	03	A1A	IH	IE	ADDITIVE	L	P-200019	AT-200019	6	8	60	120	*	15	900	6825	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1283	100	AD	113	18	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200019	0.6	1.2	60	120	*	*	900	6825	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1284	80	AD	113	18	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200019	6	8	60	120	*	15	900	6825	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1285	100	AD	113	18	06	A1A	IH	IE	ADDITIVE	L	P-200019	SMB-302002	6	8	60	120	*	15	900	6825	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1286	100	AD	113	18	07	A1A	IH	IE	ADDITIVE	L	P-200019	V-302102	6	8	60	120	*	15	900	6825	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1287	80	HOS(S)	113	18	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200019	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1288	80	HOR(S)	113	18	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200019	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1289	80	HOS(M)	113	18	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200019	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1290	80	HOS(M)	113	18	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200019	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1291	100	AD	113	18	12	A1A	IH	IE	ADDITIVE	L	P-200019	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1292	150	AD	113	19	01	A1A	IH	IE	ADDITIVE	L	SH-200020	P-200020	0.6	1.2	60	120	*	*	900	2468	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1293	100	AD	113	19	02	A1A	IH	IE	ADDITIVE	L	AT-200020	OUTGOING LI	6	8	60	120	*	*	900	2468	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1294	100	AD	113	19	05	A1A	IH	IE	ADDITIVE	L	P-200020	AT-200020	6	8	60	120	*	20	900	2468	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1295	100	AD	113	19	03	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200020	0.6	1.2	60	120	*	*	900	2468	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1296	80	AD	113	19	04	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200020	6	8	60	120	*	20	900	2468	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1297	100	AD	113	19	06	A1A	IH	IE	ADDITIVE	L	P-200020	SMB-302001	6	8	60	120	*	20	900	2468	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1298	80	HOS(S)	113	19	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200020	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1299	80	HOR(S)	113	19	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200020	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1300	80	HOS(M)	113	19	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200020	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1301	80	HOS(M)	113	19	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200020	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1302	100	AD	113	19	11	A1A	IH	IE	ADDITIVE	L	P-200020	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1303	150	AD	113	20	01	A1A	IH	IE	ADDITIVE	L	SH-200021	P-200021	0.6	1.2	60	120	*	*	900	1260	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1304	100	AD	113	20	02	A1A	IH	IE	ADDITIVE	L	AT-200021	OUTGOING LI	6	8	60	120	*	*	900	1260	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1305	100	AD	113	20	04	A1A	IH	IE	ADDITIVE	L	P-200021	AT-200021	6	8	60	120	*	20	900	1260	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1306	100	AD	113	20	03	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200021	0.6	1.2	60	120	*	*	900	1260	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1307	80	AD	113	20	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200021	6	8	60	120	*	20	900	1260	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1308	100	AD	113	20	06	A1A	IH	IE	ADDITIVE	L	P-200021	SMB-302001	6	8	60	120	*	20	900	1260	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1309	100	AD	113	20	07	A1A	IH	IE	ADDITIVE	L	P-200021	V-302101	6	8	60	120	*	20	900	1260	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1310	80	HOS(S)	113	20	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200021	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1311	80	HOR(S)	113	20	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200021	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1312	80	HOS(M)	113	20	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200021	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1313	80	HOS(M)	113	20	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200021	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1314	100	AD	113	20	12	A1A	IH	IE	ADDITIVE	L	P-200021	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1315	150	AD	113	21	01	A1A	IH	IE	ADDITIVE	L	SH-200022	P-200022	0.6	1.2	60	120	*	*	900	1208	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1316	100	AD	113	21	02	A1A	IH	IE	ADDITIVE	L	AT-200022	OUTGOING LI	6	8	60	120	*	*	900	1208	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1317	100	AD	113	21	03	A1A	IH	IE	ADDITIVE	L	P-200022	AT-200022	6	8	60	120	*	20	900	1208	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1318	100	AD	113	21	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200022	0.6	1.2	60	120	*	*	900	1208	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1319	80	AD	113	21	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200022	6	8	60	120	*	20	900	1208	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1320	100	AD	113	21	06	A1A	IH	IE	ADDITIVE	L	P-200022	SMB-302001	6	8	60	120	*	20	900	1208	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1321	100	AD	113	21	07	A1A	IH	IE	ADDITIVE	L	P-200022	V-302101	6	8	60	120	*	20	900	1208	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1322	80	HOS(S)	113	21	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200022	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1323	80	HOR(S)	113	21	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200022	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1324	80	HOS(M)	113	21	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200022	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1325	80	HOS(M)	113	21	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200022	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1326	100	AD	113	21	12	A1A	IH	IE	ADDITIVE	L	P-200022	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1327	150	AD	113	22	01	A1A	IH	IE	ADDITIVE	L	SH-200023	P-200023	0.6	1.2	60	120	*	*	900	1733	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1328	100	AD	113	22	02	A1A	IH	IE	ADDITIVE	L	AT-200023	OUTGOING LI	6	8	60	120	*	*	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1329	100	AD	113	22	03	A1A	IH	IE	ADDITIVE	L	P-200023	AT-200023	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1330	100	AD	113	22	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200023	0.6	1.2	60	120	*	*	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1331	80	AD	113	22	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200023	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1332	100	AD	113	22	06	A1A	IH	IE	ADDITIVE	L	P-200023	SMB-302001	6	8	60	120	*	20	900	1733	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1333	80	HOS(S)	113	22	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200023	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1334	80	HOR(S)	113	22	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200023	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1335	80	HOS(M)	113	22	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200023	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1336	80	HOS(M)	113	22	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200023	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1337	100	AD	113	22	11	A1A	IH	IE	ADDITIVE	L	P-200023	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1338	150	AD	113	23	01	A1A	IH	IE	ADDITIVE	L	SH-200024	P-200024	0.6	1.2	60	120	*	*	900	1680	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1339	100	AD	113	23	02	A1A	IH	IE	ADDITIVE	L	AT-200024	OUTGOING LI	6	8	60	120	*	*	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1340	100	AD	113	23	03	A1A	IH	IE	ADDITIVE	L	P-200024	AT-200024	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1341	100	AD	113	23	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200024	0.6	1.2	60	120	*	*	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1342	80	AD	113	23	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200024	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1343	100	AD	113	23	06	A1A	IH	IE	ADDITIVE	L	P-200024	SMB-302001	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1344	100	AD	113	23	07	A1A	IH	IE	ADDITIVE	L	P-200024	V-302101	6	8	60	120	*	20	900	1680	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1345	80	HOS(S)	113	23	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200024	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1346	80	HOR(S)	113	23	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200024	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1347	80	HOS(M)	113	23	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200024	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1348	80	HOS(M)	113	23	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200024	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1349	100	AD	113	23	12	A1A	IH	IE	ADDITIVE	L	P-200024	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1350	100	AD	113	24	01	A1A	-	-	ADDITIVE	L	AT-200025	P-200025	0.6	1.2	60	100	*	*	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1351	80	AD	113	24	02	A1A	-	-	ADDITIVE	L	P-200025	AT-200025	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1352	80	AD	113	24	03	A1A	-	-	ADDITIVE	L	TANK TRUCK	P-200025	0.6	1.2	60	100	*	*	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1353	80	AD	113	24	04	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-200025	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1354	80	AD	113	24	05	A1A	-	-	ADDITIVE	L	P-200025	FV-301206	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1355	80	AD	113	24	06	A1A	-	-	ADDITIVE	L	P-200025	FV-301205	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1356	80	AD	113	24	07	A1A	-	-	ADDITIVE	L	P-200025	FV-301204	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1357	80	AD	113	24	08	A1A	-	-	ADDITIVE	L	P-200025	FV-301203	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1358	80	AD	113	24	09	A1A	-	-	ADDITIVE	L	P-200025	FV-301202	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1359	80	AD	113	24	10	A1A	-	-	ADDITIVE	L	P-200025	FV-301201	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1360	80	AD	113	24	11	A1A	-	-	ADDITIVE	L	P-200025	PK-301705	6	8	60	100	*	24	900	88	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1361	80	AD	113	24	12	A1A	IH	IE	ADDITIVE	L	P-200025	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1362	100	AD	113	25	01	A1A	IH	IE	ADDITIVE	L	SH-200026	P-200026	0.6	1.2	60	120	*	*	900	851	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1363	100	AD	113	25	02	A1A	IH	IE	ADDITIVE	L	AT-200026	OUTGOING LI	6	8	60	120	*	*	900	851	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1364	80	AD	113	25	03	A1A	IH	-	ADDITIVE	L	P-200026	AT-200026	6	8	60	120	*	20	900	851	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1365	80	AD	113	25	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200026	0.6	1.2	60	120	*	*	900	851	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1366	80	AD	113	25	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FRON	P-200026	6	8	60	120	*	20	900	851	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1367	80	AD	113	25	06	A1A	IH	IE	ADDITIVE	L	P-200026	SMB-302001	6	8	60	120	*	20	900	851	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1368	80	HOS(S)	113	25	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200026	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1369	80	HOR(S)	113	25	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200026	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1370	80	HOS(M)	113	25	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200026	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1371	80	HOS(M)	113	25	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200026	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1372	80	AD	113	25	11	A1A	IH	IE	ADDITIVE	L	P-200026	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1373	150	AD	113	26	01	A1A	IH	IE	ADDITIVE	L	SH-200027	P-200027	0.6	1.2	60	120	*	*	900	1628	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1374	100	AD	113	26	02	A1A	IH	IE	ADDITIVE	L	AT-200027	OUTGOING LI	6	8	60	120	*	*	900	1628	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1375	100	AD	113	26	03	A1A	IH	IE	ADDITIVE	L	P-200027	AT-200027	6	8	60	120	*	20	900	1628	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1376	100	AD	113	26	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200027	0.6	1.2	60	120	*	*	900	1628	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1377	80	AD	113	26	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200027	6	8	60	120	*	20	900	1628	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1378	100	AD	113	26	06	A1A	IH	IE	ADDITIVE	L	P-200027	SMB-302001	6	8	60	120	*	20	900	1628	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1379	80	HOS(S)	113	26	07	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200027	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1380	80	HOR(S)	113	26	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200027	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1381	80	HOS(M)	113	26	09	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200027	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1382	80	HOS(M)	113	26	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200027	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1383	100	AD	113	26	11	A1A	IH	IE	ADDITIVE	L	P-200027	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1384	150	AD	113	27	01	A1A	IH	IE	ADDITIVE	L	SH-200028	P-200028	0.6	1.2	60	120	*	*	900	8138	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1385	100	AD	113	27	02	A1A	IH	IE	ADDITIVE	L	AT-200028	OUTGOING LI	6	8	60	120	*	*	900	8138	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1386	100	AD	113	27	03	A1A	IH	IE	ADDITIVE	L	P-200028	AT-200028	6	8	60	120	*	15	900	8138	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1387	100	AD	113	27	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200028	0.6	1.2	60	120	*	*	900	8138	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1388	80	AD	113	27	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200028	6	8	60	120	*	15	900	8138	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1389	100	AD	113	27	06	A1A	IH	IE	ADDITIVE	L	P-200028	SMB-302002	6	8	60	120	*	15	900	8138	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1390	100	AD	113	27	07	A1A	IH	IE	ADDITIVE	L	P-200028	V-302102	6	8	60	120	*	15	900	8138	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION										DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR														
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
1391	80	HOS(S)	113	27	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200028	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1392	80	HOR(S)	113	27	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200028	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1393	80	HOS(M)	113	27	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200028	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1394	80	HOS(M)	113	27	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200028	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1395	25	CA	113	27	12	G6C	-	-	OMPRESSED AI	V	*	P-200028	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1396	100	AD	113	27	13	A1A	IH	IE	ADDITIVE	L	P-200028	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1397	150	AD	113	28	01	A1A	IH	IE	ADDITIVE	L	SH-200029	P-200029	0.6	1.2	60	120	*	*	900	11700	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1398	100	AD	113	28	02	A1A	IH	IE	ADDITIVE	L	AT-200029	OUTGOING LI	6	8	60	120	*	*	900	11700	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1399	100	AD	113	28	03	A1A	IH	IE	ADDITIVE	L	P-200029	AT-200029	6	8	60	120	*	15	900	11700	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1400	100	AD	113	28	04	A1A	IH	IE	ADDITIVE	L	TANK TRUCK	P-200029	0.6	1.2	60	120	*	*	900	11700	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1401	80	AD	113	28	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE FROM	P-200029	6	8	60	120	*	15	900	11700	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1402	100	AD	113	28	06	A1A	IH	IE	ADDITIVE	L	P-200029	SMB-302002	6	8	60	120	*	15	900	11700	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1403	100	AD	113	28	07	A1A	IH	IE	ADDITIVE	L	P-200029	V-302102	6	8	60	120	*	15	900	11700	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1404	80	HOS(S)	113	28	08	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200029	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1405	80	HOR(S)	113	28	09	A1A	IH	-	HOT OIL	L	HEADER (SECONDA	AT-200029	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1406	80	HOS(M)	113	28	10	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200029	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1407	80	HOS(M)	113	28	11	A1A	IH	-	HOT OIL	L	HEADER (PRIMARY	SH-200029	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1408	25	CA	113	28	12	G6C	-	-	OMPRESSED AI	V	*	P-200029	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1409	100	AD	113	28	13	A1A	IH	IE	ADDITIVE	L	P-200029	OUTGOING LI	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-113	H1	P&ID FOR ADDITIVE TANKFARM		
1410	80	FP	117	01	01	P2A	-	-	NISHED PRODU	L	PM-500101	SCF-402002	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1411	80	FP	117	01	02	P2A	-	-	NISHED PRODU	L	PM-500101	SCF-402003	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1412	80	FP	117	01	03	P2A	-	-	NISHED PRODU	L	PM-500101	SCF-402001	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1413	80	FP	117	01	04	P2A	-	-	NISHED PRODU	L	PM-500101	PFL-402101	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1414	80	FP	117	01	05	P2A	-	-	NISHED PRODU	L	PM-500101	PFL-402103	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1415	80	FP	117	01	06	P2A	-	-	NISHED PRODU	L	PM-500101	LA-402001	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1416	80	FP	117	01	07	P2A	-	-	NISHED PRODU	L	PM-500101	LA-402002	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1417	80	FP	117	01	08	P2A	-	-	NISHED PRODU	L	PM-500101	LA-402003	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1418	80	FP	117	01	09	P2A	-	-	NISHED PRODU	L	PM-500101	LA-402004	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1419	80	FP	117	01	10	P2A	-	-	NISHED PRODU	L	PM-500101	LA-402005	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1420	80	FP	117	01	11	P2A	-	-	NISHED PRODU	L	PM-500101	DFL-402003/S	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1421	80	FP	117	01	12	P2A	-	-	NISHED PRODU	L	PM-500101	DFL-402001	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1422	80	FP	117	01	13	P2A	-	-	NISHED PRODU	L	PM-500101	DFL-402002	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1423	80	FP	117	01	14	P2A	-	-	NISHED PRODU	L	PM-500101	ABB REWORK	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1424	80	FP	117	01	15	P2A	-	-	NISHED PRODU	L	PM-500101	SPARE-1	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1425	80	FP	117	01	16	P2A	-	-	NISHED PRODU	L	PM-500101	SPARE-2	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1426	25	V	117	01	17	G6C	-	-	VAPOR	V	PM-500101	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1427	25	PA	117	01	18	G6C	-	-	PLANT AIR	V	*	PM-500101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1428	25	CA	117	01	19	G6C	-	-	VAPOR	V	*	PM-500101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1429	80	FP	117	01	20	P2A	-	-	NISHED PRODU	L	SPARE-1	PM-500101	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1430	80	FP	117	01	21	P2A	-	-	NISHED PRODU	L	SPARE-2	PM-500101	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
1431	80	FP	117	01	22	P2A	-	-	NISHED PRODU	L	SPARE-3	PM-500101	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1432	25	CA	117	02	01	G6C	-	-	VAPOR	V	*	FT-500038	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1433	80	FP	117	02	02	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500038	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1434	80	FP	117	02	03	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	FT-500038	ATM	FULL OF LIQUID	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1435	80	FP	117	02	04	B4C	-	-	NISHED PRODU	L	P-500038	FT-500038	0.6	1.2	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1436	100	FP	117	02	05	B4C	-	-	NISHED PRODU	L	FT-500038	P-500038	0.6	1.2	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1437	80	FP	117	02	06	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500038	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1438	80	FP	117	02	07	B4C	-	-	NISHED PRODU	L	P-500038	F-500038	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1439	80	FP	117	02	08	B4C	-	-	NISHED PRODU	L	F-500038	F-500102	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1440	80	FP	117	02	09	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	F-500038/F-5	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1441	80	FP	117	02	10	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF O	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1442	80	FP	117	02	11	B4C	-	-	NISHED PRODU	L	TYPE 8	LA-402001	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1443	80	FP	117	02	12	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF O	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1444	25	CA	117	03	01	G6C	-	-	OMPRESSED AI	V	*	F-500040	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1445	80	FP	117	03	02	B4C	-	-	NISHED PRODU	L	F-500154	FT-500040	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1446	80	FP	117	03	03	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500040	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1447	80	FP	117	03	04	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	FT-500040	ATM	FULL OF LIQUID	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1448	80	FP	117	03	05	B4C	-	-	NISHED PRODU	L	P-500040	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1449	100	FP	117	03	06	B4C	-	-	NISHED PRODU	L	FT-500040	P-500040	0.6	1.2	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1450	80	FP	117	03	07	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500040	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1451	80	FP	117	03	08	B4C	-	-	NISHED PRODU	L	P-500040	F-500040	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1452	80	FP	117	03	09	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	F-500040/F-5	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1453	80	FP	117	03	10	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF O	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1454	80	FP	117	03	11	B4C	-	-	NISHED PRODU	L	TYPE 8	LA-402002	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1455	80	FP	117	03	12	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF O	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1456	25	CA	117	04	01	G6C	-	-	OMPRESSED AI	V	*	F-500039A	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1457	80	FP	117	04	02	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500039A	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1458	80	FP	117	04	03	B4C	-	-	NISHED PRODU	L	F-500156	FT-500039	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1459	80	FP	117	04	04	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	FT-500039	ATM	FULL OF LIQUID	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1460	80	FP	117	04	05	B4C	-	-	NISHED PRODU	L	P-500039	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1461	100	FP	117	04	06	B4C	-	-	NISHED PRODU	L	FT-500039	P-500039	0.6	1.2	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1462	80	FP	117	04	07	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500039	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1463	80	FP	117	04	08	B4C	-	-	NISHED PRODU	L	P-500039	F-500039A	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1464	80	FP	117	04	09	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	F-500156	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1465	25	CA	117	04	10	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1466	80	FP	117	04	12	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 1	F-500039B/F-	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1467	80	FP	117	04	11	B4C	-	-	NISHED PRODU	L	PM-500101 TYPE 5	SCF-402002	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1468	80	FP	117	04	13	B4C	-	-	NISHED PRODU	L	BYPASS LINE OF O	OUTGOING LI	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1469	80	FP	117	04	14	B4C	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	DFL-402003	6	8	60	100	*	24	900	600	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1470	80	FP	117	05	01	A1A	-	-	NISHED PRODU	L	PIGGABLE MANIFO	F-500104	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1471	25	CA	117	05	02	G6C	-	-	OMPRESSED AI	V	*	F-500104	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1472	80	FP	117	05	03	A1A	-	-	NISHED PRODU	L	F-500104	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1473	80	FP	117	05	04	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	SCF-402001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1474	80	FP	117	05	05	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 6	F-500108	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1475	80	FP	117	05	06	A1A	-	-	NISHED PRODU	L	F-500108	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1476	25	CA	117	05	07	G6C	-	-	OMPRESSED AI	V	*	PM-500101 TY	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1477	25	CA	117	05	08	G6C	-	-	OMPRESSED AI	V	*	SCF-402001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1478	25	CA	117	05	09	G6C	-	-	OMPRESSED AI	V	*	SCF-402002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1479	80	FP	117	05	10	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	SCF-402002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1480	80	FP	117	05	11	A1A	-	-	NISHED PRODU	L	FT-500039	SCF-402002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1481	80	FP	117	06	01	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500010	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1482	80	FP	117	06	02	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 1	F-500010	6	8	60	100	*	24	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1483	25	CA	117	06	03	G6C	-	-	OMPRESSED AI	V	*	F-500110	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1484	25	CA	117	06	04	G6C	-	-	OMPRESSED AI	V	*	PM-500101 TY	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1485	80	FP	117	06	05	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500114	6	8	60	100	*	24	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1486	80	FP	117	06	06	A1A	-	-	NISHED PRODU	L	F-500114	PM-500101 TY	6	8	60	100	*	24	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1487	25	CA	117	06	07	G6C	-	-	OMPRESSED AI	V	*	PFL-402101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1488	25	CA	117	06	08	G6C	-	-	OMPRESSED AI	V	*	PFL-402103	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1489	80	FP	117	06	09	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	PFL-402101	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1490	80	FP	117	06	10	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	PFL-402103	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1491	80	FP	117	07	01	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 6	F-500116	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1492	25	CA	117	07	02	G6C	-	-	OMPRESSED AI	V	*	F-500116	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1493	80	FP	117	07	03	A1A	-	-	NISHED PRODU	L	F-500116	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1494	80	FP	117	07	04	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500120	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1495	80	FP	117	07	05	A1A	-	-	NISHED PRODU	L	F-500120	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1496	25	CA	117	07	06	G6C	-	-	OMPRESSED AI	V	*	F-500120	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1497	80	FP	117	07	07	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	DFL-402001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-118	H1	P&ID FOR FILLING PLANT		
1498	25	CA	117	07	08	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1499	25	CA	117	07	09	G6C	-	-	OMPRESSED AI	V	*	DFL-402002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1500	80	FP	117	07	10	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	DFL-402002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-118	H1	P&ID FOR FILLING PLANT		
1501	80	FP	117	08	01	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500124	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1502	25	CA	117	08	02	G6C	-	-	OMPRESSED AI	V	*	F-500124	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1503	80	FP	117	08	03	A1A	-	-	NISHED PRODU	L	F-500124	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1504	25	CA	117	08	04	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1505	80	FP	117	08	05	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	LA-402003	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-118	H1	P&ID FOR FILLING PLANT		
1506	25	FP	117	08	06	A1A	-	-	NISHED PRODU	L	S-500123	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-118	H1	P&ID FOR FILLING PLANT		
1507	80	FP	117	09	01	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE 5	F-500126	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1508	25	CA	117	09	02	G6C	-	-	OMPRESSED AI	V	*	F-500126	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1509	80	FP	117	09	03	A1A	-	-	NISHED PRODU	L	F-500126	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1510	25	CA	117	09	04	G6C	-	-	OMPRESSED AI	V	*	F-500128	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1511	80	FP	117	09	05	A1A	-	-	NISHED PRODU	L	PM-500101 TYPE S	F-500128	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1512	80	FP	117	09	06	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	LA-402004	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1513	80	FP	117	09	07	A1A	-	-	NISHED PRODU	L	F-500128	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1514	25	CA	117	09	08	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1515	25	CA	117	09	09	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1516	80	FP	117	09	10	A1A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	LA-402005	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1517	25	BN	117	09	11	A1A	-	-	BN EXTRACT	L	S-500125	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1518	25	BN	117	09	12	A1A	-	-	BN EXTRACT	L	S-500127	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1519	80	BN	117	10	03	P2A	-	-	BN EXTRACT	L	BN EXTRACT TYPE	TYPE 10	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1520	25	CA	117	10	04	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1521	80	FP	117	10	05	P2A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	LA-407001	6	8	60	100	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1522	80	BN	117	10	06	P2A	-	-	BN EXTRACT	L	BN EXTRACT TYPE	PIG RECEIVER	6	8	60	80	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1523	25	CA	117	10	07	G6C	-	-	OMPRESSED AI	V	*	BN EXTRACT T	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1524	80	FP	117	10	10	P2A	-	-	NISHED PRODU	L	SIMPLIFIED PIG RE	LA-407002	6	8	60	100	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1525	25	BN	117	10	11	A1A	-	-	BN EXTRACT	L	S-407001	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1526	25	BN	117	10	12	A1A	-	-	BN EXTRACT	L	S-407002	PM-500101 TY	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1527	50	TO	117	12	01	B4C	-	-	ANSFORMER C	L	DFL-404001	SLOP TANK	6	8	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1528	50	TO	117	12	02	B4C	-	-	ANSFORMER C	L	DFL-404002	SLOP TANK	6	8	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 10S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1529	25	BN	117	13	01	B4C	-	-	BN EXTRACT	L	S-411002	LA-411002	6	8	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1530	25	CL	117	14	01	B4C	-	-	COOLANT	L	S-309201	BULK	6	8	60	100	*	*	900	600	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1531	80	BN	117	16	02	A1A	-	-	BN EXTRACT	L	SIMPLIFIED PIG RE	SLOP TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1532	25	CA	117	16	03	G6C	-	-	OMPRESSED AI	V	*	SIMPLIFIED PI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1533	80	BN	117	16	04	A1A	-	-	BN EXTRACT	L	SIMPLIFIED PIG RE	DFL-413001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1534	25	CA	117	17	02	G6C	-	-	OMPRESSED AI	V	*	REX PLANT TY	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1535	80	RX	117	17	06	P2A	-	-	UBBER EXTEND	L	TYPE 10	LA-413002	6	8	60	100	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1536	25	BN	117	17	07	A1A	-	-	BN EXTRACT	L	S-413001	LA-413001	6	8	40	60	*	*	1.25	0.00001	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1537	25	BN	117	17	08	A1A	-	-	BN EXTRACT	L	S-413002	LA-413002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1538	50	BO	117	19	01	A1A	-	-	BASE OIL	L	B-400001	PROVER TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1539	25	CA	117	19	02	G6C	-	-	OMPRESSED AI	V	*	B-400001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1540	80	BO	117	19	03	A1A	-	-	BASE OIL	L	850 N GI	B-400001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1541	80	BO	117	19	04	A1A	-	-	BASE OIL	L	BS 150 GI	B-400001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1542	80	BO	117	19	05	A1A	-	-	BASE OIL	L	500 N GII	B-400001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1543	80	BO	117	19	06	A1A	-	-	BASE OIL	L	PFRCB	B-400001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1544	80	BO	117	19	07	A1A	-	-	BASE OIL	L	PFRCB	B-400002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1545	80	BO	117	19	08	A1A	-	-	BASE OIL	L	70 N G II	B-400002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1546	25	CA	117	19	09	G6C	-	-	OMPRESSED AI	V	*	B-400002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1547	50	BO	117	19	10	A1A	-	-	BASE OIL	L	B-400002	PROVER TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1548	80	BO	117	19	11	A1A	-	-	BASE OIL	L	500 N G I	B-400002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1549	80	BO	117	19	12	A1A	-	-	BASE OIL	L	500 N GI	B-400003	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1550	80	BO	117	19	13	A1A	-	-	BASE OIL	L	6 CST GIII	B-400003	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
1551	80	BO	117	19	14	A1A	-	-	BASE OIL	L	4 CST GIII	B-400003	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1552	25	CA	117	19	15	G6C	-	-	OMPRESSED AI	V	*	B-400003	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1553	50	BO	117	19	16	A1A	-	-	BASE OIL	L	B-400003	PROVER TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1554	25	BO	117	19	17	A1A	-	-	BASE OIL	L	S-400001	B-400001	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1555	25	BO	117	19	18	A1A	-	-	BASE OIL	L	S-400002	B-400002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1556	25	BO	117	19	19	A1A	-	-	BASE OIL	L	S-400003	B-400003	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1557	80	BO	117	20	01	A1A	-	-	BASE OIL	L	150 N G I	B-400004	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1558	80	BO	117	20	02	A1A	-	-	BASE OIL	L	150 N G II	B-400004	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1559	80	BO	117	20	03	A1A	-	-	BASE OIL	L	400 N G I	B-400004	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1560	50	BO	117	20	04	A1A	-	-	BASE OIL	L	B-400004	PROVER TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1561	80	BO	117	20	05	A1A	-	-	BASE OIL	L	4 CST GIII	B-400004	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1562	25	CA	117	20	06	G6C	-	-	OMPRESSED AI	V	*	B-400004	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1563	80	BO	117	20	07	A1A	-	-	BASE OIL	L	150 N GII	B-400005	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1564	80	BO	117	20	08	A1A	-	-	BASE OIL	L	500 N GII	B-400005	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1565	50	BO	117	20	09	A1A	-	-	BASE OIL	L	B-400005	PROVER TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1566	80	BO	117	20	10	A1A	-	-	BASE OIL	L	6 CST GIII	B-400005	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1567	25	CA	117	20	11	G6C	-	-	OMPRESSED AI	V	*	B-400005	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1568	80	BO	117	20	12	A1A	-	-	BASE OIL	L	70 N GII	B-400006	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1569	80	BO	117	20	13	A1A	-	-	BASE OIL	L	400 N G I	B-400006	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1570	80	BO	117	20	14	A1A	-	-	BASE OIL	L	150 N G I	B-400006	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1571	80	BO	117	20	15	A1A	-	-	BASE OIL	L	BS 150 GI	B-400006	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1572	50	BO	117	20	16	A1A	-	-	BASE OIL	L	B-400006	PROVER TANK	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1573	80	BO	117	20	17	A1A	-	-	BASE OIL	L	850 N G I	B-400006	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1574	25	CA	117	20	18	G6C	-	-	OMPRESSED AI	V	*	B-400006	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1575	25	BO	117	20	19	A1A	-	-	BASE OIL	L	S-400004	B-400004	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1576	25	BO	117	20	20	A1A	-	-	BASE OIL	L	S-400005	B-400005	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1577	25	BO	117	20	21	A1A	-	-	BASE OIL	L	S-400006	B-400006	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1578	80	BN	117	21	01	A1A	-	-	BN EXTRACT	L	BN EXTRACT	B-400007	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1579	80	BN	117	21	02	A1A	-	-	BN EXTRACT	L	BN EXTRACT	B-400008	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1580	80	RX	117	21	03	A1A	-	-	UBBER EXTEND	L	REX	B-400009	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1581	80	RX	117	21	04	A1A	-	-	UBBER EXTEND	L	REX	B-400010	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1582	80	DE	117	21	05	A1A	-	-	DIESEL EXHAUS	L	DEF	B-400011	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1583	80	DE	117	21	06	A1A	-	-	DIESEL EXHAUS	L	DEF	LA-411002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1584	80	CL	117	21	07	A1A	-	-	COOLANT	L	COOLANT	B-400012	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1585	80	CL	117	21	08	A1A	-	-	COOLANT	L	COOLANT	LA-411002	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1586	50	BN	117	21	09	P2A	-	-	BN EXTRACT	L	BN EXTRACT	IBC	6	8	60	100	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1587	50	BN	117	21	10	P2A	-	-	BN EXTRACT	L	BN EXTRACT	IBC	6	8	60	100	*	*	900	600	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1588	50	RX	117	21	12	A1A	-	-	UBBER EXTEND	L	REX	IBC	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1589	50	RX	117	21	13	A1A	-	-	UBBER EXTEND	L	REX	IBC	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1590	80	TO	117	22	01	A1A	-	-	ANSFORMER C	L	TRANSFORMER OI	SM-304402	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION										DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY															
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR	PR	TEMP	MEDIUM	PR									
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
1591	80	TO	117	22	02	A1A	-	-	ANSFORMER C	L	TRANSFORMER OI	SM-304401	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1592	80	WO	117	22	03	A1A	-	-	WHITE OIL	L	WHITE OIL	SM-305501	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1593	80	WO	117	22	04	A1A	-	-	WHITE OIL	L	WHITE OIL	SM-305502	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1594	50	WO	117	22	05	A1A	-	-	WHITE OIL	L	SM-305501	IBC	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1595	50	WO	117	22	06	A1A	-	-	WHITE OIL	L	SM-305502	IBC	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1596	25	N2	117	22	07	B4C	-	-	NITROGEN	V	*	SM-305502	6	8	40	60	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1597	50	TO	117	22	08	A1A	-	-	ANSFORMER C	L	TRANSFORMER OI	IBC	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1598	50	TO	117	22	09	A1A	-	-	ANSFORMER C	L	TRANSFORMER OI	IBC	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1599	25	N2	117	22	10	B4C	-	-	NITROGEN	V	*	SM-305501	6	8	40	60	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1600	25	N2	117	22	11	B4C	-	-	NITROGEN	V	*	SM-304402	6	8	40	60	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1601	25	N2	117	22	12	B4C	-	-	NITROGEN	V	*	SM-304401	6	8	40	60	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1602	25	N2	117	22	13	B4C	-	-	NITROGEN	V	*	SM-304401	6	8	40	60	*	*	1.25	0.00001	15	150	Water	*	1.5	SCH 40S	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1603	25	BO	117	22	14	A1A	-	-	BASE OIL	L	S-304401	SM-304401	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1604	25	BO	117	22	15	A1A	-	-	BASE OIL	L	S-305501	SM-305501	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1605	25	BO	117	22	16	A1A	-	-	BASE OIL	L	S-305502	SM-305502	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1606	25	BO	117	22	17	A1A	-	-	BASE OIL	L	S-304402	SM-304402	6	8	60	100	*	*	900	600	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1607	50	FP	117	23	01	A1A	-	-	NISHED PRODU	L	PT-500201	P-500201	0.6	1.2	40	60	*	*	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1608	50	FP	117	23	02	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500201	6	8	40	60	*	5	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1609	50	FP	117	23	03	A1A	-	-	NISHED PRODU	L	P-500201	SLOP TANK	6	8	40	60	*	5	900	600	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1610	80	V	117	24	01	G6C	-	-	VAPOR	V	DV-500101	SAFE LOCATIO	ATM	FULL OF LIQUID	40	60	*	*	900	600	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1611	50	V	117	24	02	G6C	-	-	VAPOR	V	DV-500101	DRAIN	ATM	FULL OF LIQUID	40	60	*	*	900	600	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1612	50	V	117	24	03	G6C	-	-	VAPOR	V	MANIFOLD	DV-500101	6	8	40	60	*	*	900	600	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-117	H1	P&ID FOR FILLING PLANT		
1613	100	BO	114	01	01	A1A	-	-	BASE OIL	L	ABB-1 TYPE 5	FT-500017	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1614	50	BO	114	01	02	A1A	-	-	BASE OIL	L	ST-100019/21	FT-500017	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1615	80	FP	114	01	03	A1A	-	-	NISHED PRODU	L	P-500017	FT-500017	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1616	80	HOS (S)	114	01	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500017	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1617	80	HOR(S)	114	01	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500017	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1618	80	HOS (S)	114	01	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500036	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1619	80	HOR(S)	114	01	07	A1A	IH	-	HOT OIL	L	FT-500017	HEADER	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1620	100	FP	114	01	08	A1A	-	-	NISHED PRODU	L	FT-500017	P-500017	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1621	80	FP	114	01	09	A1A	-	-	NISHED PRODU	L	P-500017	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1622	100	FP	114	01	10	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500017	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1623	80	FP	114	01	11	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500017	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1624	80	FP	114	01	12	A1A	-	-	NISHED PRODU	L	P-500017	HLDV 500017	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1625	80	LO	114	01	13	P2A	-	-	LUBE OIL	L	HLDV 500017 TYPE	PIGGED MANI	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1626	25	CA	114	01	14	G6C	-	-	OMPRESSED AI	V	*	HLDV 500017	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1627	25	V	114	01	15	G6C	-	-	VAPOR	V	HLDV 500017 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1628	25	CA	114	01	20	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1629	40	PA	114	01	17	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1630	40	PA	114	01	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500017	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1631	40	PA	114	01	19	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500017	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1632	80	FP	114	01	16	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1633	100	BO	114	02	01	A1A	-	-	BASE OIL	L	FT-500017 TYPE 9	FT-500019	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1634	50	BO	114	02	02	A1A	-	-	BASE OIL	L	FT-500017	FT-500019	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1635	80	FP	114	02	03	A1A	-	-	NISHED PRODU	L	P-500019	FT-500019	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1636	80	HOS (S)	114	02	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500019	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1637	80	HOR(S)	114	02	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500019	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1638	100	FP	114	02	07	A1A	-	-	NISHED PRODU	L	FT-500019	P-500019	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1639	80	FP	114	02	08	A1A	-	-	NISHED PRODU	L	P-500019	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1640	100	FP	114	02	09	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500019	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1641	80	FP	114	02	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500019	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1642	80	FP	114	02	11	A1A	-	-	NISHED PRODU	L	P-500019	HLDV 500019I	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1643	80	LO	114	02	12	P2A	-	-	LUBE OIL	L	HLDV 500019B TYP	PIGGED MANI	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1644	25	CA	114	02	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500019I	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1645	25	V	114	02	14	G6C	-	-	VAPOR	V	HLDV 500019B TYP	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1646	25	CA	114	02	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1647	40	PA	114	02	19	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1648	40	PA	114	02	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500019	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1649	40	PA	114	02	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500019	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1650	80	FP	114	02	16	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1651	25	V	114	02	20	G6C	-	-	VAPOR	V	HLDV 500019 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1652	25	CA	114	02	21	G6C	-	-	OMPRESSED AI	V	HLDV 500019 TYPE	*	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1653	100	BO	114	03	01	A1A	-	-	BASE OIL	L	SMB-302001 TYPE	FT-500013	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1652	50	BO	114	03	02	A1A	-	-	BASE OIL	L	FT-500019/21	FT-500013	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1653	100	FP	114	03	03	A1A	-	-	NISHED PRODU	L	SMB-302001 TYPE	FT-500015	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1654	80	FP	114	03	04	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1655	80	HOS (S)	114	03	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500013	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1656	80	HOR(S)	114	03	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500013	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1657	100	FP	114	03	08	A1A	-	-	NISHED PRODU	L	FT-500013	P-500013	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1658	100	FP	114	03	09	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500013	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1659	80	FP	114	03	10	A1A	-	-	NISHED PRODU	L	P-500013	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1660	80	FP	114	03	11	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500013	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1661	80	FP	114	03	12	A1A	-	-	NISHED PRODU	L	P-500013	HLDV 500013	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1662	80	LO	114	03	13	P2A	-	-	LUBE OIL	L	HLDV 500013 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1663	25	CA	114	03	14	G6C	-	-	OMPRESSED AI	V	*	HLDV 500013	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1664	25	V	114	03	15	G6C	-	-	VAPOR	V	HLDV 500013 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1665	25	CA	114	03	16	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1666	40	PA	114	03	17	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1667	40	PA	114	03	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500013	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1668	40	PA	114	03	19	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500013	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1669	100	FP	114	03	20	A1A	-	-	NISHED PRODU	L	FT-500015	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1670	25	CA	114	03	21	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1671	80	HOS (S)	114	03	23	A1A	IH	-	HOT OIL	L	HEADER	FT-500015	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1672	80	HOR(S)	114	03	24	A1A	IH	-	HOT OIL	L	HEADER	FT-500015	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1673	100	BO	114	03	25	A1A	-	-	BASE OIL	L	SMB-302001 TYPE	FT-500015	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1674	50	BO	114	03	26	A1A	-	-	BASE OIL	L	FT-500019/21	FT-500015	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1675	80	FP	114	03	27	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1676	80	FP	114	03	28	A1A	-	-	NISHED PRODU	L	P-500013	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1677	40	PA	114	03	29	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1678	40	PA	114	03	30	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500015	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1679	40	PA	114	03	31	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500015	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1680	100	BO	114	04	01	A1A	-	-	BASE OIL	L	FT-500013/15 TYPI	FT-500005	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1681	50	BO	114	04	02	A1A	-	-	BASE OIL	L	FT-500013/15	FT-500005	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1682	80	FP	114	04	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1683	80	HOS (S)	114	04	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500005	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1684	80	HOR(S)	114	04	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500005	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1685	100	FP	114	04	06	A1A	-	-	NISHED PRODU	L	FT-500013/15 TYPI	FT-500007	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1686	100	FP	114	04	08	A1A	-	-	NISHED PRODU	L	FT-500007	P-500007	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1687	80	FP	114	04	09	A1A	-	-	NISHED PRODU	L	P-500005	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1688	80	FP	114	04	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500005	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1689	80	FP	114	04	11	A1A	-	-	NISHED PRODU	L	P-500005	HLDV 500005	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1690	80	LO	114	04	12	P2A	-	-	LUBE OIL	L	HLDV 500005 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1691	25	CA	114	04	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500005	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1692	25	V	114	04	14	G6C	-	-	VAPOR	V	HLDV 500005 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1693	25	CA	114	04	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1694	40	PA	114	04	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1695	40	PA	114	04	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500005	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1696	40	PA	114	04	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500005	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1697	80	HOS (S)	114	04	20	A1A	IH	-	HOT OIL	L	HEADER	FT-500007	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1698	80	HOR(S)	114	04	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500007	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1699	100	BO	114	04	22	A1A	-	-	BASE OIL	L	FT-500013/15 TYPI	FT-500007	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1700	50	BO	114	04	23	A1A	-	-	BASE OIL	L	FT-500013/15	FT-500007	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1701	25	CA	114	04	24	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1702	100	FP	114	04	25	A1A	-	-	NISHED PRODU	L	FT-500007	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1703	40	PA	114	04	26	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1704	40	PA	114	04	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500007	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1705	40	PA	114	04	28	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500007	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1706	100	FP	114	04	29	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500005	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1707	80	FP	114	04	30	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1708	80	FP	114	04	31	A1A	-	-	NISHED PRODU	L	P-500005	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY													
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR	PR	TEMP	MEDIUM	PR								
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1709	100	BO	114	05	01	A1A	-	-	BASE OIL	L	FT-500006/07 TYPI	FT-500001	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1710	50	BO	114	05	02	A1A	-	-	BASE OIL	L	FT-500006/07	FT-500001	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1711	80	FP	114	05	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1712	100	FP	114	05	04	A1A	-	-	NISHED PRODU	L	FT-500006/07 TYPI	FT-500003	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1713	80	HOS (S)	114	05	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500001	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1714	80	HOR(S)	114	05	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500001	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1715	100	FP	114	05	07	A1A	-	-	NISHED PRODU	L	FT-500001	P-500001	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1716	100	FP	114	05	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500001	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1717	80	FP	114	05	09	A1A	-	-	NISHED PRODU	L	P-500001	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1718	80	FP	114	05	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500001	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1719	80	FP	114	05	11	A1A	-	-	NISHED PRODU	L	P-500001	HLDV 500001	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1720	80	LO	114	05	12	P2A	-	-	LUBE OIL	L	HLDV 500001 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1721	25	CA	114	05	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1722	25	V	114	05	14	G6C	-	-	VAPOR	V	HLDV 500001 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1723	25	CA	114	05	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1724	40	PA	114	05	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1725	40	PA	114	05	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1726	40	PA	114	05	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1727	100	BO	114	05	19	A1A	-	-	BASE OIL	L	FT-500006/07 TYPI	FT-500003	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1728	50	BO	114	05	20	A1A	-	-	BASE OIL	L	FT-500006/07	FT-500003	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1729	80	HOS (S)	114	05	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500003	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1730	80	HOR(S)	114	05	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500003	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1731	25	CA	114	05	23	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1732	100	FP	114	05	24	A1A	-	-	NISHED PRODU	L	FT-500003	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1733	40	PA	114	05	25	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1734	40	PA	114	05	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500003	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1735	40	PA	114	05	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500003	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1736	80	FP	114	05	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1737	80	FP	114	05	29	A1A	-	-	NISHED PRODU	L	P-500001	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1738	100	BO	114	06	01	A1A	-	-	BASE OIL	L	FT-500001/03 TYPI	FT-500009	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1739	50	BO	114	06	02	A1A	-	-	BASE OIL	L	FT-500001/03	FT-500009	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1740	80	FP	114	06	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1741	100	FP	114	06	04	A1A	-	-	NISHED PRODU	L	FT-500001/03 TYPI	FT-500011	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1742	80	HOS (S)	114	06	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500009	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1743	80	HOR(S)	114	06	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500009	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1744	100	FP	114	06	07	A1A	-	-	NISHED PRODU	L	FT-500009	P-500009	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1745	100	FP	114	06	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500009	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1746	80	FP	114	06	09	A1A	-	-	NISHED PRODU	L	P-500009	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1747	80	FP	114	06	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500009	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1748	80	FP	114	06	11	A1A	-	-	NISHED PRODU	L	P-500009	HLDV 500009	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1749	80	LO	114	06	12	P2A	-	-	LUBE OIL	L	HLDV 500009 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1750	25	CA	114	06	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500009	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1751	25	V	114	06	14	G6C	-	-	VAPOR	V	HLDV 500009 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1752	25	CA	114	06	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1753	40	PA	114	06	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1754	40	PA	114	06	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500009	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1755	40	PA	114	06	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500009	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1756	100	BO	114	06	19	A1A	-	-	BASE OIL	L	FT-500001/03 TYPI	FT-500011	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1757	50	BO	114	06	20	A1A	-	-	BASE OIL	L	FT-500001/03	FT-500011	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1758	80	HOS (S)	114	06	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500011	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1759	80	HOR(S)	114	06	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500011	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1760	25	CA	114	06	23	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1761	100	FP	114	06	24	A1A	-	-	NISHED PRODU	L	FT-500011	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1762	40	PA	114	06	25	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1763	40	PA	114	06	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500011	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1764	40	PA	114	06	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500011	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1765	80	FP	114	06	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1766	80	FP	114	06	29	A1A	-	-	NISHED PRODU	L	P-500009	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1767	100	BO	114	07	01	A1A	-	-	BASE OIL	L	FT-500009/11 TYPI	FT-500002	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1768	50	BO	114	07	02	A1A	-	-	BASE OIL	L	FT-500009/11	FT-500002	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1769	80	FP	114	07	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1770	100	FP	114	07	04	A1A	-	-	NISHED PRODU	L	FT-500009/11 TYPI	FT-500004	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1771	80	HOS (S)	114	07	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500002	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1772	80	HOR(S)	114	07	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500002	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1773	100	FP	114	07	08	A1A	-	-	NISHED PRODU	L	FT-500002	P-500002	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1774	100	FP	114	07	09	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500002	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1775	80	FP	114	07	10	A1A	-	-	NISHED PRODU	L	P-500002	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1776	80	FP	114	07	11	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500002	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1777	80	FP	114	07	12	A1A	-	-	NISHED PRODU	L	P-500002	HLDV 500002	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1778	80	LO	114	07	13	P2A	-	-	LUBE OIL	L	HLDV 500002 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1779	25	CA	114	07	14	G6C	-	-	OMPRESSED AI	V	*	HLDV 500002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1780	25	V	114	07	15	G6C	-	-	VAPOR	V	HLDV 500002 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1781	25	CA	114	07	16	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1782	40	PA	114	07	17	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1783	40	PA	114	07	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1784	40	PA	114	07	19	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1785	100	BO	114	07	20	A1A	-	-	BASE OIL	L	FT-500009/11 TYPI	FT-500004	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1786	50	BO	114	07	21	A1A	-	-	BASE OIL	L	FT-500009/11	FT-500004	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1787	80	HOS (S)	114	07	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500004	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1788	80	HOR(S)	114	07	23	A1A	IH	-	HOT OIL	L	HEADER	FT-500004	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1789	25	CA	114	07	25	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1790	100	FP	114	07	26	A1A	-	-	NISHED PRODU	L	FT-500004	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1791	40	PA	114	07	27	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1792	40	PA	114	07	28	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500004	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1793	40	PA	114	07	29	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500004	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1794	80	FP	114	07	30	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1795	80	FP	114	07	31	A1A	-	-	NISHED PRODU	L	P-500002	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1796	100	BO	114	08	01	A1A	-	-	BASE OIL	L	FT-500002/04 TYPI	FT-500006	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1797	50	BO	114	08	02	A1A	-	-	BASE OIL	L	FT-500002/04	FT-500006	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1798	80	FP	114	08	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1799	100	FP	114	08	04	A1A	-	-	NISHED PRODU	L	FT-500002/04 TYPI	FT-500008	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1800	80	HOS (S)	114	08	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500006	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1801	80	HOR(S)	114	08	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500006	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1802	100	FP	114	08	08	A1A	-	-	NISHED PRODU	L	FT-500006	P-500006	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1803	100	FP	114	08	09	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500006	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1804	80	FP	114	08	10	A1A	-	-	NISHED PRODU	L	P-500006	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1805	80	FP	114	08	11	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500006	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1806	80	FP	114	08	12	A1A	-	-	NISHED PRODU	L	P-500006	HLDV 500006	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1807	80	LO	114	08	13	P2A	-	-	LUBE OIL	L	HLDV 500006 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1808	25	CA	114	08	14	G6C	-	-	OMPRESSED AI	V	*	HLDV 500006	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1809	25	V	114	08	15	G6C	-	-	VAPOR	V	HLDV 500006 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1810	25	CA	114	08	16	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1811	40	PA	114	08	17	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1812	40	PA	114	08	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500006	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1813	40	PA	114	08	19	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500006	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1814	100	BO	114	08	20	A1A	-	-	BASE OIL	L	FT-500002/04 TYPI	FT-500008	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1815	50	BO	114	08	21	A1A	-	-	BASE OIL	L	FT-500002/04	FT-500008	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1816	80	HOS (S)	114	08	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500008	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1817	80	HOR(S)	114	08	23	A1A	IH	-	HOT OIL	L	HEADER	FT-500008	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1818	25	CA	114	08	25	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1819	100	FP	114	08	26	A1A	-	-	NISHED PRODU	L	FT-500008	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1820	40	PA	114	08	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500008	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1821	40	PA	114	08	28	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500008	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1822	40	PA	114	08	29	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1823	80	FP	114	08	30	A1A	-	-	NISHED PRODU	L	P-500006	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1824	80	FP	114	08	31	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1825	100	BO	114	09	01	A1A	-	-	BASE OIL	L	FT-500006/08 TYPI	FT-500010	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1826	50	BO	114	09	02	A1A	-	-	BASE OIL	L	FT-500006/08	FT-500010	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1827	80	FP	114	09	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1828	100	FP	114	09	04	A1A	-	-	NISHED PRODU	L	FT-500006/08 TYPI	FT-500012	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY													
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR	PR	TEMP	MEDIUM	PR								
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1829	80	HOS (S)	114	09	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500010	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1830	80	HOR(S)	114	09	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500010	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1831	100	FP	114	09	07	A1A	-	-	NISHED PRODU	L	FT-500010	P-500010	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1832	100	FP	114	09	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500010	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1833	80	FP	114	09	09	A1A	-	-	NISHED PRODU	L	P-500010	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1834	80	FP	114	09	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500010	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1835	80	FP	114	09	11	A1A	-	-	NISHED PRODU	L	P-500010	HLDV 500010	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1836	80	LO	114	09	12	P2A	-	-	LUBE OIL	L	HLDV 500010 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1837	25	CA	114	09	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500010	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1838	25	V	114	09	14	G6C	-	-	VAPOR	V	HLDV 500010 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1839	25	CA	114	09	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1840	40	PA	114	09	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1841	40	PA	114	09	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500010	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1842	40	PA	114	09	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500010	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1843	100	BO	114	09	19	A1A	-	-	BASE OIL	L	FT-500006/08 TYPI	FT-500012	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1844	50	BO	114	09	20	A1A	-	-	BASE OIL	L	FT-500006/08	FT-500012	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1845	80	HOS (S)	114	09	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500012	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1846	80	HOR(S)	114	09	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500012	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1847	25	CA	114	09	23	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1848	100	FP	114	09	24	A1A	-	-	NISHED PRODU	L	FT-500012	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1849	40	PA	114	09	25	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1850	40	PA	114	09	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500012	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1851	40	PA	114	09	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500012	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1852	80	FP	114	09	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1853	80	FP	114	09	29	A1A	-	-	NISHED PRODU	L	P-500010	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1854	100	BO	114	10	01	A1A	-	-	BASE OIL	L	FT-500010/12 TYPI	FT-500014	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1855	50	BO	114	10	02	A1A	-	-	BASE OIL	L	FT-500010/12	FT-500014	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1856	80	FP	114	10	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1857	80	HOS (S)	114	10	04	A1A	-	-	HOT OIL	L	HEADER	FT-500014	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1858	80	HOR(S)	114	10	05	A1A	-	-	HOT OIL	L	HEADER	FT-500014	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1859	100	FP	114	10	06	A1A	-	-	NISHED PRODU	L	FT-500010/12 TYPI	FT-500016	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1860	100	FP	114	10	07	A1A	-	-	NISHED PRODU	L	FT-500019	P-500014	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1861	100	FP	114	10	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500014	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1862	80	FP	114	10	09	A1A	-	-	NISHED PRODU	L	P-500014	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1863	80	FP	114	10	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500014	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1864	80	FP	114	10	11	A1A	-	-	NISHED PRODU	L	P-500014	HLDV 500014	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1865	80	LO	114	10	12	P2A	-	-	LUBE OIL	L	HLDV 500014 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1866	25	CA	114	10	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500014	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1867	25	V	114	10	14	G6C	-	-	VAPOR	V	HLDV 500014 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1868	25	CA	114	10	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1869	40	PA	114	10	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1870	40	PA	114	10	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500014	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1871	40	PA	114	10	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500014	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1872	100	BO	114	10	19	A1A	-	-	BASE OIL	L	FT-500010/12 TYPI	FT-500016	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1873	50	BO	114	10	20	A1A	-	-	BASE OIL	L	FT-500010/12	FT-500016	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1874	80	HOS (S)	114	10	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500016	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1875	80	HOR(S)	114	10	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500016	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1876	100	FP	114	10	23	A1A	-	-	NISHED PRODU	L	FT-500016	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1877	25	CA	114	10	24	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1878	80	FP	114	10	25	A1A	-	-	NISHED PRODU	L	P-500014	EQUALISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1879	40	PA	114	10	26	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1880	40	PA	114	10	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500016	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1881	40	PA	114	10	28	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500016	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1882	25	CA	114	10	31	G6C	-	-	OMPRESSED AI	V	*	HLDV 500016	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1883	25	V	114	10	30	G6C	-	-	VAPOR	V	HLDV 500016 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1884	80	FP	114	10	29	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1885	100	BO	114	11	01	A1A	-	-	BASE OIL	L	ABB-2 HEADER TYP	FT-500028	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1886	50	BO	114	11	02	A1A	-	-	BASE OIL	L	FT-500014/16	FT-500028	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1887	80	FP	114	11	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1888	100	FP	114	11	04	A1A	-	-	NISHED PRODU	L	ABB-2 HEADER	FT-500030	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1889	80	HOS (S)	114	11	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500028	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1890	80	HOR(S)	114	11	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500028	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1891	100	FP	114	11	07	A1A	-	-	NISHED PRODU	L	FT-500028	P-500028	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1892	100	FP	114	11	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500028	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1893	80	FP	114	11	09	A1A	-	-	NISHED PRODU	L	P-500028	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1894	80	FP	114	11	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500028	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1895	80	FP	114	11	11	A1A	-	-	NISHED PRODU	L	P-500028	HLDV 500028	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1896	80	LO	114	11	12	P2A	-	-	LUBE OIL	L	HLDV 500028 TYPE	PIGGED MANI	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1897	25	CA	114	11	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500028	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1898	25	V	114	11	14	G6C	-	-	VAPOR	V	HLDV 500028 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1899	25	CA	114	11	16	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1900	40	PA	114	11	17	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1901	40	PA	114	11	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500028	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1902	40	PA	114	11	19	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500028	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1903	100	BO	114	11	20	A1A	-	-	BASE OIL	L	ABB-2 HEADER TYP	FT-500030	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1904	50	BO	114	11	21	A1A	-	-	BASE OIL	L	FT-500014/16	FT-500030	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1905	80	HOS (S)	114	11	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500030	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1906	80	HOR(S)	114	11	23	A1A	IH	-	HOT OIL	L	HEADER	FT-500030	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1907	25	CA	114	11	24	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1908	100	FP	114	11	25	A1A	-	-	NISHED PRODU	L	FT-500030	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1909	40	PA	114	11	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500030	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1910	40	PA	114	11	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500030	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1911	40	PA	114	11	15	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1912	80	FP	114	11	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OU	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1913	80	FP	114	11	29	A1A	-	-	NISHED PRODU	L	P-500028	EQAULISATIO	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1914	100	BO	114	12	01	A1A	-	-	BASE OIL	L	FT-500028/30 TYPI	FT-500022	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1915	50	BO	114	12	02	A1A	-	-	BASE OIL	L	FT-500028/30	FT-500022	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1916	80	FP	114	12	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1917	100	FP	114	12	04	A1A	-	-	NISHED PRODU	L	FT-500028/30 TYPI	FT-500024	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1918	80	HOS (S)	114	12	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500022	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1919	80	HOR(S)	114	12	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500022	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1920	100	FP	114	12	07	A1A	-	-	NISHED PRODU	L	FT-500022	P-500022	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1921	100	FP	114	12	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500022	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1922	80	FP	114	12	09	A1A	-	-	NISHED PRODU	L	P-500022	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1923	80	FP	114	12	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500022	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1924	80	FP	114	12	11	A1A	-	-	NISHED PRODU	L	P-500022	HLDV 500022	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1925	80	LO	114	12	12	P2A	-	-	LUBE OIL	L	HLDV 500022 TYPE	PIGGED MANI	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1926	25	CA	114	12	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500022	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1927	25	V	114	12	14	G6C	-	-	VAPOR	V	HLDV 500022 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1928	25	CA	114	12	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1929	40	PA	114	12	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1930	40	PA	114	12	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500022	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1931	40	PA	114	12	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500022	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1932	100	BO	114	12	19	A1A	-	-	BASE OIL	L	FT-500028/30 TYPI	FT-500024	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1933	50	BO	114	12	20	A1A	-	-	BASE OIL	L	FT-500028/30	FT-500024	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1934	80	HOS (S)	114	12	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500024	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1935	80	HOR(S)	114	12	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500024	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1936	25	CA	114	12	23	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1937	100	FP	114	12	24	A1A	-	-	NISHED PRODU	L	FT-500024	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1938	40	PA	114	12	25	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1939	40	PA	114	12	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500024	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1940	40	PA	114	12	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500024	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1941	80	FP	114	12	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OU	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1942	80	FP	114	12	29	A1A	-	-	NISHED PRODU	L	P-500022	EQAULISATIO	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1943	100	BO	114	13	01	A1A	-	-	BASE OIL	L	FT-500022/24 TYPI	FT-500018	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1944	50	BO	114	13	02	A1A	-	-	BASE OIL	L	FT-500022/24	FT-500018	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1945	80	FP	114	13	03	A1A	-	-	NISHED PRODU	L	P-500018	FT-500018	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1946	80	HOS (S)	114	13	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500018	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1947	80	HOR(S)	114	13	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500018	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1948	40	AD	114	13	07	A1A	-	-	ADDITIVE	L	MB-500101	FT-500018	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
1949	100	FP	114	13	08	A1A	-	-	NISHED PRODU	L	FT-500018	P-500018	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1950	100	FP	114	13	09	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500018	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1951	80	FP	114	13	10	A1A	-	-	NISHED PRODU	L	P-500018	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1952	80	FP	114	13	11	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500018	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1953	80	FP	114	13	12	A1A	-	-	NISHED PRODU	L	P-500018	HLDV 500018	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1954	80	LO	114	13	13	P2A	-	-	LUBE OIL	L	HLDV 500018 TYPE	PIGGED MANI	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1955	25	CA	114	13	14	G6C	-	-	OMPRESSED AI	V	*	HLDV 500018	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1956	25	V	114	13	15	G6C	-	-	VAPOR	V	HLDV 500018 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1957	25	CA	114	13	16	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1958	40	PA	114	13	17	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1959	40	PA	114	13	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500018	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1960	40	PA	114	13	19	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500018	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1961	80	FP	114	13	20	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1962	100	BO	114	14	01	A1A	-	-	BASE OIL	L	FT-500018 TYPE 6	FT-500020	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1963	50	BO	114	14	02	A1A	-	-	BASE OIL	L	FT-500018	FT-500020	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1964	80	FP	114	14	03	A1A	-	-	NISHED PRODU	L	P-500020	FT-500020	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1965	80	HOS (S)	114	14	05	A1A	-	-	HOT OIL	L	HEADER	FT-500020	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1966	80	HOR(S)	114	14	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500020	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1967	100	FP	114	14	07	A1A	-	-	NISHED PRODU	L	FT-500018 TYPE 6	FT-500033	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1968	100	FP	114	14	08	A1A	-	-	NISHED PRODU	L	FT-500020	P-500020	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1969	100	FP	114	14	09	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500020	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1970	80	FP	114	14	10	A1A	-	-	NISHED PRODU	L	P-500020	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1971	80	FP	114	14	11	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500020	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1972	80	FP	114	14	12	A1A	-	-	NISHED PRODU	L	P-500020	HLDV 500020	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1973	25	CA	114	14	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500020	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1974	25	V	114	14	14	G6C	-	-	VAPOR	V	HLDV 500020 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1975	25	CA	114	14	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1976	40	PA	114	14	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1977	40	PA	114	14	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500020	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1978	40	PA	114	14	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500020	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1979	80	LO	114	14	19	P2A	-	-	LUBE OIL	L	HLDV 500020 TYPE	PIGGED MANI	6	8	60	100	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1980	80	FP	114	14	20	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1981	100	BO	114	15	01	A1A	-	-	BASE OIL	L	FT-500020 TYPE 6	FT-500025	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1982	50	BO	114	15	02	A1A	-	-	BASE OIL	L	FT-500020	FT-500025	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1983	80	FP	114	15	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1984	100	FP	114	15	04	A1A	-	-	NISHED PRODU	L	FT-500020 TYPE 6	FT-500035	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1985	80	HOS (S)	114	15	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500025	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1986	80	HOR(S)	114	15	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500025	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1987	100	FP	114	15	07	A1A	-	-	NISHED PRODU	L	FT-500025	P-500025	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1988	100	FP	114	15	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500025	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY													
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
1989	80	FP	114	15	09	A1A	-	-	NISHED PRODU	L	P-500025	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1990	80	FP	114	15	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500025	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1991	80	FP	114	15	11	A1A	-	-	NISHED PRODU	L	P-500025	HLDV 500025	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1992	80	LO	114	15	12	P2A	-	-	LUBE OIL	L	HLDV 500025 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1993	25	CA	114	15	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500025	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1994	25	V	114	15	14	G6C	-	-	VAPOR	V	HLDV 500025 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1995	25	CA	114	15	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1996	40	PA	114	15	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1997	40	PA	114	15	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500025	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1998	40	PA	114	15	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500025	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
1999	100	BO	114	15	19	A1A	-	-	BASE OIL	V	FT-500020 TYPE 9	FT-500026	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2000	80	HOS (S)	114	15	20	A1A	IH	-	HOT OIL	L	HEADER	FT-500026	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2001	80	HOR(S)	114	15	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500026	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2002	25	CA	114	15	22	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2003	100	FP	114	15	23	A1A	-	-	NISHED PRODU	L	FT-500026	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2004	40	PA	114	15	24	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2005	40	PA	114	15	25	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500026	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2006	40	PA	114	15	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500026	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2007	25	CA	114	15	27	G6C	-	-	OMPRESSED AI	V	*	HLDV 500026	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2008	25	V	114	15	31	G6C	-	-	VAPOR	V	HLDV 500026 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2009	80	FP	114	15	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2010	80	FP	114	15	29	A1A	-	-	NISHED PRODU	L	P-500025	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2011	50	BO	114	15	30	A1A	-	-	BASE OIL	L	FT-500020	FT-500026	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2012	100	BO	114	16	01	A1A	-	-	BASE OIL	L	SMB-302002 TYPE	FT-500021	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2013	50	BO	114	16	02	A1A	-	-	BASE OIL	L	FT-500025/26	FT-500021	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2014	80	FP	114	16	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2015	100	FP	114	16	04	A1A	-	-	NISHED PRODU	L	SMB-302002 TYPE	FT-500025	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2016	80	HOS (S)	114	16	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500021	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2017	80	HOR(S)	114	16	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500021	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2018	100	FP	114	16	07	A1A	-	-	NISHED PRODU	L	FT-500021	P-500021	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2019	100	FP	114	16	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500021	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2020	80	FP	114	16	09	A1A	-	-	NISHED PRODU	L	P-500021	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2021	80	FP	114	16	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500021	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2022	80	FP	114	16	11	A1A	-	-	NISHED PRODU	L	P-500021	HLDV 500021	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2023	80	LO	114	16	12	P2A	-	-	LUBE OIL	L	HLDV 500021 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2024	25	CA	114	16	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500021	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2025	25	V	114	16	14	G6C	-	-	VAPOR	V	HLDV 500021 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2026	25	CA	114	16	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2027	40	PA	114	16	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2028	40	PA	114	16	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500021	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2029	40	PA	114	16	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500021	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2030	100	BO	114	16	19	A1A	-	-	BASE OIL	L	SMB-302002 TYPE	FT-500023	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2031	50	BO	114	16	20	A1A	-	-	BASE OIL	L	FT-500025/26	FT-500023	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2032	100	FP	114	16	21	A1A	-	-	NISHED PRODU	L	SMB-302002 TYPE	FT-500026	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2033	80	HOS (S)	114	16	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500023	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2034	80	HOR(S)	114	16	23	A1A	IH	-	HOT OIL	L	HEADER	FT-500023	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2035	25	CA	114	16	24	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2036	100	FP	114	16	25	A1A	-	-	NISHED PRODU	L	FT-500023	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2037	40	PA	114	16	26	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2038	40	PA	114	16	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500023	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2039	40	PA	114	16	28	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500023	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2040	80	FP	114	16	29	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2041	80	FP	114	16	30	A1A	-	-	NISHED PRODU	L	P-500021	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2042	100	BO	114	17	01	A1A	-	-	BASE OIL	L	FT-500021/23 TYPI	FT-500033	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2043	50	BO	114	17	02	A1A	-	-	BASE OIL	L	FT-500021/23	FT-500033	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2044	80	FP	114	17	03	A1A	-	-	NISHED PRODU	L	P-500033	FT-500033	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2045	100	FP	114	17	04	A1A	-	-	NISHED PRODU	L	FT-500021/23 TYPI	FT-500035	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2046	80	HOS (S)	114	17	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500033	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2047	80	HOR(S)	114	17	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500033	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2048	100	FP	114	17	07	A1A	-	-	NISHED PRODU	L	FT-500033	P-500033	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2049	100	FP	114	17	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500033	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2050	80	FP	114	17	09	A1A	-	-	NISHED PRODU	L	P-500033	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2051	80	FP	114	17	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500033	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2052	80	FP	114	17	11	A1A	-	-	NISHED PRODU	L	P-500033	V-302102	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2053	80	FP	114	17	12	A1A	-	-	NISHED PRODU	L	P-500033	SMB-302002	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2054	25	CA	114	17	13	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2055	80	FP	114	17	17	A1A	-	-	NISHED PRODU	L	P-500035	FT-500035	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2056	80	HOS (S)	114	17	18	A1A	IH	-	HOT OIL	L	HEADER	FT-500035	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2057	80	HOR(S)	114	17	19	A1A	IH	-	HOT OIL	L	HEADER	FT-500035	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2058	25	CA	114	17	20	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2059	100	FP	114	17	21	A1A	-	-	NISHED PRODU	L	FT-500035	P-500035	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2060	80	FP	114	17	22	A1A	-	-	NISHED PRODU	L	P-500035	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2061	100	FP	114	17	23	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500035	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2062	80	FP	114	17	24	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500035	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2063	80	FP	114	17	25	A1A	-	-	NISHED PRODU	L	P-500035	V-302102	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2064	80	FP	114	17	26	A1A	-	-	NISHED PRODU	L	P-500035	SMB-302002	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2065	80	FP	114	17	30	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2066	80	FP	114	17	31	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2067	50	BO	114	17	32	A1A	-	-	NISHED PRODU	L	FT-500021/23	FT-500035	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2068	100	FP	114	17	33	A1A	-	-	NISHED PRODU	L	FT-500021/23 TYPI	FT-500024	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2069	100	FP	114	17	34	A1A	-	-	NISHED PRODU	L	FT-500021/23 TYPI	FT-500030	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2070	100	BO	114	17	35	A1A	-	-	BASE OIL	L	FT-500021/23	FT-500035	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2071	100	BO	114	18	01	A1A	-	-	BASE OIL	L	FT-500033/35 TYPI	FT-500027	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2072	50	BO	114	18	02	A1A	-	-	BASE OIL	L	FT-500033/35	FT-500027	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2073	80	FP	114	18	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2074	100	FP	114	18	04	A1A	-	-	NISHED PRODU	L	FT-500033/35 TYPI	FT-500029	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2075	80	HOS (S)	114	18	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500027	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2076	80	HOR(S)	114	18	06	A1A	IH	-	HOT OIL	L	HEADER	FT-500027	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2077	100	FP	114	18	07	A1A	-	-	NISHED PRODU	L	FT-500027	P-500027	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2078	100	FP	114	18	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500027	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2079	80	FP	114	18	09	A1A	-	-	NISHED PRODU	L	P-500027	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2080	80	FP	114	18	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500027	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2081	80	FP	114	18	11	A1A	-	-	NISHED PRODU	L	P-500027	HLDV 500027	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2082	80	LO	114	18	12	P2A	-	-	LUBE OIL	L	HLDV 500027 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2083	25	CA	114	18	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500027	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2084	25	V	114	18	14	G6C	-	-	VAPOR	V	HLDV 500027 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2085	25	CA	114	18	15	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2086	40	PA	114	18	16	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2087	40	PA	114	18	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500027	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2088	40	PA	114	18	18	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500027	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2089	100	BO	114	18	19	A1A	-	-	BASE OIL	L	FT-500033/35 TYPI	FT-500029	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2090	50	BO	114	18	20	A1A	-	-	BASE OIL	L	FT-500033/35	FT-500029	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2091	80	HOS (S)	114	18	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500029	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2092	80	HOR(S)	114	18	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500029	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2093	25	CA	114	18	23	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2094	100	FP	114	18	24	A1A	-	-	NISHED PRODU	L	FT-500029	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2095	40	PA	114	18	25	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2096	40	PA	114	18	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500029	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2097	40	PA	114	18	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500029	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2098	80	FP	114	18	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2099	80	FP	114	18	29	A1A	-	-	NISHED PRODU	L	P-500027	EQAULISATIO	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2100	100	BO	114	19	01	A1A	-	-	BASE OIL	L	FT-500027/29 TYPI	FT-500031	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2101	50	BO	114	19	02	A1A	-	-	BASE OIL	L	FT-500027/29	FT-500031	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2102	80	FP	114	19	03	A1A	-	-	NISHED PRODU	L	P-500031	FT-500031	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2103	80	HOS (S)	114	19	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500031	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2104	80	HOR(S)	114	19	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500031	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2105	100	FP	114	19	06	A1A	-	-	NISHED PRODU	L	FT-500031	P-500031	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2106	100	FP	114	19	07	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500031	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2107	80	FP	114	19	08	A1A	-	-	NISHED PRODU	L	P-500031	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2108	80	FP	114	19	09	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500031	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2109	80	FP	114	19	10	A1A	-	-	NISHED PRODU	L	P-500031	HLDV 500031	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2110	80	LO	114	19	11	P2A	-	-	LUBE OIL	L	HLDV 500031 TYPE	PIGGED MANI	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2111	25	CA	114	19	12	G6C	-	-	OMPRESSED AI	V	*	HLDV 500031	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2112	25	V	114	19	13	G6C	-	-	VAPOR	V	HLDV 500031 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2113	25	CA	114	19	14	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2114	40	PA	114	19	15	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2115	40	PA	114	19	16	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500031	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2116	40	PA	114	19	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500031	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2117	25	CA	114	19	20	G6C	-	-	OMPRESSED AI	V	*	HLDV 500031	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2118	25	V	114	19	19	G6C	-	-	VAPOR	V	HLDV 500031 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2119	80	FP	114	19	18	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2120	100	BO	114	20	01	A1A	-	-	BASE OIL	L	V-302101 TYPE 5	FT-500032	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2121	50	BO	114	20	02	A1A	-	-	BASE OIL	L	FT-500031	FT-500032	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2122	80	FP	114	20	03	A1A	-	-	NISHED PRODU	L	EQUALISTAION	EQUALISATIO	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2123	80	HOS (S)	114	20	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500032	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2124	80	HOR(S)	114	20	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500032	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2125	100	FP	114	20	06	A1A	-	-	NISHED PRODU	L	FT-500032	P-500032	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2126	100	FP	114	20	07	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500032	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2127	80	FP	114	20	08	A1A	-	-	NISHED PRODU	L	P-500032	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2128	80	FP	114	20	09	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500032	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2129	80	FP	114	20	10	A1A	-	-	NISHED PRODU	L	P-500032	HLDV 500032	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2130	80	LO	114	20	11	P2A	-	-	LUBE OIL	L	HLDV 500032 TYPE	DRUM FILLING	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2131	25	CA	114	20	12	G6C	-	-	OMPRESSED AI	V	*	HLDV 500032	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2132	25	V	114	20	13	G6C	-	-	VAPOR	V	HLDV 500032 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2133	25	CA	114	20	14	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2134	40	PA	114	20	15	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2135	40	PA	114	20	16	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500032	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2136	40	PA	114	20	17	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500032	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2137	100	BO	114	20	18	A1A	-	-	BASE OIL	L	V-302101 TYPE 9	FT-500034	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2138	50	BO	114	20	19	A1A	-	-	BASE OIL	L	FT-500031	FT-500034	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2139	25	CA	114	20	20	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2140	80	HOS (S)	114	20	21	A1A	IH	-	HOT OIL	L	HEADER	FT-500034	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2141	80	HOR(S)	114	20	22	A1A	IH	-	HOT OIL	L	HEADER	FT-500034	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2142	25	CA	114	20	23	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2143	100	FP	114	20	24	A1A	-	-	NISHED PRODU	L	FT-500034	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2144	40	PA	114	20	25	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2145	40	PA	114	20	26	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500034	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2146	40	PA	114	20	27	G6C	-	-	PLANT AIR	V	CONTROL BOX	FT-500034	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2147	25	CA	114	20	30	G6C	-	-	OMPRESSED AI	V	*	V-302101 TYP	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2148	25	V	114	20	31	G6C	-	-	VAPOR	V	V-302101 TYPE 9	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2149	80	FP	114	20	28	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2150	80	FP	114	20	29	A1A	-	-	NISHED PRODU	L	P-500032	EQAULISATIO	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2151	100	BO	114	21	01	A1A	-	-	BASE OIL	L	V-302103 TYPE 5	FT-500037	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2152	50	BO	114	21	02	A1A	-	-	BASE OIL	L	FT-500043	FT-500037	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2153	80	FP	114	21	03	A1A	-	-	NISHED PRODU	L	P-500037	FT-500037	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2154	80	HOS (S)	114	21	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500037	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2155	80	HOR(S)	114	21	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500037	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2156	100	FP	114	21	06	A1A	-	-	NISHED PRODU	L	FT-500037	P-500037	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2157	100	FP	114	21	07	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500037	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2158	80	FP	114	21	08	A1A	-	-	NISHED PRODU	L	P-500037	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2159	80	FP	114	21	09	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500037	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2160	80	FP	114	21	10	A1A	-	-	NISHED PRODU	L	P-500037	HLDV 500037	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2161	80	LO	114	21	11	P2A	-	-	LUBE OIL	L	HLDV 500037 TYPE	P-500037	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2162	25	CA	114	21	12	G6C	-	-	OMPRESSED AI	V	*	HLDV 500037	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2163	25	V	114	21	13	G6C	-	-	VAPOR	V	HLDV 500037 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2164	25	CA	114	21	14	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2165	80	FP	114	21	18	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2166	100	BO	114	22	01	A1A	-	-	BASE OIL	L	FT-500037 TYPE 9	FT-500036	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2167	50	BO	114	22	02	A1A	-	-	BASE OIL	L	FT-500037	FT-500036	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2168	80	FP	114	22	03	A1A	-	-	NISHED PRODU	L	P-500036	FT-500036	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2169	80	HOS (S)	114	22	04	A1A	IH	-	HOT OIL	L	HEADER	FT-500036	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2170	80	HOR(S)	114	22	05	A1A	IH	-	HOT OIL	L	HEADER	FT-500036	6	8	120	140	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2171	100	FP	114	22	07	A1A	-	-	NISHED PRODU	L	FT-500036	P-500036	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2172	100	FP	114	22	08	A1A	-	-	NISHED PRODU	L	TANK TRUCK	P-500036	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2173	80	FP	114	22	09	A1A	-	-	NISHED PRODU	L	P-500036	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2174	80	FP	114	22	10	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF P-	P-500036	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2175	80	FP	114	22	11	A1A	-	-	NISHED PRODU	L	F-500036	TYPE 10	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2176	80	LO	114	22	12	P2A	-	-	LUBE OIL	L	TYPE 10	DRUM FILLING	6	8	60	100	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2177	25	CA	114	22	13	G6C	-	-	OMPRESSED AI	V	*	HLDV 500036	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2178	25	CA	114	22	14	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2179	25	CA	114	22	21	G6C	-	-	OMPRESSED AI	V	*	HLDV 500036	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2180	25	V	114	22	19	G6C	-	-	VAPOR	V	HLDV 500036	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2181	80	FP	114	22	18	A1A	-	-	NISHED PRODU	L	BYPASS LINE OF OI	OUTGOING LI	6	8	60	100	*	30	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2182	80	LO	114	22	20	P2A	-	-	LUBE OIL	L	HLDV 500036	DRUM FILLING	6	8	60	80	*	30	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2183	40	BO	114	23	01	A1A	-	-	BASE OIL	L	DRUM ADDITIVE P	MB-500101	ATM	FULL OF LIQUID	40	80	*	*	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2184	40	AD	114	23	02	A1A	-	-	ADDITIVE	L	P-500101	MB-500101	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2185	40	AD	114	23	03	A1A	-	-	ADDITIVE	L	MB-500101	P-500101	0.6	1.2	40	80	*	*	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2186	40	AD	114	23	04	A1A	-	-	ADDITIVE	L	P-500101	OUTGOING LI	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2187	25	AD	114	23	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-500101	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2188	40	AD	114	23	06	A1A	-	-	ADDITIVE	L	P-500101	FT-500018	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2189	40	AD	114	23	07	A1A	-	-	ADDITIVE	L	P-500101	FT-500036	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2190	40	AD	114	23	08	A1A	-	-	ADDITIVE	L	P-500101	FT-500015	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2191	40	AD	114	23	09	A1A	-	-	ADDITIVE	L	P-500101	FT-500013	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2192	40	AD	114	23	10	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	OUTGOING LI	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2193	40	AD	114	23	11	A1A	-	-	ADDITIVE	L	BYPASS LINE OF OI	OUTGOING LI	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2194	100	PD	114	23	12	B4C	-	-	POWDER	S	DYE HAND ADD	MB-500101	6	8	40	80	*	0.2	900	5	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2195	50	BO	114	23	13	A1A	-	-	BASE OIL	L	MB-500101	SAFE LOCATIC	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2196	40	BO	114	24	01	A1A	-	-	BASE OIL	L	DRUM ADDITIVE P	MB-500102	ATM	FULL OF LIQUID	40	80	*	*	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2197	40	AD	114	24	02	A1A	-	-	ADDITIVE	L	P-500102	MB-500102	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2198	40	AD	114	24	03	A1A	-	-	ADDITIVE	L	MB-500102	P-500102	0.6	1.2	40	80	*	*	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2199	40	AD	114	24	04	A1A	-	-	ADDITIVE	L	P-500102	OUTGOING LI	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2200	25	AD	114	24	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-500102	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2201	40	AD	114	24	06	A1A	-	-	ADDITIVE	L	P-500102	FT-500006	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2202	40	AD	114	24	07	A1A	-	-	ADDITIVE	L	P-500102	FT-500008	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2203	40	AD	114	24	08	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	OUTGOING LI	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2204	40	AD	114	24	09	A1A	-	-	ADDITIVE	L	BYPASS LINE OF OI	OUTGOING LI	6	8	40	80	*	0.2	900	3	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2205	100	PD	114	24	10	B4C	-	-	POWDER	S	DYE HAND ADD	MB-500102	6	8	40	80	*	0.2	900	5	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2206	50	BO	114	24	11	A1A	-	-	BASE OIL	L	MB-500102	SAFE LOCATIC	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2207	40	BO	114	25	01	A1A	-	-	BASE OIL	L	DRUM ADDITIVE P	MB-500103	ATM	FULL OF LIQUID	40	80	*	*	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2208	40	AD	114	25	02	A1A	-	-	ADDITIVE	L	P-500103	MB-500103	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2209	40	AD	114	25	03	A1A	-	-	ADDITIVE	L	MB-500103	P-500103	0.6	1.2	40	80	*	*	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2210	40	AD	114	25	04	A1A	-	-	ADDITIVE	L	P-500103	OUTGOING LI	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2211	25	AD	114	25	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-500103	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2212	40	AD	114	25	06	A1A	-	-	ADDITIVE	L	P-500103	FT-500002	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2213	40	AD	114	25	07	A1A	-	-	ADDITIVE	L	P-500103	FT-500004	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2214	40	AD	114	25	08	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	OUTGOING LI	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2215	40	AD	114	25	09	A1A	-	-	ADDITIVE	L	BYPASS LINE OF OI	OUTGOING LI	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2216	100	PD	114	25	10	B4C	-	-	POWDER	S	DYE HAND ADD	MB-500103	6	8	40	80	*	0.2	900	5	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2217	50	BO	114	25	11	A1A	-	-	BASE OIL	L	MB-500103	SAFE LOCATIC	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2218	40	BO	114	26	01	A1A	-	-	BASE OIL	L	HOSE LINE	MB-500104	ATM	FULL OF LIQUID	40	80	*	*	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2219	40	AD	114	26	02	A1A	-	-	ADDITIVE	L	P-500104	MB-500104	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2220	40	AD	114	26	03	A1A	-	-	ADDITIVE	L	MB-500104	P-500104	0.6	1.2	40	80	*	*	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2221	40	AD	114	26	04	A1A	-	-	ADDITIVE	L	P-500104	OUTGOING LI	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2222	25	AD	114	26	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-500104	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2223	40	AD	114	26	06	A1A	-	-	ADDITIVE	L	P-500104	FT-500005	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2224	40	AD	114	26	07	A1A	-	-	ADDITIVE	L	P-500104	FT-500007	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2225	40	AD	114	26	08	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	OUTGOING LI	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2226	40	AD	114	26	09	A1A	-	-	ADDITIVE	L	BYPASS LINE OF OI	OUTGOING LI	6	8	40	80	*	0.2	900	10	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2227	100	PD	114	26	10	B4C	-	-	POWDER	S	DYE HAND ADD	MB-500104	6	8	40	80	*	0.2	900	5	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2228	50	BO	114	26	11	A1A	-	-	BASE OIL	L	MB-500104	SAFE LOCATIC	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2229	40	BO	114	27	01	A1A	-	-	BASE OIL	L	DRUM ADDITIVE P	MB-500105	ATM	FULL OF LIQUID	40	80	*	*	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2230	40	AD	114	27	02	A1A	-	-	ADDITIVE	L	P-500105	MB-500105	6	8	40	80	*	0.2	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2231	40	AD	114	27	03	A1A	-	-	ADDITIVE	L	MB-500105	P-500105	0.6	1.2	40	80	*	*	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2232	40	AD	114	27	04	A1A	-	-	ADDITIVE	L	P-500105	OUTGOING LI	6	8	40	80	*	0.2	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2233	25	AD	114	27	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-500105	6	8	40	80	*	0.2	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2234	40	AD	114	27	06	A1A	-	-	ADDITIVE	L	P-500105	FT-500019	6	8	40	80	*	0.2	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2235	40	AD	114	27	07	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	OUTGOING LI	6	8	40	80	*	0.2	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2236	40	AD	114	27	08	A1A	-	-	ADDITIVE	L	BYPASS LINE OF O	OUTGOING LI	6	8	40	80	*	0.2	900	15	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2237	100	PD	114	27	09	B4C	-	-	POWDER	S	DYE HAND ADD	MB-500105	6	8	40	80	*	0.2	900	5	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2238	50	BO	114	27	10	A1A	-	-	BASE OIL	L	MB-500105	SAFE LOCATIC	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-114	H1	P&ID FOR FINISHED PRODUCT TANKFARM		
2239	300	BO	112	01	01	P2A	-	-	BASE OIL	L	JETTY/CPCL	SLRB-AML-01	6	8	40	80	*	0.2	900	5	12	100	Water	*	1.5	STD	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2240	50	BO	112	01	02	A1A			BASE OIL	L	MANIFOLD 2	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2241	250	BO	112	01	03	A1B	-	-	BASE OIL	L	SLRB-AML-01 NOZ	MANIFOLD 2 T	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2242	250	BO	112	01	04	A1B	-	-	BASE OIL	L	BAR TEE	MANIFOLD 1	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2243	250	BO	112	01	05	A1B	-	-	BASE OIL	L	BAR TEE	MANIFOLD 2	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2244	250	BO	112	01	06	A1B	-	-	BASE OIL	L	BAR TEE	MANIFOLD 1/	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2245	100	BO	112	01	07	A1A	-	-	BASE OIL	L	P-100049	SO-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2246	100	BO	112	01	08	A1A	-	-	BASE OIL	L	SO-100049	P-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2247	80	BO	112	01	09	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	SO-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2248	100	BO	112	01	10	A1A	-	-	BASE OIL	L	MANIFOLD	SO-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2249	50	BO	112	01	11	A1A	-	-	BASE OIL	L	PIG LINE TRV OUTL	MANIFOLD CO	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2250	50	BO	112	01	12	A1A	-	-	BASE OIL	L	MANIFOLD 2 TRV (	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2251	50	BO	112	01	13	A1A	-	-	BASE OIL	L	MANIFOLD 1 TRV (	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2252	50	BO	112	01	14	A1A	-	-	BASE OIL	L	MANIFOLD 1 PSV	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2253	50	BO	112	01	15	A1A	-	-	BASE OIL	L	MANIFOLD 2 PSV	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2254	80	BO	112	01	16	A1A	-	-	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2255	50	BO	112	01	17	A1A	-	-	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2256	150	BO	112	01	18	A1A	-	-	BASE OIL	L	COMMOON MANI	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2257	50	BO	112	01	19	A1A	-	-	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2258	80	BO	112	01	20	A1A	-	-	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2259	50	BO	112	01	21	A1A	-	-	BASE OIL	L	MANIFOLD 1 TRV (	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2260	50	BO	112	01	22	A1A	-	-	BASE OIL	L	MANIFOLD 2 TRV (	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2261	50	BO	112	01	23	A1A	-	-	BASE OIL	L	MANIFOLD 2	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2262	50	BO	112	01	24	A1A	-	-	BASE OIL	L	MANIFOLD 1	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2263	80	BO	112	01	25	A1A	-	-	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2264	250	BO	112	01	26	A1B			BASE OIL	L	BYPASS LINE OF M	MANIFOLD 1	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2265	250	BO	112	01	27	A1B			BASE OIL	L	BYPASS LINE OF M	MANIFOLD 2	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2266	50	BO	112	01	28	A1A	-	-	BASE OIL	L	MANIFOLD 1	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2267	50	BO	112	01	29	A1A	-	-	BASE OIL	L	SAMPLING POINT	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2268	50	BO	112	01	30	A1A	-	-	BASE OIL	L	SAMPLING POINT	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION										DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2269	25	CA	112	01	31	G6C	-	-	OMPRESSED AI	V	*	TIE IN 04	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2270	25	CA	112	01	32	G6C	-	-	OMPRESSED AI	V	*	TIE IN 05	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2271	150	BO	112	01	33	A1A	-	-	BASE OIL	L	TIE IN 01	SO-100049 NO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2272	200	BO	112	01	35	A1B	-	-	BASE OIL	L	BAR TEE	OPEN	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2273	300	BO	112	02	01	P2A	IH	IE	BASE OIL	L	JETTY/CPCL	SLRB-AML-01	6	8	40	80	*	0.2	900	5	12	100	Water	*	1.5	STD	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2274	50	BO	112	02	02	A1A	IH	IE	BASE OIL	L	MANIFOLD 2	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2275	250	BO	112	02	03	A1B	IH	IE	BASE OIL	L	SLRB-AML-01 NOZ	MANIFOLD 2 T	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2276	250	BO	112	02	04	A1B	IH	IE	BASE OIL	L	BAR TEE	MANIFOLD 1	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2277	250	BO	112	02	05	A1B	IH	IE	BASE OIL	L	BAR TEE	MANIFOLD 2	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2278	250	BO	112	02	06	A1B	IH	IE	BASE OIL	L	BAR TEE	MANIFOLD 1/	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2279	100	BO	112	02	07	A1A	-	-	BASE OIL	L	P-100049	SO-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2280	100	BO	112	02	08	A1A	-	-	BASE OIL	L	SO-100049	P-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2281	80	BO	112	02	09	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	SO-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2282	100	BO	112	02	10	A1A	-	-	BASE OIL	L	MANIFOLD	SO-100049	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2283	50	BO	112	02	11	A1A	IH	IE	BASE OIL	L	PIG LINE TRV OUTL	MANIFOLD CO	6	8	40	80	*	0.2	900	5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2284	50	BO	112	02	12	A1A	IH	IE	BASE OIL	L	MANIFOLD 2 TRV (	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2285	50	BO	112	02	13	A1A	IH	IE	BASE OIL	L	MANIFOLD 1 TRV (	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2286	50	BO	112	02	14	A1A	IH	IE	BASE OIL	L	MANIFOLD 1 PSV	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2287	50	BO	112	02	15	A1A	IH	IE	BASE OIL	L	MANIFOLD 2 PSV	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2288	80	BO	112	02	16	A1A	IH	IE	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2289	50	BO	112	02	17	A1A	IH	IE	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2290	150	BO	112	02	18	A1A	IH	IE	BASE OIL	L	COMMOON MANI	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2291	50	BO	112	02	19	A1A	IH	IE	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2292	80	BO	112	02	20	A1A	IH	IE	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2293	50	BO	112	02	21	A1A	IH	IE	BASE OIL	L	MANIFOLD 1 TRV (	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2294	50	BO	112	02	22	A1A	IH	IE	BASE OIL	L	MANIFOLD 2 TRV (	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2295	50	BO	112	02	23	A1A	IH	IE	BASE OIL	L	MANIFOLD 2	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2296	50	BO	112	02	24	A1A	IH	IE	BASE OIL	L	MANIFOLD 1	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2297	80	BO	112	02	25	A1A	IH	IE	BASE OIL	L	SLRB-AML-01	COMMOON M	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2298	250	BO	112	02	26	A1B	IH	IE	BASE OIL	L	BYPASS LINE OF M	MANIFOLD 1	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2299	250	BO	112	02	27	A1B	IH	IE	BASE OIL	L	BYPASS LINE OF M	MANIFOLD 2	6	8	120	220	*	*	900	2.5	10	300	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2300	50	BO	112	02	28	A1A	IH	IE	BASE OIL	L	MANIFOLD 1	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2301	50	BO	112	02	29	A1A	IH	IE	BASE OIL	L	SAMPLING POINT	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2302	50	BO	112	02	30	A1A	IH	IE	BASE OIL	L	SAMPLING POINT	COMMON MA	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2303	25	CA	112	02	31	G6C	IH	IE	OMPRESSED AI	V	*	TIE IN 04	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2304	25	CA	112	02	32	G6C	IH	IE	OMPRESSED AI	V	*	TIE IN 05	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2305	150	BO	112	02	33	A1A	-	-	BASE OIL	L	TIE IN 01	SO-100049 NO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2306	25	CA	112	03	01	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2307	150	BO	112	03	02	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100001	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2308	150	BO	112	03	03	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100025	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2309	150	BO	112	03	04	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100016	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2310	150	BO	112	03	05	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100017	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2311	150	BO	112	03	06	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100009	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2312	150	BO	112	03	07	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100007	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2313	150	BO	112	03	08	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100026	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2314	150	BO	112	03	09	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100012	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2315	150	BO	112	03	10	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100008	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2316	150	BO	112	03	11	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100033	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2317	150	BO	112	03	12	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100004	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2318	150	BO	112	03	13	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100039/40	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2319	150	BO	112	03	14	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100031/29	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2320	150	BO	112	03	15	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100041	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2321	150	BO	112	03	16	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100037	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2322	150	BO	112	03	17	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100035	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2323	100	BO	112	03	19	A1A	-	-	BASE OIL	L	FROM/TO SLOP TA	ST-100012	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2324	150	BO	112	03	20	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100012	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2325	150	BO	112	03	21	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100026	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2326	150	BO	112	03	22	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100007	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2327	150	BO	112	03	23	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100009	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2328	150	BO	112	03	24	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100017	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2329	150	BO	112	03	25	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100016	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2330	150	BO	112	03	26	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100025	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2331	150	BO	112	03	27	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100001	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2332	25	CA	112	03	28	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD 2	6	8	60	100	*	200	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2333	150	BO	112	03	30	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100033	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2334	150	BO	112	03	31	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100035	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2335	150	BO	112	03	32	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100037	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2336	150	BO	112	03	33	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100041	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2337	150	BO	112	03	34	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100031/29	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2338	150	BO	112	03	35	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100039/40	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2339	150	BO	112	03	36	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100004	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2340	150	BO	112	03	37	A1A	-	-	BASE OIL	L	MANIFOLD 1	ST-100008	6	8	60	100	*	200	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2341	150	BO	112	04	03	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100043	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2342	150	BO	112	04	02	A1A	IH	IE	BASE OIL	L	MANIFOLD 2	ST-100048	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2343	25	N2	112	04	01	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2344	150	BO	112	04	04	A1A	IH	IE	BASE OIL	L	MANIFOLD 2	ST-100047/50	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2345	150	BO	112	04	05	A1A	-	-	BASE OIL	L	MANIFOLD 2	ST-100046	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2346	150	BO	112	04	06	A1A	-	-	BASE OIL	L	FROM/TO SLOP TA	MANIFOLD 2	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2347	150	BO	112	04	17	A1A	IH	IE	BASE OIL	L	MANIFOLD	ST-100047/50	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2348	150	BO	112	04	18	A1A	-	-	BASE OIL	L	MANIFOLD	ST-100043	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2349	150	BO	112	04	19	A1A	IH	IE	BASE OIL	L	MANIFOLD	ST-100048	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2350	150	BO	112	04	20	A1A	-	-	BASE OIL	L	MANIFOLD	ST-100046	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2351	25	CA	112	04	21	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2352	150	BO	112	05	01	A1A	-	-	BASE OIL	L	ST-100004	P-100004	0.6	1.2	60	100	*	*	900	11.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2353	100	BO	112	05	02	A1A	-	-	BASE OIL	L	ST-100004 BOTTOI	P-100004	0.6	1.2	60	100	*	*	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2354	100	BO	112	05	03	A1A	-	-	BASE OIL	L	OUTGOING LINE O	INCOMING LI	ATM	FULL OF LIQUID	60	100	*	*	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2355	150	BO	112	05	04	A1A	-	-	BASE OIL	L	ST-100005	ST-100002	ATM	FULL OF LIQUID	60	100	*	*	900	11.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2356	100	BO	112	05	05	A1A	-	-	BASE OIL	L	P-100004	OUTGOING LI	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2357	100	BO	112	05	06	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100004	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2358	100	BO	112	05	07	A1A	-	-	BASE OIL	L	P-100004	INCOMING LI	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2359	150	BO	112	05	08	A1A	-	-	BASE OIL	L	ST-100005	ST-100002	ATM	FULL OF LIQUID	60	100	*	*	900	11.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2360	100	BO	112	05	09	A1A	-	-	BASE OIL	L	ST-100005 BOTTOI	ST-100002	ATM	FULL OF LIQUID	60	100	*	*	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2361	100	BO	112	05	10	A1A	-	-	BASE OIL	L	P-100004	ST-100002	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2362	100	BO	112	05	11	A1A	-	-	BASE OIL	L	P-100004	BULK LOADIN	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2363	100	BO	112	05	12	A1A	-	-	BASE OIL	L	P-100004	TRANSFORME	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2364	100	BO	112	05	13	A1A	-	-	BASE OIL	L	P-100004	REX	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2365	100	BO	112	05	14	A1A	-	-	BASE OIL	L	P-100004	WHITE OIL	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2366	100	BO	112	05	15	A1A	-	-	BASE OIL	L	P-100004	BN EXTRACT	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2367	100	BO	112	05	16	A1A	-	-	BASE OIL	L	P-100004	PK-301705	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2368	150	BO	112	06	01	A1A	-	-	BASE OIL	L	ST-100002	P-100002	0.6	1.2	60	100	*	*	900	11.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2369	100	BO	112	06	02	A1A	-	-	BASE OIL	L	ST-100002 BOTTOI	P-100002	0.6	1.2	60	100	*	*	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2370	100	BO	112	06	03	A1A	-	-	BASE OIL	L	P-100002	ST-100002	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2371	100	BO	112	06	04	A1A	-	-	BASE OIL	L	RECIRCULATION LI	INLET OF ST-1	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2372	100	BO	112	06	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100002	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2373	100	BO	112	06	06	A1A	-	-	BASE OIL	L	P-100002	MB-302014	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2374	100	BO	112	06	07	A1A	-	-	BASE OIL	L	P-100002	V-302102	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2375	100	BO	112	06	08	A1A	-	-	BASE OIL	L	P-100002	V-302113	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2376	100	BO	112	06	09	A1A	-	-	BASE OIL	L	P-100002	V-302105	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2377	100	BO	112	06	10	A1A	-	-	BASE OIL	L	P-100002	V-302104	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2378	100	BO	112	06	11	A1A	-	-	BASE OIL	L	P-100002	V-302103	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2379	100	BO	112	06	12	A1A	-	-	BASE OIL	L	P-100002	SMB-302002	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2380	100	BO	112	06	13	A1A	-	-	BASE OIL	L	P-100002	V-302101	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2381	100	BO	112	06	14	A1A	-	-	BASE OIL	L	P-100002	DDS-302303	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2382	100	BO	112	06	15	A1A	-	-	BASE OIL	L	P-100002	DDS-302302	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2383	100	BO	112	06	16	A1A	-	-	BASE OIL	L	P-100002	SMB-302001	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2384	100	BO	112	06	17	A1A	-	-	BASE OIL	L	P-100002	TRANSFORME	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2385	100	BO	112	06	18	A1A	-	-	BASE OIL	L	P-100002	WHITE OIL	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2386	80	BO	112	06	19	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET OF ST-1	ATM	FULL OF LIQUID	60	100	*	*	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2387	80	BO	112	06	20	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100014	ATM	FULL OF LIQUID	60	80	*	*	900	11.5	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2388	100	BO	112	06	21	A1A	-	-	BASE OIL	L	P-100002	ST-100004	6	8	60	100	*	50	900	11.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2389	150	BO	112	07	01	A1A	-	-	BASE OIL	L	ST-100026	P-100026	0.6	1.2	60	100	*	*	900	160352	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2390	100	BO	112	07	02	A1A	-	-	BASE OIL	L	ST-100026 BOTTOI	P-100026	0.6	1.2	60	100	*	*	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2391	100	BO	112	07	03	A1A	-	-	BASE OIL	L	P-100026	INLET OF ST-1	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2392	100	BO	112	07	04	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100026	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2393	100	BO	112	07	05	A1A	-	-	BASE OIL	L	P-100026	ST-100026	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2394	100	BO	112	07	06	A1A	-	-	BASE OIL	L	P-100026	RUBBER EXTEI	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2395	100	BO	112	07	07	A1A	-	-	BASE OIL	L	P-100026	DDS-302303	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2396	100	BO	112	07	08	A1A	-	-	BASE OIL	L	P-100026	V-302103	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2397	100	BO	112	07	09	A1A	-	-	BASE OIL	L	P-100026	V-302104	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2398	100	BO	112	07	10	A1A	-	-	BASE OIL	L	P-100026	BN EXTRACT	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2399	100	BO	112	07	11	A1A	-	-	BASE OIL	L	P-100026	V-302101	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2400	100	BO	112	07	12	A1A	-	-	BASE OIL	L	P-100026	SMB-302001	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2401	100	BO	112	07	13	A1A	-	-	BASE OIL	L	P-100026	V-302105	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2402	100	BO	112	07	14	A1A	-	-	BASE OIL	L	P-100026	V-302113	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2403	100	BO	112	07	15	A1A	-	-	BASE OIL	L	P-100026	V-302102	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2404	100	BO	112	07	16	A1A	-	-	BASE OIL	L	P-100026	SMB-302002	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2405	100	BO	112	07	17	A1A	-	-	BASE OIL	L	P-100026	MB-302014	6	8	60	100	*	50	900	160352	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2406	150	BO	112	08	01	A1A	-	-	BASE OIL	L	ST-100014	P-100014	0.6	1.2	60	100	*	*	900	31.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2407	100	BO	112	08	02	A1A	-	-	BASE OIL	L	ST-100014	P-100014	0.6	1.2	60	100	*	*	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2408	100	BO	112	08	03	A1A	-	-	BASE OIL	L	P-100014	INLET OF ST-1	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2409	80	BO	112	08	04	A1A	-	-	BASE OIL	L	TANK TRUCK UNLC	INLET OF ST-1	ATM	FULL OF LIQUID	60	100	*	*	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2410	80	BO	112	08	05	P2A	-	-	BASE OIL	L	TANK TRUCK UNLC	ST-100044	ATM	FULL OF LIQUID	60	80	*	*	900	31.5	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2411	100	BO	112	08	06	A1A	-	-	BASE OIL	L	P-100014	OUTGOING LI	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2412	100	BO	112	08	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100014	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2413	150	BO	112	08	08	A1A	-	-	BASE OIL	L	OUTGOING LINE O	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	31.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2414	100	BO	112	08	09	A1A	-	-	BASE OIL	L	P-100014	INLET OF ST-1	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2415	100	BO	112	08	10	A1A	-	-	BASE OIL	L	P-100014	INLET OF ST-1	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2416	100	BO	112	08	11	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2417	100	BO	112	08	12	A1A	-	-	BASE OIL	L	P-100014	DDS-302302	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2418	100	BO	112	08	13	A1A	-	-	BASE OIL	L	P-100014	V-302112	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2419	100	BO	112	08	14	A1A	-	-	BASE OIL	L	P-100014	V-302102	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2420	100	BO	112	08	15	A1A	-	-	BASE OIL	L	P-100014	V-302111	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2421	100	BO	112	08	16	A1A	-	-	BASE OIL	L	P-100014	V-302101	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2422	100	BO	112	08	17	A1A	-	-	BASE OIL	L	P-100014	SMB-302002	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2423	100	BO	112	08	18	A1A	-	-	BASE OIL	L	P-100014	DDS-302301	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2424	100	BO	112	08	19	A1A	-	-	BASE OIL	L	P-100014	V-302105	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2425	100	BO	112	08	20	A1A	-	-	BASE OIL	L	P-100014	SMB-302001	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2426	100	BO	112	08	21	A1A	-	-	BASE OIL	L	P-100014	VI DISSOL.	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2427	100	BO	112	08	22	A1A	-	-	BASE OIL	L	P-100014	RUBBER EXTEI	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2428	100	BO	112	08	23	A1A	-	-	BASE OIL	L	P-100014	BN EXTRACT	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2429	100	BO	112	08	24	A1A	-	-	BASE OIL	L	P-100014	GREASE PLAN	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2430	80	HOS	112	08	25	A1A	IH	-	HOT OIL	L	HEADER	ST-100015	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2431	80	HOR	112	08	26	A1A	IH	-	HOT OIL	L	HEADER	ST-100015	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2432	80	HOS	112	08	27	A1A	IH	-	BASE OIL	L	*	ST-100050	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2433	80	HOR	112	08	28	A1A	IH	-	BASE OIL	L	ST-100015	*	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2434	40	PA	112	08	29	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2435	150	BO	112	08	30	A1A	-	-	BASE OIL	L	ST-100014	ST-100012	6	8	60	100	*	50	900	31.5	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2436	40	PA	112	08	31	G6C	-	-	PLANT AIR	V	CONTROL BOX	ST-100015	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2437	40	PA	112	08	32	G6C	-	-	PLANT AIR	V	CONTROL BOX	ST-100015	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2438	100	BO	112	08	33	A1A	-	-	BASE OIL	L	P-100014	ST-100012	6	8	40	60	*	*	1.25	0.00001	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2439	150	BO	112	09	01	A1A	-	-	BASE OIL	L	ST-100019	P-100019	0.6	1.2	60	100	*	*	900	30	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2440	100	BO	112	09	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100019	0.6	1.2	60	100	*	*	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2441	100	BO	112	09	03	A1A	-	-	BASE OIL	L	P-100019	INLET OF ST-1	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2442	150	BO	112	09	04	A1A	-	-	BASE OIL	L	ST-100021	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	30	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2443	100	BO	112	09	05	A1A	-	-	BASE OIL	L	P-100019	OUTGOING LI	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2444	100	BO	112	09	06	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100019	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2445	100	BO	112	09	07	A1A	-	-	BASE OIL	L	P-100019	INLET LINE OF	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2446	100	BO	112	09	08	A1A	-	-	BASE OIL	L	P-100019	INLET LINE OF	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2447	100	BO	112	09	09	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	OUTGOING LI	0.6	1.2	60	100	*	*	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2448	100	BO	112	09	10	A1A	-	-	BASE OIL	L	P-100019	MB-302014	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2449	100	BO	112	09	11	A1A	-	-	BASE OIL	L	P-100019	V-302104	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2450	100	BO	112	09	12	A1A	-	-	BASE OIL	L	P-100019	FP TANK	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2451	100	BO	112	09	13	A1A	-	-	BASE OIL	L	P-100019	BN EXTRACT	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2452	100	BO	112	09	14	A1A	-	-	BASE OIL	L	P-100019	RUBBER EXTEI	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2453	100	BO	112	09	15	A1A	-	-	BASE OIL	L	P-100019	VI DISSOL.	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2454	100	BO	112	09	16	A1A	-	-	BASE OIL	L	P-100019	GREASE PLAN	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2455	100	BO	112	09	17	A1A	-	-	BASE OIL	L	P-100019	SMB-302001	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2456	100	BO	112	09	18	A1A	-	-	BASE OIL	L	P-100019	DDS-302303	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2457	100	BO	112	09	19	A1A	-	-	BASE OIL	L	P-100019	DDS-302302	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2458	100	BO	112	09	20	A1A	-	-	BASE OIL	L	P-100019	V-302112	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2459	100	BO	112	09	21	A1A	-	-	BASE OIL	L	P-100019	V-302113	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2460	100	BO	112	09	22	A1A	-	-	BASE OIL	L	P-100019	DDS-302301	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2461	100	BO	112	09	23	A1A	-	-	BASE OIL	L	P-100019	V-302101	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2462	100	BO	112	09	24	A1A	-	-	BASE OIL	L	P-100019	V-302105	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2463	100	BO	112	09	25	A1A	-	-	BASE OIL	L	P-100019	SMB-302002	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2464	100	BO	112	09	26	A1A	-	-	BASE OIL	L	P-100019	V-302102	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2465	100	BO	112	09	27	A1A	-	-	BASE OIL	L	P-100019	V-302103	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2466	100	BO	112	09	28	A1A	-	-	BASE OIL	L	P-100019	V-302111	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2467	80	HOS	112	09	29	A1A	IH	-	HOT OIL	L	HEADER	ST-100021	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2468	80	HOR	112	09	30	A1A	IH	-	HOT OIL	L	HEADER	ST-100021	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2469	40	PA	112	09	33	G6C	-	-	PLANT AIR	V	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2470	40	PA	112	09	31	G6C	-	-	PLANT AIR	V	CONTROL BOX	ST-100015	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2471	40	PA	112	09	32	G6C	-	-	PLANT AIR	V	CONTROL BOX	ST-100015	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2472	150	BO	112	09	34	A1A	-	-	BASE OIL	L	ST-100019	ST-100017	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2473	100	BO	112	09	36	A1A	-	-	BASE OIL	L	P-100019	ST-100017	6	8	60	100	*	50	900	30	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2474	150	BO	112	10	01	A1A	-	-	BASE OIL	L	ST-100039	P-100039	0.6	1.2	60	100	*	*	900	20	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2475	100	BO	112	10	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100039	0.6	1.2	60	100	*	*	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2476	100	BO	112	10	03	A1A	-	-	BASE OIL	L	P-100039	INLET OF ST-1	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2477	80	BO	112	10	04	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET OF ST-1	ATM	FULL OF LIQUID	60	100	*	*	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2478	150	BO	112	10	05	A1A	-	-	BASE OIL	L	ST-100040	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	20	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2479	100	BO	112	10	06	A1A	-	-	BASE OIL	L	P-100039	OUTGOING LI	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2480	100	BO	112	10	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100039	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2481	100	BO	112	10	08	A1A	-	-	BASE OIL	L	P-100039	INLET OF ST-1	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2482	100	BO	112	10	10	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	OUTGOING LI	0.6	1.2	60	100	*	*	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2483	100	BO	112	10	11	A1A	-	-	BASE OIL	L	P-100039	V-302111	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2484	100	BO	112	10	12	A1A	-	-	BASE OIL	L	P-100039	V-302102	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2485	100	BO	112	10	13	A1A	-	-	BASE OIL	L	P-100039	V-302101	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2486	100	BO	112	10	14	A1A	-	-	BASE OIL	L	P-100039	DDS-302301	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2487	100	BO	112	10	15	A1A	-	-	BASE OIL	L	P-100039	SMB-302001	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2488	100	BO	112	10	16	A1A	-	-	BASE OIL	L	P-100039	V-302105	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2489	100	BO	112	10	17	A1A	-	-	BASE OIL	L	P-100039	SMB-302002	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2490	100	BO	112	10	18	A1A	-	-	BASE OIL	L	P-100039	GREASE PLAN	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2491	100	BO	112	10	19	A1A	-	-	BASE OIL	L	P-100039	BULK LOADIN	6	8	60	100	*	50	900	20	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2492	80	BO	112	10	20	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100029	ATM	FULL OF LIQUID	60	80	*	*	900	20	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2493	150	BO	112	11	01	A1A	-	-	BASE OIL	L	ST-100018	P-100018	0.6	1.2	60	150	*	*	900	467	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2494	100	BO	112	11	02	A1A	IH	-	BASE OIL	L	BOTTOM LINE OF S	P-100018	0.6	1.2	60	150	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2495	80	BO	112	11	03	A1A	IH	-	BASE OIL	L	P-100018	INLET OF ST-1	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2496	100	BO	112	11	04	A1A	IH	-	BASE OIL	L	ST-100020	OUTGOING LI	0.6	1.2	60	150	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2497	100	BO	112	11	05	A1A	IH	-	BASE OIL	L	P-100018	OUTGOING LI	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2498	100	BO	112	11	06	A1A	IH	-	BASE OIL	L	BYPASS LINE OF P-	P-100018	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2499	80	BO	112	11	07	A1A	IH	-	BASE OIL	L	P-100018	INCOMING LI	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2500	100	BO	112	11	08	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	OUTGOING LI	0.6	1.2	60	150	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2501	100	BO	112	11	09	A1A	IH	-	BASE OIL	L	P-100018	BITUMEN	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2502	100	BO	112	11	10	A1A	IH	-	BASE OIL	L	P-100018	RUBBER EXTEI	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2503	100	BO	112	11	11	A1A	IH	-	BASE OIL	L	P-100018	GREASE PLAN	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2504	100	BO	112	11	12	A1A	IH	-	BASE OIL	L	P-100018	BN EXTRACT	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2505	100	BO	112	11	13	A1A	IH	-	BASE OIL	L	P-100018	SMB-302001	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2506	100	BO	112	11	14	A1A	IH	-	BASE OIL	L	P-100018	DDS-302302	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2507	100	BO	112	11	15	A1A	IH	-	BASE OIL	L	P-100018	SMB-302002	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2508	100	BO	112	11	16	A1A	IH	-	BASE OIL	L	P-100018	V-302101	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR														
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2509	100	BO	112	11	17	A1A	IH	-	BASE OIL	L	P-100018	V-302102	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2510	100	BO	112	11	18	A1A	IH	-	BASE OIL	L	P-100018	DDS-302301	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2511	100	BO	112	11	19	A1A	IH	-	BASE OIL	L	P-100018	V-302105	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2512	100	BO	112	11	20	A1A	IH	-	BASE OIL	L	P-100018	V-302112	6	8	60	150	*	30	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2513	80	HOS(M)	112	11	21	A1A	IH	-	HOT OIL	L	HEADER	SH-100020	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2514	80	HOR(M)	112	11	22	A1A	IH	-	HOT OIL	L	HEADER	SH-100020	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2515	80	HOS	112	11	23	A1A	IH	-	HOT OIL	L	HEADER	ST-100020	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2516	80	HOR	112	11	24	A1A	IH	-	HOT OIL	L	HEADER	ST-100020	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2517	80	HOS(M)	112	11	25	A1A	IH	-	HOT OIL	L	HEADER	SH-100018	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2518	80	HOR(M)	112	11	26	A1A	IH	-	HOT OIL	L	HEADER	SH-100018	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2519	40	PA	112	11	29	G6C	-	-	PLANT AIR	L	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2520	40	PA	112	11	27	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100020	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2521	40	PA	112	11	28	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100020	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2522	150	BO	112	11	30	A1A	-	-	BASE OIL	L	ST-100018	ST-100016	6	8	60	150	*	*	900	467	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2523	100	BO	112	11	31	A1A	IH	-	BASE OIL	L	P-100018	ST-100016	6	8	60	150	*	*	900	467	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2524	150	BO	112	12	01	A1A	-	-	BASE OIL	L	ST-100001	P-100001	0.6	1.2	60	100	*	*	900	68	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2525	100	BO	112	12	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100001	0.6	1.2	60	100	*	*	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2526	100	BO	112	12	03	A1A	-	-	BASE OIL	L	P-100001	INLET LINE OF	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2527	100	BO	112	12	04	A1A	-	-	BASE OIL	L	P-100001	ST-100001	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2528	100	BO	112	12	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100001	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2529	100	BO	112	12	06	A1A	-	-	BASE OIL	L	P-100001	BULK LOADIN	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2530	100	BO	112	12	07	A1A	-	-	BASE OIL	L	P-100001	RUBBER EXTEI	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2531	100	BO	112	12	08	A1A	-	-	BASE OIL	L	P-100001	SMB-302001	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2532	100	BO	112	12	09	A1A	-	-	BASE OIL	L	P-100001	SMB-302002	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2533	100	BO	112	12	10	A1A	-	-	BASE OIL	L	P-100001	V-302101	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2534	100	BO	112	12	11	A1A	-	-	BASE OIL	L	P-100001	V-302102	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2535	100	BO	112	12	12	A1A	-	-	BASE OIL	L	P-100001	DDS-302302	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2536	150	BO	112	13	01	A1A	-	-	BASE OIL	L	ST-100025	P-100025	0.6	1.2	60	100	*	*	900	68	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2537	100	BO	112	13	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100025	0.6	1.2	60	100	*	*	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2538	100	BO	112	13	03	A1A	-	-	BASE OIL	L	P-100025	ST-100025	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2539	100	BO	112	13	04	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100025	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2540	100	BO	112	13	05	A1A	-	-	BASE OIL	L	P-100025	LOBS LOADIN	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2541	100	BO	112	13	06	A1A	-	-	BASE OIL	L	P-100025	RUBBER EXTEI	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2542	100	BO	112	13	07	A1A	-	-	BASE OIL	L	P-100025	SMB-302001	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2543	100	BO	112	13	08	A1A	-	-	BASE OIL	L	P-100025	V-302101	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2544	100	BO	112	13	09	A1A	-	-	BASE OIL	L	P-100025	SMB-302002	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2545	100	BO	112	13	10	A1A	-	-	BASE OIL	L	P-100025	DDS-302302	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2546	100	BO	112	13	11	A1A	-	-	BASE OIL	L	P-100025	V-302102	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2547	100	BO	112	13	12	A1A	-	-	BASE OIL	L	P-100025	V-302112	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2548	100	BO	112	13	13	A1A	-	-	BASE OIL	L	P-100025	DDS-302301	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2549	100	BO	112	13	14	A1A	-	-	BASE OIL	L	P-100025	V-302105	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2550	100	BO	112	13	15	A1A	-	-	BASE OIL	L	P-100025	INCOMING LI	6	8	60	100	*	50	900	68	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2551	150	BO	112	14	01	A1A	-	-	BASE OIL	L	ST-100031	P-100031	0.6	1.2	60	100	*	*	900	38	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2552	100	BO	112	14	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100031	0.6	1.2	60	100	*	*	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2553	100	BO	112	14	03	A1A	-	-	BASE OIL	L	P-100031	INCOMING LI	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2554	80	BO	112	14	04	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100031	ATM	FULL OF LIQUID	60	80	*	*	900	38	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2555	80	BO	112	14	05	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INCOMING LI	ATM	FULL OF LIQUID	60	100	*	*	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2556	150	BO	112	14	06	A1A	-	-	BASE OIL	L	OUTGOING LINE O	OUTGOING LI	0.6	1.2	60	100	*	*	900	38	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2557	100	BO	112	14	07	A1A	-	-	BASE OIL	L	P-100031	OUTGOING LI	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2558	100	BO	112	14	08	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100031	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2559	100	BO	112	14	09	A1A	-	-	BASE OIL	L	P-100031	INLET OF P-10	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2560	100	BO	112	14	10	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100029	ATM	FULL OF LIQUID	60	100	*	*	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2561	100	BO	112	14	11	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	OUTGOING LI	0.6	1.2	60	100	*	*	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2562	100	BO	112	14	12	A1A	-	-	BASE OIL	L	P-100031	SMB-302001	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2563	100	BO	112	14	13	A1A	-	-	BASE OIL	L	P-100031	SMB-302002	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2564	100	BO	112	14	14	A1A	-	-	BASE OIL	L	P-100031	V-302101	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2565	100	BO	112	14	15	A1A	-	-	BASE OIL	L	P-100031	V-302102	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2566	100	BO	112	14	16	A1A	-	-	BASE OIL	L	P-100031	V-302105	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2567	100	BO	112	14	17	A1A	-	-	BASE OIL	L	P-100031	DDS-302302	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2568	100	BO	112	14	18	A1A	-	-	BASE OIL	L	P-100031	V-302112	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2569	100	BO	112	14	19	A1A	-	-	BASE OIL	L	P-100031	V-302111	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2570	100	BO	112	14	20	A1A	-	-	BASE OIL	L	P-100031	GREASE PLAN	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2571	100	BO	112	14	21	A1A	-	-	BASE OIL	L	P-100031	BULK LOADIN	6	8	60	100	*	50	900	38	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2572	150	BO	112	15	01	A1A	-	-	BASE OIL	L	ST-100041	P-100041	0.6	1.2	60	100	*	*	900	32	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2573	100	BO	112	15	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100041	0.6	1.2	60	100	*	*	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2574	100	BO	112	15	03	A1A	-	-	BASE OIL	L	P-100041	INLET LINE OF	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2575	80	BO	112	15	04	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100044	ATM	FULL OF LIQUID	60	80	*	*	900	32	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2576	100	BO	112	15	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100041	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2577	100	BO	112	15	06	A1A	-	-	BASE OIL	L	P-100041	ST-100041	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2578	80	BO	112	15	07	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2579	100	BO	112	15	08	A1A	-	-	BASE OIL	L	P-100041	SMB-302001	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2580	100	BO	112	15	09	A1A	-	-	BASE OIL	L	P-100041	V-302101	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2581	100	BO	112	15	10	A1A	-	-	BASE OIL	L	P-100041	DDS-302302	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2582	100	BO	112	15	11	A1A	-	-	BASE OIL	L	P-100041	SMB-302002	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2583	100	BO	112	15	12	A1A	-	-	BASE OIL	L	P-100041	V-302102	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2584	100	BO	112	15	13	A1A	-	-	BASE OIL	L	P-100041	V-302105	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2585	100	BO	112	15	14	A1A	-	-	BASE OIL	L	P-100041	V-302112	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2586	100	BO	112	15	15	A1A	-	-	BASE OIL	L	P-100041	V-302111	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2587	100	BO	112	16	01	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2588	80	BO	112	16	02	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100010	ATM	FULL OF LIQUID	60	80	*	*	900	46	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2589	100	BO	112	16	03	A1A	-	-	BASE OIL	L	P-100044	INLET LINE OF	6	8	60	100	*	50	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2590	150	BO	112	16	04	A1A	-	-	BASE OIL	L	ST-100044	P-100044	0.6	1.2	60	100	*	*	900	46	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2591	100	BO	112	16	05	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100044	0.6	1.2	60	100	*	*	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2592	100	BO	112	16	06	A1A	-	-	BASE OIL	L	P-100044	ST-100044	6	8	60	100	*	50	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2593	100	BO	112	16	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100044	6	8	60	100	*	50	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2594	100	BO	112	16	08	A1A	-	-	BASE OIL	L	P-100044	GREASE PLAN	6	8	60	100	*	50	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2595	100	BO	112	16	09	A1A	-	-	BASE OIL	L	P-100044	SMB-302001	6	8	60	100	*	50	900	46	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2596	150	BO	112	17	01	A1A	-	-	BASE OIL	L	ST-100008	P-100008	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2597	100	BO	112	17	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100008	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2598	100	BO	112	17	03	A1A	-	-	BASE OIL	L	P-100008	ST-100008	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2599	100	BO	112	17	04	A1A	-	-	BASE OIL	L	P-100008	INCOMING LI	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2600	100	BO	112	17	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100008	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2601	100	BO	112	17	06	A1A	-	-	BASE OIL	L	P-100008	VI DISSOL.	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2602	100	BO	112	17	07	A1A	-	-	BASE OIL	L	P-100008	BN EXTRACT	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2603	100	BO	112	17	08	A1A	-	-	BASE OIL	L	P-100008	RUBBER EXTEI	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2604	100	BO	112	17	09	A1A	-	-	BASE OIL	L	P-100008	GREASE PLAN	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2605	100	BO	112	17	10	A1A	-	-	BASE OIL	L	P-100008	SMB-302001	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2606	100	BO	112	17	11	A1A	-	-	BASE OIL	L	P-100008	V-302112	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2607	100	BO	112	17	12	A1A	-	-	BASE OIL	L	P-100008	V-302102	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2608	100	BO	112	17	13	A1A	-	-	BASE OIL	L	P-100008	V-302111	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2609	100	BO	112	17	14	A1A	-	-	BASE OIL	L	P-100008	V-302101	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2610	100	BO	112	17	15	A1A	-	-	BASE OIL	L	P-100008	DDS-302302	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2611	100	BO	112	17	16	A1A	-	-	BASE OIL	L	P-100008	SMB-302002	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2612	100	BO	112	17	17	A1A	-	-	BASE OIL	L	P-100008	V-302105	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2613	100	BO	112	17	18	A1A	-	-	BASE OIL	L	P-100008	DDS-302301	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2614	100	BO	112	17	19	A1A	-	-	BASE OIL	L	P-100008	ST-100006	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2615	100	BO	112	17	20	A1A	-	-	BASE OIL	L	P-100008	ST-100010	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2616	100	BO	112	17	21	A1A	-	-	BASE OIL	L	P-100008	ST-100006	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2617	150	BO	112	18	01	A1A	-	-	BASE OIL	L	ST-100010	P-100010/ST-	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2618	100	BO	112	18	02	A1A	-	-	BASE OIL	L	BOTOM LINE OF ST	P-100010/ST-	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2619	100	BO	112	18	03	A1A	-	-	BASE OIL	L	P-100010	INLET LINE OF	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2620	80	BO	112	18	04	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100007	ATM	FULL OF LIQUID	60	80	*	*	900	70.4	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2621	100	BO	112	18	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100010	0.6	1.2	60	100	*	*	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2622	100	BO	112	18	06	A1A	-	-	BASE OIL	L	P-100010	ST-100010	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2623	100	BO	112	18	07	A1A	-	-	BASE OIL	L	P-100010	ST-100006	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2624	100	BO	112	18	08	A1A	-	-	BASE OIL	L	P-100010	SMB-302001	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2625	100	BO	112	18	09	A1A	-	-	BASE OIL	L	P-100010	V-302112	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2626	100	BO	112	18	10	A1A	-	-	BASE OIL	L	P-100010	V-302111	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2627	100	BO	112	18	11	A1A	-	-	BASE OIL	L	P-100010	V-302101	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2628	100	BO	112	18	12	A1A	-	-	BASE OIL	L	P-100010	DDS-302302	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2629	100	BO	112	18	13	A1A	-	-	BASE OIL	L	P-100010	V-302105	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2630	100	BO	112	18	14	A1A	-	-	BASE OIL	L	P-100010	DDS-302301	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2631	100	BO	112	18	15	A1A	-	-	BASE OIL	L	P-100010	V-302102	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2632	100	BO	112	18	16	A1A	-	-	BASE OIL	L	P-100010	SMB-302002	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2633	80	HOS	112	18	17	A1A	IH	-	HOT OIL	L	HEADER	ST-100010	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2634	80	HOR	112	18	18	A1A	IH	-	HOT OIL	L	HEADER	ST-100010	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2635	80	BO	112	18	24	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2636	40	PA	112	18	21	G6C	-	-	PLANT AIR	L	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2637	40	PA	112	18	19	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100010	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2638	40	PA	112	18	20	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100010	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2639	150	BO	112	18	22	A1A	-	-	BASE OIL	L	ST-100010	ST-100008	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2640	150	BO	112	18	23	A1A	-	-	BASE OIL	L	ST-100010	ST-100006	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2641	100	BO	112	18	24	A1A	-	-	BASE OIL	L	P-100010	ST-100006	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2642	100	BO	112	18	25	A1A	-	-	BASE OIL	L	P-100010	ST-100008	6	8	60	100	*	50	900	70.4	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2643	150	BO	112	19	01	A1A	-	-	BASE OIL	L	ST-100009	P-100009	0.6	1.2	60	100	*	*	900	97	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2644	100	BO	112	19	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100009	0.6	1.2	60	100	*	*	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2645	100	BO	112	19	03	A1A	-	-	BASE OIL	L	P-100009	ST-100009	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2646	100	BO	112	19	04	A1A	-	-	BASE OIL	L	P-100009	INCOMING LI	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2647	100	BO	112	19	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100009	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2648	100	BO	112	19	06	A1A	-	-	BASE OIL	L	P-100009	BITUMEN	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2649	100	BO	112	19	07	A1A	-	-	BASE OIL	L	P-100009	RUBBER EXTEI	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2650	100	BO	112	19	08	A1A	-	-	BASE OIL	L	P-100009	BN EXTRACT	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2651	100	BO	112	19	09	A1A	-	-	BASE OIL	L	P-100009	GREASE PLAN	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2652	100	BO	112	19	10	A1A	-	-	BASE OIL	L	P-100009	RINSING	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2653	100	BO	112	19	11	A1A	-	-	BASE OIL	L	P-100009	SMB-302001	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2654	100	BO	112	19	12	A1A	-	-	BASE OIL	L	P-100009	DDS-302302	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2655	100	BO	112	19	13	A1A	-	-	BASE OIL	L	P-100009	V-302111	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2656	100	BO	112	19	14	A1A	-	-	BASE OIL	L	P-100009	V-302101	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2657	100	BO	112	19	15	A1A	-	-	BASE OIL	L	P-100009	SMB-302002	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2658	100	BO	112	19	16	A1A	-	-	BASE OIL	L	P-100009	DDS-302301	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2659	100	BO	112	19	17	A1A	-	-	BASE OIL	L	P-100009	V-302112	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2660	100	BO	112	19	18	A1A	-	-	BASE OIL	L	P-100009	V-302105	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2661	100	BO	112	19	19	A1A	-	-	BASE OIL	L	P-100009	V-302102	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2662	100	BO	112	19	20	A1A	-	-	BASE OIL	L	P-100009	ST-100013	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2663	100	BO	112	19	21	A1A	-	-	BASE OIL	L	P-100009	ST-100011	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2664	150	BO	112	20	01	A1A	-	-	BASE OIL	L	ST-100011	P-100011	0.6	1.2	60	100	*	*	900	97	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2665	100	BO	112	20	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100011	0.6	1.2	60	100	*	*	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2666	100	BO	112	20	03	A1A	-	-	BASE OIL	L	P-100011	ST-100011	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2667	100	BO	112	20	04	A1A	-	-	BASE OIL	L	P-100011	INLET LINE OF	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2668	100	BO	112	20	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100011	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2669	100	BO	112	20	06	A1A	-	-	BASE OIL	L	P-100011	V-302112	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2670	100	BO	112	20	07	A1A	-	-	BASE OIL	L	P-100011	SMB-302001	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2671	100	BO	112	20	08	A1A	-	-	BASE OIL	L	P-100011	DDS-302302	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2672	100	BO	112	20	09	A1A	-	-	BASE OIL	L	P-100011	V-302111	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2673	100	BO	112	20	10	A1A	-	-	BASE OIL	L	P-100011	V-302101	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2674	100	BO	112	20	11	A1A	-	-	BASE OIL	L	P-100011	SMB-302002	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2675	100	BO	112	20	12	A1A	-	-	BASE OIL	L	P-100011	V-302105	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2676	100	BO	112	20	13	A1A	-	-	BASE OIL	L	P-100011	V-302102	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2677	100	BO	112	20	14	A1A	-	-	BASE OIL	L	P-100011	DDS-302301	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2678	80	HOS	112	20	15	A1A	IH	-	HOT OIL	L	HEADER	ST-100011	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2679	80	HOR	112	20	16	A1A	IH	-	HOT OIL	L	HEADER	ST-100011	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2680	40	PA	112	20	19	G6C	-	-	PLANT AIR	L	*	CONTROL BOX	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2681	40	PA	112	20	17	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100011	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2682	40	PA	112	20	18	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100011	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2683	150	BO	112	20	20	A1A	-	-	BASE OIL	L	ST-100011	ST-100009	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2684	150	BO	112	20	21	A1A	-	-	BASE OIL	L	ST-100011	ST-100013	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2685	100	BO	112	20	22	A1A	-	-	BASE OIL	L	P-100011	ST-100009	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2686	100	BO	112	20	23	A1A	-	-	BASE OIL	L	P-100011	ST-100013	6	8	60	100	*	50	900	97	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2687	150	BO	112	21	01	A1A	-	-	BASE OIL	L	ST-100007	P-100007	0.6	1.2	60	100	*	*	900	450	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2688	100	BO	112	21	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100007	0.6	1.2	60	100	*	*	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2689	100	BO	112	21	03	A1A	-	-	BASE OIL	L	P-100007	ST-100007	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2690	100	BO	112	21	04	A1A	-	-	BASE OIL	L	P-100007	INLET LINE OF	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2691	100	BO	112	21	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100007	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2692	100	BO	112	21	06	A1A	-	-	BASE OIL	L	P-100007	V-302105	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2693	100	BO	112	21	07	A1A	-	-	BASE OIL	L	P-100007	V-302103	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2694	100	BO	112	21	08	A1A	-	-	BASE OIL	L	P-100007	V-302102	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2695	100	BO	112	21	09	A1A	-	-	BASE OIL	L	P-100007	V-302101	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2696	100	BO	112	21	10	A1A	-	-	BASE OIL	L	P-100007	SMB-302002	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2697	100	BO	112	21	11	A1A	-	-	BASE OIL	L	P-100007	SMB-302001	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2698	100	BO	112	21	12	A1A	-	-	BASE OIL	L	P-100007	V-302104	6	8	60	100	*	30	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2699	25	CA	112	21	14	G6C	-	-	OMPRESSED AI	V	*	TANK TRUCK U	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2700	25	V	112	21	13	G6C	-	-	VAPOR	V	TANK TRUCK UNLO	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2701	80	HOS	112	21	15	A1A	IH	-	HOT OIL	L	HEADER	ST-100007	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2702	80	HOR	112	21	16	A1A	IH	-	HOT OIL	L	HEADER	ST-100007	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2703	80	BO	112	21	20	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	450	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2704	40	PA	112	21	19	G6C	-	-	PLANT AIR	L	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2705	40	PA	112	21	17	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100007	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2706	40	PA	112	21	18	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100007	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2707	150	BO	112	22	01	A1A	-	-	BASE OIL	L	ST-100022	P-100022	0.6	1.2	60	100	*	*	900	2.3	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2708	100	BO	112	22	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100022	0.6	1.2	60	100	*	*	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
2709	100	BO	112	22	03	A1A	-	-	BASE OIL	L	P-100022	ST-100022	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2710	100	BO	112	22	04	A1A	-	-	BASE OIL	L	P-100022	INLET LINE OF	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2711	100	BO	112	22	05	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2712	100	BO	112	22	06	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100022	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2713	100	BO	112	22	07	A1A	-	-	BASE OIL	L	P-100022	BULK LOADIN	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2714	100	BO	112	22	08	A1A	-	-	BASE OIL	L	P-100022	V-302103	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2715	100	BO	112	22	09	A1A	-	-	BASE OIL	L	P-100022	V-302104	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2716	100	BO	112	22	10	A1A	-	-	BASE OIL	L	P-100022	SMB-302001	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2717	100	BO	112	22	11	A1A	-	-	BASE OIL	L	P-100022	V-302102	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2718	100	BO	112	22	12	A1A	-	-	BASE OIL	L	P-100022	SMB-302002	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2719	100	BO	112	22	13	A1A	-	-	BASE OIL	L	P-100022	V-302105	6	8	60	100	*	50	900	2.3	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2720	80	BO	112	22	14	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100037	ATM	FULL OF LIQUID	60	100	*	*	900	2.3	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2721	100	BO	112	23	01	A1A	IH	IE	BASE OIL	L	ST-100043	P-100043	0.6	1.2	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2722	100	BO	112	23	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100043	0.6	1.2	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2723	100	BO	112	23	03	A1A	IH	IE	BASE OIL	L	P-100043	INLET LINE OF	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2724	100	BO	112	23	04	A1A	IH	IE	BASE OIL	L	ST-100045	OUTGOING LI	0.6	1.2	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2725	100	BO	112	23	05	A1A	IH	IE	BASE OIL	L	P-100043	OUTGOING LI	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2726	100	BO	112	23	06	A1A	IH	IE	BASE OIL	L	BYPASS LINE OF P-	P-100043	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2727	100	BO	112	23	08	A1A	IH	IE	BASE OIL	L	P-100043	OUTGOING LI	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2728	100	BO	112	23	09	A1A	-	-	BASE OIL	L	ST-100045	OUTGOING LI	0.6	1.2	60	150	*	*	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2729	100	BO	112	23	10	A1A	IH	IE	BASE OIL	L	P-100043	BITUMEN	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2730	100	BO	112	23	12	A1A	IH	IE	BASE OIL	L	P-100043	RUBBER EXTEI	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2731	100	BO	112	23	11	A1A	IH	IE	BASE OIL	L	P-100043	PK-301705	6	8	60	150	*	50	900	600	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2732	40	PA	112	23	13	G6C	-	-	PLANT AIR	V	*	ST-100045	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2733	80	HOS	112	23	22	A1A	IH	-	HOT OIL	L	HEADER	ST-100043	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2734	80	HOR	112	23	23	A1A	IH	-	HOT OIL	L	HEADER	ST-100043	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2735	80	HOS(M)	112	23	16	A1A	IH	-	HOT OIL	L	HEADER	SH-100043	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2736	80	HOR(M)	112	23	17	A1A	IH	-	HOT OIL	L	HEADER	SH-100043	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2737	80	HOS	112	23	18	A1A	IH	-	HOT OIL	L	HEADER	ST-100045	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2738	80	HOR	112	23	19	A1A	IH	-	HOT OIL	L	HEADER	ST-100045	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2739	80	HOS(M)	112	23	20	A1A	IH	-	HOT OIL	L	HEADER	SH-100045	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2740	80	HOR(M)	112	23	21	A1A	IH	-	HOT OIL	L	HEADER	SH-100045	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2741	40	PA	112	23	14	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100045	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2742	40	PA	112	23	15	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100045	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2743	100	BO	112	24	01	A1A	IH	IE	BASE OIL	L	ST-100048	P-100048	0.6	1.2	60	150	*	*	900	1300	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2744	100	BO	112	24	02	A1A	IH	IE	BASE OIL	L	BOTTOM LINE OF S	P-100048	0.6	1.2	60	150	*	*	900	1300	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2745	100	BO	112	24	03	A1A	IH	IE	BASE OIL	L	P-100048	INCOMING LI	6	8	60	150	*	*	900	1300	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2746	100	BO	112	24	04	A1A	IH	IE	BASE OIL	L	P-100048	ST-100048	6	8	60	150	*	*	900	1300	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2747	100	BO	112	24	05	A1A	IH	IE	BASE OIL	L	BYPASS LINE OF P-	P-100048	6	8	60	150	*	*	900	1300	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2748	100	BO	112	24	06	A1A	IH	IE	BASE OIL	L	P-100048	RUBBER EXTEI	6	8	60	150	*	*	900	1300	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2749	80	HOS	112	24	07	A1A	IH	-	HOT OIL	L	HEADER	ST-100048	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2750	80	HOR	112	24	08	A1A	IH	-	HOT OIL	L	HEADER	ST-100048	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2751	80	HOR(M)	112	24	09	A1A	IH	-	HOT OIL	L	HEADER	SH-100048	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2752	80	HOS(M)	112	24	10	A1A	IH	-	HOT OIL	L	HEADER	SH-100048	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2753	150	BO	112	25	01	A1A	-	-	BASE OIL	L	ST-100046	P-100046	0.6	1.2	60	100	*	*	900	298	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2754	100	BO	112	25	02	A1A	-	-	BASE OIL	L	ST-100046	P-100046	0.6	1.2	60	100	*	*	900	298	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2755	100	BO	112	25	03	A1A	-	-	BASE OIL	L	P-100046	INLET LINE OF	6	8	60	100	*	50	900	298	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2756	100	BO	112	25	04	A1A	-	-	BASE OIL	L	P-100046	ST-100046	6	8	60	100	*	50	900	298	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2757	100	BO	112	25	05	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100046	6	8	60	100	*	50	900	298	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2758	100	BO	112	25	06	A1A	-	-	BASE OIL	L	P-100046	RUBBER EXTEI	6	8	60	100	*	50	900	298	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2759	100	BO	112	26	01	A1A	IH	IE	BASE OIL	L	ST-100047	P-100047	0.6	1.2	60	150	*	*	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2760	100	BO	112	26	02	A1A	IH	IE	BASE OIL	L	BOTTOM LINE OF S	P-100047	0.6	1.2	60	150	*	*	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2761	100	BO	112	26	03	A1A	IH	IE	BASE OIL	L	P-100047	OUTGOING LI	6	8	60	150	*	30	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2762	100	BO	112	26	04	A1A	IH	IE	BASE OIL	L	ST-100050	OUTGOING LI	ATM	FULL OF LIQUID	60	100	*	*	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2763	100	BO	112	26	05	A1A	IH	IE	BASE OIL	L	P-100047	OUTGOING LI	6	8	60	150	*	30	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2764	100	BO	112	26	06	A1A	IH	IE	BASE OIL	L	BYPASS LINE OF P-	P-100047	6	8	60	150	*	30	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2765	100	BO	112	26	07	A1A	IH	IE	BASE OIL	L	P-100047	BN EXTRACT	6	8	60	150	*	30	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2766	100	BO	112	26	08	A1A	IH	IE	BASE OIL	L	P-100047	INLET OF ST-1	6	8	60	150	*	30	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2767	100	BO	112	26	09	A1A	IH	IE	BASE OIL	L	BOTTOM LINE OF S	OUTGOING LI	0.6	1.2	60	120	*	*	900	5800	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2768	80	HOS	112	26	11	A1A	IH	-	HOT OIL	L	HEADER	ST-100047	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2769	80	HOR	112	26	12	A1A	IH	-	HOT OIL	L	HEADER	ST-100047	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2770	80	HOR(M)	112	26	13	A1A	IH	-	HOT OIL	L	HEADER	SH-100047	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2771	80	HOS(M)	112	26	14	A1A	IH	-	HOT OIL	L	HEADER	SH-100047	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2772	80	HOS	112	26	15	A1A	IH	-	HOT OIL	L	HEADER	ST-100050	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2773	80	HOR	112	26	16	A1A	IH	-	HOT OIL	L	HEADER	ST-100050	6	8	60	100	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2774	80	HOR(M)	112	26	17	A1A	IH	-	HOT OIL	L	HEADER	SH-100050	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2775	80	HOS(M)	112	26	18	A1A	IH	-	HOT OIL	L	HEADER	SH-100050	6	8	160	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2776	40	PA	112	26	10	G6C	-	-	PLANT AIR	L	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2777	40	PA	112	26	19	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100047	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2778	40	PA	112	26	20	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100047	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2779	40	PA	112	26	21	G6C	-	-	PLANT AIR	L	*	CONTROL BOX	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2780	40	PA	112	26	22	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100050	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2781	40	PA	112	26	23	G6C	-	-	PLANT AIR	L	CONTROL BOX	ST-100050	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2782	150	BO	112	27	01	A1A	-	-	BASE OIL	L	ST-100037	P-100037	0.6	1.2	60	100	*	*	900	11.25	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2783	100	BO	112	27	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100037	0.6	1.2	60	100	*	*	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2784	80	BO	112	27	03	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2785	100	BO	112	27	04	A1A	-	-	BASE OIL	L	P-100037	ST-100037	6	8	60	100	*	50	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2786	100	BO	112	27	05	A1A	-	-	BASE OIL	L	P-100037	INLET LINE OF	6	8	60	100	*	50	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2787	80	BO	112	27	06	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100033	ATM	FULL OF LIQUID	60	80	*	*	900	11.25	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2788	100	BO	112	27	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100037	6	8	60	100	*	50	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2789	100	BO	112	27	08	A1A	-	-	BASE OIL	L	P-100037	TRANSFORME	6	8	60	100	*	50	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2790	100	BO	112	27	09	A1A	-	-	BASE OIL	L	P-100037	WHITE OIL	6	8	60	100	*	50	900	11.25	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2791	150	BO	112	28	01	A1A	-	-	BASE OIL	L	ST-100033	P-100033	0.6	1.2	60	100	*	*	900	20.2	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2792	100	BO	112	28	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100033	0.6	1.2	60	100	*	*	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2793	100	BO	112	28	03	A1A	-	-	BASE OIL	L	P-100033	ST-100033	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2794	100	BO	112	28	04	A1A	-	-	BASE OIL	L	P-100033	INLET LINE OF	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2795	80	BO	112	28	05	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2796	80	BO	112	28	06	P2A	-	-	BASE OIL	L	TANK TRUCK UNLO	ST-100035	ATM	FULL OF LIQUID	60	80	*	*	900	20.2	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2797	100	BO	112	28	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100033	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2798	100	BO	112	28	08	A1A	-	-	BASE OIL	L	P-100033	SMB-302001	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2799	100	BO	112	28	09	A1A	-	-	BASE OIL	L	P-100033	V-302111	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2800	100	BO	112	28	10	A1A	-	-	BASE OIL	L	P-100033	V-302101	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2801	100	BO	112	28	11	A1A	-	-	BASE OIL	L	P-100033	V-302105	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2802	100	BO	112	28	12	A1A	-	-	BASE OIL	L	P-100033	DDS-302301	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2803	100	BO	112	28	13	A1A	-	-	BASE OIL	L	P-100033	SMB-302002	6	8	60	100	*	50	900	20.2	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2804	150	BO	112	29	01	A1A	-	-	BASE OIL	L	ST-100035	P-100035	0.6	1.2	60	100	*	*	900	32	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2805	100	BO	112	29	02	A1A	-	-	BASE OIL	L	BOTTOM LINE OF S	P-100035	0.6	1.2	60	100	*	*	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2806	100	BO	112	29	03	A1A	-	-	BASE OIL	L	P-100035	ST-100035	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2807	100	BO	112	29	04	A1A	-	-	BASE OIL	L	P-100035	INLET LINE OF	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2808	80	BO	112	29	05	A1A	-	-	BASE OIL	L	TANK TRUCK UNLO	INLET LINE OF	ATM	FULL OF LIQUID	60	100	*	*	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2809	100	BO	112	29	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100035	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2810	100	BO	112	29	08	A1A	-	-	BASE OIL	L	P-100035	SMB-302001	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2811	100	BO	112	29	09	A1A	-	-	BASE OIL	L	P-100035	V-302105	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2812	100	BO	112	29	10	A1A	-	-	BASE OIL	L	P-100035	V-302101	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2813	100	BO	112	29	11	A1A	-	-	BASE OIL	L	P-100035	V-302111	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2814	100	BO	112	29	12	A1A	-	-	BASE OIL	L	P-100035	SMB-302002	6	8	60	100	*	50	900	32	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2815	25	V	112	29	13	G6C	-	-	VAPOR	V	TANK TRUCK UNLO	DV-100050	ATM	FULL OF LIQUID	40	60	*	*	900	32	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2816	25	CA	112	29	14	G6C	-	-	OMPRESSED AI	V	*	TANK TRUCK U	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2817	25	V	112	30	01	G6C	-	-	VAPOR	V	HLDV-100054 TYPE	DV-100050	ATM	FULL OF LIQUID	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2818	25	CA	112	30	02	G6C	-	-	OMPRESSED AI	V	*	HLDV-100054	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2819	80	BO	112	30	03	A1A	-	-	BASE OIL	L	P-100054	HLDV-100054	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2820	80	BO	112	30	04	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100054	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2821	100	BO	112	30	05	A1A	-	-	BASE OIL	L	TANK TRUCK	P-100054	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2822	80	BO	112	30	16	A1A	-	-	BASE OIL	L	P-100057	PIG TEE TYPE	6	8	40	60	*	*	1.25	0.00001	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2823	25	PA	112	30	07	G6C	-	-	PLANT AIR	V	*	P-100054	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2824	25	V	112	30	08	G6C	-	-	VAPOR	V	HLDV-100057 TYPE	DV-100050	ATM	FULL OF LIQUID	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2825	25	CA	112	30	09	G6C	-	-	OMPRESSED AI	V	*	HLDV-100057	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2826	80	BO	112	30	10	A1A	-	-	BASE OIL	L	HLDV-100057 TYPE	P-100057	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2827	80	BO	112	30	11	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100057	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2828	100	BO	112	30	12	A1A	-	-	BASE OIL	L	TANK TRUCK	P-100057	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2829	80	BO	112	30	15	A1A	-	-	BASE OIL	L	HLDV-100054 TYPE	HLDV-100057	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2830	25	PA	112	30	14	G6C	-	-	PLANT AIR	V	*	P-100057	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2831	80	BO	112	31	01	A1A	-	-	BASE OIL	L	P-100055	TANK TRUCK	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2832	80	BO	112	31	02	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100055	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2833	100	BO	112	31	03	A1A	-	-	BASE OIL	L	TANK TRUCK	P-100055	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2834	80	BO	112	31	11	A1A	-	-	BASE OIL	L	P-100055	PIG TEE TYPE	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2835	25	PA	112	31	05	G6C	-	-	PLANT AIR	V	*	P-100055	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2836	80	BO	112	31	06	A1A	-	-	BASE OIL	L	P-100058	TANK TRUCK	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2837	80	BO	112	31	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100058	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2838	100	BO	112	31	08	A1A	-	-	BASE OIL	L	TANK TRUCK	P-100058	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2839	80	BO	112	31	12	A1A	-	-	BASE OIL	L	P-100058	PIG TEE TYPE	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2840	25	PA	112	31	10	G6C	-	-	PLANT AIR	V	*	P-100058	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2841	80	BO	112	32	01	P2A	-	-	BASE OIL	L	P-100056	TANK TRUCK	6	8	60	100	*	40	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2842	80	BO	112	32	02	P2A	-	-	BASE OIL	L	P-100056	TANK TRUCK	6	8	60	100	*	40	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2843	80	BO	112	32	03	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100056	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2844	100	BO	112	32	04	A1A	-	-	BASE OIL	L	TANK TRUCK	P-100056	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2845	80	BO	112	32	13	A1A	-	-	BASE OIL	L	P-100056	PIG TEE TYPE	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2846	25	PA	112	32	06	G6C	-	-	PLANT AIR	V	*	P-100056	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2847	80	BO	112	32	07	A1A	-	-	BASE OIL	L	BYPASS LINE OF P-	P-100059	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2848	100	BO	112	32	08	A1A	-	-	BASE OIL	L	TANK TRUCK	P-100059	0.6	1.2	60	100	*	*	900	50	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2849	80	BO	112	32	14	A1A	-	-	BASE OIL	L	100059	PIG TEE TYPE	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2850	25	PA	112	32	10	G6C	-	-	PLANT AIR	V	*	P-100059	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2851	80	BO	112	32	11	A1A	-	-	BASE OIL	L	P-100056	UNLOADING	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2852	80	BO	112	32	12	A1A	-	-	BASE OIL	L	P-100059	UNLOADING	6	8	60	100	*	40	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2853	50	V	112	33	01	G6C	-	-	VAPOR	V	DV-100050	MANIFOLD	*	*	40	60	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2854	50	V	112	33	02	G6C	-	-	VAPOR	V	MANIFOLD	MANIFOLD	*	*	40	60	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2855	50	V	112	33	03	G6C	-	-	VAPOR	V	DV-100050	DRAIN	*	*	40	60	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2856	80	V	112	33	04	G6C	-	-	VAPOR	V	DV-100050	SAFE LOCATIO	*	*	40	60	*	*	*	*	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-112	H1	P&ID FOR BASEOIL TANKFARM		
2857	100	BO	001	01	01	A1A	-	-	BASE OIL	L	SMB-302001 BASE	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2858	100	BO	001	01	02	A1A	-	-	BASE OIL	L	SMB-302001 BASE	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2859	100	BO	001	01	03	A1A	-	-	BASE OIL	L	SMB-302001 BASE	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2860	100	BO	001	01	04	A1A	-	-	BASE OIL	L	SMB-302001 BASE	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2861	100	LO	001	01	05	P2A	-	-	LUBE OIL	L	SMB-302001 HLDV	SMB-302001	6	8	60	80	*	70	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2862	25	CA	001	01	06	G6C	-	-	OMPRESSED AI	V	*	SMB-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2863	25	V	001	01	07	G6C	-	-	VAPOR	V	SMB-302001 HLDV	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2864	25	IA	001	01	08	G6C	-	-	NSTRUMENT AI	V	*	SMB-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2865	25	PA	001	01	09	G6C	-	-	PLANT AIR	V	*	SMB-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2866	100	BO	001	01	10	A1A	-	-	BASE OIL	L	SMB-302001 BASE	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2867	100	BO	001	01	11	A1A	-	-	BASE OIL	L	SMB-302001 BASE	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2868	100	AD	001	02	01	A1A	-	-	BASE OIL	L	SMB-302001 ADDI	SMB-302001	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2869	100	AD	001	02	02	A1A	-	-	BASE OIL	L	SMB-302001 ADDI	SMB-302001 A	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2870	100	LO	001	03	01	P2A	-	-	LUBE OIL	L	SMB-302001 TYPE	FT-500013/15	6	8	60	80	*	70	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2871	100	AD	001	03	02	A1A	-	-	ADDITIVE	L	SMB-302001 ADDI	SMB-302001 A	6	8	60	100	*	70	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2872	80	LO	001	03	03	P2A	-	-	LUBE OIL	L	ABB MANIFOLD	SMB-302001 I	6	8	60	80	*	70	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2873	150	LO	001	03	04	A1A	-	-	LUBE OIL	L	SMB-302001 TYPE	P-302001A	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2874	25	CA	001	03	10	G6C	-	-	LUBE OIL	L	*	P-302001A	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2875	100	LO	001	03	06	A1A	-	-	LUBE OIL	L	BYPASS LINE OF P-	P-302001A	4	6	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2876	150	LO	001	03	07	A1A	-	-	LUBE OIL	L	P-302001A	SM-302001	6	8	60	100	*	80	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2877	25	CA	001	03	08	G6C	-	-	OMPRESSED AI	V	*	SMB-302001 I	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2878	25	V	001	03	09	G6C	-	-	VAPOR	V	SMB-302001 H VA	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2879	100	BO	001	04	01	A1A	-	-	BASE OIL	L	SMB-302002 BASE	SMB-302002 I	6	8	60	100	*	60	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2880	100	AD	001	04	02	A1A	-	-	ADDITIVE	L	SMB-302002 ADDI	SMB-302002 A	6	8	60	100	*	60	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2881	100	LO	001	04	03	P2A	-	-	LUBE OIL	L	HLDV-302002 TYPE	SMB-302002	6	8	60	80	*	60	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2882	25	V	001	04	04	G6C	-	-	VAPOR	V	SMB-302002 HLDV	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2883	25	CA	001	04	05	G6C	-	-	OMPRESSED AI	V	*	SMB-302002 I	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2884	25	IA	001	04	06	G6C	-	-	NSTRUMENT AI	V	*	SMB-302002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2885	25	PA	001	04	07	G6C	-	-	PLANT AIR	V	*	SMB-302002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2886	100	BO	001	04	08	A1A			BASE OIL	L	BASE OIL LEG -1	ADDITIVE LEG	6	8	60	100	*	60	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2887	80	LO	001	05	01	P2A	-	-	LUBE OIL	L	ABB MANIFOLD	SMB-302002 I	6	8	60	80	*	60	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2888	100	LO	001	05	02	P2A	-	-	LUBE OIL	L	SMB-302002 TYPE-	FT-500021/23	6	8	60	80	*	60	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2889	25	CA	001	05	03	G6C	-	-	OMPRESSED AI	V	*	SMB-302002 I	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2890	25	V	001	05	04	G6C	-	-	VAPOR	V	SMB-302002 H VA	DEMISTER	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2891	150	LO	001	05	05	A1A	-	-	LUBE OIL	L	SMB-302002 TYPE-	P-302002A	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2892	100	LO	001	05	06	P2A	-	-	LUBE OIL	L	BASE OIL LEG -1	SMB-302002 T	6	8	60	100	*	80	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2893	100	LO	001	05	07	A1A	-	-	LUBE OIL	L	BYPASS LINE OF P-	P-302002A	4	6	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2894	150	LO	001	05	08	A1A	-	-	LUBE OIL	L	P-302002A	SM-302002	6	8	60	100	*	80	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2895	25	CA	001	05	09	G6C	-	-	VAPOR	V	*	SMB-302002 T	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2896	100	BO	001	06	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	90	170	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2897	100	AD	001	06	02	A1A	-	-	ADDITIVE	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	90	170	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2898	25	CA	001	06	03	D5N	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	10Kg/cm2PR.RATIN	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2899	80	AD	001	06	04	P2A	-	-	ADDITIVE	L	PIGGABLE MANIFO	V-302101	6	8	60	100	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2900	25	CA	001	06	05	G6C	-	-	OMPRESSED AI	V	*	V-302101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2901	80	AD	001	06	06	P2A	-	-	ADDITIVE	L	COMMON DDU HE	V-302101	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2902	25	CA	001	06	07	G6C	-	-	OMPRESSED AI	V	*	V-302101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2903	80	AD	001	06	08	P2A	-	-	ADDITIVE	L	DEDICATED DDU H	V-302101	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2904	25	CA	001	06	09	G6C	-	-	OMPRESSED AI	V	*	V-302101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2905	25	CA	001	06	10	G6C	-	-	OMPRESSED AI	V	*	V-302101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2906	50	BO	001	06	11	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	V-302101	ATM	FULL OF LIQUID	90	170	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2907	80	HOR-M	001	06	12	A1A	IH	-	HOT OIL	L	V-302101	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2908	80	HOR-M	001	06	13	A1A	IH	-	HOT OIL	L	V-302101	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2909	80	HOR-M	001	06	14	A1A	IH	-	HOT OIL	L	V-302101	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2910	80	HOS-M	001	06	15	A1A	IH	-	HOT OIL	L	HEADER	V-302101	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2911	80	HOS-M	001	06	16	A1A	IH	-	HOT OIL	L	HEADER	V-302101	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2912	80	HOS-M	001	06	17	A1A	IH	-	HOT OIL	L	HEADER	V-302101	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2913	25	PA	001	06	18	G6C	-	-	PLANT AIR	V	*	OUTGOING O	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2914	25	IA	001	06	20	G6C	-	-	NSTRUMENT AI	V	*	OUTGOING O	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2915	150	LO	001	06	21	A1A	IH	-	LUBE OIL	L	V-302101	P-302101	0.6	1.2	90	170	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2916	100	LO	001	06	22	A1A	IH	-	LUBE OIL	L	BYPASS LINE OF P-	P-302101	6	8	90	170	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2917	100	LO	001	06	23	A1A	IH	-	LUBE OIL	L	P-302101	HLDV-313101	6	8	90	170	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2918	80	LO	001	06	24	A1A	IH	-	LUBE OIL	L	P-302101	V-302101	6	8	90	170	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2919	80	LO	001	06	25	P2A	-	-	LUBE OIL	L	HLDV-313101A TY	FT-500032/34	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2920	25	CA	001	06	26	G6C	-	-	OMPRESSED AI	V	*	HLDV-313101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2921	25	V	001	06	27	G6C	-	-	VAPOR	V	HLDV-313101A TY	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2922	80	LO	001	06	28	A1A	IH	-	LUBE OIL	L	P-302101	HLDV-313101	6	8	90	170	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2923	80	LO	001	06	29	P2A	-	-	LUBE OIL	L	HLDV-313101B TY	ABB MANIFOL	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2924	25	CA	001	06	30	G6C	-	-	OMPRESSED AI	V	*	HLDV-313101	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2925	25	V	001	06	31	G6C	-	-	VAPOR	V	HLDV-313101B TY	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2926	80	LO	001	06	32	A1A	-	-	LUBE OIL	L	P-302101	MANUAL DRU	6	8	90	170	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2927	80	LO	001	06	33	A1A	-	-	LUBE OIL	L	P-302101	MANUAL DRU	6	8	90	170	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2928	80	BO	001	07	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2929	100	AD	001	07	02	A1A	-	-	ADDITIVE	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2930	100	MB	001	07	03	A1A	-	-	ADDITIVE	L	PIGGABLE MANIFO	V-302111	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2931	25	CA	001	07	04	G6C	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2932	50	HOR-M	001	07	05	A1A	IH	-	HOT OIL	L	V-302111	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2933	100	LO	001	07	06	A1A	IH	-	LUBE OIL	L	V-302111	V-302101	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2934	50	HOS-M	001	07	07	A1A	IH	-	HOT OIL	L	*	V-302111	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2935	25	PA	001	07	08	G6C	-	-	PLANT AIR	V	*	V-302111	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2936	25	IA	001	07	09	G6C	-	-	NSTRUMENT AI	V	*	V-302111	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2937	50	BO	001	07	10	A1A	-	-	BASE OIL	L	*	V-302111	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2938	100	BO	001	08	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2939	100	AD	001	08	02	A1A	-	-	ADDITIVE	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2940	25	CA	001	08	03	G6C	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2941	80	AD	001	08	04	P2A	-	-	ADDITIVE	L	PIGGABLE MANIFO	TYPE-5 MANIF	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2942	80	AD	001	08	05	P2A	-	-	ADDITIVE	L	TYPE-5 MANIFOLD	V-302101	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2943	80	AD	001	08	06	P2A	-	-	ADDITIVE	L	TYPE-8 COMMON	V-302102	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2944	50	BO	001	08	07	A1A	-	-	BASE OIL	L	*	V-302102	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2945	80	AD	001	08	08	P2A	-	-	ADDITIVE	L	TYPE-8 COMMON	V-302102	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2946	80	AD	001	08	09	P2A	-	-	ADDITIVE	L	DEDICATED DDU H	V-302102	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2947	25	CA	001	08	10	G6C	-	-	OMPRESSED AI	V	*	PIG RECEIVER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2948	80	HOR-M	001	08	11	A1A	IH	-	HOT OIL	L	V-302102	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2949	80	HOR-M	001	08	12	A1A	IH	-	HOT OIL	L	V-302102	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2950	80	HOR-M	001	08	13	A1A	IH	-	HOT OIL	L	V-302102	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2951	80	HOS-M	001	08	14	A1A	IH	-	HOT OIL	L	*	V-302102	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2952	80	HOS-M	001	08	15	A1A	IH	-	HOT OIL	L	*	V-302102	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2953	80	HOS-M	001	08	16	A1A	IH	-	HOT OIL	L	*	V-302102	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2954	150	LO	001	08	17	A1A	IH	-	LUBE OIL	L	V-302102	P-302102	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2955	100	LO	001	08	18	A1A	IH	-	LUBE OIL	L	BYPASS LINE OF P-	P-302102	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2956	80	LO	001	08	19	A1A	IH	-	LUBE OIL	L	P-302102	HLDV-302102	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2957	80	LO	001	08	20	P2A	-	-	LUBE OIL	L	HLDV-302102 TYPE	ABB MANIFOL	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2958	100	LO	001	08	27	A1A	-	-	LUBE OIL	L	TYPE-5 MANIFOLD	V-302102	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2959	25	PA	001	08	22	G6C	-	-	PLANT AIR	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2960	80	LO	001	08	21	A1A	IH	-	LUBE OIL	L	P-302102	V-302102	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2961	25	IA	001	08	24	G6C	-	-	NSTRUMENT AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2962	25	CA	001	08	25	G6C	-	-	OMPRESSED AI	V	*	HLDV-302102	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2963	25	V	001	08	26	G6C	-	-	VAPOR	V	HLDV-302102 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2964	80	LO	001	08	23	A1A	-	-	LUBE OIL	L	P-302102	MANUAL DRU	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2965	100	LO	001	08	28	A1A	IH	-	LUBE OIL	L	V-302112	V-302102	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2966	25	CA	001	08	29	G6C	-	-	OMPRESSED AI	V	*	V-302102	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2967	80	LO	001	08	30	A1A	-	-	LUBE OIL	L	P-302102	MANUAL DRU	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2968	100	BO	001	09	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2969	100	AD	001	09	02	A1A	-	-	ADDITIVE	L	SPARE PIGGABLE M	SPARE PIGGAI	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2970	25	CA	001	09	03	G6C	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2971	100	MB	001	09	04	A1A	-	-	ADDITIVE	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2972	50	BO	001	09	05	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	V-302112	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2973	50	HOR-M	001	09	07	A1A	IH	-	HOT OIL	L	V-302112	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2974	50	HOS-M	001	09	08	A1A	IH	-	HOT OIL	L	*	V-302112	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2975	25	PA	001	09	09	G6C	-	-	PLANT AIR	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2976	25	IA	001	09	10	G6C	-	-	NSTRUMENT AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2977	100	BO	001	10	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2978	100	AD	001	10	02	A1A	-	-	ADDITIVE	L	SPARE PIGGABLE M	SPARE PIGGAI	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2979	25	CA	001	10	03	G6C	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2980	80	AD	001	10	04	P2A	-	-	ADDITIVE	L	DEDICATED DDU H	V-302103	ATM	FULL OF LIQUID	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2981	25	CA	001	10	05	G6C	-	-	OMPRESSED AI	V	*	V-302103	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2982	100	MB	001	10	06	A1A	-	-	ADDITIVE	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2983	80	HOR-M	001	10	07	A1A	IH	-	HOT OIL	L	V-302103	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2984	80	HOR-M	001	10	08	A1A	IH	-	HOT OIL	L	V-302103	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2985	80	HOR-M	001	10	09	A1A	IH	-	HOT OIL	L	V-302103	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2986	80	HOS-M	001	10	10	A1A	IH	-	HOT OIL	L	*	V-302103	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2987	80	HOS-M	001	10	11	A1A	IH	-	HOT OIL	L	*	V-302103	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2988	80	HOS-M	001	10	12	A1A	IH	-	HOT OIL	L	*	V-302103	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
2989	80	LO	001	10	13	P2A	-	-	LUBE OIL	L	F-302103 TYPE-5	FT-500037	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2990	80	LO	001	10	14	A1A	IH	-	LUBE OIL	L	BYPASS P-302103	P-302103	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2991	80	LO	001	10	15	A1A	IH	-	LUBE OIL	L	P-302103	V-302103	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2992	80	LO	001	10	16	A1A	IH	-	LUBE OIL	L	P-302103	HLDV-302103	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2993	80	LO	001	10	17	P2A	-	-	LUBE OIL	L	HLDV-302103 TYPE	HLDV-302103	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2994	100	LO	001	10	18	A1A	IH	-	LUBE OIL	L	V-302103	P-302103	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2995	80	LO	001	10	19	A1A	-	-	LUBE OIL	L	HLDV-302103 TYPE	DRUM FILLING	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2996	25	CA	001	10	20	G6C	-	-	OMPRESSED AI	V	*	HLDV-302103	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2997	25	V	001	10	21	G6C	-	-	VAPOR	V	HLDV-302103 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2998	25	PA	001	10	22	G6C	-	-	PLANT AIR	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
2999	25	CA	001	10	25	G6C	-	-	OMPRESSED AI	V	*	V-302103	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3000	25	IA	001	10	24	G6C	-	-	NSTRUMENT AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3001	80	LO	001	10	30	A1A	IH	-	LUBE OIL	L	P-302103	MANUAL DRU	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3002	50	BO	001	10	26	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	V-302103	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3003	80	BO	001	10	27	A1A	-	-	LUBE OIL	L	V-302103 TYPE 8	V-302103	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3004	80	LO	001	10	31	A1A	-	-	LUBE OIL	L	P-302103	MANUAL DRU	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3005	100	BO	001	11	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3006	25	CA	001	11	02	G6C	-	-	OMPRESSED AI	V	*	V-302113	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3007	50	LO	001	11	03	A1A	IH	-	LUBE OIL	L	P-302113	V-302113	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3008	50	BO	001	11	04	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	V-302113	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3009	200	PA	001	11	05	G6C	-	-	POWDER	S	SL-302113	SC-304101	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3010	100	LO	001	11	06	A1A	-	-	LUBE OIL	L	S-302113	V-302113	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3011	80	HOR-M	001	11	07	A1A	IH	-	HOT OIL	L	*	V-302113	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3012	80	HOS-M	001	11	08	A1A	IH	-	HOT OIL	L	V-302113	*	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3013	50	LO	001	11	09	A1A	IH	-	LUBE OIL	L	V-302113	P-302113	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3014	25	PA	001	11	10	G6C	-	-	PLANT AIR	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3015	25	IA	001	11	13	G6C	-	-	NSTRUMENT AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3016	50	LO	001	11	14	A1A	-	-	LUBE OIL	L	BYPASS LINE OF P-	P-302113	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3017	50	LO	001	11	15	A1A	IH	-	LUBE OIL	L	F-302113	V-302103	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3018	150	LO	001	11	16	A1A	-	-	LUBE OIL	L	SC-302113	SAFE LOCATIO	*	*	60	100	*	*	900	500	15	200	Water	*	2.5	6.35 MM	*	YES	90-0RG2N-PID-002	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3019	80	HOR-M	001	11	17	A1A	IH	-	HOT OIL	L	*	V-302113	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3020	80	HOS-M	001	11	18	A1A	IH	-	HOT OIL	L	V-302113	*	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3021	25	CA	001	11	19	G6C	-	-	OMPRESSED AI	V	*	V-302113	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3022	100	BO	001	12	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3023	25	CA	001	12	02	G6C	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3024	80	AD	001	12	03	A1A	-	-	ADDITIVE	L	COMMON DDU HE	V-302104	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3025	50	BO	001	12	04	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	V-302104	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3026	100	AD	001	12	05	A1A	-	-	ADDITIVE	L	SPARE PIGGABLE M	V-302104	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3027	100	LO	001	12	06	A1A	-	-	LUBE OIL	L	V-302104	P-302104	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3028	25	CA	001	12	07	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3029	80	LO	001	12	08	A1A	-	-	LUBE OIL	L	BYPASS LINE OF P-	P-302104	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3030	80	LO	001	12	09	A1A	-	-	LUBE OIL	L	P-302104	V-302104	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3031	80	LO	001	12	10	A1A	-	-	LUBE OIL	L	F-302104	HLDV-313103	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3032	80	LO	001	12	11	P2A	-	-	LUBE OIL	L	HLDV-313103 TYPE	DRUM FILLIN	6	8	60	80	*	50	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3033	25	CA	001	12	12	G6C	-	-	OMPRESSED AI	V	*	HLDV-313103	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3034	25	V	001	12	13	G6C	-	-	VAPOR	V	HLDV-313103 TYPE	DEMISTER	6	8	40	60	*	50	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3035	80	LO	001	12	15	A1A	-	-	LUBE OIL	L	P-302104	MANUAL DRU	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3036	25	CA	001	12	14	G6C	-	-	OMPRESSED AI	V	*	DRUM FILLIN	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3037	25	CA	001	12	17	G6C	-	-	OMPRESSED AI	V	*	V-302104	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3038	80	LO	001	12	16	A1A	-	-	LUBE OIL	L	P-302104	MANUAL DRU	6	8	60	100	*	50	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3039	100	BO	001	13	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3040	25	CA	001	13	02	G6C	-	-	OMPRESSED AI	V	*	PIGGABLE MA	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3041	25	CA	001	13	03	G6C	-	-	OMPRESSED AI	V	*	V-302105	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3042	100	MB	001	13	04	P2A	-	-	ADDITIVE	L	SPARE PIGGABLE M	SPARE PIGGAI	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3043	80	AD	001	13	05	A1A	-	-	ADDITIVE	L	COMMON DDU HE	V-302105	ATM	FULL OF LIQUID	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3044	50	HOR-M	001	13	07	A1A	IH	-	HOT OIL	L	V-302105	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3045	50	HOR-M	001	13	08	A1A	IH	-	HOT OIL	L	V-302105	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3046	80	LO	001	13	09	A1A	IH	-	LUBE OIL	L	P-302105	V-302105	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3047	50	HOS-M	001	13	10	A1A	IH	-	HOT OIL	L	*	V-302105	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3048	50	HOS-M	001	13	11	A1A	IH	-	HOT OIL	L	*	V-302105	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3049	80	LO	001	13	12	A1A	IH	-	LUBE OIL	L	V-302105	P-302105	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3050	50	LO	001	13	13	A1A	IH	-	BASE OIL	L	P-302105	OUTGOING LI	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3051	100	BO	001	13	14	A1A	IH	-	BASE OIL	L	BYPASS LINE OF P-	P-302105	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3052	50	LO	001	13	15	A1A	IH	-	LUBE OIL	L	P-302105	DFL-402003	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3053	25	CA	001	13	16	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3054	25	CA	001	13	18	G6C	-	-	OMPRESSED AI	V	*	OUTGOING LI	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3055	50	BO	001	13	06	A1A	-	-	BASE OIL	L	MANIFOLD	V-302105	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3056	80	LO	001	14	01	P2A	-	-	LUBE OIL	L	PM-302401	SMB-1 HEADE	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3057	100	LO	001	14	02	P2A	-	-	LUBE OIL	L	PM-302401	ABB BLEND HI	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3058	80	LO	001	14	03	P2A	-	-	LUBE OIL	L	PM-302401	SMB-2 HEADE	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3059	100	LO	001	14	04	P2A	-	-	LUBE OIL	L	PM-302401	ABB BLEND HI	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3060	80	LO	001	14	05	P2A	-	-	LUBE OIL	L	PM-302401	FILLING PIGG/	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3061	25	CA	001	14	06	G6C	-	-	OMPRESSED AI	V	*	PM-302401	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3062	25	V	001	14	08	G6C	-	-	VAPOR	V	PM-302401	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3063	80	HOR-M	001	15	01	A1A	IH	-	HOT OIL	L	DV-302201	*	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3064	80	HOS-M	001	15	02	A1A	IH	-	HOT OIL	L	*	DV-302201	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3065	25	PA	001	15	03	G6C	-	-	PLANT AIR	V	*	DV-302201	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3066	25	IA	001	15	04	G6C	-	-	NSTRUMENT AI	V	*	DV-302201	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3067	80	HOR-M	001	16	01	A1A	IH	-	HOT OIL	L	DV-302202	*	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3068	80	HOS-M	001	16	02	A1A	IH	-	HOT OIL	L	*	DV-302202	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3069	25	PA	001	16	03	G6C	-	-	PLANT AIR	V	*	DV-302202	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3070	25	IA	001	16	04	G6C	-	-	NSTRUMENT AI	V	*	DV-302202	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3071	100	BO	001	17	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3072	80	AD	001	17	02	A1A	-	-	ADDITIVE	L	HLDV-302001 TYPE	HLDV-302001	6	8	60	100	*	15	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3073	25	V	001	17	03	G6C	-	-	VAPOR	V	HLDV-302001 TYPE	DEMISTER	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3074	25	CA	001	17	04	G6C	-	-	OMPRESSED AI	V	*	HLDV-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3075	80	AD	001	17	05	A1A	-	-	ADDITIVE	L	P-302301	HLDV-302001	6	8	60	100	*	1	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3076	80	AD	001	17	06	A1A	-	-	ADDITIVE	L	DDS-302301	P-302301	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3077	80	AD	001	17	07	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302301	6	8	60	100	*	1	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3078	80	AD	001	17	08	A1A	-	-	ADDITIVE	L	DDS-302301	DDS-302301	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3079	80	AD	001	17	09	P2A	-	-	ADDITIVE	L	HLDV-302001 TYPE	V-302101	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3080	80	HOR-M	001	17	10	A1A	IH	-	HOT OIL	L	*	DDS-302301	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3081	80	HOS-M	001	17	11	A1A	IH	-	HOT OIL	L	DDS-302301	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3082	25	PA	001	17	12	G6C	-	-	PLANT AIR	V	*	SYSTEM	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3083	25	IA	001	17	13	G6C	-	-	NSTRUMENT AI	V	*	SYSTEM	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3084	25	V	001	17	14	G6C	-	-	VAPOR	V	HLDV-302001 TYPE	DEMISTER	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3085	25	CA	001	17	16	G6C	-	-	OMPRESSED AI	V	*	HLDV-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3086	80	V	001	17	15	G6C	-	-	VAPOR	V	DDS-302301	VENT	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3087	100	BO	001	18	01	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	PIGGABLE MA	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3088	80	AD	001	18	02	A1A	-	-	ADDITIVE	L	COMON ABB HEAD	HLDV-302001	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3089	25	V	001	18	03	G6C	-	-	VAPOR	V	HLDV-302001 TYPE	DEMISTER	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3090	25	CA	001	18	04	G6C	-	-	OMPRESSED AI	V	*	HLDV-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3091	80	AD	001	18	05	A1A	-	-	ADDITIVE	L	P-302302	HLDV-302001	6	8	60	100	*	1	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3092	80	AD	001	18	06	A1A	-	-	ADDITIVE	L	DDS-302302	P-302302	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3093	80	AD	001	18	07	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302302	6	8	60	100	*	1	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3094	80	AD	001	18	09	A1A	-	-	ADDITIVE	L	DDS-302302	DDS-302302	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3095	80	HOR-M	001	18	10	A1A	IH	-	HOT OIL	L	DDS-302302	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3096	80	HOS-M	001	18	11	A1A	IH	-	HOT OIL	L	*	DDS-302302	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3097	25	PA	001	18	12	G6C	-	-	PLANT AIR	V	*	SYSTEM	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3098	25	IA	001	18	13	G6C	-	-	NSTRUMENT AI	V	*	SYSTEM	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3099	80	AD	001	18	14	P2A	-	-	ADDITIVE	L	HLDV-302001 TYPE	V-302102	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3100	80	V	001	18	15	G6C	-	-	VAPOR	V	DDS-302302	VENT	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3101	80	AD	001	19	01	P2A	-	-	ADDITIVE	L	COMMON DDU HE	COMMON AB	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3102	80	AD	001	19	02	P2A	-	-	ADDITIVE	L	HLDV-302001 TYPE	V-302103	6	8	60	80	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3103	80	AD	001	19	03	P2A	-	-	ADDITIVE	L	COMMON DDU HE	HLDV-302001	6	8	60	100	*	*	900	500	12	100	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3104	25	CA	001	19	04	G6C	-	-	OMPRESSED AI	V	*	HLDV-302001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3105	25	V	001	19	05	G6C	-	-	VAPOR	V	HLDV-302001 TYPE	DEMISTER	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3106	100	BO	001	19	06	A1A	-	-	BASE OIL	L	PIGGABLE MANIFO	DDS-302303	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3107	80	AD	001	19	07	A1A	-	-	ADDITIVE	L	P-302303	DDS-302303	6	8	60	100	*	1	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3108	80	AD	001	19	08	A1A	-	-	ADDITIVE	L	DDS-302303	P-302303	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		

S.NO		LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
	PIPE SIZE (DN)	FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3109	80	AD	001	19	09	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-302303	6	8	60	100	*	1	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3110	80	AD	001	19	10	A1A	-	-	ADDITIVE	L	DDS-302303	DDS-302303	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3111	80	HOR-M	001	19	11	A1A	IH	-	HOT OIL	L	DDS-302303	RETURN	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3112	80	HOS-M	001	19	12	A1A	IH	-	HOT OIL	L	*	DDS-302303	6	8	150	180	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3113	25	PA	001	19	13	G6C	-	-	PLANT AIR	V	*	SYSTEM	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3114	25	IA	001	19	14	G6C	-	-	NSTRUMENT AI	V	*	SYSTEM	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3115	80	V	001	19	15	G6C	-	-	VAPOR	V	DDS-302303	VENT	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3116	80	AD	001	19	16	A1A	-	-	ADDITIVE	L	DDS-302303	WEIGHING SC	6	8	60	100	*	*	900	500	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3117	50	V	001	20	01	G6C	-	-	VAPOR	V	DV-302501	SAFE LOCATIC	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3118	50	V	001	20	02	G6C	-	-	VAPOR	V	DV-302501	DRAIN	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3119	50	V	001	20	03	G6C	-	-	VAPOR	V	MANIFOLD	DV-302501	6	8	40	60	*	*	900	500	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-001	H1	P&ID FOR LUBEOIL BLENDING PLANT		
3120	80	BO	008	01	01	A1A	-	-	BASE OIL	L	ST-100009 SECONI	BASE OIL HEA	6	8	60	120	*	50	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3121	80	BO	008	01	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	BASE OIL HEA	6	8	60	120	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3122	80	BO	008	02	01	A1A	-	-	BASE OIL	L	ST-100018 SECONI	BASE OIL HEA	6	8	60	120	*	30	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3123	80	BO	008	02	02	A1A	-	-	BASE OIL	L	ST-100008 SECONI	BASE OIL HEA	6	8	60	120	*	50	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3124	80	BO	008	03	01	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	60	120	*	50	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3125	40	BO	008	04	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3126	80	BO	008	04	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3127	80	BO	008	04	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3128	80	BO	008	04	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3129	40	BO	008	04	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3130	80	BO	008	04	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3131	80	BO	008	04	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3132	80	AD	008	04	08	A1A	-	-	ADDITIVE	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3133	80	BO	008	04	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3134	25	CA	008	04	10	G6C	-	-	PLANT AIR	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3135	100	BO	008	04	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3136	50	BO	008	04	12	A1A	IH	-	BASE OIL	L	RINSING	GC-301101	10	14	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3137	80	BO	008	04	13	A1A	-	-	BASE OIL	L	MANIFOLD	GC-301101	10	14	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3138	100	V	008	04	14	A1A	IH	-	VAPOR	V	GC-301101	PSV	10	14	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3139	100	V	008	04	15	A1A	IH	-	VAPOR	V	PSV	SR-301801	10	14	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3140	80	V	008	04	16	A1A	IH	-	VAPOR	V	GC-301101	SR-301801	10	14	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3141	80	V	008	04	17	A1A	IH	-	VAPOR	V	GC-301101	SECONDARY L	10	14	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3142	80	HOR(H)	008	04	18	A1B	IH	-	HOT OIL	L	GC-301101	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3143	80	COR	008	04	19	A1A	IH	-	COLD OIL	L	GC-301101	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3144	80	COR	008	04	20	A1A	IH	-	COLD OIL	L	GC-301101	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3145	80	HOR(H)	008	04	21	A1B	IH	-	HOT OIL	L	GC-301101	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3146	80	COS	008	04	22	A1A	IH	-	COLD OIL	L	*	GC-301101	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3147	80	HOS(H)	008	04	23	A1B	IH	-	HOT OIL	L	*	GC-301101	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3148	80	COS	008	04	24	A1A	IH	-	COLD OIL	L	*	GC-301101	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3149	80	HOS(H)	008	04	25	A1B	IH	-	HOT OIL	L	*	GC-301101	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3150	100	GE	008	04	26	A1A	IH	IE	GREASE	L	GC-301101	P-301101	0.6	1.2	120	220	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3151	80	GE	008	04	27	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301101	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3152	100	GE	008	04	28	A1A	IH	-	GREASE	L	P-301101	GC-301101	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3153	100	GE	008	04	29	A1A	IH	-	GREASE	L	P-301101	3 WAY VALVE	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3154	100	GE	008	04	30	A1A	IH	-	GREASE	L	P-301101	FV-301201	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3155	100	GE	008	04	31	A1A	IH	-	GREASE	L	P-301101	FV-301202	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3156	100	GE	008	04	32	A1A	IH	-	GREASE	L	P-301101	FV-301203	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3157	100	GE	008	04	33	A1A	IH	-	GREASE	L	P-301101	FV-301204	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3158	100	GE	008	04	34	A1A	IH	-	GREASE	L	P-301101	FV-301205	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3159	100	GE	008	04	35	A1A	IH	-	GREASE	L	P-301101	FV-301206	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3160	100	GE	008	04	36	A1A	IH	-	GREASE	L	P-301101	PK-301705	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3161	100	GE	008	04	37	A1A	IH	IE	GREASE	L	P-301101	BITUMEN PLA	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3162	50	TW	008	04	38	G6C	-	-	WATER	L	*	GC-301101	6	8	120	150	*	*	1000	12000	10	200	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3163	40	BO	008	05	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3164	80	BO	008	05	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3165	80	BO	008	05	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3166	80	BO	008	05	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3167	40	BO	008	05	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3168	80	BO	008	05	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3169	80	BO	008	05	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3170	80	AD	008	05	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3171	80	BO	008	05	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3172	25	CA	008	05	10	G6C	-	-	PLANT AIR	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3173	100	BO	008	05	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3174	50	BO	008	05	12	A1A	IH	-	BASE OIL	L	RINSING	GC-301102	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3175	80	BO	008	05	13	A1A	-	-	BASE OIL	L	MANIFOLD	GC-301102	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3176	100	V	008	05	14	A1A	IH	-	VAPOR	V	GC-301102	PSV	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3177	100	V	008	05	15	A1A	IH	-	VAPOR	V	PSV	SR-301801	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3178	80	V	008	05	16	A1A	IH	-	VAPOR	V	GC-301102	SR-301801	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3179	80	V	008	05	17	A1A	IH	-	VAPOR	V	GC-301102	SECONDARY L	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3180	80	HOR(H)	008	05	18	A1B	IH	-	HOT OIL	L	GC-301102	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3181	80	COR	008	05	19	A1A	IH	-	COLD OIL	L	GC-301102	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3182	80	COR	008	05	20	A1A	IH	-	COLD OIL	L	GC-301102	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3183	80	HOR(H)	008	05	21	A1B	IH	-	HOT OIL	L	GC-301102	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3184	80	COS	008	05	22	A1A	IH	-	COLD OIL	L	*	GC-301102	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3185	80	HOS(H)	008	05	23	A1B	IH	-	HOT OIL	L	*	GC-301102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3186	80	COS	008	05	24	A1A	IH	-	COLD OIL	L	*	GC-301102	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3187	80	HOS(H)	008	05	25	A1B	IH	-	HOT OIL	L	*	GC-301102	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3188	100	GE	008	05	26	A1A	IH	IE	GREASE	L	GC-301102	P-301102	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3189	80	GE	008	05	27	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301102	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3190	100	GE	008	05	28	A1A	IH	-	GREASE	L	P-301102	GC-301102	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3191	100	GE	008	05	29	A1A	IH	-	GREASE	L	P-301102	MANIFOLD	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3192	100	GE	008	05	30	A1A	IH	-	GREASE	L	P-301102	FV-301201	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3193	100	GE	008	05	31	A1A	IH	-	GREASE	L	P-301102	FV-301202	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3194	100	GE	008	05	32	A1A	IH	-	GREASE	L	P-301102	FV-301203	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3195	100	GE	008	05	33	A1A	IH	-	GREASE	L	P-301102	FV-301204	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3196	100	GE	008	05	34	A1A	IH	-	GREASE	L	P-301102	FV-301205	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3197	100	GE	008	05	35	A1A	IH	-	GREASE	L	P-301102	FV-301206	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3198	100	GE	008	05	36	A1A	IH	-	GREASE	L	P-301102	PK-301705	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3199	100	GE	008	05	37	A1A	IH	IE	GREASE	L	P-301102	BITUMEN PLA	6	8	120	150	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3200	50	TW	008	05	38	G6C	-	-	WATER	L	*	GC-301102	6	8	120	150	*	*	1000	12000	10	200	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3201	40	BO	008	06	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3202	80	BO	008	06	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3203	80	BO	008	06	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3204	80	BO	008	06	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3205	40	BO	008	06	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3206	80	BO	008	06	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3207	80	BO	008	06	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3208	80	AD	008	06	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3209	80	BO	008	06	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3210	25	CA	008	06	10	G6C	-	-	PLANT AIR	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	120	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3211	100	BO	008	06	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3212	50	BO	008	06	12	A1A	IH	-	BASE OIL	L	RINSING	FV-301201	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3213	100	V	008	06	13	A1A	IH	-	VAPOR	V	FV-301201	SCRUBBER	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3214	80	VC	008	06	14	A1A	IH	-	VACUUM	V	FV-301201	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3215	100	GE	008	06	15	A1A	IH	-	GREASE	L	P-301201	FV-301201	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3216	50	AD	008	06	16	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	FV-301201	VACUUM	VACUUM	120	150	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3217	100	GE	008	06	17	A1A	IH	IE	GREASE	L	P-301201	FV-301201	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3218	80	COR	008	06	18	A1A	IH	-	COLD OIL	L	FV-301201	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3219	80	HOR(H)	008	06	19	A1B	IH	-	HOT OIL	L	FV-301201	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3220	80	HOR(H)	008	06	20	A1B	IH	-	HOT OIL	L	FV-301201	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3221	25	CA	008	06	21	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3222	80	COS	008	06	22	A1A	IH	-	COLD OIL	L	*	FV-301201	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3223	80	HOS(H)	008	06	23	A1B	IH	-	HOT OIL	L	*	FV-301201	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3224	150	GE	008	06	24	A1A	IH	IE	GREASE	L	FV-301201	P-301201	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3225	80	GE	008	06	25	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301201	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3226	100	GE	008	06	26	A1A	IH	-	GREASE	L	P-301201	3 WAY VALVE	80	150	120	200	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3227	100	GE	008	06	27	A1A	IH	-	GREASE	L	P-301201	HOM-301301	80	150	120	200	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3228	100	GE	008	06	28	A1A	IH	-	GREASE	L	HOM-301301	3 WAY VALVE	80	150	120	200	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3229	100	GE	008	06	29	A1A	IH	-	GREASE	L	P-301201	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3230	100	GE	008	06	30	A1A	IH	-	GREASE	L	P-301201	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3231	100	GE	008	06	31	A1A	IH	-	GREASE	L	P-301201	HT-301501	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3232	100	GE	008	06	32	A1A	IH	-	GREASE	L	P-301201	HT-301502	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3233	100	GE	008	06	33	A1A	IH	-	GREASE	L	P-301201	HT-301503	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3234	100	GE	008	06	34	A1A	IH	-	GREASE	L	P-301201	HT-301504	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3235	100	GE	008	06	35	A1A	IH	-	GREASE	L	P-301201	HT-301505	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3236	100	GE	008	06	36	A1A	IH	IE	GREASE	L	P-301201	BITUMEN PLA	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3237	40	AD	008	06	37	A1A	IH	-	ADDITIVE	L	MANIFOLD	FV-301201	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3238	80	BO	008	06	38	A1A	-	-	BASE OIL	L	BASE OIL HEADER	FV-301201	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3239	40	BO	008	07	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3240	80	BO	008	07	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3241	80	BO	008	07	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3242	80	BO	008	07	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3243	40	BO	008	07	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3244	80	BO	008	07	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3245	80	BO	008	07	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3246	80	AD	008	07	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3247	80	BO	008	07	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3248	25	CA	008	07	10	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	900	0.00001	10	120	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3249	100	BO	008	07	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3250	50	BO	008	07	12	A1A	IH	-	BASE OIL	L	RINSING	FV-301202	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3251	80	VC	008	07	13	A1A	IH	-	VACUUM	V	FV-301202	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3252	100	GE	008	07	14	A1A	IH	-	GREASE	L	P-301202	FV-301202	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3253	50	AD	008	07	15	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	FV-301202	VACUUM	VACUUM	120	150	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3254	100	GE	008	07	16	A1A	IH	-	GREASE	L	P-301202	FV-301202	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3255	80	COR	008	07	17	A1A	IH	-	COLD OIL	L	FV-301202	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3256	80	HOR(H)	008	07	18	A1B	IH	-	HOT OIL	L	FV-301202	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3257	80	HOR(H)	008	07	19	A1B	IH	-	HOT OIL	L	FV-301202	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3258	25	CA	008	07	20	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3259	80	COS	008	07	21	A1A	IH	-	COLD OIL	L	*	FV-301202	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3260	80	HOS(H)	008	07	22	A1B	IH	-	HOT OIL	L	*	FV-301202	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3261	150	GE	008	07	23	A1A	IH	IE	GREASE	L	FV-301202	P-301202	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3262	80	GE	008	07	24	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301202	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3263	100	GE	008	07	25	A1A	IH	-	GREASE	L	P-301202	3 WAY VALVE	6	8	120	200	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3264	100	GE	008	07	26	A1A	IH	-	GREASE	L	P-301202	HOM-301302	6	8	120	200	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3265	100	GE	008	07	27	A1A	IH	-	GREASE	L	HOM-301302	3 WAY VALVE	80	150	120	200	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3266	100	GE	008	07	28	A1A	IH	-	GREASE	L	P-301202	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3267	100	GE	008	07	29	A1A	IH	-	GREASE	L	P-301202	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3268	100	GE	008	07	30	A1A	IH	-	GREASE	L	P-301202	HT-301501	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3269	100	GE	008	07	31	A1A	IH	-	GREASE	L	P-301202	HT-301502	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3270	100	GE	008	07	32	A1A	IH	-	GREASE	L	P-301202	HT-301503	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3271	100	GE	008	07	33	A1A	IH	-	GREASE	L	P-301202	HT-301504	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3272	100	GE	008	07	34	A1A	IH	-	GREASE	L	P-301202	HT-301505	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3273	100	GE	008	07	35	A1A	IH	IE	GREASE	L	P-301202	BITUMEN PLA	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3274	40	AD	008	07	36	A1A	IH	-	ADDITIVE	L	MANIFOLD	FV-301202	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1			
3275	100	BO	008	07	37	A1A	IH	-	BASE OIL	L	SCRUBBER	FV-301202	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1			
3276	80	BO	008	07	38	A1A	-	-	BASE OIL	L	BASE OIL HEADER	FV-301202	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1			
3277	40	BO	008	08	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3278	80	BO	008	08	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3279	80	BO	008	08	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3280	80	BO	008	08	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3281	40	BO	008	08	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3282	80	BO	008	08	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3283	80	BO	008	08	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3284	80	AD	008	08	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3285	80	BO	008	08	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3286	25	CA	008	08	10	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	900	0.00001	10	120	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3287	100	BO	008	08	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3288	50	BO	008	08	12	A1A	IH	-	BASE OIL	L	RINSING	FV-301203	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3289	80	VC	008	08	13	A1A	IH	-	VACUUM	V	FV-301203	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3290	100	GE	008	08	14	A1A	IH	-	GREASE	L	P-301203	FV-301203	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3291	50	AD	008	08	15	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	FV-301203	VACUUM	VACUUM	120	150	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3292	100	GE	008	08	16	A1A	IH	-	GREASE	L	P-301203	FV-301203	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3293	80	COR	008	08	17	A1A	IH	-	COLD OIL	L	FV-301203	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3294	80	HOR(H)	008	08	18	A1B	IH	-	HOT OIL	L	FV-301203	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3295	80	HOR(H)	008	08	19	A1B	IH	-	HOT OIL	L	FV-301203	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3296	25	CA	008	08	20	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3297	80	COS	008	08	21	A1A	IH	-	COLD OIL	L	*	FV-301203	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3298	80	HOS(H)	008	08	22	A1B	IH	-	HOT OIL	L	*	FV-301203	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3299	150	GE	008	08	23	A1A	IH	IE	GREASE	L	FV-301203	P-301203	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3300	80	GE	008	08	24	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301203	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3301	100	GE	008	08	25	A1A	IH	-	GREASE	L	P-301203	3 WAY VALVE	6	8	120	200	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3302	100	GE	008	08	26	A1A	IH	-	GREASE	L	P-301203	HOM-301303	6	8	120	200	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3303	100	GE	008	08	27	A1A	IH	-	GREASE	L	HOM-301303	3 WAY VALVE	80	150	120	200	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3304	100	GE	008	08	28	A1A	IH	-	GREASE	L	P-301203	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3305	100	GE	008	08	29	A1A	IH	-	GREASE	L	P-301203	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3306	100	GE	008	08	30	A1A	IH	-	GREASE	L	P-301203	HT-301501	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3307	100	GE	008	08	31	A1A	IH	-	GREASE	L	P-301203	HT-301502	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3308	100	GE	008	08	32	A1A	IH	-	GREASE	L	P-301203	HT-301503	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3309	100	GE	008	08	33	A1A	IH	-	GREASE	L	P-301203	HT-301504	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3310	100	GE	008	08	34	A1A	IH	-	GREASE	L	P-301203	HT-301505	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3311	100	GE	008	08	35	A1A	IH	IE	GREASE	L	P-301203	BITUMEN PLA	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3312	40	AD	008	08	36	A1A	IH	-	ADDITIVE	L	MANIFOLD	FV-301203	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3313	100	BO	008	08	37	A1A	IH	-	BASE OIL	L	SCRUBBER	FV-301203	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3314	80	BO	008	08	38	A1A	-	-	BASE OIL	L	BASE OIL HEADER	FV-301203	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3315	40	BO	008	09	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3316	80	BO	008	09	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3317	80	BO	008	09	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3318	80	BO	008	09	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3319	40	BO	008	09	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3320	80	BO	008	09	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3321	80	BO	008	09	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3322	80	AD	008	09	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3323	80	BO	008	09	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3324	25	CA	008	09	10	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	900	0.00001	10	120	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3325	100	BO	008	09	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3326	50	BO	008	09	12	A1A	IH	-	BASE OIL	L	RINSING	FV-301204	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3327	80	VC	008	09	13	A1A	IH	-	VACUUM	V	FV-301204	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3328	100	GE	008	09	14	A1A	IH	-	GREASE	L	P-301204	FV-301204	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3329	50	AD	008	09	15	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	FV-301204	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3330	100	GE	008	09	16	A1A	IH	-	GREASE	L	P-301204	FV-301204	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3331	80	COR	008	09	17	A1A	IH	-	COLD OIL	L	FV-301204	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3332	80	HOR(H)	008	09	18	A1B	IH	-	HOT OIL	L	FV-301204	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3333	80	HOR(H)	008	09	19	A1B	IH	-	HOT OIL	L	FV-301204	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3334	25	CA	008	09	20	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3335	80	COS	008	09	21	A1A	IH	-	COLD OIL	L	*	FV-301204	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3336	80	HOS(H)	008	09	22	A1B	IH	-	HOT OIL	L	*	FV-301204	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3337	150	GE	008	09	23	A1A	IH	IE	GREASE	L	FV-301204	P-301204	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3338	80	GE	008	09	24	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301204	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3339	100	GE	008	09	25	A1A	IH	-	GREASE	L	P-301204	3 WAY VALVE	6	8	120	200	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3340	100	GE	008	09	26	A1A	IH	-	GREASE	L	P-301204	HOM-301304	6	8	120	200	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3341	100	GE	008	09	27	A1A	IH	-	GREASE	L	HOM-301304	3 WAY VALVE	80	150	120	200	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3342	100	GE	008	09	28	A1A	IH	-	GREASE	L	P-301204	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3343	100	GE	008	09	29	A1A	IH	-	GREASE	L	P-301204	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3344	100	GE	008	09	30	A1A	IH	-	GREASE	L	P-301204	HT-301501	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3345	100	GE	008	09	31	A1A	IH	-	GREASE	L	P-301204	HT-301502	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3346	100	GE	008	09	32	A1A	IH	-	GREASE	L	P-301204	HT-301503	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3347	100	GE	008	09	33	A1A	IH	-	GREASE	L	P-301204	HT-301504	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3348	100	GE	008	09	34	A1A	IH	-	GREASE	L	P-301204	HT-301505	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3349	100	GE	008	09	35	A1A	IH	IE	GREASE	L	P-301204	BITUMEN PLA	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3350	40	AD	008	09	36	A1A	IH	-	ADDITIVE	L	MANIFOLD	FV-301204	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3351	100	BO	008	09	37	A1A	IH	-	BASE OIL	L	SCRUBBER	FV-301204	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3352	80	BO	008	09	38	A1A	-	-	BASE OIL	L	BASE OIL HEADER	FV-301204	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3353	40	BO	008	10	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3354	80	BO	008	10	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3355	80	BO	008	10	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3356	80	BO	008	10	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3357	40	BO	008	10	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3358	80	BO	008	10	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3359	80	BO	008	10	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3360	80	AD	008	10	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3361	80	BO	008	10	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3362	25	CA	008	10	10	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	900	0.00001	10	120	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3363	100	BO	008	10	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3364	50	BO	008	10	12	A1A	IH	-	BASE OIL	L	RINSING	FV-301205	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3365	80	VC	008	10	13	A1A	IH	-	VACUUM	V	FV-301205	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3366	100	GE	008	10	14	A1A	IH	-	GREASE	L	P-301205	FV-301205	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3367	50	AD	008	10	15	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	FV-301205	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3368	100	GE	008	10	16	A1A	IH	-	GREASE	L	P-301205	FV-301205	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3369	80	COR	008	10	17	A1A	IH	-	COLD OIL	L	FV-301205	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3370	80	HOR(H)	008	10	18	A1B	IH	-	HOT OIL	L	FV-301205	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3371	80	HOR(H)	008	10	19	A1B	IH	-	HOT OIL	L	FV-301205	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3372	25	CA	008	10	20	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3373	80	COS	008	10	21	A1A	IH	-	COLD OIL	L	*	FV-301205	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3374	80	HOS(H)	008	10	22	A1B	IH	-	HOT OIL	L	*	FV-301205	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3375	150	GE	008	10	23	A1A	IH	IE	GREASE	L	FV-301205	P-301205	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3376	80	GE	008	10	24	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301205	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3377	100	GE	008	10	25	A1A	IH	-	GREASE	L	P-301205	3 WAY VALVE	6	8	120	200	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3378	100	GE	008	10	26	A1A	IH	-	GREASE	L	P-301205	HOM-301305	6	8	120	200	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3379	100	GE	008	10	27	A1A	IH	-	GREASE	L	HOM-301305	3 WAY VALVE	80	150	120	200	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3380	100	GE	008	10	28	A1A	IH	-	GREASE	L	P-301205	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3381	100	GE	008	10	29	A1A	IH	-	GREASE	L	P-301205	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3382	100	GE	008	10	30	A1A	IH	-	GREASE	L	P-301205	HT-301501	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3383	100	GE	008	10	31	A1A	IH	-	GREASE	L	P-301205	HT-301502	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3384	100	GE	008	10	32	A1A	IH	-	GREASE	L	P-301205	HT-301503	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3385	100	GE	008	10	33	A1A	IH	-	GREASE	L	P-301205	HT-301504	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3386	100	GE	008	10	34	A1A	IH	-	GREASE	L	P-301205	HT-301505	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3387	100	GE	008	10	35	A1A	IH	IE	GREASE	L	P-301205	BITUMEN PLA	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3388	40	AD	008	10	36	A1A	-	-	ADDITIVE	L	MANIFOLD	FV-301205	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3389	100	BO	008	10	37	A1A	IH	-	BASE OIL	L	SCRUBBER	FV-302105	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3390	80	BO	008	10	38	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	FV-302105	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3391	40	BO	008	11	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3392	80	BO	008	11	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3393	80	BO	008	11	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3394	80	BO	008	11	04	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3395	40	BO	008	11	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3396	80	BO	008	11	06	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3397	80	BO	008	11	07	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3398	80	AD	008	11	08	A1A	-	-	ADDITIVE	L	HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3399	80	BO	008	11	09	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3400	25	CA	008	11	10	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	900	0.00001	10	120	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3401	100	BO	008	11	11	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3402	50	BO	008	11	12	A1A	IH	-	BASE OIL	L	RINSING	FV-301206	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3403	80	VC	008	11	13	A1A	IH	-	VACUUM	V	FV-301206	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3404	100	GE	008	11	14	A1A	IH	-	GREASE	L	P-301206	FV-301206	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3405	50	AD	008	11	15	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	FV-301206	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3406	100	GE	008	11	16	A1A	IH	-	GREASE	L	P-301206	FV-301206	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3407	80	COR	008	11	17	A1A	IH	-	COLD OIL	L	FV-301206	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3408	80	HOR(H)	008	11	18	A1B	IH	-	HOT OIL	L	FV-301206	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3409	80	HOR(H)	008	11	19	A1B	IH	-	HOT OIL	L	FV-301206	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3410	25	CA	008	11	20	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3411	80	COS	008	11	21	A1A	IH	-	COLD OIL	L	*	FV-301206	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3412	80	HOS(H)	008	11	22	A1B	IH	-	HOT OIL	L	*	FV-301206	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3413	150	GE	008	11	23	A1A	IH	IE	GREASE	L	FV-301206	P-301206	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3414	80	GE	008	11	24	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301206	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3415	100	GE	008	11	25	A1A	IH	-	GREASE	L	P-301206	3 WAY VALVE	6	8	120	200	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3416	100	GE	008	11	26	A1A	IH	-	GREASE	L	P-301206	HOM-301306	6	8	120	200	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3417	100	GE	008	11	27	A1A	IH	-	GREASE	L	HOM-301306	3 WAY VALVE	80	150	120	200	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3418	100	GE	008	11	28	A1A	IH	-	GREASE	L	P-301206	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3419	100	GE	008	11	29	A1A	IH	-	GREASE	L	P-301206	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3420	100	GE	008	11	30	A1A	IH	-	GREASE	L	P-301206	HT-301501	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3421	100	GE	008	11	31	A1A	IH	-	GREASE	L	P-301206	HT-301502	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3422	100	GE	008	11	32	A1A	IH	-	GREASE	L	P-301206	HT-301503	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3423	100	GE	008	11	33	A1A	IH	-	GREASE	L	P-301206	HT-301504	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3424	100	GE	008	11	34	A1A	IH	-	GREASE	L	P-301206	HT-301505	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3425	100	GE	008	11	35	A1A	IH	IE	GREASE	L	P-301206	BITUMEN PLA	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3426	40	AD	008	11	36	A1A	IH	-	ADDITIVE	L	MANIFOLD	FV-301206	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3427	100	BO	008	11	37	A1A	IH	-	BASE OIL	L	SCRUBBER	FV-301206	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3428	80	BO	008	11	38	A1A	-	-	BASE OIL	L	BASE OIL HEADER	FV-301206	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
													NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3429	50	BO	008	12	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	PK-301705	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3430	50	BO	008	12	02	A1A	IH	-	BASE OIL	L	RINSING	PK-301705	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3431	100	BO	008	12	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	PK-301705	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3432	100	BO	008	12	04	A1A	-	-	BASE OIL	L	FV-301206	PK-301705	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3433	25	CA	008	12	05	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3434	100	BO	008	12	22	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3435	100	GE	008	12	07	A1A	IH	IE	GREASE	L	HOM-301705 SECO	PK-301705	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3436	50	AD	008	12	08	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	PK-301705	VACUUM	VACUUM	120	150	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3437	80	COR	008	12	09	A1A	IH	-	COLD OIL	L	PK-301705	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3438	80	HOR(H)	008	12	10	A1B	IH	-	HOT OIL	L	PK-301705	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3439	80	COS	008	12	11	A1A	IH	-	COLD OIL	L	*	PK-301705	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3440	80	HOS(H)	008	12	12	A1B	IH	-	HOT OIL	L	*	PK-301705	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3441	150	GE	008	12	13	A1A	IH	IE	GREASE	L	PK-301705	P-301705	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3442	80	GE	008	12	14	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301705	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3443	100	GE	008	12	15	A1A	IH	-	GREASE	L	P-301705	DFL-401902	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3444	100	GE	008	12	16	A1A	IH	-	GREASE	L	P-301705	SR-301705	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3445	100	GE	008	12	17	A1A	IH	-	GREASE	L	SR-301705	OUTGOING LI	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3446	50	CHWR	008	12	18	A1A	IH	-	CHILLED WATEI	L	SR-301705	RETURN	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3447	50	CHWS	008	12	19	A1A	IH	-	CHILLED WATEI	L	*	SR-301705	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3448	100	GE	008	12	20	A1A	IH	-	GREASE	L	P-301705	PK-301705	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3449	100	GE	008	12	21	A1A	IH	-	GREASE	L	HOM-301705 PRIN	PK-301705	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3450	80	V	008	12	25	A1A	-	-	VACUUM	V	PK-301705	VACUUM	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1			
3451	50	AD	008	12	24	A1A	-	-	ADDITIVE	L	MANIFOLD	PK-301705	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3452	25	CA	008	12	23	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3453	80	BO	008	13	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3454	80	BO	008	13	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3455	80	BO	008	13	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3456	100	BO	008	13	04	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3457	25	CA	008	13	05	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3458	50	BO	008	13	06	A1A	IH	-	BASE OIL	L	RINSING	A-301701	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3459	100	BO	008	13	07	A1A	-	-	BASE OIL	L	MANIFOLD	A-301701	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3460	80	VC	008	13	08	A1A	-	-	VACUUM	V	A-301701	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3461	100	V	008	13	09	A6C	-	-	VAPOR	V	A-301701	PSV	6	8	60	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3462	100	V	008	13	10	A6C	-	-	VAPOR	V	PSV	SR-301802	6	8	60	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3463	80	V	008	13	11	A6C	-	-	VAPOR	V	A-301701	SR-301802	6	8	60	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3464	80	V	008	13	12	A6C	-	-	VAPOR	V	A-301701	SECONDARY L	6	8	60	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3465	50	AD	008	13	13	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	A-301701	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	3	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3466	100	GE	008	13	14	A1A	IH	IE	GREASE	L	P-301701	A-301701	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3467	80	COR	008	13	15	A1A	IH	-	COLD OIL	L	A-301701	RETURN	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3468	80	HOR(H)	008	13	16	A1B	IH	-	HOT OIL	L	A-301701	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3469	80	COS	008	13	17	A1A	IH	-	COLD OIL	L	*	A-301701	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3470	80	HOS(H)	008	13	18	A1B	IH	-	HOT OIL	L	*	A-301701	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3471	150	GE	008	13	19	A1A	IH	IE	GREASE	L	A-301701	P-301701	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3472	80	GE	008	13	20	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301701	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3473	100	GE	008	13	21	A1A	IH	-	GREASE	L	P-301701	PV-301711	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3474	80	BO	008	14	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3475	80	BO	008	14	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3476	80	BO	008	14	03	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3477	50	BO	008	14	04	A1A	IH	-	BASE OIL	L	RINSING	PV-301711	ATM	FULL OF LIQUID	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3478	100	BO	008	14	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	PV-301711	ATM	FULL OF LIQUID	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3479	80	VC	008	14	06	A1A	-	-	BASE OIL	V	PV-301711	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3480	100	GE	008	14	07	A1A	IH	IE	VACUUM	L	P-301711	PV-301711	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3481	80	COR	008	14	08	A1A	IH	-	COLD OIL	L	PV-301711	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3482	80	HOR(H)	008	14	09	A1B	IH	-	HOT OIL	L	PV-301711	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3483	80	COS	008	14	10	A1A	IH	-	COLD OIL	L	*	PV-301711	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3484	80	HOS(H)	008	14	11	A1B	IH	-	HOT OIL	L	*	PV-301711	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3485	150	GE	008	14	12	A3C	IH	IE	GREASE	L	PV-301711	P-301711	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3486	80	GE	008	14	13	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301711	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3487	100	GE	008	14	14	A1A	IH	-	GREASE	L	P-301711	CM-301605	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3488	25	CA	008	14	16	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3489	80	V	008	14	15	A6C	-	-	VAPOR	V	PV-301711	SR-301802	0.6	1.2	80	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3490	100	BO	008	14	16	A1A			BASE OIL	L	MANIFOLD	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3491	80	BO	008	15	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3492	80	BO	008	15	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3493	100	BO	008	15	03	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3494	25	CA	008	15	04	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3495	80	VC	008	15	05	A1A	-	-	VACUUM	V	PK-301702	VACUUM	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3496	50	BO	008	15	06	A1A	IH	-	BASE OIL	L	RINSING	PK-301702	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3497	80	BO	008	15	07	A1A	-	-	BASE OIL	L	BASE OIL HEADER	PK-301702	VACUUM	VACUUM	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3498	100	V	008	15	08	A6C	-	-	VAPOR	V	PSV	SR-301803	VACUUM	VACUUM	80	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3499	100	V	008	15	09	A6C	-	-	VAPOR	V	PK-301702	PSV	VACUUM	VACUUM	80	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3500	80	V	008	15	10	A6C	-	-	VAPOR	V	PK-301702	SR-301803	VACUUM	VACUUM	80	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3501	80	V	008	15	11	A6C	-	-	VAPOR	V	PK-301702	SECONDARY L	VACUUM	VACUUM	80	120	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3502	50	AD	008	15	12	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	PK-301702	VACUUM	VACUUM	120	150	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3503	100	GE	008	15	13	A1A	IH	IE	GREASE	L	P-301702	PK-301702	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3504	80	COR	008	15	14	A1A	IH	-	COLD OIL	L	PK-301702	RETURN	VACUUM	VACUUM	10	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3505	80	HOR(H)	008	15	15	A1B	IH	-	HOT OIL	L	PK-301702	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3506	80	COS	008	15	16	A1A	IH	-	COLD OIL	L	*	PK-301702	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3507	80	HOS(H)	008	15	17	A1B	IH	-	HOT OIL	L	*	PK-301702	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3508	150	GE	008	15	18	A1A	IH	IE	GREASE	L	PK-301702	P-301702	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3509	80	GE	008	15	19	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301702	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3510	100	GE	008	15	20	A1A	IH	-	GREASE	L	P-301702	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3511	100	GE	008	15	21	A1A	IH	-	GREASE	L	P-301702	PK-301706	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3512	100	GE	008	15	22	A1A	IH	-	GREASE	L	P-301702	HOM-301702	6	8	120	150	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3513	100	GE	008	15	23	A1A	IH	-	GREASE	L	HOM-301702	OUTGOING LI	80	150	120	150	*	7.5	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3514	40	BO	008	16	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3515	50	BO	008	16	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3516	25	CA	008	16	03	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3517	50	BO	008	16	05	A1A	IH	-	BASE OIL	L	RINSING	A-301703	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3518	100	V	008	16	06	A1A	IH	-	VAPOR	V	A-301703	PSV	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3519	100	V	008	16	07	A1A	IH	-	VAPOR	V	PSV	SR-301805	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3520	80	V	008	16	08	A1A	IH	-	VAPOR	V	A-301703	SR-301805	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3521	80	V	008	16	09	A1A	IH	-	VAPOR	V	A-301703	SECONDARY L	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3522	50	AD	008	16	10	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	A-301703	6	8	120	150	*	3	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3523	50	PD	008	16	11	B4C	-	-	POWDER	L	MANUAL POT	A-301703	6	8	80	120	*	*	900	12000	15	150	Water	*	1.5	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3524	80	GE	008	16	12	B4C	IH	-	GREASE	L	P-301703	A-301703	6	8	80	120	*	*	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3525	80	COR	008	16	13	A1A	IH	-	COLD OIL	L	A-301703	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3526	80	HOR(H)	008	16	14	A1B	IH	-	HOT OIL	L	A-301703	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3527	80	COS	008	16	15	A1A	IH	-	COLD OIL	L	*	A-301703	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3528	80	HOS(H)	008	16	16	A1B	IH	-	HOT OIL	L	*	A-301703	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3529	80	GE	008	16	18	B4C	IH	IE	GREASE	L	A-301703	P-301703	0.6	1.2	80	120	*	*	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3530	80	GE	008	16	19	B4C	IH	-	GREASE	L	BYPASS LINE OF P-	P-301703	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3531	80	GE	008	16	20	B4C	IH	-	GREASE	L	P-301703	PV-301713	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3532	80	GE	008	16	21	B4C	IH	-	GREASE	L	P-301703	PV-301713	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3533	50	V	008	16	04	A1A	IH	-	VACUUM	V	A-301703	VACUUM	6	8	80	120	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3534	50	BO	008	16	17	A1A	IH	-	BASE OIL	L	BASE OIL HEADER	A-301703	6	8	80	120	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3535	40	BO	008	17	01	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3536	50	BO	008	17	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3537	25	CA	008	17	03	G6C	-	-	OMPRESSED AI	V	*	MANIFOLD	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3538	50	BO	008	17	04	A1A	IH	-	BASE OIL	L	RINSING	PV-301713	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3539	50	BO	008	17	05	A1A	-	-	BASE OIL	L	BASE OIL HEADER	PV-301713	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3540	80	BO	008	17	06	A1A	-	-	BASE OIL	L	MANIFOLD	MANIFOLD	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3541	100	V	008	17	07	A1A	IH	-	VAPOR	V	PV-301713	SR-301805	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3542	80	GE	008	17	08	B4C	IH	-	GREASE	L	P-301713	PV-301713	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3543	25	AD	008	17	09	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	PV-301713	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3544	50	COR	008	17	10	A1A	IH	-	COLD OIL	L	PV-301713	RETURN	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3545	50	HOR(H)	008	17	11	A1B	IH	-	HOT OIL	L	PV-301713	RETURN	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3546	50	COS	008	17	12	A1A	IH	-	COLD OIL	L	*	PV-301713	6	8	10	25	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3547	50	HOS(H)	008	17	13	A1B	IH	-	HOT OIL	L	*	PV-301713	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3548	80	GE	008	17	14	B4C	IH	IE	GREASE	L	PV-301713	P-301713	0.6	1.2	80	120	*	*	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3549	80	GE	008	17	15	B4C	IH	-	GREASE	L	BYPASS LINE OF P-	P-301713	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3550	80	GE	008	17	16	B4C	IH	-	GREASE	L	P-301713	PFL-401902	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3551	80	GE	008	17	17	B4C	IH	-	GREASE	L	P-301713	PV-301713	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3552	80	GE	008	17	18	B4C	IH	-	GREASE	L	P-301713	SECONDARY L	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3553	80	GE	008	17	19	B4C	IH	-	GREASE	L	P-301713	CM-301604	6	8	80	120	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3554	50	V	008	17	20	A1A	IH	-	VACUUM	V	PV-301713	VACUUM	6	8	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3555	50	BO	008	18	01	A1A	IH	-	BASE OIL	L	RINSING	PK-301706	ATM	FULL OF LIQUID	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3556	80	BO	008	18	02	A1A	-	-	BASE OIL	L	BASE OIL HEADER	PK-301706	ATM	FULL OF LIQUID	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3557	25	AD	008	18	03	A1A	-	-	ADDITIVE	L	DRUM ADDITIVE P	PK-301706	ATM	FULL OF LIQUID	120	150	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3558	100	GE	008	18	04	A1A	IH	IE	GREASE	L	P-301706	PK-301706	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3559	150	GE	008	18	05	A1A	IH	IE	GREASE	L	PK-301706	P-301706	0.6	1.2	120	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3560	80	GE	008	18	06	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301706	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3561	100	GE	008	18	07	A1A	IH	IE	GREASE	L	P-301706	PK-301706	6	8	120	150	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3562	100	GE	008	18	08	A1A	IH	-	GREASE	L	P-301706	HOM-301705	6	8	120	150	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3563	100	GE	008	18	09	A1A	IH	-	GREASE	L	HOM-301705	OUTGOING LI	80	150	120	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3564	100	GE	008	18	10	A1A	IH	-	GREASE	L	P-301706	PFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3565	100	GE	008	18	11	A1A	IH	-	GREASE	L	P-301706	SCF-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3566	100	GE	008	18	12	A1A	IH	-	GREASE	L	P-301706	DFL-401901	6	8	120	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3567	50	AD	008	19	01	A1A	-	-	ADDITIVE	L	MANUAL POT	HT-301501	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3568	50	BO	008	19	03	A1A	IH	-	BASE OIL	L	RINSING	HT-301501	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3569	100	GE	008	19	04	A1A	IH	IE	GREASE	L	P-301501	HT-301501	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3570	80	COR	008	19	05	A1A	IH	-	COLD OIL	L	HT-301501	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3571	80	HOR(H)	008	19	06	A1B	IH	-	HOT OIL	L	HT-301501	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3572	80	COS	008	19	07	A1A	IH	-	COLD OIL	L	*	HT-301501	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3573	80	HOS(H)	008	19	08	A1B	IH	-	HOT OIL	L	*	HT-301501	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3574	150	GE	008	19	09	A1A	IH	IE	GREASE	L	HT-301501	P-301501	0.6	1.2	60	100	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3575	80	GE	008	19	10	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301501	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3576	100	GE	008	19	11	A1A	IH	-	GREASE	L	P-301501	SCF-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3577	100	GE	008	19	12	A1A	IH	-	GREASE	L	P-301501	PFL-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3578	100	GE	008	19	13	A1A	IH	-	GREASE	L	P-301501	DFL-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3579	100	GE	008	19	14	A1A	IH	-	GREASE	L	HOM-301501	P-301501	80	150	120	150	*	*	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3580	100	GE	008	19	15	A1A	IH	-	GREASE	L	P-301501	HOM-301501	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3581	50	AD	008	20	01	A1A	-	-	ADDITIVE	L	MANUAL POT	HT-301502	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3582	50	BO	008	20	03	A1A	IH	-	BASE OIL	L	RINSING	HT-301502	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3583	100	GE	008	20	04	A1A	IH	IE	GREASE	L	P-301502	HT-301502	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3584	80	COR	008	20	05	A1A	IH	-	COLD OIL	L	HT-301502	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3585	80	HOR(H)	008	20	06	A1B	IH	-	HOT OIL	L	HT-301502	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3586	80	COS	008	20	07	A1A	IH	-	COLD OIL	L	*	HT-301502	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3587	80	HOS(H)	008	20	08	A1B	IH	-	HOT OIL	L	*	HT-301502	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3588	150	GE	008	20	09	A1A	IH	IE	GREASE	L	HT-301502	P-301502	0.6	1.2	60	100	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3589	80	GE	008	20	10	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301502	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3590	100	GE	008	20	11	A1A	IH	-	GREASE	L	P-301502	SCF-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3591	100	GE	008	20	12	A1A	IH	-	GREASE	L	P-301502	PFL-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3592	100	GE	008	20	13	A1A	IH	-	GREASE	L	P-301502	DFL-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3593	100	GE	008	20	14	A1A	IH	-	GREASE	L	HOM-301502	P-301502	80	150	120	150	*	*	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3594	100	GE	008	20	15	A1A	IH	-	GREASE	L	P-301502	HOM-301502	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3595	50	AD	008	21	01	A1A	-	-	ADDITIVE	L	MANUAL POT	HT-301503	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3596	50	BO	008	21	03	A1A	IH	-	BASE OIL	L	RINSING	HT-301503	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3597	100	GE	008	21	04	A1A	IH	IE	GREASE	L	P-301503	HT-301503	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3598	80	COR	008	21	05	A1A	IH	-	COLD OIL	L	HT-301503	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3599	80	HOR(H)	008	21	06	A1B	IH	-	HOT OIL	L	HT-301503	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3600	80	COS	008	21	07	A1A	IH	-	COLD OIL	L	*	HT-301503	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3601	80	HOS(H)	008	21	08	A1B	IH	-	HOT OIL	L	*	HT-301503	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3602	150	GE	008	21	09	A1A	IH	IE	GREASE	L	HT-301503	P-301503	0.6	1.2	60	100	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3603	80	GE	008	21	10	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301503	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3604	100	GE	008	21	11	A1A	IH	-	GREASE	L	P-301503	SCF-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3605	100	GE	008	21	12	A1A	IH	-	GREASE	L	P-301503	PFL-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3606	100	GE	008	21	13	A1A	IH	-	GREASE	L	P-301503	DFL-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3607	100	GE	008	21	14	A1A	IH	-	GREASE	L	HOM-301503	P-301503	80	150	120	150	*	*	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3608	100	GE	008	21	15	A1A	IH	-	GREASE	L	P-301503	HOM-301503	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3609	50	AD	008	22	01	A1A	-	-	ADDITIVE	L	MANUAL POT	HT-301504	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3610	80	VC	008	22	02	A1A	IH	-	VACUUM	V	HT-301504	VACUUM	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3611	50	BO	008	22	03	A1A	IH	-	BASE OIL	L	RINSING	HT-301504	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3612	100	GE	008	22	04	A1A	IH	IE	GREASE	L	P-301504	HT-301504	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3613	80	COR	008	22	05	A1A	IH	-	COLD OIL	L	HT-301504	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3614	80	HOR(H)	008	22	06	A1B	IH	-	HOT OIL	L	HT-301504	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3615	80	COS	008	22	07	A1A	IH	-	COLD OIL	L	*	HT-301504	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3616	80	HOS(H)	008	22	08	A1B	IH	-	HOT OIL	L	*	HT-301504	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3617	150	GE	008	22	09	A1A	IH	IE	GREASE	L	HT-301504	P-301504	0.6	1.2	60	100	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3618	80	GE	008	22	10	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301504	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3619	100	GE	008	22	11	A1A	IH	-	GREASE	L	P-301504	SCF-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3620	100	GE	008	22	12	A1A	IH	-	GREASE	L	P-301504	HOM-301504	80	150	60	100	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3621	100	GE	008	22	13	A1A	IH	-	GREASE	L	HOM-301504	OUTGOING LI	80	150	60	100	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3622	25	AD	008	22	14	A1A	-	-	ADDITIVE	L	COLOR ADDITIVE	HT-301504	6	8	120	150	*	*	*	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3623	100	GE	008	22	15	A1A	IH	-	GREASE	L	P-301504	HT-301504	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3624	100	GE	008	22	16	A1A	IH	-	GREASE	L	HT-301504	HT-300502	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3625	50	AD	008	23	01	A1A	-	-	ADDITIVE	L	MANUAL POT	HT-301505	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3626	50	BO	008	23	03	A1A	IH	-	BASE OIL	L	RINSING	HT-301505	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3627	100	GE	008	23	04	A1A	IH	IE	GREASE	L	P-301505	HT-301505	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3628	80	COR	008	23	05	A1A	IH	-	COLD OIL	L	HT-301505	RETURN	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3629	80	HOR(H)	008	23	06	A1B	IH	-	HOT OIL	L	HT-301505	RETURN	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3630	80	COS	008	23	07	A1A	IH	-	COLD OIL	L	*	HT-301505	VACUUM	VACUUM	10	25	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3631	80	HOS(H)	008	23	08	A1B	IH	-	HOT OIL	L	*	HT-301505	VACUUM	VACUUM	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3632	150	GE	008	23	09	A1A	IH	IE	GREASE	L	HT-301505	P-301505	0.6	1.2	60	150	*	*	900	12000	15	200	Water	*	1.5	6.35 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3633	80	GE	008	23	10	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301505	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3634	100	GE	008	23	11	A1A	IH	-	GREASE	L	P-301505	SCF-401901	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3635	100	GE	008	23	12	A1A	IH	-	GREASE	L	P-301505	HOM-301505	6	8	60	100	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3636	100	GE	008	23	13	A1A	IH	-	GREASE	L	HOM-301505	OUTGOING LI	6	8	60	100	*	7.5	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3637	100	GE	008	23	15	A1A	IH	-	GREASE	L	P-301505	SECONDARY L	80	150	120	150	*	*	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3638	25	AD	008	23	14	A1A	-	-	ADDITIVE	L	COLOR ADDITIVE	HT-301505	VACUUM	VACUUM	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 80	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3639	100	GE	008	23	16	A1A	IH	-	GREASE	L	HT-301502	HT-301505	6	8	60	100	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3640	100	GE	008	24	01	A1A	IH	-	GREASE	L	XIC VALVE	HOM-301501	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3641	100	GE	008	24	02	A1A	IH	-	GREASE	L	HOM-301501	MANIFOLD	80	150	60	150	*	*	*	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3642	100	GE	008	24	03	A1A	IH	-	GREASE	L	3 WAY VALVE	HT-301501	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3643	100	GE	008	24	04	A1A	IH	-	GREASE	L	3 WAY VALVE	HT-301501	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3644	100	GE	008	24	05	A1A	IH	-	GREASE	L	3 WAY VALVE	HT-301502	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3645	100	GE	008	24	06	A1A	IH	-	GREASE	L	3 WAY VALVE	HT-301502	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3646	100	GE	008	24	07	A1A	IH	-	GREASE	L	3 WAY VALVE	HT-301503	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3647	100	GE	008	24	08	A1A	IH	-	GREASE	L	3 WAY VALVE	HT-301503	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3648	25	CWR	008	25	01	A1A	IH	-	COLD WATER	L	CM-301605	RETURN	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3649	25	CWS	008	25	02	A1A	IH	-	COLD WATER	L	*	CM-301605	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3650	100	GE	008	25	03	A1A	IH	-	GREASE	L	CM-301605	P-301601	0.6	1.2	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3651	100	GE	008	25	04	A1A	IH	-	GREASE	L	P-301601	PV-301711	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3652	100	GE	008	25	05	A1A	IH	-	GREASE	L	P-301601	DFL-401902	6	8	60	150	*	6	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3653	25	CWR	008	25	06	A1A	IH	-	COLD WATER	L	CM-301604	RETURN	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3654	25	CWS	008	25	07	A1A	IH	-	COLD WATER	L	*	CM-301604	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3655	80	GE	008	25	08	B4C	IH	-	GREASE	L	P-301602	PV-301713	6	8	60	150	*	3	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3656	80	GE	008	25	09	B4C	IH	-	GREASE	L	CM-301604	P-301602	0.6	1.2	60	150	*	*	900	12000	15	150	Water	*	1.5	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3657	100	GE	008	26	01	A1A	IH	-	GREASE	L	RFID READER	DU-301903	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3658	100	GE	008	26	02	A1A	IH	-	GREASE	L	BYPASS LINE	DU-301903	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3659	100	GE	008	26	03	A1A	IH	-	GREASE	L	F-301903	DFL-401901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3660	80	GE	008	26	04	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	F-301903	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3661	100	GE	008	26	05	A1A	-	-	GREASE	L	DU-301903	V-301903	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3662	25	CWS	008	26	06	A1A	-	-	COLD WATER	L	*	V-301903	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3663	25	CWR	008	26	07	A1A	-	-	COLD WATER	L	V-301903	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3664	80	GE	008	26	08	A1A	-	-	GREASE	L	V-301903	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3665	80	GE	008	26	09	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3666	100	GE	008	26	11	A1A	IH	-	GREASE	L	DU-301903	P-301903	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3667	100	GE	008	27	01	A1A	IH	-	GREASE	L	RFID READER	F-301902	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3668	100	GE	008	27	02	A1A	IH	-	GREASE	L	BYPASS LINE	DU-301902	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3669	100	GE	008	27	03	A1A	IH	-	GREASE	L	F-301902	SCF-401901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3670	80	GE	008	27	04	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	F-301902	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3671	100	GE	008	27	06	A1A	-	-	GREASE	L	DU-301902	V-301902	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3672	25	CWS	008	27	07	A1A	-	-	COLD WATER	L	*	V-301902	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3673	25	CWR	008	27	08	A1A	-	-	COLD WATER	L	V-301902	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3674	80	GE	008	27	09	A1A	-	-	GREASE	L	V-301902	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3675	80	GE	008	27	10	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3676	100	GE	008	27	11	A1A	IH	-	GREASE	L	RFID READER	F-301908	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3677	100	GE	008	27	12	A1A	IH	-	GREASE	L	BYPASS LINE	DU-301908	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3678	100	GE	008	27	13	A1A	IH	-	GREASE	L	F-301908	DFL-401902	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3679	80	GE	008	27	14	A1A	-	-	GREASE	L	BYPASS LINE OF P-	F-301908	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3680	25	CWS	008	27	16	A1A	-	-	COLD WATER	L	*	V-301908	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3681	25	CWR	008	27	17	A1A	-	-	COLD WATER	L	V-301908	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3682	100	GE	008	27	18	A1A	-	-	GREASE	L	DU-301908	V-301908	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3683	80	GE	008	27	19	A1A	-	-	GREASE	L	V-301908	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3684	80	GE	008	27	20	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3685	100	GE	008	27	21	A1A	IH	-	GREASE	L	DU-301902	P-301902	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1			
3686	100	GE	008	27	22	A1A	IH	-	GREASE	L	DU-301908	P-301908	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1			
3687	100	GE	008	28	01	A1A	IH	-	GREASE	L	RFID READER	F-301901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3688	100	GE	008	28	02	A1A	IH	-	GREASE	L	BY PASSLINE	DU-301901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3689	100	GE	008	28	03	A1A	IH	-	GREASE	L	F-301901	PFL-401901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3690	80	GE	008	28	05	A1A	-	-	GREASE	L	BYPASS LINE OF P-	F-301901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3691	100	GE	008	28	06	A1A	-	-	GREASE	L	DU-301901	V-301901	VACUUM	VACUUM	80	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3692	25	CWS	008	28	07	A1A	-	-	COLD WATER	L	*	V-301901	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3693	25	CWR	008	28	08	A1A	-	-	COLD WATER	L	V-301901	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3694	80	GE	008	28	09	A1A	-	-	GREASE	L	V-301901	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3695	80	GE	008	28	10	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3696	100	GE	008	28	11	A1A	IH	-	GREASE	L	DU-301901	P-301901	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1			
3697	100	GE	008	29	01	A1A	IH	-	GREASE	L	HT-301204	F-301905	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3698	100	GE	008	29	03	A1A	IH	-	GREASE	L	BYPASS LINE	DU-301905	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3699	80	GE	008	29	04	A1A	-	-	GREASE	L	BYPASS LINE OF P-	F-301905	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3700	100	GE	008	29	05	A1A	IH	-	GREASE	L	F-301905	DFL-401903	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3701	100	GE	008	29	06	A1A	-	-	GREASE	L	DU-301905	V-301905	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3702	25	CWS	008	29	07	A1A	-	-	COLD WATER	L	*	V-301905	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3703	25	CWR	008	29	08	A1A	-	-	COLD WATER	L	V-301905	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3704	80	GE	008	29	09	A1A	-	-	GREASE	L	V-301905	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3705	80	GE	008	29	10	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3706	80	GE	008	29	11	B4C	IH	-	GREASE	L	PV-301713	F-301907	6	8	60	150	*	3	900	12000	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3707	100	GE	008	29	13	A1A	IH	-	GREASE	L	BYPASS LINE OF DU	F-301907	6	8	60	150	*	3	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3708	80	GE	008	29	14	B4C	IH	-	GREASE	L	BYPASS LINE OF P-	F-301907	6	8	60	150	*	3	900	12000	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3709	80	GE	008	29	15	B4C	IH	-	GREASE	L	F-301907	PFL-401902	6	8	60	150	*	3	900	12000	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3710	100	GE	008	29	16	A1A	-	-	GREASE	L	DU-301907	V-301907	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3711	25	CWS	008	29	17	A1A	-	-	COLD WATER	L	*	V-301907	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3712	25	CWR	008	29	18	A1A	-	-	COLD WATER	L	V-301907	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3713	80	GE	008	29	19	A1A	-	-	GREASE	L	V-301907	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3714	80	GE	008	29	20	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3715	100	GE	008	29	21	A1A	IH	-	GREASE	L	DU-301905	P-301905	6	8	60	150	*	9	900	12000	15	200	Water	*	0	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3716	100	GE	008	29	22	A1A	IH	-	GREASE	L	DU-301907	P-301907	6	8	60	150	*	3	900	12000	15	200	Water	*	0	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3717	100	GE	008	30	01	A1A	IH	-	GREASE	L	PK-301702	P-301904	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3718	100	GE	008	30	03	A1A	IH	-	GREASE	L	BYPASS LINE OF DU	F-301904	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3719	80	GE	008	30	04	A1A	-	-	GREASE	L	BYPASS LINE OF P-	DFL-401905	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3720	100	GE	008	30	05	A1A	IH	-	GREASE	L	P-301904	DFL-401905	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3721	100	GE	008	30	06	A1A	-	-	GREASE	L	DU-301904	V-301904	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3722	25	CWS	008	30	07	A1A	-	-	COLD WATER	L	*	V-301904	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3723	25	CWR	008	30	08	A1A	-	-	COLD WATER	L	V-301904	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3724	80	GE	008	30	09	A1A	-	-	GREASE	L	V-301904	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3725	80	GE	008	30	10	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3726	100	GE	008	30	11	A1A	IH	-	GREASE	L	HT-301505	F-301906	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3727	100	GE	008	30	12	A1A	IH	-	GREASE	L	BYPASS LINE OF DU	F-301906	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3728	100	GE	008	30	14	A1A	IH	-	GREASE	L	F-301906	DFL-401904	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3729	100	GE	008	30	15	A1A	-	-	GREASE	L	DU-301906	V-301906	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3730	25	CWS	008	30	16	A1A	-	-	COLD WATER	L	*	V-301906	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3731	25	CWR	008	30	17	A1A	-	-	COLD WATER	L	V-301906	*	6	8	25	40	*	*	1000	12000	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3732	80	GE	008	30	18	A1A	-	-	GREASE	L	V-301906	OIL SEAL VAC	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3733	80	GE	008	30	19	A1A	-	-	GREASE	L	OIL SEAL VACCUM	EXHAUST	VACUUM	VACUUM	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3734	80	GE	008	30	13	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	F-301906	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3735	100	GE	008	30	21	A1A	IH	-	GREASE	L	DU-301904	P-301904	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3736	100	GE	008	30	22	A1A	IH	-	GREASE	L	DU-301906	P-301906	6	8	60	150	*	9	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3737	80	BO	008	31	01	A1A	IH	-	BASE OIL	L	HE-301001	T-301001	6	8	60	150	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3738	80	HOR(H)	008	31	03	A1B	IH	-	HOT OIL	L	HE-301001	T-301001	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3739	80	HOS(H)	008	31	04	A1B	IH	-	HOT OIL	L	*	HE-301001	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	4.78 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3740	80	BO	008	31	05	A1A	IH	-	BASE OIL	L	P-301001	HE-301001	6	8	60	100	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3741	80	BO	008	31	06	A1A	IH	-	BASE OIL	L	T-301001	P-301001	0.6	1.2	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3742	80	BO	008	31	07	A1A	IH	-	BASE OIL	L	BYPASS LINE OF P-	P-301001	6	8	60	100	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3743	80	BO	008	31	08	A1A	IH	-	BASE OIL	L	P-301001	RINSING	6	8	60	100	*	12	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3744	80	BO	008	31	09	A1A	-	-	BASE OIL	L	ST-100028	T-301001	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3745	250	V	008	32	01	A3C	-	-	VAPOR	V	SR-301801	CS-301801	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 20	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3746	50	V	008	32	02	A6C	-	-	VAPOR	V	SR-301801	P-301801	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3747	25	V	008	32	03	A6C	-	-	VAPOR	V	P-301801	SR-301801	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3748	100	V	008	32	04	A6C	-	-	VAPOR	V	B-301801A	C-302501	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3749	250	V	008	32	05	A6C	-	-	VAPOR	V	SR-301802	CS-301802	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	6.3 MM	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3750	50	V	008	32	06	A6C	-	-	VAPOR	V	SR-301802	P-301802	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3751	25	V	008	32	07	A6C	-	-	VAPOR	V	P-301802	SR-301802	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3752	100	V	008	32	08	A6C	-	-	VAPOR	V	B-301802A	C-302502	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3753	100	V	008	32	09	A3C	-	-	VAPOR	V	MANIFOLD	CS-301801	6	8	60	100	*	*	900	12000	15	150	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3754	100	V	008	32	10	A3C	-	-	VAPOR	V	MANIFOLD	CS-301802	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3755	100	V	008	33	01	A3C	-	-	VAPOR	V	MANIFOLD	CS-301803	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3756	250	V	008	33	02	A3C	-	-	VAPOR	V	SR-301803	CS-301803	*	*	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 20	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3757	100	V	008	33	03	A6C	-	-	VAPOR	V	SR-301803	B-301803A	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3758	25	V	008	33	04	A6C	-	-	VAPOR	V	P-301803	SR-301803	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3759	50	V	008	33	05	A6C	-	-	VAPOR	V	SR-301803	P-301803	*	*	60	100	*	*	900	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3760	100	V	008	33	07	A3C	-	-	VAPOR	V	MANIFOLD	CS-301805	6	8	60	100	*	*	900	12000	15	200	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3761	250	V	008	33	08	A6C	-	-	VAPOR	V	CS-301805	SR-301805	*	*	60	100	*	*	*	12000	15	150	Water	*	1.5	6.3 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3762	100	V	008	33	09	A6C	-	-	VAPOR	V	SR-301805	B-301805A	*	*	60	100	*	*	*	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3763	25	V	008	33	10	A6C	-	-	VAPOR	V	P-301805	SR-301805	*	*	60	100	*	*	*	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3764	50	V	008	33	11	A6C	-	-	VAPOR	V	SR-301805	P-301805	*	*	60	100	*	*	*	12000	15	150	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3765	40	AD	008	34	01	A1A	IH	IE	ADDITIVE	L	P-301901	MB-301901	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3766	50	AD	008	34	02	A1A	IH	IE	ADDITIVE	L	MB-301901	P-301901	0.6	1.2	60	100	*	*	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3767	50	AD	008	34	03	A1A	IH	IE	ADDITIVE	L	P-301901	OUTGOING LI	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3768	50	AD	008	34	04	A1A	IH	IE	ADDITIVE	L	HOSE CONNECTIO	P-301901	0.6	1.2	60	100	*	*	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3769	25	AD	008	34	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE OF P-	P-301901	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3770	50	AD	008	34	06	A1A	IH	IE	ADDITIVE	L	P-301901	PK-301705	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3771	50	AD	008	34	07	A1A	IH	IE	ADDITIVE	L	P-301901	FV-301206	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3772	50	AD	008	34	08	A1A	IH	IE	ADDITIVE	L	P-301901	FV-301205	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3773	50	AD	008	34	09	A1A	IH	IE	ADDITIVE	L	P-301901	FV-301204	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3774	50	AD	008	34	10	A1A	IH	IE	ADDITIVE	L	P-301901	FV-301203	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3775	50	AD	008	34	11	A1A	IH	IE	ADDITIVE	L	P-301901	FV-301202	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3776	50	AD	008	34	12	A1A	IH	IE	ADDITIVE	L	P-301901	FV-301201	6	8	60	100	*	3	900	1176	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3777	50	HOS(H)	008	34	13	A1B	IH	-	HOT OIL	L	HEADER	MB-301901	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3778	50	HOR(H)	008	34	14	A1B	IH	-	HOT OIL	L	HEADER	MB-301901	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3779	50	HOS(H)	008	34	15	A1B	IH	-	HOT OIL	L	*	MB-301901	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3780	50	HOR(H)	008	34	16	A1B	IH	-	HOT OIL	L	MB-301910	*	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3781	100	V	008	34	17	A1A	-	-	VAPOR	V	MB-301901	SAFE LOCATIC	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3782	80	AD	008	35	01	A1A	IH	IE	ADDITIVE	L	P-301902	MB-301902	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3783	100	AD	008	35	02	A1A	IH	IE	ADDITIVE	L	MB-301902	P-301902	0.6	1.2	60	100	*	*	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3784	80	AD	008	35	03	A1A	IH	IE	ADDITIVE	L	P-301902	OUTGOING LI	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3785	50	AD	008	35	04	A1A	IH	IE	ADDITIVE	L	HOSE CONNECTIO	P-301902	0.6	1.2	60	100	*	*	900	85000	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3786	80	AD	008	35	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE OF P-	P-301902	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3787	80	AD	008	35	06	A1A	IH	IE	ADDITIVE	L	P-301902	PK-301705	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3788	80	AD	008	35	07	A1A	IH	IE	ADDITIVE	L	P-301902	FV-301206	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3789	80	AD	008	35	08	A1A	IH	IE	ADDITIVE	L	P-301902	FV-301205	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3790	80	AD	008	35	09	A1A	IH	IE	ADDITIVE	L	P-301902	FV-301204	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3791	80	AD	008	35	10	A1A	IH	IE	ADDITIVE	L	P-301902	FV-301203	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3792	80	AD	008	35	11	A1A	IH	IE	ADDITIVE	L	P-301902	FV-301202	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3793	80	AD	008	35	12	A1A	IH	IE	ADDITIVE	L	P-301902	FV-301201	6	8	60	100	*	3	900	85000	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3794	50	HOS(H)	008	35	13	A1B	IH	-	HOT OIL	L	HEADER	MB-301902	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3795	50	HOR(H)	008	35	14	A1B	IH	-	HOT OIL	L	HEADER	MB-301902	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3796	100	V	008	35	15	A1A	-	-	VAPOR	V	MB-301902	SAFE LOCATIC	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3797	40	AD	008	36	01	A1A	IH	IE	ADDITIVE	L	P-301903	MB-301903	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3798	50	AD	008	36	02	A1A	IH	IE	ADDITIVE	L	MB-301903	P-301903	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3799	40	AD	008	36	03	A1A	IH	IE	ADDITIVE	L	P-301903	OUTGOING LI	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3800	50	AD	008	36	04	A1A	IH	IE	ADDITIVE	L	HOSE CONNECTIO	P-301903	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3801	25	AD	008	36	05	A1A	IH	IE	ADDITIVE	L	BYPASS LINE OF P-	P-301903	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3802	40	AD	008	36	06	A1A	IH	IE	ADDITIVE	L	P-301903	PK-301705	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3803	40	AD	008	36	07	A1A	IH	IE	ADDITIVE	L	P-301903	FV-301206	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3804	40	AD	008	36	08	A1A	IH	IE	ADDITIVE	L	P-301903	FV-301205	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3805	40	AD	008	36	09	A1A	IH	IE	ADDITIVE	L	P-301903	FV-301204	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3806	40	AD	008	36	10	A1A	IH	IE	ADDITIVE	L	P-301903	FV-301203	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3807	40	AD	008	36	11	A1A	IH	IE	ADDITIVE	L	P-301903	FV-301202	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3808	40	AD	008	36	12	A1A	IH	IE	ADDITIVE	L	P-301903	FV-301201	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3809	100	V	008	36	13	A1A	-	-	VAPOR	V	MB-301903	SAFE LOCATIC	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3810	40	AD	008	37	01	A1A	IH	-	ADDITIVE	L	P-301904	MB-301904	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3811	50	AD	008	37	02	A1A	IH	IE	ADDITIVE	L	MB-301904	P-301904	0.6	1.2	60	100	*	*	900	850	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3812	50	AD	008	37	03	A1A	-	-	ADDITIVE	L	P-301904	OUTGOING LI	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3813	50	AD	008	37	04	A1A	IH	IE	ADDITIVE	L	HOSE CONNECTIO	P-301904	0.6	1.2	60	100	*	*	900	850	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3814	25	AD	008	37	05	A1A	IH	-	ADDITIVE	L	BYPASS LINE OF P-	P-301904	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3815	40	AD	008	37	06	A1A	IH	-	ADDITIVE	L	P-301904	FV-301206	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3816	40	AD	008	37	07	A1A	IH	-	ADDITIVE	L	P-301904	FV-301205	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3817	40	AD	008	37	08	A1A	IH	-	ADDITIVE	L	P-301904	FV-301204	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3818	40	AD	008	37	09	A1A	IH	-	ADDITIVE	L	P-301904	FV-301203	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3819	40	AD	008	37	10	A1A	IH	-	ADDITIVE	L	P-301904	FV-301202	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3820	40	AD	008	37	11	A1A	IH	-	ADDITIVE	L	P-301904	FV-301201	6	8	60	100	*	3	900	850	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3821	50	HOS(H)	008	37	12	A1B	IH	-	HOT OIL	L	HEADER	MB-301904	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3822	50	HOR(H)	008	37	13	A1B	IH	-	HOT OIL	L	HEADER	MB-301904	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3823	100	V	008	37	14	A1A	-	-	VAPOR	V	MB-301904	SAFE LOCATIC	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3824	25	AD	008	38	01	A1A	-	-	ADDITIVE	L	P-301906	MB-301906	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3825	40	AD	008	38	02	A1A	-	-	ADDITIVE	L	MB-301906	P-301906	0.6	1.2	60	100	*	*	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3826	25	AD	008	38	03	A1A	-	-	ADDITIVE	L	P-301906	OUTGOING LI	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3827	50	AD	008	38	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-301906	0.6	1.2	60	100	*	*	900	100	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3828	25	AD	008	38	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-301906	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3829	25	AD	008	38	06	A1A	-	-	ADDITIVE	L	P-301906	FV-301206	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3830	25	AD	008	38	07	A1A	-	-	ADDITIVE	L	P-301906	FV-301205	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3831	25	AD	008	38	08	A1A	-	-	ADDITIVE	L	P-301906	FV-301204	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3832	25	AD	008	38	09	A1A	-	-	ADDITIVE	L	P-301906	FV-301203	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3833	25	AD	008	38	10	A1A	-	-	ADDITIVE	L	P-301906	FV-301202	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3834	25	AD	008	38	11	A1A	-	-	ADDITIVE	L	P-301906	FV-301201	6	8	60	100	*	3	900	100	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3835	100	V	008	38	13	A1A	-	-	VAPOR	V	MB-301905	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3836	40	AD	008	39	01	A1A	-	-	ADDITIVE	L	P-301907	MB-301907	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3837	50	AD	008	39	02	A1A	-	-	ADDITIVE	L	MB-301907	P-301907	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3838	40	AD	008	39	03	A1A	-	-	ADDITIVE	L	P-301907	OUTGOING LI	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3839	50	AD	008	39	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-301907	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3840	40	AD	008	39	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-301907	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3841	40	AD	008	39	06	A1A	-	-	ADDITIVE	L	P-301907	FV-301206	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3842	40	AD	008	39	07	A1A	-	-	ADDITIVE	L	P-301907	FV-301205	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3843	40	AD	008	39	08	A1A	-	-	ADDITIVE	L	P-301907	FV-301204	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3844	40	AD	008	39	09	A1A	-	-	ADDITIVE	L	P-301907	FV-301203	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3845	40	AD	008	39	10	A1A	-	-	ADDITIVE	L	P-301907	FV-301202	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3846	40	AD	008	39	11	A1A	-	-	ADDITIVE	L	P-301907	FV-301201	6	8	60	100	*	10	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3847	100	V	008	39	13	A1A	-	-	VAPOR	V	MB-301907	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3848	40	AD	008	40	01	A1A	-	-	ADDITIVE	L	P-301908	MB-301908	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3849	50	AD	008	40	02	A1A	-	-	ADDITIVE	L	MB-301908	P-301908	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3850	40	AD	008	40	03	A1A	-	-	ADDITIVE	L	P-301908	OUTGOING LI	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3851	50	AD	008	40	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-301908	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3852	40	AD	008	40	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-301908	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3853	40	AD	008	40	06	A1A	-	-	ADDITIVE	L	P-301908	PK-301705	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3854	40	AD	008	40	07	A1A	-	-	ADDITIVE	L	P-301908	FV-301206	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3855	40	AD	008	40	08	A1A	-	-	ADDITIVE	L	P-301908	FV-301205	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3856	40	AD	008	40	09	A1A	-	-	ADDITIVE	L	P-301908	FV-301204	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3857	40	AD	008	40	10	A1A	-	-	ADDITIVE	L	P-301908	FV-301203	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3858	40	AD	008	40	11	A1A	-	-	ADDITIVE	L	P-301908	FV-301202	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3859	40	AD	008	40	12	A1A	-	-	ADDITIVE	L	P-301908	FV-301201	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3860	100	V	008	40	13	A1A	-	-	VAPOR	V	MB-301908	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3861	25	AD	008	41	01	A1A	-	-	ADDITIVE	L	P-301909	MB-301909	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3862	25	AD	008	41	02	A1A	-	-	ADDITIVE	L	MB-301909	P-301909	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3863	25	AD	008	41	03	A1A	-	-	ADDITIVE	L	P-301909	OUTGOING LI	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3864	50	AD	008	41	04	A1A	-	-	ADDITIVE	L	HOSE CONNECTIO	P-301909	0.6	1.2	60	100	*	*	900	500	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3865	25	AD	008	41	05	A1A	-	-	ADDITIVE	L	BYPASS LINE OF P-	P-301909	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3866	25	AD	008	41	06	A1A	-	-	ADDITIVE	L	P-301909	PK-301705	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3867	25	AD	008	41	07	A1A	-	-	ADDITIVE	L	P-301909	FV-301206	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3868	25	AD	008	41	08	A1A	-	-	ADDITIVE	L	P-301909	FV-301205	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR												
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3869	25	AD	008	41	09	A1A	-	-	ADDITIVE	L	P-301909	FV-301204	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3870	25	AD	008	41	10	A1A	-	-	ADDITIVE	L	P-301909	FV-301203	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3871	25	AD	008	41	11	A1A	-	-	ADDITIVE	L	P-301909	FV-301202	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3872	25	AD	008	41	12	A1A	-	-	ADDITIVE	L	P-301909	FV-301201	6	8	60	100	*	3	900	500	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3873	100	V	008	41	13	A1A	-	-	VAPOR	V	MB-301909	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3874	50	BO	008	42	01	A1A	IH	-	GREASE	L	P-301910	MB-301910	6	8	60	100	*	6	900	1275	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3875	50	AD	008	42	02	A1A	IH	IE	GREASE	L	MB-301910	P-301910	0.6	1.2	60	100	*	*	900	1275	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3876	50	AD	008	42	03	A1A	IH	IE	GREASE	L	P-301910	OUTGOING LI	6	8	60	100	*	6	900	1275	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3877	50	AD	008	42	04	A1A	IH	IE	GREASE	L	HOSE CONNECTIO	P-301910	0.6	1.2	60	100	*	*	900	1275	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3878	50	AD	008	42	05	A1A	IH	-	GREASE	L	BYPASS LINE OF P-	P-301910	6	8	60	100	*	6	900	1275	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3879	50	BO	008	42	06	A1A	IH	-	GREASE	L	P-301910	BASEOIL HEAD	6	8	60	100	*	6	900	1275	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3880	50	HOS(H)	008	42	07	A1B	IH	-	GREASE	L	HEADER	MB-301910	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3881	50	HOR(H)	008	42	08	A1B	IH	-	GREASE	L	HEADER	MB-301910	6	8	200	220	*	*	900	2.5	10	300	Water	*	1.5	SCH 40	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3882	100	V	008	42	09	A1A	-	-	VAPOR	V	MB-301910	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3883	40	AD	008	43	01	A1A	-	-	GREASE	L	P-301911	MB-301911	6	8	60	100	*	3	900	47	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3884	50	AD	008	43	02	A1A	-	-	GREASE	L	MB-301911	P-301911	0.6	1.2	60	100	*	*	900	47	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3885	40	AD	008	43	03	A1A	-	-	GREASE	L	P-301911	OUTGOING LI	6	8	60	100	*	3	900	47	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3886	50	AD	008	43	04	A1A	-	-	GREASE	L	HOSE CONNECTIO	P-301911	6	8	60	100	*	3	900	47	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3887	40	AD	008	43	05	A1A	-	-	GREASE	L	BYPASS LINE OF P-	P-301911	6	8	60	100	*	*	900	47	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3888	40	AD	008	43	06	A1A	-	-	GREASE	L	P-301911	BASEOIL HEAD	6	8	60	100	*	*	900	47	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3889	100	V	008	43	07	A1A	-	-	VAPOR	V	MB-301911	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3890	40	BO	008	44	01	A1A	-	-	GREASE	L	P-301912	MB-301912	6	8	60	100	*	3	900	31	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3891	50	BO	008	44	02	A1A	-	-	GREASE	L	MB-301912	P-301912	0.6	1.2	60	100	*	*	900	31	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3892	40	BO	008	44	03	A1A	-	-	GREASE	L	P-301912	OUTGOING LI	6	8	60	100	*	3	900	31	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3893	50	AD	008	44	04	A1A	-	-	GREASE	L	HOSE CONNECTIO	P-301912	6	8	60	100	*	3	900	31	15	200	Water	*	1.5	SCH 40	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3894	40	BO	008	44	05	A1A	-	-	GREASE	L	BYPASS LINE OF P-	P-301912	6	8	60	100	*	*	900	31	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3895	40	BO	008	44	06	A1A	-	-	GREASE	L	P-301912	BASEOIL HEAD	6	8	60	100	*	*	900	31	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3896	100	V	008	44	07	A1A	-	-	VAPOR	V	MB-301912	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3897	25	AD	008	45	01	A1A	-	-	GREASE	L	P-301913	MB-301913	6	8	60	100	*	3	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3898	25	AD	008	45	02	A1A	-	-	GREASE	L	MB-301913	P-301913	0.6	1.2	60	100	*	*	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3899	25	AD	008	45	03	A1A	-	-	GREASE	L	P-301913	OUTGOING LI	6	8	60	100	*	3	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3900	25	AD	008	45	04	A1A	-	-	GREASE	L	HOSE CONNECTIO	P-301913	6	8	60	100	*	3	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3901	25	AD	008	45	05	A1A	-	-	GREASE	L	BYPASS LINE OF P-	P-301913	6	8	60	100	*	*	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3902	25	AD	008	45	06	A1A	-	-	GREASE	L	P-301913	FV-301713	6	8	60	100	*	*	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3903	25	AD	008	45	07	A1A	-	-	GREASE	L	P-301913	A-301703	6	8	60	100	*	*	900	66	15	200	Water	*	1.5	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3904	100	V	008	45	08	A1A	-	-	VAPOR	V	MB-301913	SAFE LOCATIO	6	8	120	220	*	*	900	2.5	15	200	Water	*	1.5	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3905	50	PD	008	46	02	B4C	-	-	GREASE	S	PHS-301706	SL-301706	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3906	50	PD	008	46	08	B4C	-	-	GREASE	S	SC-301706B	PK-301706	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3907	25	CA	008	46	10	G6C	-	-	GREASE	V	*	PHS-301706	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3908	25	IA	008	46	11	G6C	-	-	GREASE	V	*	PHS-301706	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR									
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO				
3909	25	PA	008	46	12	G6C	-	-	GREASE	V	*	PHS-301706	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3910	50	PD	008	47	01	B4C	-	-	GREASE	S	PHS-301701	PHS-301701	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3911	80	PD	008	47	02	B4C	-	-	GREASE	S	PHS-301701	SC-301701	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3912	100	PD	008	47	03	B4C	-	-	GREASE	S	SC-301701	PK-301701	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3913	25	CA	008	47	04	G6C	-	-	GREASE	V	*	PHS-301701	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3914	25	IA	008	47	05	G6C	-	-	GREASE	V	*	PHS-301701	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3915	25	PA	008	47	06	G6C	-	-	GREASE	V	*	PHS-301701	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3916	50	PD	008	48	01	B4C	-	-	GREASE	S	PHS-301705	SL-301705A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3917	25	CA	008	48	11	G6C	-	-	GREASE	V	*	PHS-301705	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	0	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3918	25	IA	008	48	09	G6C	-	-	GREASE	V	*	PHS-301705	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3919	25	PA	008	48	10	G6C	-	-	GREASE	V	*	PHS-301705	6	8	60	100	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3920	80	PD	008	49	01	B4C	-	-	GREASE	S	SL-301705	SC-301705	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3921	100	PD	008	49	10	B4C	-	-	GREASE	S	SC-301705	PK-301705	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3922	50	PD	008	50	01	B4C	-	-	GREASE	S	PHS-301001	SL-301001B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3923	50	PD	008	50	02	B4C	-	-	GREASE	S	PHS-301001	SL-301001A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3924	200	PD	008	50	03	B4C	-	-	GREASE	S	SC-301001A	GC-301001	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3925	200	PD	008	50	04	B4C	-	-	GREASE	S	SC-301001B	GC-301002	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3926	25	CA	008	50	05	G6C	-	-	GREASE	V	*	PHS-301001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3927	25	IA	008	50	06	G6C	-	-	GREASE	V	*	PHS-301001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3928	25	PA	008	50	07	G6C	-	-	GREASE	V	*	PHS-301001	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3929	80	PD	008	50	08	B4C	-	-	GREASE	S	SL-301001A	SC-301001A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3930	80	PD	008	50	09	B4C	-	-	GREASE	S	SL-301001B	SC-301001B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3931	50	PD	008	51	01	B4C	-	-	GREASE	S	PHS-301002	SL-301002B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3932	50	PD	008	51	02	B4C	-	-	GREASE	S	PHS-301002	SL-301002A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3933	200	PD	008	51	03	B4C	-	-	GREASE	S	SC-301002A	GC-301001	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3934	200	PD	008	51	04	B4C	-	-	GREASE	S	SC-301002B	GC-301002	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3935	25	CA	008	51	05	G6C	-	-	GREASE	V	*	PHS-301002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3936	25	IA	008	51	06	G6C	-	-	GREASE	V	*	PHS-301002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3937	25	PA	008	51	07	G6C	-	-	GREASE	V	*	PHS-301002	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3938	80	PD	008	51	08	B4C	-	-	GREASE	S	SL-301002B	SC-301002B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3939	80	PD	008	51	09	B4C	-	-	GREASE	S	SL-301002A	SC-301002A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3940	50	PD	008	52	01	B4C	-	-	GREASE	S	PHS-301003	SL-301003B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3941	50	PD	008	52	02	B4C	-	-	GREASE	S	PHS-301003	SL-301003A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3942	200	PD	008	52	03	B4C	-	-	GREASE	S	SC-301003A	GC-301001	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3943	200	PD	008	52	04	B4C	-	-	GREASE	S	SC-301003B	GC-301002	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3944	25	CA	008	52	05	G6C	-	-	GREASE	V	*	PHS-301003	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3945	25	IA	008	52	06	G6C	-	-	GREASE	V	*	PHS-301003	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3946	25	PA	008	52	07	G6C	-	-	GREASE	V	*	PHS-301003	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3947	80	PD	008	52	08	B4C	-	-	GREASE	S	SL-301003A	SC-301003A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3948	80	PD	008	52	09	B4C	-	-	GREASE	S	SL-301003B	SC-301003B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

S.NO	PIPE SIZE (DN)	LINE NUMBER							SERVICE				OPERATING CONDITION								DESIGN CONDITION		HYDRO TESTING		CORR. ALLOW.	WALL THK.	PAINTING	STRESS ANALYSIS FOR CRITICAL LINES	PID NUMBER	PID Rev	PID DESCRIPTION	SCOPE	REMARKS
													PRESSURE		TEMP		FLOW RATE		DENSITY	VISCOSITY	PR	TEMP	MEDIUM	PR										
		FLUID SYM.	PID No	PID Sheet No	Sequence No.	PIPE SPEC.	INSULATION	E TRACING	FLUID	PHASE	FROM	TO	NOR	MAX	NOR	MAX	NOR	MAX	NOR	NOR													
													kg/cm2.g		°C		m3/hr		Kg/m3	c P	kg/cm2.g	°C		kg/cm2.g	INCH	SCH. NO.		YES/NO					
3949	50	PD	008	53	01	B4C	-	-	GREASE	S	PHS-301004	SL-301004B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3950	50	PD	008	53	02	B4C	-	-	GREASE	S	PHS-301004	SL-301004A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3951	200	PD	008	53	03	B4C	-	-	GREASE	S	SC-301004A	GC-301001	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3952	200	PD	008	53	04	B4C	-	-	GREASE	S	SC-301004B	GC-301002	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3953	25	CA	008	53	05	G6C	-	-	GREASE	V	*	PHS-301004	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3954	25	IA	008	53	06	G6C	-	-	GREASE	V	*	PHS-301004	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3955	25	PA	008	53	07	G6C	-	-	GREASE	V	*	PHS-301004	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3956	80	PD	008	53	08	B4C	-	-	GREASE	S	SL-301004A	SC-301004A	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3957	80	PD	008	53	09	B4C	-	-	GREASE	S	SL-301004B	SC-301004B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3958	50	PD	008	54	03	B4C	-	-	GREASE	S	PHS-301702	SL-301702B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3959	80	PD	008	54	06	B4C	-	-	GREASE	S	SL-301702B	PK-301702	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3960	25	CA	008	54	09	G6C	-	-	GREASE	V	*	PHS-301702	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3961	25	IA	008	54	10	G6C	-	-	GREASE	V	*	PHS-301702	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3962	25	PA	008	54	11	G6C	-	-	GREASE	V	*	PHS-301702	6	8	40	60	*	*	1.25	0.00001	10	60	Water	*	1.5	HVY	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3963	80	PD	008	54	13	B4C	-	-	GREASE	S	SL-301702B	SC-301702B	*	*	60	100	*	*	*	*	15	150	Water	*	0	SCH 10S	*	YES	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3964	80	V	008	55	01	A1A	-	-	GREASE	V	MANIFOLD	P-301951	0.6	1.2	60	100	*	*	0.062	*	15	200	Water	*	0	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3965	80	V	008	55	02	A1A	-	-	GREASE	V	P-301951	CS-301951	6	8	60	100	*	300 NM3/HR	0.062	*	15	200	Water	*	0	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3966	80	V	008	55	03	A1A	-	-	GREASE	V	MANIFOLD	VENT	6	8	60	100	*	*	0.062	*	15	200	Water	*	0	4.78 MM	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3967	25	V	008	55	04	A1A	-	-	GREASE	V	MAKE UP WATER	CS-301951	6	8	60	100	*	*	900	*	15	200	Water	*	0	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		
3968	40	V	008	55	05	A1A	-	-	GREASE	V	P-301951	CS-301951	6	8	60	100	*	300 NM3/HR	0.062	*	15	200	Water	*	0	SCH 80	*	NO	90-0RG2N-PID-008	H1	P&ID FOR GREASE PLANT		

1047 SQM

A	PROCESS BUILDING	35000 SQM	250 M x 140 M
B	OFFICE BUILDING	800 SQM	40 M x 20 M
C	LABORATORY BUILDING	1047 SQM	40.73 M x 25.73M
D	UTILITY BUILDING	1402 SQM	40.73 M x 25.73M 51 M x 27.5 M

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component		Formula		Physical Properties			Operating/Blending Temp
						Name	code	% vol	% wt	SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	
1	1301	SERVO ULTRA 10W	Engine Oil	Mineral Oil base		Base Oil	400422	88.15	87.15	0.88	30	5 min	60 - 70
2	1303	SERVO ULTRA 30	Engine Oil	Mineral Oil base		Base Oil1	190412	18.35	18.00	0.88	97	10.0-12.0	60 - 70
3	1304	SERVO ULTRA 40	Engine Oil	Mineral Oil base		Base Oil1	190612	29.33	29.00	0.89	137	13.0-15.0	60 - 70
4	1305	SERVO ULTRA 50	Engine Oil	Mineral Oil base		Base Oil1	190862	59.68	58.73	0.89	216	18.0-20.0	60 - 70
5	1395	SERVO MIL C 30	Engine Oil	Mineral Oil base		Base Oil1	190612	95.814	95.384	0.88	97	10.0-12.0	60 - 70
6	1403	SERVO PRIDE 30	Engine Oil	Mineral Oil base		Base Oil1	400622	95.734	94.894	0.88	97	10.0-12.0	60 - 70
7	1404	SERVO PRIDE 40	Engine Oil	Mineral Oil base		Base Oil1	400622	29.734	29.000	0.89	137	13.0-15.0	60 - 70
8	1514	SERVO RR 520 MG	Engine Oil	Mineral Oil base		Base Oil1	190612	72.60	71.80	0.88	143	15.5-16.3	60 - 70
9	1516	SERVO RR 606 MG PLUS(II)	Engine Oil	Mineral Oil base		Base Oil1	400622	74.60	74.00	0.88	143	15.5-16.3	60 - 70
10	1517	SERVO MAGNUM RR 610	Engine Oil	Mineral Oil base		Base Oil1	400622	72.00	71.35	0.88	143	15.5-16.3	60 - 70
11	2406	TAFE AGRISTAR	Transmission Oil	Mineral Oil base		Base Oil	191502	97.40	97.05	0.9	365	25.0-28.0	60 - 70
12	2411	TAFE STEERING OIL	Transmission Oil	Mineral Oil base		Base Oil1	190612	84.26	83.95	0.9	284	24.5-28.5	60 - 70
13	2490	VST SHAKTI GEN GEAR OIL 90	Gear Oil	Mineral Oil base		Base Oil1	190612	93.29	92.95	0.89	143	14.0-16.0	60 - 70
14	2502	SERVO GEAR HP 90	Gear Oil	Mineral Oil base		Base Oil1	190612	92.58	92.20	0.88	143	14.0-16.0	60 - 70
15	2503	SERVO GEAR HP 140	Gear Oil	Mineral Oil base		Base Oil1	190612	82.79	82.20	0.88	284	24.5-28.5	60 - 70
16	2508	SERVO GEAR HP 80W-90	Gear Oil	Mineral Oil base		Base Oil1	400622	92.48	92.00	0.87	143	14.0-16.0	60 - 70
17	2515	SERVO GEAR HP 80W(T)	Gear Oil	Mineral Oil base		Base Oil1	190412	10.00	10.00	0.88	74	8.0-10.0	60 - 70
18	2518	SERVO GEAR ALT 90	Gear Oil	Mineral Oil base		Base Oil1	190862	70.80	70.00	0.89	190	16.0-18.0	60 - 70
19	2522	SERVO GEAR SUPER 80W 90(T)	Gear Oil	Mineral Oil base		Base Oil1	190412	44.50	43.00	0.89	157	14.0-15.5	60 - 70
20	2523	SERVO GEAR SUPER 85W 140(T)	Gear Oil	Mineral Oil base		Base Oil1	400622	78.42	76.98	0.89	284	24.5-28.5	60 - 70
21	2528	SERVO GEAR SUPER ALT 85W-140	Gear Oil	Mineral Oil base		Base Oil1	191502	93.40	92.50	0.9	420	28.0-32.0	60 - 70
22	2532	SERVO SYNCHRO TM 80W-90	Gear Oil	Mineral Oil base		Base Oil1	191502	52.20	51.50	0.89	157	14.0-16.0	60 - 70
23	2533	SERVO GEAR AXLE TM 85W-140	Gear Oil	Mineral Oil base		Base Oil1	191502	92.90	92.50	0.91	365	25.0-27.0	60 - 70
24	2534	SERVO GEAR AXLE ALT 85W-140	Gear Oil	Mineral Oil base		Base Oil1	191502	93.15	92.75	0.91	420	28.0-31.0	60 - 70
25	2540	MGGO 80W-90	Gear Oil	Mineral Oil base		Base Oil1	400622	74.57	73.42	0.88	142	14.0-15.5	60 - 70
26	2541	SERVO GEAR HP 80W-90(T)	Gear Oil	Mineral Oil base		Base Oil1	400622	90.83	89.93	0.88	132	14.0-16.0	60 - 70
27	2548	SERVO GEAR ALT 80W-90 LL	Gear Oil	Mineral Oil base		Base Oil1	191502	33.90	34.65	0.88	141	14.0-15.5	60 - 70
28	2549	SERVO GEAR SUPER ALT 85-140 LL	Gear Oil	Mineral Oil base		Base Oil1	191502	92.10	91.50	0.9	448	28.0-31.0	60 - 70
29	2557	HYUNDAI S.GEAR OIL 75W-85	Gear Oil	Mineral Oil base		BASE OIL 1	800452	69.00	67.95	0.87	59	11.0-12.5	60 - 70
30	2571	TMGO GO 80W-90 LL	Gear Oil	Mineral Oil base		BASE OIL 1	400622	83.97	82.18	0.87	124	15.0-16.5	60 - 70
31	2572	TMGO RA 80W-140 LL	Gear Oil	Mineral Oil base		BASE OIL 1	400622	64.72	63.29	0.87	286	26.0-28.0	60 - 70
32	2602	SERVO GEAR SUPER 90	Gear Oil	Mineral Oil base		BASE OIL 1	190612	91.08	90.44	0.88	144	14.0-16.0	60 - 70
33	2617	MAXIMILE GEAR OIL	Gear Oil	Mineral Oil base		BASE OIL 1	190412	40.50	40.00	0.89	158	14.0-16.0	60 - 70
34	2618	MAXIMILE AXLE OIL	Gear Oil	Mineral Oil base		BASE OIL 1	400622	78.42	76.98	0.88	316	24.5-28.5	60 - 70
35	2705	SERVO MLO SUPER	Speciality	Mineral Oil base		BASE OIL 1	400352	20	20.00	0.84	2.9 - 3.5		40 - 50
36	2706	SERVOCUT EGR	Grinding Oil	Water & Ester Base		Synfluid 1	650262	97.700	97.500	1			25 -35
37	2889	SERVO SCOOTOMATIC 4ST 10W-30	Engine Oil	Mineral Oil base		BASE OIL 1	400422	68.95	68.00	0.87	75	10.0-11.5	60 - 70
38	2895	SERVO SCOOTOMATIC 4ST	Engine Oil	Mineral Oil base		BASE OIL 1	400622	82.52	82.10	0.87	108	13.5-15.5	60 - 70
39	2900	SERVO 2T SUPREME-20 L	Engine Oil	Mineral Oil base		BASE OIL 1	400622	57.09	57.79	0.88	55.0-60.0	8	45 - 55
40	2909	SERVO BRAKE FLUID SUPER HD	Engine Oil	Glycol/Ether		Synfluid 1	655382	48.936	49.165	1.05		1.8	Ambient
41	2924	SERVO 4T	Engine Oil	Mineral Oil base		BASE OIL 1	400622	91.98	91.40	0.88	108	13.5-15.5	60 - 70
42	2931	SERVO 4T 20W-50	Engine Oil	Mineral Oil base		BASE OIL 1	400622	87.34	86.95	0.88	173	17.5-19.5	60 - 70
43	2936	SERVO 4T ZOOM 10W-30	Engine Oil	Mineral Oil base		BASE OIL 1	400622	24.74	25.00	0.88	73	10.0-12.0	60 - 70
44	2941	SERVO OBM 2T OIL	Engine Oil	Mineral Oil base		BASE OIL 1	400622	57.10	57.80	0.88	55.0-60.0	8	40 - 50
45	2949	SERVO 4T SYNTH	Engine Oil	Mineral Oil base		BASE OIL 1	420422	63.74	62.20	0.87	76	9.3-12.5	60 - 70
46	2953	HY.SERVO SYNTH PLUS 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	420422	28.15	27.80	0.86	69	10.5-11.5	60 - 70
47	2955	SERVO FUTURA D 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	800402	59.13	57.90	0.86	69	9.3-12.5	60 - 70

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component Name	Component code	Formula % vol	Formula % wt	Physical Properties SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	Operating/Blending Temp
48	2956	SERVO FUTURA D PLUS 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	800452	32.50	32.10	0.86	68	9.3-11.3	60 - 70
49	2959	SERVO FUTURA G 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	420422	37.70	37.10	0.86	68	11.0-12.0	60 - 70
50	2964	MGDO 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	800452	28.05	27.70	0.86	62	10.5 - 11.5	60 - 70
51	2966	SERVO 4T BULLET 15W-50	Engine Oil	Mineral Oil base		BASE OIL 1	400422	45.30	44.41	0.87	143	17.0-19.0	60 - 70
52	2967	SERVO FUTURA G PLUS 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	420422	28.15	27.80	0.86	66	10.5-11.5	60 - 70
53	2968	HY.SERVO ENGINE OIL 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	800402	59.13	57.90	0.86	67	9.3-12.5	60 - 70
54	2969	SERVO SUPERIOR K 0W-20	Engine Oil	Mineral Oil base		BASE OIL 1	800402	87.16	85.76	0.85	47	7.8-8.3	60 - 70
55	3001	SERVOSYSTEM 32	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	97.00	97.00	0.86	29.3-32.7	5.5	50 - 60
56	3002	SERVOSYSTEM 46	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	61.43	61.33	0.86	43.5-47.5	7	50 - 60
57	3006	SERVOSYSTEM 68	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	27.43	27.33	0.86	64.7-71.3	9	50 - 60
58	3007	SERVOSYSTEM 100	Hydraulic Oil	Mineral Oil base		BASE OIL 1	190862	20.43	20.33	0.87	96.0-104.0	11	50 - 60
59	3009	SERVO CASE HYDRAPOWER	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	27.45	27.38	0.87	64.7-71.3	9	50 - 60
60	3017	SERVOSYSTEM 320	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	80.18	80.21	0.89	320.0-345.0	24	50 - 60
61	3018	SERVOSYSTEM 460	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	97.18	97.21	0.89	445.0-495.0	30	50 - 60
62	3019	SERVOSYSTEM 150	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	33.43	33.33	0.88	147.0-153.0	15	50 - 60
63	3020	SERVOSYSTEM 220	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	55.43	55.33	0.88	212.0-228.0	19	50 - 60
64	3021	SERVOHYDRAULIC OIL 32	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	97.00	97.00	0.86	29.3-32.7	5.3	50 - 60
65	3022	SERVOHYDRAULIC OIL 46	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	61.43	61.33	0.86	43.5-47.5	7	50 - 60
66	3023	SERVOHYDRAULIC OIL 68	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	27.43	27.33				
67	3024	SERVOHYDRAULIC OIL 100	Hydraulic Oil	Mineral Oil base		BASE OIL 1	190862	20.43	20.33				
68	3025	SERVOHYDRAULIC OIL 150	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	33.43	33.33	0.88	147 - 153	15	50 - 60
69	3026	SERVOHYDRAULIC OIL 220	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	55.43	55.33	0.88	212.0-228.0	19	50 - 60
70	3027	SERVOHYDRAULIC OIL 320	Hydraulic Oil	Mineral Oil base		BASE OIL 1	191502	80.18	80.21				
71	3037	BEML HYDRAULIC OIL EH10CD	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	95.890	94.980	0.87	34	5.6	50 - 60
72	3057	SERVO EXCAVATOR TH 46	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400622	19.9	20	0.87	46	7.8	55 - 65
73	3064	SERVOHYDREX TH 46	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400622	19.90	20.00	0.86	46	7.8	55 - 65
74	3065	SERVOHYDREX 100	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400622	69.03	68.04	0.87	101.0-109.0	13.5	55 - 65
75	3066	SERVOHYDREX 68	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	30.00	30.00	0.87	66.0-71.0	10.5	55 - 65
76	3071	SERVOSYSTEM HLP 22	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400352	48.24	48.25	0.86	20.2-23.8	4.3	50 - 60
77	3079	SERVOSYSTEM HVLP 68	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	23.17	22.84	0.86	66.0-71.0	10	50 - 60
78	3081	SERVOSYSTEM HLP 32	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	96.06	95.88	0.86	29.3-32.7	5.3	50 - 60
79	3082	SERVOSYSTEM HLP 46 -	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	52.00	52.00	0.86	43.5-47.5	7	50 - 60
80	3084	SERVOSYSTEM HLP 68	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	26.06	25.88	0.87	64.7-71.3	9	50 - 60
81	3086	SERVOSYSTEM HLP 100	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	82.50	82.50	0.88	96.0-104.0	11	50 - 60
82	3092	SERVOSYSTEM HLP 32N	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	86.19	86.20	0.86	29.3-32.7	5.2	50 - 60
83	3093	SERVOSYSTEM HLP 46N -	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	40.19	40.20	0.86	43.5-47.5	7	50 - 60
84	3094	SERVOSYSTEM XLP 46-	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	52	52	0.86	43.5-47.5	7	50 - 60
85	3095	SERVOSYSTEM HLP 68N		Mineral Oil base		BASE OIL 1	400422	10.09	10.10	0.87	68	9	50 - 60
86	3098	SERVOSYSTEM XLP 22-210 L	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400352	40.03	40	0.86	20.2-23.8	4.3	50 - 60
87	3120	SERVOPRIME 32	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	94.63	94.58	0.86	29.3-32.7	5.3	50 - 60
88	3121	SERVOPRIME 46	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	50.00	50.00	0.86	43.5-47.5	7	50 - 60
89	3122	SERVOPRIME 57	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	33.00	33.00	0.86	57	7.9	50 - 60
90	3124	SERVOPRIME 68	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	20.00	20.00	0.87	64.7-71.3	9	50 - 60
91	3125	SERVOPRIME 76	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	22.00	22.00	0.87	74.8-79.2	9.8	50 - 60
92	3132	SERVOPRIME 32 G	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	59.58	59.53	0.86	29.3-32.7	5.3	50 - 60
93	3133	SERVOPRIME 46G	Turbine Oil	Mineral Oil base		BASE OIL 1	400622	24.58	24.53	0.86	43.5-47.5	7	50 - 60
94	3135	SERVOPRIME 32 LP	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	49.45	49.38	0.86	29.3-32.7	5.3	50 - 60

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component		Formula		Physical Properties			Operating/Blending Temp
						Name	code	% vol	% wt	SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	
95	3136	SERVOPRIME 46 LP	Turbine Oil	Mineral Oil base		BASE OIL 1	420422	65.00	65.00	0.86	43.5-47.5	7	50 - 60
96	3139	SERVOPRIME 32 LL	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	84.00	84.00	0.86	29.3-32.7	5.3	50 - 60
97	3141	SERVOPRIME 32T	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	94.63	94.58	0.86	29.3-32.7	5.3	50 - 60
98	3142	SERVOPRIME 46T	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	50.00	50.00	0.86	43.5-47.5	7	50 - 60
99	3144	SERVOPRIME 68 LL	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	10.00	10.00	0.86	64.7-71.3	8.8	50 - 60
100	3146	SERVOPRIME 76T	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	5.00	5.00	0.87	74.8-79.2	9.5	50 - 60
101	3161	SERVOCONVAL 46	Circ Oil	Mineral Oil base		BASE OIL 1	400422	40.00	40.00	0.86	43.5-47.5	7	50 - 60
102	3204	SERVOLINE 100	Gen Mach Oil	Mineral Oil base		BASE OIL 1	190522	74.000	74.000	0.87	96-104	11	50 - 60
103	3206	SERVOLINE 220 -	Gen Mach Oil	Mineral Oil base		BASE OIL 1	191502	26.970	26.970	0.89	212-228	19	50 - 60
104	3210	SERVOFILM 12 -	Gen Mach Oil	Mineral Oil base		BASE OIL	290352	100.000	100.000	0.87	9.0-13	3.2	50 - 60
105	3212	SERVOFILM 32 -	Gen Mach Oil	Mineral Oil base		BASE OIL 1	190412	96.000	96.000	0.87	29.3-32.7	5.1	50 - 60
106	3216	SERVOFILM 100 -	Gen Mach Oil	Mineral Oil base		BASE OIL 1	190522	52.000	52.000				
107	3221	SERVOFILM 68 -	Gen Mach Oil	Mineral Oil base		BASE OIL	190522	100.000	100.000	0.88	64.7-71.3	9	Ambient
108	3225	SERVOFILM 460 -	Gen Mach Oil	Mineral Oil base		BASE OIL 1	190862	2.000	2.000	0.9	445-475	30	50 - 60
109	3245	SERVOLOOM J150	Gen Mach Oil	Mineral Oil base		BASE OIL 1	190412	67.500	67.500	0.92	147-153	12	50 - 60
110	3246	SERVOLOOM J220 -	Gen Mach Oil	Mineral Oil base		BASE OIL 1	190412	52.800	52.800	0.92	212-228	15	50 - 60
111	3247	SERVOLOOM J320	Gen Mach Oil	Mineral Oil base		BASE OIL 1	110522	44.600	44.600	0.94	305-345	18	50 - 60
112	3248	SERVOLOOM J 460 -	Gen Mach Oil	Mineral Oil base		BASE OIL 1	110612	73.500	73.500	0.95	425-495	22	50 - 60
113	3251	SERVOFILM 12 PLUS-	Bearing Oils	Mineral Oil base		BASE OIL	400352	100.000	100.000	0.86	9.0-13	3.3	Ambient
114	3252	SERVOFILM 32 PLUS-	Bearing Oils	Mineral Oil base		BASE OIL	400422	100.000	100.000	0.86	29.3-32.7	5.4	Ambient
115	3253	SERVOFILM 100 PLUS-	Bearing Oils	Mineral Oil base		BASE OIL	400622	100.000	100.000	0.87	80	10	Ambient
116	3301	SERVOWAY 32	Machine Way Oils	Mineral Oil base		BASE OIL 1	190412	97.650	97.480	0.88	29.3-32.7	5.1	50 - 60
117	3302	SERVOWAY 68	Machine Way Oils	Mineral Oil base		BASE OIL 1	190522	97.650	97.480	0.89	64.7-71.3	8.5	50 - 60
118	3304	SERVOWAY 220	Machine Way Oils	Mineral Oil base		BASE OIL 1	190612	45.500	45.000	0.89	212-228	19	
119	3311	SERVOWAY H 32	Machine Way Oils	Mineral Oil base		BASE OIL 1	190412	89.980	90.030	0.88	29.3-32.7	5.2	50 - 60
120	3312	SERVOWAY H 57	Machine Way Oils	Mineral Oil base		BASE OIL 1	190412	11.000	11.000	0.88	54.5-59.5	7.5	50 - 60
121	3313	SERVOWAY H 68	Machine Way Oils	Mineral Oil base		BASE OIL 1	190522	94.780	94.530	0.88	70.5-74.5	9.1	50 - 60
122	3401	SERVOSPIN 2	Textile Oils	Mineral Oil base		BASE OIL 1	585962	99.640	99.590	0.84	2.0-2.5		40 -50
123	3402	SERVOSPIN 12	Textile Oils	Mineral Oil base		BASE OIL 1	290352	99.300	99.140	0.86	11.2-13.8	3.3	50 - 60
124	3403	SERVOSPIN 22	Textile Oils	Mineral Oil base		BASE OIL 1	290352	35.000	35.000	0.86	21-23	4.2	50 - 60
125	3404	SERVOSPIN EE 10	Textile Oils	Mineral Oil base		BASE OIL 1	400352	83.000	83.000	0.85	9.0-11	3.1	50 - 60
126	3411	SERVOSPIN 5	Textile Oils	Mineral Oil base		BASE OIL 1	290352	55.000	55.000	0.85	4.5-5		40 - 50
127	3501	SERVOPRESS C 100	Compressor Oil	Mineral Oil base		BASE OIL 1	190612	97.250	97.140	0.88	96-104	10.5	50 - 60
128	3502	SERVOPRESS C 220	Compressor Oil	Mineral Oil base		BASE OIL 1	190612	43.000	42.720		231-239	19	50 - 60
129	3512	SERVOPRESS 68	Compressor Oil	Mineral Oil base		BASE OIL 1	400622	65.320	65.230		65-71	8.8	50 - 60
130	3513	SERVOPRESS 100	Compressor Oil	Mineral Oil base		BASE OIL 1	400622	50.000	50.000	0.88	94-104	11	50 - 60
131	3514	SERVOPRESS 150	Compressor Oil	Mineral Oil base		BASE OIL 1	400622	10.000	10.000	0.89	146-154	15	50 - 60
132	3516	SERVOPRESS 220	Compressor Oil	Mineral Oil base		BASE OIL 1	190862	72.000	72.000	0.89	212-228	19	50 - 60
133	3519	SERVOPRESS 100 RR	Compressor Oil	Mineral Oil base		BASE OIL 1	400622	64.320	64.230	0.88	96-104	11	50 - 60
134	3520	SERVOPRESS 150 RR	Compressor Oil	Mineral Oil base		BASE OIL 1	400622	70.000	70.000	0.89	146-154	15	50 - 60
135	3523	SERVOPRESS T 57	Compressor Oil	Mineral Oil base		BASE OIL 1	400422	30.000	30.000	0.86	55.6-59.4	7.8	50 - 60
136	3524	SERVOPRESS T 32	Compressor Oil	Mineral Oil base		BASE OIL 1	400422	84.000	84.000	0.86	29.3-32.7	5.3	50 - 60
137	3525	SERVOPRESS T 46	Compressor Oil	Mineral Oil base		BASE OIL 1	400422	50.000	50.000	0.86	43.5-47.5	7	50 - 60
138	3611	SERVOMESH SP 68	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190412	30.000	30.000	0.88	64.7-71.3	8.5	50 - 60
139	3612	SERVOMESH SP 150	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190612	68.000	68.000	0.89	147-153	14.8	50 - 60
140	3613	SERVOMESH SP 257	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190612	32.000	32.000	0.89	255-275	20.5	50 - 60
141	3614	SERVOMESH SP 320 -	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190612	18.000	18.000	0.89	320-345	24	50 - 60

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component Name	Component code	Formula % vol	Formula % wt	Physical Properties SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	Operating/Blending Temp
142	3615	SERVOMESH SP 460	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190612	4.000	4.000	0.9	425-495	29	50 - 60
143	3616	SERVOMESH SP 680	Ind Gear Oils	Mineral Oil base		BASE OIL 1	191502	98.990	98.790	0.91	645-695	38.9	55 -65
144	3617	SERVOMESH SP 220 -	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190612	41.000	41.000	0.9	212-228	19	50 - 60
145	3618	SERVOMESH SP 100	Ind Gear Oils	Mineral Oil base		BASE OIL 1	190612	96.990	96.790	0.89	96-104	10.5	50 - 60
146	3620	SERVOMESH SP 1000	Ind Gear Oils	Mineral Oil base		BASE OIL 1	191502	88.490	88.290	0.88	985-1045	49	55 - 65
147	3711	SERVO ULTRA 40 RR	Engine Oil	Mineral Oil base		BASE OIL 1	190862	88.650	88.040	0.9	143	14-15	60 - 70
148	3712	SERVO ULTRA KB 10	Engine Oil	Mineral Oil base		BASE OIL 1	190412	95.840	94.930	0.88	34-36	6	60 - 70
149	3713	SERVO ULTRA KB 30	Engine Oil	Mineral Oil base		BASE OIL 1	190612	96.000	95.080	0.88	80-120	11	60 - 70
150	3723	SERVO PUMP SET OIL 20W-40	Pumpset Oil	Mineral Oil base		BASE OIL 1	190612	78.114	78.504	0.9	132	13.5-15.5	60 - 70
151	3725	SERVO SUPER PUMP SO 40	Pumpset Oil	Mineral Oil base		BASE OIL 1	190612	35.394	34.994	0.89	139	13-15	60 - 70
152	3805	SERVONEUM 100	Pneumatic tools Oils	Mineral Oil base		BASE OIL 1	190612	92.000	92.000	0.88	96-104	10.8	50 - 60
153	3809	SERVONEUM 181		Mineral Oil base		BASE OIL 1	190612	12.000	12.000	0.89	176-184	16.5	50 - 60
154	4001	SERVOCYL C 460	Cylinder Oils	Mineral Oil base		BASE OIL 1	191502	92.200	92.000	0.89	455-495	29	50 - 60
155	4003	SERVO CYL C 800	Cylinder Oils	Mineral Oil base		BASE OIL 1	191502	90.700	90.600	0.89	800	42	55 - 65
156	4004	SERVOCYL C 680	Cylinder Oils	Mineral Oil base		BASE OIL 1	191502	91.000	92.000	0.89	645-675	38	55 - 65
157	4006	SERVOCYL M 680	Cylinder Oils	Mineral Oil base		BASE OIL 1	191502	96.5	96.5	0.89	645-675	38	55 - 65
158	4007	SERVOCYL M 1000	Cylinder Oils	Mineral Oil base		BASE OIL 1	191502	89.500	89.500	0.9	985-1045	49	55 - 65
159	4008	SERVOCYL M 1500	Cylinder Oils	Mineral Oil base		BASE OIL 1	191502	84.200	84.200	0.92	1460-1540	63	55 - 65
160	4101	SERVO SM 85	Open Gear Oils	**Bituminus**		BASE OIL 1	585242	20.000	20.000	0.97	800	38.5-44.5	70 - 80
161	4102	SERVO SM 175	Open Gear Oils	Mineral Oil base		BASE OIL 1	585242	34.850	34.640	0.97	1800	79-87	70 - 80
162	4103	SERVO SM 260	Open Gear Oils	Mineral Oil base		BASE OIL 1	585242	45.000	45.000	0.97	2400	121-129	70 - 80
163	4202	SERVOCOAT 120	Open Gear Oils	Mineral Oil base		BASE OIL 1	110612	50.000	50.000	0.97	4000	205-245	70 - 80
164	4203	SERVOCOAT 140	Open Gear Oils	Mineral Oil base		BASE OIL 1	585242	64.500	64.500	0.97	8000	400	70 - 80
165	4206	SERVOCOAT 190	Open Gear Oils	Mineral Oil base		BASE OIL 1	110612	18.000	18.000	0.98	20000	910-1090	70 - 80
166	4211	SERVOCOAT 170T	Open Gear Oils	Mineral Oil base		BASE OIL 1	110612	24.000	24.000	0.98	16000	715-755	70 - 80
167	4215	SERVO TAK 40	Open Gear Oils	Mineral Oil base		BASE OIL 1	191502	20.000	17.200	0.86	1410-1590	62	70 - 80
168	4302	SERVORUBBER 24	Rubber extender	Mineral Oil base		BASE OIL 1	110612	29.100	26.800	0.88	20-24	4	55 - 65
169	4303	SERVORUBBESS 12 -	Rubber extender	Mineral Oil base		BASE OIL 1	290352	20.070	20.000	0.87	12	3.5-7.5	50 - 60
170	4305	SERVORUBBESS 32	Rubber extender	Mineral Oil base		BASE OIL	400422	100.000	100.000	0.86	27-34	5.3	ambient
171	4305	SERVORUBBESS 32-	Rubber extender	Mineral Oil base		BASE OIL	400422	100.000	100.000	0.86	27-34	5.3	ambient
172	4309	SERVORUBBESS 150	Rubber extender	Mineral Oil base		BASE OIL	190862	100.000	100.000	0.88	140-165	16	ambient
173	4318	SERVORUBBER EF-4 -	Rubber extender	Mineral Oil base		BASE OIL 1	111502	93.000	93.500	0.9	3600	55-80	55 - 65
174	4319	SERVORUBBER EF-5	Rubber extender	Mineral Oil base		BASE OIL	811402	100.000	100.000	0.91	630-770	26	55 - 65
175	4320	SERVORUBBER EF-4(M)-	Rubber extender	Mineral Oil base		BASE OIL 1	111502	59.300	63.000	0.9	650	24-32	55 - 65
176	4326	IOC PROCESS OIL MEDIUM -	Rubber extender	Mineral Oil base		BASE OIL	110612	100.000	100.000	0.99	600	20.5-25.5	55 - 65
177	4328	IOC PROCESS OIL EXTRA HEAVY	Rubber extender	Mineral Oil base		BASE OIL	110862	100.000	100.000	0.99	750	33.5-39.5	55 - 65
178	4342	SERVOLOOM MCWM-	Gen Pur Mach Oils	Mineral Oil base			110612	100.000	100.000	0.99	600	20.5 - 25.5	55 - 65
179	4401	SERVOQUENCH 11	Quenching Oil	Mineral Oil base		BASE OIL	190412	100.000	100.000	0.89	27-33	5	ambient
180	4402	SERVOQUENCH C 11	Quenching Oil	Mineral Oil base		BASE OIL 1	190412	95.900	96.000	0.89	27-35	5	50 - 60
181	4408	SERVOQUENCH 107	Quenching Oil	Mineral Oil base		BASE OIL 1	190412	50.700	51.000	0.89	20-24	4.3	50 - 60
182	4409	SERVOQUENCH 707	Quenching Oil	Mineral Oil base		BASE OIL 1	290352	43.954	43.100	0.89	20-24	4.3	50 - 60
183	4422	SERVOMARTEMP 170	Quenching Oil	Mineral Oil base		BASE OIL 1	190862	99.350	99.390	0.88	160.-180.0	15.3	50 - 60
184	4501	SERVOCUT S	Sol Cutiing Oils	Mineral Oil base		BASE OIL 1	290352	80.340	78.000	0.89	20-40		ambient
185	4524	SERVOCUT 945	Neat cutting Oil	Mineral Oil base		BASE OIL 1	190412	98.800	98.650	0.89	28-34	5	55 -65
186	4530	SERVOCUT ANC	Neat cutting Oil	Mineral Oil base		BASE OIL 1	290352	93.440	93.330	0.85	11.0-14	3	50 - 60
187	4532	SERVOCUT GR -	Grinding Oil	Mineral Oil base		BASE OIL 1	585962	99.933	99.930	0.82	2-2.5		40 - 45
188	4538	SERVOCUT 253	Neat cutting Oil	Mineral Oil base		BASE OIL 1	290352	79.525	79.284	0.89	9.5-12.5	2.5	55 -65

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component		Formula		Physical Properties			Operating/Blending Temp
						Name	code	% vol	% wt	SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	
189	4539	SERVOCUT 353	Neat cutting Oil	Mineral Oil base		BASE OIL 1	400352	68.610	66.200	0.88	11.5 - 14.5	3.5	55 -65
190	4551	SERVOHONE 7	Honing Oils	Mineral Oil base		BASE OIL 1	290352	-	-	0.88	4.0-6		45 - 55
191	4571	SERVOGRIND 12	Grinding Oil	Mineral Oil base		BASE OIL 1	400352	98.240	98.100	0.89	11.0-15	3	45 - 55
192	4713	SERVO GUARD 30	Engine Oil	Mineral Oil base		BASE OIL 1	190522	92.000	91.200	0.9	86	10.0-12	60 - 70
193	4715	SERVO GUARD 50	Engine Oil	Mineral Oil base		BASE OIL 1	190862	80.000	80.000	0.9	241	19.5-20.5	60 - 70
194	4731	SERVO SHOCAB	Shock Absorber Oil	Mineral Oil base		BASE OIL 1	400352	72.000	72.000	0.88	17.5-18.5	4	50 - 60
195	4732	SERVO TELESHOCAB -	Shock Absorber Oil	Mineral Oil base		BASE OIL 1	400422	45.960	45.880	0.87	35-40	6	50 - 60
196	4750	SERVOCONMOULD 22	Moulding Oil	Mineral Oil base		BASE OIL 1	110612	20.000	20.000	0.91	21-23	3	50 - 60
197	4801	SERVOTHERM LIGHT	Heat Transfer Oil	Mineral Oil base		BASE OIL 1	290352	100.000	100.000	0.85	11.0-14	3.2	Ambient
198	4802	SERVOTHERM MEDIUM	Heat Transfer Oil	Mineral Oil base		BASE OIL 1	190412	100.000	100.000	0.88	28.5-33.5	5.2	Ambient
199	4803	SERVOTHERM SPECIAL	Heat Transfer Oil	Mineral Oil base		BASE OIL 1	420422	99.630	99.600	0.88	28.5-33.5	5.4	45 - 55
200	4902	SERVO TORQUE 10	Hydraulic Oil	Mineral Oil base		BASE OIL 1	190412	20.000	20.000	0.89	33	4.5-5.7	50 - 60
201	4906	SERVO TRANSFLUID A	Transmission Oil	Mineral Oil base		BASE OIL 1	400422	79.005	78.594	0.85	41	7.4-7.8	50 - 60
202	4913	BEML TRANSMSN OIL ET30CD	Transmission Oil	Mineral Oil base		BASE OIL 1	400622	96.020	95.110	0.88	98	9.0-13	60 - 70
203	4914	BEML ALS TRANMISN OIL	Transmission Oil	Mineral Oil base		BASE OIL 1	400622	94.900	94.200	0.88	98	9.3-12.5	60 - 70
204	4920	TAFE TRANSHYDRO OIL	Transmission Oil	Mineral Oil base		BASE OIL 1	400622	95.030	93.980	0.88	98	10.0-12	60 - 70
205	4922	SERVO TRANSMISSION XL 439	Transmission Oil	Mineral Oil base		BASE OIL 1	400622	96.120	95.300	0.89	98	9.3-12.5	60 - 70
206	4924	SERVO TRANSMISSION C4 SAE 10W	Transmission Oil	Mineral Oil base		BASE OIL 1	400422	96.400	95.500	0.88	34	5.6-12.5	60 - 70
207	4925	SERVO TRANSMISSION C4 SAE 30	Transmission Oil	Mineral Oil base		BASE OIL 1	400622	96.120	95.300	0.89	98	9.3-12.5	60 - 70
208	4938	SERVO TRANSMISSION TO4 SAE 50	Transmission Oil	Mineral Oil base		BASE OIL 1	190412	29.550	29.000	0.91	188	16.3-21.9	60 - 70
209	4941	SERVO TRANSDEX II	Transmission Oil	Mineral Oil base		BASE OIL 1	400422	94.684	94.194	0.88	42	7.0-8	55 - 65
210	4942	SERVO TRANSDEX III	Transmission Oil	Mineral Oil base		BASE OIL 1	400352	25.883	25.630	0.89	42	7.0-7.5	55 - 65
211	4947	SERVO STEER TM	Transmission Oil	Mineral Oil base		BASE OIL 1	400422	45.140	45.490	0.88	42	7.0-8	55 - 65
212	4950	SERVO OIB OIL	Tractor OIB Oil	Mineral Oil base		BASE OIL 1	191502	33.960	35.000	0.89	93	10.3-11	55 - 65
213	4955	SERVO VST SHAKTI OIB OIL-	Tractor OIB Oil	Mineral Oil base		BASE OIL 1	400622	96.110	95.470	0.87	94	10.0-11	55 - 65
214	5002	SERVOSTEEL 680	Bearing Oil	Mineral Oil base		BASE OIL 1	190612	6.000	5.985	0.9	655-695	38.5	55 - 65
215	5004	SERVOSTEEL 460	Bearing Oil	Mineral Oil base		BASE OIL 1	190862	2.000	2.000	0.89	445-475	30	55 - 65
216	5005	SERVOSTEEL EP 150 PLUS	Bearing Oil	Mineral Oil base		BASE OIL 1	400622	53.500	53.440	0.9	147-153	16	50 - 60
217	5010	SERVOSTEEL EP 220 PLUS	Bearing Oil	Mineral Oil base		BASE OIL 1	400622	24.500	24.400	0.9	212-228	19	50 - 60
218	5024	SERVOSTEEROL 22	Rolling Oil	Mineral Oil base		BASE OIL 1	190412	19.456	20.000	0.89	18-22	4.3	50 - 60
219	5032	SERVO NG 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	190412	78.344	78.184	0.88	102	14.0-15.5	60 - 70
220	5033	SERVO NG 40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	42.860	42.000	0.88	138	13.0-15	60 - 70
221	5034	SERVO GAS ENGINE OIL	Engine Oil	Mineral Oil base		BASE OIL 1	400622	68.920	68.360	0.89	181	17.5-19.5	60 - 70
222	5803	SERVO MARINE 0530	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	88.320	87.880	0.89	83	10 - 12	55 - 65
223	5823	SERVO MARINE 1030	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	81.000	80.000	0.89	83	11.0-12	55 - 65
224	5824	SERVO MARINE 1040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	27.000	27.000	0.9	129	13.0-14	55 - 65
225	5825	SERVO MARINE 3040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	53.760	52.000	0.9	128	13.0-14.5	55 - 65
226	5826	SERVO MARINE 4040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	62.310	60.000	0.9	127	13.0-14.5	55 - 65
227	5828	SERVO MARINE 3030	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	58.500	57.180	0.89	83	10.0-11	55 - 65
228	5829	SERVO MARINE 2030	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	71.510	70.000	0.89	83	10.0-11	55 - 65
229	5830	SERVO MARINE 2040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	26.600	26.000	0.89	128	13.0-14.5	55 - 65
230	5831	SERVO MARINE 7050-	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190862	68.480	65.520	0.93	214	19.0-21.5	55 - 65
231	5833	SERVO MARINE K 4040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	76.300	73.400	0.89	130	13.0-14.5	55 - 65
232	5836	SERVO MARINE K 2040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	30.730	30.000	0.89	133	13.5-14	55 - 65
233	5837	SERVO MARINE K 1040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	13.150	12.800	0.88	134	13.5-14	55 - 65
234	5839	SERVO MARINE K 4030	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	65.600	63.400	0.89	110	11.3-12.4	55 - 65
235	5840	SERVO MARINE K 5040	Marine Engine Oil	Mineral Oil base		BASE OIL 1	190612	75.000	71.900	0.91	120	13.0-13.5	

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component		Formula		Physical Properties			Operating/Blending Temp
						Name	code	% vol	% wt	SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	
236	5901	SERVO MARINE T 32	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	97.000	97.000	0.86	29.3-32.7	5.3	50 - 60
237	5902	SERVO MARINE T 68	Turbine Oil	Mineral Oil base		BASE OIL 1	400422	8.000	8.000	0.87	64.7-71.3	9	50 - 60
238	5922	SERVO MARINE HYD 32	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	96.650	96.490	0.86	29.3-32.7	5.3	50 - 60
239	5923	SERVO MARINE HYD 46	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	50.000	50.000	0.87	43.5-47.5	7	50 - 60
240	5924	SERVO MARINE HYD 68	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	23.000	23.000	0.87	64.7-71.3	9	50 - 60
241	5925	SERVO MARINE HYD 100	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400622	96.650	96.490	0.88	96-104	11	50 - 60
242	6001	SERVO MARINE GEAR EP 150	Gear oil	Mineral Oil base		BASE OIL 1	190612	68.000	68.000	0.88	147-153	15	50 - 60
243	6004	SERVO MARINE GEAR EP 220	Gear oil	Mineral Oil base		BASE OIL 1	190612	41.000	41.000	0.89	212-228	19.2	50 - 60
244	6005	SERVO MARINE GEAR EP 320	Gear oil	Mineral Oil base		BASE OIL 1	190612	18.000	18.000	0.89	320-345	24.4	50 - 60
245	6114	SERVO MARINE STERN 220	Stern Tube	Mineral Oil base		BASE OIL 1	191502	55.000	55.000	0.9	210-230	19	50 - 60
246	6318	SERVODEDUSTING OIL-	Dedusting	Mineral Oil base		BASE OIL 1	110612	42.240	45.000	0.9	685-725	29	50 - 60
247	6325	IOC ROEL 280	Axle Oil	Mineral Oil base		BASE OIL 1	110612	91.800	91.800	0.9	280	17	50 - 60
248	6326	IOC ROEL 320	Axle Oil	Mineral Oil base		BASE OIL 1	191502	20.000	20.000	0.9	320	19	50 - 60
249	6352	LUBREX 22	Gen Pur Oil	Mineral Oil base		BASE OIL 1	400352	80.000	80.000	0.87	19.2-21.8	4	50 - 60
250	6360	LUBREX 320	Gen Pur Oil	Mineral Oil base		BASE OIL 1	400622	37.000	37.000	0.89	305-345	22	50 - 60
251	6372	IOC N 32/ Film 32	Base Oil	Mineral Oil base		BASE OIL 1	190412	100.000	100.000	0.88	27-34	5	Ambient
252	6373	IOC N 68	Base Oil	Mineral Oil base		BASE OIL 1	190522	100.000	100.000	0.89	68	8.0-10	Ambient
253	6374	IOC N 100 / Film 100	Base Oil	Mineral Oil base		BASE OIL 1	190612	100.000	100.000	0.89	94	10.0-12	Ambient
254	6375	IOC N 150	Base Oil	Mineral Oil base		BASE OIL 1	190862	100.000	100.000	0.89	165	13.5-18	Ambient
255	6376	IOC N 460	Base Oil	Mineral Oil base		BASE OIL 1	191502	100.000	100.000	0.91	460	28-33	Ambient
256	6381	IOC H 12	Base Oil	Mineral Oil base		BASE OIL 1	400352	100.000	100.000	0.84	9.5-15.0	3	Ambient
257	6382	IOC H 32 / Film 32 Plus	Base Oil	Mineral Oil base		BASE OIL 1	400422	100.000	100.000	0.85	28.0-34	5.4	Ambient
258	6384	IOC H 100 / Film 100 Plus	Base Oil	Mineral Oil base		BASE OIL 1	400622	100.000	100.000	0.86	78	10	Ambient
259	6726	OIL OM 33	Hydraulic oil	Mineral Oil base		BASE OIL 1	400422	88.110	87.940	0.88	32	5.4	50 - 60
260	6730	OIL OM 65	Gen Pur Mach Oil	Mineral Oil base		BASE OIL 1	400422	76.200	76.200	0.87	60-65	8.4	50 - 60
261	6733	OIL OM 100	Turbine Oil	Mineral Oil base		BASE OIL 1	400622	84.630	84.580	0.88	72-81	9.8	50 - 60
262	6738	OIL OM 750	Gen Pur Mach Oil	Mineral Oil base		BASE OIL 1	191502	57.000	57.000	0.92	612-748	38	50 - 60
263	6745	OIL C 600	Gear Oil	Mineral Oil base		BASE OIL 1	191502	94.000	94.000	0.90	450	32	50 - 60
264	6750	OIL OC 300	Gear Oil	Mineral Oil base		BASE OIL 1	191502	82.220	82.000		260.0-300.0	20.0-24.0	50 - 60
265	6753	OIL OX 52	Cleaning Oil	Mineral Oil base		BASE OIL 1	290352	47.630	46.615	0.88	41.0-52.0	6	50 - 60
266	6754	OIL MS 20	Gen Pur Mach Oil	Mineral Oil base		BASE OIL 1	191502	65.200	65.300	0.89	210	20.0-22.0	50 - 60
267	6769	OIL OMD (N) 50	Engine Oil	Mineral Oil base		BASE OIL 1	190862	56.000	56.000	0.89	210	18.5	60 - 70
268	6780	OIL OMD (N) 1240	Engine Oil	Mineral Oil base		BASE OIL 1	190862	91.460	89.960	0.89	138	14	60 - 70
269	6783	OIL OMD 113 ISSUE 4	Engine Oil	Mineral Oil base		BASE OIL 1	190862	54.260	53.970	0.89	120	12.5-16.3	60 - 70
270	6786	SERVO OEP 220	Gear oil	Mineral Oil base		BASE OIL 1	190412	37.000	37.000	0.89	130	14-14.5	50 - 60
271	6791	SERVOSTERN NDH	Stern Tube Oil	Mineral Oil base		BASE OIL 1	191502	41.000	41.000	0.89	158	15 - 16	50 - 60
272	7403	SERVOELECTRA	Transfromer Oil	Mineral Oil base		BASE OIL 1	400352	90.020	90.000	0.88	7.0-15.0	3	80 - 90
273	7403	SERVOELECTRA	Transfromer Oil	Mineral Oil base		BASE OIL 1	400352	90.020	90.000	0.88			80 - 90
274	7412	SERVO ELECTRA PLUS	Transfromer Oil	Mineral Oil base		BASE OIL 1	400352	90.020	90.000	0.88	7.0-15.0	3	80 - 90
275	7439	TMGO DEO CH4	Engine Oil	Mineral Oil base		BASE OIL 1	400622	36.593	35.993	0.87	111	14.0-15	60 - 70
276	7440	TMGO DEO CI4 PLUS	Engine Oil	Mineral Oil base		BASE OIL 1	400622	28.420	28.450	0.88	108	14 - 15	60 - 70
277	7442	ECSTAR NEXA 0W-20 PETROL	Engine Oil	Mineral Oil base		BASE OIL 1	800402	82.110	80.900	0.85	47	7.8-8.38	60 - 70
278	7443	ECSTAR NEXA 5W-30 DIESEL	Engine Oil	Mineral Oil base		BASE OIL 1	800452	28.050	27.700	0.86	72	10.5-11.5	60 - 70
279	7444	TAFE MILES	Engine Oil	Mineral Oil base		BASE OIL 1	400622	95.080	93.980	0.87	88	10.5-12	60 - 70
280	7447	SERVO XEE SM 10W-30	Engine Oil	Mineral Oil base		BASE OIL 1	800402	20.350	20.000	0.87			60 - 70
281	7459	SERVO PRIDE TM PLUS 15W40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	28.408	28.388	0.88	108	14.0-15	60 - 70
282	7471	TGO PLUS	Engine Oil	Mineral Oil base		BASE OIL 1	190412	48.200	47.400	0.89	108	14.0-15	60 - 70

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component		Formula		Physical Properties			Operating/Blending Temp
						Name	code	% vol	% wt	SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	
283	7472	SERVO FLT CF4 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	190612	15.950	16.000	0.89	110	13-15	60 - 70
284	7473	SERVO FLT XTRA CH4 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	43.200	42.050	0.88	110	14.0-15	60 - 70
285	7474	SERVO FLEET SUPREME CI4	Engine Oil	Mineral Oil base		BASE OIL 1	400622	31.140	31.100	0.88	106	14.0-15	60 - 70
286	7475	MGO 0W-20	Engine Oil	Mineral Oil base		BASE OIL 1	800402	82.110	80.900	0.85	47	7.8-8.38	60 - 70
287	7477	HY.SERVO PRM.EO SM 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	420422	69.340	68.630	0.86	69	9.3-12.5	60 - 70
288	7481	HYUNDAI SERVO COOLANT LL	Coolant	Glycol/Water Base		Syn Fluid 1	655322		87.770	1.12			35 - 45
289	7485	SERVO SUPER 20W-40 MG	Engine Oil	Mineral Oil base		BASE OIL 1	190612	74.274	74.004	0.89	125	13.5-15.5	60 - 70
290	7487	MAXIMILE PLUS ENGINE OIL	Engine Oil	Mineral Oil base		BASE OIL 1	400622	28.700	28.600	0.87	106	14.0-15	60 - 70
291	7488	SERVO PRIDE TM 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	36.693	35.993	0.87	106	14.0-15	60 - 70
292	7489	SERVO PRIDE XL PLUS 15W40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	31.140	31.100	0.88	99	14.0-15.0	60 - 70
293	7494	SERVO SUP.MULT.GRD 20W-40	Engine Oil	Mineral Oil base		BASE OIL 1	190612	72.994	72.864	0.89	127	13.5-15.5	60 - 70
294	7505	SERVO AUTORIKSHW OIL 20W40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	73.414	72.804	0.89	127	13.5-15.5	60 - 70
295	7506	VST SHAKTI GEN ENGINE OIL	Engine Oil	Mineral Oil base		BASE OIL 1	190612	73.275	72.994	0.89	106	13.5-15.5	60 - 70
296	7511	SERVO GEAR SUPER 80W-90	Gear oil	Mineral Oil base		BASE OIL 1	400622	68.100	67.200	0.88	130	14-14.5	50 - 60
297	7513	SERVO SUPER MG 20W-40	Engine Oil	Mineral Oil base		BASE OIL 1	190612	72.994	72.864	0.89	127	13.5-15.5	60 - 70
298	7514	SERVO SUPER MG 20W50	Engine Oil	Mineral Oil base		Base Oil	400622	89.944	89.624	0.89	175	18	60 - 70
299	7515	SERVO GEAR SUP 85W-140	Gear Oil	Mineral Oil base		Base Oil 1	400622	80.400	79.490	0.87	421	28	60 - 70
300	7521	SERVOVACUUM VM 4	Vacuum Oil	Mineral Oil base		Base Oil	890652	100.000	1000.000	0.88		10	40-50
301	7527	SERVO SUPER 20W-40	Engine Oil	Mineral Oil base		Base Oil 1	190612	78.114	78.004	0.89	127	13.5-15.5	60 - 70
302	7546	TAFE GENUINE OIL-FE	Engine Oil	Mineral Oil base		Base Oil	400622	95.080	93.980	0.87	78	10	60 - 70
303	7552	SERVO PREMIUM TM 15W-40	Engine Oil	Mineral Oil base		Base Oil 1	190612	15.480	15.500	0.89	106	13.5-15.5	60 - 70
304	7554	SERVO TRCTR OIL PLS 15W40	Engine Oil	Mineral Oil base		Base Oil 1	400622	31.300	31.180	0.88	106	13.5-15.5	60 - 70
305	7556	SERVO PRIDE TC 15W-40	Engine Oil	Mineral Oil base		Base Oil 1	400622	68.070	67.800	0.88	106	13.5-15.5	60 - 70
306	7557	SERVOKOOL ST	Coolant	Glycol/Water base		Syn Fluid 1	655322		87.770	1.12			35 - 40
307	7561	SERVO PREMIUM 20W-40	Engine Oil	Mineral Oil base		BASE OIL 1	**190612**	74.000	73.900				
308	7563	SERVO TRACTOR OIL MG 20W-40	Engine Oil	Mineral Oil base		Base Oil	400622	90.930	89.680	0.88	125	13.5-15.5	60 - 70
309	7571	SERVO CALIB 5	Calib Oil	Mineral Oil base		Base Oil 1	585962	34.154	33.000	0.85	5		40 - 50
310	7575	SERVO PRIDE SUPER 15W-40	Engine Oil	Mineral Oil base		Base Oil 1	190412	52.000	51.000	0.89	125	13.5-15.5	60 - 70
311	7576	TAFE GENUINE OIL	Engine Oil	Mineral Oil base		Base Oil 1	400622	90.930	89.680	0.89	125	13.5-15.5	60 - 70
312	7577	SERVO PREMIUM CF 4 15W 40	Engine Oil	Mineral Oil base		Base Oil 1	190612	15.540	15.500	0.89	108	13.0-15.0	60 - 70
313	7584	SERVO KOOL PLUS	Coolant	Glycol/Water base		Synfluid 1	655322		87.990	1.11			35 - 40
314	7587	SERVO RUBBER SPRAY OIL	Spray oil	Mineral Oil base		Base Oil 1	400352	9.84	10.000	0.85	2.7		40 - 50
315	7600	MGO 20W-40	Engine Oil	Mineral Oil base		Base Oil 1	190612	73.900	73.800	0.89	138	13.5 - 15.5	60 - 70
316	7607	SERVO KOOL READY	Coolant	Glycol/Water base		Synfluid 1	655322		87.990	1.08			35 - 40
317	7612	SERVO PRIDE XL 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	43.2	42.05	0.89	109	14.0-15.0	60 - 70
318	7614	SERVO PREMIUM (N) 15W40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	57.700	57.400	0.88	109	14 - 15	60 - 70
319	7620	SERVO BOAT ENGINE OIL 15W-40	Engine Oil	Mineral Oil base		Base Oil 1	400422	46.480	45.300	0.88	109	14 - 15	60 - 70
320	7624	SERVO ANTICORR BF	Coolant	Glycol/Water base		synfluid 1	650262		80.250	1.05			35 - 40
321	7629	SERVO KOOL ALT	Coolant	Glycol/Water base		Syn Fluid 1	655322		87.770	1.11			35 - 40
322	7630	SERVO PRIDE ALT 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400422	36.4	36	0.88	110	14.0-15.0	60 - 70
323	7637	SERVO SUPERIOR 0W-20	Engine Oil	Mineral Oil base		BASE OIL 1	800402	86.6	85.06	0.85	47	7.8-8.3	60 - 70
324	7638	HYUNDAI SERVO CRDi PREM.ENG.OIL	Engine Oil	Mineral Oil base		BASE OIL 1	400622	36.4	36	0.88	110	14.0-15.0	60 - 70
325	7648	SERVO XEE SL 0W-20	Engine Oil	Mineral Oil base		BASE OIL 1	800402	82.11	80.9	0.85	47	7.8-8.3	60 - 70
326	7655	SERVO XEE SM 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	400422	52.22	51.95	0.86	73	10.0-12.0	60 - 70
327	7660	SERVO FUTURA D	Engine Oil	Mineral Oil base		BASE OIL 1	400622	33.3	33	0.87	108	14.0-15.0	60 - 70
328	7663	SERVO PRIDE SUPREME 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400422	43.2	42.05	0.88	108	14.0-15.0	60 - 70
329	7664	SERVO KOOL TM	Coolant	Glycol/Water base		Syn Fluid 1	655322		87.990	1.11			35 - 40

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component Name	Component code	Formula % vol	Formula % wt	Physical Properties SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	Operating/Blending Temp
330	7670	MGDO 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	36.40	36.00	0.88	108	14.0-15.0	60 - 70
331	7677	SERVO PRIDE ALT PLUS 15W-40	Engine Oil	Mineral Oil base		BASE OIL 1	400622	45.300	45.000	0.88	108	14.0-15.0	60 - 70
332	7678	SERVO KOOL 5050	Coolant	Glycol/Water base		Syn Fluid 1	655322		43.885	1.08			35 - 40
333	7684	SERVO KOOL HWC	Coolant	Glycol/Water base		Syn Fluid 1	655322		87.770	1.12			35 - 40
334	7686	SERVO KOOL ALT 50	Coolant	Glycol/Water base		Syn Fluid 2	650262		51.000	1.05			35 - 40
335	7687	SERVO PRIDE SU PLUS 15W40	Engine Oil	Mineral Oil base		BASE OIL 1	400422	47.000	46.500	0.88	110	14.0.-15.0	60 - 70
336	7688	MGO 5W-30	Engine Oil	Mineral Oil base		BASE OIL 1	420422	40.500	40.200				
337	8124	SERVO ACT AF 100	Hydraulic Oil	Mineral Oil base		BASE OIL 1	400422	2.000	2.000	0.87	96-104	11	50 - 60
338	7707	SERVO GREASE MOLEX	Grease	Mineral base Oils		Base Oil 1	490622		59.60				120 -210
339	7754	SERVO GREASE CVJ	Grease	Mineral base Oils		Base Oil 1	190612		46.00				
340	7734	SERVO GREASE REP 1											
341	7818	SERVO PLEX 3M	Grease	Mineral base Oils		Base Oil 1	190612		38.90				120 -225
342	7812	SERVOPLEX TM 3	Grease	Mineral base Oils		Base Oil 1	810902		15.00				120 -225
343	7837	SERVO GREASE SUPER 3	Grease	Mineral base Oils		Base Oil 1	190612		16.00				120 -225
344	7701	SERVO GREASE MP	Grease	Mineral base Oils		Base Oil 1	190612		48.30				120 -225
345	7700	SERVO GREASE ALT	Grease	Mineral base Oils		Base Oil 1	190612		65.00				120 -225
346	7705	SERVO GREASE MP 3	Grease	Mineral base Oils		Base Oil 1	190612		81.25				120 -225
347	7823	SERVO GREASE TM	Grease	Mineral base Oils		Base Oil 1	190612		80.95				120 -225
348	7727	SERVO GEM RR 3	Grease	Mineral base Oils		Base Oil 1	190612		49.70				120 -225
349	7849	SERVO LONG LIFE GREASE	Grease	Mineral base Oils		Base Oil 1	190612		48.56				120 -225
350	7822	SERVO GEM TM 3	Grease	Mineral base Oils		Base Oil 1	191502		78.60				120 -225
351	7699	SERVO GEM ALT	Grease	Mineral base Oils		Base Oil 1	190612		48.05				120 -225
352	7847	SERVO PLEX ALT 3	Grease	Mineral base Oils		Base Oil 1	810902		15.00				120 -225
353	7722	SERVOGEM 2	Grease	Mineral base Oils		Base Oil 1	190612		86.00				120 -225
354	7723	SERVOGEM 3	Grease	Mineral base Oils		Base Oil 1	190612		83.95				120 -225
355	7741	SERVOGEM EP 1	Grease	Mineral base Oils		Base Oil 1	190412		42.60				120 -225
356	7742	SERVOGEM EP 2	Grease	Mineral base Oils		Base Oil 1	190612		55.50				120 -225
357	7888	SERVOGEM EP 2X	Grease	Mineral base Oils		Base Oil 1	400622		54.00				120 -225
358	7743	SERVOGEM HT	Grease	Mineral base Oils		Base Oil 1	191502		87.80				120 -225
359	7744	SERVOGEM HTXX	Grease	Mineral base Oils		Base Oil 1	191502		86.60				120 -225
360	7745	SERVOGEM SUPER HT	Grease	Mineral base Oils		Base Oil 1	190612		18.75				120 -225
361	7764	SERVOGEM SUPER HTXX	Grease	Mineral base Oils		Base Oil 1	191502		38.75				120 -225
362	8128	SERVOGREASE WR2	Grease	Mineral base Oils		Base Oil 1	190612		21.50				120 -225
363	8152	SERVOGREASE WR2 (460)	Grease	Mineral base Oils		Base Oil 1	190612		14.50				120 -225
364	7801	SERVOPLEX LC 1	Grease	Mineral base Oils		Base Oil 1	190612		41.40				120 -225
365	7810	SERVOPLEX LC 2	Grease	Mineral base Oils		Base Oil 1	190612		40.20				120 -225
366	7842	SERVOPLEX PU 2	Grease	Mineral base Oils		Base Oil 1	400622		50.00				120 -225
367	7848	SERVOPLEX PU 1(S)	Grease	Mineral base Oils		Base Oil 1	191502		79.10				120 -225
368	8168	SERVO SYNTHPLEX 2	Grease	Mineral base Oils		Base Oil 1	190612		51.85				120 -225
369	7763	SERVOGREASE CG 10	Grease	Mineral base Oils		Base Oil 1	191502		71.50				120 -225
370	7768	SERVOGREASE CG 50	Grease	Mineral base Oils		Base Oil 1	191502		58.15				120 -225
371	7748	SERVOGREASE GRAPHITED 4080	Grease	Mineral base Oils		Base Oil 1	110612		30.20				120 -225
372	8155	SERVOPLEX SHC 120	Grease	PAO (synthetic base Oils)		Base Oil 1	555702		49.00				120 -225
373	8167	SERVOPLEX SHC 220	Grease	PAO (synthetic Base Oils)		Base Oil 1	555722		39.00				120 -225
374	8154	SERVOPLEX SHC 460	Grease	PAO (synthetic Base Oils)		Base Oil 1	555702		19.85				120 -225
375	7832	SERVOGREASE FG EP2	Food Grade Grease	PAO (synthetic Base Oils)		Base Oil 1	555762		79.45				120 -225
376	7359	SERVOWHITE POL	SPECIALITY OIL	Mineral Oil base		Base Oil	400352	100.00	100.00	0.83	12	2.5	70-90

PHYSICAL PROPERTIES FOR FINISHED PRODUCTS

Sr No.	Product Code	Product Description	Family	Type of Product	Dyed/Undyed	Component Name	Component code	Formula % vol	Formula % wt	Physical Properties SP.Gr @15°C	Viscosity (in cSt) @40°C	Viscosity (in cSt) @100°C	Operating/Blending Temp
377	2913	SERVO BRAKE FLUID DOT 3 PLUS				Base Oil 1	655382		34.175			2.0	
378	2914	SERVO BRAKE FLUID DOT 4				Base Oil 1	655382		23.000				
379	7330	SERVO AUS-32				Additive 1	150002		32.5	1.09	1.4		
380	4319	SERVO RUBBER EF-5				Additive 1	811402		100.00				
381	7760	SERVOGEM TXX					130902		3.9				
382	7758	SERVOGEM TX					130382		3.7				

PHYSICAL PROPERTIES FOR BASE OILS

S. NO	ST.NO	PRODUCT	API Cat	Imported/ Indegenous	KV@100OC, (cst)	KV@40OC, (cst)	Density@15.C	FP OC, (Min.)
1	110412	Light Neutral Extract	Extract	Indigenous	9.79	182.71	1.024	242
2	110522	Inter Neutral Extract	Extract	Indigenous	16	298	1.02	250
3	110612	500 Neutral Extract	Extract	Indigenous	21	600	1.01	220
4	110862	Heavy Neutral Extract	Extract	Indigenous	34	1300	1.02	230
5	111502	BN Extract	Extract	Indigenous	75	5800	1.0019	250
6	160352	70 N (LVI)	API G I	Indigenous	3.11	14.5	0.858	170
7	190412	150 N	API G I	Indigenous	5.3	30	0.877	210
8	190522	400 N	API G I	Indigenous	9	68	0.8882	228
9	190612	500 N	API G I	Indigeneous	11	97	0.8901	232
10	190862	850 N	API G I	Indigeneous	16	162	0.8994	240
11	191502	BS 150	API G I	Indigeneous	31	467	0.9055	290
12	200352	70 N	API G II	Indigeneous	3.11	11.5	0.8411	180
13	200442	150 N	API G II	Indigeneous	5.6	31	0.8487	205
14	200622	500 N	API G II	Indigeneous	11	87	0.8611	232
15	290352	70 N	API G I	Indigeneous	3.1	11.5	0.8517	160
16	290432	150 N	API G I	Indigeneous	5.2	29	0.8775	210
17	290632	500 N	API G I	Indigeneous	11	89	0.8919	232
18	290962	1300 N	API G I	Indigeneous	19	238	0.898	240
19	291502	BS 150	API G I	Indigeneous	31	485	0.906	290
20	300352	65 N	API G II	Indigeneous	3.13	11	0.841	180
21	300422	150 N	API G II	Indigeneous	5.6	31	0.8487	210
22	300622	500 N	API G II	Indigeneous	11.25	86.9	0.8611	220
23	400352	70 N	API G II	Indigeneous	3.16	11.5	0.8393	180
24	400392	90 N	API G II	Indigeneous	4.05	20	0.84	210
25	400422	150 N	API G II	Indigeneous	5.4	31.5	0.8551	200
26	400622	500 N	API G II	Indigeneous	11.5	70.4	0.8722	232
27	420422	6 cSt	API G III	Indigeneous	6.3	38	0.8476	220
28	555522	PAO 4 CST	API G IV	Import	4	17	0.82	204
29	555542	LAB	Base fluid	Import		3.8 min	0.86	120
30	555682	PAO 2 CST	API G IV	Import	1.95	6	0.805	145
31	555702	PAO 6 CST	API G IV	Import	5.85	31	0.83	225
32	555722	PAO 8 CST	API G IV	Import	7.95	47.5	0.832	232
33	555762	PAO 10 CST	Base fluid	Import	10	68	0.836	250

PHYSICAL PROPERTIES FOR BASE OILS

S. NO	ST.NO	PRODUCT	API Cat	Imported/ Indegenous	KV@100OC, (cst)	KV@40OC, (cst)	Density@15.C	FP OC, (Min.)
34	555782	PAO 40 CST	API G IV	Import	55	628	0.85	240
35	555802	PAO 100 CST	API G IV	Import	133	1320	0.85	230
36	555822	PAO 1000 cSt	API G IV	Import	1000	10000	0.855	265
37	555862	PAO 150 cSt	API G IV	Import	155	1825	0.849	280
38	585662	HAB	Base fluid	Indigeneous		30	0.88	175
39	585682	JBO (C)	Base fluid	Indigeneous		15	0.85	100
40	585902	HF HSD	Base fluid	Indigeneous		3.5	0.84	66
41	585962	PFRCB	Base fluid	Indigeneous		2.3	0.81	96
42	585982	HAB (light)	Base fluid	Indigeneous		15.5	0.875	175
43	586002	60 N	API G II	Import	2.15	7.79	0.822	150
44	600432	150 N	API G II	Import	5.4	32	0.86	210
45	600622	500 N	API G II	Import	11	97	0.87	230
46	690402	130 SN	API G I	Import	4.6	25	0.875	200
47	690432	150 N	API G I	Import	5.2	31	0.88	210
48	690702	600 N	API G I	Import	12.1	112.5	0.889	246
49	691502	150 BS	API G I	Import	31.5	472	0.902	294
50	700402	4 cSt	API G III	Import	4.2	19.4	0.836	200
51	700542	8 cSt	API G III	Import	7.25	45	0.847	240
52	700662	600 N	API G II	Import	12	104	0.8664	200
53	800392	4 cSt (NESTE)	API G III	Import	4.3	20	0.837	220
54	800402	4 cSt Base Oil	API G III	Import	4.3	20	0.835	200
55	800412	4 Cst. (Yubase 4)	API G III	Import	4.3	20.2	0.8368	200
56	800422	White Mineral Base Oil - 5 cSt	API G III	Import	5	29.5	0.853	205
57	800432	6 cSt (NESTE)	API G III	Import	6	32	0.843	220
58	800452	6 cSt	API G III	Import	6.1	32	0.842	220
59	800472	6 cSt (Yubase 6)	API G III	Import	6.1	32	0.842	210
60	800482	White Mineral Oil - 6 cSt	API G III	Import	6	36	0.842	210
61	800512	8 cSt	API G III	Import	7.75	46	0.847	220
62	800522	8 cSt (NESTE)	API G II	Import	8	46	0.847	220
63	800532	Yubase 8	API G III	Import	8	46	0.847	240
64	890322	60 N (NESTE)	API G II	Import	2.2	7.1	0.818	155
65	890352	70 N	API G I	Import	2.85	11.25	0.86	160
66	890432	150 N	API G I	Import	5.2	30	0.87	200

PHYSICAL PROPERTIES FOR BASE OILS

S. NO	ST.NO	PRODUCT	API Cat	Imported/ Indegenous	KV@100OC, (cst)	KV@40OC, (cst)	Density@15.C	FP OC, (Min.)
67	890642	500 N	API G I	Import	11	98.1	0.89	232
68	891502	150 BS	API G I	Import	31	472	0.9	288
69	895402	110 N	API G II	Import	4.5	22.25	0.853	220
70	895482	230 N	API G II	Import	6.75	45	0.8617	220
71	895642	500 N	API G II	Import	11	93	0.865	232
72	895652	500 N GPII	API G II	Import	11.5	94	0.866	230
73	898432	150 N GPII	API G II	Import	5.3	32	0.854	210
74	982002	BS 220 (Power Oil Sapphire H 5	API GP I	Indigeneous	42	934	0.91	280

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
1	110502	Liquid	1.0675	35.0	368	145	Local	1 Month	Drums/Bulk
2	110522	Liquid	0.9800	150.0	1575	150	Local	1 Month	Drums/Bulk
3	112042	Liquid	1.1050	8.0	84	145	Obsolete		
4	112062	Liquid	0.9540	115.0	1208	150	Local	1 Month	Drums/Bulk
5	112082	Liquid	0.9470	53.0	557	150	Local	1 Month	Drums/Bulk
6	112282	Liquid	1.0040	15.5	163	172	Obsolete		
7	112302	Liquid	1.0400	28.0	294	120	Not in approved list		
8	112312	Liquid	1.0550	24.0	252	115	Local	1 Month	Drums/Bulk
9	112322	Liquid	0.9950	90.0	945	150	Not in approved list		
10	112332	Liquid	1.0200	23.0	242	120	Local	1 Month	Drums/Bulk
11	112362	Liquid	1.0456	20.0	210	170	Local	1 Month	Drums/Bulk
12	112422	Liquid	0.9330	27.0	170	115	Imported	3 Months	Drums/Bulk
13	112432	Liquid	0.9360	124.0	1036	135	Imported	3 Months	Drums/Bulk
14	112442	Liquid	0.9320	178.0	1486	131	Imported	3 Months	Drums/Bulk
15	112452	Liquid	0.9050	180.0	1503	105	Imported	3 Months	Drums/Bulk
16	112482	Liquid	1.0600	75.0	788	160	Obsolete		
17	112582	Liquid	1.0350	95.0	998	140	Obsolete		
18	112802	Liquid	1.0300	95.0	998	140	Not in approved list		
19	112822	Liquid	1.0800	90.0	945	145	Not in approved list		
20	112882	Liquid	1.0130	122.0	1281	182	Obsolete		
21	112922	Liquid	1.0070	185.0	1943	130	Not in approved list		
22	112962	Liquid	1.1200	30.0	315	150	Imported	3 Months	Drums/Bulk
23	112982	Liquid	0.9800	200.0	2100	150	Not in approved list		
24	114022	Liquid	1.0340	200.0	2100	150	Not in approved list		
25	114042	Liquid	0.9500	150.0	1575	100	Not in approved list		
26	114082	Liquid	0.9410	60.0	650	145	Not in approved list		
27	114122	Liquid	1.0010	340.0	3500	150	Obsolete		
28	114142	Liquid	0.9800	260.0	3200	150	Local	1 Month	Drums/Bulk
29	114172	Liquid	1.1300	70.0	735	180	Imported	3 Months	Drums/Bulk
30	114182	Liquid	1.1200	65.0	800	145	Local	1 Month	Drums/Bulk
31	114192	Liquid	1.1800	65.0	683	160	Local	1 Month	Drums/Bulk
32	114202	Liquid	1.0410	180.0	2310	145	Obsolete		
33	114242	Liquid	1.0160	150.0	1575	100	Not in approved list		
34	114262	Liquid	1.1140	68.0	714	160	Local	1 Month	Drums/Bulk
35	114312	Liquid	1.0040	68.0	714	158	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
36	114362	Liquid	1.0000	20.0	210	170	Local	1 Month	Drums/Bulk
37	114382	Liquid	1.0110	205.0	2153	145	Not in approved list		
38	114402	Liquid	0.9550	28.0	294	140	Not in approved list		
39	114462	Liquid	0.9710	55.0	578	115	Local	1 Month	Drums/Bulk
40	114482	Liquid	0.9563	160.0	1680	180	Local	1 Month	Drums/Bulk
41	114502	Liquid	0.9800	75.0	788	150	Imported	3 Months	Drums/Bulk
42	114522	Liquid	0.9500	83.0	872	154	Local	1 Month	Drums/Bulk
43	114562	Liquid	0.9320	160.0	1680	145	Not in approved list		
44	114582	Liquid	0.9730	280.0	2940	140	Obsolete		
45	114602	Liquid	1.0150	125.0	1313	135	Not in approved list		
46	114612	Liquid	0.9700	70.0	735	155	Not in approved list		
47	114622	Liquid	0.9800	70.0	735	135	Local	1 Month	Drums/Bulk
48	114642	Liquid	0.9650	72.0	756	150	Not in approved list		
49	114682	Liquid	1.0510	45.0	473	145	Local	1 Month	Drums/Bulk
50	114692	Liquid	0.9000	170.0	1785	120	Obsolete		
51	114702	Liquid	1.1060	60.0	630	150	Not in approved list		
52	114712	Liquid	0.9000	210.0	2205	120	Obsolete		
53	114722	Liquid	0.9800	225.0	2363	145	Obsolete		
54	114732	Liquid	0.9000	700.0	7350	120	Obsolete		
55	114742	Liquid	0.9300	205.0	2153	135	Imported	3 Months	Drums/Bulk
56	114752	Liquid	0.9520	235.0	2468	180	Local	1 Month	Drums/Bulk
57	114772	Liquid	1.0150	28.0	294	214	Local	1 Month	Drums/Bulk
58	114782	Liquid	1.0950	200.0	2100	165	Local	1 Month	Drums/Bulk
59	114802	Liquid	1.0150	95.0	998	140	Obsolete		
60	114822	Liquid	1.2000	120.0	1260	170	Local	1 Month	Drums/Bulk
61	114832	Liquid	1.2000	80.0	840	180	Imported	3 Months	Drums/Bulk
62	114842	Liquid	0.9870	280.0	2940	145	Obsolete		
63	114862	Liquid	1.0130	220.0	2310	140	Obsolete		
64	114882	Liquid	1.1500	112.0	1176	120	Not in approved list		
65	114902	Liquid	0.9720	120.0	1260	145	Local	1 Month	Drums/Bulk
66	114912	Liquid	0.9800	140.0	1470	144	Obsolete		
67	114922	Liquid	0.9390	165.0	1733	150	Local	1 Month	Drums/Bulk
68	114942	Liquid	1.1270	38.0	399	812	Local	1 Month	Drums/Bulk
69	114962	Liquid	0.9050	95.0	998	145	Local	1 Month	Drums/Bulk
70	115062	Liquid	0.9180	7.5	40	120	Obsolete		

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
71	115122	Liquid	0.9300	275.0	2888	145	Local	1 Month	Drums/Bulk
72	115142	Liquid	0.9200	350.0	3675	150	Imported	3 Months	Drums/Bulk
73	115162	Liquid	0.9200	360.0	3780	145	Local	1 Month	Drums/Bulk
74	115182	Liquid	0.9300	178.0	1869	180	Obsolete		
75	115192	Liquid	0.9651	79.0	830	180	Obsolete		
76	115202	Liquid	1.0520	90.0	945	135	Not in approved list		
77	115222	Liquid	1.0100	140.0	1470	135	Not in approved list		
78	115232	Liquid	1.0550	60.0	630	135	Obsolete		
79	115242	Liquid	0.9300	34.0	357	148	Obsolete		
80	115262	Liquid	1.0860	65.0	683	150	Local	1 Month	Drums/Bulk
81	115272	Liquid	0.9250	110.0	1155	125	Local	1 Month	Drums/Bulk
82	115282	Liquid	0.9190	145.0	1523	130	Not in approved list		
83	115292	Liquid	0.9250	159.0	1670	94	Local	1 Month	Drums/Bulk
84	115302	Liquid	0.9300	175.0	1838	70	Obsolete		
85	115312	Liquid	0.9200	220.0	2310	105	Obsolete		
86	115322	Liquid	1.0300	19.0	165	150	Not in approved list		
87	115332	Liquid	0.9300	100.0	1000	104	Obsolete		
88	115342	Liquid	1.0350	50.0	525	130	Local	1 Month	Drums/Bulk
89	115362	Liquid	0.9200	80.0	840	180	Not in approved list		
90	115382	Liquid	1.0120	200.0	2100	182	Obsolete		
91	115392	Liquid	0.9130	250.0	2625	120	Imported	3 Months	Drums/Bulk
92	115402	Liquid	1.0190	255.0	2678	145	Obsolete		
93	115412	Liquid	0.9850	200.0	2100	163	Local	1 Month	Drums/Bulk
94	115422	Liquid	0.9230	105.0	1103	135	Imported	3 Months	Drums/Bulk
95	115432	Liquid	0.9880	170.0	1785	165	Local	1 Month	Drums/Bulk
96	115442	Liquid	0.9504	62.0	651	154	Local	1 Month	Drums/Bulk
97	115452	Liquid	0.9890	200.0	2100	166	Local	1 Month	Drums/Bulk
98	115462	Liquid	1.0620	115.0	1208	140	Local	1 Month	Drums/Bulk
99	115472	Liquid	0.9700	60.0	630	152	Local	1 Month	Drums/Bulk
100	115482	Liquid	0.9940	125.0	1313	180	Obsolete		
101	115502	Liquid	1.0420	145.0	1523	182	Not in approved list		
102	115522	Liquid	1.0230	115.0	1208	130	Local	1 Month	Drums/Bulk
103	115542	Liquid	1.0510	80.0	840	135	Imported	3 Months	Drums/Bulk
104	115552	Liquid	1.0770	29.0	305	135	Imported	3 Months	Drums/Bulk
105	115562	Liquid	1.0610	120.0	1260	130	Obsolete		

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
106	115602	Liquid	1.0800	190.0	1995	150	Local	1 Month	Drums/Bulk
107	115622	Liquid	0.9920	135.0	1418	135	Obsolete		
108	115642	Liquid	0.9400	17.0	179	150	Not in approved list		
109	115662	Liquid	1.0130	210.0	2205	182	Local	1 Month	Drums/Bulk
110	115672	Liquid	1.0860	105.0	1103	180	Local	1 Month	Drums/Bulk
111	115682	Liquid	1.0950	320.0	3360	180	Local	1 Month	Drums/Bulk
112	115702	Liquid	0.9800	96.0	1008	200	Obsolete		
113	115722	Liquid	1.0440	135.0	1418	172	Obsolete		
114	115732	Liquid	1.0420	102.0	1071	172	Local	1 Month	Drums/Bulk
115	115742	Liquid	1.1050	205.0	2153	174	Local	1 Month	Drums/Bulk
116	115762	Liquid	0.9930	110.0	1155	135	Not in approved list		
117	115772	Liquid	0.9800	75.0	788	150	Obsolete		
118	115782	Liquid	0.9920	135.0	1418	135	Obsolete		
119	115792	Liquid	0.9980	137.0	1439	135	Obsolete		
120	115802	Liquid	0.9380	75.0	788	120	Obsolete		
121	115812	Liquid	0.9870	185.0	3500	151	Local	1 Month	Drums/Bulk
122	115822	Liquid	1.0490	228.0	2394	182	Local	1 Month	Drums/Bulk
123	115832	Liquid	1.0050	105.0	1103	180	Local	1 Month	Drums/Bulk
124	115842	Liquid	0.9780	175.0	1838	158	Obsolete		
125	115862	Liquid	1.1370	120.0	1260	160	Local	1 Month	Drums/Bulk
126	115872	Liquid	0.9700	115.0	1208	133	Local	1 Month	Drums/Bulk
127	115882	Liquid	1.0020	58.0	609	172	Obsolete		
128	115902	Liquid	0.9970	53.0	557	130	Not in approved list		
129	115912	Liquid	1.1800	90.0	850	142	Not in approved list		
130	115922	Liquid	0.9050	185.0	1943	105	Obsolete		
131	115932	Liquid	1.1770	90.0	850	142	Obsolete		
132	115942	Liquid	0.9379	662.0	6951	180	Local	1 Month	Drums/Bulk
133	115952	Liquid	0.9800	88.0	924	130	Obsolete		
134	115962	Liquid	0.9710	155.0	1628	165	Local	1 Month	Drums/Bulk
135	115972	Liquid	0.9710	160.0	2800	148	Local	1 Month	Drums/Bulk
136	115982	Liquid	0.9690	9.0	95	130	Not in approved list		
137	115992	Liquid	1.0021	165.0	1733	150	Local	1 Month	Drums/Bulk
138	116012	Liquid	0.9760	115.0	1208	118	Local	1 Month	Drums/Bulk
139	116022	Liquid	0.9790	105.0	1103	180	Local	1 Month	Drums/Bulk
140	116042	Liquid	0.9700	5.5	58	170	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
141	116052	Liquid	1.0160	140.0	1470	180	Local	1 Month	Drums/Bulk
142	116062	Liquid	0.9730	100.0	1650	100	Local	1 Month	Drums/Bulk
143	116072	Liquid	1.0400	45.0	473	154	Obsolete		
144	116082	Liquid	1.0010	30.0	315	150	Local	1 Month	Drums/Bulk
145	116092	Liquid	1.0650	82.0	1340	152	Local	1 Month	Drums/Bulk
146	116102	Liquid	0.9410	200.0	2100	160	Not in approved list		
147	116122	Liquid	0.9950	180.0	1890	150	Not in approved list		
148	116142	Liquid	0.9610	89.0	935	152	Imported	3 Months	Drums/Bulk
149	116182	Liquid	0.9420	109.0	1818	171	Local	1 Month	Drums/Bulk
150	116202	Liquid	0.9990	152.5	1601	165	Local	1 Month	Drums/Bulk
151	116212	Liquid	1.1200	22.0	230	130	Imported	3 Months	Drums/Bulk
152	116222	Liquid	1.0150	123.0	1313	180	Local	1 Month	Drums/Bulk
153	116232	Liquid	0.9943	170.0	2100	180	Local	1 Month	Drums/Bulk
154	116242	Liquid	0.9580	99.0	1050	145	Local	1 Month	Drums/Bulk
155	116282	Liquid	0.9770	116.0	1750	149	Obsolete		
156	116302	Liquid	0.9720	83.0	872	135	Obsolete		
157	116322	Liquid	0.9700	150.0	2600	142	Local	1 Month	Drums/Bulk
158	116342	Liquid	1.0000	160.0	1680	133	Not in approved list		
159	116422	Liquid	0.9800	160.0	1680	171	Local	1 Month	Drums/Bulk
160	116432	Liquid	1.0005	160.0	1680	140	Local	1 Month	Drums/Bulk
161	116442	Liquid	0.9820	250.0	2625	170	Local	1 Month	Drums/Bulk
162	116482	Liquid	1.0630	75.0	788	135	Imported	3 Months	Drums/Bulk
163	116502	Liquid	0.9730	110.0	1155	150	Local	1 Month	Drums/Bulk
164	116522	Liquid	0.9590	130.0	1365	174	Local	1 Month	Drums/Bulk
165	116532	Liquid	1.0470	185.0	1943	154	Local	1 Month	Drums/Bulk
166	116542	Liquid	1.0310	95.0	998	135	Imported	3 Months	Drums/Bulk
167	116552	Liquid	0.9920	80.0	840	135	Imported	3 Months	Drums/Bulk
168	116572	Liquid	0.9992	127.0	1334	180	Local	1 Month	Drums/Bulk
169	116582	Liquid	0.9721	210.0	2205	182	Local	1 Month	Drums/Bulk
170	116592	Liquid	0.9590	141.0	1481	168	Local	1 Month	Drums/Bulk
171	116602	Liquid	0.9902	85.0	893	180	Local	1 Month	Drums/Bulk
172	116622	Liquid	0.9720	180.0	1890	156	Local	1 Month	Drums/Bulk
173	116742	Liquid	0.9780	190.0	1995	158	Local	1 Month	Drums/Bulk
174	116822	Liquid	0.9750	155.0	1628	152	Local	1 Month	Drums/Bulk
175	116882	Liquid	0.9850	140.0	1470	135	Imported	3 Months	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
176	116902	Liquid	0.9187	187.0	3000	135	Imported	3 Months	Drums/Bulk
177	116922	Liquid	0.9828	135.0	1418	150	Obsolete		
178	116942	Liquid	0.9930	81.0	851	152	Local	1 Month	Drums/Bulk
179	117002	Liquid	0.9830	170.0	1785	168	Local	1 Month	Drums/Bulk
180	117022	Liquid	0.9932	155.0	1628	150	Local	1 Month	Drums/Bulk
181	117082	Liquid	0.9676	164.0	1722	180	Local	1 Month	Drums/Bulk
182	117102	Liquid	0.9530	77.0	809	150	Local	1 Month	Drums/Bulk
183	117122	Liquid	0.9694	215.0	2258	180	Local	1 Month	Drums/Bulk
184	117142	Liquid	0.9480	108.0	1134	135	Imported	3 Months	Drums/Bulk
185	117162	Liquid	0.9540	139.0	1460	135	Not in approved list		
186	117182	Liquid	0.9500	200.0	2100	174	Not in approved list		
187	120022	Liquid	1.0700	56.0	588	160	Local	1 Month	Drums/Bulk
188	120042	Liquid	1.0300	9.5	100	150	Local	1 Month	Drums/Bulk
189	120062	Liquid	1.0600	6.0	70	50	Local	1 Month	Drums/Bulk
190	120072	Liquid	1.1400	11.0	95	80	Imported	3 Months	Drums/Bulk
191	120102	Liquid	1.1300	25.0	215	160	Local	1 Month	Drums/Bulk
192	120142	Liquid	1.1100	5.0	45	62	Local	1 Month	Drums/Bulk
193	120322	Liquid	1.1850	45.0	473	90	Local	1 Month	Drums/Bulk
194	120332	Liquid	1.5000	18.0	189	90	Local	1 Month	Drums/Bulk
195	120402	Liquid	1.0900	20.0	210	130	Local	1 Month	Drums/Bulk
196	120462	Liquid	0.9250	300.0	3150	120	Local	1 Month	Drums/Bulk
197	120502	Liquid	1.5000		0		Local	1 Month	Drums/Bulk
198	120612	Liquid	1.0240	21.3	224	120	Local	1 Month	Drums/Bulk
199	120622	Liquid	1.0550	7.5	70	55	Not in approved list		
200	120632	Liquid	0.9480	182.0	1911	130	Local	1 Month	Drums/Bulk
201	120642	Liquid	0.9780	13.0	80	150	Local	1 Month	Drums/Bulk
202	120652	Liquid	1.0350	10.0	93	100	Local	1 Month	Drums/Bulk
203	120672	Liquid	0.9100	6.0	53	141	Local	1 Month	Drums/Bulk
204	120682	Liquid	0.9570	15.0	180	192	Local	1 Month	Drums/Bulk
205	120702	Liquid	1.0040	11.0	100	100	Imported	3 Months	Drums/Bulk
206	120782	Liquid	1.1000	21.0	190	150	Not in approved list		
207	120822	Liquid	1.0150	6.0	52	70	Local	1 Month	Drums/Bulk
208	120842	Liquid	0.9600	35.0	298	90	Not in approved list		
209	120862	Liquid	0.9768	27.0	285	170	Not in approved list		
210	120882	Liquid	1.1200	1.0	5	70	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
211	120892	Liquid	1.1118	9.5	91	162	Local	1 Month	Drums/Bulk
212	120902	Liquid	0.9700	28.0	260	150	Imported	3 Months	Drums/Bulk
213	120922	Liquid	0.9700	19.0	170	150	Local	1 Month	Drums/Bulk
214	120932	Liquid	0.9200	3.0	12	110	Local	1 Month	Drums/Bulk
215	120942	Liquid	1.0820	9.0	88	76	Obsolete		
216	120962	Liquid	0.9150	4.0	35	135	Local	1 Month	Drums/Bulk
217	120982	Liquid	1.0000	6.2	57	100	Local	1 Month	Drums/Bulk
218	121002	Liquid	1.0750	8.0	88	100	Imported	3 Months	Drums/Bulk
219	121012	Liquid	1.0750	11.0	108	90	Local	1 Month	Drums/Bulk
220	121022	Liquid	1.0000	5.7	61	165	Obsolete		
221	121042	Powder	POWDER				Local	1 Month	Bags
222	121062	Liquid	0.9780	3.0	9	55	Not in approved list		
223	121082	Liquid	1.0250	5.5	53	80	Not in approved list		
224	121102	Liquid	0.9900	6.0	45	170	Not in approved list		
225	121162	Liquid	1.0940	25.0	260	130	Local	1 Month	Drums/Bulk
226	121182	Liquid	0.9950	7.0	45	90	Local	1 Month	Drums/Bulk
227	121202	Liquid	1.0270	15.0	160	82	Local	1 Month	Drums/Bulk
228	121222	Liquid	0.9650	2.2	9	55	Not in approved list		
229	121242	Liquid	0.9690	4.4	45	80	Local	1 Month	Drums/Bulk
230	121262	Liquid	1.0000	13.0	138	97	Imported	3 Months	Drums/Bulk
231	121282	Liquid	1.0410	19.0	180	65	Obsolete		
232	121302	Liquid	0.9000	14.0	125	105	Local	1 Month	Drums/Bulk
233	121312	Liquid	1.0100	1.7	16	76	Local	1 Month	Drums/Bulk
234	121322	Liquid	1.1000	8.4	75	76	Imported	3 Months	Drums/Bulk
235	121332	Liquid	1.0600	8.8	92	75	Obsolete		
236	121342	Liquid	0.9740	2.6	11	96	Imported	3 Months	Drums/Bulk
237	121352	Liquid	1.0050	12.5	131	80	Imported	3 Months	Drums/Bulk
238	121362	Liquid	1.0060	9.3	98	85	Imported	3 Months	Drums/Bulk
239	121382	Liquid	1.0900	210.0	16000	145	Imported	3 Months	Drums/Bulk
240	121402	Liquid	0.9020	198.0	2079	125	Imported	3 Months	Drums/Bulk
241	121422	Liquid	0.9400	1500.0	12750	120	Not in approved list		
242	121442	Liquid	1.0100	29.0	285	170	Obsolete		
243	121462	Liquid	0.9460	66.0	645	122	Local	1 Month	Drums/Bulk
244	121482	Liquid	1.0750	35.0	325	200	Local	1 Month	Drums/Bulk
245	121502	Liquid	0.9980	2.0	8	90	Imported	3 Months	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
246	121522	Liquid	0.8980	5.4	58	165	Local	1 Month	Drums/Bulk
247	121542	Liquid	0.9940	11.4	108	124	Local	1 Month	Drums/Bulk
248	121562	Liquid	1.0220	11.0	116	85	Imported	3 Months	Drums/Bulk
249	121582	Liquid	1.0010	2.5	21	76	Imported	3 Months	Drums/Bulk
250	121602	Liquid	1.0040	11.7	100	82	Local	1 Month	Drums/Bulk
251	121622	Liquid	1.0140	6.0	62	82	Imported	3 Months	Drums/Bulk
252	121642	Liquid	0.9700	8.0	60	76	Local	1 Month	Drums/Bulk
253	121702	Liquid	0.9320	8.0	94	128	Local	1 Month	Drums/Bulk
254	130012	Liquid	0.9260	6.0	45	220	Local	1 Month	Drums/Bulk
255	130022	Liquid	POWDER				Local	1 Month	Bags
256	130062	Liquid	0.9150	4.0	39	270	Local	1 Month	Drums/Bulk
257	130082	Liquid	0.9660	6.0	45		Local	1 Month	Drums/Bulk
258	130102	Liquid	0.9200	3.0	27	240	Local	1 Month	Drums/Bulk
259	130122	Liquid	0.9600	5.2	45	240	Local	1 Month	Drums/Bulk
260	130142	Liquid	0.9200	6.2	40	290	Local	1 Month	Drums/Bulk
261	130162	Liquid	0.9180	3.2	25	90	Local	1 Month	Drums/Bulk
262	130262	Liquid	0.9600	18.0	183	280	Local	1 Month	Drums/Bulk
263	130282	Semi solid	SEMI SOLID				Local	1 Month	Drums/Bulk
264	130292	Liquid	0.8980	7.0	60	280	Local	1 Month	Drums/Bulk
265	130302	Liquid	0.9860	7.5	60	280	Local	1 Month	Drums/Bulk
266	130382	Powder	FLLAKES				Local	1 Month	Bags
267	130682	Liquid	1.1600	10.5	98	120	Local	1 Month	Drums/Bulk
268	130742	Liquid	0.8600	1.5	7	170	Local	1 Month	Drums/Bulk
269	130782	Powder	POWDER				Local	1 Month	Bags
270	130802	Liquid	0.9000	45.0	460	250	Local	1 Month	Drums/Bulk
271	130842	Powder	POWDER				Local	1 Month	Bags
272	130862	Powder	POWDER				Local	1 Month	Bags
273	130882	Powder	POWDER				Local	1 Month	Bags
274	130922	Powder	POWDER				Local	1 Month	Bags
275	130982	Powder	FLAKES				Local	1 Month	Bags
276	131022	Liquid	0.9100	7.0	45	280	Local	1 Month	Drums/Bulk
277	131042	Semi solid	SEMI SOLID				Local	1 Month	Bags
278	131062	Liquid	0.9100	6.0	39	112	Imported	3 Months	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
279	131082	Liquid	0.9300	4.0	27		Imported	3 Months	Drums/Bulk
280	135122		JELLY			200	Local	1 Month	
281	135142	Powder	POWDER				Local	1 Month	Bags
282	13518C2	Solid	SOLID			220	Obsolete		
283	135202	Liquid	1.0400	108.0	1176		Local	1 Month	Drums/Bulk
284	135222	Solid	SOLID			250	Local	1 Month	Bags
285	135242	Solid					Local	1 Month	Bags
286	135422	Liquid	0.9620	4.8	30	150	Imported	3 Months	Drums/Bulk
287	135442	Liquid	0.9900	108.0	1155		Local	1 Month	Drums/Bulk
288	135462	Solid	SOLID			80	Not in approved list		
289	135522	Solid					Local	1 Month	Bags
290	135542	Solid					Local	1 Month	Bags
291	135562	Liquid	1.0200	6.0	45		Local	1 Month	Drums/Bulk
292	135582	Liquid	0.9400	7.0	48	280	Local	1 Month	Drums/Bulk
293	135642	Liquid	1.0300	6.4	38	258	Local	1 Month	Drums/Bulk
294	135662	Liquid	0.9150	168.0	1600	260	Local	1 Month	Drums/Bulk
295	135682	Solid	SOLID				Local	1 Month	Bags
296	135702	Liquid	0.9300	8.0	50	200	Local	1 Month	Drums/Bulk
297	135722	Liquid	0.9450	4.0	26	200	Local	1 Month	Drums/Bulk
298	135742	Liquid	0.9000	2.5	18	190	Local	1 Month	Drums/Bulk
299	135762	Liquid	0.9935	23.0	225	100	Obsolete		
300	135802	Powder					Local	1 Month	Bags
301	140252	Powder	POWDER			150	Local	1 Month	Bags
302	140262	Liquid	0.9800	28.0	260	185	Local	1 Month	Drums/Bulk
303	140272	Powder	POWDER			186	Local	1 Month	Bags
304	140282	Liquid	0.9150		2	80	Local	1 Month	Drums/Bulk
305	140292	Liquid	0.9000	40.0	500	120	Local	1 Month	Drums/Bulk
306	140302	Liquid	0.8600	3.0	13	160	Local	1 Month	Drums/Bulk
307	140312	Liquid	0.9920	1.8	8	140	Imported	3 Months	Drums/Bulk
308	140322	Liquid	0.9400	1.8	9	80	Local	1 Month	Drums/Bulk
309	140332	Liquid	0.8620	3.3	11	160	Local	1 Month	Drums/Bulk
310	140342	Liquid	0.7550		3	38	Local	1 Month	Drums/Bulk
311	140362	Liquid	0.9490	25.0	243	110	Local	1 Month	Drums/Bulk
312	140382	Liquid	0.9100		58	35	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
313	140402	Liquid	1.0000	10.0	102	180	Obsolete		
314	140432	Liquid	1.0100	32.0	3000	108	Local	1 Month	Drums/Bulk
315	140442	Liquid	1.0500	8.0	80	140	Local	1 Month	Drums/Bulk
316	140452	Liquid		12.0	100	125	Local	1 Month	Drums/Bulk
317	140462	Liquid	1.0000	13.0	120	200	Not in approved list		
318	140472	Powder				280	Local	1 Month	Bags
319	140482	Liquid	0.9700	13.0	123		Local	1 Month	Drums/Bulk
320	140502	Powder	POWDER				Local	1 Month	Bags
321	140512	Liquid	0.9890	4.0	28	95	Imported	3 Months	Drums/Bulk
322	140532	Liquid	1.0000	6.5	75	130	Local	1 Month	Drums/Bulk
323	140542	Liquid	0.9400	11.0		180	Not in approved list		
324	140562	Liquid	0.9400	23.0	230	150	Not in approved list		
325	140572	Liquid	1.0600	15.0	162	150	Local	1 Month	Drums/Bulk
326	140582	Liquid	1.0000	5.6	44	120	Local	1 Month	Drums/Bulk
327	140622	Liquid	0.9750	31.0	325	185	Local	1 Month	Drums/Bulk
328	140642	Liquid	0.9500	1.0	6	92	Local	1 Month	Drums/Bulk
329	140662	Liquid	1.0600	18.0	189	170	Local	1 Month	Drums/Bulk
330	140702	Liquid	0.9400	3.5	36	120	Local	1 Month	Drums/Bulk
331	140882	Liquid	1.0100	1.0	6	210	Local	1 Month	Drums/Bulk
332	140902	Powder	POWDER				Not in approved list		
333	140962	Powder	FLAKES				Local	1 Month	Bags
334	140982	Powder	POWDER				Local	1 Month	Bags
335	141002	Liquid	0.9700		2	72	Local	1 Month	Drums/Bulk
336	141022	Liquid	0.9700	13.0	120	140	Local	1 Month	Drums/Bulk
337	145102	Liquid	0.9906	14.0	200	170	Local	1 Month	Drums/Bulk
338	145222	Liquid	0.9500	34.0	930	90	Local	1 Month	Drums/Bulk
339	145262	Powder					Local	1 Month	Bags
340	145272	Liquid	1.0600	70.0	700	135	Local	1 Month	Drums/Bulk
341	145282	Liquid	1.1800	16.5	170	120	Imported	3 Months	Drums/Bulk
342	145292	Liquid	1.0300	11.0	97	125	Imported	3 Months	Drums/Bulk
343	145312	Liquid	1.0000	5.0	45	85	Imported	3 Months	Drums/Bulk
344	145332	Liquid	1.0180	25.0	272	92	Obsolete		
345	145342	Liquid	1.1000	19.6			Imported	3 Months	Drums/Bulk
346	145352	Liquid	1.0600	65.0	950	130	Local	1 Month	Drums/Bulk
347	145362	Liquid	1.0800	55.0	800		Imported	3 Months	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
348	145382	Liquid	0.9800	12.0	115	98	Local	1 Month	Drums/Bulk
349	145402	Liquid	1.2000	15.0	180	101	Local	1 Month	Drums/Bulk
350	145422	Liquid	1.0300	8.0	75	110	Imported	3 Months	Drums/Bulk
351	145452	Liquid	0.9700				Local	1 Month	Drums/Bulk
352	145462	Liquid	1.0050	5.8	29	90	Local	1 Month	Drums/Bulk
353	145482	Liquid	1.0370	7.0	63	100	Not in approved list		
354	145502	Liquid	1.1118	11.0	95	95	Imported	3 Months	Drums/Bulk
355	145542	Liquid	1.1080	8.5	79	100	Local	1 Month	Drums/Bulk
356	145542	Liquid	1.1300	9.0	88	100	Local	1 Month	Drums/Bulk
357	145562	Powder	POWDER				Local	1 Month	Bags
358	145572	Liquid	1.1770	3.5	22	120	Imported	3 Months	Drums/Bulk
359	145582	Liquid	1.1350	6.0	44	255	Local	1 Month	Drums/Bulk
360	145592	Liquid	1.1600	4.8	32	250	Imported	3 Months	Drums/Bulk
361	145612	Liquid	1.1550	5.1	42	255	Imported	3 Months	Drums/Bulk
362	145622	Liquid	1.1800	14.0	145	101	Local	1 Month	Drums/Bulk
363	145632	Liquid	1.1250	11.0	120	255	Imported	3 Months	Drums/Bulk
364	145642	Liquid	0.8750	6.7	49	150	Imported	3 Months	Drums/Bulk
365	145652	Liquid	1.0300	12.0	130	112	Imported	3 Months	Drums/Bulk
366	145662	Liquid	0.9420	11.8	127	120	Local	1 Month	Drums/Bulk
367	145692	Liquid	1.1100	15.5	160	125	Local	1 Month	Drums/Bulk
368	145822	Liquid	1.1200	24.0	229	180	Not in approved list		
369	145842	Powder	FLAKES				Not in approved list		
370	145852	Liquid	0.9518	9.0	80	90	Obsolete		
371	145862	Liquid	0.9850	8.0	55	90	Local	1 Month	Drums/Bulk
372	1458622	Liquid	1.0700	7.0	21	120	Local	1 Month	Drums/Bulk
373	1458632	Liquid	0.9500	8.8	65	180	Local	1 Month	Drums/Bulk
374	1458642	Liquid	0.9500	8.5	65	150	Local	1 Month	Drums/Bulk
375	145882	Liquid	1.0800	8.7	95	160	Local	1 Month	Drums/Bulk
376	145902	Liquid	0.9200	240.0	2390	180	Local	1 Month	Drums/Bulk
377	145942	Liquid	0.9700	13.0	135	160	Local	1 Month	Drums/Bulk
378	145962	Liquid	0.9000	6.0	46	150	Imported	3 Months	Drums/Bulk
379	145972	Liquid	1.4450	5.0	46	260	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
380	145982	Liquid	1.0300	4.0	39	120	Local	1 Month	Drums/Bulk
381	146022	Liquid	0.9500	16.0	158	158	Local	1 Month	Drums/Bulk
382	146032	Liquid	0.9700	46.0	388	80	Local	1 Month	Drums/Bulk
383	146042	Liquid	0.9135	28.0	310	135	Local	1 Month	Drums/Bulk
384	146082	Liquid	1.0800	55.0	488	150	Imported	3 Months	Drums/Bulk
385	146122	Liquid	0.9010	3.2	18	116	Local	1 Month	Drums/Bulk
386	150042	Powder	FLAKE				Local	1 Month	Bags
387	150102	Powder	POWDER				Local	1 Month	Bags
388	150112	Powder	POWDER				Local	1 Month	Bags
389	150122	Powder	POWDER				Local	1 Month	Bags
390	150142	Powder	POWDER				Local	1 Month	Bags
391	150162	Powder	POWDER				Local	1 Month	Bags
392	150182	Powder	POWDER				Local	1 Month	Bags
393	150202	Powder	POWDER				Not in approved list		
394	150212	Powder	POWDER				Local	1 Month	Bags
395	150222	Powder	FLAKE				Local	1 Month	Bags
396	150282	Solid	SOLID				Local	1 Month	Bags
397	150382	Powder	POWDER				Local	1 Month	Bags
398	150442	Powder	POWDER				Local	1 Month	Bags
399	150482	Powder	POWDER				Local	1 Month	Bags
400	150522	Powder	POWDER				Local	1 Month	Bags
401	150542	Powder	POWDER				Local	1 Month	Bags
402	150562	Powder	GRANULES				Local	1 Month	Bags
403	150582	Solid	SOLID				Local	1 Month	Bags
404	150602	Solid	SOLID				Local	1 Month	Bags
405	150622	Powder	POWDER				Local	1 Month	Bags
406	150642	Powder	POWDER				Local	1 Month	Bags
407	150662	Powder	POWDER				Local	1 Month	Bags
408	150682	Powder	POWDER				Local	1 Month	Bags
409	150702	Powder	SOLID				Local	1 Month	Bags
410	150722	Powder	POWDER				Local	1 Month	Bags
411	15072A2	Powder	POWDER				Local	1 Month	Bags
412	150742	Powder	SOLID				Local	1 Month	Bags
413	150762	Liquid	1.1300		44	120	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
414	150822	Liquid	1.1450		45	120	Local	1 Month	Drums/Bulk
415	150842	Liquid	1.1360		45	120	Local	1 Month	Drums/Bulk
416	155142	Liquid	1.1270		18	90	Local	1 Month	Drums/Bulk
417	160002	Solid	SOLID				Local	1 Month	Bags
418	160012	Liquid	0.8970	3900.0	38000	180	Imported	3 Months	Drums/Bulk
419	160022	Liquid	0.9390	750.0	6375	180	Obsolete		
420	160032	Liquid	0.8810	1175.0	12338	170	Local	1 Month	Drums/Bulk
421	1600312	Liquid	0.8810	1175.0	12338	170	Local	1 Month	Drums/Bulk
422	160042	Liquid	0.8930	850.0	8925	120	Local	1 Month	Drums/Bulk
423	160052	Solid	SOLID				Local	1 Month	Bags
424	160062	Liquid	0.8720	1300.0	11050	197	Local	1 Month	Drums/Bulk
425	1600612	Liquid	0.8720	1300.0	11050	197	Local	1 Month	Drums/Bulk
426	160072	Liquid	0.9350	550.0	4675	60	Local	1 Month	Drums/Bulk
427	160082	Liquid	0.9330	335.0	3465	100	Local	1 Month	Drums/Bulk
428	160092	Liquid	0.9350	425.0	4463	170	Local	1 Month	Drums/Bulk
429	160102	Liquid	0.9400	1500.0	15750	170	Imported	3 Months	Drums/Bulk
430	160112	Liquid	0.9400	650.0	6825	150	Imported	3 Months	Drums/Bulk
431	160122	Liquid	0.9400	650.0	6825	150	Imported	3 Months	Drums/Bulk
432	160142	Liquid	0.8650	650.0	6825	175	Not in approved list		
433	16014A2	Liquid	0.8800	622.5	6563	150	Local	1 Month	Drums/Bulk
434	16014B2	Liquid	0.8650	675.0	7088	170	Local	1 Month	Drums/Bulk
435	160152	Liquid	0.9450	1240.0	13020	120	Imported	3 Months	Drums/Bulk
436	160162	Liquid	0.8750	1350.0	14175	150	Not in approved list		
437	160172	Liquid	0.8600	1140.0	11970	155	Imported	3 Months	Drums/Bulk
438	160202	Liquid	0.8660	1165.0	12233	135	Obsolete		
439	160222	Liquid	0.8660	1240.0	13020	135	Not in approved list		
440	160232	Liquid	0.8650	1240.0	13020	135	Imported	3 Months	Drums/Bulk
441	160242	Liquid	0.9210	740.0	7770	150	Local	1 Month	Drums/Bulk
442	160252	Liquid	0.9200	1200.0	12600	150	Local	1 Month	Drums/Bulk
443	160272	Liquid	0.8900	800.0	8400	180	Not in approved list		
444	160282	Liquid	0.8720	1300.0	13650	197	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
445	160292	Liquid	0.8700	600.0	6300	70	Local	1 Month	Drums/Bulk
446	160302	Liquid	0.9050	825.0	8663	145	Local	1 Month	Drums/Bulk
447	160342	Liquid	0.9400	775.0	8138	114	Imported	3 Months	Drums/Bulk
448	160382	Liquid	0.8670	1125.0	11813	200	Local	1 Month	Drums/Bulk
449	160392	Liquid	0.9300	500.0	5250	110	Local	1 Month	Drums/Bulk
450	160402	Liquid	0.8550	2950.0	30975	170	Local	1 Month	Drums/Bulk
451	160442	Liquid	0.8650	1060.0	13000	170	Not in approved list		
452	160472	Liquid	0.8680	1150.0	12075	135	Imported	3 Months	Drums/Bulk
453	160502	Liquid	0.8700	1125.0	9713	180	Local	1 Month	Drums/Bulk
454	160662	Liquid	0.8700	1000.0	10500	210	Obsolete		
455	160702	Liquid	0.8700	6000.0	51000	170	Imported	3 Months	Drums/Bulk
456	160762	Liquid	0.9000	1800.0	18900	180	Local	1 Month	Drums/Bulk
457	160782	Liquid	0.9150	600.0	6300	60	Local	1 Month	Drums/Bulk
458	160882	Liquid	0.8720	2950.0	30975	170	Not in approved list		
459	160922	Liquid	0.8980	410.0	11700	165	Local	1 Month	Drums/Bulk
460	161022	Liquid	0.8810	1000.0	10500	170	Local	1 Month	Drums/Bulk
461	161042	Liquid	0.8800	10000.0	85000	170	Local	1 Month	Drums/Bulk
462	161052	Liquid	0.8900	2950.0	30975	120	Obsolete		
463	161062	Liquid	0.8600	1150.0	12075	170	Not in approved list		
464	161082	Liquid	0.8810	1175.0	12338	170	Local	1 Month	Drums/Bulk
465	162022	Liquid	0.8660	1135.0	11918	135	Local	1 Month	Drums/Bulk
466	162042	Liquid	0.8680	1300.0	13650	193	Local	1 Month	Drums/Bulk
467	165522	Liquid	0.9100	325.0	3413	140	Local	1 Month	Drums/Bulk
468	165542	Liquid	0.9100	325.0	3413	120	Local	1 Month	Drums/Bulk
469	165582	Liquid	0.9120	400.0	4200	164	Local	1 Month	Drums/Bulk
470	165652	Liquid	0.9240	5.0	42	46	Obsolete		
471	165662	Liquid	0.9200	637.5	6563	120	Imported	3 Months	Drums/Bulk
472	165762	Liquid	0.8900	100.0	600	120	Local	1 Month	Drums/Bulk
473	165842	Liquid	0.9000	350.0	3675	150	Not in approved list		
474	165872	Liquid	0.9100	375.0	3938	120	Local	1 Month	Drums/Bulk
475	165882	Liquid	0.8700	200.0	2100	170	Local	1 Month	Drums/Bulk
476	165892	Liquid	0.9150	360.0	3780	150	Imported	3 Months	Drums/Bulk
477	165902	Liquid	0.9030	120.0	1260	116	Imported	3 Months	Drums/Bulk
478	165912	Liquid	0.9000	176.0	1848	170	Not in approved list		
479	165922	Liquid	0.9100	375.0	3938	150	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
480	165982	Liquid	0.9100	300.0	3150	150	Local	1 Month	Drums/Bulk
481	170062	Liquid	0.8500	6.0	54	85	Imported	3 Months	Drums/Bulk
482	170082	Liquid	1.0100	4000.0	36488	130	Imported	3 Months	Drums/Bulk
483	170092	Liquid	0.8970	11.0	120	110	Local	1 Month	Drums/Bulk
484	170112	Liquid	0.8970	50.0	530	100	Local	1 Month	Drums/Bulk
485	170122	Liquid	0.8060	2.0	9	54	Local	1 Month	Drums/Bulk
486	170142	Liquid	0.9730	3500.0	36488	260	Local	1 Month	Drums/Bulk
487	170152	Liquid	0.8000		4	96	Imported	3 Months	Drums/Bulk
488	170162	Liquid	0.8200		4	92	Not in approved list		
489	170282	Liquid	1.0320	110.0	1000	135	Local	1 Month	Drums/Bulk
490	170292	Liquid	1.0000	1.0	8	90	Local	1 Month	Drums/Bulk
491	170322	Liquid	0.9900	25.5	77	130	Not in approved list		
492	170502	Liquid	0.8500	200.0	2100	145	Imported	3 Months	Drums/Bulk
493	180022	Liquid	0.9800	45.0	500	180	Imported	3 Months	Drums/Bulk
494	180182	Liquid	0.9600	16.0	165	148	Not in approved list		
495	180222	Liquid	0.9850	33.0	347	150	Local	1 Month	Drums/Bulk
496	18022A2	Liquid	1.0000	23.0	242	150	Local	1 Month	Drums/Bulk
497	18022B2	Liquid	0.9600	22.5	242	150	Local	1 Month	Drums/Bulk
498	18022C2	Liquid	0.9850	120.0	1100	160	Local	1 Month	Drums/Bulk
499	1802242	Liquid	0.9850	35.0	380	150	Local	1 Month	Drums/Bulk
500	180262	Liquid	0.9400	6-11	88	140	Local	1 Month	Drums/Bulk
501	180272	Liquid	0.9650	130.0	1350	78	Local	1 Month	Drums/Bulk
502	180382	Liquid	1.0150	54.0	560	160	Local	1 Month	Drums/Bulk
503	18038A2	Liquid	1.0100	54.0	560	150	Local	1 Month	Drums/Bulk
504	180402	Liquid	1.1400	8.5	75	130	Local	1 Month	Drums/Bulk
505	180482	Liquid	0.9700	350.0	3600	165	Local	1 Month	Drums/Bulk
506	180502	Liquid	1.0200	150.0	1600	190	Local	1 Month	Drums/Bulk
507	180522	Liquid	1.0150	150.0	1600	190	Local	1 Month	Drums/Bulk
508	180742	Liquid	0.9920	80.0	870	120	Local	1 Month	Drums/Bulk
509	180752	Liquid	0.9920	85.0	865	120	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
510	180762	Liquid	0.9750	37.0	380	166	Not in approved list		
511	180862	Liquid	0.9810	80.0	850	160	Local	1 Month	Drums/Bulk
512	180882	Liquid	1.0250	60.0	620	150	Local	1 Month	Drums/Bulk
513	180892	Liquid	1.0114	105.0	1100	190	Local	1 Month	Drums/Bulk
514	181182	Liquid	1.0400	3.5	28	210	Imported	3 Months	Drums/Bulk
515	181192	Liquid	0.9850	28.0	300	150	Local	1 Month	Drums/Bulk
516	181202	Liquid	0.9920	23.0	225	150	Local	1 Month	Drums/Bulk
517	181212	Liquid	1.0240	30.0	290	150	Local	1 Month	Drums/Bulk
518	181222	Liquid	1.0270	23.0	245	150	Local	1 Month	Drums/Bulk
519	181232	Liquid	1.0000	23.0	225	150	Obsolete		
520	181262	Liquid	1.0050	33.0	320	150	Local	1 Month	Drums/Bulk
521	181282	Liquid	1.0450	38.0	400	150	Not in approved list		
522	181292	Liquid	0.9800	55.0	540	150	Local	1 Month	Drums/Bulk
523	181362	Semi solid	waxy solid			150	Not in approved list		
524	181442	Powder	powder				Local	1 Month	Bags
525	181552	Liquid	0.9200	8.0	60	100	Local	1 Month	Drums/Bulk
526	181562	Liquid	0.8700	6.0	58	100	Local	1 Month	Drums/Bulk
527	18156A2	Liquid	0.9060	6.0	58	100	Local	1 Month	Drums/Bulk
528	181572	Liquid	0.9410	6.0	58	100	Local	1 Month	Drums/Bulk
529	181602	Liquid	1.0000	36.0	350	150	Not in approved list		
530	181622	Liquid	0.9450	35.0	380	100	Imported	3 Months	Drums/Bulk
531	181642	Powder	powder				Local	1 Month	Bags
532	181662	Liquid	1.0000	41.0	390	110	Local	1 Month	Drums/Bulk
533	181682	Liquid	0.9670	80.0	795	150	Local	1 Month	Drums/Bulk
534	181712	Liquid	0.9000	35.0	370	150	Local	1 Month	Drums/Bulk
535	181742	Liquid	0.9600	3.8	20	120	Not in approved list		
536	181762	Liquid	1.0200	5.0	24	62	Local	1 Month	Drums/Bulk
537	181782	Liquid	1.0200	200.0	2100		Local	1 Month	Drums/Bulk
538	181802	Liquid	0.9600	120.0	1300	170	Local	1 Month	Drums/Bulk
539	181822	Liquid	0.9455	140.0	1500		Local	1 Month	Drums/Bulk
540	181842	Liquid	soild				Imported	3 Months	Drums/Bulk
541	181862	Liquid	solid				Local	1 Month	Drums/Bulk
542	185282	Liquid	0.8900	3.0	15	85	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
543	185302	Powder	powder				Local	1 Month	Bags
544	185342	Powder	powder				Local	1 Month	Bags
545	185382	Powder	powder				Local	1 Month	Bags
546	1853812	Powder	powder				Local	1 Month	Bags
547	185402	Liquid	0.9950		3	100	Local	1 Month	Drums/Bulk
548	185602	Liquid	0.9200		5	67	Local	1 Month	Drums/Bulk
549	185942	Liquid	0.9600		5	28 (Abel)	Local	1 Month	Drums/Bulk
550	185982	Liquid	0.9650		5	28 (Abel)	Local	1 Month	Drums/Bulk
551	186042	Liquid	0.9700		5		Local	1 Month	Drums/Bulk
552	1860412	Liquid	0.9700		5	28 (Abel)	Local	1 Month	Drums/Bulk
553	186062	Powder	powder				Local	1 Month	Bags
554	186082	Powder	powder				Local	1 Month	Bags
555	187042	Liquid	0.9650	40.0	4800	250	Not in approved list		
556	555522	Liquid	0.8200	4.0	17	204	Imported	3 Months	Drums/Bulk
557	555542	Liquid	0.8600		4	120	Local	1 Month	Drums/Bulk
558	555682	Liquid	0.8001	1.9	6	145	Imported	3 Months	Drums/Bulk
559	555702	Liquid	0.8270	5.9	31	225	Imported	3 Months	Drums/Bulk
560	555722	Liquid	0.8320	7.9	47	232	Imported	3 Months	Drums/Bulk
561	555762	Liquid	0.8360	9.8	66	250	Imported	3 Months	Drums/Bulk
562	555782	Liquid	0.8450	40.0	405	266	Imported	3 Months	Drums/Bulk
563	555802	Liquid	0.8500	100.0	1275	270	Imported	3 Months	Drums/Bulk
564	555922	Liquid	0.8750	8.0	65	220	Local	1 Month	Drums/Bulk
565	560182	Liquid	0.8950	630.0	6615	220	Local	1 Month	Drums/Bulk
566	560322	Liquid	0.8900	4700.0	49350	250	Imported	3 Months	Drums/Bulk
567	560362	Liquid	0.9020	4350.0	45675	245	Local	1 Month	Drums/Bulk
568	560582	Liquid	0.8950	255.0	2678	180	Local	1 Month	Drums/Bulk
569	560602	Liquid	0.9200	19.0	180	>55	Local	1 Month	Drums/Bulk
570	560642	Liquid	0.9420	480.0	5040	225	Local	1 Month	Drums/Bulk
571	585042	Liquid	0.8150		4	133	Not in approved list		
572	585062	Liquid	0.8150		6	150	Local	1 Month	Drums/Bulk
573	585242	Liquid	0.9800	2500.0	26000	230	Local	1 Month	Drums/Bulk
574	585362	Liquid	0.9800	3000.0	32000	250	Not in approved list		
575	585662	Liquid	0.8800	5.0	30	175	Local	1 Month	Drums/Bulk

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
576	585902	Liquid	0.8400		5	66	Local	1 Month	Drums/Bulk
577	585962	Liquid	0.8100		5	96	Local	1 Month	Drums/Bulk
578	585982	Liquid	0.8700	3.0	16	175	Local	1 Month	Drums/Bulk
579	586002	Liquid	0.8220	2.2	8	150	Imported	3 Months	Drums/Bulk
580	635522	Liquid	0.9100	9.8	50	150	Imported	3 Months	Drums/Bulk
581	586062	Liquid	0.9830	5.0	32	150	Imported	3 Months	Drums/Bulk
582	635522	Liquid	0.9100	6.0	50	280	Imported	3 Months	Drums/Bulk
583	635582	Liquid	0.9430	11.0	85	200	Not in approved list		
584	635622	Liquid	0.9200	8.7	41	150	Not in approved list		
585	635642	Liquid	0.9180	5.0	45	180	Local	1 Month	Drums/Bulk
586	650262	Liquid	1.0000		1		Local	1 Month	Drums/Bulk
587	655242	Liquid	1.1300	7.1	75	150	Local	1 Month	Drums/Bulk
588	655322	Liquid	1.1150		3	140	Local	1 Month	Drums/Bulk
589	655382	Liquid	1.1400	4.1	44	150	Local	1 Month	Drums/Bulk
590	655462	Liquid	1.1250			280	Not in approved list		
591	655482	Liquid	1.0650	172.5	1050	230	Local	1 Month	Drums/Bulk
592	116982	Liquid	0.969	140		135			
593	121322	Liquid	1.1	8.4		76			
594	140522	Liquid	0.980			140			
595	145442	Liquid	1.06			130			
596	185942	Liquid	0.98			28			
597	401502	Liquid		30.1		290			
598	600312	Liquid	0.881	1175		170			
599	655502	Liquid	0.99	1.025		90			
600	890652	Liquid		11		230			
601	1165882	Liquid				170			
602	1560582	Liquid	0.895	255		180			
603	1585962	Liquid	0.81		2.35	96			
604	1650262	Liquid							
605	1655822	Thin Fluid				215			
606	1656042	Liquid			110				
607	121142	Liquid	1.025			150			
608	130392	Flakes	0.957						
609	145432	Fine Powder							

PHYSICAL PROPERTIES FOR ADDITIVES

S.No.	STOCK NO.	Type	Sp.Gravity @ 15.6 C	KV@ 100 °C	KV@ 40° C	Flash Point PMCC/COC	Local/ Imported	Inventory Norm	ISO tanks/ Flexibags / IBCs/ DRUMS
610	145561	Fine Powder							
611	150502	Liquid	1.20±0.02			130			
612	155162	Liquid							
613	155182	Crystalline solid							
614	155202	Liquid	1.221						
615	155222	Liquid to paste							
616	155282	Liquid							
617	155422	Liquid	0.79						
618	155602	Liquid							
619	155902	Liquid				61			
620	180142	Liquid	0.96 ±0.02						
621	181082								
622	181422	Solid							
623	181702	Liquid	0.90±0.02			150			
624	185922	Dark green powder							
625	490621	Liquid		11.5		232			
626	490622	Liquid		11.5		230			
627	810902	Liquid	0.926	18.5		240			
628	1458622	Liquid	1.07		20.61	120			
629	1657062	Liquid	0.925		150	235			

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ELECTRICAL DESIGN CONSIDERATIONS – FOR PACKAGE EQUIPMENT & SYSTEM

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

		Total No. of Sheets			Page	
		10 (Incl, cover sheet)			Page 1 of 10	
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TITLE:
ELECTRICAL DESIGN CONSIDERATIONS – FOR PACKAGE EQUIPMENT & SYSTEM

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO. HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)	DOCUMENT NO.	SIZE	REV.
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PREAMBLE

Scope of work

The purpose of this document is to provide guidelines to be used by EPC Contractor for the design of electrical services required for Package equipment & Skids that will be supplied by EPC Contractors.
The requirements of this document are to be considered as minimum requirements and Contractor may submit to the CLIENT / PMC, for agreement of any possible solutions, with supporting calculation notes that could give technical and / or economic improvement.
Any omission in this requirement shall not relieve the Contractor / Subcontractor / Vendor of his responsibility to deliver the equipment along with other associated items, which are complete, of proven design, and conforms to the Performance Requirements.
Broad concept of services:

The services for the project have been conceptualized based on past experience and acceptable International design standards. All electrical services shall be designed for ease of maintenance with keeping in view functionalities, aesthetic & provision for accommodating changes in requirement in future. Conservation of energy, optimization of resources, eco-friendliness and State of the art technology shall be the key factors in the design concept to ensure least downtime and reduce maintenance hassles.
Every effort shall be made to design, layout and install equipment in locations which will tend to encourage routine preventive maintenance by providing easy access for operation personnel. Manual isolation will be provided to enable servicing, expansion or renovation of any part of the system without interrupting the services in adjacent areas.

ELECTRICAL SYSTEM

1. Reference Documents

The latest revision of following project specification / documents shall referred.

1. Electrical design basis. Document No.: 80-0RG2N-DB-001.
2. Electrical Equipment Specifications. Document No.: 80-0RG2N-IS-001.

The electrical systems designed and equipment installed shall be in accordance with the latest edition (including addenda / errata) of applicable codes and standards listed below.

- IS-1180 : Specifications for Outdoor/Indoor Type Oil Immersed Distribution Transformers
- IS-1646 : Code of practice for the fire safety of buildings - Electrical Installations
- IS-1554 : PVC Insulated cables for working voltages up to 1100 V
- IS-2026 : Specifications for Power Transformers / Dry Type Transformers
- IS-2309 : Code of practice for the protection of buildings and allied structures against lightning
- IS-3043 : Code of practice for earthing.
- IS-3646 : Code of practice for interior illumination.
- IS-5571 : Code of selection of electrical equipment for hazardous areas
- IS 7752 : Guide for improvement of power factor in consumer
- IS 7098 : XLPE Insulated thermoplastic sheathed cables – specification
- IS-8130 : Specification for conductors for insulated electric cables and flexible cords
- IS-10028 : Code of practice for selection, installation and maintenance of transformer
- IS-10118 : Code of practice for selection, installation and maintenance of switchgear and control gear.
- OISD-RP-110 : Recommended practices on static electricity
- OISD-RP-147 : Inspection and Safe practices during electrical installations
- OISD-RP-149 : Design aspects for safety in electrical systems
- OISD-STD-173 : Fire Prevention & Protection System for Electrical Installations.
- OISD-STD-180 : Lightning Protection
- OISD-234 : Oil Industries Safety Directorate Regulations
- CEA : Central Electricity Authority Regulations
- CPCB : Central Pollution Control Board of India & Chennai Pollution Control Board
- NBC : National Building Code of India – 2016
- ECBC : Energy Conservation Building Codes 2007
- NEC : National Electric Code of India
- IEC : International Electric Code

Rules, Regulations and Directives of the Statutory Bodies listed below shall be adhered to while designing the electrical system:

- Local By – laws
- Indian Electricity Rules
- CEA : Central Electricity Authority Regulations
- CPCB : Central Pollution Control Board of India & Chennai Pollution Control Board
- State Electrical Inspectorate

Note:
The reference shall be made to the relevant codes and standards as applicable though it is not listed above.
2. General Design Requirements

2.1 Electrical System Design Philosophy

Electrical system shall be designed to ensure:

- Safety to personnel, plant and equipment during operation and maintenance
- Reliability of service
- Flexibility of operation
- Minimum fire risk
- Automatic protection of all electrical equipment through selective relaying system
- Ease of operation & maintenance
- Fail safe feature
- Provision for future extension and modification
- Interchange ability of equipment
- Load management and energy monitoring systems

2.2 General Safety

The electrical system shall incorporate safety margins to ensure that the installation is safe under all operating conditions, including those associated with the start-up and shut down of equipment and through intervening shut down periods. The emphasis in equipment specification shall be on operability, prevention of accident / fault and functionality for the intended design life. All insulating material specified for the equipment shall be non-toxic.

2.3 Fire Integrity

All cable penetrations through switchgear room walls, control room walls and between safe and hazardous area shall be sealed by waterproof, weatherproof & fireproof sealing methods. Cable penetrations if any, between a safe area and a classified hazardous area, shall be sealed using multi cable transits (MCT) to maintain fire integrity and prevent gas migration.

2.4 Degree of Ingress Protection for Enclosures

Degree of protection against ingress of dust and water, necessary for electrical equipment, is determined by the equipment duty, environment and location. All equipment shall be suitable for installation under site conditions specified in the equipment specifications and shall be protected against dust, humidity and designed to prevent the entry of vermin and insects.
Equipment located outdoors or those subject to deluge from fire water system / rain shall be specified as outdoor weatherproof construction and protected against the most adverse conditions anticipated. Such enclosures shall be specified to have a minimum IP-55 degree of protection, with the degree increased as necessary where the location / situation demands.
Degree of ingress protection for enclosures shall be defined according to applicable standards.

2.6 System Design Standards

Voltage Levels

- Power & Motor Control Centre : 415 V ± 5%, 50 Hz ± 3%, 3 Phase, 4 Wire
- Lighting & Small Power Distribution : 240 V ± 5%, 50 Hz ± 3%, 1 Phase, 3 Wire
- UPS system (for control & protection circuits, PLC, instrumentation, critical lighting) : 240 V ± 1%, 50 Hz ± 3%, 1 Phase, 3 Wire

Allowable Voltage Drops

The maximum allowable voltage drop in any feeder under steady state condition at full load is to be maintained as follows:

- From Transformer to Motor : 5%

2.7 Electrical Equipment Protection

Protection system for electrical equipment shall be selected and coordinated for the protection of equipment and operating personnel against internal / external faults, overloading of equipment, earth faults, and other abnormal conditions such as lightning surges, switching surges etc., and to ensure continuity of operation of those parts not affected by the fault.
System protective devices (relays, fuses, breaker trip units, etc.) shall be selected and coordinated to ensure that the closest interrupter to the point of short circuit (or high overload) shall open first and minimize disturbances on the rest of the system. The protection and metering schemes shall be selected in accordance with guidelines provided by OISD-RP-149 'Design Aspects for Safety in Electrical systems'.

3. Sources of Power Supply

Equipment power shall be provided from nearby IMCC panel as per requirements. Project has dedicated HT power source from TNEB as well as emergency back power DG sets. DG set power shall be provided for emergency process loads only.

4. LT Power Distribution Scheme

LT Power from the Intelligent Motor Control Centres (IMCC) shall be transferred to various equipments through LT armoured cables on cable tray mounted on pipe racks. Single point power source for each equipment skid shall be provided by electrical contractor. Further distribution of power from particular equipment control panel as well as incoming cable termination shall be done by mechanical contractor in considering this document reference.
Sub main protection shall be by circuit breakers, which will discriminate with upstream protective devices. All switch boards switching of incoming & outgoing circuits up to 630 amps shall be moulded case circuit breakers and above 630 amps shall be air circuit breakers. EC grade Aluminium bus bar shall be provided for all power & control panels.
Intelligent type motor control centres are proposed for entire operation of plant. This IMCC will have a intelligent controller in each starter which will control starters with soft signal. Similarly all equipment control panels are also should be provided with intelligent controller which will control starters with soft signal. These entire controllers, control panels as well as IMCC will be inter-connected with one defined protocol.
All intelligent relays & Variable Frequency Drive for motor management functions shall includes Protection, Monitoring, Control and Diagnostics functions. These devices monitoring shall support for maintenance management system & shall have the following features:

a) Construction
   i. The intelligent motor management & control device shall be modular in construction. The device shall have integrated communication port for direct link to higher level, with open protocol bus system like Profibus without any routers or gateways in between.
   ii. The protection system should be independent /autonomous of the automation system. i.e. motor protection & control should continue to be available even in the event of a communication or automation system failure.

b) The device should have a control voltage of 240V AC/24 V DC.

c) The basic unit shall have the following minimum status display:
   i. Device Readiness.
   ii. Status of control supply
   iii. Feeder fault indication
   iv. Status of communication with PLC

d) The device shall have facility to test/reset as standard. It shall be possible to select between manual reset (at the device) or remote reset (through PLC/SCADA from control room).

e) The basic device shall have following minimum no. of Digital Inputs & Outputs as standard which should be freely assignable.
   i. 4 Digital Inputs
   ii. 3 Digital outputs (Potential free)

Optionally, it shall be possible to expand the no. of digital inputs / outputs if required by using add-on modules. The system of basic device with add-on modules should provide maximum 12 Inputs and 7 Outputs.

f) The IMCC fuseless feeder shall have Type 2 coordinated feeder components and tested for short circuit rating of 50kA.

g) Functional requirements

i. Protection Functions

1. Overload protection with trip class 5E,10E,20E,30E & 40E.
2. It shall be possible to adjust the reset time after an overload.
3. Provision for connecting thermistor inputs.
4. Phase failure, Phase imbalance & Phase sequence protection.
5. Stall protection with adjustable time delay (Rotor Blocking).
6. Over current protection with adjustable time delay.
7. Under current protection with adjustable time delay.
8. Internal earth fault protection with adjustable time delay.

ii. Monitoring Functions

1. Motor Operating Hours monitoring.
2. Motor Stop time monitoring.
3. No. of motor starts monitoring.
4. No. of overload trips
5. Comments stored in the device
6. Device operating hours

iii. Diagnostics data like early warning signals, fault messages, Device-internal faults logs (minimum 20 logs) & value of last tripping currents.

iv. Operating data

1. Motor switching state (ON,OFF, Direction of rotation, Right, Slow fast etc)
2. Current in phase 1, 2, 3 and maximum current
3. Phase unbalance in %
4. Time remaining to trip in seconds
5. Motor heating model
6. Remaining motor cooling time of motor

Optionally where power monitoring is specified the following additional data should be available:

1. Consumed energy in kWh
2. Active Power in Watts
3. Apparent power in VA
4. Phase voltages in V
5. Power factor
6. Phase sequence

v. When threshold value of above settings exceed, it shall be possible to set the type of response (Warning, Tripping, Signaling and Disable).

vi. Additionally, the following monitoring functions should be made available as optional features:

1. Under voltage monitoring
2. Power factor monitoring
3. Active Power Monitoring
4. Phase sequence monitoring
5. Temperature monitoring via PT100 / PT 1000, NTC, KTY 83/84 inputs
6. Recording the curves of measured values like current, voltage, power etc.
7. Monitoring of analog values using Analog I/O modules
8. External earth fault monitoring with adjustable time delay

h) Control Functions - In order to minimize the wiring & interlocks, the device shall have built-in software logics to achieve the various control functions like Direct on line starter (DOL) & Automatic star delta starter.

i) Logic Function Modules - Protection Device shall offer the following in- built logic functions:

i. Truth table
ii. Timer
iii. Counter
iv. Flashing function
v. Flickering function

j) Standard function blocks - Protection Device shall have standard control function control blocks as follows:

i. Check back signal
ii. Test Position feedback
iii. External fault
iv. Emergency start
v. Communication Error
vi. Testing the device including tripping of the feeder
vii. Testing the device without tripping the feeder

k) Configuring & Communication

i. It shall be possible to configure a device either individually or over a communication network. For this purpose, provision for connecting a laptop should be available at the motor management device.
ii. The device shall have integrated communication facility to communicate directly on a bus system without any additional components.
iii. The device shall be able to transmit cyclic as well as acyclic data to higher- level automation. The data defined as cyclic shall be fetched by master PLC in a sequential manner. i.e. one slave after other. The critical events like tripping of device or alarm shall be sent to master PLC on priority basis (acyclic transfer) irrespective of the slave sequence. In order to achieve minimum response time, the device shall be capable of transmitting data at a speed of 12Mbps, suitable for operation with high-speed bus systems.

iv. In case DCS and IMCC are of different make then Basic Device should support Master Class 2 in order to access all the devices from single location. This Master Class 2 should enable parameterization and data archiving of basic device.

5. System Earthing

External earthing system & earthing grid shall be provided by electrical contractor in accordance with CEA Regulations, IE Rules, IS: 3043/ BS 7430 & OISD Standard 149 for earthing system. Dedicated maintenance free chemical type earthing pits shall be provided for critical equipment earthing. Interconnected Earthing pits shall be provided for body earthing of equipment etc.
Number of earth connections and size of earth conductor connections for each type / rating of equipment is defined in 'Typical Earthing Scheme Drawings'. Vendor shall provide earth connections of all equipments as per standards provided. All earth connections shall be visible for inspection.

6. Cables & Cable Installation

Electrical power, lighting, control and signal cables shall comply with relevant standards and requirements specified in project specification. Cables shall specifically comply with the standards listed in clause 1 of this design basis.

MV cables shall be copper / aluminium conductor, multi-strand, extruded XLPE insulated, extruded PVC inner sheathed, armoured, overall extruded PVC outer sheathed. The PVC outer jacket shall be black in colour, flame retardant low smoke (FRLS), heat & oil resistant and UV resistant.

MV cable size for power / lighting purposes shall be minimum 2.5 sq mm. & cable size for control / signal purposes shall be minimum 1.5 sq mm. Voltage grade for power & control cables shall be 600 / 1100 V. MV cable for power / lighting shall be copper conductor till 16 Sqmm size & beyond 16 Sqmm i.e. 25 Sqmm & above shall be aluminium conductor.

Maximum cross section area shall be 300 sq mm for multi core cables. All cables shall be of single continuous lengths, as far as possible. Power cables shall be colour coded for core identification. For control cables, core shall be identified by numbers printed on them. Control cables shall have 20% spare cores.

Cables shall be sized based on maximum continuous load current, voltage drop, system short-circuit withstand capability and laying conditions. Derating factor due to ambient air temperature, ground temperature, grouping and proximity of cables with each other, thermal resistivity of soil, etc. shall be taken into account. Derating factor shall be 0.7 or less.

Special cables e.g. twisted pair or shielded control cables etc. shall be used as applicable for numerical relays, VFDs etc. to suit selected equipment as per equipment supplier's recommendations. Cables connected in parallel shall be of the same type, cross-section and terminations.

All power and control cables shall be in continuous lengths without any splices or intermediate joints. Cables used for lighting and wires in conduits shall have appropriate junction boxes with adequately sized terminals. Cable joints in hazardous areas shall not be permitted.

Cables shall be installed on well supported cable trays as per requirements. Cable tray and accessories shall be made of Aluminium.

Minimum distances as per relevant codes shall be maintained between power and control circuits, between electrical and instrument system cables over prolonged route lengths.

7. Socket Outlets

Small equipments which may require local power socket which also shall be powered from particular equipment skid control panel. Such socket shall have ELCB + MCB & each receptacle circuit shall be protected by MCB. Not more than three socket outlets shall be supplied by one circuit.

Convenience socket outlets shall have necessary mechanical interlocks and earthing facilities. The enclosure shall have suitable protection for site conditions specified (flame proof, weather proof, dust proof, corrosion resistant, etc.).

8. AC Uninterruptable Power Supply (UPS) System

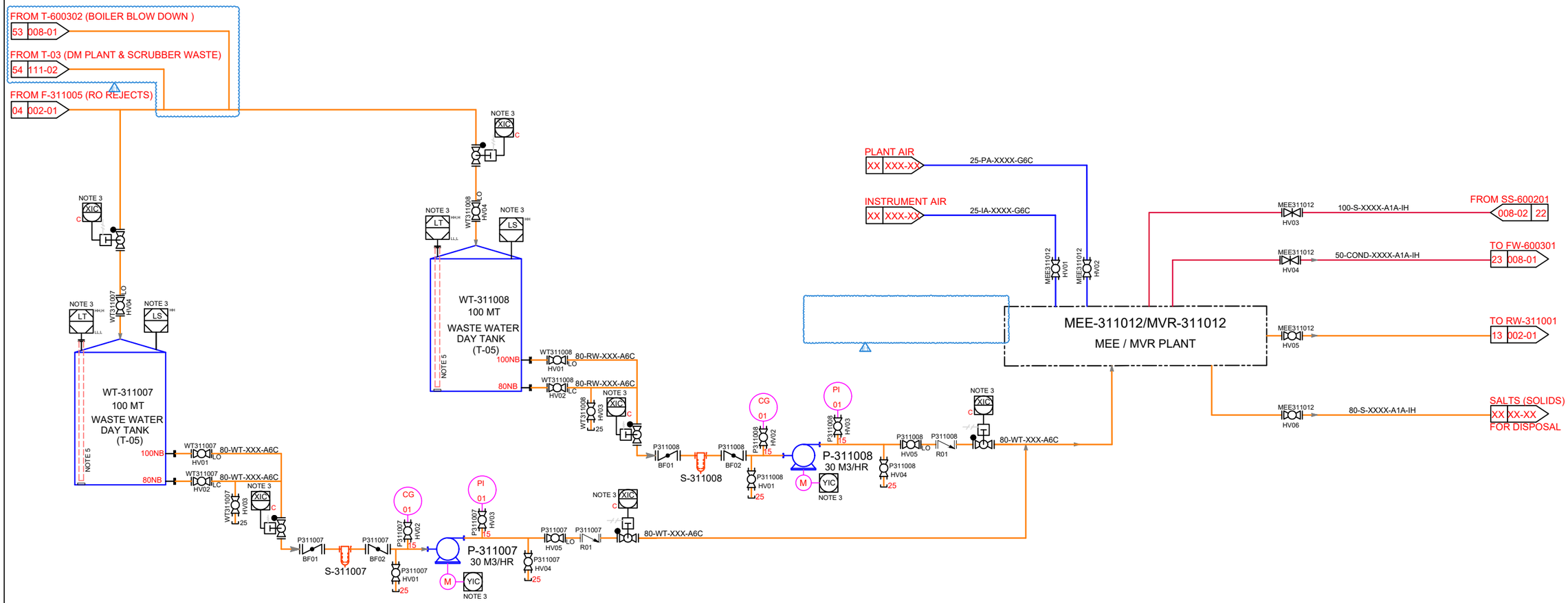
AC Uninterruptible Power Supply system for individual equipment skid shall be supplied by vendor only. This comprises of converter, inverter, static automatic transfer switch, battery bank, constant voltage by-pass transformer with maintenance bypass switch and feeder to ACUPS Distribution Board. ACUPS shall be parallel redundant type with bypass.

The system output shall be 240 V ± 1 %, 50 Hz ± 1 %, for single phase & 415 V ± 1 %, 50 Hz ± 1 %, for three phase. Battery back-up time shall be 20 minutes unless otherwise specified.

Under normal working conditions, the converter will feed the inverter load requirement through static switch and keep the battery under charged condition. On failure of supply to converter, the battery will start feeding the inverter load requirements automatically without interruption. Reverse transfer on resumption of normal power will also be automatic. The inverter will be operating in synchronism with the bypass mains within specified voltage and frequency limits. UPS shall be fed from individual equipment control panel.

RECOMMENDED TYPE & MAKE OF INSTRUMENTS & ERECTION HARDWARES

Sr no.	Item	Type	Recommended make
1	Level transmitter	Non contact type radar/ contact type radar	Endress and hauser, Emerson, Vega
2	Level transmitter, Differential	Differential pressure type	Endress and hauser, Emerson, Vega
3	Level Switch-	Vibrating tunning fork type	Endress and hauser, Emerson, Vega, forbes marshall.
4	RTD element with Temprature indicator transmitter	RTD elenment (Pt-100) with temprature transmitter	Invensis, Endress and hauser, Emerson, ABB, Wika, honeywell
5	Temprature Gauge-	Bimetalic type	Baumer, General instruments, Waree, wika. Radix
6	Pressure transmitter	CAPACITATIVE / PEIZORESISTIVE TYPE with Diphragm seal	Invensis, Endress and hauser, Emerson, ABB, Wika, honeywell
7	Differential Pressure transmitter - Diaphragm	CAPACITATIVE / PEIZORESISTIVE TYPE with Diphragm seal	Invensis, Endress and hauser, Emerson, ABB, Wika, honeywell
8	Pressure Gauge	Bourdon tube type	Baumer, General instruments, Waree, wika. Radix
9	Compound gauge - Diaphragm type f	Bourdon tube type	Baumer, General instruments, Waree, wika. Radix
10	ON-OFF Valve	(Solenoid Operated Single / Double acting actuator)	IPC valves, Micropneumatics, virgo, L & T, Rotex, Microfinish.
11	Flush Bottom ball valve	Solenoid operated songle/ double acting actuator	IPC valves, Micropneumatics, virgo, L & T, Rotex, Microfinish.
13	Motor operated ON-OFF valve ()	Electrical motor operated actuator	IPC valves, Micropneumatics, virgo, L & T, Rotex, Microfinish.
14	Pressure Control Valve-	Modulating type	Pneucon valves, Forbes marshall, Dembla valves, Uniflow flow valves.
15	Temperature Control Valve -	Modulating type	Pneucon valves, Forbes marshall, Dembla valves, Uniflow flow valves.
16	Flow Control valve	Modulating type	Pneucon valves, Forbes marshall, Dembla valves, Uniflow flow valves.
17	Mass flow Meter	Coriolis type	Emerson, Endress and hauser, Krone, Forbes marshall
18	Self Regulating Pressure control valve	Pilot operated	Nirmal industries, samson, forbes marshall, Laser india.
19	Weight Indicator Transmitter (WIT)	Strain gauge type.	Precia molen, Mettler toledo.
20	Flow switch	Paddle type	Kobold, danfoce, Emerson, Endress and hauser.
21	Pressure Safety/Relief valve-Mechnical spring type	Spring loaded	Nirmal industries, samson, forbes marshall, Leser india.
22	Pressure Safety valve	Spring loaded	Nirmal industries, samson, forbes marshall, Leser india.
23	Thermal Safety/Relief valve-Mechnical pre loaded spring type	Spring loaded	Nirmal industries, samson, forbes marshall, Leser india.
24	Proximity sensor	Inductive type	P & F, Turck, emerson.
	Erection Hardware		
1	Instrumentation cables		KEI, Thermocable, Udey, Polycab, Pyrocable
2	Tubing for valves		Wesmech, ChampAstec
3	Cable trays		Indiana, Profab, Reliance
4	Control Panels		Rittal make,Pyrotech, Hoffman
5	PLC		Siemenc, GE , ABB, Honeywell
6	Desktop computers		panasonic,Sony, Toshiba,LG, Hitachi.
7	HMI		P & F, Honeywell, Siemens



NOTE :-

1. SUPPLY SCOPE OF VENDOR.
2. UFD FOR WATER TREATMENT PLANT PROPOSED IS ONLY REPRESENTATIVE & SHOWS MINIMUM REQUIREMENTS FOR INTEGRATION WITH MAIN PLANT CONTROL SYSTEM. THIS WILL BE REPLACED BY VENDORS P&ID WHICH WILL BE HAZOP COMPATIBLE – AT THE END OF AFC DESIGN FREEZE.
3. SUPPLY BY CPE VENDER & INTEGRATED INTO MAIN PLANT CONTROL SYSTEM.
4. ESD – TO BE INTEGRATED WITH THE MAIN PLANT CONTROL SYSTEM THROUGH NEARBY RIO PANEL.
5. STILLING WELL SLOTS OR HOLES AS PER REQUIREMENT OF INSTRUMENT SUPPLIER.

EQUIPMENT TAG NO.	DESCRIPTION	CAPACITY	DIMENSION	MOC	PRESSURE OPR / DESIGN	TEMP. OPR / DESIGN	PUMP TEMP.(MIN/NOR/MAX)	PUMP PRE.(MIN/NOR/MAX)	INSULATION (Y/N)	TRACING (ST/ET)
WT–311007/08	WATER DAY TANK	100 MT								
P–311007/08	PUMP FOR WATER DAY TANK	50 M3/HR								

REV	DATE	DESCRIPTION	DRAW'N	CHK'D	APPR'D
E1	19.08.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
E	05.06.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
D	17.12.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
C	13.10.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
B	15.05.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
A	22.01.20	FOR APPROVAL	N.B.M.	A.P.G	D.R.

CLIENT: INDIAN OIL CORPORATION LIMITED
MUMBAI–400051
INDIA

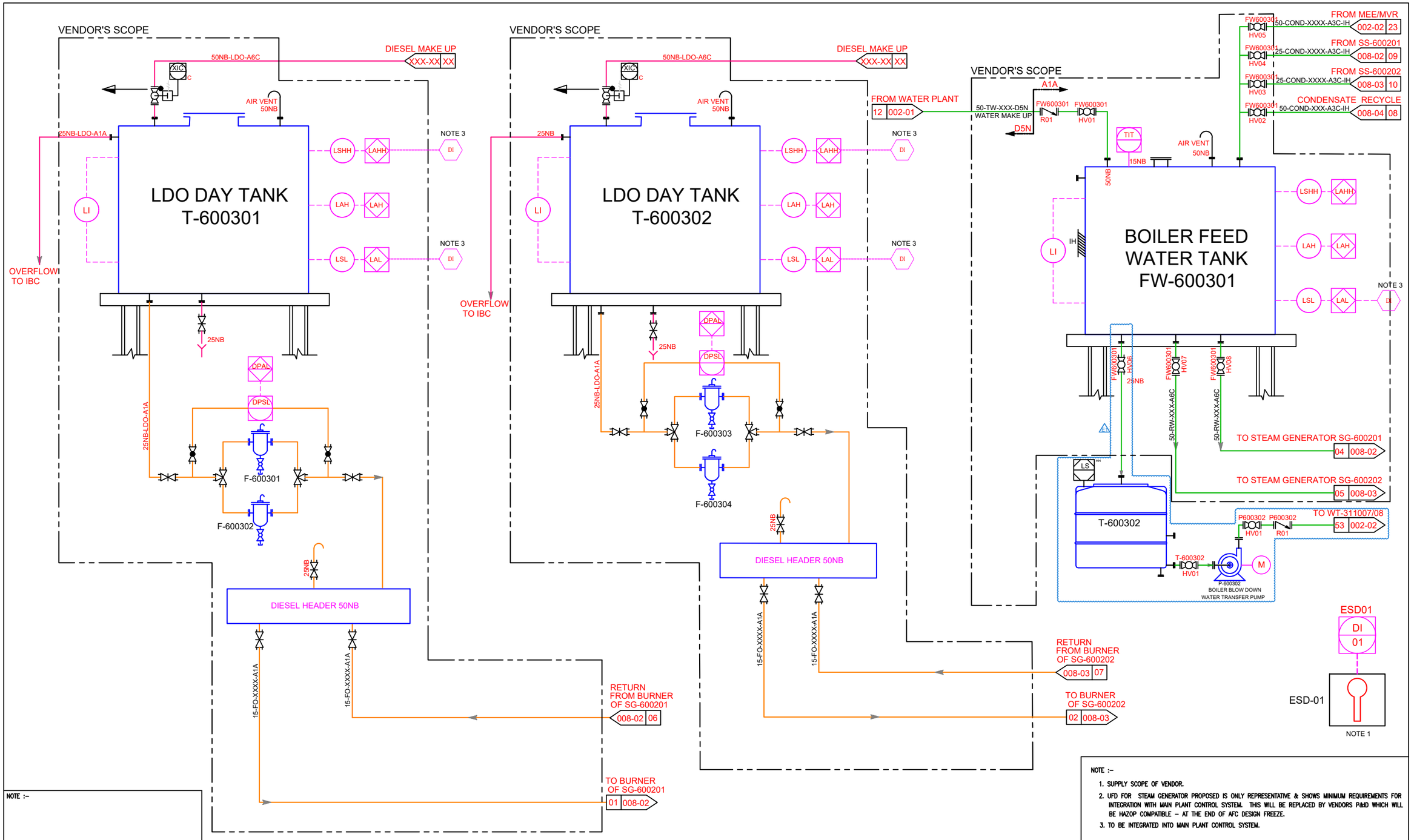
CONSULTANT: PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT: GRASSROOT LUBE COMPLEX

TITLE: UFD FOR TREATED WATER

DWG NO. 90–ORG2N–UFD–002–02 REV E1

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NOTE :-

NOTE :-

1. SUPPLY SCOPE OF VENDOR.
2. UFD FOR STEAM GENERATOR PROPOSED IS ONLY REPRESENTATIVE & SHOWS MINIMUM REQUIREMENTS FOR INTEGRATION WITH MAIN PLANT CONTROL SYSTEM. THIS WILL BE REPLACED BY VENDORS P&ID WHICH WILL BE HAZOP COMPATIBLE – AT THE END OF AFC DESIGN FREEZE.
3. TO BE INTEGRATED INTO MAIN PLANT CONTROL SYSTEM.

EQUIPMENT TAG NO.	DESCRIPTION	CAPACITY	DIMENSION	MOC	PRESSURE OPR / DESIGN	TEMP. OPR / DESIGN	PUMP TEMP.(MIN/NOR/MAX)	PUMP PRE.(MIN/NOR/MAX)	INSULATION (Y/N)	TRACING (ST/ET)
T-600301/02	LDO DAY TANK	990 L								
FW-600302	BOILER FEED WATER TANK	1 KL								

REV	DATE	DESCRIPTION	DRAW'N	CHK'D	APPR'D
E1	19.08.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
E	05.06.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
D	18.12.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
C	13.10.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
B	15.05.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
A	14.03.20	FOR APPROVAL	N.B.M.	S.B.S.	D.R.

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MUMBAI–400051
INDIA

CONSULTANT: PEC LTD.
21 Shipyard Road,
Singapore 628144

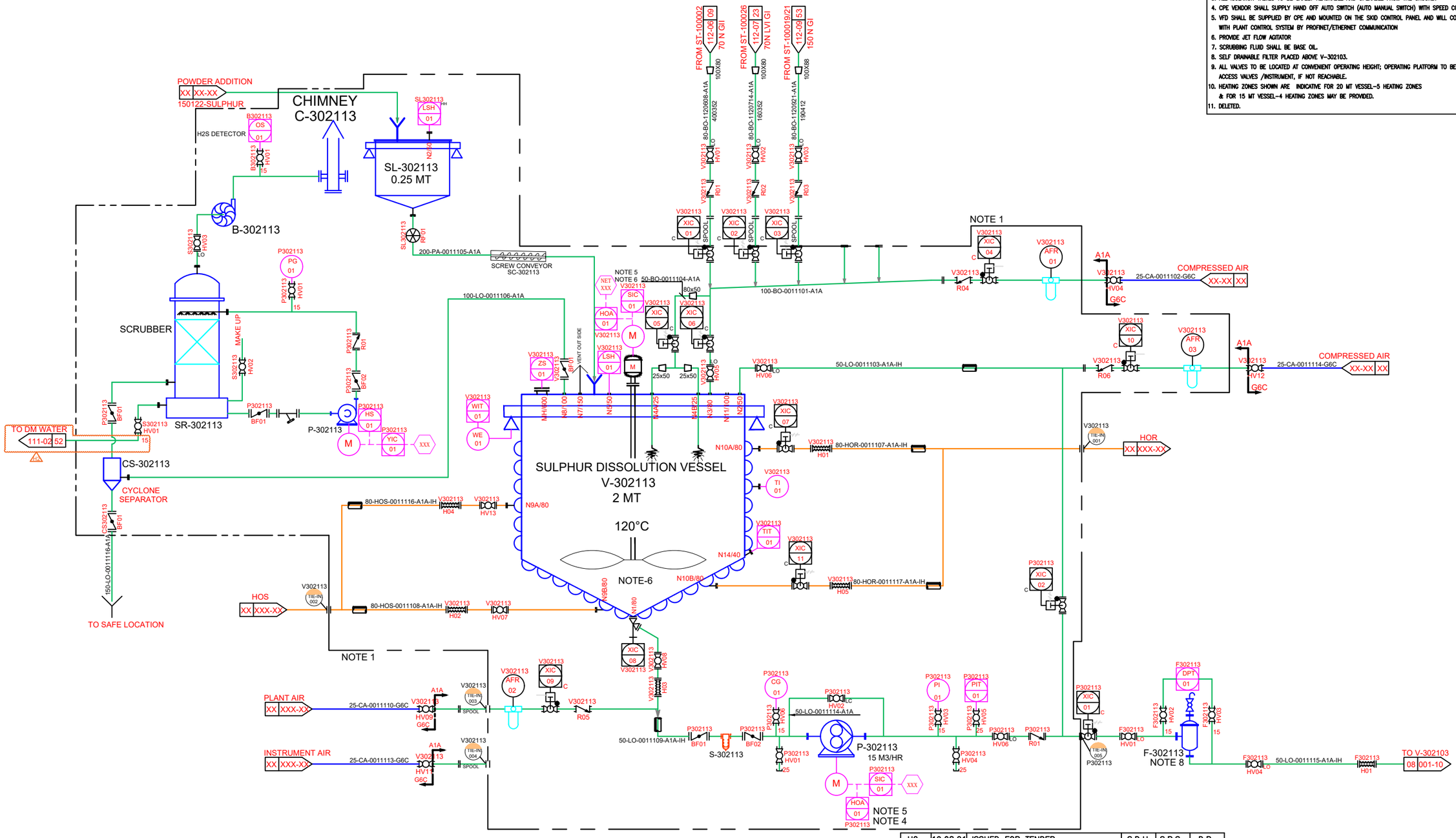
PROJECT: GRASSROOT LUBE COMPLEX

TITLE: UFD FOR STEAM GENERATION

DWG NO. 90–ORG2N–UFD–008–01 REV E1

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NOTE :-
1. SUPPLY SCOPE OF CPE VENDOR (SKID).
2. P&ID FOR V-302113 WILL BE PROVIDED BY VENDOR AND SHALL BE INSERTED INTO THIS DRAWING AT THE END OF THE DESIGN FREEZE.
3. ALL ISOLATION VALVES TO BE SAFELY REACHABLE AND OPERABLE FROM THE GROUND.
4. CPE VENDOR SHALL SUPPLY HAND OFF AUTO SWITCH (AUTO MANUAL SWITCH) WITH SPEED CONTROLLER.
5. VFD SHALL BE SUPPLIED BY CPE AND MOUNTED ON THE SKID CONTROL PANEL AND WILL COMMUNICATE WITH PLANT CONTROL SYSTEM BY PROFINET/ETHERNET COMMUNICATION
6. PROVIDE JET FLOW AGITATOR
7. SCRUBBING FLUID SHALL BE BASE OIL.
8. SELF DRAINABLE FILTER PLACED ABOVE V-302103.
9. ALL VALVES TO BE LOCATED AT CONVENIENT OPERATING HEIGHT; OPERATING PLATFORM TO BE PROVIDED TO ACCESS VALVES /INSTRUMENT, IF NOT REACHABLE.
10. HEATING ZONES SHOWN ARE INDICATIVE FOR 20 MT VESSEL-5 HEATING ZONES & FOR 15 MT VESSEL-4 HEATING ZONES MAY BE PROVIDED.
11. DELETED.



EQUIPMENT TAG NO.	DESCRIPTION	CAPACITY	DIMENSION (M)	MOC	PRESSURE OPR / DESIGN	TEMP. OPR / DESIGN	PUMP TEMP.(MIN/NOR/MAX)	PUMP PRE.(MIN/NOR/MAX)	INSULATION (Y/N)	TRACING (ST/ET)
V-302113	SULPHUR DISSOLVER VESSEL	2 MT	VTA	SS	ATM	90/120	NA	NA	YES	NO
P-302113	PUMP FOR SULPHUR DISSOLVER VESSEL	15 M3/HR	VTA	CI/CS	TBD	90/120	60/120/170	TBD	NO	NO
S-302113	BASKET TYPE STRAINER FOR P-302113	15 M3/HR	VTA	CS	TBD	90/120	NA	TBD	NO	NO
SR-302113	SCRUBBER	VTA	VTA	VTA	VTA	VTA	VTA	VTA	VTA	VTA
F-302113	PRODUCT BAG FILTER	15 M3/HR	VTA	SS	TBD	90/120	NA	TBD	NO	NO

REV	DATE	DESCRIPTION	DRAW'N	CHK'D	APPR'D
H2	19.08.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
H1	07.07.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
H	07.03.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
G	04.11.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
F	18.09.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
E	20.07.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
D	11.06.20	REVISED AS PER COMENTS	S.D.U	S.B.S.	D.R.
C	06.04.20	REVISED AS PER COMENTS	S.D.U	S.B.S.	D.R.
B	02.03.20	REVISED AS PER COMENTS	S.D.U	S.B.S.	D.R.
A	07.0120	FOR APPROVAL	S.D.U	S.B.S.	D.R.

CLIENT:
INDIAN OIL CORPORATION LIMITED
MUMBAI-400051
INDIA

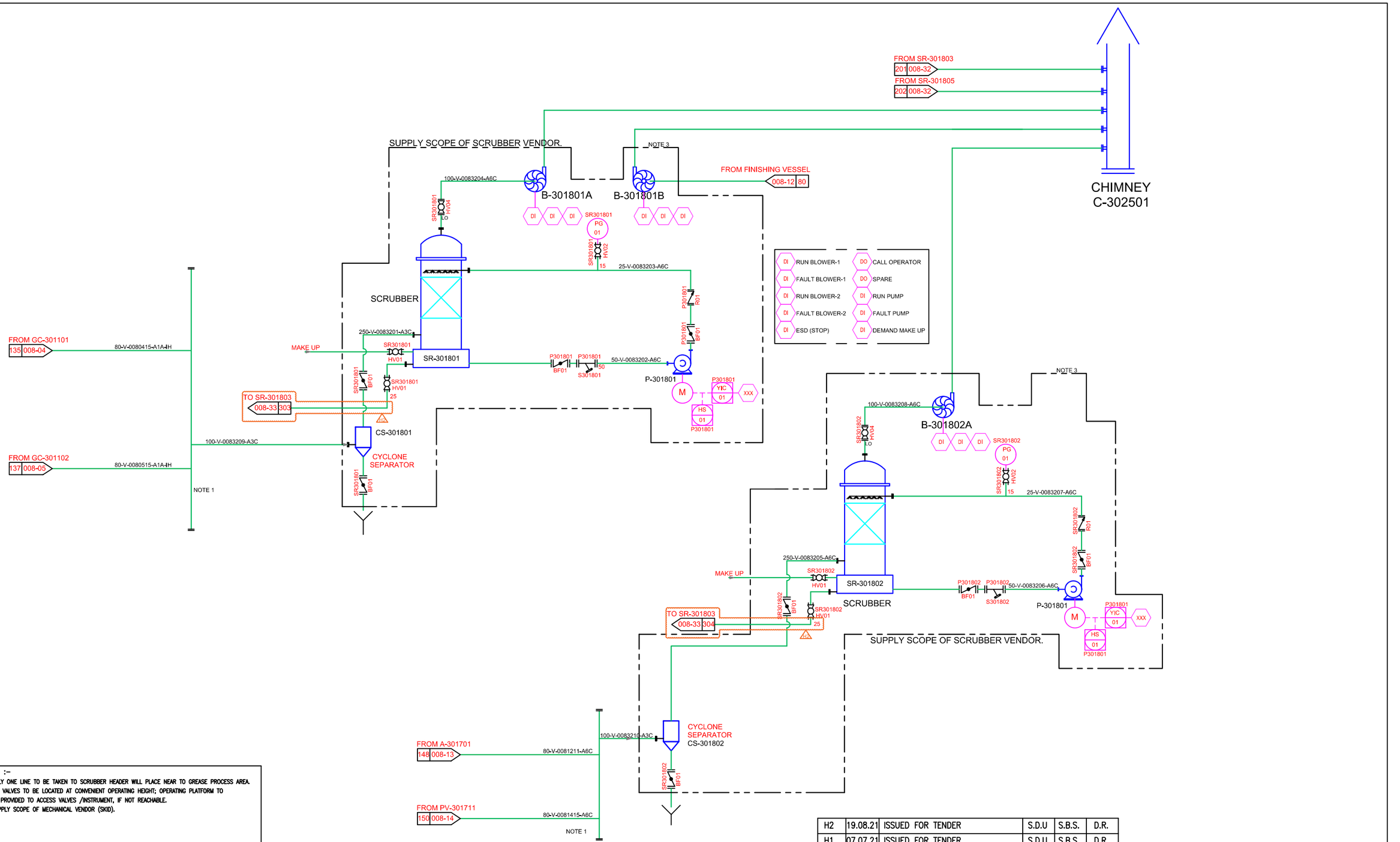
CONSULTANT:
PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT: GRASSROOT LUBE COMPLEX

TITLE: P&ID FOR SULPHUR DISSOLUTION VESSEL V-302113

DWG NO. 90-ORG2N-PID-001-11 REV H2

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NOTE :-

1. ONLY ONE LINE TO BE TAKEN TO SCRUBBER HEADER WILL PLACE NEAR TO GREASE PROCESS AREA.
2. ALL VALVES TO BE LOCATED AT CONVENIENT OPERATING HEIGHT; OPERATING PLATFORM TO BE PROVIDED TO ACCESS VALVES /INSTRUMENT, IF NOT REACHABLE.
3. SUPPLY SCOPE OF MECHANICAL VENDOR (SKID).

EQUIPMENT TAG NO.	DESCRIPTION	CAPACITY	DIMENSION	MOC	PRESSURE OPR / DESIGN	TEMP. OPR / DESIGN	PUMP TEMP.(MIN/NOR/MAX)	PUMP PRE.(MIN/NOR/MAX)	INSULATION (Y/N)	TRACING (ST/ET)
SR-301801	SCRUBBER FOR LI GREASE	6000 NM^3/HR	VTA	CS/STD	VTA	200°/260°	-	-	-	-
SR-301802	SCRUBBER FOR PU GREASE	5000 NM^3/HR	VTA	CS/STD	VTA	200°/260°	-	-	-	-

REV	DATE	DESCRIPTION	DRAW'N	CHK'D	APPR'D
H2	19.08.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
H1	07.07.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
H	07.03.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
G	04.11.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
F	18.09.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
E	20.07.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
D	11.06.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
C	06.04.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
B	02.03.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
A	07.01.20	FOR APPROVAL	S.D.U	S.B.S.	D.R.

CLIENT: INDIAN OIL CORPORATION LIMITED
MUMBAI-400051
INDIA

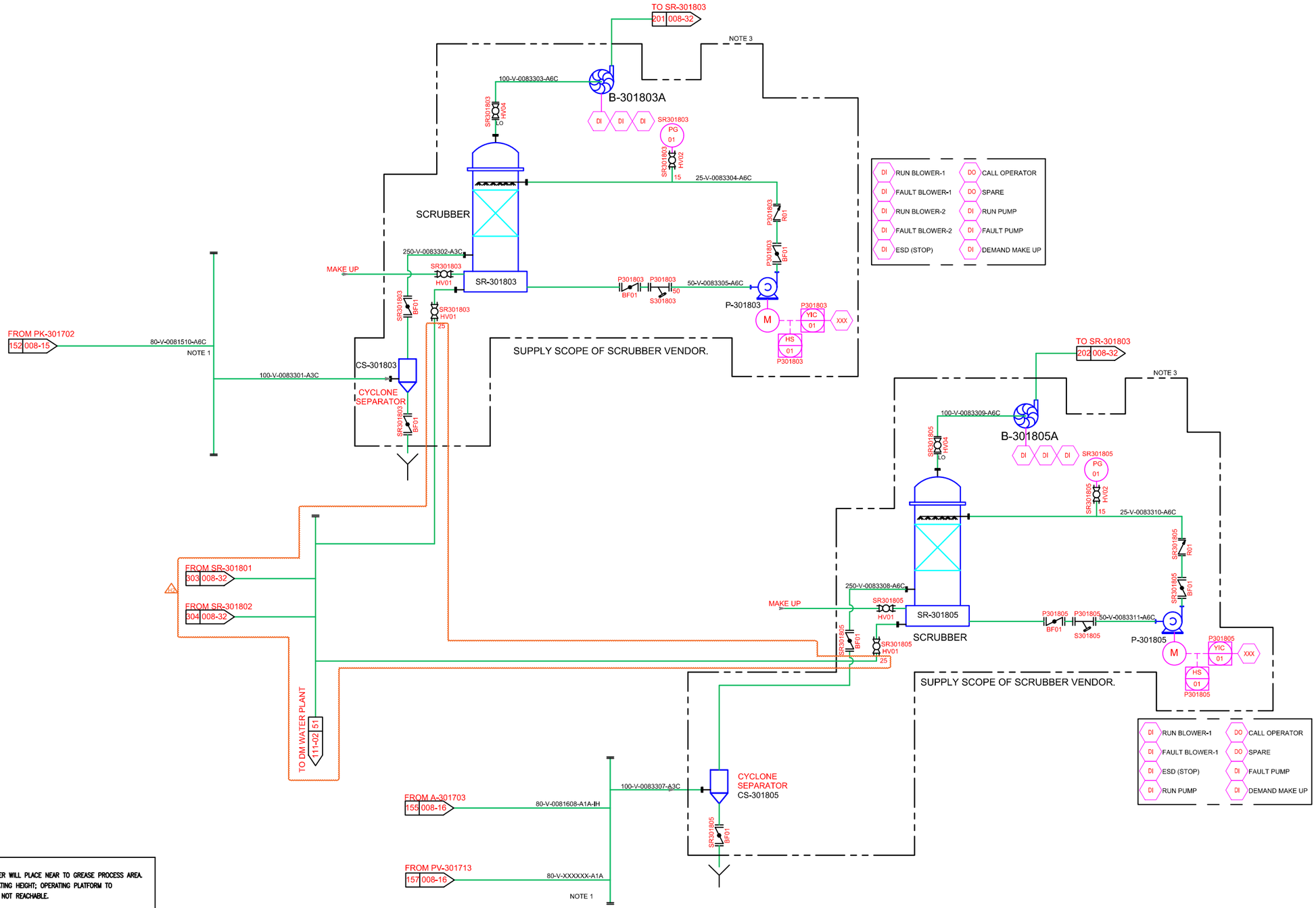
CONSULTANT: PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT: GRASSROOT LUBE COMPLEX

TITLE: P&ID FOR GREASE PROCESS

DWG NO. 90-ORG2N-PID-008-32 REV H2

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1. ONLY ONE LINE TO BE TAKEN TO SCRUBBER HEADER WILL PLACE NEAR TO GREASE PROCESS AREA.
2. ALL VALVES TO BE LOCATED AT CONVENIENT OPERATING HEIGHT; OPERATING PLATFORM TO BE PROVIDED TO ACCESS VALVES /INSTRUMENT, IF NOT REACHABLE.
3. SUPPLY SCOPE OF MECHANICAL VENDOR (SKID).

EQUIPMENT TAG NO.	DESCRIPTION	CAPACITY	DIMENSION	MOC	PRESSURE OPR / DESIGN	TEMP. OPR / DESIGN	PUMP TEMP.(MIN/NOR/MAX)	PUMP PRE.(MIN/NOR/MAX)	INSULATION (Y/N)	TRACING (ST/ET)
SR-301803	SCRUBBER FOR CA-S GREASE	3000 NM^3/HR	VTA	CS/STD	VTA	200°/260°	-	-	-	-
SR-301805	SCRUBBER FOR FG GREASE	1000 NM^3/HR	VTA	CS/STD	VTA	200°/260°	-	-	-	-

REV	DATE	DESCRIPTION	DRAW'N	CHK'D	APPR'D
H2	19.08.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
H1	07.07.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
H	07.03.21	ISSUED FOR TENDER	S.D.U	S.B.S.	D.R.
G	04.11.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
F	18.09.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
E	20.07.20	REVISED AS PER REVIEW COMMENTS	S.D.U	S.B.S.	D.R.
D	11.06.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
C	06.04.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
B	02.03.20	REVISED AS PER COMMENTS	S.D.U	S.B.S.	D.R.
A	07.01.20	FOR APPROVAL	S.D.U	S.B.S.	D.R.

CLIENT: INDIAN OIL CORPORATION LIMITED
MUMBAI-400051
INDIA

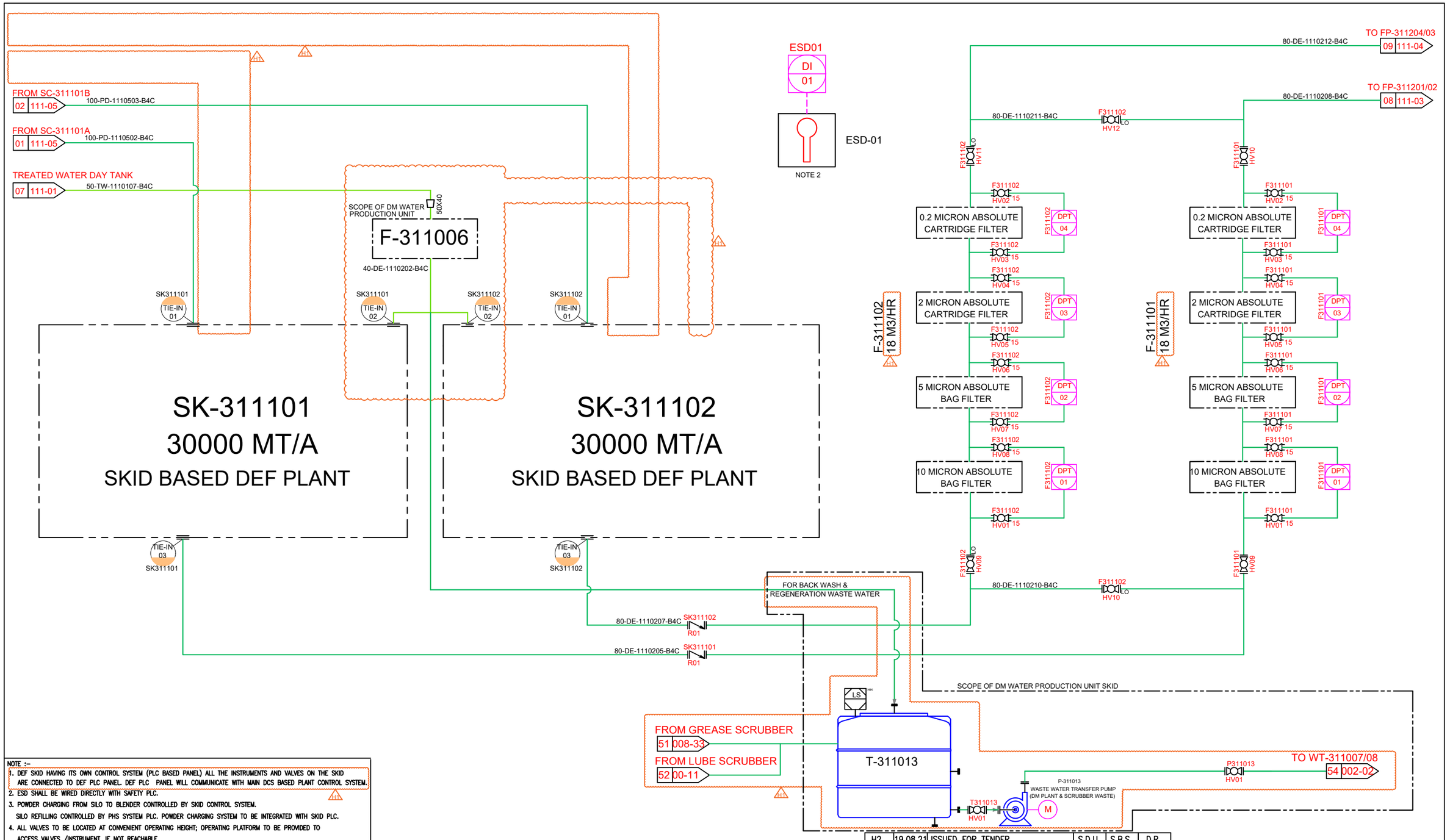
CONSULTANT: PEC LTD.
21 Shipyard Road,
Singapore 628144

PROJECT: GRASSROOT LUBE COMPLEX

TITLE: P&ID FOR GREASE PROCESS

DWG NO. 90-ORG2N-PID-008-33 REV H2

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NOTE :-

1. DEF SKID HAVING ITS OWN CONTROL SYSTEM (PLC BASED PANEL) ALL THE INSTRUMENTS AND VALVES ON THE SKID ARE CONNECTED TO DEF PLC PANEL. DEF PLC PANEL WILL COMMUNICATE WITH MAIN DCS BASED PLANT CONTROL SYSTEM.
2. ESD SHALL BE WIRED DIRECTLY WITH SAFETY PLC.
3. POWDER CHARGING FROM SILO TO BLENDER CONTROLLED BY SKID CONTROL SYSTEM. SILO REFILLING CONTROLLED BY PHS SYSTEM PLC. POWDER CHARGING SYSTEM TO BE INTEGRATED WITH SKID PLC.
4. ALL VALVES TO BE LOCATED AT CONVENIENT OPERATING HEIGHT; OPERATING PLATFORM TO BE PROVIDED TO ACCESS VALVES /INSTRUMENT, IF NOT REACHABLE.
5. SCOPE OF MECHANICAL VENDOR.

EQUIPMENT TAG NO.	DESCRIPTION	CAPACITY	DIMENSION (M)	MOC	PRESSURE OPR / DESIGN	TEMP. OPR / DESIGN	PUMP TEMP.(MIN/NOR/MAX)	PUMP PRE.(MIN/NOR/MAX)	INSULATION (Y/N)	TRACING (ST/ET)
SK-311101/02	SKID BASED DEF PLANT	30000 MT/A	VTA	SS	TBD	50/120	NA	TBD	NO	NO
F-311101/02	FILTRATION SYSTEM	18 M3/HR	VTC	–	TBD	50/120	NA	NA	NO	NO
T-311013	SUMP TANK	VTA	VTA	–	TBD	–	–	–	–	–
P-311013	PUMP FOR WASTE WATER	VTA	VTA	–	TBD	–	–	–	–	–

REV	DATE	DESCRIPTION	DRAW'N	CHK'D	APPR'D
H2	19.08.21	ISSUED FOR TENDER	S.D.U.	S.B.S.	D.R.
H1	07.07.21	ISSUED FOR TENDER	S.D.U.	S.B.S.	D.R.
H	07.03.21	ISSUED FOR TENDER	S.D.U.	S.B.S.	D.R.
G	04.11.20	REVISED AS PER REVIEW COMMENTS	S.D.U.	S.B.S.	D.R.
F	18.09.20	REVISED AS PER REVIEW COMMENTS	S.D.U.	S.B.S.	D.R.
E	20.07.20	REVISED AS PER REVIEW COMMENTS	S.D.U.	S.B.S.	D.R.
D	11.06.20	REVISED AS PER COMMENTS	S.D.U.	S.B.S.	D.R.
C	06.04.20	REVISED AS PER COMMENTS	S.D.U.	S.B.S.	D.R.
B	02.03.20	REVISED AS PER COMMENTS	S.D.U.	S.B.S.	D.R.
A	26.08.19	FOR APPROVAL	S.D.U	S.B.S.	D.R.

CLIENT: INDIAN OIL CORPORATION LIMITED, MUMBAI–400051, INDIA

CONSULTANT: PEC LTD., 21 Shipyard Road, Singapore 628144

PROJECT: GRASSROOT LUBE COMPLEX

TITLE: P&ID FOR DIESEL EXHAUST PLANT

DWG NO. 90–ORG2N–PID–111–02 REV H2

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